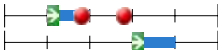
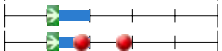
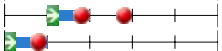
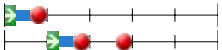
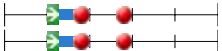
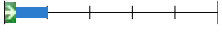
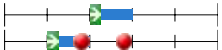
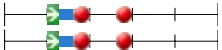
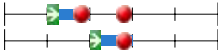
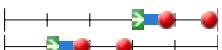
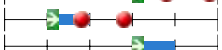
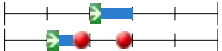
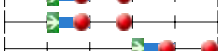
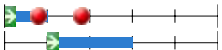
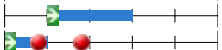
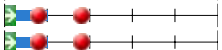
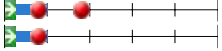
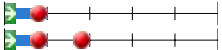
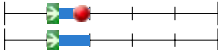
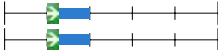
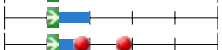
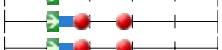
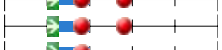
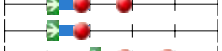
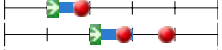
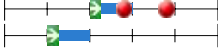
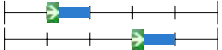
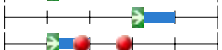


Program overview

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Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master EE 2023							
Master Electrical Engineering 2023							
Track Electrical Power Engineering 2023							
Main core EPE 2023							
CESE4010	Advanced Computing Systems	5					
EE4C01	Profile Orientation and Academic Skills	3					
EE4C03	Statistical Digital Signal Processing and Modeling	5					
EE4C04	Control System Design	5					
EE4C05	Electromagnetics	5					
EE4C06	Networking	5					
EE4C08	Measurement and Instrumentation	5					
EE4C10	Analog Circuit Design Fundamentals	5					
EE4C11	Systems Engineering	5					
EE4C12	Machine Learning for Electrical Engineering	5					
EE4C13	Wireless Systems for Electrical Engineering Applications	5					
Track core EPE 2023							
EE4585	Semiconductor Device Physics	5					
ET4103	High Voltage Technology	4					
ET4108	Transients in Power Systems	4					
ET4117	Electrical Machines and Drives	4					
ET4119	Electronic Power Conversion	4					
ET4376	Photovoltaic Basics	4					
Suggested specialization courses EPE 2023							
EE4109	Structured Electronic Design	5					
EE4111	High-Voltage DC	4					
EE4114	Power System Protection and Control	4					
EE4375	Finite Element Modeling for Electrical Energy Applications	4					
EE4410	Cyber Security of Power Grids	4					
EE4415	Photovoltaics	4					
EE4515	Advanced Power Electronics	4					
EE4536	DC and AC Microgrids	4					
EE4545	Electrical Power Systems of the Future	4					
EE4625	High Voltage Cable System	3					
EE4655	Co-simulation of Energy Systems	4					
EE4665	Uncertainty modelling and risk assessment in electrical power systems	4					
EE4680	Photovoltaic Modelling	4					
EE4755	Reliable Power Electronic Components and Systems	5					
ET4107	Power Systems Analysis II	4					
ET4113	Power System Dynamics	4					
ET4116	Power Electronics	4					
ET4121	A.C. Machines	4					
ET4291	Control of Electrical Drives	5					
ET4377	Photovoltaic Technologies	4					
ET4378	Photovoltaic Systems	4					
ET4379	Photovoltaic Lab Course	4					
ET8020	High Voltage Testing and Diagnostics	4					
SET3100	Electric vehicle and charging technology	4					
Suggested specialization courses EPE, Other departments 2023							
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AE4W09	Wind Turbine Design	5					
EE4530	Applied Convex Optimization	5					
ET4277	Microelectronics Reliability	4					
SET3065	Intelligent Electrical Power Grids	4					
Suggested profiles EPE 2023							
High Voltage 2023							
EE4111	High-Voltage DC	4					
EE4114	Power System Protection and Control	4					
EE4375	Finite Element Modeling for Electrical Energy Applications	4					
EE4545	Electrical Power Systems of the Future	4					
EE4585	Semiconductor Device Physics	5					
EE4625	High Voltage Cable System	3					
ET4103	High Voltage Technology	4					
ET4108	Transients in Power Systems	4					

ET4116	Power Electronics	4	
ET8020	High Voltage Testing and Diagnostics	4	
Solar Energy 2023			
EE4415	Photovoltaics	4	
EE4536	DC and AC Microgrids	4	
EE4545	Electrical Power Systems of the Future	4	
EE4585	Semiconductor Device Physics	5	
EE4670	PV Materials Processing and Characterization	4	
EE4680	Photovoltaic Modelling	4	
ET4119	Electronic Power Conversion	4	
ET4376	Photovoltaic Basics	4	
ET4377	Photovoltaic Technologies	4	
ET4378	Photovoltaic Systems	4	
ET4379	Photovoltaic Lab Course	4	
Power Electronics and Electrical Machines 2023			
EE4375	Finite Element Modeling for Electrical Energy Applications	4	
EE4515	Advanced Power Electronics	4	
EE4585	Semiconductor Device Physics	5	
EE4755	Reliable Power Electronic Components and Systems	5	
ET4116	Power Electronics	4	
ET4117	Electrical Machines and Drives	4	
ET4119	Electronic Power Conversion	4	
ET4121	A.C. Machines	4	
ET4291	Control of Electrical Drives	5	
SET3100	Electric vehicle and charging technology	4	
Intelligent Electric Power Systems 2023			
EE4410	Cyber Security of Power Grids	4	
EE4530	Applied Convex Optimization	5	
EE4536	DC and AC Microgrids	4	
EE4545	Electrical Power Systems of the Future	4	
EE4655	Co-simulation of Energy Systems	4	
EE4665	Uncertainty modelling and risk assessment in electrical power systems	4	
EE4C12	Machine Learning for Electrical Engineering	5	
ET4107	Power Systems Analysis II	4	
ET4108	Transients in Power Systems	4	
ET4113	Power System Dynamics	4	
ET4116	Power Electronics	4	
ET4117	Electrical Machines and Drives	4	
ET4119	Electronic Power Conversion	4	
ET4378	Photovoltaic Systems	4	
Free Elective Space EPE 2022			
Thesis project 2023			
ET4300	Master Thesis	45	
Track Microelectronics 2023			
Main core ME 2023			
CESE4010	Advanced Computing Systems	5	
EE4C01	Profile Orientation and Academic Skills	3	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
EE4C04	Control System Design	5	
EE4C05	Electromagnetics	5	
EE4C06	Networking	5	
EE4C08	Measurement and Instrumentation	5	
EE4C10	Analog Circuit Design Fundamentals	5	
EE4C11	Systems Engineering	5	
EE4C12	Machine Learning for Electrical Engineering	5	
EE4C13	Wireless Systems for Electrical Engineering Applications	5	
Track core ME 2023			
CESE4085	Modern Computer Architectures	5	
EE4520	Analog CMOS design I	3	
EE4585	Semiconductor Device Physics	5	
EE4605	Integrated Circuits for RF/Wireless Applications	5	
EE4610	Digital IC Design	3	
ET4257	Sensors and Actuators	4	
ET4289	Integrated Circuits and MEMS Technology	4	
ET4362	High Speed Digital Design for Embedded Systems	5	
Suggested specialisation courses ME 2023			
BM41055	Anatomy and Physiology	4	
CESE4035	Computer Arithmetic	5	
CESE4040	Processor Design Project	5	
EE4016	Antenna Systems	5	
EE4109	Structured Electronic Design	5	
EE4525	Analog CMOS design II	3	
EE4530	Applied Convex Optimization	5	
EE4585	Semiconductor Device Physics	5	
EE4595	Wavefield Imaging	5	

EE4615	Digital IC Design II	3
EE4650	Advanced Magnetic Resonance Imaging	5
EE4690	Hardware Architectures for Artificial Intelligence	5
EE4705	Solid State Physics	3
EE4710	Solid State Physics with Quantum and Nano Electronics	5
EE4736	Introduction Imaging Sensors	4
ET4127	Themes in Biomedical Electronics	4
ET4173	Introduction to UWB technology, systems and applications	4
ET4252	Analog Integrated Circuit Design	4
ET4260	Microsystem Integration	4
ET4277	Microelectronics Reliability	4
ET4278	Over-Sampled Data Converters	4
ET4351	Digital VLSI Systems on Chip	4
ET4362	High Speed Digital Design for Embedded Systems	5
ET4369	Nyquist-Rate Data Converters	4
ET4371	Mixed-mode Wireless transceivers	4
ET4376	Photovoltaic Basics	4
ET4377	Photovoltaic Technologies	4
ET4378	Photovoltaic Systems	4
ET4379	Photovoltaic Lab Course	4
ET4382	Power conversion techniques in CMOS technology	3
ET4391	Advanced Microelectronics packaging	3
ET4icp	IC-Technology Lab	2
ET8011MSC	Structured Electronic Design Laboratory	3
QIST4400	Quantum Computer Architecture	5
TPM027A	Cyber Risk Management	5

Suggested profiles 2023

Digital Systems 2023

CESE4010	Advanced Computing Systems	5
CESE4035	Computer Arithmetic	5
CESE4040	Processor Design Project	5
EE4610	Digital IC Design	3
EE4615	Digital IC Design II	3
EE4690	Hardware Architectures for Artificial Intelligence	5
EE4695	Hardware Dependability	5
ET4351	Digital VLSI Systems on Chip	4
ET4362	High Speed Digital Design for Embedded Systems	5
ET4371	Mixed-mode Wireless transceivers	4

Microsystems & Technology 2023

EE4520	Analog CMOS design I	3
EE4585	Semiconductor Device Physics	5
EE4610	Digital IC Design	3
EE4C08	Measurement and Instrumentation	5
ET4257	Sensors and Actuators	4
ET4260	Microsystem Integration	4
ET4277	Microelectronics Reliability	4
ET4289	Integrated Circuits and MEMS Technology	4
ET4391	Advanced Microelectronics packaging	3
ET4icp	IC-Technology Lab	2

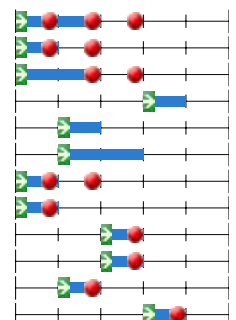
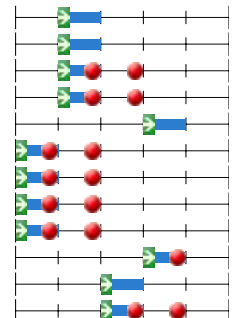
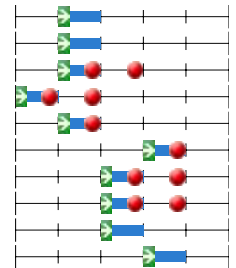
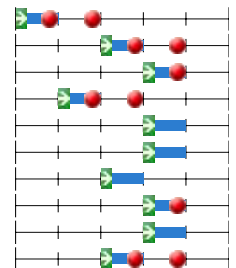
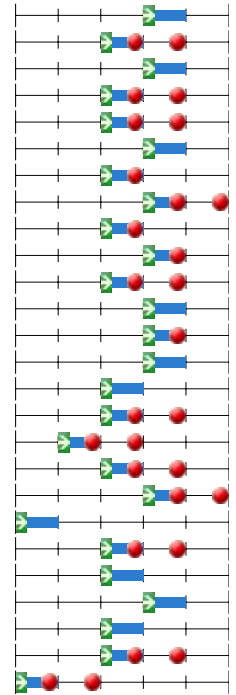
RF/Wireless Design 2023

EE4520	Analog CMOS design I	3
EE4585	Semiconductor Device Physics	5
EE4605	Integrated Circuits for RF/Wireless Applications	5
EE4610	Digital IC Design	3
EE4615	Digital IC Design II	3
EE4C05	Electromagnetics	5
EE4C08	Measurement and Instrumentation	5
EE4C10	Analog Circuit Design Fundamentals	5
EE4C13	Wireless Systems for Electrical Engineering Applications	5
ET4351	Digital VLSI Systems on Chip	4
ET4369	Nyquist-Rate Data Converters	4
ET4371	Mixed-mode Wireless transceivers	4

BioSemi 2023

Biomedical Devices, Biosensors, BioMEMS and Microsystem Integration (BioSemi) 2023

BM41055	Anatomy and Physiology	4
BM41095	Medical Devices: Past, Present, Future	3
BM41141	Fundamentals of Biomedical Engineering	5
EE4555	Active Implantable Biomedical Microsystems	5
EE4585	Semiconductor Device Physics	5
EE4C01	Profile Orientation and Academic Skills	3
EE4C08	Measurement and Instrumentation	5
EE4C11	Systems Engineering	5
ET4127	Themes in Biomedical Electronics	4
ET4130	Bioelectricity	3
ET4257	Sensors and Actuators	4
ET4260	Microsystem Integration	4



ET4277	Microelectronics Reliability	4	
ET4289	Integrated Circuits and MEMS Technology	4	
ET4391	Advanced Microelectronics packaging	3	
ET4icp	IC-Technology Lab	2	
WM1402TU	Ethics of Healthcare Technologies	5	
BioCAS 2023			
Biomedical Devices, Biomedical Circuits and Systems (BioCAS) 2023			
BM41055	Anatomy and Physiology	4	
BM41095	Medical Devices: Past, Present, Future	3	
BM41141	Fundamentals of Biomedical Engineering	5	
EE4109	Structured Electronic Design	5	
EE4520	Analog CMOS design I	3	
EE4555	Active Implantable Biomedical Microsystems	5	
EE4610	Digital IC Design	3	
EE4615	Digital IC Design II	3	
EE4C01	Profile Orientation and Academic Skills	3	
EE4C08	Measurement and Instrumentation	5	
EE4C10	Analog Circuit Design Fundamentals	5	
EE4C11	Systems Engineering	5	
ET4127	Themes in Biomedical Electronics	4	
ET4130	Bioelectricity	3	
ET4252	Analog Integrated Circuit Design	4	
ET4257	Sensors and Actuators	4	
ET4278	Over-Sampled Data Converters	4	
ET4351	Digital VLSI Systems on Chip	4	
ET4369	Nyquist-Rate Data Converters	4	
WM1402TU	Ethics of Healthcare Technologies	5	
Analog Circuit Design 2023			
EE4109	Structured Electronic Design	5	
EE4520	Analog CMOS design I	3	
EE4525	Analog CMOS design II	3	
EE4585	Semiconductor Device Physics	5	
EE4C08	Measurement and Instrumentation	5	
EE4C10	Analog Circuit Design Fundamentals	5	
ET4252	Analog Integrated Circuit Design	4	
ET4278	Over-Sampled Data Converters	4	
ET4369	Nyquist-Rate Data Converters	4	
ET4382	Power conversion techniques in CMOS technology	3	
Sub-THz Electronic Systems ME 2023			
EE4510	Advanced Electromagnetics	5	
EE4520	Analog CMOS design I	3	
EE4605	Integrated Circuits for RF/Wireless Applications	5	
EE4620	Spectral Domain Methods in Electromagnetics	4	
EE4725	Quasi Optical Systems	5	
EE4730	High Frequency Wireless Architectures	3	
EE4C05	Electromagnetics	5	
EE4C10	Analog Circuit Design Fundamentals	5	
EE4C13	Wireless Systems for Electrical Engineering Applications	5	
ET4169	Radar I: From Basic Principles to Applications	5	
ET4369	Nyquist-Rate Data Converters	4	
Quantum Electronics 2023			
AP3421-PR	Quantum Information Project	2	
AP3432	Quantum Hardware 1 - Theoretical Concepts	4	
AP3442	Quantum Hardware 2 - Experimental State of the Art	4	
EE4520	Analog CMOS design I	3	
EE4610	Digital IC Design	3	
EE4710	Solid State Physics with Quantum and Nano Electronics	5	
EE4C08	Measurement and Instrumentation	5	
EE4C10	Analog Circuit Design Fundamentals	5	
QIST4310	Fundamentals of Quantum Information	4	
QIST4400	Quantum Computer Architecture	5	
Thesis project 2023			
ET4300	Master Thesis	45	
Track Wireless Communication and Sensing 2023			
Main core WiCoS 2023			
CESE4010	Advanced Computing Systems	5	
EE4C01	Profile Orientation and Academic Skills	3	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
EE4C04	Control System Design	5	
EE4C05	Electromagnetics	5	
EE4C06	Networking	5	
EE4C08	Measurement and Instrumentation	5	
EE4C10	Analog Circuit Design Fundamentals	5	
EE4C11	Systems Engineering	5	
EE4C12	Machine Learning for Electrical Engineering	5	

EE4C13	Wireless Systems for Electrical Engineering Applications	5		
Track core WiCoS 2023				
EE4510	Advanced Electromagnetics	5		
EE4565	Propagation and Scattering of EM waves	5		
ET4358	Fundamentals of Wireless Communications	5		
ET4386	Estimation and Detection	5		
IN4341	Performance Analysis	5		
Suggested specialisation courses WiCoS 2023				
CESE4045	High-performance data networking	5		
CESE4050	Measuring and Simulating the Internet	5		
CESE4055	Ad hoc and Sensor Networks	5		
CESE4060	Wireless IoT and Local Area Networks	5		
CESE4065	Advanced Practical I.o.T. and Seminar	5		
CESE4120	Smart Phone Sensing	5		
CS4160	Blockchain Engineering	5		
CS4195	Modeling and Data Analysis in Complex Networks	5		
CS4430	Network Security	5		
EE4016	Antenna Systems	5		
EE4396	Mobile Networks	5		
EE4530	Applied Convex Optimization	5		
EE4595	Wavefield Imaging	5		
EE4605	Integrated Circuits for RF/Wireless Applications	5		
EE4620	Spectral Domain Methods in Electromagnetics	4		
EE4630	Telecommunication Network Architectures	3		
EE4670	PV Materials Processing and Characterization	4		
EE4675	Object classification with radar	4		
EE4685	Machine Learning, a Bayesian Perspective	5		
EE4715	Array Processing	5		
EE4725	Quasi Optical Systems	5		
EE4730	High Frequency Wireless Architectures	3		
EE4740	Data Compression: Entropy and Sparsity Perspectives	5		
EE4745	Superconducting Astronomical Instrumentation	5		
EE5020	Sensor Signal and Data Processing	4		
ET4030	Error Correcting Codes	4		
ET4034	Telecom Business Architectures and Models	4		
ET4169	Radar I: From Basic Principles to Applications	5		
ET4173	Introduction to UWB technology, systems and applications	4		
ET4175	Radar II: Theory and System Design	4		
IN4150	Distributed Algorithms	6		
IN4191	Security and Cryptography	5		
SC42101	Networked and Distributed Control Systems	4		
Suggested profiles WiCoS 2023				
Applied Electromagnetics 2023				
EE4510	Advanced Electromagnetics	5		
EE4565	Propagation and Scattering of EM waves	5		
EE4595	Wavefield Imaging	5		
EE4620	Spectral Domain Methods in Electromagnetics	4		
EE4725	Quasi Optical Systems	5		
EE4730	High Frequency Wireless Architectures	3		
EE4745	Superconducting Astronomical Instrumentation	5		
EE4C05	Electromagnetics	5		
ET4169	Radar I: From Basic Principles to Applications	5		
Astronomical Instrumentation 2023				
AP3701	Submm and terahertz physics and applications	3		
EE4510	Advanced Electromagnetics	5		
EE4565	Propagation and Scattering of EM waves	5		
EE4620	Spectral Domain Methods in Electromagnetics	4		
EE4725	Quasi Optical Systems	5		
EE4730	High Frequency Wireless Architectures	3		
EE4745	Superconducting Astronomical Instrumentation	5		
EE4C05	Electromagnetics	5		
ET4289	Integrated Circuits and MEMS Technology	4		
ET4386	Estimation and Detection	5		
Microwave Sensing and Radar 2023				
EE4016	Antenna Systems	5		
EE4510	Advanced Electromagnetics	5		
EE4565	Propagation and Scattering of EM waves	5		
EE4620	Spectral Domain Methods in Electromagnetics	4		
EE4675	Object classification with radar	4		
EE4C05	Electromagnetics	5		
EE5020	Sensor Signal and Data Processing	4		
ET4169	Radar I: From Basic Principles to Applications	5		
ET4173	Introduction to UWB technology, systems and applications	4		
ET4175	Radar II: Theory and System Design	4		
Physical Layer Signal Processing for Communication 2023				

EE4396	Mobile Networks	5	
EE4530	Applied Convex Optimization	5	
EE4685	Machine Learning, a Bayesian Perspective	5	
EE4715	Array Processing	5	
EE4740	Data Compression: Entropy and Sparsity Perspectives	5	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
ET4358	Fundamentals of Wireless Communications	5	
ET4386	Estimation and Detection	5	
Network Architectures and Services 2023			
CESE4045	High-performance data networking	5	
CESE4050	Measuring and Simulating the Internet	5	
CS4195	Modeling and Data Analysis in Complex Networks	5	
CS4220	Machine Learning 1	5	
EE4396	Mobile Networks	5	
EE4530	Applied Convex Optimization	5	
EE4630	Telecommunication Network Architectures	3	
EE4740	Data Compression: Entropy and Sparsity Perspectives	5	
EE4C06	Networking	5	
ET4034	Telecom Business Architectures and Models	4	
ET4358	Fundamentals of Wireless Communications	5	
IN4341	Performance Analysis	5	
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5	
Electives: Internet of Things 2023			
CESE4045	High-performance data networking	5	
CESE4050	Measuring and Simulating the Internet	5	
CESE4055	Ad hoc and Sensor Networks	5	
CESE4060	Wireless IoT and Local Area Networks	5	
CESE4065	Advanced Practical I.o.T. and Seminar	5	
CESE4120	Smart Phone Sensing	5	
CS4430	Network Security	5	
EE4396	Mobile Networks	5	
EE4630	Telecommunication Network Architectures	3	
EE4C06	Networking	5	
EE5020	Sensor Signal and Data Processing	4	
ET4173	Introduction to UWB technology, systems and applications	4	
ET4358	Fundamentals of Wireless Communications	5	
IN4341	Performance Analysis	5	
Thesis project 2023			
ET4300	Master Thesis	45	
Track Signals & Systems 2023			
Main core S&S 2023			
CESE4010	Advanced Computing Systems	5	
EE4C01	Profile Orientation and Academic Skills	3	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
EE4C04	Control System Design	5	
EE4C05	Electromagnetics	5	
EE4C06	Networking	5	
EE4C08	Measurement and Instrumentation	5	
EE4C10	Analog Circuit Design Fundamentals	5	
EE4C11	Systems Engineering	5	
EE4C12	Machine Learning for Electrical Engineering	5	
EE4C13	Wireless Systems for Electrical Engineering Applications	5	
Track core S&S 2023			
EE4530	Applied Convex Optimization	5	
EE4595	Wavefield Imaging	5	
EE4740	Data Compression: Entropy and Sparsity Perspectives	5	
ET4169	Radar I: From Basic Principles to Applications	5	
ET4358	Fundamentals of Wireless Communications	5	
ET4386	Estimation and Detection	5	
Suggested specialisation courses S&S 2023			
AP3132	Advanced Digital Image Processing	6	
AP3232	Medical Imaging Signals and Systems	6	
BM41055	Anatomy and Physiology	4	
BM41100	Medical Devices: Designing for CE-marking	3	
CESE4055	Ad hoc and Sensor Networks	5	
CS4220	Machine Learning 1	5	
CS4240	Deep Learning	5	
EE4016	Antenna Systems	5	
EE4530	Applied Convex Optimization	5	
EE4595	Wavefield Imaging	5	
EE4650	Advanced Magnetic Resonance Imaging	5	
EE4675	Object classification with radar	4	
EE4685	Machine Learning, a Bayesian Perspective	5	
EE4715	Array Processing	5	
EE4740	Data Compression: Entropy and Sparsity Perspectives	5	

EE4750	Tensor Networks for Green AI and Signal Processing	4	
EE5020	Sensor Signal and Data Processing	4	
ET4030	Error Correcting Codes	4	
ET4127	Themes in Biomedical Electronics	4	
ET4130	Bioelectricity	3	
ET4169	Radar I: From Basic Principles to Applications	5	
ET4173	Introduction to UWB technology, systems and applications	4	
ET4175	Radar II: Theory and System Design	4	
ET4257	Sensors and Actuators	4	
ET4358	Fundamentals of Wireless Communications	5	
ET4386	Estimation and Detection	5	
IN4191	Security and Cryptography	5	
IN4307	Medical Visualization	5	
IN4314	Seminar Selected Topics in Multimedia Computing	5	
RO47019	Intelligent Control Systems	4	
SC42015	Control Theory	6	
SC42035	Integration Project Systems and Control	5	
SC42075	Modeling and Control of Hybrid Systems	3	
SC42125	Model Predictive Control	4	
Free elective courses S&S 2023			
TUD4040	Joint Interdisciplinary Project	15	
Suggested profiles S&S 2023			
Array Processing 2023			
EE4530	Applied Convex Optimization	5	
EE4540	Distributed Signal Processing	5	
EE4595	Wavefield Imaging	5	
EE4685	Machine Learning, a Bayesian Perspective	5	
EE4715	Array Processing	5	
EE4740	Data Compression: Entropy and Sparsity Perspectives	5	
EE4750	Tensor Networks for Green AI and Signal Processing	4	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
EE4C12	Machine Learning for Electrical Engineering	5	
ET4358	Fundamentals of Wireless Communications	5	
ET4386	Estimation and Detection	5	
Biomedical Signal Processing 2023			
AP3232	Medical Imaging Signals and Systems	6	
AP3531	Acoustical Imaging	6	
BM41055	Anatomy and Physiology	4	
BM41095	Medical Devices: Past, Present, Future	3	
EE4555	Active Implantable Biomedical Microsystems	5	
EE4595	Wavefield Imaging	5	
EE4650	Advanced Magnetic Resonance Imaging	5	
EE4685	Machine Learning, a Bayesian Perspective	5	
EE4715	Array Processing	5	
EE4750	Tensor Networks for Green AI and Signal Processing	4	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
EE4C12	Machine Learning for Electrical Engineering	5	
ET4127	Themes in Biomedical Electronics	4	
ET4130	Bioelectricity	3	
ET4386	Estimation and Detection	5	
IN4307	Medical Visualization	5	
Signal Processing for Radar Systems and Remote Sensing 2023			
EE4530	Applied Convex Optimization	5	
EE4540	Distributed Signal Processing	5	
EE4565	Propagation and Scattering of EM waves	5	
EE4595	Wavefield Imaging	5	
EE4675	Object classification with radar	4	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
EE4C12	Machine Learning for Electrical Engineering	5	
EE5020	Sensor Signal and Data Processing	4	
ET4169	Radar I: From Basic Principles to Applications	5	
ET4173	Introduction to UWB technology, systems and applications	4	
ET4175	Radar II: Theory and System Design	4	
ET4386	Estimation and Detection	5	
Systems and Control for EE 2023			
EE4530	Applied Convex Optimization	5	
EE4540	Distributed Signal Processing	5	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
ET4386	Estimation and Detection	5	
RO47002	Machine Learning for Robotics	5	
RO47019	Intelligent Control Systems	4	
SC42015	Control Theory	6	
SC42025	Filtering & Identification	6	
SC42056	Optimization for Systems and Control	3	
SC42075	Modeling and Control of Hybrid Systems	3	
		4	

SC42101	Networked and Distributed Control Systems		
SC42125	Model Predictive Control	4	
SC42130	Fault Diagnosis and Fault Tolerant Control	4	
Thesis project 2023			
ET4300	Master Thesis	45	
Suggested Free elective space 2023			
General Elective Courses and Academic Skills 2023			
AP3841	Systems Engineering Company Project	15	
EE5010	Internship 10-15 ECTS	10	
ET4300-I	Thesis Preparation at Industry	0	
ET4399	Extra Project (max. 15EC)	15	
IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15	
TUD4040	Joint Interdisciplinary Project	15	
Laboratories 2023			
ET4icp	IC-Technology Lab	2	
Business Engineering & Management 2023			
TPM416A	Turning Technology into Business	6	
Courses in Control Systems Engineering 2023			
ME46085-23	Mechatronic System Design	4	
RO47019	Intelligent Control Systems	4	
SC42025	Filtering & Identification	6	
SC42075	Modeling and Control of Hybrid Systems	3	
Language courses and skills 2023			
TPM301B	Spoken English for Academic Purposes Intermediate	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM303A	Intermediate Writing in English for the University	2	
TPM304A	Advanced Writing in English for the University	2	
TPM305A	Writing a Masters Thesis in English	2	
WM1115TU	Dutch Elementary 1	3	
WM1116TU	Dutch Elementary 2	3	
WM1117TU	Dutch Intermediate 1	3	
WM1135TU	Advanced English for the University	3	
Free elective courses SS 2023			
Entrepreneurship and Innovation Education 2023			
MOT9611	Project Entrepreneurship Thesis related	5	
TPM401A	Technology Entrepreneurship and Innovation	5	
TPM411A	Idea to Start-up IT & AI	5	
Quantum Technology 2022			
AP3432	Quantum Hardware 1 - Theoretical Concepts	4	
AP3442	Quantum Hardware 2 - Experimental State of the Art	4	
CS4090	Quantum Communication and Cryptography	5	
QIST4310	Fundamentals of Quantum Information	4	
QIST4400	Quantum Computer Architecture	5	
Other 2023			
EE4700	Modelling, Algorithms and Data Structures	5	
Specialisation BD 2023		Specialisation Biomedical Devices (BD) 2023	
Main core BD 2023			
CESE4010	Advanced Computing Systems	5	
EE4C01	Profile Orientation and Academic Skills	3	
EE4C03	Statistical Digital Signal Processing and Modeling	5	
EE4C04	Control System Design	5	
EE4C05	Electromagnetics	5	
EE4C06	Networking	5	
EE4C08	Measurement and Instrumentation	5	
EE4C10	Analog Circuit Design Fundamentals	5	
EE4C11	Systems Engineering	5	
EE4C12	Machine Learning for Electrical Engineering	5	
EE4C13	Wireless Systems for Electrical Engineering Applications	5	
Suggested profiles BD 2022			
BioSemi 2023			
Track core BD Bio Semi 2023			
EE4585	Semiconductor Device Physics	5	
ET4257	Sensors and Actuators	4	
ET4289	Integrated Circuits and MEMS Technology	4	
Profile core BD BioSemi 2023			
BM41055	Anatomy and Physiology	4	
BM41095	Medical Devices: Past, Present, Future	3	
EE4555	Active Implantable Biomedical Microsystems	5	
ET4127	Themes in Biomedical Electronics	4	
Suggested free electives BioSemi 2023		Suggested free electives BD BioSemi 2023	
BM41141	Fundamentals of Biomedical Engineering	5	
ET4130	Bioelectricity	3	

ET4260	Microsystem Integration	4	
ET4277	Microelectronics Reliability	4	
ET4391	Advanced Microelectronics packaging	3	
ET4icp	IC-Technology Lab	2	
WM1402TU	Ethics of Healthcare Technologies	5	
BioCAS 2023			
Track core BD BioCAS 2023			
EE4520	Analog CMOS design I	3	
EE4610	Digital IC Design	3	
ET4257	Sensors and Actuators	4	
Profile core BD BioCAS 2023			
BM41055	Anatomy and Physiology	4	
BM41095	Medical Devices: Past, Present, Future	3	
EE4555	Active Implantable Biomedical Microsystems	5	
ET4127	Themes in Biomedical Electronics	4	
Suggested free electives BioCAS 2023			
Suggested free electives BD BioCAS 2023			
BM41141	Fundamentals of Biomedical Engineering	5	
EE4109	Structured Electronic Design	5	
EE4615	Digital IC Design II	3	
ET4130	Bioelectricity	3	
ET4252	Analog Integrated Circuit Design	4	
ET4278	Over-Sampled Data Converters	4	
ET4351	Digital VLSI Systems on Chip	4	
ET4369	Nyquist-Rate Data Converters	4	
WM1402TU	Ethics of Healthcare Technologies	5	
BioSP 2023			
Track core BD BioSP 2023			
EE4595	Wavefield Imaging	5	
ET4386	Estimation and Detection	5	
BD BioSP 2023 (select 1 out of 4)			
EE4530	Applied Convex Optimization	5	
EE4740	Data Compression: Entropy and Sparsity Perspectives	5	
ET4169	Radar I: From Basic Principles to Applications	5	
ET4358	Fundamentals of Wireless Communications	5	
Profile core BD BioSP 2023			
BM41055	Anatomy and Physiology	4	
BM41095	Medical Devices: Past, Present, Future	3	
EE4555	Active Implantable Biomedical Microsystems	5	
ET4127	Themes in Biomedical Electronics	4	
Suggested free electives BioSP 2023			
Suggested free electives BD BioSP 2023			
AP3232	Medical Imaging Signals and Systems	6	
BM41141	Fundamentals of Biomedical Engineering	5	
EE4650	Advanced Magnetic Resonance Imaging	5	
EE4685	Machine Learning, a Bayesian Perspective	5	
EE4715	Array Processing	5	
ET4130	Bioelectricity	3	
IN4307	Medical Visualization	5	
SC42025	Filtering & Identification	6	
WM1402TU	Ethics of Healthcare Technologies	5	
Specialisation European Wind Energy Master - Electrical Power Systems within the track Electrical Power Engineering 2023			
ET4108	Transients in Power Systems	4	
EWEM Electives/Specialisation Courses 2023			
AE4W09	Wind Turbine Design	5	
EE4111	High-Voltage DC	4	
EE4114	Power System Protection and Control	4	
EE4375	Finite Element Modeling for Electrical Energy Applications	4	
EE4515	Advanced Power Electronics	4	
EE4536	DC and AC Microgrids	4	
EE4545	Electrical Power Systems of the Future	4	
EE4665	Uncertainty modelling and risk assessment in electrical power systems	4	
EE4755	Reliable Power Electronic Components and Systems	5	
ET4113	Power System Dynamics	4	
ET4116	Power Electronics	4	
ET4121	A.C. Machines	4	
ET4291	Control of Electrical Drives	5	
ET4378	Photovoltaic Systems	4	
ET4379	Photovoltaic Lab Course	4	
ET8020	High Voltage Testing and Diagnostics	4	
SET3065	Intelligent Electrical Power Grids	4	

Suggested profiles EWEM 2023			
High voltage 2023			
EE4111	High-Voltage DC	4	
EE4114	Power System Protection and Control	4	
EE4375	Finite Element Modeling for Electrical Energy Applications	4	
EE4545	Electrical Power Systems of the Future	4	
ET4108	Transients in Power Systems	4	
ET4116	Power Electronics	4	
ET8020	High Voltage Testing and Diagnostics	4	
Electronics and Electrical Machines 2023			
EE4375	Finite Element Modeling for Electrical Energy Applications	4	
EE4515	Advanced Power Electronics	4	
EE4755	Reliable Power Electronic Components and Systems	5	
ET4116	Power Electronics	4	
ET4117	Electrical Machines and Drives	4	
ET4121	A.C. Machines	4	
ET4291	Control of Electrical Drives	5	
Smart AC Grids and DC Grids 2023			
EE4536	DC and AC Microgrids	4	
EE4545	Electrical Power Systems of the Future	4	
ET4108	Transients in Power Systems	4	
ET4113	Power System Dynamics	4	
ET4116	Power Electronics	4	
ET4117	Electrical Machines and Drives	4	
SET3065	Intelligent Electrical Power Grids	4	
Free electives 2023			
EE5015	Internship EWEM	6	
WM1115TU	Dutch Elementary 1	3	
Thesis project 2023			
ET4300	Master Thesis	45	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Master EE 2023

Introduction 1	An IEP consists of the following parts, 1. The program main core of 18 credits, 2. A track core of three courses, 3. Specialisation courses worth an amount of credits required to obtain the minimum of the sum of 60 credits in combination with the main core and the track core, 4. A thesis project worth 45 credits, 5. If the total number of credits under 1 to 4 is lower than 120 credits the IEP should be completed with free electives to a minimum of 120 credits. Free electives should be master level courses, except that a maximum of 10 credits in the free electives can be BSc-level homologation courses.
Introduction 2	The MSc EE program in Electrical Engineering (EE) is provided by the Electrical Engineering, Mathematics and Computer Science (EEMCS) faculty at TU Delft. The groups of the faculty contributing to these programs are all involved in cutting edge EE related research. The program contains specialized contributions from other faculties throughout the university. The EE program offers the following tracks of study: Electrical Power Engineering (EPE) Microelectronics (ME) Signals and Systems (S&S) Wireless Communication and Sensing (WiCoS) The program takes 24 months and is fully conducted in English. It offers challenging high-level education and research to talented students holding a suitable BSc degree in technology or science. These tracks share a common core from which all students select certain courses to provide a solid foundation of knowledge and skills. Subsequently, the program branches out to offer either deep specialization possibilities or breadth and multi-disciplinarity. The fundamental goals of this MSc program is to train students to become independently thinking and broadly developed professionals that are able to solve future problems using novel concepts and tools, by offering thorough fundamental knowledge and by involving them in cutting-edge research programs in a stimulating environment. This guide first presents a description of the background and motivation of the tracks. For each of the tracks, a list of laboratories/sections that are involved in the track is given. As such, students in a certain track will execute their MSc project under supervision of a professor from one of the associated laboratories. The track descriptions will be followed by a description of the structure of the program and a detailed program description including an overview of the main courses of the program.
Program Structure 1	The total program takes 2 years, each year being divided in 4 quarters. The study load (credits) is measured in European Credits (EC), whereby the complete program amounts to 120 EC. A typical course size is 3 to 5 EC; the thesis project is 45 EC. The program allows substantial freedom in courses to be followed in agreement with the personal learning goals of all students. As such, all students draw up an Individual Exam Program (IEP) in the beginning of their studies. They do this in consultation with the coordinator of the track and (if known) their (tentative) supervisor. The Masters degree in Electrical Engineering will be awarded if all units of the approved IEP have been passed (i.e. with a mark that is greater than or equal to 6.0). In that case of the EE program, their the track of studies (one of the four listed above) will be formally stated on the degree certificate.
Program Structure 2	All tracks/programs have the same multi-tier structure. In each tier, the student selects a number of courses subject to some minimum and maximum numbers of courses or sum of credits associated with these courses. Some prerequisite requirements may apply. Also, the (tentative) thesis supervisor may require or recommend certain courses. The overall study program of choice is formally agreed upon via the Individual Exam Program (IEP), to be co-signed by the MSc coordinator, the tentative thesis supervisor (if already known) and the responsible professor. The tiers are called the common core, the track core, the specialization space and the free electives space. Schematically, the program is structured as in the figure below, where the track core is not explicitly indicated but is the row with multiple columns (one column for each track). The program is concluded with a thesis project.

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Track Electrical Power Engineering 2023

Program Title	Track Electrical Power Engineering
EC Program	120
Introduction 1	Electricity is the most flexible and efficient source of energy available to society. If we improve the exploitation of sustainable energy sources, electricity is sure to play a leading role throughout the foreseeable future. In the decades to come, a revolution will unfold in the generation and delivery of electrical power, driven by the large-scale introduction of renewable and distributed power generation. These changes demand environmentally friendly and sustainable components and the liberalization of the energy markets. Electrical sustainable energy engineers are needed in order to design new components with an eye towards sustainable materials and efficient conversion processes, as well as to integrate them into a smart and adaptable electricity infrastructure. They should be able to solve problems of power systems under current and future characteristics and challenges. Those challenges include having less inertia in the system (because of introducing smaller machines, reducing stability), reactive power support and temporal variability of renewable energy sources. The laboratories involved in the MSc Program Electrical Power Engineering are: DC Systems, Energy Conversion & Storage Intelligent Electrical Power Grids Photovoltaic Materials and Devices The EPE track covers a wide range of topics, including renewable energy, smart grids, DC systems, environmentally friendly material technologies, the decentralisation of power generation, power-systems operations, power electronics and electrical machines. The Department of Electrical Power Engineering will provide you with a number of outstanding research facilities of the Electrical Sustainable Energy Laboratory (ESE lab), including the High-Voltage Laboratory, PV lab, DENlab, a Real-Time Digital Simulator and the DC Systems lab with Electrical Machines and Electronic Conversion laboratory.
Program Goals	Program Goals The fundamental goal of this MSc program is to train students to become independently thinking and broadly developed professionals that are able to solve problems of power systems under current and future characteristics/challenges (e.g. less inertia and reactive power support, variability of RES), using novel concepts and tools, by offering thorough fundamental knowledge and by involving them in cutting-edge research programs in a stimulating environment.

Main core EPE 2023	
Introduction 1	This is the common core. It is common to all programs and contains two general engineering oriented courses being compulsory for all students. These courses are Profile Orientation and Academic skills and Systems Engineering. The common core further contains 9 fundamental courses introducing all students to the breadth of EE and CE. Of these 8 fundamentals, all students must select two courses. Common core courses are fundamental and general in nature, and should be attractive to a relevant fraction of the students in multiple tracks. These courses are programmed in the first quarter of the academic year. Typically, they have three parts. These are (1) an introductory part that might be seen as a quick repeat of the expected prerequisite knowledge of incoming students (or as a quick introduction if some knowledge is still missing), (2) the main body, and (3) a part that makes a specific connection to the different tracks in the program. This last part may actually be track-specific. Common core EE EE4C01 Profile Orientation and Academic Skills EE4C11 System Engineering Common core: Select 2 out of 9 EE4C03 Statistical Digital Signal Processing and Modeling EE4C04 Control Theory EE4C05 Electromagnetics EE4C06 Networking CESE4010 Advanced Computing Systems EE4C08 Measurement and Instrumentation EE4C10 Analog Circuit Design Fundamentals EE4C12 Machine Learning for Electrical Engineering Applications EE4C13 Wireless Systems for Electrical Engineering Applications

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Instructor	Ir. J.W. Peltenburg	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = $0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3)$ L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1] In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C01	Profile Orientation and Academic Skills	3
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Responsible Instructor	L.C. Schroten	
Instructor	Dr. C.A. Benitez Avila	
Instructor	L.C. Schroten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: basic knowledge as acquired with a BSc in Physics, EE, Material Science. Strongly advised: Information Literacy 1 and 2 (e.g. Master Programme Information Literacy 1 & 2 or Electrical Engineering Information Literacy 1 and 2, both available in Brightspace).	
Course Contents	- Introduction to the MScEE tracks (Microelectronics, Electrical Power Engineering, Signals & Systems, Wireless Communication and Sensing) - Introduction to the research and expertise of the research groups taking part in the selected master track - Knowledge of ethics and intellectual property right - Skills in academic writing - Skills in oral presentation of scientific topics	
Study Goals	With this course the students - obtain relevant information on the MScEE study and the chosen track, become aware of cultural differences and how this affects intercultural communication, form a network with fellow master EE students. - acquire an overview of the main challenges in and potential applications of electrical engineering - become familiar with research topics addressed within the EE department - master basic skills in oral presentation of scientific topics - master basic skills in academic writing - acquire basic principle of ethics and intellectual property right - are able to extract the main information from a scientific paper or report - are able to write a paper (mostly a literature review) on a scientific question	
Education Method	The course includes lectures, reading materials and exercises. During the course, students work on a writing assignment, an ethics case and give several presentations. Attendance is mandatory as all class activities are highly interactive and peer feedback is a necessity.	
Literature and Study Materials	TU Present & TU Write. Both are available in Brightspace.	
Assessment	The pass/fail for this course is determined by the evaluation of the final writing assignment, the oral presentation and attending the Ethics seminar(s). All three components should be evaluated with a pass to pass the whole EE4C01 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C04	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential Realization of transfer function models by state space descriptions, coordinate changes, canonical forms Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback Application of LQ regulator Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle Reference signal modeling, integral action for zero steady-state error. Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design and - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	P.P. Sundaramoorthy	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C13	Wireless Systems for Electrical Engineering Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	C. Gao	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	M. Babaie	
Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	M. Spirito	
Instructor	Prof. M. Alavi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE4605 'Integrated Circuits and Systems for Wireless Applications' and EE4371 'Advanced Wireless Transceivers.'	
Course Contents	In this course, you will learn about the commonly used architectures in wireless systems, with the requirements on their building blocks. The system key building blocks are addressed at their functional level regarding electrical properties, such as transfer/gain, noise, linearity, frequency selectivity, bandwidth, and other parameters. Physical constraints of practical components and circuits are used to identify the achievable signal quality, selectivity, sensitivity, and link budget of a wireless system. The lectures will explain the causes and impact of device/circuit non-linearities and noise on wireless systems' sensitivity and dynamic range. The course will introduce the concepts of frequency generation, modulation/demodulation, and up/down conversion. To teach you to work at RF frequencies with wireless systems, the principles of impedance matching are explained, and tools are provided to analyze/design/characterize RF (matching) circuitry/systems. The common part of the course also includes an overview of the figure of merits (FoMs) of various TX/Rx architectures/system blocks and the required instrumentation to characterize them. This course belongs to the RF/wireless profile and is an excellent preparation for the course; EE4605, Integrated Circuits for RF/Wireless Applications, which focuses on the design and realization of analog intensive building blocks for wireless transceivers as well as for ET4371, Advanced Digital Wireless transceivers which is addressing the latest trends in RF/wireless design aiming to fully utilize the integration and signal processing strengths of advanced CMOS. The course is completed by four short specialization topics which can be followed based on your personal preference, namely: (1) Introduction to wireless power transfer technology, (2) Beamforming basics: concepts and architectures, (3) Neural Networks for Wireless & Radar Systems, (4) Hands-on experience with RF characterization equipment on practical circuits. (The latter (4) has an attendee limit of 15 people.) Only one of these specialization topics is required to complete the overall course.	
Study Goals	Understand wireless system architectures with their subblocks to such a level that you can analyze their various performance parameters in terms of; link budget, sensitivity, spectral purity, P1dB, IP3, DR, SFDR, noise, phase noise, selectivity, EVM, APCR, and efficiency. In addition, you will master a basic understanding of S-parameters, impedance matching, and RF instrumentation.	
Education Method	This course will be delivered with complementary exercise lectures to familiarize students with the concepts and calculations described during the lectures.	
Literature and Study Materials	These lectures are (in part) based on the book of Behzad Razavi, RF microelectronics 2nd edition, Chapters 1 to 4.	
Assessment	Examination and grading: Your overall grade will be based on your performance in the written exam (80%), compulsory homework, and the mark obtained in your track-specific project (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year

Organization

Education

2023/2024

Electrical Engineering, Mathematics and Computer Science

Master Electrical Engineering

Track core EPE 2023	
Introduction 1	This is the track core, offering the first level of branching between tracks. The track cores typically contain some 8 courses, of which the students select three courses.

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4103	High Voltage Technology	4
Responsible Instructor	Prof.ir. P.T.M. Vaessen	
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Practical Coordinator	I.M. Splinter	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course presents the basic principles of High Voltage technology and practical application of this knowledge in the power industry. Proper design of High Voltage components requires knowledge and insight of electric fields and breakdown mechanisms.	
Study Goals	By the end of the course students will be able to: - Demonstrate insight in the relation between charge, electric field and voltage - Identify and calculate capacitive voltage distributions - Calculate and analyze electric fields using analytical and numerical methods - Describe the breakdown mechanisms in different insulation materials and the phenomena leading to this - Explain methods of field grading and field control - Describe impulse breakdown phenomena (for example lightning) - Evaluate the effect of combinations of materials and interfaces on electric fields - Remember maximum permissible field strengths in different materials and configurations	
Education Method	Lectures, demonstrations	
Literature and Study Materials	Kreuger, F.H., Industrial High Voltage. Vol.1, 2, 3, ISBN 90-6275-561-5 (available for purchase at the course), hand-outs (available at the course)	
Prerequisites	Electromagnetics (EE3P11) or equivalent courses	
Assessment	Written exam (70%) and project (30%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4108	Transients in Power Systems	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	EE2E21, ET4107 - knowledge of symmetrical component is highly recommended	
Course Contents	This course deals with the rapid phenomena in power systems caused by switching actions and atmospheric discharges. Some of the topics covered are circuit breakers, arc-circuit interaction, short-line fault, interruption of small inductive capacitive currents, travelling waves, atmospheric discharges, simulation of transients, circuit breaker testing.	
Course Contents Continuation	The subject deals with power system transients. It is divided in a few sections the goals of which are the students to be introduced by the basic concepts of switching transients and transient analysis as well as travelling wave phenomena. The subject also deals with the current interruption phenomena and circuit breakers as well as circuit breaker testing and insulation coordination. Additional material on Ferroresonance and surge arrester selections is being studied as a supplement to this project. All results are validated by computer simulations in Electromagnetic Transient Program (EMTP), which is used as a tool for detailed analysis of different cases like switching small inductive and capacitive currents, arc interruption, thermal and dielectric breakdown of the circuit breakers, travelling wave phenomena. Each chapter is also supported by several simplified examples (which are solvable on the brightspace)	
Study Goals	After following this course the student should be able to: - model power system components for transient analysis by means of lumped elements. - to understand the current interruption process and gained insight in the testing of switchgear in the High Power Lab. - to use symmetrical components during transient processes in a network, especially when one pole of the circuit breaker opens and the other two remain in closed position. - to understand under which circumstances the worst case scenario (highest, steepest TRV) takes place. - to be able to assess the behavior of the lightning stroke when arriving and junction points connecting one or more components with different characteristic impedance Based on aforementioned learning objectives, the student should be able to simulate at least simplified cases of switching and lighting transient phenomena in Electromagnetic Transient Program ATP-EMTP or other EMTP-based software environment.	
Education Method	Lectures (*). Students should spent approximately ten hours per week in order to achieve the learning goals well. Each chapter, if suitable, is also supported by solvable examples on the board and simplified/detailed simulation cases performed in Electromagnetic Transient Program (EMTP) environment. Students also get opportunity to obtain the EMTP royalty free software and to learn EMTP so that they can use it as a tool for different project cases or M.Sc thesis. The subject also provides opportunity for extra projects in which students may obtain credits depending on the amount of work done. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	L. vander Sluis, Transients in Power System, Wiley, ISBN 0 471 486 396. JA Martinez-Velasco, M Popov, Circuit breakers: Power System Transients Parameter determination, CRC Press, ISBN-13: 978-1420065299, pp. 447-556.	
Assessment	Additional material on ferroresonance phenomena will be provided by the lecturer Oral (closed book examination) which can be taken as written in case of more than 10 students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular transient case that occurs in practice. The amount of work required for particular problem solutions is in line with the number of points. The exam is passed when the student scores more than 57 %. Resit of the examination is possible according to the examination schedule. The content of the course material is divided into 9 chapters. All chapters weight equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Responsible Instructor	H. Vahedi	
Instructor	G.R. Chandra Mouli	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4119	Electronic Power Conversion	4
Responsible Instructor	S.A. Rivera Iunnissi	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	EE1P21, EE1C11, EE2E11, EE2M21	
Course Contents	This course covers the principles of power conversion with switch-mode circuits. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated and isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain the circuit diagram and their waveforms; to be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practical	
Literature and Study Materials	N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition. (Available at ETV at reduced price)	
Assessment	Written exam (closed book) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carriers, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovoltaic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested specialization courses EPE 2023

Introduction 1	This is the specialization space. In essence, all of the MSc EE and MSc CE courses can be chosen as a specialization course, including those from the main core and the track core that were not yet chosen. Suggested specialization courses ESE: - ET4107 Power Systems Analysis II - EE4111 High-Voltage DC - ET4113 Power System Dynamics - EE4114 Power System Protection and Control - ET4116 Power Electronics - ET4121 A.C. Machines - EE4755 Reliable Power Electronic Components and Systems - ET4291 Control of Electrical Drives - EE4375 Finite Element Modeling for Electrical Energy Applications - ET4377 Photovoltaic Technologies - ET4378 Photovoltaic Systems - ET4379 Photovoltaic Lab Course - EE4680 Photovoltaic Modelling - ET8020 High Voltage Testing and Diagnostics - EE4515 Advanced Power Electronics - EE4625 High Voltage Cable System - EE4536 DC and AC Microgrids - EE4545 Electrical Power Systems of the Future - EE4655 Co-simulation of Energy Systems - EE4109 Structured Electronic Design - EE4665 Uncertainty modelling and risk assessment in electrical power systems - SET3100 Electric Vehicle & Charging Technology - EE4410 Cyber Security of Power Grids - EE4415 Photovoltaics Suggested specialization courses EPE, Other departments: - AE4W02TU Introduction to wind turbines: physics and technology - AE4W09 Wind Turbine Design - ET4277 Microelectronics Reliability - EE4530 Applied Convex Optimization - SET3065 Intelligent Electrical Power Grids
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EE4109	Structured Electronic Design	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Instructor	A.J.M. Montagne	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	A grade of a minimal 8 for this course is required to obtain access to the masterclass EE8011MSC	
Expected prior knowledge	This course required the knowledge and skills of MSc course EE4C10 and BSc course EE3C11. This course uses the same book used for BSc course EE3C11. A free download of the book is available via: https://analog-electronics.tudelft.nl Chapters that contain prior knowledge are: 1, 2, 7,8, 9,10 (excluding 10.3.5, 10.3.7, 10.3.8) , 11 (excluding 11.4.4, 11.4.6, 11.4.7, 11.5.1) and 12.1 and 12.2 Students that did not follow BSc course EE3C11 should (could) follow the homologation program from: https://analog-electronics.tudelft.nl/Homologation/courseWebSite/index.html Here you can find lecture material and videos that cover all electronics topics of EE3C11. The teaching staff can assist in studying the homologation program. On request, the lecturers can provide extra lectures for external/international students on topics treated in EE3C11 on background knowledge, like poles and zeros.	
Course Contents	This course deals with the structured design of analog electronic circuits using CMOS(and Bipolar) transistors. The design method presented in this course is based on network theory, control theory, signal processing, and physics. Many people consider the design of analog electronic circuits to be complex. This is mainly because designers have to deal with many relevant performance aspects that can be achieved in many different ways. In other words, there are many degrees of freedom for obtaining the desired performance of an electronic circuit. Theoretical concepts, circuit topologies, electronic devices, their operating conditions, and the physical layout of a system constitute an enormous design space in which it is easy to get lost. For this reason, analog electronics is often regarded as an art rather than a solid discipline. At first glance, there doesn't seem to be a straightforward way to design analog circuits and systems: experienced designers intuitively use all these degrees of freedom to modify and combine known solutions into new ones. However, intuition is knowledge, the origin of which has become unclear. It cannot easily be shared with novices, and therefore, it cannot be a basis for the education of designers. However, the design of electronic circuits can be shared and understood if it is presented in a structured way. This requires a precise formulation of design goals and strategies and a clear distinction between theoretical concepts and their physical implementations. Instead of taking numerous existing solutions to known problems as a starting point, it is far more effective to start a new design with clear distinction between theoretical concepts and study how they can be implemented in such a way that the possibilities of the implementation technology are maximally exploited. Such an approach results in well-defined and reliable designs with predictable performance and provides a basis for sharing and developing knowledge. During the engineering phase, designers need to find or create real-world components, determine their operating conditions, and fix them by design. To this end, they need to formulate and solve so-called design equations for the circuit. Design equations relate the properties and the operating conditions of the electronic devices to the performance requirements and the cost factors of the circuit. For example, to determine a budget for the gain-bandwidth product of an operational amplifier, we must find relations between this gain-bandwidth product and the performance requirements of the circuit that comprises this operational amplifier. I Setting up design equations for analog circuits, especially at transistor-level is a difficult and cumbersome task. It requires knowledge about technology, modeling techniques, and interactions between the behavior of the circuit and the transistor properties. It goes hand in hand with symbolic analysis. Performing symbolic calculations by hand is sensitive to errors, and in practice, only a limited number of circuit performance aspects can be considered. For this reason, we will use SLiCAP (Symbolic Linear Circuit Analysis Program) during this course. SLiCAP (https://analog-electronics.tudelft.nl/SLiCAP.html)	
Study Goals	The main goal of this course is to provide the student with the essential knowledge and skills to design an amplifier in BiCMOS or CMOS technology according to given specifications and to motivate and fully understand all design steps involved. After the course the student is able to: - work in a design team in a complete CMOS design - translate an application description of an amplifier into a specification comprising, a functional specification, specification of performance measures, operational conditions and cost factors. - design a proper amplifier concept based on source and load specification. - evaluate the amplifying capabilities of basic amplifier stages such as CS and CE stage for a given technology - identify performance aspects such as noise, signal handling capability and bandwidth of basic amplifier stages such as CS and CE stage for a given technology - scale devices and select their operating conditions to influence these performance aspects - to apply error reduction techniques, such as compensation, balancing and (local) negative feedback to improve the performance of these basic amplifying stages - to motivate the decision to use multi-stage amplifiers - to design (high-performance) multi-stage amplifiers using error reduction techniques such as compensation, balancing and feedback. - setup design considerations for noise and take adequate design decisions	

	- setup design considerations for signal handling capability and power efficiency and take adequate design decisions - setup design considerations for frequency response and take adequate design decisions - setup design considerations for weak distortion and take adequate design decisions - setup design considerations for biasing and design and implement proper biasing circuitry - evaluate the performance versus requirements of the final design and propose options for improvement - select proper models and design verification tools for each design step - write a design report in which all design steps are motivated and the performance is evaluated and explained.
Education Method	Lectures and Colstructions: The systematic electronics design method will be illustrated during the lectures and colstructions with a design example made in SLiCAP. The design example closely relates to the hearing loop system designed during the BSc course EE3C11. This BSc design was based on operational amplifiers as controllers. In this course, the focus will be on the design of the controllers as transistor circuits. During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design. The design steps are documented in a report using SLiCAP.
Computer Use	The design report can be efficiently used as a basis for the exam, which is organized as a design review for the design team. A computer with SliCAP and the free circuit simulator LT-Spice is necessary. Help for installing and using SLiCAP will be available during the lectures. SLiCAP is an acronym for Symbolic Linear Circuit Analysis Program.
Literature and Study Materials	See: https://analog-electronics.tudelft.nl/SLiCAP.html Book (also used during BSc course EE3C11): Structured Electronic Design Edition downloadable from: https://analog-electronics.tudelft.nl
Assessment	During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design used in every lecture to illustrate the application of the theory from the lectures. At the end of the lecture period, the student team makes an appointment review of their design. The design review consists of two sessions: 1) A design preview, where the students present and defend their design in a relaxed atmosphere. They can use the feedback to make improvements. 2) A design review, where the students present and defend their design again. Each team member has to give substantial input during the session. After this session, each student is graded individually. The design report, created with SLiCAP during the lectures by the team, can be efficiently used as a basis for the review. If there are logistic or personal reasons that make it impossible for a student to join a design team, it is possible to opt for an individual design review (oral exam). Please contact the educators about this. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4111	High-Voltage DC	4
Responsible Instructor	Prof.ir. P.T.M. Vaessen	
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	ET4103 High Voltage Technology (High Voltage Constructions)	
Course Contents	This course presents the physical challenges faced by HVDC power components. Electrical insulation behaves quite differently under DC voltage compared to AC. This will influence the entire cycle from design to testing. Knowledge and insight of electric field behavior under DC stress is given theoretically and illustrated by practical applications from power industry (hvdc cable, converter transformers, bushings). The course contents are: Introduction to HVDC Electric fields under different DC conditions Surface charge, origin, effect and measurement Space charge, origin, effect and measurement Breakdown mechanisms Partial discharges and DC tests	
Study Goals	Note:the information provided in this page may change depending on the developments around the coronavirus By the end of the course students will be able to: - Know the difference between the electrical behavior of AC and DC insulation - Calculate the charge distribution and field strength in different situations - Elaborate on the behavior of materials under DC stress - Formulate design requirement for DC insulation - Describe the differences in partial discharges between AC and DC - Describe methods for surface and space charge measurements - Comment on DC testing	
Education Method	Lectures Demonstration(s)	
Literature and Study Materials	book: F.H. Kreuger. Industrial DC High Voltage. Delft University Press, 1995. hand-outs at lectures	
Assessment	Written examination (80% final written exam, 20% course project) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4114	Power System Protection and Control	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Instructor	Dr. A. Lekic	
Contact Hours / Week x/x/x/x	0,0,0,4 (32 hr. in total)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course deals with the important aspects of power system protection with respect to fault detection and isolation and it is organized as follows: - System grounding - types, characteristics, and the effects on fault current behavior - Instrument transformers - Fundamentals of power system protection - Overcurrent protection - Transformer protection - Distance protection - Fault locators based on point-of-wave and synchrophasor measurements - Testing of power system protection - Control strategies for the type-3 and type-4 wind turbines - Control of HVDC-based connections	
Study Goals	After the course, the student can learn: - typical grounding practices and how the grounding system influences applied protection principles - important aspects of current transformers and procedures and guidelines on how to choose current transformers for particular protection function - the effect of the saturation and DC offset on the choice of the instrument transformer - how the microprocessor protection operates - the difference between overcurrent, differential, and distance protection and the advantages/disadvantages of each - protection of electric grids with inverter-based generation (fault currents, relay performances) - control strategies for type-4 and type-3 wind turbines - HVDC MMC (full/half bridge) control for monopolar and bipolar connection - to use RTDS for different protection and MTDC operation scenarios The students should also be able to: - deal at least with basic settings of overcurrent protection - deal with the choice of primary and backup protection for simplified networks consisting of a generator, a transformer, and a line. - understand how protective relays are being tested by RTDS and Omicron	
Education Method	Lectures, contact hours (*). The students are expected to spend about 8h a week on this exam. The students will work on the experimental verification of the studied protection and control approaches using RTDS.	
Literature and Study Materials	G. Ziegler: Numerical Distance Protection: Principles and Applications G. Ziegler: Numerical Differential Protection: Principles and Applications Network Protection and Automation Guide, Protective Relays, Measurements and Control, General Electric S. Du, A. Dekka, B. Wu, N. Zargari, Modular multilevel converters: analysis, control, and applications, John Wiley & Sons, 2017 R. Wachal, A. Jindal, S. Denetiere, H. Saad, HVDC Converters in a HVDC Grid, December, 2014	
Assessment	Oral (closed book examination) can be taken as written in case more than ten students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular protection function that occurs in practice or a solvable problem that is in line with what has been studied during the course. The work required for particular problem solutions aligns with the number of points. The exam is passed when the student scores more than 57 %. Resit is possible according to the examination schedule. The content of the course material is divided into 8 chapters. All chapters weigh equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Remarks Consulting hours according to the appointment	

EE4375	Finite Element Modeling for Electrical Energy Applications	4
Responsible Instructor	Dr. D.J.P. Lahaye	
Contact Hours / Week x/x/x/x	0/0/2/0 and 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	We expect the following prior knowledge 1) linear algebra: concepts of vector, matrix, matrix-vector multiplication, solving linear systems, computing eigenvalues and eigenvectors of a matrix; 2) calculus of functions of single variable: concepts of derivative of a function, finite difference approximation, integral of a function, quadrature, ordinary differential equations; 3) calculus of functions of more variables: concepts of partial derivative of a function, surface and line integral, quadrature, partial differential equations; 4) electric and magnetic fields: concepts of electric field, magnetic field, Maxwell equations, electrical machines and transformers;	
Course Contents	This course consists of the following three blocks: 1) Block 1 (Week 1 and Week 2): In this block, the finite difference method for the Poisson equation on the interval (Week 1) and the square (Week 2) will be discussed. The differential equation and the boundary conditions will be discretized on a mesh by a finite difference scheme. This will result in linear equations for the grid unknowns. A stencil notation for these linear equations will be introduced. A linear system will be formed and solved. The discrete solution will be visualized on the mesh. Properties of the discretization method and the linear system will be discussed. 2) Block 2 (Week 3 and Week 4): In this block, the Galerkin finite element method for the one-dimensional (Week 3) and two-dimensional (Week 4) Poisson problem will be introduced. The problem in so-called strong form will be transformed into the problem in so-called weak or variational form. The problem in weak form will then be discretized in space using Lagrangian basis or shape functions. A linear system represents the Poisson equation on a discrete level will be assembled and solved. The finite difference and finite element solution methods will be compared. 3) Block 3 (Week 5, Week 6 and Week 7): In this block, the application of the finite element will be extended to treat more complex problems. Partial differential equations for the scalar and vector potential for the magnetic flux or field will be derived from the Maxwell equations. More complex geometries that represent transformers, actuators or rotating extrical machines will be meshed using unstructured triangular meshes. The Maxwell equations will be discretized on these meshes. The discrete linear system will be assembled by a element-by-element procedure and solved. The computed vector potential solution will be processed to compute the magnetic flux and the magnetic field.	
Study Goals	After successfully completing the course, the student will be able to: * formulate various boundary value problems for the one-dimensional and two-dimensional Poisson equation; * derive these boundary value problems from the Maxwell equations for the magnetic field by introducing the scalar and vector potential; * apply the finite difference and finite element method to solve these boundary value problems on a scale of problems of increasing complexity; * state properties of the finite difference and finite element method applied to Poisson boundary value problems; * describe the implementation of the finite difference and the finite element method in software; * give a physical interpretation of the numerical solution obtained for magnetic field problems;	
Education Method	Weekly lectures will be complemented by weekly practical lab sessions and three home work assignments. During the lab sessions, algorithms discussed in the lectures will be implemented, tested and further refined. Each homework assignment will correspond to a course block. The set of three assignment will form a portfolio.	
Reader	see http://ta.twi.tudelft.nl/nw/users/domenico/electro_fem/electro_fem.html	
Assessment	During the course, the student will build a portfolio of three home work assignments, each correspond to a block in the course. The objective of these assignments is to guide the students towards the implementation of a proper finite element code. Each assignment consist of an elective and a compulsory part. This portfolio will be assessed during the oral examination. During this examination, the student will be assessed on the theoretical foundation of finite difference and finite element method, on the application of these methods and their implementation in software. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4410	Cyber Security of Power Grids	4
Responsible Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	The energy system transition towards a sustainable, low carbon future is led by decarbonisation, decentralisation, and digitalisation. Power grids are undergoing a fast-paced process of digitalisation for enhanced monitoring and control capabilities and grid intelligence. However, the increased integration of digital technologies, such as the next generation of Operational Technologies (OTs) and digital substations, implies a new risk as Information Technology (IT) OT systems for smart grids are vulnerable to cyber attacks. Furthermore, the combination of heterogeneous, co-existing smart and legacy technologies generates significant vulnerabilities and cyber security challenges. Examples of cyber security incidents related to power grids already exist around the world. On December 23, 2015 cyber attacks were conducted on the power grid in Ukraine that resulted in power outages, which affected 225,000 customers. More sophisticated cyber attacks on the Ukrainian power grid followed on December 17, 2016 resulting in a power outage in the distribution network where 200 MW of load was unsupplied. The complexity of cyber attacks on power systems is on the rise. They can initiate cascading failures and result in a catastrophic blackout. A power outage can disrupt the entire energy chain including water supply, heating, and gas networks. Without power, hospitals and other critical services are severely affected. A disruption of service may lead to financial loss, damages, chaos, or even a loss of lives. With respect to security of supply and reliability of the future energy system provision, special attention is needed for new vulnerabilities and cyber threats that come with digitalisation. Accordingly, the cyber security aspects are critical for a further power grid digitalisation.	
Course Contents	This course aims to assess and improve the cyber security of smart grids. It discusses the stakeholders of and complexities in critical infrastructures such as electrical power systems. It analyses the typical IT-OT system architectures for grid monitoring and control and explains the main cyber security concepts and policies. The course provides essential knowledge to analyse cyber security threats, threat vectors, and vulnerabilities of IT OT systems for power grids. It examines the kill chain stages of the cyber attack process on industrial control systems. The course analyses the impact of cyber attacks on power system operation in terms of loss of load, equipment damage, cascading effects, and risk of a blackout. The course studies the state-of-the-art security controls and best practices for threat intelligence management and incident response to defend against cyber attacks and effectively mitigate their impact on power system operation. This course focuses on cyber security of Information Technology (IT) Operational Technologies (OT) networks used for power system operation. It starts with an overview of the smart grid and its industrial control systems (EMS/SCADA) and introduces computer networks and cyber security. Next, the course will introduce the cyber kill chain method for cyber security investigation and intelligence-driven defence. Kill chains are used to understand, anticipate, recognise, and combat cyber attacks and advanced persistent threats. The course uses the kill chains of the 2015/2016 cyber attacks on Ukrainian power grid as real-world study cases throughout the course. The main IT-OT system vulnerabilities and cyber threats are analysed. The course covers threats such as passive and active reconnaissance, social engineering, and malware. It investigates various threat vectors on host and network security and advanced cyber attacks targeting power grids. The impact on power system operation is analysed. The course discusses the state-of-the-art security controls to defend and mitigate cyber attacks on power grids such as access controls, authentication management, cryptography, firewalls, secure communication protocols, network segmentation, and intrusion detection and prevention systems. The best practices to improve power grid resilience to cyber attacks are discussed, i.e., threat intelligence management, incident response, and wide area monitoring, protection and control. This course provides a deep dive into core cyber security concepts for power grids such as cryptography (symmetric and asymmetric encryption), hash functions, digital signatures, and cyber-physical anomaly detection using machine learning techniques. Cyber security demonstrations on power grids will be conducted for students in Control Room of the Future (CRoF) technology centre.	
Study Goals	At the end of the course, the students will be able to: LO1: Analyse cyber security threats, threat vectors, and vulnerabilities of Information Technology (IT) Operational Technology (OT) systems for power grids. LO2: Examine kill chain stages of the cyber attack process on cyber-physical power systems. LO3: Analyse the impact of cyber attacks on power system operation. LO4: Recommend appropriate security controls to detect, prevent and mitigate cyber attacks and cyber secure the future power grids.	
Education Method	Lectures: theory and real-world study cases (8 lectures x 2h/lecture) Practicums: hands-on computer labs (5 practicums x 2h/practicum)	
Literature and Study Materials	Chwan-Hwa Wu & J. David Irwin. (2013). Introduction to computer networks and cybersecurity. CRC Press. ISBN: 9781498760133 Nigel P. Smart. (2016). Cryptography made simple. Springer, 2nd Edition.	
Assessment	The grade is formed based on the following elements: 40% Assignment reports (five assignment reports submitted in time on Brightspace) 60% Final digital (computer) exam All assignments and reports are mandatory. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4415	Photovoltaics	4
Responsible Instructor	P. Manganiello	
Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Photovoltaic Basics course ET4376	
Course Contents	This course will focus on the new research field of Photovoltaics and how to create multi-functional PV-based intelligent energy agents (PV-IEAs), able to optimally generate electricity and actively participate to communication. It will start with a general introduction to Photovoltaics and an overview of its different research areas. The issue of partial shading and non-uniform operating conditions of PV modules and systems will be discussed, and solutions at cell (low-breakdown voltage PV cells and integrated power electronic components), module (shade-tolerant architectures) and system (distributed maximum power point tracking) level will be presented. Methods and algorithms to optimize the operation of such solutions will be introduced. The dynamic behavior of solar cells and how it can be exploited to design PV-IEAs able to serve as receivers of information in optical wireless communication systems will be discussed, and equivalent DC and AC models of PV devices will be presented.	
Study Goals	By the end of the course, the student should be able to: Explain the effect of non-uniform operating conditions on the performance of PV modules and systems. Analyze shade-tolerant PV modules and systems. Analyze different control algorithms for shade-tolerant PV modules and systems. Understand the dynamic behavior of PV cells under real-world operating conditions and the underlying physical processes. Analyze PV-IEAs that use solar cells as data receivers in optical wireless communication links.	
Education Method	Lectures and Instructions	
Literature and Study Materials	Course Handouts Research articles Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	Written exam (closed book) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4515	Advanced Power Electronics	4
Responsible Instructor	Dr. Z. Qin	
Instructor	Dr. T. Batista Soeiro	
Contact Hours / Week x/x/x/x	0/0/0/3 + lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is expected that the student has taken all the previous courses in the power electronics teaching line: ET4119, ET4116, ET4145	
Course Contents	The focus of this course is on providing the theoretical background of various design related aspects of power converters not covered in the previous courses. The topics covered include: switching behavior and loss modelling, soft switching vs hard switching, passive components, thermal management and control. The lab projects integrate the knowledge from this course and the preceding courses in a design assignment of designing, building and testing an efficient and small power converter.	
Study Goals	After this course, the students will be able to: - design a transformer/inductor - design the control - design the cooling - analyse and improve the efficiency - design, build, and test a converter with small size and high efficiency	
Education Method	Lectures and lab project	
Literature and Study Materials	Lecture handouts N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition.	
Assessment	projects (80%) and oral exam (20%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4536	DC and AC Microgrids	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	L.M. Ramirez Elizondo	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatziaargyriou	
Assessment	Written exam 70%, Assignments 30% In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTS) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

EE4625	High Voltage Cable System	3
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Contact Hours / Week x/x/x/x	0/0/3/0 lecture and workshop	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	High Voltage Technology	
Course Contents	Course aims and objectives: The course is aimed to introduce and discuss technical and scientific aspects of high voltage cable systems. This involves design and manufacturing of cables, cable insulation material, electric and magnetic field distribution, cable bonding, concerns related to cable termination and cable joints, diagnostics and monitoring, failure modes of the cable systems, and cable testing. Course contents and description: The course starts with an introduction to cable systems, their evolution, and current state of the art. Various cables used in practice (e.g. land cables, submarine cables, superconducting cables), the insulation material and their properties will be discussed. An overview of the cable manufacturing process is given. Gas Insulated transmission line (GIL), as an alternative for cable systems, will be overviewed. Technical and economic comparison between the two options will be made. The following topics will be discusses in detail during the course: Modeling of electric, magnetic, and thermal behavior of cable systems by means of circuit and FEM models. Concerns related to cable joint and termination Wave propagation in cables, cable bonding methods and cable installation Calculation of cable parameters. Partial discharge measurement and diagnostics and monitoring techniques used for cable systems. Aging mechanisms and life time evaluation of cables and cable systems. Failure modes of cable systems and cable testing procedures. Additional construction elements (mechanical issues, watertight, pulling facilities,).	
Study Goals	Note:the information provided in this page may change depending on the developments around the coronavirus To know the position of cables in electric transmission grids. To be able to describe different type of cable insulation, semiconducting and conducting materials and the manufacturing process of plastic and oil-paper impregnated cables To be able to calculate electric, magnetic and thermal characteristics of a cable. To be able to calculate cable parameters. To know the principal of wave propagation through cables To know different cable bonding techniques and their consequences. To be able to recall different cable testing and their purpose. To know the diagnostics techniques used for cable monitoring. To know the most important failure modes of cable systems and how to evaluate remaining lifetime of a cable system To be able to evaluate electric field distribution for a cable joint or termination using a FEM model	
Education Method	The course consists of three parts, i.e. lectures, computer modeling session and laboratory demonstration.	
Literature and Study Materials	The following references is used to construct the material of the course. Handout will be given after each session. [1] Cable systems for high and extra high voltage, Pirelli, E. Peschke and R. von Olshausen, 1999 [2] Electric cables handbook, third edition, BICC Cables, 1997 [3] The history of electric wires and cables, R.M. Black, 1983 [4] Power cable engineering lectures, Nexan, Dietmar Meurer, 2014 [5] Power cables, KEMA, E.F. Steennis, 2007 [6] Power and Communication Cables Theory and Applications. R. Bartnikas** [7] Rating of electric power cables in unfavorable thermal environment, George J. Anders.** [8] Electrical Power Cable Engineering, William A. Thue [9] Electrical cables for power and signal transmission, Oswald I. Gilbertson, 2000 [10] Extruded cables for high-voltage direct-current transmission, Giovanni Mazzanti, et al. [11] Submarine power cables, Thomas Worzyk [12] CIGRE Technical Brochures (89, 338, 531,) [13] Advanced power cable technology Volume I, Toshikatsu Tanaka, ALLAN Greenwood, 2000 [14] Advanced power cable technology Volume II, Toshikatsu Tanaka, ALLAN Greenwood, 2000 [15] Cable system transients, theory modeling and simulation, Akihiro Ametani, et al [16] Application of high temperature superconductor to electric power equipment, Swarn singh kalsi [17] Condition Assessment of High Voltage Insulation in Power System Equipment, R.E. James and Q. Su [18] Power cables and their application Part I, Lothar Heinhold, 1993 [19] Power cables and their application Part II, Lothar Heinhold, 1993 [20] Underground transmission of electric power, B. M. Weedy, 1980 [21] Rating of electric power cables, ampacity computations for transmission, distribution, and industrial applications, George J. Anders, 1997 [22] Underground power cables, King and Halfter, 1982 [23] Electromagnetic transients in power cables, Filipe Faria, et al	
Assessment	Written exam (50%), Project and assignment (50%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4655	Co-simulation of Energy Systems	4
Responsible Instructor	Dr. M. Cvetkovic	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Some experience with programming in languages such as Matlab, Python, Java or C/C++ is needed. Basic knowledge of power systems is preferred although not necessary.	
Course Contents	The course introduces simulation methods and models that are used in the field of energy system integration and shows how these simulation methods and models can be combined into so called co-simulation in order to answer energy integration questions. The course consists of three themes which run in parallel and complement each other. In the first theme, we review the simulation methods (time-domain simulation, discrete simulation, equilibrium simulation, etc.) and modeling approaches (optimization models, agent-based models, etc.). We explain when and how these methods and models can be applied to answer engineering questions in the field of power and energy systems (stability, reliability, control, regulatory analysis, market design, etc.). Each method is supported with the examples in the common engineering tools (Matlab, Modelica, etc.) In the second theme, we introduce the co-simulation methods which can be used to link the simulation methods together. The important aspects are discussed, such as synchronization between simulators (serial-Jacobi, parallel-Gauss-Seidel) and the choice of information to be exchanged between simulators. The examples of these methods are made using the educational co-simulation environments. These environments are Matlab and Python based which gives students the opportunity to learn the basics of Python programming and to strengthen their Matlab skills. In the third theme, the students work in groups on a co-simulation project. The main goal of the project is to create design specs and implement a co-simulation for one energy system integration challenge. In this way the students get practical knowledge in interpreting an energy system integration challenge in terms of simulation tools and models needed to analyze it. The instructor provides a list of available project topics. The students have sufficient freedom to develop their own co-simulation solutions following good design practice introduced in the class.	
Study Goals	The aim of the course is to educate the MSc students on how to approach simulation and modeling of energy integration challenges. The integration challenges are problems that require system of systems thinking and often involve composition of diverse subsystems and corresponding simulation tools. For example, the impact of electric vehicles on the electricity grid can be accurately simulated only if both, the electricity grid and the drivers behavior, are modeled and simulated. Another example is development of management algorithms for the combined heat and power (CHP) plants which feed both electricity grid with electric energy and the heat grid with excessive heat. The third example comes from smart grids, where the latency of wide-area controllers might impact the proper operation of remedial action schemes. Each of these examples is a cross-domain example that requires: 1) a system of systems thinking and the compositional approach to model/simulation development; 2) awareness of available simulation environments and the means of interconnecting them. The objective of the course is to 1) introduce different integration challenges, 2) bring system thinking closer to students, and 3) give students practical knowledge on model/simulation composition. Learning objectives: By the end of the course the students should be able to: 1) Describe some of the most important energy integration challenges, 2) Model these integration challenges by composition of existing modeling concepts and tools 3) Have basic practical knowledge on composition of adequate simulations tools using co-simulation	
Education Method	The course consists of lectures, assignments and a project. The lectures introduce different topics while the project asks of students to apply the material studied in the lectures. The exercise assignments are given each week to practice the concepts from the lectures before integrating them in the project. Project milestones are organized several times during the course in order to monitor and reflect on the progress of the project. Disclaimer: keep in mind that the course might go through the adjustments depending on the developments around the coronavirus. In that case, the study guide information may be adjusted.	
Literature and Study Materials	Supporting scientific papers, tutorials, and notes are published on Brightspace each week.	
Assessment	The grade is formed based on the student and team performance. Five aspects of the project are evaluated to obtain the final project mark: project context, co-simulation requirements, energy system/component models, co-simulation architecture and the co-simulation results. This mark is complemented with student's performance in the oral exam at the end of the course, upon which, the final mark is formed. In order to successfully complete the course, the students need to score a passing grade on all project aspects and during the oral exam. All students must participate in completion of milestones. Final report is a document that combines content of all intermediate milestones. Submission of the final report is mandatory for the passing grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4665	Uncertainty modelling and risk assessment in electrical power systems	4
Responsible Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ir. J.L. Rueda Torres	
Co-responsible for assignments	E. Sharifnia	
Co-responsible for assignments	K. Bölat	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Elementary knowledge of probability theory and statistics (discrete and continuous probability distributions, expectation/variance/standard deviation, quantiles, least squares regression). Basic programming skills (examples code samples will be provided in Python, but other languages may be used for hand-in exercises). Working knowledge of power system structure, components and actors.	
Course Contents	This course is aimed at MSc EE students in the EPE track and MSc SET students with the Power profile. This course considers the challenges faced by operators when operating and planning electrical power systems in the face of uncertainty balancing cost, reliability and environmental objectives. The course specifically focuses on quantitative modelling of uncertainty and performing quantitative risk assessments to support decision making. The course starts with a description of uncertainties that are important to power system operation and planning, and common metrics for system performance. Case studies in power system reliability and system capacity planning are introduced. In this context, various approaches to modelling uncertainties are introduced. Historical data is used to create models for component reliability. Students will learn how to compute system level reliability measures from bottom-up reliability models, using enumeration approaches and Monte Carlo sampling. Special attention is given to the estimation sample variation using the normal assumption and bootstrap approaches. Finally, the impact of risk metrics and decision criteria on decision making is discussed, thereby giving students insight into the whole chain from measurements to models to model-based decision making.	
Study Goals	After completing this course, students will be able to: 1) Calculate commonly used reliability measures and interpret their values in the context of Western European power systems. 2) Construct and use Markov models for component reliability. 3) Follow prescribed steps to construct reliability models from failure and repair data. 4) Calculate the reliability of model systems on the basis of constituent component models. 5) Implement Monte Carlo sampling and bootstrapping methods to estimate risk measures and the accuracy of such estimates. 6) Compare model representations of uncertainty, and illustrate these in the context of power system reliability and planning. 7) Evaluate different grid development options using quantitative risk assessment and discuss how a decision for one option depends on the choice of risk measure and decision criterion.	
Education Method	Lectures (8x 2 hours); Computer labs (8x 2 hours); additional exercises for self study.	
Literature and Study Materials	1) Book, available as e-book through TU Delft (no additional cost) or physical copy: Bart W. Tuinema, José L. Rueda Torres, Alexandru I. Stefanov, Francisco M. Gonzalez-Longatt, and Mart A. M. van der Meijden, "Probabilistic Reliability Analysis of Power Systems," Springer Nature Switzerland AG 2020, May 2020. DOI: https://doi.org/10.1007/978-3-030-43498-4. 2) Supplementary notes, videos, selected papers, reports and book chapters, available through Brightspace.	
Assessment	Completion with 'pass' mark of a small group assignment and computer lab exercise (with basic report) is required. The final grade is determined by a written exam (80%) and computer exercise with accompanying report (20%). A minimum mark of 5.0 must be attained for each component to pass. Component marks are given on a scale 0-10, rounded to the nearest 0.1 increment; the final grade is rounded to the nearest 0.5 increment. One resit opportunity is provided for the written exam. One repair opportunity is offered for the pass/fail assignments (two weeks after the initial deadline, unless agreed otherwise). A repair opportunity will also be offered for the graded exercise if the initial mark was between 4.0 and 5.7 (out of 10), with a deadline 3 weeks after the initial grade is announced. A maximum mark of 6.0 out of 10 can be obtained for the exercise through this route. If a student passes only part of the course in this academic year, all parts need to be retaken. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Elective	Yes	
Tags	Energy Modelling Programming Stochastics	
Maximum number of participants	30.	

EE4680	Photovoltaic Modelling	4
Responsible Instructor	Dr.ir. R. Santbergen	
Instructor	Prof.dr.ir. O. Isabella	
Responsible for assignments	Dipl.-Ing. Y. Blom	
Co-responsible for assignments	F.S. Saitta	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Modelling is an important tool for gaining deeper insight in the working principles and loss mechanisms of solar cells and photovoltaic systems. Researchers use this insight to develop more efficient solar cells and to maximize the energy yield of photovoltaic systems. In this course we will cover the entire photovoltaic modelling chain: modelling of photovoltaic materials, devices (solar cells), modules and systems. The focus will be on understanding the working principles of these models, and using them to analyse the effects of design parameters on solar cell and photovoltaic system performance. The online course material consists of two parts, each consisting of several phases: 1) PV materials and device modelling 1.1) PV materials and device characterization 1.2) optical modelling of PV devices 1.3) electrical modelling of PV devices 2) PV systems modelling 2.1) modelling of irradiance 2.2) modelling of energy yield In addition there will be software demos, given by expert users, allowing you to familiarize yourself with the advanced modelling software commonly used by PV researchers. The prior knowledge from the courses PV basics and PV systems is required. This course PV modelling will prepare you for an MSc thesis project that involves modelling of solar cells or photovoltaic systems, as well as a future career in photovoltaics research.	
Study Goals	The online course material consists of two parts: 1) PV materials and device modelling, 2) PV systems modelling. Each consist of several phases: At the end of phase 1.1 (characterization of PV materials and devices) you should: - Be familiar with the main techniques used in the characterization of PV materials and devices - Understand the importance of a correct characterization for the modelling of PV - Be able to extract useful and important information (trends and properties) from the characterization of both PV materials and devices At the end of phase 1.2 (optical modelling of PV devices) you will; - have a complete overview of the modelling approaches commonly used for photovoltaic optical modelling, and which are their strength and limitations - get a thorough understanding of different optical modelling methods and which structures they are best suited to model - improve and practice your skills on optical modelling, by carrying out activities with our own GenPro4 software At the end of phase 1.3 (electrical modelling of PV devices) you should be able to: - have a complete overview of the modelling approaches commonly for photovoltaics, both optical and electrical - get a thorough understanding of different electrical modelling methods - improve and practice your skills on optical and electrical modelling, by carrying out activities with our own advanced semiconductor analysis (ASA) software. At the end of phase 2.1 (modelling of irradiance) you should be able to: - Gather and analyse meteorological data, including the utilization of geographic information systems, - Obtain information on horizon and shading at a specific location, and derive the relevant location parameters, - Combine meteorological data and location properties to calculate the irradiance on the angle of incidence with different levels of detail, and - Optimize the orientation of a module to maximize the irradiance on the angle of incidence. At the end of phase 2.2 (modelling of the energy yield) you should be able to: - Model the operating temperature of modules, - Obtain information from module datasheet, - Combine module properties with temperature and irradiance modelling to calculate DC power (and yield), - Model the performance of inverters (DC to AC) and batteries, - Analyse load profiles, and use them to size a PV system (module and batteries), and - Calculate the AC power and energy yield of an entire system, and analyse its performance from an economic point of view. After the software demos you will be able to explain the input, working principles and output of the following commonly used software packages: - Complete EASE (extracting optical constants from spectroscopic ellipsometry measurement of PV materials) - HFSS (rigorous solving of Maxwell equation for EM-wave (light) propagation in PV devices) - Sentaurus (modelling charge transport in PV devices by solving semiconductor equations) - Radiance (PV module plane of array irradiance calculation using backward ray-tracing) - PV Syst (PV system energy yield calculation)	
Education Method	This course consists of recorded video lectures, interactive lectures and instructions. The video lectures provide you with the basic information. The interactive lectures give you the opportunity to ask questions to the teacher and discuss selected topics in depth. In addition you will work on modeling assignments. During the instruction sessions the modeling assignments will be explained and feedback will be given on your (preliminary) results.	
Assessment	Every week you make and submit modeling assignments, which are graded. The weighted average counts for 70% of the final grade. In addition, every week you write a short report about the software demo, which is graded as well. The average of this counts for the remaining 30% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Remarks	The on-campus Q&A session is not mandatory. Because the course material (video lectures, live demos) is given online, this course can also be taken by students who cannot physically come to the campus. Because the assessment is done through weekly assignments, without an exam, the final grade (and credits) can be obtained already before the exam period (end of October). This can be relevant for those who have not yet obtained sufficient credits to start their graduation project.
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EE4755	Reliable Power Electronic Components and Systems	5
Responsible Instructor	A. Shekhar	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Following this course is only possible with course ET4119 or SET3095 in advance	
Course Contents	Understanding of Power electronic components: diodes, BJT's, MOSFET's, thyristors, IGBT's; Gate drive circuits; High frequency inductors; Semiconductor's thermal management and heatsinks. The course also deals with conducted electromagnetic interference in power electronic circuits. Finally, the course will discuss component reliability and its corresponding influence on the power electronic systems	
Study Goals	THEORY: Understanding the theoretical and practical operation of power electronics components (semiconductors and passives) based on the physical structure, the V-I characteristics, time and thermal behavior. PRACTICAL: Analysis of Boost-type Converter and the Design of components, including semiconductors, inductor and capacitor. The ability to perform (differential) voltage-, current and temperature measurements on these components via a remote controlled test setup and explain these measurements at the hand of theory. Students will be given a demo on reliability tests using power cycler in the PEARL lab and asked to discuss their observations on the obtained test result data.	
Education Method	Lectures, practicals	
Literature and Study Materials	TEXT BOOK: Mohan, Undeland and Robbins, Power Electronics, Converters, Applications and Design, ISBN:978-0-471-22693-2, 824 pages, 3rd edition, John Wiley & Sons, 2003. ADDITIONAL READING: Provided via Brightspace DATA SHEETS: Provided via Brightspace and hard-copy in the lab. HARDWARE: Provided in the lab sessions.	
Assessment	Written EXAM (40%) DESIGN REPORT of your inductors DUO PRESENTATION on design- and practical results (20%) FINAL REPORT on practical (40%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4107	Power Systems Analysis II	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Instructor	Dr. A. Lekic	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The students who obtained their B.Sc degree at TU Delft are expected to have taken the subject EE2E21 - Sustainable Energy Supply. Other students, with previous education at other universities and abroad are expected to have passed a similar subject and gained prior knowledge about transformers, generators, lines and cables, three phase circuits and per unit value calculation.	
Course Contents	The course is divided in three major parts: - Fault computation in electrical power systems - Economic operation and control of power systems - Introduction to power system stability The first part deals with the analysis of symmetrical and asymmetrical faults based on the application of symmetrical components by using power (in)variant transformation. In order to apply symmetrical components, particular time will be spent on sequence circuits of system components. A special attention will be paid on sequence network derivations for star/delta loads, transformers, overhead lines and generators. The second part, the economic operation involves the minimization of fuel costs, while the power system control deals with the primary and secondary control on the generating units to enable stable operation and maintaining constant grid frequency. Finally, in the third part, the stability of the power system after faults is studied through the method of the equal-area criterion. In the course, comprehensive exercises are being solved, which are related to realistic networks, which can later be utilized in other subjects in the electrical power systems track.	
Study Goals	By the end of this course, the student should be able: - to compute all types of fault currents that may occur in an arbitrary power systems by applying symmetrical components with a power variant and invariant transform. - to assess the values of specific fault currents depending on how the neutral transformer points are grounded. - to build the Zbus matrix of a system and from the Zbus matrix, at least to determine the amplitude of the fault currents at any bus of the network corresponding the built/pre-defined Zbus matrix. - to investigate transient stability by using the equal area criterion	
Education Method	Lectures (*). The number of hours that the students need to spend on this subject is about 14h a week. Experience reveals that this time is particularly related to the fact that students should devote more time on solving more exercises in order to master the topics of symmetrical components and transient stability. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	John. J. Grainger and William D. Stevenson: Power System Analysis, ISBN 0-07035958	
Assessment	Oral. Students normally get theoretical questions and problems. Problem solutions are important to see if the students govern the theory well. In case when more than 10 students take the exam, the assessment is done in a written form. Resit of the exam is possible according to the examination schedule. As the subject is divided in three major parts, the weights of part 1, 2 and 3 are 50%, 25% and 25% respectively, which is in line with the number of contact hours devoted for each part. These weights are also applied in the examination questions. The exam is passed when the student score more than 57%. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consulting hours according to the appointment	

ET4113	Power System Dynamics	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Required: ET4107 Power System Analysis II (Master SET / Master EE / Master EWEM) SET3065 Intelligent Electrical Power Grids (Master SET / Master EE / Master EWEM) Optional: ET4116 Power Electronics (Master SET / Master EE / Master EWEM) ET4117 Electrical Machines and Drives (Master SET / Master EE / Master EWEM)	
Parts	Theory lectures: Lecture 1: Introduction to power system stability and synchronous generator Lecture 2: Controllers of the synchronous generator Lecture 3: The voltage source converter Lecture 4: Controllers of the voltage source converter Lecture 5: Applications of voltage source converter in electrical power systems Lecture 6: Rotor angle stability: transient stability and small-signal stability Lecture 7: Small-disturbance (small-signal) rotor angle stability (measurement based methods) Lab sessions: Session 1: Tutorial for PowerFactory (Single-Machine Infinite Bus system) Session 2: Modelling of an interconnected system (steady-state performance) Session 3: Dynamic models - Part 1 Session 4: Dynamic models - Part 2 Session 5: Long-term voltage stability assessment (including tutorial for Python) Session 6: Rotor angle stability assessment Session 7: Sensitivity analysis	
Course Contents	The course "Power Systems Dynamics" is devoted to the study behaviour of electrical power systems in non-steady state. Knowledge of the dynamic behaviour of electrical power systems and their components is important in order to predict whether the system (or a part thereof) remains in a stable state after a disturbance. The time frame of observation ranges from milliseconds to several minutes, depending on the kind of disturbance and the influence of applied controls. For each time frame, special models of the network components are derived. In this course, the model of the synchronous generator, the voltage source converter, and their associated control systems receive special attention due to their dominant influence on the dynamic behaviour of the power system. In the computer assignments, the theory is put into practice by means of software based simulations in DigSILENT PowerFactory as well as scripting in Python and Matlab. Subject to investigation are: modelling of synchronous generator and voltage source converter for RMS simulations, frequency and voltage controls, model-based and measurement-based eigenvalue analysis, wide-area monitoring, damping control, and RMS simulation of interconnected power systems.	
Study Goals	At the end of the course the students will be able to: - Describe the modeling aspects of power system components for computer aided simulation of electro-mechanical transients - Apply and interpret stability criteria for the analysis of systems dynamic behavior - Create a model of a power system in a simulation software package - Assess the system performance following typical disturbances	
Education Method	Lectures and computer assignments. Contact hours (8 weeks: 7 theory lectures, 7 lab sessions): 2 hours/theory lecture and 4 hours/lab session. Total: 42 hours. Self-study: 70 hours (9 hours/week).	
Computer Use	DigSILENT PowerFactory, Python, Matlab	
Literature and Study Materials	P. Kundur, Power System Stability and Control, Mc.Graw Hill, inc. 1994, ISBN-0-07-035958.	
Assessment	Lab report and final exam/re-exam including a theoretical questions component and a computer-based component. The computer-based part of the test requires familiarity with the models built during the lab assignments. The students are expected to run time-domain simulations and give insight into the observed phenomena. The final grade comprises: 1) Exam or re-exam (5 points). 2) Lab report and implementations (5 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Digital exam in the exam period of Q4. Duration: 3 hours. The resit is in the exam period of summer period. Duration: 3 hours.	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Study load: 112 hours.	
maximum aantal deelnemers	Limited to 25 students due to software availability	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalar - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteresis modulation - carrier modulation with feedback - scalar and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET4121	A.C. Machines	4
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	Control of Electrical Drives (ET4291)	
Expected prior knowledge	Electrical machines and drives (ET4117)	
Course Contents	The following subjects are discussed: - winding function and inductances; - time and spatial harmonics; - skin effect in electrical machines; - phasor, space vector and transformation in electrical machines; - the salient pole synchronous machine; - modeling and simulation of the dynamic performance of DC and AC machines.	
Study Goals	After following this course the students should be able to - describe and calculate the steady-state performance of AC machines; - calculate spatial harmonics and inductances of given winding distributions; - derive the dynamic models of DC and AC machines using space vectors; - model and simulate the dynamic performance of DC and AC machines; - evaluate the limitations of the dynamic models.	
Education Method	Lecture / Computer Simulation / Discussion	
Literature and Study Materials	- A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill. - lecture notes on Brightspace	
Assessment	Oral exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	- Presence at the computer simulation session is obliged. - Submission of assignments on Brightspace is obliged.	

ET4291	Control of Electrical Drives	5
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	practicum 0/0/0/4 vanaf week 2, instructie 0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Course Contents	- Dynamic modelling of DC and AC Machines supplied with power electronic converters. - Modelling of mechanical systems and thermal effects in drives. - Modelling and simulation of digital control of electrical drives. - Controller design for cascaded motor controller. - Scalar and vector control, field weakening control. - Practical design of an electrical drive system.	
Study Goals	Students should be able to: - Model the dynamic behaviour of DC and AC machines using state space representation. - Integrate ideal power electronic converter models and cascade control systems in the drive model. - Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive. - Calculate the steady state performance of DC and AC drives. - Implement scalar control system and vector control system for induction machines. - Implement vector control system for synchronous machines. - Implement field weakening control for DC and AC drives. - Implement the dynamic simulation models in MATLAB/Simulink. - Explain the practical design of electrical drives. - Evaluate the performance of an electrical drive system. - Design the controllers in electrical drive.	
Education Method	Lecture / Self-study / Simulation practical / Discussion	
Literature and Study Materials	- Rik De Doncker, Duco W.J. Pule, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: https://www.springer.com/gp/book/9789400701816 - Self-study materials on Brightspace.	
Reader	"Reader Control of Electrical Drives ET4291 Version maart 2009"	
Prerequisites	Electrical Machines and drives (ET4117) Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Assessment	Individual assignment and oral exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Instructor	Dr. L. Mazzarella	
Instructor	Prof.dr. I.M.F. Gordon	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Note: In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	

ET4379	Photovoltaic Lab Course	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Responsible Instructor	Dr.ir. R. Vismara	
Contact Hours / Week x/x/x/x	4/0/0/0 Q1=Q5! Also possible online in Q6, Q7, Q8!	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The aim is to offer you an advanced practical course on solar cells and photovoltaic systems. The following areas of practical training will be part of the course: (1) Characteristics of artificial and natural light (2) Characterization of the performance of lab-scale solar cells (3) Behavior of photovoltaic modules in different temperature and shading conditions (4) Charge and discharge properties of batteries (5) Performance of power electronic components (charge controllers, inverters, maximum power point trackers) (6) Operation of entire photovoltaic systems	
Study Goals	(1) Experiment with photovoltaic systems and all of their components under lab and real-life conditions. (2) Describe the working principle behind each component of a photovoltaic system. (3) Analyze the operation of all the elements of photovoltaic systems and of the system as a whole. (4) Compare the performance of different types of components of photovoltaic systems.	
Education Method	Laboratory sessions	
Literature and Study Materials	Manuals and templates are available on BrightSpace.	
Prerequisites	This course is meant for students who have already passed at least the PV Basics course (ET4376) and preferably also PV Technologies (ET4377) and PV Systems (ET4378). This course is also meant for 1st-year masters students with a bachelor degree in electrical engineering.	
Assessment	The assessment is based on reporting results from the individual tasks, using an Excel template provided on Brightspace. Two-person teams will be formed at the plenary session, the attendance of which is compulsory. The final grade is per team. The final grade is the average of the six technical reports. A sufficient grade (greater than or equal to 5.8) in all six technical reports is required to complete the course successfully. In case of failure of a technical report, you can use buffer sessions on week 1.7 and 1.8 to repeat the experiments. --- A virtual version of this course is made available for students to attend in Q2, Q3, and Q4 of the academic year 2023-2024. Each of the six tasks of the on-campus course will run in the internet browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, using numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to pass the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. Attending one of these instances of the course, no teams will be formed and the final grade is per person. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	The on-campus course can accommodate up to 48 students. Anyone interested in attending this course must apply during Q4 of the immediately previous academic year. A selection will then take place and, before the beginning of the instance in Q5, the line-up of students will be announced. Student(s) who failed on-campus in Q5 can re-take the virtual version of the course. There is no limit to the number of students who can attend the course's virtual version.	
Remarks	This practical course will be given on campus only in Q1. Conversely, a virtual version will be available for students to attend in Q2, Q3, and Q4. The enrolment is open on Brightspace from Q1 of the academic year 2023-2024.	

ET8020	High Voltage Testing and Diagnostics	4
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Instructor	Ir. A.G.A. Lathouwers	
Instructor	C.S. Engelbrecht	
Practical Coordinator	I.M. Splinter	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The quality of the insulation of high voltage components (assets) e.g. transformers, cables, switch-gear, generators, is of large importance for the availability and reliability of power supply. Prior to service operation and in accordance with international standards all these components are heavily tested at different voltage stresses (HVDC, AC, Impulse). During the service life of 30 or more years local insulation defects (on mm-scale) occurring as a result of aging and degradation may produce a total blackout e.g. of a 3 km long power cable connection. To provide during continuous service high reliability level, insulation condition assessment by means of advanced and non-destructive diagnostics is very important. In particular, by means of advanced sensor techniques, signal processing, knowledge rules, the condition of the components has to be continuously or periodically monitored. The liberalization of energy sector, requests from the network providers an economic responsible technical management of the assets. In particular, based on the actual condition, the maintenance strategies have to be implemented. This course, supported by demonstrations, learns about the different voltage generation, diagnostics and monitoring techniques, weakest link statistics, knowledge rules generation, diagnosis support systems for asset management are shown.	
Study Goals	To obtain insight in the testing, diagnostics of HV components and be able to understand the processes of insulation-coordination and quality insurance by maintenance strategies. Within power utility, to be able to recognize and to understand the AM processes and to be able to implement methods to control these processes.	
Education Method	Lectures, demonstrations, practical	
Literature and Study Materials	Kreuger, F.H., Industrial High Voltage Vol. 4, 5, 6; ISBN 90-6275-562-3 (available for purchase at the course); Hand-outs (available at the course) Weibull analysis: handouts replace Ch.6 R. Ross, Reliability Analysis for Asset Management of Electric Power Grids, Wiley-IEEE, 2019, ISBN 978-1-119-12517-4, DOI:10.1002/9781119125204.	
Assessment	Oral, closed book, hand outs about Weibull Analysis- substituting Ch.6, and hand-outs about Asset Management Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3100	Electric vehicle and charging technology	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Strongly advised: SET3095/ET4119 Electronic Power Conversion ET4117 Electric machines & drives	
Course Contents	This course focusses on the technology of (hybrid) electric vehicles and its charging technology Tank to Wheel, Well to Tank, Well to Wheel efficiencies and emission estimation for conventional, hybrid and fully electric vehicles based on energy source for propulsion. Types, operational principles, advantages and disadvantages of vehicle architectures: internal combustion engine; micro, mild and full hybrids (series, parallel, series-parallel), plug-in hybrids, full electrics, fuel cell vehicles; Vehicle load forces & dynamics, mechanical transmission components, power and energy consumption, drive cycles, battery sizing and electric vehicle range, MPGe, planetary gears as a power-split, regenerative braking. Principles of battery operation, battery types, the key performance parameters of batteries, from cell to battery pack, charge-discharge curves, battery and thermal management systems. Fundamentals and types of electric machine drives used in electric cars: its characteristics, efficiency, basics of motor control. Basics of power electronic converters used in electric cars and chargers, topologies and control. Charging methods of EVs: AC charging, DC charging, vehicle-to-grid technology, wireless charging, battery swap; charging time estimation, effects of fast charging on battery; Smart charging of EVs and vehicle-to-grid, communication protocols for smart charging. Modelling and investigating the performance of the electric vehicle drivetrain using quasi-static simulation in MATLAB/Simulink. Case study of EV: Toyota Prius HEV, Tesla Model S, Porsche Taycan, Audi e-tron, Hyundai Ioniq 5 The course will have a design project where students have to form teams to design their own EV by making choices for the vehicle drivetrain and corresponding charging technology. Emphasis will be on justifying and quantifying the technological design choices including drivetrain architecture, well-to-wheel efficiency and emissions, energy storage, power electronics, type of motor and charging method.	
Study Goals	The goal of this course is to educate students to design and evaluate various (hybrid) electric vehicles with respect to their performance, efficiency, sustainability, charging methodologies and potential for grid support. At the end of the course, the students should be able to: Evaluate the performance of hybrids (series, parallel, series-parallel), plug-in hybrid and fully electric drivetrain architectures based on power and energy consumption, efficiency, driving range, well-to-wheel emissions and charging time. Explain the design trade-offs in the energy sources (battery, fuel cell, super capacitors) and motors used in electric vehicles based on the key performance indicators like efficiency, energy and/or power density, lifetime, and control. Analyse various charging technologies for electric vehicles (conductive, inductive or battery swap) based on power electronics converters used, type of the charger (AC Vs DC), connector types, power levels, and modes. Investigate the performance of the (hybrid) electric vehicle drivetrain using quasi-static modelling and simulation in MATLAB/Simulink. Design a (hybrid) electric drivetrain by sizing and making the relevant choice of the electric machine, internal combustion engine, energy source (batteries, fuel cells, etc.), power electronic converters, transmissions components, and charging technology for a given vehicle specification.	
Education Method	Classroom lectures and problem solving workshops, Discussion and case study, Demonstrations using tools and animation in the classroom, Online exercises with self-evaluation, Design project	
Literature and Study Materials	- Lecture slides & handouts - Primary reading - Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, Third Edition, 2018, Mehrdad Ehsani, Yimin Gao, Ali Emadi, CRC Press, ISBN 9781138330498 - Supplementary reading - Chapter 1 & 14 of Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, ISBN: 978-1-119-06364-3, 2018 - Supplementary reading Electric and Hybrid Vehicles: Design Fundamentals, Second Edition, 2010, Iqbal Husain, CRC Press, ISBN 9781439811757 The enlisted books are available via the TU Delft library (free e-books), via the ETV desk & bookstore, via studystore.nl and the publisher website	
Assessment	The final grade of the course consists of the following components: - Written exam (70% weightage) - Closed book, Scientific (non-graphic) calculator can be used. No formula sheets - Design assignment (30% weightage) - Group assignment with Written report + oral presentation. Students must register in groups for the design assignment within the first two weeks of the course. Final grade calculation: $(0.7 * \text{Written exam} + 0.3 * \text{Design assignment})$ In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Written exam: Resit in next quarter Assignment: Re-submit assignment report in the next quarter based on feedback received (Grade is capped) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Follow up course: ET4291SET Digital modelling of electric powertrain/ ET4291 Control of Electrical Drives	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested specialization courses EPE, Other departments 2023

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: - describe the components and configurations of wind turbines - describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine - describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design - explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. - identify the drivers of dynamic behaviour and calculate their typical values - describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure - describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control The use of standards for load calculations Other optional material (tbd)	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): - AE4W13 Site conditions for wind turbine design - AE4135 Rotor / wake aerodynamics - AE4W21-14 Wind turbine aeroelasticity - AE4W31 Floating offshore wind energy - ET4117 Electrical machines and drives - OE44135 Offshore wind support structures	

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4277	Microelectronics Reliability	4
Responsible Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Prof.dr. G.Q. Zhang	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course aims to provide the students with a thorough understanding of the reliability issues involved in electronic components and systems. The following subjects will be treated. - Basic reliability definition and lifetime distributions. - Reliability prediction methods. - Physical failure mechanisms in electronic components - Package related failures - Reliability screening and reliability testing - Failure analysis methods - Reliability data handling - Design considerations - System reliability	
Study Goals	After the course the students: - should see reliability as a basic requirement that should receive attention throughout a products complete lifecycle: specification, design, production, exploitation and disposal. - should have a sound understanding of the physical background of failures and the reliability test and failure analysis methods available. - should be able to make a proper reliability prediction based on available reliability (test) data. - should be able to evaluate reliability and availability figures of complex systems based on component reliability data. - should be able to make the proper design and maintenance choices to optimize these figures.	
Education Method	Lecture	
Assessment	written exam or project (project only possible when homework assignments are passed with average grade 6 or more) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to LO1: Describe the functions and topologies of cyber-physical energy systems LO2: Implement data driven modelling and numerical simulation of energy systems LO3: Calculate variable power flows based on optimization/probability theory LO4: Synthesize functional requirements of energy management systems LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1: The energy transition and future energy systems (Peter) 2: Analysis of steady-state performance in variable operating conditions (Jose) 3: Assessment and control of power system dynamic performance (Jose) 4: Lab 1 Digital and data driven modelling of interconnected systems 5: Lab 2 Simulation and impact analysis of renewables, storage, and demand 6: Data acquisition and communication for energy automation (Peter) 7: Artificial intelligence-based functionalities for situational awareness (Pedro & Jochen) 8: Cyber security of smart grids (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The final grade of the course consists of the following components: - Assignment reports (in time!) (50%) - Multiple choice Computer Exam (50%) In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Final grade calculation: $(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Assignment reports (in time!): resubmit deliverable - Multiple choice Computer Exam: computer resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested profiles EPE 2023

Introduction 1	The tracks in EE typically have an informal sub-structure called profiles. A profile is a more or less canned study path. It typically prepares for a thesis subject in a certain area and typically with a specific professor or small set of professors of one section, overseeing that research area. For example, the Microelectronics track offers the Digital Systems profile, the RF/Analog Circuit Design profile, the Microsystems & Technology profile and the Biomedical Electronics profile. Other tracks offer different profiles. For the purposes of composing an IEP, a profile presents a short list of coherent courses (in all tiers) that together leads to a coherent program of study that prepare for a thesis topic in the area of the profile. As such, a profile can specify (at most) two from three courses in the common core, two from three courses in the track core, and at most 12 EC of additional specialization courses. Note, however, that profiles are nothing more than convenient selections of courses that go well together, and do not imply any formal restriction whatsoever. Based on individual interest and subject to approval of the IEP by the (tentative) thesis supervisor and the MSc coordinator and the responsible professor, a much wider variety of r programs is possible. Students are free to compose their individual program within the rules as specified in FIXME.
Introduction 2	The track offers the following specialisations or profiles: - High Voltage - Smart AC and DC Grids - Power Electronics and Electrical Machines - Solar Energy Examples of graduation projects - Optimisation of on-site partial discharge measurements and evaluation of diagnostic parameters for assessing the condition of distribution cable systems - Control of a multi-terminal voltage source converter HVDC for offshore wind power - Sustainable off-grid power for rural areas - Wireless power transfer for e-mobility - Directional relay coordination in ungrounded radial distribution networks using a real-time digital simulator - Design of fault-tolerant energy conversion systems for increasing wind turbine reliability.

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

High Voltage 2023

Introduction 1

Profile: High Voltage
 High Voltage Track core for this profile:
 ET4103 High Voltage Technology (C)
 ET4108 Transients in Power Systems (R)
 EE4585 Semiconductor Device Physics (R)

High Voltage Specialisation core (select at least 12 EC):
 ET4111 High-Voltage DC (C)
 ET4114 Power System Grounding and Protection (R)
 ET4116 Power Electronics (R)
 ET8020 High Voltage Testing and Diagnostics (C)
 ET4375 Finite Element modelling for Electrical Energy applications (R)
 EE4545 Electrical Power Systems of the Future (C)
 EE4625 High Voltage Cable System (C)

(C)= compulsory
 (R)= recommended

Introduction 2

This profile focuses on High Voltage Transmission Networks for large scale implementation of Renewable Energy Sources (solar, wind, wave), development of intelligent materials, monitoring of HV systems, and the optimization and controllability of HV transmission grids.

Responsible professor/head of the research group:

- Dr.ir. Pavol Bauer

EE4111	High-Voltage DC	4
Responsible Instructor	Prof.ir. P.T.M. Vaessen	
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	ET4103 High Voltage Technology (High Voltage Constructions)	
Course Contents	This course presents the physical challenges faced by HVDC power components. Electrical insulation behaves quite differently under DC voltage compared to AC. This will influence the entire cycle from design to testing. Knowledge and insight of electric field behavior under DC stress is given theoretically and illustrated by practical applications from power industry (hvdc cable, converter transformers, bushings). The course contents are: Introduction to HVDC Electric fields under different DC conditions Surface charge, origin, effect and measurement Space charge, origin, effect and measurement Breakdown mechanisms Partial discharges and DC tests	
Study Goals	Note:the information provided in this page may change depending on the developments around the coronavirus By the end of the course students will be able to: - Know the difference between the electrical behavior of AC and DC insulation - Calculate the charge distribution and field strength in different situations - Elaborate on the behavior of materials under DC stress - Formulate design requirement for DC insulation - Describe the differences in partial discharges between AC and DC - Describe methods for surface and space charge measurements - Comment on DC testing	
Education Method	Lectures Demonstration(s)	
Literature and Study Materials	book: F.H. Kreuger. Industrial DC High Voltage. Delft University Press, 1995. hand-outs at lectures	
Assessment	Written examination (80% final written exam, 20% course project) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4114	Power System Protection and Control	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Instructor	Dr. A. Lekic	
Contact Hours / Week x/x/x/x	0,0,0,4 (32 hr. in total)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course deals with the important aspects of power system protection with respect to fault detection and isolation and it is organized as follows: - System grounding - types, characteristics, and the effects on fault current behavior - Instrument transformers - Fundamentals of power system protection - Overcurrent protection - Transformer protection - Distance protection - Fault locators based on point-of-wave and synchrophasor measurements - Testing of power system protection - Control strategies for the type-3 and type-4 wind turbines - Control of HVDC-based connections	
Study Goals	After the course, the student can learn: - typical grounding practices and how the grounding system influences applied protection principles - important aspects of current transformers and procedures and guidelines on how to choose current transformers for particular protection function - the effect of the saturation and DC offset on the choice of the instrument transformer - how the microprocessor protection operates - the difference between overcurrent, differential, and distance protection and the advantages/disadvantages of each - protection of electric grids with inverter-based generation (fault currents, relay performances) - control strategies for type-4 and type-3 wind turbines - HVDC MMC (full/half bridge) control for monopolar and bipolar connection - to use RTDS for different protection and MTDC operation scenarios The students should also be able to: - deal at least with basic settings of overcurrent protection - deal with the choice of primary and backup protection for simplified networks consisting of a generator, a transformer, and a line. - understand how protective relays are being tested by RTDS and Omicron	
Education Method	Lectures, contact hours (*). The students are expected to spend about 8h a week on this exam. The students will work on the experimental verification of the studied protection and control approaches using RTDS.	
Literature and Study Materials	G. Ziegler: Numerical Distance Protection: Principles and Applications G. Ziegler: Numerical Differential Protection: Principles and Applications Network Protection and Automation Guide, Protective Relays, Measurements and Control, General Electric S. Du, A. Dekka, B. Wu, N. Zargari, Modular multilevel converters: analysis, control, and applications, John Wiley & Sons, 2017 R. Wachal, A. Jindal, S. Denetiere, H. Saad, HVDC Converters in a HVDC Grid, December, 2014	
Assessment	Oral (closed book examination) can be taken as written in case more than ten students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular protection function that occurs in practice or a solvable problem that is in line with what has been studied during the course. The work required for particular problem solutions aligns with the number of points. The exam is passed when the student scores more than 57 %. Resit is possible according to the examination schedule. The content of the course material is divided into 8 chapters. All chapters weigh equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Remarks Consulting hours according to the appointment	

EE4375	Finite Element Modeling for Electrical Energy Applications	4
Responsible Instructor	Dr. D.J.P. Lahaye	
Contact Hours / Week x/x/x/x	0/0/2/0 and 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	We expect the following prior knowledge 1) linear algebra: concepts of vector, matrix, matrix-vector multiplication, solving linear systems, computing eigenvalues and eigenvectors of a matrix; 2) calculus of functions of single variable: concepts of derivative of a function, finite difference approximation, integral of a function, quadrature, ordinary differential equations; 3) calculus of functions of more variables: concepts of partial derivative of a function, surface and line integral, quadrature, partial differential equations; 4) electric and magnetic fields: concepts of electric field, magnetic field, Maxwell equations, electrical machines and transformers;	
Course Contents	This course consists of the following three blocks: 1) Block 1 (Week 1 and Week 2): In this block, the finite difference method for the Poisson equation on the interval (Week 1) and the square (Week 2) will be discussed. The differential equation and the boundary conditions will be discretized on a mesh by a finite difference scheme. This will result in linear equations for the grid unknowns. A stencil notation for these linear equations will be introduced. A linear system will be formed and solved. The discrete solution will be visualized on the mesh. Properties of the discretization method and the linear system will be discussed. 2) Block 2 (Week 3 and Week 4): In this block, the Galerkin finite element method for the one-dimensional (Week 3) and two-dimensional (Week 4) Poisson problem will be introduced. The problem in so-called strong form will be transformed into the problem in so-called weak or variational form. The problem in weak form will then be discretized in space using Lagrangian basis or shape functions. A linear system represents the Poisson equation on a discrete level will be assembled and solved. The finite difference and finite element solution methods will be compared. 3) Block 3 (Week 5, Week 6 and Week 7): In this block, the application of the finite element will be extended to treat more complex problems. Partial differential equations for the scalar and vector potential for the magnetic flux or field will be derived from the Maxwell equations. More complex geometries that represent transformers, actuators or rotating extrical machines will be meshed using unstructured triangular meshes. The Maxwell equations will be discretized on these meshes. The discrete linear system will be assembled by a element-by-element procedure and solved. The computed vector potential solution will be processed to compute the magnetic flux and the magnetic field.	
Study Goals	After successfully completing the course, the student will be able to: * formulate various boundary value problems for the one-dimensional and two-dimensional Poisson equation; * derive these boundary value problems from the Maxwell equations for the magnetic field by introducing the scalar and vector potential; * apply the finite difference and finite element method to solve these boundary value problems on a scale of problems of increasing complexity; * state properties of the finite difference and finite element method applied to Poisson boundary value problems; * describe the implementation of the finite difference and the finite element method in software; * give a physical interpretation of the numerical solution obtained for magnetic field problems;	
Education Method	Weekly lectures will be complemented by weekly practical lab sessions and three home work assignments. During the lab sessions, algorithms discussed in the lectures will be implemented, tested and further refined. Each homework assignment will correspond to a course block. The set of three assignment will form a portfolio.	
Reader	see http://ta.twi.tudelft.nl/nw/users/domenico/electro_fem/electro_fem.html	
Assessment	During the course, the student will build a portfolio of three home work assignments, each correspond to a block in the course. The objective of these assignments is to guide the students towards the implementation of a proper finite element code. Each assignment consist of an elective and a compulsory part. This portfolio will be assessed during the oral examination. During this examination, the student will be assessed on the theoretical foundation of finite difference and finite element method, on the application of these methods and their implementation in software. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTS) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4625	High Voltage Cable System	3
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Contact Hours / Week x/x/x/x	0/0/3/0 lecture and workshop	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	High Voltage Technology	
Course Contents	Course aims and objectives: The course is aimed to introduce and discuss technical and scientific aspects of high voltage cable systems. This involves design and manufacturing of cables, cable insulation material, electric and magnetic field distribution, cable bonding, concerns related to cable termination and cable joints, diagnostics and monitoring, failure modes of the cable systems, and cable testing. Course contents and description: The course starts with an introduction to cable systems, their evolution, and current state of the art. Various cables used in practice (e.g. land cables, submarine cables, superconducting cables), the insulation material and their properties will be discussed. An overview of the cable manufacturing process is given. Gas Insulated transmission line (GIL), as an alternative for cable systems, will be overviewed. Technical and economic comparison between the two options will be made. The following topics will be discusses in detail during the course: Modeling of electric, magnetic, and thermal behavior of cable systems by means of circuit and FEM models. Concerns related to cable joint and termination Wave propagation in cables, cable bonding methods and cable installation Calculation of cable parameters. Partial discharge measurement and diagnostics and monitoring techniques used for cable systems. Aging mechanisms and life time evaluation of cables and cable systems. Failure modes of cable systems and cable testing procedures. Additional construction elements (mechanical issues, watertight, pulling facilities,).	
Study Goals	Note:the information provided in this page may change depending on the developments around the coronavirus To know the position of cables in electric transmission grids. To be able to describe different type of cable insulation, semiconducting and conducting materials and the manufacturing process of plastic and oil-paper impregnated cables To be able to calculate electric, magnetic and thermal characteristics of a cable. To be able to calculate cable parameters. To know the principal of wave propagation through cables To know different cable bonding techniques and their consequences. To be able to recall different cable testing and their purpose. To know the diagnostics techniques used for cable monitoring. To know the most important failure modes of cable systems and how to evaluate remaining lifetime of a cable system To be able to evaluate electric field distribution for a cable joint or termination using a FEM model	
Education Method	The course consists of three parts, i.e. lectures, computer modeling session and laboratory demonstration.	
Literature and Study Materials	The following references is used to construct the material of the course. Handout will be given after each session. [1] Cable systems for high and extra high voltage, Pirelli, E. Peschke and R. von Olshausen, 1999 [2] Electric cables handbook, third edition, BICC Cables, 1997 [3] The history of electric wires and cables, R.M. Black, 1983 [4] Power cable engineering lectures, Nexan, Dietmar Meurer, 2014 [5] Power cables, KEMA, E.F. Steennis, 2007 [6] Power and Communication Cables Theory and Applications. R. Bartnikas** [7] Rating of electric power cables in unfavorable thermal environment, George J. Anders.** [8] Electrical Power Cable Engineering, William A. Thue [9] Electrical cables for power and signal transmission, Oswald I. Gilbertson, 2000 [10] Extruded cables for high-voltage direct-current transmission, Giovanni Mazzanti, et al. [11] Submarine power cables, Thomas Worzyk [12] CIGRE Technical Brochures (89, 338, 531,) [13] Advanced power cable technology Volume I, Toshikatsu Tanaka, ALLAN Greenwood, 2000 [14] Advanced power cable technology Volume II, Toshikatsu Tanaka, ALLAN Greenwood, 2000 [15] Cable system transients, theory modeling and simulation, Akihiro Ametani, et al [16] Application of high temperature superconductor to electric power equipment, Swarn singh kalsi [17] Condition Assessment of High Voltage Insulation in Power System Equipment, R.E. James and Q. Su [18] Power cables and their application Part I, Lothar Heinhold, 1993 [19] Power cables and their application Part II, Lothar Heinhold, 1993 [20] Underground transmission of electric power, B. M. Weedy, 1980 [21] Rating of electric power cables, ampacity computations for transmission, distribution, and industrial applications, George J. Anders, 1997 [22] Underground power cables, King and Halfter, 1982 [23] Electromagnetic transients in power cables, Filipe Faria, et al	
Assessment	Written exam (50%), Project and assignment (50%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4103	High Voltage Technology	4
Responsible Instructor	Prof.ir. P.T.M. Vaessen	
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Practical Coordinator	I.M. Splinter	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course presents the basic principles of High Voltage technology and practical application of this knowledge in the power industry. Proper design of High Voltage components requires knowledge and insight of electric fields and breakdown mechanisms.	
Study Goals	By the end of the course students will be able to: - Demonstrate insight in the relation between charge, electric field and voltage - Identify and calculate capacitive voltage distributions - Calculate and analyze electric fields using analytical and numerical methods - Describe the breakdown mechanisms in different insulation materials and the phenomena leading to this - Explain methods of field grading and field control - Describe impulse breakdown phenomena (for example lightning) - Evaluate the effect of combinations of materials and interfaces on electric fields - Remember maximum permissible field strengths in different materials and configurations	
Education Method	Lectures, demonstrations	
Literature and Study Materials	Kreuger, F.H., Industrial High Voltage. Vol.1, 2, 3, ISBN 90-6275-561-5 (available for purchase at the course), hand-outs (available at the course)	
Prerequisites	Electromagnetics (EE3P11) or equivalent courses	
Assessment	Written exam (70%) and project (30%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4108	Transients in Power Systems	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	EE2E21, ET4107 - knowledge of symmetrical component is highly recommended	
Course Contents	This course deals with the rapid phenomena in power systems caused by switching actions and atmospheric discharges. Some of the topics covered are circuit breakers, arc-circuit interaction, short-line fault, interruption of small inductive capacitive currents, travelling waves, atmospheric discharges, simulation of transients, circuit breaker testing.	
Course Contents Continuation	The subject deals with power system transients. It is divided in a few sections the goals of which are the students to be introduced by the basic concepts of switching transients and transient analysis as well as travelling wave phenomena. The subject also deals with the current interruption phenomena and circuit breakers as well as circuit breaker testing and insulation coordination. Additional material on Ferroresonance and surge arrester selections is being studied as a supplement to this project. All results are validated by computer simulations in Electromagnetic Transient Program (EMTP), which is used as a tool for detailed analysis of different cases like switching small inductive and capacitive currents, arc interruption, thermal and dielectric breakdown of the circuit breakers, travelling wave phenomena. Each chapter is also supported by several simplified examples (which are solvable on the brightspace)	
Study Goals	After following this course the student should be able to: - model power system components for transient analysis by means of lumped elements. - to understand the current interruption process and gained insight in the testing of switchgear in the High Power Lab. - to use symmetrical components during transient processes in a network, especially when one pole of the circuit breaker opens and the other two remain in closed position. - to understand under which circumstances the worst case scenario (highest, steepest TRV) takes place. - to be able to assess the behavior of the lightning stroke when arriving and junction points connecting one or more components with different characteristic impedance Based on aforementioned learning objectives, the student should be able to simulate at least simplified cases of switching and lighting transient phenomena in Electromagnetic Transient Program ATP-EMTP or other EMTP-based software environment.	
Education Method	Lectures (*). Students should spent approximately ten hours per week in order to achieve the learning goals well. Each chapter, if suitable, is also supported by solvable examples on the board and simplified/detailed simulation cases performed in Electromagnetic Transient Program (EMTP) environment. Students also get opportunity to obtain the EMTP royalty free software and to learn EMTP so that they can use it as a tool for different project cases or M.Sc thesis. The subject also provides opportunity for extra projects in which students may obtain credits depending on the amount of work done. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	L. vander Sluis, Transients in Power System, Wiley, ISBN 0 471 486 396. JA Martinez-Velasco, M Popov, Circuit breakers: Power System Transients Parameter determination, CRC Press, ISBN-13: 978-1420065299, pp. 447-556.	
Assessment	Additional material on ferroresonance phenomena will be provided by the lecturer Oral (closed book examination) which can be taken as written in case of more than 10 students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular transient case that occurs in practice. The amount of work required for particular problem solutions is in line with the number of points. The exam is passed when the student scores more than 57 %. Resit of the examination is possible according to the examination schedule. The content of the course material is divided into 9 chapters. All chapters weight equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalair - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteretic modulation - carrier modulation with feedback - scalair and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET8020	High Voltage Testing and Diagnostics	4
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Instructor	Ir. A.G.A. Lathouwers	
Instructor	C.S. Engelbrecht	
Practical Coordinator	I.M. Splinter	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The quality of the insulation of high voltage components (assets) e.g. transformers, cables, switch-gear, generators, is of large importance for the availability and reliability of power supply. Prior to service operation and in accordance with international standards all these components are heavily tested at different voltage stresses (HVDC, AC, Impulse). During the service life of 30 or more years local insulation defects (on mm-scale) occurring as a result of aging and degradation may produce a total blackout e.g. of a 3 km long power cable connection. To provide during continuous service high reliability level, insulation condition assessment by means of advanced and non-destructive diagnostics is very important. In particular, by means of advanced sensor techniques, signal processing, knowledge rules, the condition of the components has to be continuously or periodically monitored. The liberalization of energy sector, requests from the network providers an economic responsible technical management of the assets. In particular, based on the actual condition, the maintenance strategies have to be implemented. This course, supported by demonstrations, learns about the different voltage generation, diagnostics and monitoring techniques, weakest link statistics, knowledge rules generation, diagnosis support systems for asset management are shown.	
Study Goals	To obtain insight in the testing, diagnostics of HV components and be able to understand the processes of insulation-coordination and quality insurance by maintenance strategies. Within power utility, to be able to recognize and to understand the AM processes and to be able to implement methods to control these processes.	
Education Method	Lectures, demonstrations, practical	
Literature and Study Materials	Kreuger, F.H., Industrial High Voltage Vol. 4, 5, 6; ISBN 90-6275-562-3 (available for purchase at the course); Hand-outs (available at the course) Weibull analysis: handouts replace Ch.6 R. Ross, Reliability Analysis for Asset Management of Electric Power Grids, Wiley-IEEE, 2019, ISBN 978-1-119-12517-4, DOI:10.1002/9781119125204.	
Assessment	Oral, closed book, hand outs about Weibull Analysis- substituting Ch.6, and hand-outs about Asset Management Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Solar Energy 2023

Introduction 1	Profile: Solar Energy Solar Energy Track core for this profile: ET4119 Electronic Power Conversion (R) ET4376 Photovoltaic Basics (C) EE4585 Semiconductor Device Physics (C) Solar Energy Specialisation core (select at least 12 EC): ET4377 Photovoltaic Technologies (C) ET4378 Photovoltaic Systems (C) ET4379 Photovoltaic Lab Course (R) EE4670 PV materials processing & characterization (R) EE4680 Photovoltaic Modelling (R) EE4545 Electrical Power Systems of the Future (R) EE4536 DC and AC Microgrids (R) (C)= compulsory (R)= recommended
Introduction 2	The profile Solar Energy addresses all aspects of photovoltaic technology, from the design of a PV cell to the implementation of PV systems in the electricity grid. Students are free to compose their own specialisations by choosing from a list of courses, in consultation with their thesis supervisors. Responsible professor: - Prof. dr.ir. Miro Zeman

EE4415	Photovoltaics	4
Responsible Instructor	P. Manganiello	
Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Photovoltaic Basics course ET4376	
Course Contents	This course will focus on the new research field of Photovoltaics and how to create multi-functional PV-based intelligent energy agents (PV-IEAs), able to optimally generate electricity and actively participate to communication. It will start with a general introduction to Photovoltaics and an overview of its different research areas. The issue of partial shading and non-uniform operating conditions of PV modules and systems will be discussed, and solutions at cell (low-breakdown voltage PV cells and integrated power electronic components), module (shade-tolerant architectures) and system (distributed maximum power point tracking) level will be presented. Methods and algorithms to optimize the operation of such solutions will be introduced. The dynamic behavior of solar cells and how it can be exploited to design PV-IEAs able to serve as receivers of information in optical wireless communication systems will be discussed, and equivalent DC and AC models of PV devices will be presented.	
Study Goals	By the end of the course, the student should be able to: Explain the effect of non-uniform operating conditions on the performance of PV modules and systems. Analyze shade-tolerant PV modules and systems. Analyze different control algorithms for shade-tolerant PV modules and systems. Understand the dynamic behavior of PV cells under real-world operating conditions and the underlying physical processes. Analyze PV-IEAs that use solar cells as data receivers in optical wireless communication links.	
Education Method	Lectures and Instructions	
Literature and Study Materials	Course Handouts Research articles Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	Written exam (closed book) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4536	DC and AC Microgrids	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	L.M. Ramirez Elizondo	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatziaargyriou	
Assessment	Written exam 70%, Assignments 30% In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTS) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4670	PV Materials Processing and Characterization	4
Responsible Instructor	Prof.dr. A.W. Weeber	
Instructor	Dr. L. Mazzarella	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3 Different, to be announced	
Course Language	English	
Course Contents	The course is a follow-up of PV Technologies. Details of processing and characterization techniques of PV absorber materials and the solar cells functional coating will be discussed. Examples of lecture topics are advanced light management structures, novel surface passivating layers and contacting schemes. Background of deposition technologies such as Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD) and Atomic Layer Deposition (ALD) will be given. Furthermore, working principles of special coatings and its functionality when applied to different solar cell concepts, including tandem devices, will be presented. This course is an elective one following the solar profile. PV Basics and PV Technologies should have been concluded successfully.	
Study Goals	After this course students are able to understand processing mechanisms, material properties, and functionality and working principle of the novel functional coating applied to different solar cell concepts. Students are expected to be able to explain these working principles and have a solid background on different deposition technologies and characterization techniques, and will be able to describe the effect on the solar cells output. Students will be able to analyse and evaluate recent technology developments and the background of it, and summarize them during an oral presentation for the entire group. As we can see, these learning objectives are specific about the content, and they are phrased in terms of observable behaviour: understand, explain, evaluate, present and describe.	
Education Method	This course consists of eight 2-hour lectures. In the same week a 2 hour work session / instruction will be organized for a deeper understanding of the technologies discussed in the lectures. During the whole course the students will work on assignments which need to be submitted as reports. All assignments/reports should be assessed as sufficiently good (no specific mark). In week 9 (final schedule will be given at the beginning of the course) you will present results of a literature study on a special topic covered by the course for the entire group.	
Literature and Study Materials	Book (not obligatory, expensive): Photovoltaic Solar Energy: from fundamentals to applications by Angèle Reinders et al. Material from online course: PVCDROM, PV-manufacturing.org Slides during lectures Articles (referred to in lectures)	
Assessment	The final grade consists of the following parts: presentation (team work) (50% of final grade); written exam (individual) (50%); Both marks will be expressed on a scale of 1 to 10 and should be 5.80 or higher. Reports on the assignments shall be completed as well, and assessed as sufficiently good (no specific mark). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
maximum aantal deelnemers	Expected number of student: Maximum of 30. Use Brightspace and inform our secretary Mark Vielvoije (m.vielvoije@tudelft.nl) for attending the course.	

EE4680	Photovoltaic Modelling	4
Responsible Instructor	Dr.ir. R. Santbergen	
Instructor	Prof.dr.ir. O. Isabella	
Responsible for assignments	Dipl.-Ing. Y. Blom	
Co-responsible for assignments	F.S. Saitta	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Modelling is an important tool for gaining deeper insight in the working principles and loss mechanisms of solar cells and photovoltaic systems. Researchers use this insight to develop more efficient solar cells and to maximize the energy yield of photovoltaic systems. In this course we will cover the entire photovoltaic modelling chain: modelling of photovoltaic materials, devices (solar cells), modules and systems. The focus will be on understanding the working principles of these models, and using them to analyse the effects of design parameters on solar cell and photovoltaic system performance. The online course material consists of two parts, each consisting of several phases: 1) PV materials and device modelling 1.1) PV materials and device characterization 1.2) optical modelling of PV devices 1.3) electrical modelling of PV devices 2) PV systems modelling 2.1) modelling of irradiance 2.2) modelling of energy yield In addition there will be software demos, given by expert users, allowing you to familiarize yourself with the advanced modelling software commonly used by PV researchers. The prior knowledge from the courses PV basics and PV systems is required. This course PV modelling will prepare you for an MSc thesis project that involves modelling of solar cells or photovoltaic systems, as well as a future career in photovoltaics research.	
Study Goals	The online course material consists of two parts: 1) PV materials and device modelling, 2) PV systems modelling. Each consist of several phases: At the end of phase 1.1 (characterization of PV materials and devices) you should: - Be familiar with the main techniques used in the characterization of PV materials and devices - Understand the importance of a correct characterization for the modelling of PV - Be able to extract useful and important information (trends and properties) from the characterization of both PV materials and devices At the end of phase 1.2 (optical modelling of PV devices) you will; - have a complete overview of the modelling approaches commonly used for photovoltaic optical modelling, and which are their strength and limitations - get a thorough understanding of different optical modelling methods and which structures they are best suited to model - improve and practice your skills on optical modelling, by carrying out activities with our own GenPro4 software At the end of phase 1.3 (electrical modelling of PV devices) you should be able to: - have a complete overview of the modelling approaches commonly for photovoltaics, both optical and electrical - get a thorough understanding of different electrical modelling methods - improve and practice your skills on optical and electrical modelling, by carrying out activities with our own advanced semiconductor analysis (ASA) software. At the end of phase 2.1 (modelling of irradiance) you should be able to: - Gather and analyse meteorological data, including the utilization of geographic information systems, - Obtain information on horizon and shading at a specific location, and derive the relevant location parameters, - Combine meteorological data and location properties to calculate the irradiance on the angle of incidence with different levels of detail, and - Optimize the orientation of a module to maximize the irradiance on the angle of incidence. At the end of phase 2.2 (modelling of the energy yield) you should be able to: - Model the operating temperature of modules, - Obtain information from module datasheet, - Combine module properties with temperature and irradiance modelling to calculate DC power (and yield), - Model the performance of inverters (DC to AC) and batteries, - Analyse load profiles, and use them to size a PV system (module and batteries), and - Calculate the AC power and energy yield of an entire system, and analyse its performance from an economic point of view. After the software demos you will be able to explain the input, working principles and output of the following commonly used software packages: - Complete EASE (extracting optical constants from spectroscopic ellipsometry measurement of PV materials) - HFSS (rigorous solving of Maxwell equation for EM-wave (light) propagation in PV devices) - Sentaurus (modelling charge transport in PV devices by solving semiconductor equations) - Radiance (PV module plane of array irradiance calculation using backward ray-tracing) - PV Syst (PV system energy yield calculation)	
Education Method	This course consists of recorded video lectures, interactive lectures and instructions. The video lectures provide you with the basic information. The interactive lectures give you the opportunity to ask questions to the teacher and discuss selected topics in depth. In addition you will work on modeling assignments. During the instruction sessions the modeling assignments will be explained and feedback will be given on your (preliminary) results.	
Assessment	Every week you make and submit modeling assignments, which are graded. The weighted average counts for 70% of the final grade. In addition, every week you write a short report about the software demo, which is graded as well. The average of this counts for the remaining 30% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Remarks	The on-campus Q&A session is not mandatory. Because the course material (video lectures, live demos) is given online, this course can also be taken by students who cannot physically come to the campus. Because the assessment is done through weekly assignments, without an exam, the final grade (and credits) can be obtained already before the exam period (end of October). This can be relevant for those who have not yet obtained sufficient credits to start their graduation project.
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ET4119	Electronic Power Conversion	4
Responsible Instructor	S.A. Rivera Iunnissi	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	EE1P21, EE1C11, EE2E11, EE2M21	
Course Contents	This course covers the principles of power conversion with switch-mode circuits. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated and isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain the circuit diagram and their waveforms; to be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practical	
Literature and Study Materials	N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition. (Available at ETV at reduced price)	
Assessment	Written exam (closed book) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carriers, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovoltaic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Instructor	Dr. L. Mazzarella	
Instructor	Prof.dr. I.M.F. Gordon	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Note: In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	

ET4379	Photovoltaic Lab Course	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Responsible Instructor	Dr.ir. R. Vismara	
Contact Hours / Week x/x/x/x	4/0/0/0 Q1=Q5! Also possible online in Q6, Q7, Q8!	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The aim is to offer you an advanced practical course on solar cells and photovoltaic systems. The following areas of practical training will be part of the course: (1) Characteristics of artificial and natural light (2) Characterization of the performance of lab-scale solar cells (3) Behavior of photovoltaic modules in different temperature and shading conditions (4) Charge and discharge properties of batteries (5) Performance of power electronic components (charge controllers, inverters, maximum power point trackers) (6) Operation of entire photovoltaic systems	
Study Goals	(1) Experiment with photovoltaic systems and all of their components under lab and real-life conditions. (2) Describe the working principle behind each component of a photovoltaic system. (3) Analyze the operation of all the elements of photovoltaic systems and of the system as a whole. (4) Compare the performance of different types of components of photovoltaic systems.	
Education Method	Laboratory sessions	
Literature and Study Materials	Manuals and templates are available on BrightSpace.	
Prerequisites	This course is meant for students who have already passed at least the PV Basics course (ET4376) and preferably also PV Technologies (ET4377) and PV Systems (ET4378). This course is also meant for 1st-year masters students with a bachelor degree in electrical engineering.	
Assessment	The assessment is based on reporting results from the individual tasks, using an Excel template provided on Brightspace. Two-person teams will be formed at the plenary session, the attendance of which is compulsory. The final grade is per team. The final grade is the average of the six technical reports. A sufficient grade (greater than or equal to 5.8) in all six technical reports is required to complete the course successfully. In case of failure of a technical report, you can use buffer sessions on week 1.7 and 1.8 to repeat the experiments. --- A virtual version of this course is made available for students to attend in Q2, Q3, and Q4 of the academic year 2023-2024. Each of the six tasks of the on-campus course will run in the internet browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, using numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to pass the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. Attending one of these instances of the course, no teams will be formed and the final grade is per person. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	The on-campus course can accommodate up to 48 students. Anyone interested in attending this course must apply during Q4 of the immediately previous academic year. A selection will then take place and, before the beginning of the instance in Q5, the line-up of students will be announced. Student(s) who failed on-campus in Q5 can re-take the virtual version of the course. There is no limit to the number of students who can attend the course's virtual version.	
Remarks	This practical course will be given on campus only in Q1. Conversely, a virtual version will be available for students to attend in Q2, Q3, and Q4. The enrolment is open on Brightspace from Q1 of the academic year 2023-2024.	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Power Electronics and Electrical Machines 2023

Introduction 1	Power Electronics and Electrical Machines Track core for this profile: EE4585 Semiconductor Device Physics (R) ET4117 Electrical Machines and Drives (C) ET4119 Electronic Power Conversion (C) Power Electronics and Electrical Machines Specialisation core (select at least 12 EC): ET4121 A.C. Machines (R) EE4375 Finite Element Modeling for Electrical Energy Applications (C) EE4515 Advanced Power Electronics (R) ET4145 Power Electronic Components (R) ET4116 Power Electronics (C) ET4291 Control of Electrical Drives (R) (C)= compulsory (R)= recommended Introduction 2 This specialisation profile covers electrical power processing and conversion, with an emphasis on improving efficiency in industrial processes, offices, homes and e-mobility. It also covers the operation and design of electrical machines. Responsible professor/head of research group: - Dr.ir. Pavol Bauer
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EE4375	Finite Element Modeling for Electrical Energy Applications	4
Responsible Instructor	Dr. D.J.P. Lahaye	
Contact Hours / Week x/x/x/x	0/0/2/0 and 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	We expect the following prior knowledge 1) linear algebra: concepts of vector, matrix, matrix-vector multiplication, solving linear systems, computing eigenvalues and eigenvectors of a matrix; 2) calculus of functions of single variable: concepts of derivative of a function, finite difference approximation, integral of a function, quadrature, ordinary differential equations; 3) calculus of functions of more variables: concepts of partial derivative of a function, surface and line integral, quadrature, partial differential equations; 4) electric and magnetic fields: concepts of electric field, magnetic field, Maxwell equations, electrical machines and transformers;	
Course Contents	This course consists of the following three blocks: 1) Block 1 (Week 1 and Week 2): In this block, the finite difference method for the Poisson equation on the interval (Week 1) and the square (Week 2) will be discussed. The differential equation and the boundary conditions will be discretized on a mesh by a finite difference scheme. This will result in linear equations for the grid unknowns. A stencil notation for these linear equations will be introduced. A linear system will be formed and solved. The discrete solution will be visualized on the mesh. Properties of the discretization method and the linear system will be discussed. 2) Block 2 (Week 3 and Week 4): In this block, the Galerkin finite element method for the one-dimensional (Week 3) and two-dimensional (Week 4) Poisson problem will be introduced. The problem in so-called strong form will be transformed into the problem in so-called weak or variational form. The problem in weak form will then be discretized in space using Lagrangian basis or shape functions. A linear system represents the Poisson equation on a discrete level will be assembled and solved. The finite difference and finite element solution methods will be compared. 3) Block 3 (Week 5, Week 6 and Week 7): In this block, the application of the finite element will be extended to treat more complex problems. Partial differential equations for the scalar and vector potential for the magnetic flux or field will be derived from the Maxwell equations. More complex geometries that represent transformers, actuators or rotating extrical machines will be meshed using unstructured triangular meshes. The Maxwell equations will be discretized on these meshes. The discrete linear system will be assembled by a element-by-element procedure and solved. The computed vector potential solution will be processed to compute the magnetic flux and the magnetic field.	
Study Goals	After successfully completing the course, the student will be able to: * formulate various boundary value problems for the one-dimensional and two-dimensional Poisson equation; * derive these boundary value problems from the Maxwell equations for the magnetic field by introducing the scalar and vector potential; * apply the finite difference and finite element method to solve these boundary value problems on a scale of problems of increasing complexity; * state properties of the finite difference and finite element method applied to Poisson boundary value problems; * describe the implementation of the finite difference and the finite element method in software; * give a physical interpretation of the numerical solution obtained for magnetic field problems;	
Education Method	Weekly lectures will be complemented by weekly practical lab sessions and three home work assignments. During the lab sessions, algorithms discussed in the lectures will be implemented, tested and further refined. Each homework assignment will correspond to a course block. The set of three assignment will form a portfolio.	
Reader	see http://ta.twi.tudelft.nl/nw/users/domenico/electro_fem/electro_fem.html	
Assessment	During the course, the student will build a portfolio of three home work assignments, each correspond to a block in the course. The objective of these assignments is to guide the students towards the implementation of a proper finite element code. Each assignment consist of an elective and a compulsory part. This portfolio will be assessed during the oral examination. During this examination, the student will be assessed on the theoretical foundation of finite difference and finite element method, on the application of these methods and their implementation in software. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4515	Advanced Power Electronics	4
Responsible Instructor	Dr. Z. Qin	
Instructor	Dr. T. Batista Soeiro	
Contact Hours / Week x/x/x/x	0/0/0/3 + lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is expected that the student has taken all the previous courses in the power electronics teaching line: ET4119, ET4116, ET4145	
Course Contents	The focus of this course is on providing the theoretical background of various design related aspects of power converters not covered in the previous courses. The topics covered include: switching behavior and loss modelling, soft switching vs hard switching, passive components, thermal management and control. The lab projects integrate the knowledge from this course and the preceding courses in a design assignment of designing, building and testing an efficient and small power converter.	
Study Goals	After this course, the students will be able to: - design a transformer/inductor - design the control - design the cooling - analyse and improve the efficiency - design, build, and test a converter with small size and high efficiency	
Education Method	Lectures and lab project	
Literature and Study Materials	Lecture handouts N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition.	
Assessment	projects (80%) and oral exam (20%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4755	Reliable Power Electronic Components and Systems	5
Responsible Instructor	A. Shekhar	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Following this course is only possible with course ET4119 or SET3095 in advance	
Course Contents	Understanding of Power electronic components: diodes, BJT's, MOSFET's, thyristors, IGBT's; Gate drive circuits; High frequency inductors; Semiconductor's thermal management and heatsinks. The course also deals with conducted electromagnetic interference in power electronic circuits. Finally, the course will discuss component reliability and its corresponding influence on the power electronic systems	
Study Goals	THEORY: Understanding the theoretical and practical operation of power electronics components (semiconductors and passives) based on the physical structure, the V-I characteristics, time and thermal behavior. PRACTICAL: Analysis of Boost-type Converter and the Design of components, including semiconductors, inductor and capacitor. The ability to perform (differential) voltage-, current and temperature measurements on these components via a remote controlled test setup and explain these measurements at the hand of theory. Students will be given a demo on reliability tests using power cycler in the PEARL lab and asked to discuss their observations on the obtained test result data.	
Education Method	Lectures, practicals	
Literature and Study Materials	TEXT BOOK: Mohan, Undeland and Robbins, Power Electronics, Converters, Applications and Design, ISBN:978-0-471-22693-2, 824 pages, 3rd edition, John Wiley & Sons, 2003. ADDITIONAL READING: Provided via Brightspace DATA SHEETS: Provided via Brightspace and hard-copy in the lab. HARDWARE: Provided in the lab sessions.	
Assessment	Written EXAM (40%) DESIGN REPORT of your inductors DUO PRESENTATION on design- and practical results (20%) FINAL REPORT on practical (40%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalair - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteric modulation - carrier modulation with feedback - scalair and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Responsible Instructor	H. Vahedi	
Instructor	G.R. Chandra Mouli	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4119	Electronic Power Conversion	4
Responsible Instructor	S.A. Rivera Iunnissi	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	EE1P21, EE1C11, EE2E11, EE2M21	
Course Contents	This course covers the principles of power conversion with switch-mode circuits. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated and isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain the circuit diagram and their waveforms; to be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practical	
Literature and Study Materials	N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition. (Available at ETV at reduced price)	
Assessment	Written exam (closed book) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4121	A.C. Machines	4
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	Control of Electrical Drives (ET4291)	
Expected prior knowledge	Electrical machines and drives (ET4117)	
Course Contents	The following subjects are discussed: - winding function and inductances; - time and spatial harmonics; - skin effect in electrical machines; - phasor, space vector and transformation in electrical machines; - the salient pole synchronous machine; - modeling and simulation of the dynamic performance of DC and AC machines.	
Study Goals	After following this course the students should be able to - describe and calculate the steady-state performance of AC machines; - calculate spatial harmonics and inductances of given winding distributions; - derive the dynamic models of DC and AC machines using space vectors; - model and simulate the dynamic performance of DC and AC machines; - evaluate the limitations of the dynamic models.	
Education Method	Lecture / Computer Simulation / Discussion	
Literature and Study Materials	- A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill. - lecture notes on Brightspace	
Assessment	Oral exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	- Presence at the computer simulation session is obliged. - Submission of assignments on Brightspace is obliged.	

ET4291	Control of Electrical Drives	5
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	practicum 0/0/0/4 vanaf week 2, instructie 0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Course Contents	- Dynamic modelling of DC and AC Machines supplied with power electronic converters. - Modelling of mechanical systems and thermal effects in drives. - Modelling and simulation of digital control of electrical drives. - Controller design for cascaded motor controller. - Scalar and vector control, field weakening control. - Practical design of an electrical drive system.	
Study Goals	Students should be able to: - Model the dynamic behaviour of DC and AC machines using state space representation. - Integrate ideal power electronic converter models and cascade control systems in the drive model. - Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive. - Calculate the steady state performance of DC and AC drives. - Implement scalar control system and vector control system for induction machines. - Implement vector control system for synchronous machines. - Implement field weakening control for DC and AC drives. - Implement the dynamic simulation models in MATLAB/Simulink. - Explain the practical design of electrical drives. - Evaluate the performance of an electrical drive system. - Design the controllers in electrical drive.	
Education Method	Lecture / Self-study / Simulation practical / Discussion	
Literature and Study Materials	- Rik De Doncker, Duco W.J. Pulle, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: https://www.springer.com/gp/book/9789400701816 - Self-study materials on Brightspace.	
Reader	"Reader Control of Electrical Drives ET4291 Version maart 2009"	
Prerequisites	Electrical Machines and drives (ET4117) Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Assessment	Individual assignment and oral exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3100	Electric vehicle and charging technology	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Strongly advised: SET3095/ET4119 Electronic Power Conversion ET4117 Electric machines & drives	
Course Contents	This course focusses on the technology of (hybrid) electric vehicles and its charging technology Tank to Wheel, Well to Tank, Well to Wheel efficiencies and emission estimation for conventional, hybrid and fully electric vehicles based on energy source for propulsion. Types, operational principles, advantages and disadvantages of vehicle architectures: internal combustion engine; micro, mild and full hybrids (series, parallel, series-parallel), plug-in hybrids, full electrics, fuel cell vehicles; Vehicle load forces & dynamics, mechanical transmission components, power and energy consumption, drive cycles, battery sizing and electric vehicle range, MPGe, planetary gears as a power-split, regenerative braking. Principles of battery operation, battery types, the key performance parameters of batteries, from cell to battery pack, charge-discharge curves, battery and thermal management systems. Fundamentals and types of electric machine drives used in electric cars: its characteristics, efficiency, basics of motor control. Basics of power electronic converters used in electric cars and chargers, topologies and control. Charging methods of EVs: AC charging, DC charging, vehicle-to-grid technology, wireless charging, battery swap; charging time estimation, effects of fast charging on battery; Smart charging of EVs and vehicle-to-grid, communication protocols for smart charging. Modelling and investigating the performance of the electric vehicle drivetrain using quasi-static simulation in MATLAB/Simulink. Case study of EV: Toyota Prius HEV, Tesla Model S, Porsche Taycan, Audi e-tron, Hyundai Ioniq 5 The course will have a design project where students have to form teams to design their own EV by making choices for the vehicle drivetrain and corresponding charging technology. Emphasis will be on justifying and quantifying the technological design choices including drivetrain architecture, well-to-wheel efficiency and emissions, energy storage, power electronics, type of motor and charging method.	
Study Goals	The goal of this course is to educate students to design and evaluate various (hybrid) electric vehicles with respect to their performance, efficiency, sustainability, charging methodologies and potential for grid support. At the end of the course, the students should be able to: Evaluate the performance of hybrids (series, parallel, series-parallel), plug-in hybrid and fully electric drivetrain architectures based on power and energy consumption, efficiency, driving range, well-to-wheel emissions and charging time. Explain the design trade-offs in the energy sources (battery, fuel cell, super capacitors) and motors used in electric vehicles based on the key performance indicators like efficiency, energy and/or power density, lifetime, and control. Analyse various charging technologies for electric vehicles (conductive, inductive or battery swap) based on power electronics converters used, type of the charger (AC Vs DC), connector types, power levels, and modes. Investigate the performance of the (hybrid) electric vehicle drivetrain using quasi-static modelling and simulation in MATLAB/Simulink. Design a (hybrid) electric drivetrain by sizing and making the relevant choice of the electric machine, internal combustion engine, energy source (batteries, fuel cells, etc.), power electronic converters, transmissions components, and charging technology for a given vehicle specification.	
Education Method	Classroom lectures and problem solving workshops, Discussion and case study, Demonstrations using tools and animation in the classroom, Online exercises with self-evaluation, Design project	
Literature and Study Materials	- Lecture slides & handouts - Primary reading - Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, Third Edition, 2018, Mehrdad Ehsani, Yimin Gao, Ali Emadi, CRC Press, ISBN 9781138330498 - Supplementary reading - Chapter 1 & 14 of Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, ISBN: 978-1-119-06364-3, 2018 - Supplementary reading Electric and Hybrid Vehicles: Design Fundamentals, Second Edition, 2010, Iqbal Husain, CRC Press, ISBN 9781439811757 The enlisted books are available via the TU Delft library (free e-books), via the ETV desk & bookstore, via studystore.nl and the publisher website	
Assessment	The final grade of the course consists of the following components: - Written exam (70% weightage) - Closed book, Scientific (non-graphic) calculator can be used. No formula sheets - Design assignment (30% weightage) - Group assignment with Written report + oral presentation. Students must register in groups for the design assignment within the first two weeks of the course. Final grade calculation: $(0.7 * \text{Written exam} + 0.3 * \text{Design assignment})$ In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Written exam: Resit in next quarter Assignment: Re-submit assignment report in the next quarter based on feedback received (Grade is capped) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Follow up course: ET4291SET Digital modelling of electric powertrain/ ET4291 Control of Electrical Drives	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Intelligent Electric Power Systems 2023

Introduction 1

Intelligent Electric Power Systems Track core for this profile:

ET4119 Electronic Power Conversion (C)
ET4117 Electrical Machines and Drives (R)
ET4108 Transients in Power Systems (C)
EE4C12 Machine Learning for Electrical Engineering Applications (R)

Intelligent Electric Power Systems Specialisation core (select at least 12 EC):

ET4378 Photovoltaic Systems (R)
ET4116 Power Electronics (R)
ET4113 Power System Dynamics (R)
ET4107 Power Systems Analysis II (R)
EE4536 DC and AC Microgrids (C)
EE4545 Electrical Power Systems of the Future (C)
EE4530 Applied Convex Optimization
EE4655 Co-simulation of Energy Systems
EE4665 Uncertainty modelling and risk assessment in electrical power systems

(C)= compulsory
(R)= recommended

Introduction 2

This profile focuses on the generation, transmission and distribution of electrical energy, and it explores technological options for the design and operation of future intelligent power grids, which will be subject to sustainability and reliability constraints. The topics of microgrids, e-mobility and smart cities are also covered within this profile.

Responsible professor/head of research group:

- Prof. dr. Peter Palensky
- Dr.ir. Pavol Bauer

EE4410	Cyber Security of Power Grids	4
Responsible Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	The energy system transition towards a sustainable, low carbon future is led by decarbonisation, decentralisation, and digitalisation. Power grids are undergoing a fast-paced process of digitalisation for enhanced monitoring and control capabilities and grid intelligence. However, the increased integration of digital technologies, such as the next generation of Operational Technologies (OTs) and digital substations, implies a new risk as Information Technology (IT) OT systems for smart grids are vulnerable to cyber attacks. Furthermore, the combination of heterogeneous, co-existing smart and legacy technologies generates significant vulnerabilities and cyber security challenges. Examples of cyber security incidents related to power grids already exist around the world. On December 23, 2015 cyber attacks were conducted on the power grid in Ukraine that resulted in power outages, which affected 225,000 customers. More sophisticated cyber attacks on the Ukrainian power grid followed on December 17, 2016 resulting in a power outage in the distribution network where 200 MW of load was unsupplied. The complexity of cyber attacks on power systems is on the rise. They can initiate cascading failures and result in a catastrophic blackout. A power outage can disrupt the entire energy chain including water supply, heating, and gas networks. Without power, hospitals and other critical services are severely affected. A disruption of service may lead to financial loss, damages, chaos, or even a loss of lives. With respect to security of supply and reliability of the future energy system provision, special attention is needed for new vulnerabilities and cyber threats that come with digitalisation. Accordingly, the cyber security aspects are critical for a further power grid digitalisation.	
Course Contents	This course aims to assess and improve the cyber security of smart grids. It discusses the stakeholders of and complexities in critical infrastructures such as electrical power systems. It analyses the typical IT-OT system architectures for grid monitoring and control and explains the main cyber security concepts and policies. The course provides essential knowledge to analyse cyber security threats, threat vectors, and vulnerabilities of IT OT systems for power grids. It examines the kill chain stages of the cyber attack process on industrial control systems. The course analyses the impact of cyber attacks on power system operation in terms of loss of load, equipment damage, cascading effects, and risk of a blackout. The course studies the state-of-the-art security controls and best practices for threat intelligence management and incident response to defend against cyber attacks and effectively mitigate their impact on power system operation. This course focuses on cyber security of Information Technology (IT) Operational Technologies (OT) networks used for power system operation. It starts with an overview of the smart grid and its industrial control systems (EMS/SCADA) and introduces computer networks and cyber security. Next, the course will introduce the cyber kill chain method for cyber security investigation and intelligence-driven defence. Kill chains are used to understand, anticipate, recognise, and combat cyber attacks and advanced persistent threats. The course uses the kill chains of the 2015/2016 cyber attacks on Ukrainian power grid as real-world study cases throughout the course. The main IT-OT system vulnerabilities and cyber threats are analysed. The course covers threats such as passive and active reconnaissance, social engineering, and malware. It investigates various threat vectors on host and network security and advanced cyber attacks targeting power grids. The impact on power system operation is analysed. The course discusses the state-of-the-art security controls to defend and mitigate cyber attacks on power grids such as access controls, authentication management, cryptography, firewalls, secure communication protocols, network segmentation, and intrusion detection and prevention systems. The best practices to improve power grid resilience to cyber attacks are discussed, i.e., threat intelligence management, incident response, and wide area monitoring, protection and control. This course provides a deep dive into core cyber security concepts for power grids such as cryptography (symmetric and asymmetric encryption), hash functions, digital signatures, and cyber-physical anomaly detection using machine learning techniques. Cyber security demonstrations on power grids will be conducted for students in Control Room of the Future (CRoF) technology centre.	
Study Goals	At the end of the course, the students will be able to: LO1: Analyse cyber security threats, threat vectors, and vulnerabilities of Information Technology (IT) Operational Technology (OT) systems for power grids. LO2: Examine kill chain stages of the cyber attack process on cyber-physical power systems. LO3: Analyse the impact of cyber attacks on power system operation. LO4: Recommend appropriate security controls to detect, prevent and mitigate cyber attacks and cyber secure the future power grids.	
Education Method	Lectures: theory and real-world study cases (8 lectures x 2h/lecture) Practicums: hands-on computer labs (5 practicums x 2h/practicum)	
Literature and Study Materials	Chwan-Hwa Wu & J. David Irwin. (2013). Introduction to computer networks and cybersecurity. CRC Press. ISBN: 9781498760133 Nigel P. Smart. (2016). Cryptography made simple. Springer, 2nd Edition.	
Assessment	The grade is formed based on the following elements: 40% Assignment reports (five assignment reports submitted in time on Brightspace) 60% Final digital (computer) exam All assignments and reports are mandatory. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4536	DC and AC Microgrids	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	L.M. Ramirez Elizondo	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatziaargyriou	
Assessment	Written exam 70%, Assignments 30% In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTS) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

EE4655	Co-simulation of Energy Systems	4
Responsible Instructor	Dr. M. Cvetkovic	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Some experience with programming in languages such as Matlab, Python, Java or C/C++ is needed. Basic knowledge of power systems is preferred although not necessary.	
Course Contents	The course introduces simulation methods and models that are used in the field of energy system integration and shows how these simulation methods and models can be combined into so called co-simulation in order to answer energy integration questions. The course consists of three themes which run in parallel and complement each other. In the first theme, we review the simulation methods (time-domain simulation, discrete simulation, equilibrium simulation, etc.) and modeling approaches (optimization models, agent-based models, etc.). We explain when and how these methods and models can be applied to answer engineering questions in the field of power and energy systems (stability, reliability, control, regulatory analysis, market design, etc.). Each method is supported with the examples in the common engineering tools (Matlab, Modelica, etc.) In the second theme, we introduce the co-simulation methods which can be used to link the simulation methods together. The important aspects are discussed, such as synchronization between simulators (serial-Jacobi, parallel-Gauss-Seidel) and the choice of information to be exchanged between simulators. The examples of these methods are made using the educational co-simulation environments. These environments are Matlab and Python based which gives students the opportunity to learn the basics of Python programming and to strengthen their Matlab skills. In the third theme, the students work in groups on a co-simulation project. The main goal of the project is to create design specs and implement a co-simulation for one energy system integration challenge. In this way the students get practical knowledge in interpreting an energy system integration challenge in terms of simulation tools and models needed to analyze it. The instructor provides a list of available project topics. The students have sufficient freedom to develop their own co-simulation solutions following good design practice introduced in the class.	
Study Goals	The aim of the course is to educate the MSc students on how to approach simulation and modeling of energy integration challenges. The integration challenges are problems that require system of systems thinking and often involve composition of diverse subsystems and corresponding simulation tools. For example, the impact of electric vehicles on the electricity grid can be accurately simulated only if both, the electricity grid and the drivers behavior, are modeled and simulated. Another example is development of management algorithms for the combined heat and power (CHP) plants which feed both electricity grid with electric energy and the heat grid with excessive heat. The third example comes from smart grids, where the latency of wide-area controllers might impact the proper operation of remedial action schemes. Each of these examples is a cross-domain example that requires: 1) a system of systems thinking and the compositional approach to model/simulation development; 2) awareness of available simulation environments and the means of interconnecting them. The objective of the course is to 1) introduce different integration challenges, 2) bring system thinking closer to students, and 3) give students practical knowledge on model/simulation composition. Learning objectives: By the end of the course the students should be able to: 1) Describe some of the most important energy integration challenges, 2) Model these integration challenges by composition of existing modeling concepts and tools 3) Have basic practical knowledge on composition of adequate simulations tools using co-simulation	
Education Method	The course consists of lectures, assignments and a project. The lectures introduce different topics while the project asks of students to apply the material studied in the lectures. The exercise assignments are given each week to practice the concepts from the lectures before integrating them in the project. Project milestones are organized several times during the course in order to monitor and reflect on the progress of the project. Disclaimer: keep in mind that the course might go through the adjustments depending on the developments around the coronavirus. In that case, the study guide information may be adjusted.	
Literature and Study Materials	Supporting scientific papers, tutorials, and notes are published on Brightspace each week.	
Assessment	The grade is formed based on the student and team performance. Five aspects of the project are evaluated to obtain the final project mark: project context, co-simulation requirements, energy system/component models, co-simulation architecture and the co-simulation results. This mark is complemented with student's performance in the oral exam at the end of the course, upon which, the final mark is formed. In order to successfully complete the course, the students need to score a passing grade on all project aspects and during the oral exam. All students must participate in completion of milestones. Final report is a document that combines content of all intermediate milestones. Submission of the final report is mandatory for the passing grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4665	Uncertainty modelling and risk assessment in electrical power systems	4
Responsible Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ir. J.L. Rueda Torres	
Co-responsible for assignments	E. Sharifnia	
Co-responsible for assignments	K. Bölat	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Elementary knowledge of probability theory and statistics (discrete and continuous probability distributions, expectation/variance/standard deviation, quantiles, least squares regression). Basic programming skills (examples code samples will be provided in Python, but other languages may be used for hand-in exercises). Working knowledge of power system structure, components and actors.	
Course Contents	This course is aimed at MSc EE students in the EPE track and MSc SET students with the Power profile. This course considers the challenges faced by operators when operating and planning electrical power systems in the face of uncertainty balancing cost, reliability and environmental objectives. The course specifically focuses on quantitative modelling of uncertainty and performing quantitative risk assessments to support decision making. The course starts with a description of uncertainties that are important to power system operation and planning, and common metrics for system performance. Case studies in power system reliability and system capacity planning are introduced. In this context, various approaches to modelling uncertainties are introduced. Historical data is used to create models for component reliability. Students will learn how to compute system level reliability measures from bottom-up reliability models, using enumeration approaches and Monte Carlo sampling. Special attention is given to the estimation sample variation using the normal assumption and bootstrap approaches. Finally, the impact of risk metrics and decision criteria on decision making is discussed, thereby giving students insight into the whole chain from measurements to models to model-based decision making.	
Study Goals	After completing this course, students will be able to: 1) Calculate commonly used reliability measures and interpret their values in the context of Western European power systems. 2) Construct and use Markov models for component reliability. 3) Follow prescribed steps to construct reliability models from failure and repair data. 4) Calculate the reliability of model systems on the basis of constituent component models. 5) Implement Monte Carlo sampling and bootstrapping methods to estimate risk measures and the accuracy of such estimates. 6) Compare model representations of uncertainty, and illustrate these in the context of power system reliability and planning. 7) Evaluate different grid development options using quantitative risk assessment and discuss how a decision for one option depends on the choice of risk measure and decision criterion.	
Education Method	Lectures (8x 2 hours); Computer labs (8x 2 hours); additional exercises for self study.	
Literature and Study Materials	1) Book, available as e-book through TU Delft (no additional cost) or physical copy: Bart W. Tuinema, José L. Rueda Torres, Alexandru I. Stefanov, Francisco M. Gonzalez-Longatt, and Mart A. M. van der Meijden, "Probabilistic Reliability Analysis of Power Systems," Springer Nature Switzerland AG 2020, May 2020. DOI: https://doi.org/10.1007/978-3-030-43498-4 . 2) Supplementary notes, videos, selected papers, reports and book chapters, available through Brightspace.	
Assessment	Completion with 'pass' mark of a small group assignment and computer lab exercise (with basic report) is required. The final grade is determined by a written exam (80%) and computer exercise with accompanying report (20%). A minimum mark of 5.0 must be attained for each component to pass. Component marks are given on a scale 0-10, rounded to the nearest 0.1 increment; the final grade is rounded to the nearest 0.5 increment. One resit opportunity is provided for the written exam. One repair opportunity is offered for the pass/fail assignments (two weeks after the initial deadline, unless agreed otherwise). A repair opportunity will also be offered for the graded exercise if the initial mark was between 4.0 and 5.7 (out of 10), with a deadline 3 weeks after the initial grade is announced. A maximum mark of 6.0 out of 10 can be obtained for the exercise through this route. If a student passes only part of the course in this academic year, all parts need to be retaken. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Energy Modelling Programming Stochastics	
Maximum number of participants	30.	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4107	Power Systems Analysis II	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Instructor	Dr. A. Lekic	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The students who obtained their B.Sc degree at TU Delft are expected to have taken the subject EE2E21 - Sustainable Energy Supply. Other students, with previous education at other universities and abroad are expected to have passed a similar subject and gained prior knowledge about transformers, generators, lines and cables, three phase circuits and per unit value calculation.	
Course Contents	The course is divided in three major parts: - Fault computation in electrical power systems - Economic operation and control of power systems - Introduction to power system stability The first part deals with the analysis of symmetrical and asymmetrical faults based on the application of symmetrical components by using power (in)variant transformation. In order to apply symmetrical components, particular time will be spent on sequence circuits of system components. A special attention will be paid on sequence network derivations for star/delta loads, transformers, overhead lines and generators. The second part, the economic operation involves the minimization of fuel costs, while the power system control deals with the primary and secondary control on the generating units to enable stable operation and maintaining constant grid frequency. Finally, in the third part, the stability of the power system after faults is studied through the method of the equal-area criterion. In the course, comprehensive exercises are being solved, which are related to realistic networks, which can later be utilized in other subjects in the electrical power systems track.	
Study Goals	By the end of this course, the student should be able: - to compute all types of fault currents that may occur in an arbitrary power systems by applying symmetrical components with a power variant and invariant transform. - to assess the values of specific fault currents depending on how the neutral transformer points are grounded. - to build the Zbus matrix of a system and from the Zbus matrix, at least to determine the amplitude of the fault currents at any bus of the network corresponding the built/pre-defined Zbus matrix. - to investigate transient stability by using the equal area criterion	
Education Method	Lectures (*). The number of hours that the students need to spend on this subject is about 14h a week. Experience reveals that this time is particularly related to the fact that students should devote more time on solving more exercises in order to master the topics of symmetrical components and transient stability. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	John. J. Grainger and William D. Stevenson: Power System Analysis, ISBN 0-07035958	
Assessment	Oral. Students normally get theoretical questions and problems. Problem solutions are important to see if the students govern the theory well. In case when more than 10 students take the exam, the assessment is done in a written form. Resit of the exam is possible according to the examination schedule. As the subject is divided in three major parts, the weights of part 1, 2 and 3 are 50%, 25% and 25% respectively, which is in line with the number of contact hours devoted for each part. These weights are also applied in the examination questions. The exam is passed when the student score more than 57%. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consulting hours according to the appointment	

ET4108	Transients in Power Systems	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	EE2E21, ET4107 - knowledge of symmetrical component is highly recommended	
Course Contents	This course deals with the rapid phenomena in power systems caused by switching actions and atmospheric discharges. Some of the topics covered are circuit breakers, arc-circuit interaction, short-line fault, interruption of small inductive capacitive currents, travelling waves, atmospheric discharges, simulation of transients, circuit breaker testing.	
Course Contents Continuation	The subject deals with power system transients. It is divided in a few sections the goals of which are the students to be introduced by the basic concepts of switching transients and transient analysis as well as travelling wave phenomena. The subject also deals with the current interruption phenomena and circuit breakers as well as circuit breaker testing and insulation coordination. Additional material on Ferroresonance and surge arrester selections is being studied as a supplement to this project. All results are validated by computer simulations in Electromagnetic Transient Program (EMTP), which is used as a tool for detailed analysis of different cases like switching small inductive and capacitive currents, arc interruption, thermal and dielectric breakdown of the circuit breakers, travelling wave phenomena. Each chapter is also supported by several simplified examples (which are solvable on the brightspace)	
Study Goals	After following this course the student should be able to: - model power system components for transient analysis by means of lumped elements. - to understand the current interruption process and gained insight in the testing of switchgear in the High Power Lab. - to use symmetrical components during transient processes in a network, especially when one pole of the circuit breaker opens and the other two remain in closed position. - to understand under which circumstances the worst case scenario (highest, steepest TRV) takes place. - to be able to assess the behavior of the lightning stroke when arriving and junction points connecting one or more components with different characteristic impedance Based on aforementioned learning objectives, the student should be able to simulate at least simplified cases of switching and lighting transient phenomena in Electromagnetic Transient Program ATP-EMTP or other EMTP-based software environment.	
Education Method	Lectures (*). Students should spent approximately ten hours per week in order to achieve the learning goals well. Each chapter, if suitable, is also supported by solvable examples on the board and simplified/detailed simulation cases performed in Electromagnetic Transient Program (EMTP) environment. Students also get opportunity to obtain the EMTP royalty free software and to learn EMTP so that they can use it as a tool for different project cases or M.Sc thesis. The subject also provides opportunity for extra projects in which students may obtain credits depending on the amount of work done. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	L. vander Sluis, Transients in Power System, Wiley, ISBN 0 471 486 396. JA Martinez-Velasco, M Popov, Circuit breakers: Power System Transients Parameter determination, CRC Press, ISBN-13: 978-1420065299, pp. 447-556.	
Assessment	Additional material on ferroresonance phenomena will be provided by the lecturer Oral (closed book examination) which can be taken as written in case of more than 10 students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular transient case that occurs in practice. The amount of work required for particular problem solutions is in line with the number of points. The exam is passed when the student scores more than 57 %. Resit of the examination is possible according to the examination schedule. The content of the course material is divided into 9 chapters. All chapters weight equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4113	Power System Dynamics	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Required: ET4107 Power System Analysis II (Master SET / Master EE / Master EWEM) SET3065 Intelligent Electrical Power Grids (Master SET / Master EE / Master EWEM) Optional: ET4116 Power Electronics (Master SET / Master EE / Master EWEM) ET4117 Electrical Machines and Drives (Master SET / Master EE / Master EWEM)	
Parts	Theory lectures: Lecture 1: Introduction to power system stability and synchronous generator Lecture 2: Controllers of the synchronous generator Lecture 3: The voltage source converter Lecture 4: Controllers of the voltage source converter Lecture 5: Applications of voltage source converter in electrical power systems Lecture 6: Rotor angle stability: transient stability and small-signal stability Lecture 7: Small-disturbance (small-signal) rotor angle stability (measurement based methods) Lab sessions: Session 1: Tutorial for PowerFactory (Single-Machine Infinite Bus system) Session 2: Modelling of an interconnected system (steady-state performance) Session 3: Dynamic models - Part 1 Session 4: Dynamic models - Part 2 Session 5: Long-term voltage stability assessment (including tutorial for Python) Session 6: Rotor angle stability assessment Session 7: Sensitivity analysis	
Course Contents	The course "Power Systems Dynamics" is devoted to the study behaviour of electrical power systems in non-steady state. Knowledge of the dynamic behaviour of electrical power systems and their components is important in order to predict whether the system (or a part thereof) remains in a stable state after a disturbance. The time frame of observation ranges from milliseconds to several minutes, depending on the kind of disturbance and the influence of applied controls. For each time frame, special models of the network components are derived. In this course, the model of the synchronous generator, the voltage source converter, and their associated control systems receive special attention due to their dominant influence on the dynamic behaviour of the power system. In the computer assignments, the theory is put into practice by means of software based simulations in DigSILENT PowerFactory as well as scripting in Python and Matlab. Subject to investigation are: modelling of synchronous generator and voltage source converter for RMS simulations, frequency and voltage controls, model-based and measurement-based eigenvalue analysis, wide-area monitoring, damping control, and RMS simulation of interconnected power systems.	
Study Goals	At the end of the course the students will be able to: - Describe the modeling aspects of power system components for computer aided simulation of electro-mechanical transients - Apply and interpret stability criteria for the analysis of systems dynamic behavior - Create a model of a power system in a simulation software package - Assess the system performance following typical disturbances	
Education Method	Lectures and computer assignments. Contact hours (8 weeks: 7 theory lectures, 7 lab sessions): 2 hours/theory lecture and 4 hours/lab session. Total: 42 hours. Self-study: 70 hours (9 hours/week).	
Computer Use	DigSILENT PowerFactory, Python, Matlab	
Literature and Study Materials	P. Kundur, Power System Stability and Control, Mc.Graw Hill, inc. 1994, ISBN-0-07-035958.	
Assessment	Lab report and final exam/re-exam including a theoretical questions component and a computer-based component. The computer-based part of the test requires familiarity with the models built during the lab assignments. The students are expected to run time-domain simulations and give insight into the observed phenomena. The final grade comprises: 1) Exam or re-exam (5 points). 2) Lab report and implementations (5 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Digital exam in the exam period of Q4. Duration: 3 hours. The resit is in the exam period of summer period. Duration: 3 hours.	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Study load: 112 hours.	
maximum aantal deelnemers	Limited to 25 students due to software availability	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalar - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteresis modulation - carrier modulation with feedback - scalar and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Responsible Instructor	H. Vahedi	
Instructor	G.R. Chandra Mouli	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4119	Electronic Power Conversion	4
Responsible Instructor	S.A. Rivera Iunnissi	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	EE1P21, EE1C11, EE2E11, EE2M21	
Course Contents	This course covers the principles of power conversion with switch-mode circuits. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated and isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain the circuit diagram and their waveforms; to be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practical	
Literature and Study Materials	N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition. (Available at ETV at reduced price)	
Assessment	Written exam (closed book) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Note: In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Free Elective Space EPE 2022

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Thesis project 2023

Introduction 1

This is the thesis project, this being the concluding study unit of the program. The project amounts to 45 EC and involves a research or design task with sufficient academic level. The project may be executed within a research program at TU Delft, or in a suitable research institute or company. The project must be carried out with a systematic approach and should include all phases of a research or design project: analysis, modelling, implementation/construction and validation/evaluation. The student executes the thesis project independently, with guidance of a thesis supervisor and under the responsibility of the full professor who approved the IEP.

ET4300	Master Thesis	45
Responsible Instructor	Prof. M. Alavi	
Responsible Instructor	Dr. D. Cavallo	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	M. Babaie	
Responsible Instructor	Dr. A. Lekic	
Responsible Instructor	M.A. Kitsak	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MScEE and MScCE programs. It is an individual in-depth research or expert design assignment with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. The thesis project may be executed within a research program at TU Delft, or in a suitable research institute or company. The nominal duration of a thesis project is 30 weeks (45 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. With regard to the thesis project, the thesis supervisor may impose additional conditions on the IEP to be met before starting the thesis project. If desired, an internship or an extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in scope and a separate deliverable such that the thesis project can be clearly evaluated in isolation. The thesis project is governed by the EEMCS Graduation Policy. Below you find a short description of the different steps to be taken during the thesis work: 1. Orientation 2. Kick-off meeting 3. Midterm meeting 4. Green-light review 5. Graduation Day (defence, assessment and graduation ceremony)	
Course Contents Continuation	Declaration of Credits Obtained Before you can start your Masters Thesis Project you need to have a by the Board of Examiners approved Individual Exam Programme (IEP) and at least 60 credits of the Master's degree course work as stated in the IEP should be completed. You can prove this by a form called the Declaration of Credits Obtained (Verklaring behaalde studiepunten) (60 EC). You can request this form from the service desk in the EEMCS building 36 (Mekelweg 4) at the servicedesk with your campuscard or send an email to servicepuntEWI@tudelft.nl (use your with your TU Delft e-mail!) You have to show this form to your thesis advisor.	
Study Goals	The master thesis project serves to prove that the student has acquired the academic competencies of a Master of Science. The thesis project is an individual in-depth research or expert design assignment independently carried out by the student with a systematic approach including all phases of a research or design project: - problem formulation and planning of the work - analysis, - modeling, - implementation/construction, - validation/evaluation, - presentation of the obtained results, orally and in writing. The achievements and resultant thesis report should comprise original work carried out by the student.	
Education Method	Orientation After having obtained a DCO, the thesis work starts with a thesis proposal, which is defined together with the supervisor. The proposal, of at most 2 pages A4, is written according to the format which can be downloaded from the Brightspace page of ET4300. The thesis proposal contains: - a working title, - a preliminary problem definition, - intended outcomes of the thesis project, - intended milestones, - expected duration and (preliminary) date of the thesis defense. The (preliminary) thesis committee is also indicated on this proposal. If desired, an internship or extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in activity and a separate deliverable such that the thesis project can be clearly evaluated in isolation. This is indicated in the thesis proposal. The thesis related work done during an internship or extra project should be described in the thesis report (typically one chapter). Kick-off meeting The Orientation phase indicated above results in a thesis proposal written by the student. The objective of the kick-off meeting is the formal approval of this thesis proposal by the supervisor(s). In this meeting, the research question(s), research methodology and planning are presented by the student and discussed with the supervisor(s). After approval, the "Thesis Proposal form" is filled out. Midterm meeting After 3-4 months, the student presents (by an oral presentation and a written report) the work performed so far to the supervisor(s). On the basis of the work done, it is decided whether the goals, as formulated in the thesis proposal, are still feasible. If not, they are reconsidered and adapted. After the mid-term meeting, the student has a clear idea of how to proceed in order to finish the thesis in time. Thesis committee This committee is responsible for the assessment of the thesis work and the thesis presentation. In order to be able to defend the thesis, the student must fill out the thesis committee form, which needs to be signed by the responsible full or associate professor. This form needs to be handed in to the Board of Examiners of the study programme at least 3 months prior to the MSc thesis presentation (the defence). Green-light review In the final phase of the thesis project, there will be a green-light review. During this meeting it is decided whether the student is	

ready to proceed with the thesis defence, resulting in a "go/no-go". It is compulsory to have a (draft) thesis ready for the green-light review. In case of a "go", the "Final Examination form" is handed in, which will trigger a check on the fulfillment of all requirements for getting the MScEE or MScCE diploma.

Graduation day

The final step of the graduation process is the presentation and defence of the thesis work. The candidate may present and defend the thesis work only after having finished all courses as stated in the approved IEP. The defence consists of two parts. First, the work is presented in a public session in front of the thesis committee, family, friends and fellow students. The presentation is followed by a closed session where the candidate will defend the work in front of the thesis committee. Immediately following the assessment, provided that there is a positive outcome, the diploma ceremony starts and the diploma will be handed out.

Assessment

The thesis project is assessed by the thesis committee based on the actual thesis work carried out, the final report, the public presentation, and the closed defense session and based on the learning objectives of the thesis. The items assessed in the final grading are:

Quality of the work: research method/design, application/development of theory, interpretation of results, scientific significance;

Performance: critical attitude, creativity, initiative, interaction with peers/superiors, planning;

Report: content, structure, quality of writing, independence in writing;

Presentation & defense: content, form, performance, scientific discussion.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Track Microelectronics 2023

Program Title	Track Microelectronics
EC Program	120
Introduction 1	Microelectronics is a very multi-disciplinary field, both in terms of its fundamentals (e.g. the physics, materials and chemistry required to make devices work) and with regard to its wide variety of application areas. It also is a particularly innovative field, fuelling the ever-increasing capabilities of processing power, sensing capabilities and miniaturization. As such, it is at the basis of most of the rapid changes in almost all of society, including advanced health care, social networks, traffic and mobility, telecommunications and smart grids. The MSc track in Microelectronics provides electronics engineers with the highly specialized knowledge and experience that they need in order to design, fabricate and test such devices, circuits and systems. The department is internationally renowned for its research on smart sensors, microsystems, semiconductor fabrication, solar cells, signal processing, RF transceivers and RF components. Students have the opportunity to transform their own ideas from theory to application in the departments own clean room as well as the various state-of-the-art IC foundries that the department has access to. The range of facilities also encourages fruitful discussions between design-oriented and technology-oriented students. The laboratories involved in the MSc Program Microelectronics are: - Signal Processing Systems (SPS) - Electronic Instrumentation (EI) - Electronic Circuits and Architectures (ELCA) - Bioelectronics (BE) - Electronic Components, Technology and Materials (ECTM) - Terahertz sensing group (TS) - Microwave Sensing, Signals and Systems (MS3) The job prospects for microelectronics graduates are excellent, and graduates of the programme have a wide choice of career paths. Many of our graduates are working in leading companies such as Philips, NXP, Broadcom, ASML, Nokia, Texas Instruments, Maxim, Analog Devices, Philips Healthcare and BioMetrix. The combination of a solid engineering background and the wide application field of microelectronics makes many other career perspectives possible as well.
Program Goals	The fundamental goal of this MSc program is to train students to become independently thinking and broadly developed professionals that are able to solve future problems, yet unknown, using novel concepts, yet unknown, by offering them thorough fundamental knowledge and by involving them in cutting-edge research programs in a stimulating environment.

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Main core ME 2023	
Introduction 1	This is the common core. It is common to all programs and contains two general engineering oriented courses being compulsory for all students. These courses are Profile Orientation and Academic skills and Systems Engineering. The common core further contains 9 fundamental courses introducing all students to the breadth of EE and CE. Of these 9 fundamentals, all students must select two courses. Common core courses are fundamental and general in nature, and should be attractive to a relevant fraction of the students in multiple tracks. These courses are programmed in the first quarter of the academic year. Typically, they have three parts. These are (1) an introductory part that might be seen as a quick repeat of the expected prerequisite knowledge of incoming students (or as a quick introduction if some knowledge is still missing), (2) the main body, and (3) a part that makes a specific connection to the different tracks in the program. This last part may actually be track-specific. Common core EE EE4C01 Profile Orientation and Academic Skills EE4C11 System Engineering Common core: Select 2 out of 9 EE4C03 Statistical Digital Signal Processing and Modeling EE4C04 Control Theory EE4C05 Electromagnetics EE4C06 Networking CESE4010 Advanced Computing Systems EE4C08 Measurement and Instrumentation EE4C10 Analog Circuit Design Fundamentals EE4C12 Machine Learning for Electrical Engineering Applications EE4C13 Wireless Systems for Electrical Engineering Applications

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Instructor	Ir. J.W. Peltenburg	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = $0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3)$ L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1] In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C01	Profile Orientation and Academic Skills	3
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Responsible Instructor	L.C. Schroten	
Instructor	Dr. C.A. Benitez Avila	
Instructor	L.C. Schroten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: basic knowledge as acquired with a BSc in Physics, EE, Material Science. Strongly advised: Information Literacy 1 and 2 (e.g. Master Programme Information Literacy 1 & 2 or Electrical Engineering Information Literacy 1 and 2, both available in Brightspace).	
Course Contents	- Introduction to the MScEE tracks (Microelectronics, Electrical Power Engineering, Signals & Systems, Wireless Communication and Sensing) - Introduction to the research and expertise of the research groups taking part in the selected master track - Knowledge of ethics and intellectual property right - Skills in academic writing - Skills in oral presentation of scientific topics	
Study Goals	With this course the students - obtain relevant information on the MScEE study and the chosen track, become aware of cultural differences and how this affects intercultural communication, form a network with fellow master EE students. - acquire an overview of the main challenges in and potential applications of electrical engineering - become familiar with research topics addressed within the EE department - master basic skills in oral presentation of scientific topics - master basic skills in academic writing - acquire basic principle of ethics and intellectual property right - are able to extract the main information from a scientific paper or report - are able to write a paper (mostly a literature review) on a scientific question	
Education Method	The course includes lectures, reading materials and exercises. During the course, students work on a writing assignment, an ethics case and give several presentations. Attendance is mandatory as all class activities are highly interactive and peer feedback is a necessity.	
Literature and Study Materials	TU Present & TU Write. Both are available in Brightspace.	
Assessment	The pass/fail for this course is determined by the evaluation of the final writing assignment, the oral presentation and attending the Ethics seminar(s). All three components should be evaluated with a pass to pass the whole EE4C01 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C04	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential Realization of transfer function models by state space descriptions, coordinate changes, canonical forms Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback Application of LQ regulator Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle Reference signal modeling, integral action for zero steady-state error. Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design and - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanegue Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	P.P. Sundaramoorthy	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C13	Wireless Systems for Electrical Engineering Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	C. Gao	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	M. Babaie	
Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	M. Spirito	
Instructor	Prof. M. Alavi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE4605 'Integrated Circuits and Systems for Wireless Applications' and EE4371 'Advanced Wireless Transceivers.'	
Course Contents	In this course, you will learn about the commonly used architectures in wireless systems, with the requirements on their building blocks. The system key building blocks are addressed at their functional level regarding electrical properties, such as transfer/gain, noise, linearity, frequency selectivity, bandwidth, and other parameters. Physical constraints of practical components and circuits are used to identify the achievable signal quality, selectivity, sensitivity, and link budget of a wireless system. The lectures will explain the causes and impact of device/circuit non-linearities and noise on wireless systems' sensitivity and dynamic range. The course will introduce the concepts of frequency generation, modulation/demodulation, and up/down conversion. To teach you to work at RF frequencies with wireless systems, the principles of impedance matching are explained, and tools are provided to analyze/design/characterize RF (matching) circuitry/systems. The common part of the course also includes an overview of the figure of merits (FoMs) of various TX/Rx architectures/system blocks and the required instrumentation to characterize them. This course belongs to the RF/wireless profile and is an excellent preparation for the course; EE4605, Integrated Circuits for RF/Wireless Applications, which focuses on the design and realization of analog intensive building blocks for wireless transceivers as well as for ET4371, Advanced Digital Wireless transceivers which is addressing the latest trends in RF/wireless design aiming to fully utilize the integration and signal processing strengths of advanced CMOS. The course is completed by four short specialization topics which can be followed based on your personal preference, namely: (1) Introduction to wireless power transfer technology, (2) Beamforming basics: concepts and architectures, (3) Neural Networks for Wireless & Radar Systems, (4) Hands-on experience with RF characterization equipment on practical circuits. (The latter (4) has an attendee limit of 15 people.) Only one of these specialization topics is required to complete the overall course.	
Study Goals	Understand wireless system architectures with their subblocks to such a level that you can analyze their various performance parameters in terms of; link budget, sensitivity, spectral purity, P1dB, IP3, DR, SFDR, noise, phase noise, selectivity, EVM, APCR, and efficiency. In addition, you will master a basic understanding of S-parameters, impedance matching, and RF instrumentation.	
Education Method	This course will be delivered with complementary exercise lectures to familiarize students with the concepts and calculations described during the lectures.	
Literature and Study Materials	These lectures are (in part) based on the book of Behzad Razavi, RF microelectronics 2nd edition, Chapters 1 to 4.	
Assessment	Examination and grading: Your overall grade will be based on your performance in the written exam (80%), compulsory homework, and the mark obtained in your track-specific project (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year

Organization

Education

2023/2024

Electrical Engineering, Mathematics and Computer Science

Master Electrical Engineering

Track core ME 2023	
Introduction 1	This is the track core, offering the first level of branching between tracks. The track core contains 8 courses, of which the students select three courses.

CESE4085	Modern Computer Architectures	5
Responsible Instructor	Dr. S.D. Cotofana	
Responsible Instructor	Dr.ir. J.S.S.M. Wong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	The course emphasises on theoretical aspects and practical trade-offs of computer architecture and organization of the newest microprocessors currently on the market, and the latest developments in computer architecture research. Quantifying design decisions in terms of performance and cost. Subjects: basic principles, instruction set architectures, pipelining and pipelining consequences, multiple-issue (superscalar and VLIW) processors, multimedia SIMD extensions, out-of-order execution, branch prediction, speculative execution, advanced memory hierarchies, pre-fetching, multithreaded processors and multiprocessors, energy consumption, and reliability.	
Study Goals	1. The student can operate with concepts and notions related to: - Instruction sets: characteristics, functions, formats, addressing modes; - Processor structure, functions, and pipelining; - Instruction level parallelism and its static and dynamic exploitation; - Distributed memory hierarchy; - Multiprocessors. 2. Given a set of functional and non-functional requirements, the student can select the most appropriate architecture among a set of different architectures of modern microprocessors. 3. She/he can optimize code for a particular processor using, e.g., code scheduling and loop unrolling. 4. She/he can perform design space exploration and quantify design decisions in terms of performance, energy consumption, cost, flexibility, programmability, predictability for various processor and multiprocessors building blocks and architecture features, e.g., instruction set, predication vs. speculation, cache vs. scratch-pad memory, in-order/out-of-order execution, message passing vs. shared memory, etc. 5. She/he can study recent advances in computer architecture, classify research papers, and report about them.	
Education Method	Lectures, lab work, and reading assignment.	
Literature and Study Materials	J.L. Hennessy and D.A. Patterson, Computer Architecture: A Quantitative Approach, 4th edition, Morgan Kaufman, 2007, ISBN 0123704901. A set of high-impact articles and presentations on computer architecture trends and practices available on Brightspace. IEEE Explore	
Assessment	The lab work and assignments contribute to the final grade as follows: Lab work 40% of the final grade, Reading assignment 60 % of the final grade. Repair option: Re-submit deliverable Individual Assignments Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effetc of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cacoded amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4605	Integrated Circuits for RF/Wireless Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Responsible Instructor	M. Spirito	
Instructor	Ing. M.J. Pelk	
Contact Hours / Week x/x/x/x	0,3,0,0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Common core course EE4C13. Other useful courses to consider as part of your preparation for RF design (not mandatory) are EE4C05 and EE4C10.	
Course Contents	In this course, you will learn to understand and design the circuit blocks of wireless systems. You can design and analyze circuits/ICs for wireless systems when completed. To do so, fundamental knowledge of RF/microwave design is required. To achieve such an insight, we discuss; optimum bias, device scaling, circuit configurations, and harmonic matching techniques for active devices (BJTs & FETS) operating at high frequencies to reach optimum gain, noise, linearity, and efficiency. The outline of the course is as follows; - Design of RF/microwave amplifiers, stability, gain, and impedance matching (HW1a) - Design of RF/microwave hybrids and transmission line-based passives, as well as on-chip (high Q) passive components; capacitors, varactors, inductors, transformers, and transmission lines (HW1b) - LNA design for simultaneous noise & impedance matching and LNA noise cancellation techniques (HW2). - Analysis and design techniques for circuit linearity enhancements (HW3) - RF mixer design for linearity - Design of energy-efficient & wideband power amplifiers & transmitters (HW4). A detailed course description addressing all topics covered in this lecture series can be found on Brightspace. Note that this EE4605, Integrated Circuits for RF/Wireless Applications course is part of the RF/wireless track and is preceded by EE4C13, Wireless Systems for Electrical Engineering Applications, which discusses wireless applications at the system level. This course (EE4605) can be followed up by ET4371 (strongly recommended), Advanced Digital Wireless transceivers, which addresses the latest trends in RF/wireless design which aim to fully utilize the high integration and signal processing strengths of advanced CMOS. Background reading for the various topics can be found in the recommended books given in the Literature and Study Materials section. This course will have three to four homework (to be handed in) that address the design of the various circuit blocks (RF amplifier & microwave passives, LNA, mixer, PA/transmitter) and their sub-components. For this homework, use will be made of Keysights ADS Advanced Design System and Cadence. Aspects of packaging, CAD, and RF testing relevant to wireless / mm-wave design are also emphasized.	
Study Goals	Examination and grading: Your overall grade will be based on your performance in the homework and a written exam. - Understand how to (optimally) design RF/wireless components and circuits. This includes: - How to design, dimension, and layout passive components like: PCB and on-chip transmission lines, capacitors, inductors, and transformers. - How to bias and scale active devices while understanding their basic operation and equivalent models. - How to stabilize active devices while maintaining their RF performance. - How to design matching networks for maximum gain, low noise, and high linearity. - How to design for maximum efficiency, bandwidth, ACPR, and lowest EVM. - Know how to use RF/Microwave design tools like Keysights ADS, Momentum (in addition to Cadence) to design RF integrated circuits and systems. - Understand the RF/microwave-specific data representations and analysis tools. - To understand the full design cycle from RF/wireless concept, RFIC design, to the final testing. - Being able to analyze and design integrated RF transceiver front-end circuits, like LNAs, mixers, and PAs.	
Education Method	Lectures + design related homeworks	
Literature and Study Materials	Hand-outs + recommended background books * Microwave Transistor Amplifiers: Analysis and Design (2nd edition) by Gonzalez, Guillermo * RF Microelectronics (2nd edition), by Behzad Razavi * Microwave Circuit Design, using Linear and Nonlinear Techniques, by George D. Vendelin, Ulrich L Rohde, Anthony M. Pavio * RF Power Amplifiers for wireless Communications (2nd edition), by Steve C. Cripps.	
Assessment	RF design related homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	More details are available on Bright Space. To participate in this course, please enroll.	

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4289	Integrated Circuits and MEMS Technology	4
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Instructor	Dr. C.M.F. Viellard-Boutry	
Instructor	Dr.ing. H.W. van Zeijl	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics (EE 4585)	
Course Contents	This lecture is an introduction in the fabrication technologies used for Integrated Circuits (ICs) and Micro Electro-Mechanical Systems (MEMS). The aim of this lecture is to build a bridge between the physical world (the device physics) and the electronic world (the device characteristics). In this way, students gain more insight in the critical aspects and challenges in IC fabrication. The series of lectures starts with the discussion of basic microelectronic devices with a brief review of the physics involved. The associated basic fabrication technology, required to fabricate these devices is also discussed. After that, the mainstream IC fabrication technology (CMOS) will be discussed in more detail. In separate chapters, silicon crystal growth, epitaxy, ion implantation, chemical vapor deposition, etching and photolithography will be introduced. Their physical motivation and the impact on the device characteristics will be discussed. Finally, silicon bulk and surface micromachining technologies are introduced to illustrate the potential of 3D micro structuring in the development of Microsystems or MEMS. These IC compatible technologies are employed to realize multi-function systems for many scientific and industrial application areas.	
Study Goals	You know the definition of the basic process steps to fabricate a device (IC or MEMS) You can explain the basic physics and chemistry of these fabrication steps in an IC /MEMS process flow You can analyse the relationship between the series of process steps necessary to build the device and the device performance. You can evaluate how device requirements are translated into a process flow You are able to propose solutions to specific device technological questions	
Education Method	lectures	
Literature and Study Materials	R.C. Jager introduction to Microelectronic Fabrication & course slides	
Assessment	written exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. S. Vollebregt	

ET4362	High Speed Digital Design for Embedded Systems	5
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The course covers many important topics computer engineers and embedded system designers should be aware of. First, the physical effects of high-frequency signal propagation and transport are discussed. Next, the state of the art for high bandwidth/low latency digital communication and memory systems is presented. Special attention is given to parallel memory systems. Thereafter, low level circuit design and analysis principles relevant to high-speed digital design are explained. Clock distribution, metastability and skew-tolerant design are emphasized. Advanced techniques for high-speed combinational and sequential logic structures are considered in detail. More precisely, approaches like domino logic, pulsed latches and dual-rail logic will be discussed. The next part discusses advanced design techniques such as globally asynchronous locally synchronous (GALS) systems, systolic arrays, streaming dataflow, wave-pipelined circuits and fully asynchronous design. The last part concentrates on understanding the true theoretical bounds in, e.g., the energy consumption, of computers. The necessary theory and some practically realizable adiabatic logic design styles will be introduced. When time allows, brief introduction to the Gödel's incompleteness theorem and its impact on computers will be presented.	
Study Goals	The purpose of this course is to familiarize students with the latest developments in the field of high-speed digital design. The emphasis is on novel, non-traditional design concepts, techniques and methodologies.	
Education Method	Lectures and individual assignments.	
Literature and Study Materials	Presentations and selected research papers.	
Assessment	Written report on a selected research topic and oral exam on appointment.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested specialisation courses ME 2023

Introduction 1	This is the specialization space. In this tier, students have to select courses totalling to at least 24 EC . In essence, all of the MSc EE and MSc CE courses can be chosen as a specialization course, including those from the main core and the track core that were not yet chosen.
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BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

CESE4035	Computer Arithmetic	5
Responsible Instructor	Dr. S.D. Cotofana	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course emphasises on theoretical aspects of computer arithmetic. It covers concepts and various topics related to fixed and floating-point number systems, algorithms and implementations for addition, multiplication, division, square root, and other high order arithmetic operations.	
Study Goals	The study goals for the course are as follows: 1. The student can operate with concepts and notions related to: (i) number representation systems; (ii) algorithms and implementations for basic integer arithmetic operations, i.e., addition/subtraction, multiplication, and division. 2. Assuming certain processor architecture and requirements she/he can perform design space exploration and select the most appropriate algorithms for the implementation of the basic functional units. 3. She/he can operate with concepts and notions related to floating point systems and operations, elementary function evaluation, e.g., CORDIC, and error analysis. 4. She/he can design arithmetic units and application specific (co-)processors, optimized for speed, area, power consumption, or combinations of those. 5. She/he can study recent advances in computer arithmetic, classify research papers, and report about them.	
Education Method	Lectures and homework assignments.	
Literature and Study Materials	Text book: Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000, ISBN 0-19-512583-5. Example exams with solutions available on Brightspace.	
Assessment	Exam and homework assignments contribute to the final grade as follows: written open book exam - 60% of the final grade; homework assignments - 40% of the final grade; Repair option: Written exam = resit Individual Assignment = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Exam is open book. Books on computer arithmetic, computer architecture, and logic design, and lecture slides are allowed during the exams.	

CESE4040	Processor Design Project	5
Responsible Instructor	Dr. S.D. Cotofana	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	To be able to complete the project the student has to have Digital Systems and VHDL knowledge. Moreover before embarking in this course she/he has to successfully complete the following courses: CESE4085 Modern Computer Architecture and CESE4035 Computer Arithmetic courses.	
Course Contents	This is an integration course that is aiming to let the students utilize computer architecture and arithmetic theoretical knowledge in the design of a general purpose processor. In this line of reasoning they have to improve the performance of a MIPS-based computing platform by focusing on computer architecture and arithmetic relevant parts of it. As starting point students receive two archives: one which includes the MIPS System on Chip VHDL code and a set of benchmarks selected from the GMPbench and MiBench suites, as well as all the necessary files required for simulation and FPGA implementation, and a second archive with the MIPS cross-compiler. To evaluate their new design, students are given a Zynq-7000 FPGA board per group.	
Study Goals	The course study goals are as follows: 1. The student can perform, assuming certain processor architecture and requirements, a design space exploration and select the most appropriate algorithms for the implementation of the basic functional units. 2. The student can integrate computer arithmetic and computer architecture knowledge and utilize them in order to design/optimize processors. 3. The student is able to evaluate processor performance based on benchmarks and take the appropriate architectural decisions in order to improve on various design metrics. 4. The student can design arithmetic units and processors and implement them using FPGA technology. 5. The student is able to operate in a small team and collaborate for the successful completion of a design assignment. 6. The student can communicate his/her proposal, experimental results, and conclusions in English using the appropriate technical language in written as well as orally.	
Education Method	Lectures and Project.	
Literature and Study Materials	There is no dedicated study material but the students are advised however to utilize as supporting study material Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000, and Computer Architecture: A Quantitative Approach, J.L. Hennessy and D.A. Patterson, 4th edition. Moreover we encourage you to utilize the IEEE Explore library to find research publications that may help you in the successful project completion.	
Assessment	To complete the course students have to submit the project archive of the new core they developed, a report describing their approach and findings, and to give a short presentation of their project in the context of a symposium. The new core functionality is verified and all projects are checked for (between groups) plagiarism. Non functional designs do not pass the project. Plagiarism can also make you fail. Your final score for the project is determined based on the following criteria: The performance of your design (DP). The benchmark scores you report are important. The higher the benchmark score you get the better but this is not the only relevant aspect. In the performance evaluation we also take into consideration the other metrics with more emphasis on the compound ones. The technical merit of your approach (TM). Aspects as innovation level and implementation quality are considered. The report (R). Report organization, content, and language are important aspects at this point. The presentation (P). Here we also look at the capability to ask questions and to answer questions from the auditorium. The ET4 171 final grade is computed as: $\text{Grade} = 0.35 \cdot \text{DP} + 0.35 \cdot \text{TM} + 0.20 \cdot \text{R} + 0.10 \cdot \text{P} - \text{C},$ where C can assume values between 0 and 1 and reflects: (i) the lack of collaboration in the group; (ii) the amount of consultancy you asked me for during the project completion. If the group is functioning like a team and you do not ask too much help C is 0, otherwise it can get values up to 1. Repair option: Individual Assignment or Group Assignment = Repair: Re-submit deliverable Individual Project or Group Project = Repair: Re-submit deliverable Individual Presentation or Group Presentation = Resit (new presentation) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4016	Antenna Systems	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	Y. Aslan	
Instructor	Ir. P.J. Aubry	
Co-responsible for assignments	Ir. P.J.F. Swart	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4288, ET4595	
Expected prior knowledge	EE4C05	
Course Contents	The aim of this course is to introduce the students to the Antenna Techniques and Antenna Systems that are instrumental for applications such as telecommunications, remote sensing and radar systems. After introducing basic antenna concepts the basic antenna parameters are defined. A detailed description of the antenna gain, directivity, polarization, radiation pattern, bandwidth, beamwidth, efficiency, temperature, effective length, effective area and input impedance is given. The reflection and radiation from a dipole are detailed and analysed. Microstrip antenna is introduced and its performance is analysed. The radiation from linear, circular and planar array antennas is discussed in detail while stressing their importance in radar and remote sensing systems. The beamforming in arrays and antenna array synthesis are studied. Characterisation of Antenna System performance is discussed and derivation of Antenna System specifications from overall wireless system performance requirements is studied. Some basic concepts about antenna measurements techniques are introduced. The theoretical studies are complimented by computer simulations and measurements.	
Study Goals	By the end of this course, you should be able to: LO1: Determine (theoretically and experimentally) the fundamental antenna system performance metrics (such as directivity, gain, bandwidth etc.). LO2: Design basic antenna element/array structures (i.e. linear wire, microstrip, aperture) for the given criteria, by using CST and MATLAB tools. LO3: Assess the impact of a given antenna on the wireless system performance (in terms of the signal and noise power). LO4: Develop a suitable antenna model for a given application with specified requirements (e.g. communication, broadcasting, radar).	
Education Method	Lectures, Instructions, computer simulations, antenna measurements	
Literature and Study Materials	Course books (Balanis and Orfanidis), Lectures handouts, simulation and measurement assignments.	
Assessment	Written exam (problems solving) plus home works, simulation and measurement assignments In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4109	Structured Electronic Design	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Instructor	A.J.M. Montagne	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	A grade of a minimal 8 for this course is required to obtain access to the masterclass EE8011MSC	
Expected prior knowledge	This course required the knowledge and skills of MSc course EE4C10 and BSc course EE3C11. This course uses the same book used for BSc course EE3C11. A free download of the book is available via: https://analog-electronics.tudelft.nl Chapters that contain prior knowledge are: 1, 2, 7,8, 9,10 (excluding 10.3.5, 10.3.7, 10.3.8) , 11 (excluding 11.4.4, 11.4.6, 11.4.7, 11.5.1) and 12.1 and 12.2 Students that did not follow BSc course EE3C11 should (could) follow the homologation program from: https://analog-electronics.tudelft.nl/Homologation/courseWebSite/index.html Here you can find lecture material and videos that cover all electronics topics of EE3C11. The teaching staff can assist in studying the homologation program. On request, the lecturers can provide extra lectures for external/international students on topics treated in EE3C11 on background knowledge, like poles and zeros.	
Course Contents	This course deals with the structured design of analog electronic circuits using CMOS(and Bipolar) transistors. The design method presented in this course is based on network theory, control theory, signal processing, and physics. Many people consider the design of analog electronic circuits to be complex. This is mainly because designers have to deal with many relevant performance aspects that can be achieved in many different ways. In other words, there are many degrees of freedom for obtaining the desired performance of an electronic circuit. Theoretical concepts, circuit topologies, electronic devices, their operating conditions, and the physical layout of a system constitute an enormous design space in which it is easy to get lost. For this reason, analog electronics is often regarded as an art rather than a solid discipline. At first glance, there doesn't seem to be a straightforward way to design analog circuits and systems: experienced designers intuitively use all these degrees of freedom to modify and combine known solutions into new ones. However, intuition is knowledge, the origin of which has become unclear. It cannot easily be shared with novices, and therefore, it cannot be a basis for the education of designers. However, the design of electronic circuits can be shared and understood if it is presented in a structured way. This requires a precise formulation of design goals and strategies and a clear distinction between theoretical concepts and their physical implementations. Instead of taking numerous existing solutions to known problems as a starting point, it is far more effective to start a new design with clear distinction between theoretical concepts and study how they can be implemented in such a way that the possibilities of the implementation technology are maximally exploited. Such an approach results in well-defined and reliable designs with predictable performance and provides a basis for sharing and developing knowledge. During the engineering phase, designers need to find or create real-world components, determine their operating conditions, and fix them by design. To this end, they need to formulate and solve so-called design equations for the circuit. Design equations relate the properties and the operating conditions of the electronic devices to the performance requirements and the cost factors of the circuit. For example, to determine a budget for the gain-bandwidth product of an operational amplifier, we must find relations between this gain-bandwidth product and the performance requirements of the circuit that comprises this operational amplifier. I Setting up design equations for analog circuits, especially at transistor-level is a difficult and cumbersome task. It requires knowledge about technology, modeling techniques, and interactions between the behavior of the circuit and the transistor properties. It goes hand in hand with symbolic analysis. Performing symbolic calculations by hand is sensitive to errors, and in practice, only a limited number of circuit performance aspects can be considered. For this reason, we will use SLiCAP (Symbolic Linear Circuit Analysis Program) during this course. SLiCAP (https://analog-electronics.tudelft.nl/SLiCAP.html)	
Study Goals	The main goal of this course is to provide the student with the essential knowledge and skills to design an amplifier in BiCMOS or CMOS technology according to given specifications and to motivate and fully understand all design steps involved. After the course the student is able to: - work in a design team in a complete CMOS design - translate an application description of an amplifier into a specification comprising, a functional specification, specification of performance measures, operational conditions and cost factors. - design a proper amplifier concept based on source and load specification. - evaluate the amplifying capabilities of basic amplifier stages such as CS and CE stage for a given technology - identify performance aspects such as noise, signal handling capability and bandwidth of basic amplifier stages such as CS and CE stage for a given technology - scale devices and select their operating conditions to influence these performance aspects - to apply error reduction techniques, such as compensation, balancing and (local) negative feedback to improve the performance of these basic amplifying stages - to motivate the decision to use multi-stage amplifiers - to design (high-performance) multi-stage amplifiers using error reduction techniques such as compensation, balancing and feedback. - setup design considerations for noise and take adequate design decisions	

	- setup design considerations for signal handling capability and power efficiency and take adequate design decisions - setup design considerations for frequency response and take adequate design decisions - setup design considerations for weak distortion and take adequate design decisions - setup design considerations for biasing and design and implement proper biasing circuitry - evaluate the performance versus requirements of the final design and propose options for improvement - select proper models and design verification tools for each design step - write a design report in which all design steps are motivated and the performance is evaluated and explained.
Education Method	Lectures and Colstructions: The systematic electronics design method will be illustrated during the lectures and colstructions with a design example made in SLiCAP. The design example closely relates to the hearing loop system designed during the BSc course EE3C11. This BSc design was based on operational amplifiers as controllers. In this course, the focus will be on the design of the controllers as transistor circuits. During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design. The design steps are documented in a report using SLiCAP. The design report can be efficiently used as a basis for the exam, which is organized as a design review for the design team.
Computer Use	A computer with SliCAP and the free circuit simulator LT-Spice is necessary. Help for installing and using SLiCAP will be available during the lectures. SLiCAP is an acronym for Symbolic Linear Circuit Analysis Program.
Literature and Study Materials	See: https://analog-electronics.tudelft.nl/SLiCAP.html Book (also used during BSc course EE3C11): Structured Electronic Design Edition downloadable from: https://analog-electronics.tudelft.nl
Assessment	During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design used in every lecture to illustrate the application of the theory from the lectures. At the end of the lecture period, the student team makes an appointment review of their design. The design review consists of two sessions: 1) A design preview, where the students present and defend their design in a relaxed atmosphere. They can use the feedback to make improvements. 2) A design review, where the students present and defend their design again. Each team member has to give substantial input during the session. After this session, each student is graded individually. The design report, created with SLiCAP during the lectures by the team, can be efficiently used as a basis for the review. If there are logistic or personal reasons that make it impossible for a student to join a design team, it is possible to opt for an individual design review (oral exam). Please contact the educators about this. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4525	Analog CMOS design II	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10; EE4520	
Course Contents	The goal of this course is to introduce students to the design of advanced analog circuits in modern CMOS technology. Students will learn how to use basic building blocks, such as current mirrors and differential pairs, to build more complex CMOS circuits, such as two-stage amplifiers. At the end of the course, the students will be able to design and optimize a complex CMOS circuit given its specifications. No new topic will be introduced in this course, and students should have a good understanding of basic circuits block from EE4C10 and EE4520. This course will focus on the students designing a complex analog circuit given a set of specifications, and the interactions with the lecturers and the teaching assistants will focus on discussing the on-going design and answering questions from the students.	
Study Goals	- Students can derive the performance of complex analog CMOS circuits (such as switched-capacitor amplifiers and class-AB amplifiers) in terms of amplification, noise and distortion. - Given the specifications for a CMOS circuit, students can chose the best architecture for the circuit and design and optimize the circuit to meet the specifications, by means of first-order hand calculations and verification by a Spice-like simulator.	
Education Method	Lectures, teaching-assistant session	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Individual project	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4615	Digital IC Design II	3
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will consolidate the knowledge acquired in EE4610 by practicing the design of an advanced complex digital VLSI design, where high-level specifications are translated to transistor sizes, going through several abstraction levels. At each level, the opportunities and limitations of the physical implementation will be considered for finding better solutions and tradeoffs, by means optimization and the search of new topologies, while keeping complexity to a minimum to achieve the desired functionality. Parasitics and the other effects deriving from the analog behavior of digital circuits will be taken into account using four-corner analysis, as well as simplifications and modeling at each abstraction level and other semi-automatic tools.	
Study Goals	The main goal of the course is to give students the tools to successfully design complex, scalable circuits and systems. The students will also learn how to work in and coordinate teams of designers around a complex project from high-level specifications to the actual transistor-level implementation and post-layout verification.	
Education Method	The course will entirely be based in the lab where the target design will be implemented by students in groups of 2-3. Only based on students/groups wish, the final design will be laid out in a commercial CMOS process and possibly fabricated and tested.	
Assessment	The course will be graded based on the target design. The following criteria will be taken into account: 1. Functionality achieved; 2. High-level specifications met; 3. Four-corner analysis completed; Bonus points will be given to those students that will exceed the specifications and create a robust, reproducible design. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4650	Advanced Magnetic Resonance Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	A. Webb	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Vector algebra and vector calculus, linear circuit theory, signals and systems (Fourier transform, steady-state analysis), linear algebra and differential equations, electromagnetics (BSc level or higher)	
Course Contents	General introduction, history of MRI, general description of an MR scanner, general principles of MRI, applications and current research themes. General and brief description of spin, restriction to particles with spin ½ (proton), equilibrium magnetization for spin ½ particles (proton), the Bloch equation, general description of the gyromagnetic ratio, T1 and T2 relaxation times. Electromagnetic fields: Maxwells equations, the electromagnetic boundary conditions, the temporal and spatial Fourier transform, time-harmonic fields, the polarization state of an electromagnetic field (linear, circular, elliptical), Poyntings theorem (power balance), and reciprocity. Flipping the magnetization: received signal in terms of the electromagnetic receive field. Solutions of Blochs equation for various magnetic fields, the most important formula in MRI: formula for the Larmor frequency, chemical shift, scalar coupling, basics of 1-DNMR spectroscopy, elementary measurement sequences, spin-echo, inversion recovery, measuring T1 and T2, introducing T2*. Using gradients to spatially encode signals, k-space formalism, image reconstruction via multi-dimensional Fourier transforms, simple gradient echo and spin echo pulse sequences, three-dimensional imaging, angiography. Background knowledge for hardware design: Maxwells equations and quasi-static fields, the law of Biot-Savart, spherical harmonics. Interaction of RF with tissue as a function of frequency, different media/tissue types and the constitutive relations. RF coil and array design: design of RF coils and RF arrays, the Specific Absorption Rate (SAR). The electromagnetic field due to an electric dipole (small current-carrying piece of wire) and a magnetic dipole (small current-carrying loop). Gradient coil design and the static background field: Realizing gradient fields, Helmholtz and Maxwell coils, advanced gradient coil design using stream functions, consequences of nonuniform static background fields. Numerical modeling, software packages, body models, coil models. Measurements and imaging in practice, SNR and other practical aspects. Current/future research topics, parallel imaging, sparse sampling, electrical properties tomography	
Study Goals	After completing this course, the students will: 1. Understand the principles of magnetic resonance from both a quantum and classical mechanics point-of-view, 2. Understand the interaction of strong electromagnetic fields at high, medium and low frequencies, how these can be manipulated, and the safety features associated with these fields, 3. Understand how spatial information is encoded into magnetic resonance images, the origins of image artifacts, and the signal processing algorithms involved in image reconstruction, 4. Understand the relaxation phenomena associated with magnetic resonance, and how these are important in determining image contrast, 5. Understand how the different components of magnetic resonance systems are designed using Maxwells equations, spherical harmonics, and the Biot-Savart law, 6. Understand the current state-of-the-art of MRI hardware and image processing, and the new opportunities for future developments.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes. Will be made available via Brightspace.	
Assessment	Assignment and oral exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4690	Hardware Architectures for Artificial Intelligence	5
Responsible Instructor	Dr.ing. R.K. Bishnoi	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basics of Digital design	
Course Contents	Recent advancements in the field of Artificial Intelligence (AI) have demonstrated its great potential in solving exceptionally complex problems, that can even surpass humans. To support such AI algorithms, the demand of developing new and efficient hardware is growing with rapid pace. In particular, for a variety of edge/IoT devices, hardware platform locally deploys these algorithms that decreases the exchange of information with the cloud which significantly reduces the networking costs and boost the overall performance of the devices. Hardware architectures play a critical role in executing AI algorithms. This course gives an overview of AI from hardware perspective and discusses close to the state-of-the-art architectures and training mechanisms. In this course, you will learn the concept of advanced forms of AI (i.e. Deep Neural Networks) and how they can solve the complex problems. You will also learn, what are the challenges associated with the underlying hardware and how new architectures can address those. Moreover, the course will teach you the concept of brain-inspired computing using in-memory processing hardware platform. Technically, this course covers following topics: 1) Machine inference models (i.e. Support Vector Machine, Decision tree, Regression analysis, clustering, etc.), machine learning algorithms (i.e., Supervised, Unsupervised, Semi-supervised, Reinforcement learning models) and their associated challenges. 2) Deep Neural Network (DNN), Classification of DNNs, feed-forward, back propagation and learning mechanism (such as Gradient decent and Stochastic learning). 3) Spiking Neural Network, Spike Time Dependent Plasticity training algorithms and some existing hardware architectures (such as Intels Loihi, SpinNaker, IBMs TrueNorth, Neurogrid, etc.). 4) Emerging technologies for neuromorphic computing (using memristor devices such as RRAM, MRAM & PCRAM) and non-conventional computing paradigm (such as computing-in-memory, hybrid-design approach, crossbar arrays).	
Study Goals	The main aims of the course are: 1. Describe the concept of AI, inference and training models as well as their associated challenges from hardware perspective. 2. Understand the hardware AI requirements for the data-centers as well as edge devices. 3. Discuss the neuromorphic computing and its advantages with respect to conventional computing paradigms. 4. State details about Artificial Neural Network and their training algorithms. 5. Develop understanding about Spiking Neural Network and their training mechanisms and also, comparison with Artificial Neural Network and their evaluations based on design metrics such as accuracy, power, speed, size, etc. 6. Describe state-of-the-art neuromorphic hardware architectures and discuss their design implementation details. 7. Study the advanced topics of neural networks based-on computation-in-memory concept using emerging non-volatile memories. 8. Become an AI engineer with good understanding about hardware implementation.	
Education Method	Lectures, assignments and lab-work	
Assessment	Oral and lab In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Artificial intelligence	

EE4705	Solid State Physics	3
Responsible Instructor	Dr. R. Ishihara	
Instructor	Dr. S. Nur	
Contact Hours / Week x/x/x/x	0,0,4,0 week 1-4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In electrical engineering, solid-state materials play an essential role in many applications. A thorough understanding of the basic quantum mechanics and electrical and physical properties of semiconductors, metals and insulators are essential for design/fabrication of new electronic devices and circuits, such as, field-effect transistors, solar-cells, light emitting diodes, sensors and quantum computer devices. After introduction of IC fabrication process, the course starts with the crystallography and then will be followed by the quantum mechanics, the quantum theory of solid, semiconductor physics. In short, you will be acquainted with the properties of individual atoms and what happens when they meet and become the form of a solid. If the solid -state physics are thoroughly understood, the physics of the semiconductor devices for IC, photovoltaic, LED, sensors and quantum computer can be understood efficiently.	
Study Goals	The student can 1) determine the crystal structure, the density of atoms and the Miller indices of a crystal, 2) apply Schrodinger's wave equation to various potential functions and derive a probability of finding electrons, 3) find the eigenvalues and eigenfunctions of the Hamiltonian of the system by perturbation methods, 4) discuss the concepts of energy band formation using nearly free electron model and tight-binding approximation, and difference of material properties in terms of the band, 5) derive the concentrations of electron and holes with a given temperature in terms of Fermi energy, and 6) can discuss drift, diffusion and scattering of carriers in a semiconductor under various temperature and impurity concentrations.	
Education Method	Lecture + exercise sessions	
Course Relations	Student cannot enroll both EE4705 and EE4710 and should select one of those two courses. EE4710 has additional contents and credits of quantum and nanoelectronics to this course.	
Literature and Study Materials	Nanotechnology for Microelectronics and Optoelectronics, 2nd edition by Raul J. Martin-Palmer and José Manuel Martínez-Duart Elsevier Science	
Assessment	Written exam + Homework	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	3 hour	

EE4710	Solid State Physics with Quantum and Nano Electronics	5
Responsible Instructor	Dr. R. Ishihara	
Instructor	Dr. S. Nur	
Contact Hours / Week x/x/x/x	0,0,4,0 week 1-7	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In electrical engineering, solid-state materials play an essential role in many applications. When solids are reduced in the size to nanometer scale by nanofabrication, quantum effect becomes evident, and they exhibit exiting behaviors. A thorough understanding of the basic quantum mechanics and electrical and physical properties of semiconductors, metals and insulators are essential for design and fabrication of novel nanoelectronic and photonic devices for the applications in classical and quantum computers, memories, displays, solar-cells, light emitting diodes, and sensors. The course consists of two parts. The first part deals with solid-state physics; After introduction of IC fabrication process, the course starts with the crystallography and then will be followed by the quantum mechanics, the quantum theory of solid, semiconductor physics. In short, you will be acquainted with the properties of individual atoms and what happens when they meet and become the form of a solid. In the second part, we will discuss (1) properties of low-dimensional structures, with examples of nanomaterials such as graphene, (2) electric field transport in nanostructures, (3) the quantum Hall effect, (4) electrical and optical process in quantum wells and quantum dots, and (5) selected applications using the nanomaterial and nanostructures, such as, single-electron transistors and photonic crystals.	
Study Goals	For the first part, the student can 1) determine the crystal structure, the density of atoms and the Miller indices of a crystal, 2) apply Schrodingers wave equation to various potential functions and derive a probability of finding electrons, 3) find the eigenvalues and eigenfunctions of the Hamiltonian of the system by perturbation methods, 4) discuss the concepts of energy band formation using nearly free electron model and tight-binding approximation, and difference of material properties in terms of the band, 5) derive the concentrations of electron and holes with a given temperature in terms of Fermi energy, and 6) can discuss drift, diffusion and scattering of carriers in a semiconductor under various temperature and impurity concentrations. For the second part, the student can 1) analyze, explain and evaluate a specific application of an (nano)electronic or photonic device/material using the obtained knowledge, and 2) create idea for device(s)/material(s) beyond the state-of-the-art for a specific application.	
Education Method	Lecture + exercise sessions	
Course Relations	Student cannot enroll both EE4705 and EE4710 and should select one of those two courses. EE4705 has only the first part of this course.	
Literature and Study Materials	Nanotechnology for Microelectronics and Optoelectronics, 2nd edition by Raul J. Martin-Palmer and José Manuel Martínez-Duart Elsevier Science	
Assessment	Written exam, homework, minitest and presentation In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	3 hour for written exam	

EE4736	Introduction Imaging Sensors	4
Responsible Instructor	Prof.dr. K.A.A. Makinwa	
Responsible Instructor	P. Ramachandra Rao	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	The course will provide necessary background/ references, but a knowledge of the following will be helpful: 1. Basic knowledge of semiconductor physics (bandgap, doping), photons, electromagnetic spectrum 2. Basic knowledge of semiconductor devices (pn junctions, mosfet basic operation) 3. Basic calculus (integration, differentiation) 4. General interest in image sensors & photon detectors	
Course Contents	Welcome to EE4736, Introduction imaging sensors. This course aims at providing you with the essentials of CMOS image sensors. The topic of CMOS image sensors is very active/hot these days: in consumer market, in the industrial arena, as well as in science & engineering. Thus, the job market & opportunities surrounding the field of image sensors is very high and expected to grow in future. This course aims at providing you with a solid foundation on this topic, and to serve as a stepping stone for you to progress towards more advanced image sensor topics. Broadly, this course covers the following: 1. Overview of the history and application of image sensors 2. Semiconductor device physics of photodiodes (from basics) 3. 3-Transistor and 4-Transistor CMOS image sensor pixels 4. Design, layout & fabrication of image sensors 5. Characterization of image sensors 6. Advanced image sensor topics 7. Introduction to Spectra - a device physics simulator tool	
Study Goals	Be able to understand 1. Semiconductor physics behind p-n junctions when used for photon detection 2. How a CMOS image sensor pixel works and how to measure their properties 3. The design and fabrication process of CMOS image sensor pixels 4. Application of image sensors/camera systems	
Education Method	Oral lectures using PowerPoint presentations Some hands on demonstrations using a camera and SPECTRA simulation	
Books	1. S. M. Sze, Physics of Semiconductor Devices, 2nd ed. New York: John Wiley & Sons, Inc., 1981. 2. A. J. P. Theuwissen, Solid-State Imaging with Charge-Coupled Devices, Kluwer Academic Publishers, 1995. 3. J. Nakamura, Image Sensors and Signal Processing for Digital Still Cameras, CRC Press, 2006. 4. P. Seitz, A. J. P. Theuwissen, Single-Photon Imaging, Springer, 2011	
Assessment	Mandatory student presentation followed by an oral examination The student presentation is not evaluated, but is necessary to appear for the oral exam. The oral exam is based on the course content as well as a few questions from the student presentation	
Co-Instructor	S. Kanjirakkat Raveendran	

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4173	Introduction to UWB technology, systems and applications	4
Responsible Instructor	Prof.dr. O. Yarovy	
Instructor	O.A. Krasnov	
Responsible for assignments	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0,0,0,6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Ultra-wideband (UWB) technology is becoming of greater importance as the pressure for signal bandwidth increases. This course introduces students to UWB technology as it will be used in future communications, radar and navigation systems, and gives detailed attention to both UWB wireless systems and applications in the frequency (FD) and time (TD) domains. The course consists of three major parts. In Part 1 the fundamentals of UWB technology such as waveforms and spectra of short pulses, basic UWB signal processing, radiation and propagation of short electromagnetic pulses are discussed. In Part 2 of the course is dedicated to realizations of UWB technology and systems. General approach to UWB wireless system design is introduced. UWB system design implementation of the key facets of modern UWB systems in both the frequency (FD) and time (TD) domains is discussed. In Part 3 applications of UWB technology in telecommunications, radar, imaging and positioning are discussed. General discussion of the system-level trade offs of such UWB systems is complemented by a detailed analysis of practical systems designed and implemented at TU-Delft. Lectures in the course are complemented by laboratories, at which students get hands-on experience in UWB signal processing and get in touch with wireless applications of UWB systems.	
Study Goals	This course introduces students to UWB technology as it is used in telecommunications, radar and navigation systems, and gives detailed attention to UWB wireless systems. After finishing this course, students will understand UWB signal properties in time and frequency domains, impact of UWB signal properties on wireless system architecture and performance, become familiar with available UWB technologies and UWB wireless system designs, and understand advantages and limitations of UWB technology for different applications. Students will also get hands-on experience with UWB signals and systems.	
Education Method	The course consists of lectures, supervised laboratories with experimental UWB equipment and the final assignment. The course finishes with a symposium, at which the final assignments are presented.	
Literature and Study Materials	Lecture handouts; course books, research papers. M.-G. Di Benedetto, G. Giancola Understanding Ultra Wide Band Radio Fundamentals, Prentice Hall PTR, 2004, ISBN-0-13-148003-0; T. Zwick et al., Ultra-wideband RF system engineering, Cambridge University Press, 2013, ISBN 978-1-107-01555-5	
Assessment	The total mark is based on evaluations of laboratory assignments reports, final assignment essay and presentation at the closing symposium. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4252	Analog Integrated Circuit Design	4
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	This course is recommended for students intending to take: - Structured Electronic Design Laboratory (ET8011MSC), - Analog CMOS design II (EE4525), - Oversampled data converters (ET4278), - Nyquist-rate data converters (ET4369), - Introduction to Power Conversion Technology (ET4382). This course is obligatory for students that wish to do their final project in the area of Biomedical Circuits (and Systems) (BioCAS).	
Expected prior knowledge	This course assumes (though it is not obligatory) that you have completed your BSc studies in electrical engineering and taken our MSc courses Analog Circuit Design Fundamentals (EE4C10) and Structured Electronic Design (EE4109) Having attended Measurement and Instrumentation (EE4C08) and Analog CMOS Design I (EE4520) is a plus.	
Course Contents	An intermediate course in analog circuit synthesis for microelectronic designers. Topics include: - review of analog design basics (i.e., frequency response, feedback and stability, noise analysis); - MOS models for short-channel modern devices; - gm/ID design methodology for modern analog circuits; - physical layout (e.g., device matching) for robust analog circuits; - design of voltage sources ranging from simple voltage dividers to high-performance bandgaps, and - current source implementations from a single resistor to high-quality references based on negative-feedback structures; - biasing techniques; - high-performance analog filters, from impulse response or transfer function down to circuit-level design; - quality aspects, such as accuracy, noise and impedance levels are treated within the context of power and area consumption requirements.	
Study Goals	To know, understand and be able to analyze and design (synthesize): - high-performance analog integrated filters - voltage references - bandgap references - current sources - amplifiers in all IC technologies (CMOS, bipolar, ...) and regions of operation (weak, moderate and strong inversion; saturation and triode/conduction)	
Education Method	Interactive lectures, flipped classroom (discussing lecture recordings and answering questions), homework assignments, class project.	
Literature and Study Materials	Reader, course notes, and video recordings of the lectures - all available via Brightspace.	
Assessment	Written examination (70%; the other 30% comes from a successfully completed class project). Students are allowed to bring 1 handwritten piece (A4) of paper and a pocket calculator to the exam. The results of the homework assignments may add up to one bonus point to the exam, only for the exam in April. The re-sit written exam in June will count for 100% of the final grade. It is not possible to re-sit the class project. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4260	Microsystem Integration	4
Responsible Instructor	T. Manzanque Garcia	
Instructor	K.M. Dowling	
Instructor	F. Arroyo Cardoso	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Instrumentation (EE4C08) Integrated circuits and MEMS technology (ET4289) Sensors & Actuators (ET4257)	
Course Contents	A microsystem is a complete instrument on a chip that can involve different domains of physics (electrical, mechanical, magnetic, etc.). The challenges associated with the combination of elements of different nature, including transducers and circuits into a single-chip integrated system, are more than compensated by the opportunities this concept offers in a wide range of applications. In this course, general concepts related to microsystems as sensors are first discussed. Secondly, approaches for multi-domain modeling are introduced and applied to the modelling electro-mechanical systems. Then, different transduction mechanisms, that allow for the conversion of signals between different physical domains, are introduced. Finally, all the modelling concepts are extended to the design of microsystem-based sensors. Throughout the course, the software package Comsol is introduced and used in several tutorials. The course includes theory (~8 lectures), Comsol tutorials, and tutorials for the realization of a final project, in which a microsystem is designed and modelled in groups.	
Study Goals	By the end of the course, you will be able to: - Identify parts of microsystems and their function. - Identify equations that govern the behaviour of different microsystem components, from different domains of physics. - Use electrical domain models for multidomain microsystems. - Use the finite element method (FEM) to model multidomain microsystems, using Comsol. - Analyze the dynamic behaviour of linear microsystems. - Design microsystems that meet given specifications, using both electrical domain modelling and FEM.	
Education Method	Lectures, tutorials and hands-on practical sessions with Comsol.	
Computer Use	The student requires access to computer systems for carrying out the Comsol tutorials and final assignment.	
Literature and Study Materials	Literature plus lecture notes.	
Assessment	The final grade of the course consists of the following components: - Final assignment report (50%) - Oral exam (50%) Final grade calculation: $(0.5 * \text{Final assignment report} + 0.5 * \text{Oral exam})$ In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4277	Microelectronics Reliability	4
Responsible Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Prof.dr. G.Q. Zhang	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course aims to provide the students with a thorough understanding of the reliability issues involved in electronic components and systems. The following subjects will be treated. - Basic reliability definition and lifetime distributions. - Reliability prediction methods. - Physical failure mechanisms in electronic components - Package related failures - Reliability screening and reliability testing - Failure analysis methods - Reliability data handling - Design considerations - System reliability	
Study Goals	After the course the students: - should see reliability as a basic requirement that should receive attention throughout a products complete lifecycle: specification, design, production, exploitation and disposal. - should have a sound understanding of the physical background of failures and the reliability test and failure analysis methods available. - should be able to make a proper reliability prediction based on available reliability (test) data. - should be able to evaluate reliability and availability figures of complex systems based on component reliability data. - should be able to make the proper design and maintenance choices to optimize these figures.	
Education Method	Lecture	
Assessment	written exam or project (project only possible when homework assignments are passed with average grade 6 or more) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4278	Over-Sampled Data Converters	4
Responsible Instructor	Prof.dr. K.A.A. Makinwa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10), Analog CMOS Design I (ET4295) and Nyquist-rate Analog-to-Digital Converters (ET4369). A good knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce students to the design of over-sampled converters. Students will learn about the basic functionality of sigma-delta converters, their architecture, how to simulate and design these architectures and how to map them to actual circuit blocks. Covered topics: - First-order sigma-delta: oversampling; noise-shaping; NTF/STF; decimation; overload; leakage. - Second-order sigma delta: linear model; stability; quantizer model; overload stability. - Sigma-delta architectures: bandpass; continuous-time; multi-bit; cascaded/MASH; feedback; feedforward; local feedback - Building blocks: SC integrator; RC integrator; loopfilter; current DAC; SC DAC; clock jitter - Case studies.	
Study Goals	- Students can explain the basic functionality of sigma delta modulators (over-sampling, noise-shaping, decimation, stability). - Students can describe the circuit implementations of various building blocks (integrators, DAC, quantizer, feed-forward and feedback coefficients). - Students can identify and compare the advantages and disadvantages of different architectures for sigma delta modulators (feedback, feed-forward, MASH, multi-bit, single-bit, continuous time and discrete time modulators). - Students can derive the STF and NTF from the linear model of a sigma delta modulator. - Students can design and simulate the architecture of a sigma delta modulator (continuous and discrete time, first and second order modulators, single-bit and multi-bit quantizer, MASH). - Students can simulate the effect of non-idealities of various blocks (such integrators limited gain, DAC non-linearity) on the performance of a sigma delta modulator. - Students can propose a (non-optimized) circuit implementation of a sigma delta modulator. - Students can identify the main techniques used in state-of-the-art designs of sigma delta modulators and assess their weakness and strengths. - Students can propose improvements to existing designs of sigma delta modulators.	
Education Method	Lectures and practical sessions	
Literature and Study Materials	hand-outs provided during the lectures; optional reference: R. Schreier, S. Pavan, and G.C.Temes, "Understanding Delta-Sigma Data Converters", Second Edition, published by John Wiley & Sons Inc., New Jersey, 2017.	
Assessment	Homework (30%), a final assignment discussed during the oral exam (50%), and an oral exam (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Textbook	
Remarks	Guest Instructor: Dr.M.Bolat kale, NXP Semiconductors	

ET4351	Digital VLSI Systems on Chip	4
Responsible Instructor	Dr.ing. C.P. Frenkel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of digital design (logic gates, combinational logic and sequential logic, finite state machines). Knowledge of hardware description languages (Verilog or VHDL), including how to write proper synthesizable code, is necessary. Previous exposure to microprocessor architecture is helpful but not necessary.	
Course Contents	Today's large-scale digital integrated circuits and systems-on-a-chip, such as modern general processing units (GPUs), can >50 billion transistors. Two things were instrumental in enabling this massive scalability: electronic design automation (EDA) tools and intellectual property (IP) blocks allowing for hierarchical design. In this course, you will learn how to use such EDA tools and IP blocks to go from the register-transfer level (RTL) description of your digital design to a final layout that is clean for tapeout and meets your specifications. To do so, you will also learn about power/performance/area tradeoff optimization, hardware-algorithm co-design, and memory hierarchy. These tools and concepts will be practiced and applied in a project, in which you will explore and apply your own design choices to meet a well-defined design target in a digital system-on-chip.	
Study Goals	After successful completion of the course, you will be able to: LO1. Design modern digital systems-on-a-chip with automated synthesis and place-and-route tools by - identifying the key design targets and tradeoffs, - selecting a proper IP to meet design targets, - translating the design constraints and tradeoffs to tool commands, - applying the tools to meet design targets. LO2. Optimize, in a rigorous and creative manner, the power/performance/area tradeoff associated with a given application use case. LO3. Link algorithmic requirements to hardware constraints. LO4. Extract the key performance metrics of a digital design by - verifying that simulation results can be trusted and that the design is clean for tapeout, - communicating these results effectively and rigorously, - discussing their implications within a broader societal context.	
Education Method	Lectures, labs	
Literature and Study Materials	* Course material (slides will be provided) * (Recommended) Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, by Erik Brunvand, Addison-Wesley, ISBN-10: 0321547993	
Assessment	The final course grade consists of two parts: * Final design report (50%) * Individual written exam (50%) The results of the two parts must have been obtained within the same academic year, and students must have a grade of 5.8 or higher for both the final design report and the individual written exam to pass the course. In case either of the two parts is failed, a resit must be taken in the form of an individual written exam that will account for 100% of the grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	C. Gao	

ET4362	High Speed Digital Design for Embedded Systems	5
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The course covers many important topics computer engineers and embedded system designers should be aware of. First, the physical effects of high-frequency signal propagation and transport are discussed. Next, the state of the art for high bandwidth/low latency digital communication and memory systems is presented. Special attention is given to parallel memory systems. Thereafter, low level circuit design and analysis principles relevant to high-speed digital design are explained. Clock distribution, metastability and skew-tolerant design are emphasized. Advanced techniques for high-speed combinational and sequential logic structures are considered in detail. More precisely, approaches like domino logic, pulsed latches and dual-rail logic will be discussed. The next part discusses advanced design techniques such as globally asynchronous locally synchronous (GALS) systems, systolic arrays, streaming dataflow, wave-pipelined circuits and fully asynchronous design. The last part concentrates on understanding the true theoretical bounds in, e.g., the energy consumption, of computers. The necessary theory and some practically realizable adiabatic logic design styles will be introduced. When time allows, brief introduction to the Gödel's incompleteness theorem and its impact on computers will be presented.	
Study Goals	The purpose of this course is to familiarize students with the latest developments in the field of high-speed digital design. The emphasis is on novel, non-traditional design concepts, techniques and methodologies.	
Education Method	Lectures and individual assignments.	
Literature and Study Materials	Presentations and selected research papers.	
Assessment	Written report on a selected research topic and oral exam on appointment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4369	Nyquist-Rate Data Converters	4
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10) and Analog CMOS Design I (ET4295)	
Course Contents	Introduction, sampling theory, quantization, and reconstruction; Digital-to-analog conversion: resistive, capacitive, and current steering; Sampling circuits: theory, non-idealities, and advanced techniques; Comparators: circuits, non-idealities, and design examples; Analog-to-digital conversion: flash, SAR, pipeline, time-interleaving.	
Study Goals	By the end of this course, you will be able to: - Explain the fundamentals of sampling, quantisation and reconstruction in data converters - Analyse the effect of non-idealities in sampling circuits and discuss advanced techniques to deal with them (bottom plate sampling, bootstrapping, oversampling, etc.) - Analyse the architectures and circuits commonly used in D/A interfaces (capacitive, resistive, current steering) and discuss their pros and cons - Analyse the architectures and circuits commonly used in A/D interfaces (flash, SAR, pipeline, time-interleaving) and discuss their pros and cons - read, understand and critically analyse state-of-the-art IEEE papers on Nyquist-rate data converters - Select the appropriate ADC architecture for a given set of performance specifications	
Education Method	4 hours of lectures plus a 4 hour tutorial	
Literature and Study Materials	MJM Pelgrom: Analog-to-Digital Conversion, 3rd Ed. Springer 2017	
Reader	Pdf versions of the slides and the homework will be provided	
Assessment	Weekly homework (formative assessment) + oral exam based on a selected ADC publication (summative assessment for 100% of the final grade). Resit oral exam based on a selected ADC publication possible in Q4 for 100% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4371	Mixed-mode Wireless transceivers	4
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0,0,4,0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	EE4C13 Wireless Systems for Electrical Engineering Applications EE4520, Analog CMOS Design I Basic signal processing; basic knowledge of Matlab	
Course Contents	The past several years have successfully brought all-digital techniques to radio frequency (RF) frequency synthesizers and transmitters, as well as digitally-intensive techniques to receivers. In addition, digital assistance is applied to RF circuits to improve performance and power consumption. This course will introduce the concepts of the digital RF approach and walk through the major building blocks that comprise new RF transceiver architectures, emphasis is placed on the: - Conventional charge-pump PLL and all-digital phase-locked loop (ADPLL) comprising: digitally-controlled oscillator (DCO), time-to-digital converter (TDC), and digital loop filter (HW1); - All-digital transmitters featuring polar or Cartesian signal processing in digitally-controlled power amplifiers (DPA) with wideband modulation capabilities possibly including Quadrature clock generation (HW2); - Direct-sampling discrete-time receivers comprising switched-cap circuits that perform various FIR and IIR filter operations (HW3). Exercises and projects use Matlab/ADS to model and simulate the above three subsystems.	
Study Goals	In-depth knowledge of all-digital PLL, advanced transmitter and discrete-time receiver	
Education Method	Lectures + design related homework	
Literature and Study Materials	-R. B. Staszewski and P. T. Balsara, All-Digital Frequency Synthesizer in Deep-Submicron CMOS, New Jersey: John Wiley & Sons, Inc., Sept. 2006. ISBN: 978-0471772552 -Babaie, M., Shahmohammadi, M., & Staszewski, R.B. (2019). RF CMOS Oscillators for Modern Wireless Applications (1st ed.). River Publishers. https://doi.org/10.1201/9781003339311 -M. Alavi, J. Mehta, and R. B. Staszewski, Radio-Frequency Digital-to-Analog Converters Implementation in Nanoscale CMOS, Elsevier/Academic Press, 302 pages, ISBN: 978-0-12-802263-4 (eBook: 978-0-12-802503-1), 26 Oct 2016. -Several related JSSC, TMTT, TCAS papers	
Assessment	RF design related homeworks (40-50% final grade) + written exam (60-50% final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Analysis Chips Circuits Design Integrated Intensive Matlab Project Semiconductors Signals and Systems Technology Telecommunication	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Instructor	Dr. L. Mazzarella	
Instructor	Prof.dr. I.M.F. Gordon	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Note: In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	

ET4379	Photovoltaic Lab Course	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Responsible Instructor	Dr.ir. R. Vismara	
Contact Hours / Week x/x/x/x	4/0/0/0 Q1=Q5! Also possible online in Q6, Q7, Q8!	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The aim is to offer you an advanced practical course on solar cells and photovoltaic systems. The following areas of practical training will be part of the course: (1) Characteristics of artificial and natural light (2) Characterization of the performance of lab-scale solar cells (3) Behavior of photovoltaic modules in different temperature and shading conditions (4) Charge and discharge properties of batteries (5) Performance of power electronic components (charge controllers, inverters, maximum power point trackers) (6) Operation of entire photovoltaic systems	
Study Goals	(1) Experiment with photovoltaic systems and all of their components under lab and real-life conditions. (2) Describe the working principle behind each component of a photovoltaic system. (3) Analyze the operation of all the elements of photovoltaic systems and of the system as a whole. (4) Compare the performance of different types of components of photovoltaic systems.	
Education Method	Laboratory sessions	
Literature and Study Materials	Manuals and templates are available on BrightSpace.	
Prerequisites	This course is meant for students who have already passed at least the PV Basics course (ET4376) and preferably also PV Technologies (ET4377) and PV Systems (ET4378). This course is also meant for 1st-year masters students with a bachelor degree in electrical engineering.	
Assessment	The assessment is based on reporting results from the individual tasks, using an Excel template provided on Brightspace. Two-person teams will be formed at the plenary session, the attendance of which is compulsory. The final grade is per team. The final grade is the average of the six technical reports. A sufficient grade (greater than or equal to 5.8) in all six technical reports is required to complete the course successfully. In case of failure of a technical report, you can use buffer sessions on week 1.7 and 1.8 to repeat the experiments. --- A virtual version of this course is made available for students to attend in Q2, Q3, and Q4 of the academic year 2023-2024. Each of the six tasks of the on-campus course will run in the internet browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, using numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to pass the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. Attending one of these instances of the course, no teams will be formed and the final grade is per person. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	The on-campus course can accommodate up to 48 students. Anyone interested in attending this course must apply during Q4 of the immediately previous academic year. A selection will then take place and, before the beginning of the instance in Q5, the line-up of students will be announced. Student(s) who failed on-campus in Q5 can re-take the virtual version of the course. There is no limit to the number of students who can attend the course's virtual version.	
Remarks	This practical course will be given on campus only in Q1. Conversely, a virtual version will be available for students to attend in Q2, Q3, and Q4. The enrolment is open on Brightspace from Q1 of the academic year 2023-2024.	

ET4382	Power conversion techniques in CMOS technology	3
Responsible Instructor	Dr.drs. Q. Fan	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	No modern electronic systems could work without power conversion technology. For instance, a laptop, a mobile phone, or an electric car has only one battery, which offers only one supply voltage. However, there are thousands of sub-circuits in these devices, analog and digital, high voltage and low voltage, CMOS and bipolar, and they require various supply voltages to operate properly. How to generate different supplies efficiently and accurately for these sub-circuits is a crucial and challenging task. This course will cover the most common DC-DC power conversion techniques, starting from the simplest resistor divider, to more sophisticated voltage regulator such as LDOs, switched-capacitor DC-DC converter and inductive DC-DC converter (buck converter). In addition, basics of class D power amplifiers will be introduced in this course, which shares similar principle as a buck converter. A class D power amplifier converts power from a DC supply to a high energy (>1W) AC audio signal, which can be found in a lot of headphones and high-end sound systems.	
Study Goals	Learning objectives Understanding the basic working principle of a switched-capacitor DC-DC converter Understanding the basic working principle of a buck converter Understanding negative feedback loops in DC-DC converters Understanding the basic working principle of a class D power amplifier	
Education Method	Lectures	
Literature and Study Materials	not applicable	
Assessment	30% homework, 70% project In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Guest Instructors: Dr. Marco Berkhout and Dr. Gerard Villar Pique from NXP Semiconductors	

ET4391	Advanced Microelectronics packaging	3
Responsible Instructor	Prof.dr. G.Q. Zhang	
Responsible Instructor	M. Mastrangeli	
Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Dr.ir. R.H. Poelma	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This is a multi-disciplinary course, dealing with scientifically challenging and economically important technology issues. MSc students from but not limited to electrical engineering, mechanical engineering, material science and physics are encouraged to attend.	
Course Contents	1. Introduction of micro/nanoelectronics packaging - Basics and state of the art of advanced front-end and back-end technologies - International roadmaps and strategic research agenda - Major application trends 2. Interconnect technologies 3. MEMS & sensor packaging 4. 3D and wafer-level system integration 5. Application specific packaging -1 (IC packaging, etc.) 6. Application specific packaging -2 (LED, etc) 7. Application specific packaging -3 (RF, Power, automotive, healthcare, etc.)	
Course Contents Continuation	As the bridge between devices and various multi-functional electronics systems, microelectronic system integration and packaging control more than 90% of the size, 60% of the cost, and largely the system performance and reliability. It is one of the most fascinating and rapidly developing technology and business fields in semiconductors, playing a dominant role in the development of future micro/nanoelectronics and systems.	
Study Goals	At the end of the course, the student should be able to: 1) Describe and assess basics and state-of-the-art in microelectronic packaging and reliability; 2) Describe and assess application-specific packaging and associated requirements, constraints and solutions for reliability; 3) Design, implement and characterize a standalone electronic and/or microelectromechanical system.	
Education Method	Lectures and field visit.	
Assessment	Assessment is based on mini-research projects, conducted preferably in teams. The projects are proposed by the instructors, and can also be proposed in agreement with the instructors by the students based on the content of the course or personal interests. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ing. H.W. van Zeijl	
Co-Instructor	M. Mastrangeli	

ET4icp	IC-Technology Lab	2
Responsible Instructor	Dr.ir. S. Vollebregt	
Contact Hours / Week x/x/x/x	28 hrs.	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics as in EE4585	
Course Contents	During seven half days the students, in groups of 3, will have the opportunity to gain experience in the process- and device-simulation of a MOSFET transistor. They will perform several processing steps in the Else Kooi Lab clean room and will measure and characterize the fabricated device. Enrollment through Brightspace, max course capacity is 36 students!	
Study Goals	After this course, students are able to: 1. Explain the different process steps of the IC fabrication process 2. Perform calculations on the oxide thickness of silicon after thermal oxidation, and the implantation depth and diffusion depth of dopants 3. Explain the relation between threshold voltage and implantations, and relate this to the measured threshold voltages from transistors 4. Select dopant atoms and masking material for silicon devices based on the simulations performed in the course 5. Identify the different implanted regions and junctions in cross-sections obtained from doping profile simulations 6. Argue on the specific advantages and disadvantages of wet and dry etching as used during the lab course. 7. Analyse the measurements obtained from different test structures described in the course guide 8. Determine the doping levels from sheet resistance measurements	
Education Method	Lab course	
Literature and Study Materials	Lecture slides and a course manual which will be supplied during the compulsory lecture	
Assessment	Group report. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) 28 hrs, first session is scheduled, after that on individual appointment. Students have to attend all parts of the course. The course can only accommodate 30 students.	

ET8011MSC	Structured Electronic Design Laboratory	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Responsible Instructor	A.J.M. Montagne	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Grade 8 or higher for one of the courses Structured Electronic Design EE4109. During the oral exam of EE4109 it is decided if the student is granted access to the masterclass.	
Course Contents	This lab is a master class in electronic design. Selected students design and fully simulate an amplifier. Design reviews are organised every week. In a small group (5-6) results are discussed, special topics addressed and more complicated design problems tackled. This course offers the students the opportunity to refine their skills as an electronic designer. Enrollment is limited and restricted. Applicants are evaluated before acceptance, and should have sufficient standing (grade 8 or higher) in course "Structured Electronic Design", EE4109.	
Study Goals	To become an expert analog expert designer that is able to create optimal amplifiers while fully understanding every design decision and being able to defend them in front of other experts.	
Education Method	Master class. Regular design reviews with expert designers with industrial experience.	
Literature and Study Materials	All study material, including the book (free downloadable PDF) can be found on the companion website: https://analog-electronics.tudelft.nl/	
Assessment	The assessment is done via the design report made during the masterclasses with SLiCAP in which the design is fully documented and the oral defense thereof. The grade is based on the quality of the work (70%) and the performance during the oral defense (30%) of the design.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Master classes start when enough students (4-5) have qualified. A session schedule is made with the students involved.	

QIST4400	Quantum Computer Architecture	5
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr. S. Feld	
Instructor	Dr.ir. T.H. Oosterkamp	
Instructor	Dr. R. Ishihara	
Contact Hours / Week x/x/x/x	0/0/4/0/	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The course will assume knowledge of fundamentals of quantum information. Basic knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce the students the overall system of a quantum computer, focusing on the classical hardware and software infrastructure required to build a quantum computer together with the quantum hardware. Covered topics include: - Quantum computer architecture; quantum languages; compilers; QISA; microarchitecture. - Quantum simulator (QX simulator); - Quantum error correction: quantum error correction codes; encoding; logical operations. - Mapping of quantum circuits; placement of qubits; scheduling of operations; routing of qubits; - Classical hardware for quantum computing: classical controller and quantum processor; qubit hardware and the need for cryogenic electronics; - Receiver and transmitter architectures for controlling qubits: frequency up/down-conversion, modulation schemes, frequency generation. - Hardware building blocks: amplifiers, analog-to-digital converters, digital-to-analog converters, digital processing, FPGA - Cryo-CMOS: device characteristics, state-of-the-art, design principles.	
Study Goals	By the end of this course, students are able to: - Explain the different layers that are required to build a quantum computer. - Discuss the basic functioning of quantum error correction (QEC). - Carry out the mapping step of simple quantum circuits onto specific physical device layouts. - Interpret the results of different quantum circuits written in language QASM and executed using simulator QX, including the following: Basic quantum algorithms; Fundamental QEC codes; Surface code, especially for a Ninja star. - Identify and describe the different hardware blocks in the classical controller in a quantum computer. - Derive the specifications of the individual components in a classical controller from the requirements of the whole controller. - Compare different hardware blocks based on their specifications and can select the appropriate component for target functionality and specifications. - Describe the main features and advantages/disadvantages of the adoption of cryogenic electronics in general and cryo-CMOS specifically.	
Education Method	- Slide decks including small exercises - Videos including quizzes - Interactive lectures including activities - Homework assignments - Paper reading including processing its content	
Assessment	The final grade consists of two parts: - Part 1: Weekly homework, will be submitted electronically. Individual submission. - Part 2: Final assignment, preparation of presentation of 20 minutes with 10 minutes Q&A. Assignment will be handled by pairs of two, there will be individual grades. The global course grade is an average of each parts grade (50% part 1 + 50% part 2). Students are allowed to omit one homework. There is no resit.	

TPM027A	Cyber Risk Management	5
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. A.P. Afghari	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	MOTIVATION: The challenge of selecting the optimal technical, organisational, legal, and other preventive and repressive measures to reduce cyber risks to acceptable levels can only be understood in the context of the application of Cyber Risk Management. Risk Management is about analysing the relationships between threats, incidents and risks (here in the complex world of cyberspace), based on which an adequate set of countermeasures can be designed. SYNOPSIS: Risk (= the potential lo losing something of value) can manifest itself in cyberspace in all kinds of ways: values at stake are financial wealth, health, physical condition (of people, materials, goods, infrastructures, etc.), well-being, reputation, privacy, trust, etc. Based on a conceptualisation of cyberspace and its various sub-domains (discussed in the project week of year 1), we introduce risk assessment approaches, both of qualitative and quantitative manner, illustrated with case studies. CONTENT: Cyberspace Risk concepts Probabilistic and adversarial risk System and attacker models Requirements and policies Risk assessment methods Qualitative and quantitative methods Measuring security and effectiveness	
Study Goals	After this course, students are able to: Understand and apply risk concepts to (possible) incidents in cyberspace Understand the theoretical principles of cyber risk management Explain the role of adversarial risk models, system models, and attacker profiles in cyber risk assessment Analyze and apply state-of-the-art cyber risk assessment methodologies to (complex, multi-step) cyber incidents Analyze the (expected) effect of measures that help to prevent the occurrence of cyber incidents Justify investments in cyber security	
Education Method	LECTURES: There are two lecture slots per week EXAM: There is a written exam at the end of the course. LANGUAGE: The course is taught in English. LECTURERS: Prof. dr. ir. Pieter v. Gelder (TUD/TPM) and guest lecturers.	
Assessment	#### Assessment Grading will be based on the individual written exam at the end of the course. Final mark of this course is determined by the grade of the individual exam (100%). The minimum grade for the individual exam to pass this course is 6.0. The grade for the individual exam is 2 years valid. The individual exam is summative. Detailed feedback on the individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within the study year 2023-2024.	
Elective	Yes	
Targetgroup	MSc	
Category	MSc level	

Suggested profiles 2023	
Introduction 1	The tracks in EE typically have an informal sub-structure called profiles. A profile is a more or less canned study path. It typically prepares for a thesis subject in a certain area and typically with a specific professor or small set of professors of one section, overseeing that research area. For example, the Microelectronics track offers the Digital Systems profile, the RF/Analog Circuit Design profile, the Microsystems & Technology profile and the Biomedical Electronics profile. Other tracks offer different profiles. For the purposes of composing an IEP, a profile presents a short list of coherent courses (in all tiers) that together leads to a coherent program of study that prepare for a thesis topic in the area of the profile. As such, a profile can specify (at most) two from three courses in the common core, two from three courses in the track core, and at most 12 EC of additional specialization courses. Note, however, that profiles are nothing more than convenient selections of courses that go well together, and do not imply any formal restriction whatsoever. Based on individual interest and subject to approval of the IEP by the (tentative) thesis supervisor and the MSc coordinator and the responsible professor, a much wider variety of r programs is possible. Students are free to compose their individual program within the rules as specified in FIXME.

Digital Systems 2023	
Introduction 1	This profile covers theory and design methodologies of digital circuits and systems, with application areas in wireless communication, biomedical imaging based on single-photon technology, quantum computing, 3D IC design. Current areas of interest are (1) Quantum imaging sensors, (2) high-level electronic (VLSI) system level design (power optimization and reduction techniques, SoC design methodology, communication architectures and protocols), and (3) physical modeling and verification of large VLSI circuits (parasitics and temperature modeling). Responsible professors: Dr. Charlotte Frenkel Dr. Chan Gao
Introduction 2	Compulsory EE4610 Digital IC Design I EE4615 Digital IC Design II Suggested EE4C07 Advanced Computing Systems EE4C08 Measurement and Instrumentation ET4054 Methods and Algorithms for System Design ET4170 Computer Arithmetic ET4351 VLSI Systems on Chip ET4371 Advanced Wireless transceivers

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Instructor	Ir. J.W. Peltenburg	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = $0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3)$ L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1] In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4035	Computer Arithmetic	5
Responsible Instructor	Dr. S.D. Cotofana	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course emphasises on theoretical aspects of computer arithmetic. It covers concepts and various topics related to fixed and floating-point number systems, algorithms and implementations for addition, multiplication, division, square root, and other high order arithmetic operations.	
Study Goals	The study goals for the course are as follows: 1. The student can operate with concepts and notions related to: (i) number representation systems; (ii) algorithms and implementations for basic integer arithmetic operations, i.e., addition/subtraction, multiplication, and division. 2. Assuming certain processor architecture and requirements she/he can perform design space exploration and select the most appropriate algorithms for the implementation of the basic functional units. 3. She/he can operate with concepts and notions related to floating point systems and operations, elementary function evaluation, e.g., CORDIC, and error analysis. 4. She/he can design arithmetic units and application specific (co-)processors, optimized for speed, area, power consumption, or combinations of those. 5. She/he can study recent advances in computer arithmetic, classify research papers, and report about them.	
Education Method	Lectures and homework assignments.	
Literature and Study Materials	Text book: Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000, ISBN 0-19-512583-5. Example exams with solutions available on Brightspace.	
Assessment	Exam and homework assignments contribute to the final grade as follows: written open book exam - 60% of the final grade; homework assignments - 40% of the final grade; Repair option: Written exam = resit Individual Assignment = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Exam is open book. Books on computer arithmetic, computer architecture, and logic design, and lecture slides are allowed during the exams.	

CESE4040	Processor Design Project	5
Responsible Instructor	Dr. S.D. Cotofana	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	To be able to complete the project the student has to have Digital Systems and VHDL knowledge. Moreover before embarking in this course she/he has to successfully complete the following courses: CESE4085 Modern Computer Architecture and CESE4035 Computer Arithmetic courses.	
Course Contents	This is an integration course that is aiming to let the students utilize computer architecture and arithmetic theoretical knowledge in the design of a general purpose processor. In this line of reasoning they have to improve the performance of a MIPS-based computing platform by focusing on computer architecture and arithmetic relevant parts of it. As starting point students receive two archives: one which includes the MIPS System on Chip VHDL code and a set of benchmarks selected from the GMPbench and MiBench suites, as well as all the necessary files required for simulation and FPGA implementation, and a second archive with the MIPS cross-compiler. To evaluate their new design, students are given a Zynq-7000 FPGA board per group.	
Study Goals	The course study goals are as follows: 1. The student can perform, assuming certain processor architecture and requirements, a design space exploration and select the most appropriate algorithms for the implementation of the basic functional units. 2. The student can integrate computer arithmetic and computer architecture knowledge and utilize them in order to design/optimize processors. 3. The student is able to evaluate processor performance based on benchmarks and take the appropriate architectural decisions in order to improve on various design metrics. 4. The student can design arithmetic units and processors and implement them using FPGA technology. 5. The student is able to operate in a small team and collaborate for the successful completion of a design assignment. 6. The student can communicate his/her proposal, experimental results, and conclusions in English using the appropriate technical language in written as well as orally.	
Education Method	Lectures and Project.	
Literature and Study Materials	There is no dedicated study material but the students are advised however to utilize as supporting study material Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000, and Computer Architecture: A Quantitative Approach, J.L. Hennessy and D.A. Patterson, 4th edition. Moreover we encourage you to utilize the IEEE Explore library to find research publications that may help you in the successful project completion.	
Assessment	To complete the course students have to submit the project archive of the new core they developed, a report describing their approach and findings, and to give a short presentation of their project in the context of a symposium. The new core functionality is verified and all projects are checked for (between groups) plagiarism. Non functional designs do not pass the project. Plagiarism can also make you fail. Your final score for the project is determined based on the following criteria: The performance of your design (DP). The benchmark scores you report are important. The higher the benchmark score you get the better but this is not the only relevant aspect. In the performance evaluation we also take into consideration the other metrics with more emphasis on the compound ones. The technical merit of your approach (TM). Aspects as innovation level and implementation quality are considered. The report (R). Report organization, content, and language are important aspects at this point. The presentation (P). Here we also look at the capability to ask questions and to answer questions from the auditorium. The ET4 171 final grade is computed as: $\text{Grade} = 0.35 \cdot \text{DP} + 0.35 \cdot \text{TM} + 0.20 \cdot \text{R} + 0.10 \cdot \text{P} - \text{C},$ where C can assume values between 0 and 1 and reflects: (i) the lack of collaboration in the group; (ii) the amount of consultancy you asked me for during the project completion. If the group is functioning like a team and you do not ask too much help C is 0, otherwise it can get values up to 1. Repair option: Individual Assignment or Group Assignment = Repair: Re-submit deliverable Individual Project or Group Project = Repair: Re-submit deliverable Individual Presentation or Group Presentation = Resit (new presentation) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

EE4615	Digital IC Design II	3
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will consolidate the knowledge acquired in EE4610 by practicing the design of an advanced complex digital VLSI design, where high-level specifications are translated to transistor sizes, going through several abstraction levels. At each level, the opportunities and limitations of the physical implementation will be considered for finding better solutions and tradeoffs, by means optimization and the search of new topologies, while keeping complexity to a minimum to achieve the desired functionality. Parasitics and the other effects deriving from the analog behavior of digital circuits will be taken into account using four-corner analysis, as well as simplifications and modeling at each abstraction level and other semi-automatic tools.	
Study Goals	The main goal of the course is to give students the tools to successfully design complex, scalable circuits and systems. The students will also learn how to work in and coordinate teams of designers around a complex project from high-level specifications to the actual transistor-level implementation and post-layout verification.	
Education Method	The course will entirely be based in the lab where the target design will be implemented by students in groups of 2-3. Only based on students/groups wish, the final design will be laid out in a commercial CMOS process and possibly fabricated and tested.	
Assessment	The course will be graded based on the target design. The following criteria will be taken into account: 1. Functionality achieved; 2. High-level specifications met; 3. Four-corner analysis completed; Bonus points will be given to those students that will exceed the specifications and create a robust, reproducible design. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4690	Hardware Architectures for Artificial Intelligence	5
Responsible Instructor	Dr.ing. R.K. Bishnoi	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basics of Digital design	
Course Contents	Recent advancements in the field of Artificial Intelligence (AI) have demonstrated its great potential in solving exceptionally complex problems, that can even surpass humans. To support such AI algorithms, the demand of developing new and efficient hardware is growing with rapid pace. In particular, for a variety of edge/IoT devices, hardware platform locally deploys these algorithms that decreases the exchange of information with the cloud which significantly reduces the networking costs and boost the overall performance of the devices. Hardware architectures play a critical role in executing AI algorithms. This course gives an overview of AI from hardware perspective and discusses close to the state-of-the-art architectures and training mechanisms. In this course, you will learn the concept of advanced forms of AI (i.e. Deep Neural Networks) and how they can solve the complex problems. You will also learn, what are the challenges associated with the underlying hardware and how new architectures can address those. Moreover, the course will teach you the concept of brain-inspired computing using in-memory processing hardware platform. Technically, this course covers following topics: 1) Machine inference models (i.e. Support Vector Machine, Decision tree, Regression analysis, clustering, etc.), machine learning algorithms (i.e., Supervised, Unsupervised, Semi-supervised, Reinforcement learning models) and their associated challenges. 2) Deep Neural Network (DNN), Classification of DNNs, feed-forward, back propagation and learning mechanism (such as Gradient decent and Stochastic learning). 3) Spiking Neural Network, Spike Time Dependent Plasticity training algorithms and some existing hardware architectures (such as Intels Loihi, SpinNaker, IBMs TrueNorth, Neurogrid, etc.). 4) Emerging technologies for neuromorphic computing (using memristor devices such as RRAM, MRAM & PCRAM) and non-conventional computing paradigm (such as computing-in-memory, hybrid-design approach, crossbar arrays).	
Study Goals	The main aims of the course are: 1. Describe the concept of AI, inference and training models as well as their associated challenges from hardware perspective. 2. Understand the hardware AI requirements for the data-centers as well as edge devices. 3. Discuss the neuromorphic computing and its advantages with respect to conventional computing paradigms. 4. State details about Artificial Neural Network and their training algorithms. 5. Develop understanding about Spiking Neural Network and their training mechanisms and also, comparison with Artificial Neural Network and their evaluations based on design metrics such as accuracy, power, speed, size, etc. 6. Describe state-of-the-art neuromorphic hardware architectures and discuss their design implementation details. 7. Study the advanced topics of neural networks based-on computation-in-memory concept using emerging non-volatile memories. 8. Become an AI engineer with good understanding about hardware implementation.	
Education Method	Lectures, assignments and lab-work	
Assessment	Oral and lab In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Artificial intelligence	

EE4695	Hardware Dependability	5
Responsible Instructor	M. Taouil	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	0,0,4,0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A course on VHDL/Verilog design such as one of the following: - Digital Systems B (EE1D21) - System Design with HDLs (ET4272)	
Course Contents	Integrated circuits (ICs) are integral part of any electronics system we see around. Developing, designing and deploying of such chips is by far more than just getting your SPICE simulation showing a correct behavior. Ensuring high quality, reliable and secure manufactured chips is a critical and crucial step in the whole IC process. A chip targeting children toys could cost 0.5 while a similar chip (in terms of functionality) could cost 100X to 1000X more when used for critical application such as military. Hardware dependability (e.g., quality, reliability, security) is a key differentiator. This course will introduce you to hardware dependability concepts and aspects. You will learn that ensuring such dependability needs the design of appropriate countermeasures, both during the design phase (i.e., part of the design) and in field/ during the lifetime of the application (i.e., continue monitoring and acting when needed). The course will teach you how to guarantee that a chip that may consists of billions of transistors and connections can be tested in less than one second, why the warranty you get for your phone is just two years rather than e.g., 5 years, how hackers can retrieve secret information from your device or smart phone and how to prevent it from happening, etc. Technically, the topics that will be covered are: a) Test/Quality: manufacturing defects, fault modeling, test development, design -for-testability, memory test, build-in-self test, etc; b) Reliability: reliability failure mechanism, modeling, design-for-reliability, testing, etc.; c) Security: hardware attacks and countermeasures, IP attacks, functionality Attacks, Data Attack, design-for-Security, etc.	
Study Goals	The main aims of the course are: 1. Describe the importance of Hardware Dependability (i.e., testability, reliability, security, etc.) and understand its relevance to electronic design and its overall impact. . 2. Point out the strong correlation between VLSI Design, Test, Reliability and security 3. State the trends and challenges in test, reliability, security domain 4. Describe transistor and interconnects defect mechanisms and the way they behave at the electrical/functional level and how they are tested using fault models and test algorithms 5. Examine different test methodologies for logic and sequential circuits, their advantages, disadvantages, cost, limitations, etc and how they impact the security 6. Analyze and propose different test algorithms and Design-for-Testability DFT methodologies, their advantages, disadvantages, cost and limitations and how they impact the security 7. Implement block ciphers in hardware and evaluate their weaknesses 8. Analyse hardware attacks and evaluate the countermeasures that can be used against them 9. Develop fault injection and side channel attacks and design countermeasures for them 10. Become a better VLSI designer, a better test engineer/ product engineer, and a better hardware security engineer.	
Education Method	Lectures, homework assignments and lab	
Literature and Study Materials	- Lecture Notes - Lab manual - Book: Essentials of Electronic Testing (by M.L. Bushnell and V.D. Agrawal, ISBN 0-7923-799-1-8) - Syllabus containing material from online resources and material from the following books: - Introduction to Hardware Security and Trust - Trustworthy Reconfigurable Systems - Trusted Computing for Embedded Systems - Security in Embedded Devices	
Assessment	Assignments + lab + "group" oral examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4351	Digital VLSI Systems on Chip	4
Responsible Instructor	Dr.ing. C.P. Frenkel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of digital design (logic gates, combinational logic and sequential logic, finite state machines). Knowledge of hardware description languages (Verilog or VHDL), including how to write proper synthesizable code, is necessary. Previous exposure to microprocessor architecture is helpful but not necessary.	
Course Contents	Today's large-scale digital integrated circuits and systems-on-a-chip, such as modern general processing units (GPUs), can >50 billion transistors. Two things were instrumental in enabling this massive scalability: electronic design automation (EDA) tools and intellectual property (IP) blocks allowing for hierarchical design. In this course, you will learn how to use such EDA tools and IP blocks to go from the register-transfer level (RTL) description of your digital design to a final layout that is clean for tapeout and meets your specifications. To do so, you will also learn about power/performance/area tradeoff optimization, hardware-algorithm co-design, and memory hierarchy. These tools and concepts will be practiced and applied in a project, in which you will explore and apply your own design choices to meet a well-defined design target in a digital system-on-chip.	
Study Goals	After successful completion of the course, you will be able to: LO1. Design modern digital systems-on-a-chip with automated synthesis and place-and-route tools by - identifying the key design targets and tradeoffs, - selecting a proper IP to meet design targets, - translating the design constraints and tradeoffs to tool commands, - applying the tools to meet design targets. LO2. Optimize, in a rigorous and creative manner, the power/performance/area tradeoff associated with a given application use case. LO3. Link algorithmic requirements to hardware constraints. LO4. Extract the key performance metrics of a digital design by - verifying that simulation results can be trusted and that the design is clean for tapeout, - communicating these results effectively and rigorously, - discussing their implications within a broader societal context.	
Education Method	Lectures, labs	
Literature and Study Materials	* Course material (slides will be provided) * (Recommended) Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, by Erik Brunvand, Addison-Wesley, ISBN-10: 0321547993	
Assessment	The final course grade consists of two parts: * Final design report (50%) * Individual written exam (50%) The results of the two parts must have been obtained within the same academic year, and students must have a grade of 5.8 or higher for both the final design report and the individual written exam to pass the course. In case either of the two parts is failed, a resit must be taken in the form of an individual written exam that will account for 100% of the grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	C. Gao	

ET4362	High Speed Digital Design for Embedded Systems	5
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The course covers many important topics computer engineers and embedded system designers should be aware of. First, the physical effects of high-frequency signal propagation and transport are discussed. Next, the state of the art for high bandwidth/low latency digital communication and memory systems is presented. Special attention is given to parallel memory systems. Thereafter, low level circuit design and analysis principles relevant to high-speed digital design are explained. Clock distribution, metastability and skew-tolerant design are emphasized. Advanced techniques for high-speed combinational and sequential logic structures are considered in detail. More precisely, approaches like domino logic, pulsed latches and dual-rail logic will be discussed. The next part discusses advanced design techniques such as globally asynchronous locally synchronous (GALS) systems, systolic arrays, streaming dataflow, wave-pipelined circuits and fully asynchronous design. The last part concentrates on understanding the true theoretical bounds in, e.g., the energy consumption, of computers. The necessary theory and some practically realizable adiabatic logic design styles will be introduced. When time allows, brief introduction to the Gödel's incompleteness theorem and its impact on computers will be presented.	
Study Goals	The purpose of this course is to familiarize students with the latest developments in the field of high-speed digital design. The emphasis is on novel, non-traditional design concepts, techniques and methodologies.	
Education Method	Lectures and individual assignments.	
Literature and Study Materials	Presentations and selected research papers.	
Assessment	Written report on a selected research topic and oral exam on appointment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4371	Mixed-mode Wireless transceivers	4
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0,0,4,0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	EE4C13 Wireless Systems for Electrical Engineering Applications EE4520, Analog CMOS Design I Basic signal processing; basic knowledge of Matlab	
Course Contents	The past several years have successfully brought all-digital techniques to radio frequency (RF) frequency synthesizers and transmitters, as well as digitally-intensive techniques to receivers. In addition, digital assistance is applied to RF circuits to improve performance and power consumption. This course will introduce the concepts of the digital RF approach and walk through the major building blocks that comprise new RF transceiver architectures, emphasis is placed on the: - Conventional charge-pump PLL and all-digital phase-locked loop (ADPLL) comprising: digitally-controlled oscillator (DCO), time-to-digital converter (TDC), and digital loop filter (HW1); - All-digital transmitters featuring polar or Cartesian signal processing in digitally-controlled power amplifiers (DPA) with wideband modulation capabilities possibly including Quadrature clock generation (HW2); - Direct-sampling discrete-time receivers comprising switched-cap circuits that perform various FIR and IIR filter operations (HW3). Exercises and projects use Matlab/ADS to model and simulate the above three subsystems.	
Study Goals	In-depth knowledge of all-digital PLL, advanced transmitter and discrete-time receiver	
Education Method	Lectures + design related homework	
Literature and Study Materials	-R. B. Staszewski and P. T. Balsara, All-Digital Frequency Synthesizer in Deep-Submicron CMOS, New Jersey: John Wiley & Sons, Inc., Sept. 2006. ISBN: 978-0471772552 -Babaie, M., Shahmohammadi, M., & Staszewski, R.B. (2019). RF CMOS Oscillators for Modern Wireless Applications (1st ed.). River Publishers. https://doi.org/10.1201/9781003339311 -M. Alavi, J. Mehta, and R. B. Staszewski, Radio-Frequency Digital-to-Analog Converters Implementation in Nanoscale CMOS, Elsevier/Academic Press, 302 pages, ISBN: 978-0-12-802263-4 (eBook: 978-0-12-802503-1), 26 Oct 2016. -Several related JSSC, TMTT, TCAS papers	
Assessment	RF design related homeworks (40-50% final grade) + written exam (60-50% final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Analysis Chips Circuits Design Integrated Intensive Matlab Project Semiconductors Signals and Systems Technology Telecommunication	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Microsystems & Technology 2023	
Introduction 1	This profile is directed to the design and development of innovative devices and microsystems for various applications. It comprises technology development, hybrid device integration and application specific packaging. Responsible professors: Prof. Lina Sarro Prof: Paddy French Prof: Edoardo Charbon
Introduction 2	Compulsory EE4610 Digital IC Design I ET4289 Integrated Circuits and MEMS Technology ET4257 Sensors and Actuators Suggested EE4C08 Measurement and Instrumentation EE4520 Analog CMOS design I EE4585 Semiconductor Device Physics ET4391 Advanced Microelectronics packaging ET4260 Microsystem Integration ET4icp IC-Technology Lab

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effetc of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cacoded amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4260	Microsystem Integration	4
Responsible Instructor	T. Manzanque Garcia	
Instructor	K.M. Dowling	
Instructor	F. Arroyo Cardoso	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Instrumentation (EE4C08) Integrated circuits and MEMS technology (ET4289) Sensors & Actuators (ET4257)	
Course Contents	A microsystem is a complete instrument on a chip that can involve different domains of physics (electrical, mechanical, magnetic, etc.). The challenges associated with the combination of elements of different nature, including transducers and circuits into a single-chip integrated system, are more than compensated by the opportunities this concept offers in a wide range of applications. In this course, general concepts related to microsystems as sensors are first discussed. Secondly, approaches for multi-domain modeling are introduced and applied to the modelling electro-mechanical systems. Then, different transduction mechanisms, that allow for the conversion of signals between different physical domains, are introduced. Finally, all the modelling concepts are extended to the design of microsystem-based sensors. Throughout the course, the software package Comsol is introduced and used in several tutorials. The course includes theory (~8 lectures), Comsol tutorials, and tutorials for the realization of a final project, in which a microsystem is designed and modelled in groups.	
Study Goals	By the end of the course, you will be able to: - Identify parts of microsystems and their function. - Identify equations that govern the behaviour of different microsystem components, from different domains of physics. - Use electrical domain models for multidomain microsystems. - Use the finite element method (FEM) to model multidomain microsystems, using Comsol. - Analyze the dynamic behaviour of linear microsystems. - Design microsystems that meet given specifications, using both electrical domain modelling and FEM.	
Education Method	Lectures, tutorials and hands-on practical sessions with Comsol.	
Computer Use	The student requires access to computer systems for carrying out the Comsol tutorials and final assignment.	
Literature and Study Materials	Literature plus lecture notes.	
Assessment	The final grade of the course consists of the following components: - Final assignment report (50%) - Oral exam (50%) Final grade calculation: $(0.5 * \text{Final assignment report} + 0.5 * \text{Oral exam})$ In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4277	Microelectronics Reliability	4
Responsible Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Prof.dr. G.Q. Zhang	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course aims to provide the students with a thorough understanding of the reliability issues involved in electronic components and systems. The following subjects will be treated. - Basic reliability definition and lifetime distributions. - Reliability prediction methods. - Physical failure mechanisms in electronic components - Package related failures - Reliability screening and reliability testing - Failure analysis methods - Reliability data handling - Design considerations - System reliability	
Study Goals	After the course the students: - should see reliability as a basic requirement that should receive attention throughout a products complete lifecycle: specification, design, production, exploitation and disposal. - should have a sound understanding of the physical background of failures and the reliability test and failure analysis methods available. - should be able to make a proper reliability prediction based on available reliability (test) data. - should be able to evaluate reliability and availability figures of complex systems based on component reliability data. - should be able to make the proper design and maintenance choices to optimize these figures.	
Education Method	Lecture	
Assessment	written exam or project (project only possible when homework assignments are passed with average grade 6 or more) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4289	Integrated Circuits and MEMS Technology	4
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Instructor	Dr. C.M.F. Viellard-Boutry	
Instructor	Dr.ing. H.W. van Zeijl	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics (EE 4585)	
Course Contents	This lecture is an introduction in the fabrication technologies used for Integrated Circuits (ICs) and Micro Electro-Mechanical Systems (MEMS). The aim of this lecture is to build a bridge between the physical world (the device physics) and the electronic world (the device characteristics). In this way, students gain more insight in the critical aspects and challenges in IC fabrication. The series of lectures starts with the discussion of basic microelectronic devices with a brief review of the physics involved. The associated basic fabrication technology, required to fabricate these devices is also discussed. After that, the mainstream IC fabrication technology (CMOS) will be discussed in more detail. In separate chapters, silicon crystal growth, epitaxy, ion implantation, chemical vapor deposition, etching and photolithography will be introduced. Their physical motivation and the impact on the device characteristics will be discussed. Finally, silicon bulk and surface micromachining technologies are introduced to illustrate the potential of 3D micro structuring in the development of Microsystems or MEMS. These IC compatible technologies are employed to realize multi-function systems for many scientific and industrial application areas.	
Study Goals	You know the definition of the basic process steps to fabricate a device (IC or MEMS) You can explain the basic physics and chemistry of these fabrication steps in an IC /MEMS process flow You can analyse the relationship between the series of process steps necessary to build the device and the device performance. You can evaluate how device requirements are translated into a process flow You are able to propose solutions to specific device technological questions	
Education Method	lectures	
Literature and Study Materials	R.C. Jager introduction to Microelectronic Fabrication & course slides	
Assessment	written exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. S. Vollebregt	

ET4391	Advanced Microelectronics packaging	3
Responsible Instructor	Prof.dr. G.Q. Zhang	
Responsible Instructor	M. Mastrangeli	
Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Dr.ir. R.H. Poelma	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This is a multi-disciplinary course, dealing with scientifically challenging and economically important technology issues. MSc students from but not limited to electrical engineering, mechanical engineering, material science and physics are encouraged to attend.	
Course Contents	1. Introduction of micro/nanoelectronics packaging - Basics and state of the art of advanced front-end and back-end technologies - International roadmaps and strategic research agenda - Major application trends 2. Interconnect technologies 3. MEMS & sensor packaging 4. 3D and wafer-level system integration 5. Application specific packaging -1 (IC packaging, etc.) 6. Application specific packaging -2 (LED, etc) 7. Application specific packaging -3 (RF, Power, automotive, healthcare, etc.)	
Course Contents Continuation	As the bridge between devices and various multi-functional electronics systems, microelectronic system integration and packaging control more than 90% of the size, 60% of the cost, and largely the system performance and reliability. It is one of the most fascinating and rapidly developing technology and business fields in semiconductors, playing a dominant role in the development of future micro/nanoelectronics and systems.	
Study Goals	At the end of the course, the student should be able to: 1) Describe and assess basics and state-of-the-art in microelectronic packaging and reliability; 2) Describe and assess application-specific packaging and associated requirements, constraints and solutions for reliability; 3) Design, implement and characterize a standalone electronic and/or microelectromechanical system.	
Education Method	Lectures and field visit.	
Assessment	Assessment is based on mini-research projects, conducted preferably in teams. The projects are proposed by the instructors, and can also be proposed in agreement with the instructors by the students based on the content of the course or personal interests. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ing. H.W. van Zeijl	
Co-Instructor	M. Mastrangeli	

ET4icp	IC-Technology Lab	2
Responsible Instructor	Dr.ir. S. Vollebregt	
Contact Hours / Week x/x/x/x	28 hrs.	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics as in EE4585	
Course Contents	During seven half days the students, in groups of 3, will have the opportunity to gain experience in the process- and device-simulation of a MOSFET transistor. They will perform several processing steps in the Else Kooi Lab clean room and will measure and characterize the fabricated device. Enrollment through Brightspace, max course capacity is 36 students!	
Study Goals	After this course, students are able to: 1. Explain the different process steps of the IC fabrication process 2. Perform calculations on the oxide thickness of silicon after thermal oxidation, and the implantation depth and diffusion depth of dopants 3. Explain the relation between threshold voltage and implantations, and relate this to the measured threshold voltages from transistors 4. Select dopant atoms and masking material for silicon devices based on the simulations performed in the course 5. Identify the different implanted regions and junctions in cross-sections obtained from doping profile simulations 6. Argue on the specific advantages and disadvantages of wet and dry etching as used during the lab course. 7. Analyse the measurements obtained from different test structures described in the course guide 8. Determine the doping levels from sheet resistance measurements	
Education Method	Lab course	
Literature and Study Materials	Lecture slides and a course manual which will be supplied during the compulsory lecture	
Assessment	Group report. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	28 hrs, first session is scheduled, after that on individual appointment. Students have to attend all parts of the course. The course can only accommodate 30 students.	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

RF/Wireless Design 2023

Introduction 1	This profile has two focus points: - Analog circuit design concentrating on low-power, low voltage amplifier and AD converter design for sensor interface electronics. - RF Circuit design concentrating on circuit design and optimization for RF . Info: RF: Prof: L de Vreede Analog: Prof. K Makinwa
Introduction 2	Compulsory EE4C05 Electromagnetics EE4C08 Measurement and Instrumentation EE4C10 Analog Circuit Design Fundamentals EE4520 Analog CMOS design I EE4600 RF / Wireless Concepts and Systems Suggested EE4610 Digital IC Design I EE4585 Semiconductor Device Physics ET4369 Nyquist-Rate Data Converters EE4525 Analog CMOS design II EE4605 Integrated Circuits for RF/Wireless Applications ET4371 Advanced Wireless transceivers

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effetc of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cacoded amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4605	Integrated Circuits for RF/Wireless Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Responsible Instructor	M. Spirito	
Instructor	Ing. M.J. Pelk	
Contact Hours / Week x/x/x/x	0,3,0,0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Common core course EE4C13. Other useful courses to consider as part of your preparation for RF design (not mandatory) are EE4C05 and EE4C10.	
Course Contents	In this course, you will learn to understand and design the circuit blocks of wireless systems. You can design and analyze circuits/ICs for wireless systems when completed. To do so, fundamental knowledge of RF/microwave design is required. To achieve such an insight, we discuss; optimum bias, device scaling, circuit configurations, and harmonic matching techniques for active devices (BJTs & FETS) operating at high frequencies to reach optimum gain, noise, linearity, and efficiency. The outline of the course is as follows; - Design of RF/microwave amplifiers, stability, gain, and impedance matching (HW1a) - Design of RF/microwave hybrids and transmission line-based passives, as well as on-chip (high Q) passive components; capacitors, varactors, inductors, transformers, and transmission lines (HW1b) - LNA design for simultaneous noise & impedance matching and LNA noise cancellation techniques (HW2). - Analysis and design techniques for circuit linearity enhancements (HW3) - RF mixer design for linearity - Design of energy-efficient & wideband power amplifiers & transmitters (HW4). A detailed course description addressing all topics covered in this lecture series can be found on Brightspace. Note that this EE4605, Integrated Circuits for RF/Wireless Applications course is part of the RF/wireless track and is preceded by EE4C13, Wireless Systems for Electrical Engineering Applications, which discusses wireless applications at the system level. This course (EE4605) can be followed up by ET4371 (strongly recommended), Advanced Digital Wireless transceivers, which addresses the latest trends in RF/wireless design which aim to fully utilize the high integration and signal processing strengths of advanced CMOS. Background reading for the various topics can be found in the recommended books given in the Literature and Study Materials section. This course will have three to four homework (to be handed in) that address the design of the various circuit blocks (RF amplifier & microwave passives, LNA, mixer, PA/transmitter) and their sub-components. For this homework, use will be made of Keysights ADS Advanced Design System and Cadence. Aspects of packaging, CAD, and RF testing relevant to wireless / mm-wave design are also emphasized.	
Study Goals	Examination and grading: Your overall grade will be based on your performance in the homework and a written exam. - Understand how to (optimally) design RF/wireless components and circuits. This includes: - How to design, dimension, and layout passive components like: PCB and on-chip transmission lines, capacitors, inductors, and transformers. - How to bias and scale active devices while understanding their basic operation and equivalent models. - How to stabilize active devices while maintaining their RF performance. - How to design matching networks for maximum gain, low noise, and high linearity. - How to design for maximum efficiency, bandwidth, ACPR, and lowest EVM. - Know how to use RF/Microwave design tools like Keysights ADS, Momentum (in addition to Cadence) to design RF integrated circuits and systems. - Understand the RF/microwave-specific data representations and analysis tools. - To understand the full design cycle from RF/wireless concept, RFIC design, to the final testing. - Being able to analyze and design integrated RF transceiver front-end circuits, like LNAs, mixers, and PAs.	
Education Method	Lectures + design related homeworks	
Literature and Study Materials	Hand-outs + recommended background books * Microwave Transistor Amplifiers: Analysis and Design (2nd edition) by Gonzalez, Guillermo * RF Microelectronics (2nd edition), by Behzad Razavi * Microwave Circuit Design, using Linear and Nonlinear Techniques, by George D. Vendelin, Ulrich L Rohde, Anthony M. Pavio * RF Power Amplifiers for wireless Communications (2nd edition), by Steve C. Cripps.	
Assessment	RF design related homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	More details are available on Bright Space. To participate in this course, please enroll.	

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

EE4615	Digital IC Design II	3
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will consolidate the knowledge acquired in EE4610 by practicing the design of an advanced complex digital VLSI design, where high-level specifications are translated to transistor sizes, going through several abstraction levels. At each level, the opportunities and limitations of the physical implementation will be considered for finding better solutions and tradeoffs, by means optimization and the search of new topologies, while keeping complexity to a minimum to achieve the desired functionality. Parasitics and the other effects deriving from the analog behavior of digital circuits will be taken into account using four-corner analysis, as well as simplifications and modeling at each abstraction level and other semi-automatic tools.	
Study Goals	The main goal of the course is to give students the tools to successfully design complex, scalable circuits and systems. The students will also learn how to work in and coordinate teams of designers around a complex project from high-level specifications to the actual transistor-level implementation and post-layout verification.	
Education Method	The course will entirely be based in the lab where the target design will be implemented by students in groups of 2-3. Only based on students/groups wish, the final design will be laid out in a commercial CMOS process and possibly fabricated and tested.	
Assessment	The course will be graded based on the target design. The following criteria will be taken into account: 1. Functionality achieved; 2. High-level specifications met; 3. Four-corner analysis completed; Bonus points will be given to those students that will exceed the specifications and create a robust, reproducible design. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovyi	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovyi	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanegue Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C13	Wireless Systems for Electrical Engineering Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	C. Gao	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	M. Babaie	
Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	M. Spirito	
Instructor	Prof. M. Alavi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE4605 'Integrated Circuits and Systems for Wireless Applications' and EE4371 'Advanced Wireless Transceivers.'	
Course Contents	In this course, you will learn about the commonly used architectures in wireless systems, with the requirements on their building blocks. The system key building blocks are addressed at their functional level regarding electrical properties, such as transfer/gain, noise, linearity, frequency selectivity, bandwidth, and other parameters. Physical constraints of practical components and circuits are used to identify the achievable signal quality, selectivity, sensitivity, and link budget of a wireless system. The lectures will explain the causes and impact of device/circuit non-linearities and noise on wireless systems' sensitivity and dynamic range. The course will introduce the concepts of frequency generation, modulation/demodulation, and up/down conversion. To teach you to work at RF frequencies with wireless systems, the principles of impedance matching are explained, and tools are provided to analyze/design/characterize RF (matching) circuitry/systems. The common part of the course also includes an overview of the figure of merits (FoMs) of various TX/Rx architectures/system blocks and the required instrumentation to characterize them. This course belongs to the RF/wireless profile and is an excellent preparation for the course; EE4605, Integrated Circuits for RF/Wireless Applications, which focuses on the design and realization of analog intensive building blocks for wireless transceivers as well as for ET4371, Advanced Digital Wireless transceivers which is addressing the latest trends in RF/wireless design aiming to fully utilize the integration and signal processing strengths of advanced CMOS. The course is completed by four short specialization topics which can be followed based on your personal preference, namely: (1) Introduction to wireless power transfer technology, (2) Beamforming basics: concepts and architectures, (3) Neural Networks for Wireless & Radar Systems, (4) Hands-on experience with RF characterization equipment on practical circuits. (The latter (4) has an attendee limit of 15 people.) Only one of these specialization topics is required to complete the overall course.	
Study Goals	Understand wireless system architectures with their subblocks to such a level that you can analyze their various performance parameters in terms of; link budget, sensitivity, spectral purity, P1dB, IP3, DR, SFDR, noise, phase noise, selectivity, EVM, APCR, and efficiency. In addition, you will master a basic understanding of S-parameters, impedance matching, and RF instrumentation.	
Education Method	This course will be delivered with complementary exercise lectures to familiarize students with the concepts and calculations described during the lectures.	
Literature and Study Materials	These lectures are (in part) based on the book of Behzad Razavi, RF microelectronics 2nd edition, Chapters 1 to 4.	
Assessment	Examination and grading: Your overall grade will be based on your performance in the written exam (80%), compulsory homework, and the mark obtained in your track-specific project (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4351	Digital VLSI Systems on Chip	4
Responsible Instructor	Dr.ing. C.P. Frenkel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of digital design (logic gates, combinational logic and sequential logic, finite state machines). Knowledge of hardware description languages (Verilog or VHDL), including how to write proper synthesizable code, is necessary. Previous exposure to microprocessor architecture is helpful but not necessary.	
Course Contents	Today's large-scale digital integrated circuits and systems-on-a-chip, such as modern general processing units (GPUs), can >50 billion transistors. Two things were instrumental in enabling this massive scalability: electronic design automation (EDA) tools and intellectual property (IP) blocks allowing for hierarchical design. In this course, you will learn how to use such EDA tools and IP blocks to go from the register-transfer level (RTL) description of your digital design to a final layout that is clean for tapeout and meets your specifications. To do so, you will also learn about power/performance/area tradeoff optimization, hardware-algorithm co-design, and memory hierarchy. These tools and concepts will be practiced and applied in a project, in which you will explore and apply your own design choices to meet a well-defined design target in a digital system-on-chip.	
Study Goals	After successful completion of the course, you will be able to: LO1. Design modern digital systems-on-a-chip with automated synthesis and place-and-route tools by - identifying the key design targets and tradeoffs, - selecting a proper IP to meet design targets, - translating the design constraints and tradeoffs to tool commands, - applying the tools to meet design targets. LO2. Optimize, in a rigorous and creative manner, the power/performance/area tradeoff associated with a given application use case. LO3. Link algorithmic requirements to hardware constraints. LO4. Extract the key performance metrics of a digital design by - verifying that simulation results can be trusted and that the design is clean for tapeout, - communicating these results effectively and rigorously, - discussing their implications within a broader societal context.	
Education Method	Lectures, labs	
Literature and Study Materials	* Course material (slides will be provided) * (Recommended) Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, by Erik Brunvand, Addison-Wesley, ISBN-10: 0321547993	
Assessment	The final course grade consists of two parts: * Final design report (50%) * Individual written exam (50%) The results of the two parts must have been obtained within the same academic year, and students must have a grade of 5.8 or higher for both the final design report and the individual written exam to pass the course. In case either of the two parts is failed, a resit must be taken in the form of an individual written exam that will account for 100% of the grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	C. Gao	

ET4369	Nyquist-Rate Data Converters	4
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10) and Analog CMOS Design I (ET4295)	
Course Contents	Introduction, sampling theory, quantization, and reconstruction; Digital-to-analog conversion: resistive, capacitive, and current steering; Sampling circuits: theory, non-idealities, and advanced techniques; Comparators: circuits, non-idealities, and design examples; Analog-to-digital conversion: flash, SAR, pipeline, time-interleaving.	
Study Goals	By the end of this course, you will be able to: - Explain the fundamentals of sampling, quantisation and reconstruction in data converters - Analyse the effect of non-idealities in sampling circuits and discuss advanced techniques to deal with them (bottom plate sampling, bootstrapping, oversampling, etc.) - Analyse the architectures and circuits commonly used in D/A interfaces (capacitive, resistive, current steering) and discuss their pros and cons - Analyse the architectures and circuits commonly used in A/D interfaces (flash, SAR, pipeline, time-interleaving) and discuss their pros and cons - read, understand and critically analyse state-of-the-art IEEE papers on Nyquist-rate data converters - Select the appropriate ADC architecture for a given set of performance specifications	
Education Method	4 hours of lectures plus a 4 hour tutorial	
Literature and Study Materials	MJM Pelgrom: Analog-to-Digital Conversion, 3rd Ed. Springer 2017	
Reader	Pdf versions of the slides and the homework will be provided	
Assessment	Weekly homework (formative assessment) + oral exam based on a selected ADC publication (summative assessment for 100% of the final grade). Resit oral exam based on a selected ADC publication possible in Q4 for 100% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4371	Mixed-mode Wireless transceivers	4
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0,0,4,0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	EE4C13 Wireless Systems for Electrical Engineering Applications EE4520, Analog CMOS Design I Basic signal processing; basic knowledge of Matlab	
Course Contents	The past several years have successfully brought all-digital techniques to radio frequency (RF) frequency synthesizers and transmitters, as well as digitally-intensive techniques to receivers. In addition, digital assistance is applied to RF circuits to improve performance and power consumption. This course will introduce the concepts of the digital RF approach and walk through the major building blocks that comprise new RF transceiver architectures, emphasis is placed on the: - Conventional charge-pump PLL and all-digital phase-locked loop (ADPLL) comprising: digitally-controlled oscillator (DCO), time-to-digital converter (TDC), and digital loop filter (HW1); - All-digital transmitters featuring polar or Cartesian signal processing in digitally-controlled power amplifiers (DPA) with wideband modulation capabilities possibly including Quadrature clock generation (HW2); - Direct-sampling discrete-time receivers comprising switched-cap circuits that perform various FIR and IIR filter operations (HW3). Exercises and projects use Matlab/ADS to model and simulate the above three subsystems.	
Study Goals	In-depth knowledge of all-digital PLL, advanced transmitter and discrete-time receiver	
Education Method	Lectures + design related homework	
Literature and Study Materials	-R. B. Staszewski and P. T. Balsara, All-Digital Frequency Synthesizer in Deep-Submicron CMOS, New Jersey: John Wiley & Sons, Inc., Sept. 2006. ISBN: 978-0471772552 -Babaie, M., Shahmohammadi, M., & Staszewski, R.B. (2019). RF CMOS Oscillators for Modern Wireless Applications (1st ed.). River Publishers. https://doi.org/10.1201/9781003339311 -M. Alavi, J. Mehta, and R. B. Staszewski, Radio-Frequency Digital-to-Analog Converters Implementation in Nanoscale CMOS, Elsevier/Academic Press, 302 pages, ISBN: 978-0-12-802263-4 (eBook: 978-0-12-802503-1), 26 Oct 2016. -Several related JSSC, TMTT, TCAS papers	
Assessment	RF design related homeworks (40-50% final grade) + written exam (60-50% final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Analysis Chips Circuits Design Integrated Intensive Matlab Project Semiconductors Signals and Systems Technology Telecommunication	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

BioSemi 2023	
Responsible Program Employee	Prof.dr.ir. W.A. Serdijn
Introduction 1	The BioSemi sub-profile concentrates on microfabrication of biosensors, bio-micro-electro-mechanical systems, micro-assembly, and complete biomedical devices on chip for, e.g., organs on-chip, smart catheters and flexible implants.
Introduction 2	Compulsory: EE4C01 Profile Orientation and Academic Skills EE4C11 System Engineering EE4C08 Measurement and Instrumentation ET4257 Sensors and Actuators (4) EE4585 Semiconductor Device Physics (5) ET4289 Integrated Circuits and MEMS Technology (4) ET4127 Themes in Biomedical Electronics (4) EE4555 Active Implantable Biomedical Microsystems (5) BM41055 Anatomy and Physiology (4) BM41140 Fundamentals of Biomedical Engineering (6) Suggested free electives: ET4130 Bioelectricity (3) ET4391 Advanced Microelectronics Packaging (3) ET4277 Microelectronics Reliability (4) ET4260 Microsystem Integration (4) ET4icp IC Technology Lab (2) BM41095 Medical devices: past, present, future (3) BM41100 Medical devices: designing for CE marking (3) WM1402TU Ethics of Healthcare Technologies (5)

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41095	Medical Devices: Past, Present, Future	3
Responsible Instructor	Dr. D. Dodou	
Instructor	Prof.dr. J.J. van den Dobbelsteen	
Instructor	Prof.dr. J. Dankelman	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. History of surgery 2. Minimally invasive techniques 3. Single-port procedures & NOTES 4. Percutaneous interventions 5. Robotic surgery 6. In vivo robots 7. Medical safety 8. Maintenance of surgical instruments	
Study Goals	The student will be able to describe the function, effectiveness, and challenges of surgical instruments in minimally invasive procedures. More specifically, the student will be able to: 1. illustrate advantages and disadvantages of minimally invasive surgery. 1. illustrate advantages and disadvantages of single-port procedures and NOTES. 3. describe instruments used during minimally invasive surgery, single-port procedures, and NOTES. 4. formulate criteria that surgical instruments have to fulfill in order to prevent sterilization problems. 5. indicate problems with quality control of surgical instruments. 6. describe the fields where robotic surgery is used and indicate the possibilities and limitations of the use of robotics. 7. classify technologies developed to train surgeons outside the operating theater. 8. discuss safety issues in surgery. 9. select information from the literature related to a certain issue and evaluate new developments.	
Education Method	Lectures. 4 hours per week	
Assessment	Homework assignments	
Department	3mE Department Biomechanical Engineering	

BM41141	Fundamentals of Biomedical Engineering	5
Responsible Instructor	N. Tümer	
Instructor	Ir. A. Sakes	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, there will be: 1) lectures, through which you will become familiar with the field of biomedical engineering (BME) and the stakeholders playing essential roles within, 2) workshops on engineering role play, team dynamics, peer-review, and debate. Through these workshops, you will practice some of the mentioned skills, e.g., being able to work in teams, and having good communication skills, 3) two assignments: (1) self-reflective journey (mostly individual), and (2) challenge-based assignment (group work). These two assignments are connected with each other. In the first assignment, you will explore who you are and what your role would be in relation to your future career in the BME field. In the second assignment, you will have a challenging context. Together with your team, you will define a solution to a challenging problem considering your engineering roles.	
Study Goals	Knowledge, insight, and skills By the end of this course, you should be able to: 1. Define the problems that need to be addressed in the health field, considering crucial societal aspects, and the roles biomedical engineers and stakeholders can play in solving these issues. 2. Apply physical and engineering terms and principles in building a strategy within the multidisciplinary context of biomedical engineering. 3. Outline the steps needed to be taken in order to validate or implement biomedical techniques, devices, and methods in the health system. Transferable skills By the end of this course, you should be able to: 4. Analyse the impact and consequences of your work when intervening in the field of biomedical engineering. 5. Develop evaluative judgment on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback. 6. Reflect on your own interests, role, and personal development in relation to your future professional self in the field of biomedical engineering.	
Education Method	Lectures, workshops and assignments	
Assessment	Exam (40%) and assignments (60%)	
Department	3mE Department Biomechanical Engineering	

EE4555	Active Implantable Biomedical Microsystems	5
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Responsible Instructor	Dr.ir. C. Strydis	
Responsible Instructor	Dr. V. Giagka	
Responsible Instructor	T.M. Lopes Marta da Costa	
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none none	
Course Language	English	
Required for	- MSc Biomedical Engineering, Medical Devices track; - MSc Electrical Engineering, Biomedical Devices profile; - MSc thesis projects in the Section Bioelectronics.	
Expected prior knowledge	It is recommended but not required to have completed/attended the following courses: 1. Anatomy and Physiology (BM41055) 2. Fundamentals of Biomedical Engineering (BM41140) 3. Bioelectricity (ET4130) 4. Themes in Biomedical Electronics (ET4127) 5. Sensors and actuators (ET4257) 6. Measurement and Instrumentation (EE4C08) 7. Systems Engineering (EE4C02)	
Course Contents	Active Implantable Biomedical Microsystems (AIBM) in the human body are now becoming more widely acceptable and available since the development of cardiac pacemakers. Today, miniaturized wireless implantable systems are changing the face of biomedical research and clinical practices through the development of cochlear implants, neuroprostheses, braincomputer interfaces, deep organ pressure sensors, precise drug delivery units, bioelectronic medicine and electroceuticals. With a greater life expectancy and an increasing demand for medical healthcare to guarantee a sufficient health-related quality of life, there is a greater demand on technology and electrical and biomedical engineers to develop active implantable systems for a wide variety of medical diagnostics, treatments and therapies. This course provides a background on the application of AIBMs, an introduction to the latest techniques used and examples of existing medical devices. It can be used as a guide to the design of AIBMs and also as a reference for existing medical devices. The course is aimed at MSc students in Electrical Engineering (tracks Microelectronics, and Signals&Systems) and in Biomedical Engineering (track Medical Devices) who have an interest in the research and design of AIBMs. The following topics are treated: 1. Electrophysiology 2. Electrodes, optodes, sensors and actuators 3. Electrode characterization and modeling 4. Electronic system design 5. Stimulation electronics 6. Front-end electronics (signal conditioning, amplifiers, filters, ADCs, DACs, etc.) 7. Wireless data transfer 8. Energy harvesting and power delivery 9. Applications: cochlear implants and cardiac pacemakers 10. Digital design 11. Biocompatibility, materials and fabrication considerations 12. Assembly, system integration and packaging 13. Application: spinal cord stimulator 14. Reliability and security 15. Risk management and/or 16. Ethics	
Study Goals	After successful completion of the course, you: 1. Know, understand and can apply (3) the functionality, construction and application of active implantable biomedical microsystems (AIBMs) and their sub-systems and parts; 2. Know, understand and can apply (3) the fundamental principles and approaches for designing an implantable system; 3. Know, understand, can apply, analyse and can reason about (evaluate; 5) the design requirements, application-specific implementation challenges, and appropriate design choices of existing AIBMs, incl. cardiac pacemakers, cochlear implants and spinal cord stimulators; 4. Know and understand (2) the most important implantability, biocompatibility, reliability, safety and security issues of AIBMs; 5. Know and understand (2) the most important regulatory and clinical constraints of AIBMs 6. Can reason (5) about future developments of AIBMs, a.o. electroceuticals and bioelectronic medicine; NB. The numbers correspond with Bloom's taxonomy.	
Education Method	1. Lectures 2. Self study 3. Case study 4. Group project	
Literature and Study Materials	1. a collection of book chapters, thesis chapters and review/overview papers; 2. handouts of the slides used during the lectures, 3. video recordings of the lectures all made available via the online learning system (Brightspace). 4. scientific papers (IEEE Xplore, PubMed); 5. patents; 6. user manuals; 7. patient manuals; 8. manufacturer websites; 9. the Internet in general, to be consulted by the students themselves.	
Assessment	The student has to: - Form a group (6 students/group) in the first week, based on affinity with the topics below. - As a group, perform a case study of a complete device, which will be announced in Wk. 4.2.	

- Investigate 6 topics:
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- Prepare an annotated slide set on the first 3 topics in Wk. 4.5
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Max. 12 slides (4 per group of 2 students), using a prescribed template
 - *To be uploaded on Brightspace before Wk 4.6
 - *To be commented on in written form and handed in by another group during the 1st lecture in Wk. 4.7
 - *Sub-group (of 2 students) on topic A comment on topic B; B -> C; C -> A
- Prepare a group presentation on the second 3 topics:
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- One student, to be randomly selected by the teachers on the day of the presentation, will do the presentation (max. 12 minutes) for the entire group in Wk. 4.10 or 4.11.
- Other groups will discuss the presentations (4 minutes) and ask questions and will comment briefly (max. 2 minutes per group / presentation) on all presentations

The teachers will:

- Organize/manage the course
- Give the lectures
- Provide relevant material and background knowledge
- Define the case study
- Give the templates for the annotated slides and the comments
- Grade the slide sets per group of 2 students
- Grade the comments on the slide sets per group of 2 students
- Return the commented slide sets to the authors
- Buy the T-shirts to wear during the final presentations
- Grade the presentations per group of 6 students
- Grade the comments made on the presentations per group of 6 students
- Decide on the final marks

The final mark is based on the grade received for the annotated and commented slide sets (37.5% + 12.5% (bonus/malus), respectively), and the grade received for the group presentation (45%) and the comments made on another group presentation (5%).

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

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EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C01	Profile Orientation and Academic Skills	3
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Responsible Instructor	L.C. Schroten	
Instructor	Dr. C.A. Benitez Avila	
Instructor	L.C. Schroten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: basic knowledge as acquired with a BSc in Physics, EE, Material Science. Strongly advised: Information Literacy 1 and 2 (e.g. Master Programme Information Literacy 1 & 2 or Electrical Engineering Information Literacy 1 and 2, both available in Brightspace).	
Course Contents	- Introduction to the MScEE tracks (Microelectronics, Electrical Power Engineering, Signals & Systems, Wireless Communication and Sensing) - Introduction to the research and expertise of the research groups taking part in the selected master track - Knowledge of ethics and intellectual property right - Skills in academic writing - Skills in oral presentation of scientific topics	
Study Goals	With this course the students - obtain relevant information on the MScEE study and the chosen track, become aware of cultural differences and how this affects intercultural communication, form a network with fellow master EE students. - acquire an overview of the main challenges in and potential applications of electrical engineering - become familiar with research topics addressed within the EE department - master basic skills in oral presentation of scientific topics - master basic skills in academic writing - acquire basic principle of ethics and intellectual property right - are able to extract the main information from a scientific paper or report - are able to write a paper (mostly a literature review) on a scientific question	
Education Method	The course includes lectures, reading materials and exercises. During the course, students work on a writing assignment, an ethics case and give several presentations. Attendance is mandatory as all class activities are highly interactive and peer feedback is a necessity.	
Literature and Study Materials	TU Present & TU Write. Both are available in Brightspace.	
Assessment	The pass/fail for this course is determined by the evaluation of the final writing assignment, the oral presentation and attending the Ethics seminar(s). All three components should be evaluated with a pass to pass the whole EE4C01 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanegue Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	P.P. Sundaramoorthy	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4260	Microsystem Integration	4
Responsible Instructor	T. Manzanque Garcia	
Instructor	K.M. Dowling	
Instructor	F. Arroyo Cardoso	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Instrumentation (EE4C08) Integrated circuits and MEMS technology (ET4289) Sensors & Actuators (ET4257)	
Course Contents	A microsystem is a complete instrument on a chip that can involve different domains of physics (electrical, mechanical, magnetic, etc.). The challenges associated with the combination of elements of different nature, including transducers and circuits into a single-chip integrated system, are more than compensated by the opportunities this concept offers in a wide range of applications. In this course, general concepts related to microsystems as sensors are first discussed. Secondly, approaches for multi-domain modeling are introduced and applied to the modelling electro-mechanical systems. Then, different transduction mechanisms, that allow for the conversion of signals between different physical domains, are introduced. Finally, all the modelling concepts are extended to the design of microsystem-based sensors. Throughout the course, the software package Comsol is introduced and used in several tutorials. The course includes theory (~8 lectures), Comsol tutorials, and tutorials for the realization of a final project, in which a microsystem is designed and modelled in groups.	
Study Goals	By the end of the course, you will be able to: - Identify parts of microsystems and their function. - Identify equations that govern the behaviour of different microsystem components, from different domains of physics. - Use electrical domain models for multidomain microsystems. - Use the finite element method (FEM) to model multidomain microsystems, using Comsol. - Analyze the dynamic behaviour of linear microsystems. - Design microsystems that meet given specifications, using both electrical domain modelling and FEM.	
Education Method	Lectures, tutorials and hands-on practical sessions with Comsol.	
Computer Use	The student requires access to computer systems for carrying out the Comsol tutorials and final assignment.	
Literature and Study Materials	Literature plus lecture notes.	
Assessment	The final grade of the course consists of the following components: - Final assignment report (50%) - Oral exam (50%) Final grade calculation: $(0.5 * \text{Final assignment report} + 0.5 * \text{Oral exam})$ In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4277	Microelectronics Reliability	4
Responsible Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Prof.dr. G.Q. Zhang	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course aims to provide the students with a thorough understanding of the reliability issues involved in electronic components and systems. The following subjects will be treated. - Basic reliability definition and lifetime distributions. - Reliability prediction methods. - Physical failure mechanisms in electronic components - Package related failures - Reliability screening and reliability testing - Failure analysis methods - Reliability data handling - Design considerations - System reliability	
Study Goals	After the course the students: - should see reliability as a basic requirement that should receive attention throughout a products complete lifecycle: specification, design, production, exploitation and disposal. - should have a sound understanding of the physical background of failures and the reliability test and failure analysis methods available. - should be able to make a proper reliability prediction based on available reliability (test) data. - should be able to evaluate reliability and availability figures of complex systems based on component reliability data. - should be able to make the proper design and maintenance choices to optimize these figures.	
Education Method	Lecture	
Assessment	written exam or project (project only possible when homework assignments are passed with average grade 6 or more) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4289	Integrated Circuits and MEMS Technology	4
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Instructor	Dr. C.M.F. Viellard-Boutry	
Instructor	Dr.ing. H.W. van Zeijl	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics (EE 4585)	
Course Contents	This lecture is an introduction in the fabrication technologies used for Integrated Circuits (ICs) and Micro Electro-Mechanical Systems (MEMS). The aim of this lecture is to build a bridge between the physical world (the device physics) and the electronic world (the device characteristics). In this way, students gain more insight in the critical aspects and challenges in IC fabrication. The series of lectures starts with the discussion of basic microelectronic devices with a brief review of the physics involved. The associated basic fabrication technology, required to fabricate these devices is also discussed. After that, the mainstream IC fabrication technology (CMOS) will be discussed in more detail. In separate chapters, silicon crystal growth, epitaxy, ion implantation, chemical vapor deposition, etching and photolithography will be introduced. Their physical motivation and the impact on the device characteristics will be discussed. Finally, silicon bulk and surface micromachining technologies are introduced to illustrate the potential of 3D micro structuring in the development of Microsystems or MEMS. These IC compatible technologies are employed to realize multi-function systems for many scientific and industrial application areas.	
Study Goals	You know the definition of the basic process steps to fabricate a device (IC or MEMS) You can explain the basic physics and chemistry of these fabrication steps in an IC /MEMS process flow You can analyse the relationship between the series of process steps necessary to build the device and the device performance. You can evaluate how device requirements are translated into a process flow You are able to propose solutions to specific device technological questions	
Education Method	lectures	
Literature and Study Materials	R.C. Jager introduction to Microelectronic Fabrication & course slides	
Assessment	written exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. S. Vollebregt	

ET4391	Advanced Microelectronics packaging	3
Responsible Instructor	Prof.dr. G.Q. Zhang	
Responsible Instructor	M. Mastrangeli	
Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Dr.ir. R.H. Poelma	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This is a multi-disciplinary course, dealing with scientifically challenging and economically important technology issues. MSc students from but not limited to electrical engineering, mechanical engineering, material science and physics are encouraged to attend.	
Course Contents	1. Introduction of micro/nanoelectronics packaging - Basics and state of the art of advanced front-end and back-end technologies - International roadmaps and strategic research agenda - Major application trends 2. Interconnect technologies 3. MEMS & sensor packaging 4. 3D and wafer-level system integration 5. Application specific packaging -1 (IC packaging, etc.) 6. Application specific packaging -2 (LED, etc) 7. Application specific packaging -3 (RF, Power, automotive, healthcare, etc.)	
Course Contents Continuation	As the bridge between devices and various multi-functional electronics systems, microelectronic system integration and packaging control more than 90% of the size, 60% of the cost, and largely the system performance and reliability. It is one of the most fascinating and rapidly developing technology and business fields in semiconductors, playing a dominant role in the development of future micro/nanoelectronics and systems.	
Study Goals	At the end of the course, the student should be able to: 1) Describe and assess basics and state-of-the-art in microelectronic packaging and reliability; 2) Describe and assess application-specific packaging and associated requirements, constraints and solutions for reliability; 3) Design, implement and characterize a standalone electronic and/or microelectromechanical system.	
Education Method	Lectures and field visit.	
Assessment	Assessment is based on mini-research projects, conducted preferably in teams. The projects are proposed by the instructors, and can also be proposed in agreement with the instructors by the students based on the content of the course or personal interests. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ing. H.W. van Zeijl	
Co-Instructor	M. Mastrangeli	

ET4icp	IC-Technology Lab	2
Responsible Instructor	Dr.ir. S. Vollebregt	
Contact Hours / Week x/x/x/x	28 hrs.	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics as in EE4585	
Course Contents	During seven half days the students, in groups of 3, will have the opportunity to gain experience in the process- and device-simulation of a MOSFET transistor. They will perform several processing steps in the Else Kooi Lab clean room and will measure and characterize the fabricated device. Enrollment through Brightspace, max course capacity is 36 students!	
Study Goals	After this course, students are able to: 1. Explain the different process steps of the IC fabrication process 2. Perform calculations on the oxide thickness of silicon after thermal oxidation, and the implantation depth and diffusion depth of dopants 3. Explain the relation between threshold voltage and implantations, and relate this to the measured threshold voltages from transistors 4. Select dopant atoms and masking material for silicon devices based on the simulations performed in the course 5. Identify the different implanted regions and junctions in cross-sections obtained from doping profile simulations 6. Argue on the specific advantages and disadvantages of wet and dry etching as used during the lab course. 7. Analyse the measurements obtained from different test structures described in the course guide 8. Determine the doping levels from sheet resistance measurements	
Education Method	Lab course	
Literature and Study Materials	Lecture slides and a course manual which will be supplied during the compulsory lecture	
Assessment	Group report. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	28 hrs, first session is scheduled, after that on individual appointment. Students have to attend all parts of the course. The course can only accommodate 30 students.	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (5ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

BioCAS 2023	
Responsible Program Employee	Prof.dr.ir. W.A. Serdijn
Introduction 1	The BioCAS sub-profile concentrates on circuits and systems for biomedical applications, such as CMOS microfluidic multi-electrode and sensor arrays, 3D trans-esophageal echography, autonomous vital sign monitoring, transcutaneous and implantable closed-loop neurostimulators, and brain-machine interfaces.
Introduction 2	Compulsory: - EE4C01 Profile Orientation and Academic Skills - EE4C11 System Engineering - EE4C08 Measurement and Instrumentation - EE4C10 Analog Circuit Design Fundamentals - ET4257 Sensors and Actuators (4) - EE4520 Analog CMOS design I (3) - EE4610 Digital IC Design (3) - ET4127 Themes in Biomedical Electronics (4) - EE4555 Active Implantable Biomedical Microsystems (5) - BM41005 Anatomy and Physiology (4) - BM41140 Fundamentals of Biomedical Engineering (6) Suggested free electives: - ET4130 Bioelectricity (3) - EE4109 Structured Electronic Design (5) - EE4615 Digital IC Design II (3) - ET4252 Analog Integrated Circuit Design (4) - ET4278 Over-Sampled Data Converters (4) - ET4351 Digital VLSI Systems on Chip (4) - ET4369 Nyquist-Rate Data Converters (4) - BM41095 Medical devices: past, present, future (3) - BM41100 Medical devices: designing for CE marking (3) - WM1402TU Ethics of Healthcare Technologies (5)

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41095	Medical Devices: Past, Present, Future	3
Responsible Instructor	Dr. D. Dodou	
Instructor	Prof.dr. J.J. van den Dobbelsteen	
Instructor	Prof.dr. J. Dankelman	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. History of surgery 2. Minimally invasive techniques 3. Single-port procedures & NOTES 4. Percutaneous interventions 5. Robotic surgery 6. In vivo robots 7. Medical safety 8. Maintenance of surgical instruments	
Study Goals	The student will be able to describe the function, effectiveness, and challenges of surgical instruments in minimally invasive procedures. More specifically, the student will be able to: 1. illustrate advantages and disadvantages of minimally invasive surgery. 1. illustrate advantages and disadvantages of single-port procedures and NOTES. 3. describe instruments used during minimally invasive surgery, single-port procedures, and NOTES. 4. formulate criteria that surgical instruments have to fulfill in order to prevent sterilization problems. 5. indicate problems with quality control of surgical instruments. 6. describe the fields where robotic surgery is used and indicate the possibilities and limitations of the use of robotics. 7. classify technologies developed to train surgeons outside the operating theater. 8. discuss safety issues in surgery. 9. select information from the literature related to a certain issue and evaluate new developments.	
Education Method	Lectures. 4 hours per week	
Assessment	Homework assignments	
Department	3mE Department Biomechanical Engineering	

BM41141	Fundamentals of Biomedical Engineering	5
Responsible Instructor	N. Tümer	
Instructor	Ir. A. Sakes	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, there will be: 1) lectures, through which you will become familiar with the field of biomedical engineering (BME) and the stakeholders playing essential roles within, 2) workshops on engineering role play, team dynamics, peer-review, and debate. Through these workshops, you will practice some of the mentioned skills, e.g., being able to work in teams, and having good communication skills, 3) two assignments: (1) self-reflective journey (mostly individual), and (2) challenge-based assignment (group work). These two assignments are connected with each other. In the first assignment, you will explore who you are and what your role would be in relation to your future career in the BME field. In the second assignment, you will have a challenging context. Together with your team, you will define a solution to a challenging problem considering your engineering roles.	
Study Goals	Knowledge, insight, and skills By the end of this course, you should be able to: 1. Define the problems that need to be addressed in the health field, considering crucial societal aspects, and the roles biomedical engineers and stakeholders can play in solving these issues. 2. Apply physical and engineering terms and principles in building a strategy within the multidisciplinary context of biomedical engineering. 3. Outline the steps needed to be taken in order to validate or implement biomedical techniques, devices, and methods in the health system. Transferable skills By the end of this course, you should be able to: 4. Analyse the impact and consequences of your work when intervening in the field of biomedical engineering. 5. Develop evaluative judgment on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback. 6. Reflect on your own interests, role, and personal development in relation to your future professional self in the field of biomedical engineering.	
Education Method	Lectures, workshops and assignments	
Assessment	Exam (40%) and assignments (60%)	
Department	3mE Department Biomechanical Engineering	

EE4109	Structured Electronic Design	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Instructor	A.J.M. Montagne	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	A grade of a minimal 8 for this course is required to obtain access to the masterclass EE8011MSC	
Expected prior knowledge	This course required the knowledge and skills of MSc course EE4C10 and BSc course EE3C11. This course uses the same book used for BSc course EE3C11. A free download of the book is available via: https://analog-electronics.tudelft.nl Chapters that contain prior knowledge are: 1, 2, 7,8, 9,10 (excluding 10.3.5, 10.3.7, 10.3.8) , 11 (excluding 11.4.4, 11.4.6, 11.4.7, 11.5.1) and 12.1 and 12.2 Students that did not follow BSc course EE3C11 should (could) follow the homologation program from: https://analog-electronics.tudelft.nl/Homologation/courseWebSite/index.html Here you can find lecture material and videos that cover all electronics topics of EE3C11. The teaching staff can assist in studying the homologation program. On request, the lecturers can provide extra lectures for external/international students on topics treated in EE3C11 on background knowledge, like poles and zeros.	
Course Contents	This course deals with the structured design of analog electronic circuits using CMOS(and Bipolar) transistors. The design method presented in this course is based on network theory, control theory, signal processing, and physics. Many people consider the design of analog electronic circuits to be complex. This is mainly because designers have to deal with many relevant performance aspects that can be achieved in many different ways. In other words, there are many degrees of freedom for obtaining the desired performance of an electronic circuit. Theoretical concepts, circuit topologies, electronic devices, their operating conditions, and the physical layout of a system constitute an enormous design space in which it is easy to get lost. For this reason, analog electronics is often regarded as an art rather than a solid discipline. At first glance, there doesn't seem to be a straightforward way to design analog circuits and systems: experienced designers intuitively use all these degrees of freedom to modify and combine known solutions into new ones. However, intuition is knowledge, the origin of which has become unclear. It cannot easily be shared with novices, and therefore, it cannot be a basis for the education of designers. However, the design of electronic circuits can be shared and understood if it is presented in a structured way. This requires a precise formulation of design goals and strategies and a clear distinction between theoretical concepts and their physical implementations. Instead of taking numerous existing solutions to known problems as a starting point, it is far more effective to start a new design with clear distinction between theoretical concepts and study how they can be implemented in such a way that the possibilities of the implementation technology are maximally exploited. Such an approach results in well-defined and reliable designs with predictable performance and provides a basis for sharing and developing knowledge. During the engineering phase, designers need to find or create real-world components, determine their operating conditions, and fix them by design. To this end, they need to formulate and solve so-called design equations for the circuit. Design equations relate the properties and the operating conditions of the electronic devices to the performance requirements and the cost factors of the circuit. For example, to determine a budget for the gain-bandwidth product of an operational amplifier, we must find relations between this gain-bandwidth product and the performance requirements of the circuit that comprises this operational amplifier. I Setting up design equations for analog circuits, especially at transistor-level is a difficult and cumbersome task. It requires knowledge about technology, modeling techniques, and interactions between the behavior of the circuit and the transistor properties. It goes hand in hand with symbolic analysis. Performing symbolic calculations by hand is sensitive to errors, and in practice, only a limited number of circuit performance aspects can be considered. For this reason, we will use SLiCAP (Symbolic Linear Circuit Analysis Program) during this course. SLiCAP (https://analog-electronics.tudelft.nl/SLiCAP.html)	
Study Goals	The main goal of this course is to provide the student with the essential knowledge and skills to design an amplifier in BiCMOS or CMOS technology according to given specifications and to motivate and fully understand all design steps involved. After the course the student is able to: - work in a design team in a complete CMOS design - translate an application description of an amplifier into a specification comprising, a functional specification, specification of performance measures, operational conditions and cost factors. - design a proper amplifier concept based on source and load specification. - evaluate the amplifying capabilities of basic amplifier stages such as CS and CE stage for a given technology - identify performance aspects such as noise, signal handling capability and bandwidth of basic amplifier stages such as CS and CE stage for a given technology - scale devices and select their operating conditions to influence these performance aspects - to apply error reduction techniques, such as compensation, balancing and (local) negative feedback to improve the performance of these basic amplifying stages - to motivate the decision to use multi-stage amplifiers - to design (high-performance) multi-stage amplifiers using error reduction techniques such as compensation, balancing and feedback. - setup design considerations for noise and take adequate design decisions	

- setup design considerations for signal handling capability and power efficiency and take adequate design decisions
- setup design considerations for frequency response and take adequate design decisions
- setup design considerations for weak distortion and take adequate design decisions
- setup design considerations for biasing and design and implement proper biasing circuitry
- evaluate the performance versus requirements of the final design and propose options for improvement
- select proper models and design verification tools for each design step
- write a design report in which all design steps are motivated and the performance is evaluated and explained.

Education Method

Lectures and Colstructions:

The systematic electronics design method will be illustrated during the lectures and colstructions with a design example made in SLiCAP.

The design example closely relates to the hearing loop system designed during the BSc course EE3C11. This BSc design was based on operational amplifiers as controllers. In this course, the focus will be on the design of the controllers as transistor circuits.

During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design. The design steps are documented in a report using SLiCAP.

The design report can be efficiently used as a basis for the exam, which is organized as a design review for the design team.

Computer Use

A computer with SLiCAP and the free circuit simulator LT-Spice is necessary. Help for installing and using SLiCAP will be available during the lectures.

SLiCAP is an acronym for Symbolic Linear Circuit Analysis Program.

See: <https://analog-electronics.tudelft.nl/SLiCAP.html>

Literature and Study Materials

Book (also used during BSc course EE3C11):

Structured Electronic Design Edition downloadable from:

<https://analog-electronics.tudelft.nl>

Assessment

During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design used in every lecture to illustrate the application of the theory from the lectures.

At the end of the lecture period, the student team makes an appointment review of their design. The design review consists of two sessions:

- 1) A design preview, where the students present and defend their design in a relaxed atmosphere. They can use the feedback to make improvements.
- 2) A design review, where the students present and defend their design again. Each team member has to give substantial input during the session. After this session, each student is graded individually.

The design report, created with SLiCAP during the lectures by the team, can be efficiently used as a basis for the review.

If there are logistic or personal reasons that make it impossible for a student to join a design team, it is possible to opt for an individual design review (oral exam). Please contact the educators about this.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effetc of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cacoded amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4555	Active Implantable Biomedical Microsystems	5
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Responsible Instructor	Dr.ir. C. Strydis	
Responsible Instructor	Dr. V. Giagka	
Responsible Instructor	T.M. Lopes Marta da Costa	
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none none	
Course Language	English	
Required for	- MSc Biomedical Engineering, Medical Devices track; - MSc Electrical Engineering, Biomedical Devices profile; - MSc thesis projects in the Section Bioelectronics.	
Expected prior knowledge	It is recommended but not required to have completed/attended the following courses: 1. Anatomy and Physiology (BM41055) 2. Fundamentals of Biomedical Engineering (BM41140) 3. Bioelectricity (ET4130) 4. Themes in Biomedical Electronics (ET4127) 5. Sensors and actuators (ET4257) 6. Measurement and Instrumentation (EE4C08) 7. Systems Engineering (EE4C02)	
Course Contents	Active Implantable Biomedical Microsystems (AIBM) in the human body are now becoming more widely acceptable and available since the development of cardiac pacemakers. Today, miniaturized wireless implantable systems are changing the face of biomedical research and clinical practices through the development of cochlear implants, neuroprostheses, braincomputer interfaces, deep organ pressure sensors, precise drug delivery units, bioelectronic medicine and electroceuticals. With a greater life expectancy and an increasing demand for medical healthcare to guarantee a sufficient health-related quality of life, there is a greater demand on technology and electrical and biomedical engineers to develop active implantable systems for a wide variety of medical diagnostics, treatments and therapies. This course provides a background on the application of AIBMs, an introduction to the latest techniques used and examples of existing medical devices. It can be used as a guide to the design of AIBMs and also as a reference for existing medical devices. The course is aimed at MSc students in Electrical Engineering (tracks Microelectronics, and Signals&Systems) and in Biomedical Engineering (track Medical Devices) who have an interest in the research and design of AIBMs. The following topics are treated: 1. Electrophysiology 2. Electrodes, optodes, sensors and actuators 3. Electrode characterization and modeling 4. Electronic system design 5. Stimulation electronics 6. Front-end electronics (signal conditioning, amplifiers, filters, ADCs, DACs, etc.) 7. Wireless data transfer 8. Energy harvesting and power delivery 9. Applications: cochlear implants and cardiac pacemakers 10. Digital design 11. Biocompatibility, materials and fabrication considerations 12. Assembly, system integration and packaging 13. Application: spinal cord stimulator 14. Reliability and security 15. Risk management and/or 16. Ethics	
Study Goals	After successful completion of the course, you: 1. Know, understand and can apply (3) the functionality, construction and application of active implantable biomedical microsystems (AIBMs) and their sub-systems and parts; 2. Know, understand and can apply (3) the fundamental principles and approaches for designing an implantable system; 3. Know, understand, can apply, analyse and can reason about (evaluate; 5) the design requirements, application-specific implementation challenges, and appropriate design choices of existing AIBMs, incl. cardiac pacemakers, cochlear implants and spinal cord stimulators; 4. Know and understand (2) the most important implantability, biocompatibility, reliability, safety and security issues of AIBMs; 5. Know and understand (2) the most important regulatory and clinical constraints of AIBMs 6. Can reason (5) about future developments of AIBMs, a.o. electroceuticals and bioelectronic medicine; NB. The numbers correspond with Bloom's taxonomy.	
Education Method	1. Lectures 2. Self study 3. Case study 4. Group project	
Literature and Study Materials	1. a collection of book chapters, thesis chapters and review/overview papers; 2. handouts of the slides used during the lectures, 3. video recordings of the lectures all made available via the online learning system (Brightspace). 4. scientific papers (IEEE Xplore, PubMed); 5. patents; 6. user manuals; 7. patient manuals; 8. manufacturer websites; 9. the Internet in general, to be consulted by the students themselves.	
Assessment	The student has to: - Form a group (6 students/group) in the first week, based on affinity with the topics below. - As a group, perform a case study of a complete device, which will be announced in Wk. 4.2.	

- Investigate 6 topics:
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- Prepare an annotated slide set on the first 3 topics in Wk. 4.5
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Max. 12 slides (4 per group of 2 students), using a prescribed template
 - *To be uploaded on Brightspace before Wk 4.6
 - *To be commented on in written form and handed in by another group during the 1st lecture in Wk. 4.7
 - *Sub-group (of 2 students) on topic A comment on topic B; B -> C; C -> A
- Prepare a group presentation on the second 3 topics:
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- One student, to be randomly selected by the teachers on the day of the presentation, will do the presentation (max. 12 minutes) for the entire group in Wk. 4.10 or 4.11.
- Other groups will discuss the presentations (4 minutes) and ask questions and will comment briefly (max. 2 minutes per group / presentation) on all presentations

The teachers will:

- Organize/manage the course
- Give the lectures
- Provide relevant material and background knowledge
- Define the case study
- Give the templates for the annotated slides and the comments
- Grade the slide sets per group of 2 students
- Grade the comments on the slide sets per group of 2 students
- Return the commented slide sets to the authors
- Buy the T-shirts to wear during the final presentations
- Grade the presentations per group of 6 students
- Grade the comments made on the presentations per group of 6 students
- Decide on the final marks

The final mark is based on the grade received for the annotated and commented slide sets (37.5% + 12.5% (bonus/malus), respectively), and the grade received for the group presentation (45%) and the comments made on another group presentation (5%).

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

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EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

EE4615	Digital IC Design II	3
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will consolidate the knowledge acquired in EE4610 by practicing the design of an advanced complex digital VLSI design, where high-level specifications are translated to transistor sizes, going through several abstraction levels. At each level, the opportunities and limitations of the physical implementation will be considered for finding better solutions and tradeoffs, by means optimization and the search of new topologies, while keeping complexity to a minimum to achieve the desired functionality. Parasitics and the other effects deriving from the analog behavior of digital circuits will be taken into account using four-corner analysis, as well as simplifications and modeling at each abstraction level and other semi-automatic tools.	
Study Goals	The main goal of the course is to give students the tools to successfully design complex, scalable circuits and systems. The students will also learn how to work in and coordinate teams of designers around a complex project from high-level specifications to the actual transistor-level implementation and post-layout verification.	
Education Method	The course will entirely be based in the lab where the target design will be implemented by students in groups of 2-3. Only based on students/groups wish, the final design will be laid out in a commercial CMOS process and possibly fabricated and tested.	
Assessment	The course will be graded based on the target design. The following criteria will be taken into account: 1. Functionality achieved; 2. High-level specifications met; 3. Four-corner analysis completed; Bonus points will be given to those students that will exceed the specifications and create a robust, reproducible design. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C01	Profile Orientation and Academic Skills	3
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Responsible Instructor	L.C. Schroten	
Instructor	Dr. C.A. Benitez Avila	
Instructor	L.C. Schroten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: basic knowledge as acquired with a BSc in Physics, EE, Material Science. Strongly advised: Information Literacy 1 and 2 (e.g. Master Programme Information Literacy 1 & 2 or Electrical Engineering Information Literacy 1 and 2, both available in Brightspace).	
Course Contents	- Introduction to the MScEE tracks (Microelectronics, Electrical Power Engineering, Signals & Systems, Wireless Communication and Sensing) - Introduction to the research and expertise of the research groups taking part in the selected master track - Knowledge of ethics and intellectual property right - Skills in academic writing - Skills in oral presentation of scientific topics	
Study Goals	With this course the students - obtain relevant information on the MScEE study and the chosen track, become aware of cultural differences and how this affects intercultural communication, form a network with fellow master EE students. - acquire an overview of the main challenges in and potential applications of electrical engineering - become familiar with research topics addressed within the EE department - master basic skills in oral presentation of scientific topics - master basic skills in academic writing - acquire basic principle of ethics and intellectual property right - are able to extract the main information from a scientific paper or report - are able to write a paper (mostly a literature review) on a scientific question	
Education Method	The course includes lectures, reading materials and exercises. During the course, students work on a writing assignment, an ethics case and give several presentations. Attendance is mandatory as all class activities are highly interactive and peer feedback is a necessity.	
Literature and Study Materials	TU Present & TU Write. Both are available in Brightspace.	
Assessment	The pass/fail for this course is determined by the evaluation of the final writing assignment, the oral presentation and attending the Ethics seminar(s). All three components should be evaluated with a pass to pass the whole EE4C01 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	P.P. Sundaramoorthy	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4252	Analog Integrated Circuit Design	4
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	This course is recommended for students intending to take: - Structured Electronic Design Laboratory (ET8011MSC), - Analog CMOS design II (EE4525), - Oversampled data converters (ET4278), - Nyquist-rate data converters (ET4369), - Introduction to Power Conversion Technology (ET4382). This course is obligatory for students that wish to do their final project in the area of Biomedical Circuits (and Systems) (BioCAS).	
Expected prior knowledge	This course assumes (though it is not obligatory) that you have completed your BSc studies in electrical engineering and taken our MSc courses Analog Circuit Design Fundamentals (EE4C10) and Structured Electronic Design (EE4109) Having attended Measurement and Instrumentation (EE4C08) and Analog CMOS Design I (EE4520) is a plus.	
Course Contents	An intermediate course in analog circuit synthesis for microelectronic designers. Topics include: - review of analog design basics (i.e., frequency response, feedback and stability, noise analysis); - MOS models for short-channel modern devices; - gm/ID design methodology for modern analog circuits; - physical layout (e.g., device matching) for robust analog circuits; - design of voltage sources ranging from simple voltage dividers to high-performance bandgaps, and - current source implementations from a single resistor to high-quality references based on negative-feedback structures; - biasing techniques; - high-performance analog filters, from impulse response or transfer function down to circuit-level design; - quality aspects, such as accuracy, noise and impedance levels are treated within the context of power and area consumption requirements.	
Study Goals	To know, understand and be able to analyze and design (synthesize): - high-performance analog integrated filters - voltage references - bandgap references - current sources - amplifiers in all IC technologies (CMOS, bipolar, ...) and regions of operation (weak, moderate and strong inversion; saturation and triode/conduction)	
Education Method	Interactive lectures, flipped classroom (discussing lecture recordings and answering questions), homework assignments, class project.	
Literature and Study Materials	Reader, course notes, and video recordings of the lectures - all available via Brightspace.	
Assessment	Written examination (70%; the other 30% comes from a successfully completed class project). Students are allowed to bring 1 handwritten piece (A4) of paper and a pocket calculator to the exam. The results of the homework assignments may add up to one bonus point to the exam, only for the exam in April. The re-sit written exam in June will count for 100% of the final grade. It is not possible to re-sit the class project. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4278	Over-Sampled Data Converters	4
Responsible Instructor	Prof.dr. K.A.A. Makinwa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10), Analog CMOS Design I (ET4295) and Nyquist-rate Analog-to-Digital Converters (ET4369). A good knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce students to the design of over-sampled converters. Students will learn about the basic functionality of sigma-delta converters, their architecture, how to simulate and design these architectures and how to map them to actual circuit blocks. Covered topics: - First-order sigma-delta: oversampling; noise-shaping; NTF/STF; decimation; overload; leakage. - Second-order sigma delta: linear model; stability; quantizer model; overload stability. - Sigma-delta architectures: bandpass; continuous-time; multi-bit; cascaded/MASH; feedback; feedforward; local feedback - Building blocks: SC integrator; RC integrator; loopfilter; current DAC; SC DAC; clock jitter - Case studies.	
Study Goals	- Students can explain the basic functionality of sigma delta modulators (over-sampling, noise-shaping, decimation, stability). - Students can describe the circuit implementations of various building blocks (integrators, DAC, quantizer, feed-forward and feedback coefficients). - Students can identify and compare the advantages and disadvantages of different architectures for sigma delta modulators (feedback, feed-forward, MASH, multi-bit, single-bit, continuous time and discrete time modulators). - Students can derive the STF and NTF from the linear model of a sigma delta modulator. - Students can design and simulate the architecture of a sigma delta modulator (continuous and discrete time, first and second order modulators, single-bit and multi-bit quantizer, MASH). - Students can simulate the effect of non-idealities of various blocks (such integrators limited gain, DAC non-linearity) on the performance of a sigma delta modulator. - Students can propose a (non-optimized) circuit implementation of a sigma delta modulator. - Students can identify the main techniques used in state-of-the-art designs of sigma delta modulators and assess their weakness and strengths. - Students can propose improvements to existing designs of sigma delta modulators.	
Education Method	Lectures and practical sessions	
Literature and Study Materials	hand-outs provided during the lectures; optional reference: R. Schreier, S. Pavan, and G.C.Temes, "Understanding Delta-Sigma Data Converters", Second Edition, published by John Wiley & Sons Inc., New Jersey, 2017.	
Assessment	Homework (30%), a final assignment discussed during the oral exam (50%), and an oral exam (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Textbook	
Remarks	Guest Instructor: Dr.M.Bolatkale, NXP Semiconductors	

ET4351	Digital VLSI Systems on Chip	4
Responsible Instructor	Dr.ing. C.P. Frenkel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of digital design (logic gates, combinational logic and sequential logic, finite state machines). Knowledge of hardware description languages (Verilog or VHDL), including how to write proper synthesizable code, is necessary. Previous exposure to microprocessor architecture is helpful but not necessary.	
Course Contents	Today's large-scale digital integrated circuits and systems-on-a-chip, such as modern general processing units (GPUs), can >50 billion transistors. Two things were instrumental in enabling this massive scalability: electronic design automation (EDA) tools and intellectual property (IP) blocks allowing for hierarchical design. In this course, you will learn how to use such EDA tools and IP blocks to go from the register-transfer level (RTL) description of your digital design to a final layout that is clean for tapeout and meets your specifications. To do so, you will also learn about power/performance/area tradeoff optimization, hardware-algorithm co-design, and memory hierarchy. These tools and concepts will be practiced and applied in a project, in which you will explore and apply your own design choices to meet a well-defined design target in a digital system-on-chip.	
Study Goals	After successful completion of the course, you will be able to: LO1. Design modern digital systems-on-a-chip with automated synthesis and place-and-route tools by - identifying the key design targets and tradeoffs, - selecting a proper IP to meet design targets, - translating the design constraints and tradeoffs to tool commands, - applying the tools to meet design targets. LO2. Optimize, in a rigorous and creative manner, the power/performance/area tradeoff associated with a given application use case. LO3. Link algorithmic requirements to hardware constraints. LO4. Extract the key performance metrics of a digital design by - verifying that simulation results can be trusted and that the design is clean for tapeout, - communicating these results effectively and rigorously, - discussing their implications within a broader societal context.	
Education Method	Lectures, labs	
Literature and Study Materials	* Course material (slides will be provided) * (Recommended) Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, by Erik Brunvand, Addison-Wesley, ISBN-10: 0321547993	
Assessment	The final course grade consists of two parts: * Final design report (50%) * Individual written exam (50%) The results of the two parts must have been obtained within the same academic year, and students must have a grade of 5.8 or higher for both the final design report and the individual written exam to pass the course. In case either of the two parts is failed, a resit must be taken in the form of an individual written exam that will account for 100% of the grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	C. Gao	

ET4369	Nyquist-Rate Data Converters	4
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10) and Analog CMOS Design I (ET4295)	
Course Contents	Introduction, sampling theory, quantization, and reconstruction; Digital-to-analog conversion: resistive, capacitive, and current steering; Sampling circuits: theory, non-idealities, and advanced techniques; Comparators: circuits, non-idealities, and design examples; Analog-to-digital conversion: flash, SAR, pipeline, time-interleaving.	
Study Goals	By the end of this course, you will be able to: - Explain the fundamentals of sampling, quantisation and reconstruction in data converters - Analyse the effect of non-idealities in sampling circuits and discuss advanced techniques to deal with them (bottom plate sampling, bootstrapping, oversampling, etc.) - Analyse the architectures and circuits commonly used in D/A interfaces (capacitive, resistive, current steering) and discuss their pros and cons - Analyse the architectures and circuits commonly used in A/D interfaces (flash, SAR, pipeline, time-interleaving) and discuss their pros and cons - read, understand and critically analyse state-of-the-art IEEE papers on Nyquist-rate data converters - Select the appropriate ADC architecture for a given set of performance specifications	
Education Method	4 hours of lectures plus a 4 hour tutorial	
Literature and Study Materials	MJM Pelgrom: Analog-to-Digital Conversion, 3rd Ed. Springer 2017	
Reader	Pdf versions of the slides and the homework will be provided	
Assessment	Weekly homework (formative assessment) + oral exam based on a selected ADC publication (summative assessment for 100% of the final grade). Resit oral exam based on a selected ADC publication possible in Q4 for 100% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (5ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Analog Circuit Design 2023

Introduction 1	This profile has two focus points: - Analog circuit design concentrating on low-power, low voltage amplifier and AD converter design for sensor interface electronics. - RF Circuit design concentrating on circuit design and optimization for RF . Info: RF: Prof: L de Vreede Analog: Prof. K Makinwa
Introduction 2	Compulsory EE4C08 Measurement and Instrumentation EE4520 Analog CMOS design I EE4525 Analog CMOS design II EE4C10 Analog Circuit Design Fundamentals Suggested ET4252 Analog Integrated Circuit Design EE4109 Structured Electronic Design EE4585 Semiconductor Device Physics ET4369 Nyquist-Rate Data Converters ET4278 Over-Sampled Data Converters ET4382 Power conversion techniques in CMOS technology

EE4109	Structured Electronic Design	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Instructor	A.J.M. Montagne	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	A grade of a minimal 8 for this course is required to obtain access to the masterclass EE8011MSC	
Expected prior knowledge	This course required the knowledge and skills of MSc course EE4C10 and BSc course EE3C11. This course uses the same book used for BSc course EE3C11. A free download of the book is available via: https://analog-electronics.tudelft.nl Chapters that contain prior knowledge are: 1, 2, 7,8, 9,10 (excluding 10.3.5, 10.3.7, 10.3.8) , 11 (excluding 11.4.4, 11.4.6, 11.4.7, 11.5.1) and 12.1 and 12.2 Students that did not follow BSc course EE3C11 should (could) follow the homologation program from: https://analog-electronics.tudelft.nl/Homologation/courseWebSite/index.html Here you can find lecture material and videos that cover all electronics topics of EE3C11. The teaching staff can assist in studying the homologation program. On request, the lecturers can provide extra lectures for external/international students on topics treated in EE3C11 on background knowledge, like poles and zeros.	
Course Contents	This course deals with the structured design of analog electronic circuits using CMOS(and Bipolar) transistors. The design method presented in this course is based on network theory, control theory, signal processing, and physics. Many people consider the design of analog electronic circuits to be complex. This is mainly because designers have to deal with many relevant performance aspects that can be achieved in many different ways. In other words, there are many degrees of freedom for obtaining the desired performance of an electronic circuit. Theoretical concepts, circuit topologies, electronic devices, their operating conditions, and the physical layout of a system constitute an enormous design space in which it is easy to get lost. For this reason, analog electronics is often regarded as an art rather than a solid discipline. At first glance, there doesn't seem to be a straightforward way to design analog circuits and systems: experienced designers intuitively use all these degrees of freedom to modify and combine known solutions into new ones. However, intuition is knowledge, the origin of which has become unclear. It cannot easily be shared with novices, and therefore, it cannot be a basis for the education of designers. However, the design of electronic circuits can be shared and understood if it is presented in a structured way. This requires a precise formulation of design goals and strategies and a clear distinction between theoretical concepts and their physical implementations. Instead of taking numerous existing solutions to known problems as a starting point, it is far more effective to start a new design with clear distinction between theoretical concepts and study how they can be implemented in such a way that the possibilities of the implementation technology are maximally exploited. Such an approach results in well-defined and reliable designs with predictable performance and provides a basis for sharing and developing knowledge. During the engineering phase, designers need to find or create real-world components, determine their operating conditions, and fix them by design. To this end, they need to formulate and solve so-called design equations for the circuit. Design equations relate the properties and the operating conditions of the electronic devices to the performance requirements and the cost factors of the circuit. For example, to determine a budget for the gain-bandwidth product of an operational amplifier, we must find relations between this gain-bandwidth product and the performance requirements of the circuit that comprises this operational amplifier. I Setting up design equations for analog circuits, especially at transistor-level is a difficult and cumbersome task. It requires knowledge about technology, modeling techniques, and interactions between the behavior of the circuit and the transistor properties. It goes hand in hand with symbolic analysis. Performing symbolic calculations by hand is sensitive to errors, and in practice, only a limited number of circuit performance aspects can be considered. For this reason, we will use SLiCAP (Symbolic Linear Circuit Analysis Program) during this course. SLiCAP (https://analog-electronics.tudelft.nl/SLiCAP.html)	
Study Goals	The main goal of this course is to provide the student with the essential knowledge and skills to design an amplifier in BiCMOS or CMOS technology according to given specifications and to motivate and fully understand all design steps involved. After the course the student is able to: - work in a design team in a complete CMOS design - translate an application description of an amplifier into a specification comprising, a functional specification, specification of performance measures, operational conditions and cost factors. - design a proper amplifier concept based on source and load specification. - evaluate the amplifying capabilities of basic amplifier stages such as CS and CE stage for a given technology - identify performance aspects such as noise, signal handling capability and bandwidth of basic amplifier stages such as CS and CE stage for a given technology - scale devices and select their operating conditions to influence these performance aspects - to apply error reduction techniques, such as compensation, balancing and (local) negative feedback to improve the performance of these basic amplifying stages - to motivate the decision to use multi-stage amplifiers - to design (high-performance) multi-stage amplifiers using error reduction techniques such as compensation, balancing and feedback. - setup design considerations for noise and take adequate design decisions	

	- setup design considerations for signal handling capability and power efficiency and take adequate design decisions - setup design considerations for frequency response and take adequate design decisions - setup design considerations for weak distortion and take adequate design decisions - setup design considerations for biasing and design and implement proper biasing circuitry - evaluate the performance versus requirements of the final design and propose options for improvement - select proper models and design verification tools for each design step - write a design report in which all design steps are motivated and the performance is evaluated and explained.
Education Method	Lectures and Colstructions: The systematic electronics design method will be illustrated during the lectures and colstructions with a design example made in SLiCAP. The design example closely relates to the hearing loop system designed during the BSc course EE3C11. This BSc design was based on operational amplifiers as controllers. In this course, the focus will be on the design of the controllers as transistor circuits. During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design. The design steps are documented in a report using SLiCAP.
Computer Use	The design report can be efficiently used as a basis for the exam, which is organized as a design review for the design team. A computer with SLiCAP and the free circuit simulator LT-Spice is necessary. Help for installing and using SLiCAP will be available during the lectures. SLiCAP is an acronym for Symbolic Linear Circuit Analysis Program.
Literature and Study Materials	See: https://analog-electronics.tudelft.nl/SLiCAP.html Book (also used during BSc course EE3C11): Structured Electronic Design Edition downloadable from: https://analog-electronics.tudelft.nl
Assessment	During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design used in every lecture to illustrate the application of the theory from the lectures. At the end of the lecture period, the student team makes an appointment review of their design. The design review consists of two sessions: 1) A design preview, where the students present and defend their design in a relaxed atmosphere. They can use the feedback to make improvements. 2) A design review, where the students present and defend their design again. Each team member has to give substantial input during the session. After this session, each student is graded individually. The design report, created with SLiCAP during the lectures by the team, can be efficiently used as a basis for the review. If there are logistic or personal reasons that make it impossible for a student to join a design team, it is possible to opt for an individual design review (oral exam). Please contact the educators about this. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effect of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cased amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4525	Analog CMOS design II	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10; EE4520	
Course Contents	The goal of this course is to introduce students to the design of advanced analog circuits in modern CMOS technology. Students will learn how to use basic building blocks, such as current mirrors and differential pairs, to build more complex CMOS circuits, such as two-stage amplifiers. At the end of the course, the students will be able to design and optimize a complex CMOS circuit given its specifications. No new topic will be introduced in this course, and students should have a good understanding of basic circuits block from EE4C10 and EE4520. This course will focus on the students designing a complex analog circuit given a set of specifications, and the interactions with the lecturers and the teaching assistants will focus on discussing the on-going design and answering questions from the students.	
Study Goals	- Students can derive the performance of complex analog CMOS circuits (such as switched-capacitor amplifiers and class-AB amplifiers) in terms of amplification, noise and distortion. - Given the specifications for a CMOS circuit, students can chose the best architecture for the circuit and design and optimize the circuit to meet the specifications, by means of first-order hand calculations and verification by a Spice-like simulator.	
Education Method	Lectures, teaching-assistant session	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Individual project	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanegue Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4252	Analog Integrated Circuit Design	4
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	This course is recommended for students intending to take: - Structured Electronic Design Laboratory (ET8011MSC), - Analog CMOS design II (EE4525), - Oversampled data converters (ET4278), - Nyquist-rate data converters (ET4369), - Introduction to Power Conversion Technology (ET4382). This course is obligatory for students that wish to do their final project in the area of Biomedical Circuits (and Systems) (BioCAS).	
Expected prior knowledge	This course assumes (though it is not obligatory) that you have completed your BSc studies in electrical engineering and taken our MSc courses Analog Circuit Design Fundamentals (EE4C10) and Structured Electronic Design (EE4109) Having attended Measurement and Instrumentation (EE4C08) and Analog CMOS Design I (EE4520) is a plus.	
Course Contents	An intermediate course in analog circuit synthesis for microelectronic designers. Topics include: - review of analog design basics (i.e., frequency response, feedback and stability, noise analysis); - MOS models for short-channel modern devices; - gm/ID design methodology for modern analog circuits; - physical layout (e.g., device matching) for robust analog circuits; - design of voltage sources ranging from simple voltage dividers to high-performance bandgaps, and - current source implementations from a single resistor to high-quality references based on negative-feedback structures; - biasing techniques; - high-performance analog filters, from impulse response or transfer function down to circuit-level design; - quality aspects, such as accuracy, noise and impedance levels are treated within the context of power and area consumption requirements.	
Study Goals	To know, understand and be able to analyze and design (synthesize): - high-performance analog integrated filters - voltage references - bandgap references - current sources - amplifiers in all IC technologies (CMOS, bipolar, ...) and regions of operation (weak, moderate and strong inversion; saturation and triode/conduction)	
Education Method	Interactive lectures, flipped classroom (discussing lecture recordings and answering questions), homework assignments, class project.	
Literature and Study Materials	Reader, course notes, and video recordings of the lectures - all available via Brightspace.	
Assessment	Written examination (70%; the other 30% comes from a successfully completed class project). Students are allowed to bring 1 handwritten piece (A4) of paper and a pocket calculator to the exam. The results of the homework assignments may add up to one bonus point to the exam, only for the exam in April. The re-sit written exam in June will count for 100% of the final grade. It is not possible to re-sit the class project. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4278	Over-Sampled Data Converters	4
Responsible Instructor	Prof.dr. K.A.A. Makinwa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10), Analog CMOS Design I (ET4295) and Nyquist-rate Analog-to-Digital Converters (ET4369). A good knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce students to the design of over-sampled converters. Students will learn about the basic functionality of sigma-delta converters, their architecture, how to simulate and design these architectures and how to map them to actual circuit blocks. Covered topics: - First-order sigma-delta: oversampling; noise-shaping; NTF/STF; decimation; overload; leakage. - Second-order sigma delta: linear model; stability; quantizer model; overload stability. - Sigma-delta architectures: bandpass; continuous-time; multi-bit; cascaded/MASH; feedback; feedforward; local feedback - Building blocks: SC integrator; RC integrator; loopfilter; current DAC; SC DAC; clock jitter - Case studies.	
Study Goals	- Students can explain the basic functionality of sigma delta modulators (over-sampling, noise-shaping, decimation, stability). - Students can describe the circuit implementations of various building blocks (integrators, DAC, quantizer, feed-forward and feedback coefficients). - Students can identify and compare the advantages and disadvantages of different architectures for sigma delta modulators (feedback, feed-forward, MASH, multi-bit, single-bit, continuous time and discrete time modulators). - Students can derive the STF and NTF from the linear model of a sigma delta modulator. - Students can design and simulate the architecture of a sigma delta modulator (continuous and discrete time, first and second order modulators, single-bit and multi-bit quantizer, MASH). - Students can simulate the effect of non-idealities of various blocks (such integrators limited gain, DAC non-linearity) on the performance of a sigma delta modulator. - Students can propose a (non-optimized) circuit implementation of a sigma delta modulator. - Students can identify the main techniques used in state-of-the-art designs of sigma delta modulators and assess their weakness and strengths. - Students can propose improvements to existing designs of sigma delta modulators.	
Education Method	Lectures and practical sessions	
Literature and Study Materials	hand-outs provided during the lectures; optional reference: R. Schreier, S. Pavan, and G.C.Temes, "Understanding Delta-Sigma Data Converters", Second Edition, published by John Wiley & Sons Inc., New Jersey, 2017.	
Assessment	Homework (30%), a final assignment discussed during the oral exam (50%), and an oral exam (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Textbook	
Remarks	Guest Instructor: Dr.M.Bolat kale, NXP Semiconductors	

ET4369	Nyquist-Rate Data Converters	4
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10) and Analog CMOS Design I (ET4295)	
Course Contents	Introduction, sampling theory, quantization, and reconstruction; Digital-to-analog conversion: resistive, capacitive, and current steering; Sampling circuits: theory, non-idealities, and advanced techniques; Comparators: circuits, non-idealities, and design examples; Analog-to-digital conversion: flash, SAR, pipeline, time-interleaving.	
Study Goals	By the end of this course, you will be able to: - Explain the fundamentals of sampling, quantisation and reconstruction in data converters - Analyse the effect of non-idealities in sampling circuits and discuss advanced techniques to deal with them (bottom plate sampling, bootstrapping, oversampling, etc.) - Analyse the architectures and circuits commonly used in D/A interfaces (capacitive, resistive, current steering) and discuss their pros and cons - Analyse the architectures and circuits commonly used in A/D interfaces (flash, SAR, pipeline, time-interleaving) and discuss their pros and cons - read, understand and critically analyse state-of-the-art IEEE papers on Nyquist-rate data converters - Select the appropriate ADC architecture for a given set of performance specifications	
Education Method	4 hours of lectures plus a 4 hour tutorial	
Literature and Study Materials	MJM Pelgrom: Analog-to-Digital Conversion, 3rd Ed. Springer 2017	
Reader	Pdf versions of the slides and the homework will be provided	
Assessment	Weekly homework (formative assessment) + oral exam based on a selected ADC publication (summative assessment for 100% of the final grade). Resit oral exam based on a selected ADC publication possible in Q4 for 100% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4382	Power conversion techniques in CMOS technology	3
Responsible Instructor	Dr.drs. Q. Fan	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	No modern electronic systems could work without power conversion technology. For instance, a laptop, a mobile phone, or an electric car has only one battery, which offers only one supply voltage. However, there are thousands of sub-circuits in these devices, analog and digital, high voltage and low voltage, CMOS and bipolar, and they require various supply voltages to operate properly. How to generate different supplies efficiently and accurately for these sub-circuits is a crucial and challenging task. This course will cover the most common DC-DC power conversion techniques, starting from the simplest resistor divider, to more sophisticated voltage regulator such as LDOs, switched-capacitor DC-DC converter and inductive DC-DC converter (buck converter). In addition, basics of class D power amplifiers will be introduced in this course, which shares similar principle as a buck converter. A class D power amplifier converts power from a DC supply to a high energy (>1W) AC audio signal, which can be found in a lot of headphones and high-end sound systems.	
Study Goals	Learning objectives - Understanding the basic working principle of a switched-capacitor DC-DC converter - Understanding the basic working principle of a buck converter - Understanding negative feedback loops in DC-DC converters - Understanding the basic working principle of a class D power amplifier	
Education Method	Lectures	
Literature and Study Materials	not applicable	
Assessment	30% homework, 70% project In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Guest Instructors: Dr. Marco Berkhout and Dr. Gerard Villar Pique from NXP Semiconductors	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Sub-THz Electronic Systems ME 2023

Contact for students Dr. M. Spirito

Introduction 1

Compulsory

EE4C05 Electromagnetics (Q1)

EE4C13 Wireless Systems for Electrical Engineering Applications (Q1)

EE4605 Integrated Circuits for RF/Wireless Applications (Q2)

EE4510 Advanced Electromagnetics (Q2)

EE4725 Quasi Optical Systems (Q3)

EE4730 High-Frequency Wireless Architectures (Q4)

Suggested

ET4169 Radar I: From Basic Principles to Applications (Q3)

EE4C10 Analog Circuit Design Fundamentals (Q1)

EE4520 Analog CMOS design I (Q2)

ET4369 Nyquist-Rate Data Converters ()

EE4620 Spectral Domain Methods in Electromagnetics

EE4510	Advanced Electromagnetics	5
Responsible Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	1) Ohmic losses at high frequencies 2) Potentials and Green's functions 3) Image theorem 4) Fields radiated by sources in the far field region 5) Equivalence and reciprocity theorems 6) Antennas in transmission and reception 7) Green's function of infinite line source 3) Application case: quasi-analytic analysis of transmission lines at high frequencies	
Study Goals	1) EM analysis of wireless links Basic characterization/understanding of a wireless link between a transmitting and receiving antenna 2) EM analysis of transmission lines Basic understanding of the impact of radiation and ohmic losses at high frequencies	
Education Method	The course consists on 1) Theoretical lectures on the basic physical mechanisms and how to analyse them via EM analytical techniques 2) Practical Instructions to get familiar with such techniques 3) Study of an application case	
Assessment	The assessment of the course will be the following. The final mark will be the average between the home assignments, written exam and an oral exam (depending on the number of students). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effect of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cascoded amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4605	Integrated Circuits for RF/Wireless Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Responsible Instructor	M. Spirito	
Instructor	Ing. M.J. Pelk	
Contact Hours / Week x/x/x/x	0,3,0,0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Common core course EE4C13. Other useful courses to consider as part of your preparation for RF design (not mandatory) are EE4C05 and EE4C10.	
Course Contents	In this course, you will learn to understand and design the circuit blocks of wireless systems. You can design and analyze circuits/ICs for wireless systems when completed. To do so, fundamental knowledge of RF/microwave design is required. To achieve such an insight, we discuss; optimum bias, device scaling, circuit configurations, and harmonic matching techniques for active devices (BJTs & FETS) operating at high frequencies to reach optimum gain, noise, linearity, and efficiency. The outline of the course is as follows; - Design of RF/microwave amplifiers, stability, gain, and impedance matching (HW1a) - Design of RF/microwave hybrids and transmission line-based passives, as well as on-chip (high Q) passive components; capacitors, varactors, inductors, transformers, and transmission lines (HW1b) - LNA design for simultaneous noise & impedance matching and LNA noise cancellation techniques (HW2). - Analysis and design techniques for circuit linearity enhancements (HW3) - RF mixer design for linearity - Design of energy-efficient & wideband power amplifiers & transmitters (HW4). A detailed course description addressing all topics covered in this lecture series can be found on Brightspace. Note that this EE4605, Integrated Circuits for RF/Wireless Applications course is part of the RF/wireless track and is preceded by EE4C13, Wireless Systems for Electrical Engineering Applications, which discusses wireless applications at the system level. This course (EE4605) can be followed up by ET4371 (strongly recommended), Advanced Digital Wireless transceivers, which addresses the latest trends in RF/wireless design which aim to fully utilize the high integration and signal processing strengths of advanced CMOS. Background reading for the various topics can be found in the recommended books given in the Literature and Study Materials section. This course will have three to four homework (to be handed in) that address the design of the various circuit blocks (RF amplifier & microwave passives, LNA, mixer, PA/transmitter) and their sub-components. For this homework, use will be made of Keysights ADS Advanced Design System and Cadence. Aspects of packaging, CAD, and RF testing relevant to wireless / mm-wave design are also emphasized.	
Study Goals	Examination and grading: Your overall grade will be based on your performance in the homework and a written exam. - Understand how to (optimally) design RF/wireless components and circuits. This includes: - How to design, dimension, and layout passive components like: PCB and on-chip transmission lines, capacitors, inductors, and transformers. - How to bias and scale active devices while understanding their basic operation and equivalent models. - How to stabilize active devices while maintaining their RF performance. - How to design matching networks for maximum gain, low noise, and high linearity. - How to design for maximum efficiency, bandwidth, ACPR, and lowest EVM. - Know how to use RF/Microwave design tools like Keysights ADS, Momentum (in addition to Cadence) to design RF integrated circuits and systems. - Understand the RF/microwave-specific data representations and analysis tools. - To understand the full design cycle from RF/wireless concept, RFIC design, to the final testing. - Being able to analyze and design integrated RF transceiver front-end circuits, like LNAs, mixers, and PAs.	
Education Method	Lectures + design related homeworks	
Literature and Study Materials	Hand-outs + recommended background books * Microwave Transistor Amplifiers: Analysis and Design (2nd edition) by Gonzalez, Guillermo * RF Microelectronics (2nd edition), by Behzad Razavi * Microwave Circuit Design, using Linear and Nonlinear Techniques, by George D. Vendelin, Ulrich L Rohde, Anthony M. Pavio * RF Power Amplifiers for wireless Communications (2nd edition), by Steve C. Cripps.	
Assessment	RF design related homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	More details are available on Bright Space. To participate in this course, please enroll.	

EE4620	Spectral Domain Methods in Electromagnetics	4
Responsible Instructor	Dr. D. Cavallo	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System	
Course Contents	1) Green's functions of stratified media (6 hours) 2) Dominant spectral contributions and far-field (4 hours) 2) Surface waves and leaky waves (6 hours) 3) Artificial dielectric materials (4 hours) 4) Spectral representations for planar Tx-lines (4 hours) 5) Connected arrays (10 hours)	
Study Goals	The students will be able to - apply spectral domain methods for the analysis of a number of electromagnetic problems (planar transmission lines, artificial dielectrics, stratified media problems) - create functional MATLAB scripts that implement the studied methods - present and critically analyze the results obtained with the MATLAB codes	
Education Method	- Lectures through PowerPoint slide projection - Practical MATLAB sessions with implementation of the analysis methods - Analysis of the results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	Home assignments (50% weight on the final grade) Final project presentation (50% weight on the final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4725	Quasi Optical Systems	5
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. D. Cavallo	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is required knowledge of - Electromagnetics - Transmission lines - Antennas	
Course Contents	The exam 'AP3701 Sub-mm and terahertz physics and applications' is recommended. This course includes the 5 topics listed hereafter: 1) Spectral domain Green's function and Far Field evaluation (4 hours) 2) Physical Optics for reflector and lens antennas (8 hours) 3) Focal Plane Arrays and Fourier Optics (4 hours) 4) Diffraction from half-plane (2 hours) 5) Periodic structures and arrays (8 hours)	
Study Goals	The student will be able to 1. derive and use the spectral Green's function method for EM problems (diffraction, optics, evaluation of far-fields); 2. apply electromagnetic methods (physical optics, Fourier optics) for quasi-optical systems and analyze focal plane arrays; 3. apply the mathematical tools for describing the behavior of periodic arrays;	
Education Method	Course slides, support material, exercises and examples in Matlab, homework assignments (all available in brightspace).	
Assessment	Exam 33% from Matlab exercises during the course (one assignment per instruction) 33% from a 15min individual oral presentation (mandatory) 33% from oral examination (mandatory) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4730	High Frequency Wireless Architectures	3
Responsible Instructor	Dr.ir. M. Alonso Del Pino	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - Advanced Electromagnetics - RF Wireless Systems - Quasi-Optical Systems	
Course Contents	1) Time-domain systems with photo-conductive antennas 2) Line of Sight communication systems with directive antennas 3) Imaging systems at high frequencies	
Study Goals	The students will be able to - Understand time-domain sensing systems and photoconductive pulsed source modeling techniques - Analyze spectroscopic measurements obtained on a time-domain system - Understand high-frequency communication scenarios and their trade-offs - Apply quasi-optical system concepts on over-the-air high-frequency antenna measurements - Understand high-frequency imaging systems and their trade-offs - Apply quasi-optical system concepts on a real high-frequency imaging system	
Education Method	- Lectures through PowerPoint slides - Practical laboratory sessions - Analysis of the experimental results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	The final mark will be the average between: 50% from the reports of each laboratory session 50% from a final presentation and QA session In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C13	Wireless Systems for Electrical Engineering Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	C. Gao	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	M. Babaie	
Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	M. Spirito	
Instructor	Prof. M. Alavi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE4605 'Integrated Circuits and Systems for Wireless Applications' and EE4371 'Advanced Wireless Transceivers.'	
Course Contents	In this course, you will learn about the commonly used architectures in wireless systems, with the requirements on their building blocks. The system key building blocks are addressed at their functional level regarding electrical properties, such as transfer/gain, noise, linearity, frequency selectivity, bandwidth, and other parameters. Physical constraints of practical components and circuits are used to identify the achievable signal quality, selectivity, sensitivity, and link budget of a wireless system. The lectures will explain the causes and impact of device/circuit non-linearities and noise on wireless systems' sensitivity and dynamic range. The course will introduce the concepts of frequency generation, modulation/demodulation, and up/down conversion. To teach you to work at RF frequencies with wireless systems, the principles of impedance matching are explained, and tools are provided to analyze/design/characterize RF (matching) circuitry/systems. The common part of the course also includes an overview of the figure of merits (FoMs) of various TX/Rx architectures/system blocks and the required instrumentation to characterize them. This course belongs to the RF/wireless profile and is an excellent preparation for the course; EE4605, Integrated Circuits for RF/Wireless Applications, which focuses on the design and realization of analog intensive building blocks for wireless transceivers as well as for ET4371, Advanced Digital Wireless transceivers which is addressing the latest trends in RF/wireless design aiming to fully utilize the integration and signal processing strengths of advanced CMOS. The course is completed by four short specialization topics which can be followed based on your personal preference, namely: (1) Introduction to wireless power transfer technology, (2) Beamforming basics: concepts and architectures, (3) Neural Networks for Wireless & Radar Systems, (4) Hands-on experience with RF characterization equipment on practical circuits. (The latter (4) has an attendee limit of 15 people.) Only one of these specialization topics is required to complete the overall course.	
Study Goals	Understand wireless system architectures with their subblocks to such a level that you can analyze their various performance parameters in terms of; link budget, sensitivity, spectral purity, P1dB, IP3, DR, SFDR, noise, phase noise, selectivity, EVM, APCR, and efficiency. In addition, you will master a basic understanding of S-parameters, impedance matching, and RF instrumentation.	
Education Method	This course will be delivered with complementary exercise lectures to familiarize students with the concepts and calculations described during the lectures.	
Literature and Study Materials	These lectures are (in part) based on the book of Behzad Razavi, RF microelectronics 2nd edition, Chapters 1 to 4.	
Assessment	Examination and grading: Your overall grade will be based on your performance in the written exam (80%), compulsory homework, and the mark obtained in your track-specific project (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovyj	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4369	Nyquist-Rate Data Converters	4
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10) and Analog CMOS Design I (ET4295)	
Course Contents	Introduction, sampling theory, quantization, and reconstruction; Digital-to-analog conversion: resistive, capacitive, and current steering; Sampling circuits: theory, non-idealities, and advanced techniques; Comparators: circuits, non-idealities, and design examples; Analog-to-digital conversion: flash, SAR, pipeline, time-interleaving.	
Study Goals	By the end of this course, you will be able to: - Explain the fundamentals of sampling, quantisation and reconstruction in data converters - Analyse the effect of non-idealities in sampling circuits and discuss advanced techniques to deal with them (bottom plate sampling, bootstrapping, oversampling, etc.) - Analyse the architectures and circuits commonly used in D/A interfaces (capacitive, resistive, current steering) and discuss their pros and cons - Analyse the architectures and circuits commonly used in A/D interfaces (flash, SAR, pipeline, time-interleaving) and discuss their pros and cons - read, understand and critically analyse state-of-the-art IEEE papers on Nyquist-rate data converters - Select the appropriate ADC architecture for a given set of performance specifications	
Education Method	4 hours of lectures plus a 4 hour tutorial	
Literature and Study Materials	MJM Pelgrom: Analog-to-Digital Conversion, 3rd Ed. Springer 2017	
Reader	Pdf versions of the slides and the homework will be provided	
Assessment	Weekly homework (formative assessment) + oral exam based on a selected ADC publication (summative assessment for 100% of the final grade). Resit oral exam based on a selected ADC publication possible in Q4 for 100% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year

Organization

Education

2023/2024

Electrical Engineering, Mathematics and Computer Science

Master Electrical Engineering

Quantum Electronics 2023	
Introduction 2	Compulsory EE4C08 Measurement and Instrumentation EE4C10 Analog Circuit Design Fundamentals EE4610 Digital IC Design I Suggested EE4520 Analog CMOS design I EE4585 Semiconductor Device Physics AP3421 Fundamentals of quantum information AP3292 Quantum hardware CESE4080 Electronics for Quantum Computation ET8027 Solid State Physics

AP3421-PR	Quantum Information Project	2
Responsible Instructor	Dr. L. di Carlo	
Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	Different	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	This is an optional additional module for students who have taken the QIST4310 (Fundamentals of Quantum Information) course.	
Course Contents	Working in groups, students will perform an independent project (to be defined by students with approval/help from instructors). The goal is to simulate/implement one of the protocols and algorithms covered in the lecture course using one of several available simulators and quantum processors. Examples of protocols or algorithms would be: Shors algorithm on a few qubits, Bell-inequality violations, tomography experiment of a one/two-qubit state, process tomography experiment of one/two-qubit gate, implementation of a small quantum error correction decoder, etc. Possible simulation platforms include: Microsoft Liquid, QuTech QX, QuTech QuantumSim, QuTech Network Simulator, QuTip. Possible experimental platforms include: IBM Quantum Experience, QuTech superconducting quantum processor.	
Study Goals	a) to apply knowledge from the Fundamentals of Quantum Information course; b) to present and defend the content to specialists in this field of research.	
Education Method	Supported self-study in groups of three students.	
Literature and Study Materials	See QIST4310	
Prerequisites	The Fundamentals of Quantum Information (QIST4310) course is required for this project. Credits for this project are not valid without passing the QIST4310 course.	
Assessment	Assessment: Each group will (1) hand in the source code if applicable and (2) make a final presentation of the project at a final meeting including the instructors and students. The instructors will assign a grade based on quality, originality, creativity, and effort.	
Enrolment / Application	Participation is limited to a maximum of 30 students (10 groups). Selection is based on a proposal/motivation. Proposals (including names of all participants) must be received before Januari 23. The selection will be announced before Februari 1.	

AP3432	Quantum Hardware 1 - Theoretical Concepts	4
Responsible Instructor	C.K. Andersen	
Instructor	M.F. Rimbach-Russ	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Undergraduate electricity and magnetism; AP3421 Fundamentals of quantum information; Bachelor Quantum Mechanics	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to control, couple, transmit and read out the fragile state of quantum systems with great precision, and in a technologically viable way. Quantum Hardware I is focused on teaching theoretical physics concepts for understanding this Hamiltonian engineering challenge in various quantum hardware platforms. The material will be taught using example systems such as superconducting qubits, atomic qubits, defect centers, and trapped-ion qubits.	
Study Goals	This course should enable you to: L01: Discuss single-qubit and two-qubit gate dynamics, qubit measurement, Rabi oscillations, dephasing & relaxation times on an abstract level. L02: Analyse quantum hardware systems by obtaining effective dynamics using approximations to the full Lindbladian of the system. L03: Evaluate performance of quantum hardware by identifying sources of noise and quantifying the effect of these. L04: Establish various quantum hardware systems relative to each other by identifying challenges and opportunities of the systems for quantum information processing.	
Education Method	The course will consist of weekly lectures and tutorial exercise sessions. Every week there will be hand-in assignments, which are factored in the final evaluation of the course.	
Course Relations	This course forms part of the curriculum on Quantum Technologies offered at TU Delft, which at present consists of: AP3421 Fundamentals of quantum information CS4090 Quantum communication and cryptography AP3432 Quantum Hardware 1 - Theoretical Concepts AP3442 Quantum Hardware 2 - Experimental State of the Art EE4575 Electronics for quantum computation Other Related Courses: AP3112 Quantum Optics and Lasers	
Literature and Study Materials	In this course, we will make use of: 1) 0Lecture notes And some selected material from the textbooks: a) Nielsen and Chuang, Quantum computation and information, Cambridge University Press. b) Gerry and Knight, "Introductory Quantum Optics", Cambridge University Press	
Assessment	Both hand-in assignments during the course and a final written exam will contribute to the final assessment of the course. Specifically, hand-in assignments will contribute 30% and the final written exam 70% of the final grade. Feedback on the hand-in assignments will be provided on a weekly basis and should help prepare students for the final exam.	
Continuing Courses	AP3442 Quantum Hardware 2 - Experimental State of the Art AP3472 Modeling of Superconducting Devices AP3662 Special Topics in Quantum Technology	

AP3442	Quantum Hardware 2 - Experimental State of the Art	4
Responsible Instructor	Prof.dr.ir. L.M.K. Vandersypen	
Instructor	Prof.dr.ir. R. Hanson	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course requires an understanding of undergraduate electricity and magnetism and the following concepts from quantum information theory: - Hamiltonian of a spin-1/2 particle and of a harmonic oscillator, and tensor products - quantum states (pure and mixed states, including density matrices) - unitary evolution (generated from the above Hamiltonians) - standard quantum operations (X, Y, Z rotations, Hadamard, CNOT, CPhase, SWAP, ...) - decoherence and non-unitary operations (e.g. through the product operator formalism) - simple quantum circuits and algorithms (e.g. teleportation, Deutsch-Jozsa algorithm, Grover's algorithm) - basics of quantum error correction (3-qubit bit flip/phase flip code, notion of fault-tolerance threshold) - simple quantum communication protocols (e.g. BB84, entanglement swapping for quantum repeaters) - quantum measurement (projective measurement in different bases)	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to initialize, control, couple, transmit and read out the fragile state of quantum systems with great precision and in a technologically viable way. While Quantum Hardware I is focused on teaching the underpinning theoretical tools, Quantum Hardware II will give you an overview of the experimental state-of-the-art. You will learn about several promising approaches for realizing quantum hardware, and critically assess the strengths and weaknesses of each approach. You will also get insight in the conceptual similarities and differences between the various technologies. Specifically, the course will cover general concepts and considerations of qubit hardware, trapped ions, superconducting circuits, quantum dots, impurities, cold atoms, photonic circuits, single-photon sources, single-photon detectors and quantum repeaters.	
Study Goals	- To acquire a good understanding of the requirements of quantum hardware for computing, communication and sensing, both at the conceptual level and at the practical level. - To acquire conceptual insight in the operation, opportunities, and challenges of various qubit realisations. - To obtain a good overview of the state-of-the-art.	
Education Method	In this course, lectures are combined with homework assignments as well as student presentations of recent research papers.	
Literature and Study Materials	Review papers to accompany the lecture slides.	
Assessment	Grades will be established based on homework assignments (40%), paper presentations (25%), and a final exam (35%). The lowest homework grade does not count. This means you can miss one homework without having points deducted. Two missed homeworks will affect your grade, regardless of why you missed them. In unusual cases, a retake exam--which will be oral--will be offered by appointment.	
Continuing Courses	This course forms part of the QuTech Academy curriculum on Quantum Technologies offered at TU Delft, which at present consists of: - AP3421 Fundamentals of quantum information - CS4090 Quantum communication and cryptography - AP3432 Quantum Hardware 1 - Theoretical Concepts - AP3442 Quantum Hardware 2 - Experimental State of the Art - EE4575 Electronics for quantum computation	

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effect of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cased amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

EE4710	Solid State Physics with Quantum and Nano Electronics	5
Responsible Instructor	Dr. R. Ishihara	
Instructor	Dr. S. Nur	
Contact Hours / Week x/x/x/x	0,0,4,0 week 1-7	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In electrical engineering, solid-state materials play an essential role in many applications. When solids are reduced in the size to nanometer scale by nanofabrication, quantum effect becomes evident, and they exhibit exiting behaviors. A thorough understanding of the basic quantum mechanics and electrical and physical properties of semiconductors, metals and insulators are essential for design and fabrication of novel nanoelectronic and photonic devices for the applications in classical and quantum computers, memories, displays, solar-cells, light emitting diodes, and sensors. The course consists of two parts. The first part deals with solid-state physics; After introduction of IC fabrication process, the course starts with the crystallography and then will be followed by the quantum mechanics, the quantum theory of solid, semiconductor physics. In short, you will be acquainted with the properties of individual atoms and what happens when they meet and become the form of a solid. In the second part, we will discuss (1) properties of low-dimensional structures, with examples of nanomaterials such as graphene, (2) electric field transport in nanostructures, (3) the quantum Hall effect, (4) electrical and optical process in quantum wells and quantum dots, and (5) selected applications using the nanomaterial and nanostructures, such as, single-electron transistors and photonic crystals.	
Study Goals	For the first part, the student can 1) determine the crystal structure, the density of atoms and the Miller indices of a crystal, 2) apply Schrodingers wave equation to various potential functions and derive a probability of finding electrons, 3) find the eigenvalues and eigenfunctions of the Hamiltonian of the system by perturbation methods, 4) discuss the concepts of energy band formation using nearly free electron model and tight-binding approximation, and difference of material properties in terms of the band, 5) derive the concentrations of electron and holes with a given temperature in terms of Fermi energy, and 6) can discuss drift, diffusion and scattering of carriers in a semiconductor under various temperature and impurity concentrations. For the second part, the student can 1) analyze, explain and evaluate a specific application of an (nano)electronic or photonic device/material using the obtained knowledge, and 2) create idea for device(s)/material(s) beyond the state-of-the-art for a specific application.	
Education Method	Lecture + exercise sessions	
Course Relations	Student cannot enroll both EE4705 and EE4710 and should select one of those two courses. EE4705 has only the first part of this course.	
Literature and Study Materials	Nanotechnology for Microelectronics and Optoelectronics, 2nd edition by Raul J. Martin-Palmer and José Manuel Martínez-Duart Elsevier Science	
Assessment	Written exam, homework, minitest and presentation In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	3 hour for written exam	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

QIST4400	Quantum Computer Architecture	5
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr. S. Feld	
Instructor	Dr.ir. T.H. Oosterkamp	
Instructor	Dr. R. Ishihara	
Contact Hours / Week x/x/x/x	0/0/4/0/	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The course will assume knowledge of fundamentals of quantum information. Basic knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce the students the overall system of a quantum computer, focusing on the classical hardware and software infrastructure required to build a quantum computer together with the quantum hardware. Covered topics include: - Quantum computer architecture; quantum languages; compilers; QISA; microarchitecture. - Quantum simulator (QX simulator); - Quantum error correction: quantum error correction codes; encoding; logical operations. - Mapping of quantum circuits; placement of qubits; scheduling of operations; routing of qubits; - Classical hardware for quantum computing: classical controller and quantum processor; qubit hardware and the need for cryogenic electronics; - Receiver and transmitter architectures for controlling qubits: frequency up/down-conversion, modulation schemes, frequency generation. - Hardware building blocks: amplifiers, analog-to-digital converters, digital-to-analog converters, digital processing, FPGA - Cryo-CMOS: device characteristics, state-of-the-art, design principles.	
Study Goals	By the end of this course, students are able to: - Explain the different layers that are required to build a quantum computer. - Discuss the basic functioning of quantum error correction (QEC). - Carry out the mapping step of simple quantum circuits onto specific physical device layouts. - Interpret the results of different quantum circuits written in language QASM and executed using simulator QX, including the following: Basic quantum algorithms; Fundamental QEC codes; Surface code, especially for a Ninja star. - Identify and describe the different hardware blocks in the classical controller in a quantum computer. - Derive the specifications of the individual components in a classical controller from the requirements of the whole controller. - Compare different hardware blocks based on their specifications and can select the appropriate component for target functionality and specifications. - Describe the main features and advantages/disadvantages of the adoption of cryogenic electronics in general and cryo-CMOS specifically.	
Education Method	- Slide decks including small exercises - Videos including quizzes - Interactive lectures including activities - Homework assignments - Paper reading including processing its content	
Assessment	The final grade consists of two parts: - Part 1: Weekly homework, will be submitted electronically. Individual submission. - Part 2: Final assignment, preparation of presentation of 20 minutes with 10 minutes Q&A. Assignment will be handled by pairs of two, there will be individual grades. The global course grade is an average of each parts grade (50% part 1 + 50% part 2). Students are allowed to omit one homework. There is no resit.	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Thesis project 2023

EC Program

This is the thesis project, this being the concluding study unit of the program. The project amounts to 45 EC and involves a research or design task with sufficient academic level. The project may be executed within a research program at TU Delft, or in a suitable research institute or company. The project must be carried out with a systematic approach and should include all phases of a research or design project: analysis, modelling, implementation/construction and validation/evaluation. The student executes the thesis project independently, with guidance of a thesis supervisor and under the responsibility of the full professor who approved the IEP.

ET4300	Master Thesis	45
Responsible Instructor	Prof. M. Alavi	
Responsible Instructor	Dr. D. Cavallo	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	M. Babaie	
Responsible Instructor	Dr. A. Lekic	
Responsible Instructor	M.A. Kitsak	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MScEE and MScCE programs. It is an individual in-depth research or expert design assignment with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. The thesis project may be executed within a research program at TU Delft, or in a suitable research institute or company. The nominal duration of a thesis project is 30 weeks (45 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. With regard to the thesis project, the thesis supervisor may impose additional conditions on the IEP to be met before starting the thesis project. If desired, an internship or an extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in scope and a separate deliverable such that the thesis project can be clearly evaluated in isolation. The thesis project is governed by the EEMCS Graduation Policy. Below you find a short description of the different steps to be taken during the thesis work: 1. Orientation 2. Kick-off meeting 3. Midterm meeting 4. Green-light review 5. Graduation Day (defence, assessment and graduation ceremony)	
Course Contents Continuation	Declaration of Credits Obtained Before you can start your Masters Thesis Project you need to have a by the Board of Examiners approved Individual Exam Programme (IEP) and at least 60 credits of the Master's degree course work as stated in the IEP should be completed. You can prove this by a form called the Declaration of Credits Obtained (Verklaring behaalde studiepunten) (60 EC). You can request this form from the service desk in the EEMCS building 36 (Mekelweg 4) at the servicedesk with your campuscard or send an email to servicepuntEWI@tudelft.nl (use your with your TU Delft e-mail!) You have to show this form to your thesis advisor.	
Study Goals	The master thesis project serves to prove that the student has acquired the academic competencies of a Master of Science. The thesis project is an individual in-depth research or expert design assignment independently carried out by the student with a systematic approach including all phases of a research or design project: - problem formulation and planning of the work - analysis, - modeling, - implementation/construction, - validation/evaluation, - presentation of the obtained results, orally and in writing. The achievements and resultant thesis report should comprise original work carried out by the student.	
Education Method	Orientation After having obtained a DCO, the thesis work starts with a thesis proposal, which is defined together with the supervisor. The proposal, of at most 2 pages A4, is written according to the format which can be downloaded from the Brightspace page of ET4300. The thesis proposal contains: - a working title, - a preliminary problem definition, - intended outcomes of the thesis project, - intended milestones, - expected duration and (preliminary) date of the thesis defense. The (preliminary) thesis committee is also indicated on this proposal. If desired, an internship or extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in activity and a separate deliverable such that the thesis project can be clearly evaluated in isolation. This is indicated in the thesis proposal. The thesis related work done during an internship or extra project should be described in the thesis report (typically one chapter). Kick-off meeting The Orientation phase indicated above results in a thesis proposal written by the student. The objective of the kick-off meeting is the formal approval of this thesis proposal by the supervisor(s). In this meeting, the research question(s), research methodology and planning are presented by the student and discussed with the supervisor(s). After approval, the "Thesis Proposal form" is filled out. Midterm meeting After 3-4 months, the student presents (by an oral presentation and a written report) the work performed so far to the supervisor(s). On the basis of the work done, it is decided whether the goals, as formulated in the thesis proposal, are still feasible. If not, they are reconsidered and adapted. After the mid-term meeting, the student has a clear idea of how to proceed in order to finish the thesis in time. Thesis committee This committee is responsible for the assessment of the thesis work and the thesis presentation. In order to be able to defend the thesis, the student must fill out the thesis committee form, which needs to be signed by the responsible full or associate professor. This form needs to be handed in to the Board of Examiners of the study programme at least 3 months prior to the MSc thesis presentation (the defence). Green-light review In the final phase of the thesis project, there will be a green-light review. During this meeting it is decided whether the student is	

ready to proceed with the thesis defence, resulting in a "go/no-go". It is compulsory to have a (draft) thesis ready for the green-light review. In case of a "go", the "Final Examination form" is handed in, which will trigger a check on the fulfillment of all requirements for getting the MScEE or MScCE diploma.

Graduation day

The final step of the graduation process is the presentation and defence of the thesis work. The candidate may present and defend the thesis work only after having finished all courses as stated in the approved IEP. The defence consists of two parts. First, the work is presented in a public session in front of the thesis committee, family, friends and fellow students. The presentation is followed by a closed session where the candidate will defend the work in front of the thesis committee. Immediately following the assessment, provided that there is a positive outcome, the diploma ceremony starts and the diploma will be handed out.

Assessment

The thesis project is assessed by the thesis committee based on the actual thesis work carried out, the final report, the public presentation, and the closed defense session and based on the learning objectives of the thesis. The items assessed in the final grading are:

Quality of the work: research method/design, application/development of theory, interpretation of results, scientific significance;

Performance: critical attitude, creativity, initiative, interaction with peers/superiors, planning;

Report: content, structure, quality of writing, independence in writing;

Presentation & defense: content, form, performance, scientific discussion.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

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Year
Organization
Education

2023/2024
Electrical Engineering, Mathematics and Computer Science
Master Electrical Engineering

Track Wireless Communication and Sensing 2023

Program Title	Track Wireless Communication and Sensing
EC Program	120
Introduction 1	Wireless Communication and Sensing systems are an integral part of our modern society. It is because of these systems that we can live more safely, securely, healthy, comfortably and more sustainably. They shape society in profound ways and their impact will become only bigger. Offering you education in those fields, the WiCoS track is a multidisciplinary master level educational program in Electrical Engineering (EE) at TU Delft. Related to faculty expertise, it focuses on wireless technology as well as network design & management. The program provides fundamental knowledge for societally important developments, some of which are the following: 5G (and beyond) communications Autonomous driving Environmental and medical sensing Internet of Things Each of the themes above offers a manifold of practical applications. For example, sensing encompasses applications in medical imaging, weather forecast and weather nowcast, traffic control, Earth monitoring, (deep) space observations and security applications. The program takes 2 years and is fully conducted in English. It offers challenging high-level education and research to talented students holding a BSc degree in technology or science.
Program Goals	The scope of the program can be summarized as covering wireless communication, wireless observation and sensing and communication networks. Wireless communication systems, e.g. cellular networks (4G, LTE, 5G) and Wi-Fi (2.4 GHz, 5 GHz, 60 GHz), are continuously evolving towards further improved services (higher data rates, better coverage, serving more users) at lower cost (more efficient spectrum use and low power techniques). New developments in wireless communication systems include the use of highly directive beams (Massive MIMO), ad hoc networking, dynamic spectrum access, and (low power) sensor networks. New standards such as 5G and beyond will integrate Wi-Fi and support wireless sensor networks, e.g. the Internet of Things (IoT) and vehicle sensor communication. All of them evolve toward achieving a higher efficiency, more flexibility, increased robustness and safety and security. Wireless sensing and observation technologies are developed in parallel and based on similar techniques. They are widely deployed in a broad spectrum of applications from health care to space. Microwave and Terahertz sensors and remote sensing systems are used to collect up-to-date information required e.g. to enable autonomous driving, to support reliable weather observations and forecast, to monitor climate changes for example by measuring the sea level, NO_x and CO₂ emissions, to diagnose and treat human diseases, to manage crop health on large areas of the Earth surface, and to provide real-time observations for security and disaster prediction applications. The needs for this kind of information is vastly increasing to efficiently manage and safeguard our complex society. Communication networks (as well as other complex networks) rely to a large extent on decentralization and self-organization. While decentralization removes obvious vulnerabilities related to single points of failure, it requires the management of a higher complexity and sophisticated design to optimize performance. The Internet is a clear example: there is no global control center, and obtaining complete information on its global state is an illusion. Additionally, a complex type of vulnerability appears in such systems. For instance, denial of service attacks, power failures and computer viruses are imminent threats to all computer networks. The MSc EE track Wireless Communication and Sensing covers radio communication systems, networking and complex networks, signal processing, wave propagation and detection, remote sensing systems and electromagnetics, with several opportunities for specialization. Wireless Communication and Sensing is a vivid field with research institutes closely cooperating with the industry. The laboratories involved in this MSc Program (all from the EEMCS faculty) are: Terahertz Sensing (TS) Microwave Sensing, Signals and Systems (MS3) Signal Processing Systems (SPS) Network Architectures and Services (NAS) These laboratories are related to many partners, be it commercial companies, research laboratories and other universities. This is also materialized in externally sponsored professors and other researchers at the laboratories. For students this implies that internships and thesis projects at companies during the two-years program can easily be arranged.
Program Structure 1	The WiCoS program is spread over a period of 2 years with 4 quarters per year. Your courses will evolve from more general ones (common core) in the beginning of the program to more specialized ones in the second quarter and beyond (track core and specialization), leading to your thesis project. An Individual Exam Program (IEP) of the track Wireless Communication and Sensing consists of the parts as indicated below. Note that course loads are expressed in EC, with 1 EC indicating nominally 28 hours of active study, including class and self-study.

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Main core WiCoS 2023	
Introduction 1	This is the common core. It is common to all programs and contains two general engineering oriented courses being compulsory for all students. These courses are Profile Orientation and Academic skills and Systems Engineering. The common core further contains 9 fundamental courses introducing all students to the breadth of EE and CE. Of these 9 fundamentals, all students must select two courses. Common core courses are fundamental and general in nature, and should be attractive to a relevant fraction of the students in multiple tracks. These courses are programmed in the first quarter of the academic year. Typically, they have three parts. These are (1) an introductory part that might be seen as a quick repeat of the expected prerequisite knowledge of incoming students (or as a quick introduction if some knowledge is still missing), (2) the main body, and (3) a part that makes a specific connection to the different tracks in the program. This last part may actually be track-specific. Common core EE EE4C01 Profile Orientation and Academic Skills EE4C11 System Engineering Common core: Select 2 out of 9 EE4C03 Statistical Digital Signal Processing and Modeling EE4C04 Control Theory EE4C05 Electromagnetics EE4C06 Networking CESE4010 Advanced Computing Systems EE4C08 Measurement and Instrumentation EE4C10 Analog Circuit Design Fundamentals EE4C12 Machine Learning for Electrical Engineering Applications EE4C13 Wireless Systems for Electrical Engineering Applications

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Instructor	Ir. J.W. Peltenburg	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = $0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3)$ L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1] In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C01	Profile Orientation and Academic Skills	3
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Responsible Instructor	L.C. Schroten	
Instructor	Dr. C.A. Benitez Avila	
Instructor	L.C. Schroten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: basic knowledge as acquired with a BSc in Physics, EE, Material Science. Strongly advised: Information Literacy 1 and 2 (e.g. Master Programme Information Literacy 1 & 2 or Electrical Engineering Information Literacy 1 and 2, both available in Brightspace).	
Course Contents	- Introduction to the MScEE tracks (Microelectronics, Electrical Power Engineering, Signals & Systems, Wireless Communication and Sensing) - Introduction to the research and expertise of the research groups taking part in the selected master track - Knowledge of ethics and intellectual property right - Skills in academic writing - Skills in oral presentation of scientific topics	
Study Goals	With this course the students - obtain relevant information on the MScEE study and the chosen track, become aware of cultural differences and how this affects intercultural communication, form a network with fellow master EE students. - acquire an overview of the main challenges in and potential applications of electrical engineering - become familiar with research topics addressed within the EE department - master basic skills in oral presentation of scientific topics - master basic skills in academic writing - acquire basic principle of ethics and intellectual property right - are able to extract the main information from a scientific paper or report - are able to write a paper (mostly a literature review) on a scientific question	
Education Method	The course includes lectures, reading materials and exercises. During the course, students work on a writing assignment, an ethics case and give several presentations. Attendance is mandatory as all class activities are highly interactive and peer feedback is a necessity.	
Literature and Study Materials	TU Present & TU Write. Both are available in Brightspace.	
Assessment	The pass/fail for this course is determined by the evaluation of the final writing assignment, the oral presentation and attending the Ethics seminar(s). All three components should be evaluated with a pass to pass the whole EE4C01 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C04	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential Realization of transfer function models by state space descriptions, coordinate changes, canonical forms Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback Application of LQ regulator Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle Reference signal modeling, integral action for zero steady-state error. Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design and - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	P.P. Sundaramoorthy	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C13	Wireless Systems for Electrical Engineering Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	C. Gao	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	M. Babaie	
Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	M. Spirito	
Instructor	Prof. M. Alavi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE4605 'Integrated Circuits and Systems for Wireless Applications' and EE4371 'Advanced Wireless Transceivers.'	
Course Contents	In this course, you will learn about the commonly used architectures in wireless systems, with the requirements on their building blocks. The system key building blocks are addressed at their functional level regarding electrical properties, such as transfer/gain, noise, linearity, frequency selectivity, bandwidth, and other parameters. Physical constraints of practical components and circuits are used to identify the achievable signal quality, selectivity, sensitivity, and link budget of a wireless system. The lectures will explain the causes and impact of device/circuit non-linearities and noise on wireless systems' sensitivity and dynamic range. The course will introduce the concepts of frequency generation, modulation/demodulation, and up/down conversion. To teach you to work at RF frequencies with wireless systems, the principles of impedance matching are explained, and tools are provided to analyze/design/characterize RF (matching) circuitry/systems. The common part of the course also includes an overview of the figure of merits (FoMs) of various TX/Rx architectures/system blocks and the required instrumentation to characterize them. This course belongs to the RF/wireless profile and is an excellent preparation for the course; EE4605, Integrated Circuits for RF/Wireless Applications, which focuses on the design and realization of analog intensive building blocks for wireless transceivers as well as for ET4371, Advanced Digital Wireless transceivers which is addressing the latest trends in RF/wireless design aiming to fully utilize the integration and signal processing strengths of advanced CMOS. The course is completed by four short specialization topics which can be followed based on your personal preference, namely: (1) Introduction to wireless power transfer technology, (2) Beamforming basics: concepts and architectures, (3) Neural Networks for Wireless & Radar Systems, (4) Hands-on experience with RF characterization equipment on practical circuits. (The latter (4) has an attendee limit of 15 people.) Only one of these specialization topics is required to complete the overall course.	
Study Goals	Understand wireless system architectures with their subblocks to such a level that you can analyze their various performance parameters in terms of; link budget, sensitivity, spectral purity, P1dB, IP3, DR, SFDR, noise, phase noise, selectivity, EVM, APCR, and efficiency. In addition, you will master a basic understanding of S-parameters, impedance matching, and RF instrumentation.	
Education Method	This course will be delivered with complementary exercise lectures to familiarize students with the concepts and calculations described during the lectures.	
Literature and Study Materials	These lectures are (in part) based on the book of Behzad Razavi, RF microelectronics 2nd edition, Chapters 1 to 4.	
Assessment	Examination and grading: Your overall grade will be based on your performance in the written exam (80%), compulsory homework, and the mark obtained in your track-specific project (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year

Organization

Education

2023/2024

Electrical Engineering, Mathematics and Computer Science

Master Electrical Engineering

Track core WiCoS 2023	
Introduction 1	This is the track core, offering the first level of branching between tracks. The track core contains 5 courses, of which the students select three courses.

EE4510	Advanced Electromagnetics	5
Responsible Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	1) Ohmic losses at high frequencies 2) Potentials and Green's functions 3) Image theorem 4) Fields radiated by sources in the far field region 5) Equivalence and reciprocity theorems 6) Antennas in transmission and reception 7) Green's function of infinite line source 3) Application case: quasi-analytic analysis of transmission lines at high frequencies	
Study Goals	1) EM analysis of wireless links Basic characterization/understanding of a wireless link between a transmitting and receiving antenna 2) EM analysis of transmission lines Basic understanding of the impact of radiation and ohmic losses at high frequencies	
Education Method	The course consists on 1) Theoretical lectures on the basic physical mechanisms and how to analyse them via EM analytical techniques 2) Practical Instructions to get familiar with such techniques 3) Study of an application case	
Assessment	The assessment of the course will be the following. The final mark will be the average between the home assignments, written exam and an oral exam (depending on the number of students). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4565	Propagation and Scattering of EM waves	5
Responsible Instructor	Prof.dr. O. Yarovyj	
Responsible Instructor	O.A. Krasnov	
Instructor	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	ET4169,	
Expected prior knowledge	Basic knowledge of Electromagnetics, EE4C05 Electromagnetics	
Course Contents	The Propagation and Scattering of electromagnetic (EM) waves course provides an overview of the essential aspects of the physical layer for a wide range of applications that rely on electromagnetic waves for the information transmission and sensing (wireless communications, radar, and navigation). Both the propagation medium and environment change the characteristics of the electromagnetic waves and affects the quality of the wireless link. The course focuses on physical models of propagation and scattering phenomena with emphasis on practical estimations of these effects influence on operation and design of telecommunication systems. The following aspects will be covered in the course: * Overview of the radio spectrum, services, and frequency allocation, propagation/scattering environment, frequency band designation for EM waves * Characterization of propagation media * Antenna and noise concepts * EM waves interaction with material objects * EM wave scattering on an ensemble of discrete particles * EM wave scattering on a rough surface * EM wave scattering on a complex and distributed objects, the concept of radar cross-section as a characteristic observed with radar objects, * Direct transmission, * Power balance for communication and radar systems in cases of direct transmission, atmospheric gas attenuation, signal attenuation, and scattering in rain * Refraction - wave propagation through a stratified medium * Propagation near the Earth's surface * Diffraction on terrain, Fresnel zones, propagation over a plane/spherical Earth, practical propagation path analysis * Empirical Path Loss Models * Models of fading effects * Ionospheric propagation/scattering of electromagnetic waves Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	After this course students have to be able * to recognize and explain different mechanisms of electromagnetic waves (EMW) propagation/scattering and relations between them; * to assess different mechanisms of EMW propagation/scattering in relation to characteristics of EMW and media; * to estimate quantitatively the influence of the propagation and scattering mechanisms on trade-offs that have to be made when designing a wireless communication or sensing system. * to analyze real-world scenarios and identify the involved propagation and scattering mechanisms and their relevant aspects;	
Education Method	Lectures, instruction lectures, self-study recommended references, home exercises, practicum	
Literature and Study Materials	Levis, C.A. Radiowave propagation: physics and applications, Wiley, 2010 - 301 pages. ISBN-13: 978-0470542958 Barclay, L.W. Propagation of Radiowaves, IET, 1996, 2003. - 352 pages. ISBN-13: 978-0852961025	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: 10% of averaged grades for homeworks (max 1 point); 20% of averaged grades for practicum reports (max 2 points); 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Elektricity & Magnetism Information & Communication Practicals Signals and Systems Technology Telecommunication	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formulary is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year

Organization

Education

2023/2024

Electrical Engineering, Mathematics and Computer Science

Master Electrical Engineering

Suggested specialisation courses WiCoS 2023	
Introduction 1	This is the specialization space. In this tier, students have to select courses totalling to at least 24 EC . In essence, all of the MSc EE and MSc CE courses can be chosen as a specialization course, including those from the main core and the track core that were not yet chosen.

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
Study Goals	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment (and hence 100% of the grade) will be based on an exam that covers both the theory from the slides as well as the content from the reader. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0 (Lectures(2x2h), Pract)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Strongly advised: (Advanced) Networking course (e.g., CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved).	

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ----- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. ----- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) - o Definition of ad-hoc networks - o Comparison with infrastructure based systems - o Typical applications - o Advantages and challenges - o Radio technologies for ad-hoc networks - o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks - o Propagation models - o Topology models based on graph theory - o Degree and hopcount - o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) - o Introduction to MAC protocols - o Issues and design goals - o Classification - o Directional, multi-channel MAC protocols - o Energy efficiency in MAC protocols - o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) - o Flooding - o Node discovery, neighbour discovery - o Route establishment - o Topology maintenance, localisation - o Proactive, reactive and hybrid routing - o Typical protocols - o Energy efficiency in routing - o Broadcast and multicast - o Effects of mobility on connectivity and capacity - o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks - o Wireless sensor networks (Chapter 12 of the textbook and papers) - o Cooperation (Reference papers) - o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) - o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for	

textbooks, research papers, etc).
3. Some recent journal papers

4. Optional Reference Books

- 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms)
- 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library.
- 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library.
- 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer.
5. OPNET/ns-2 web pages, tutorials and video lectures

Books

Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoh, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.

Assessment

1. There will be totally three written/oral tests/examinations for this course (including the final exam).
2. The students will carry out a project in a group and submit a short report.
3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis.

Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class.

Breakup:

2 Tests + Final Exam = 55%

Project 40%

Self assessment + Reflection 3%

Activities on Feedback Fruits or any Online platform 2%

Resit: written/oral exam [+ carry over from projects].

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Changes due to COVID19 in 2021 is below;
in case there is a similar situation, we use the following methods.

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There are three chapters

Chapter 1: Network modelling

Chapter 2: MAC protocol

Chapter 3: Routing

+

Lab: Simple experiments using laptops (individual) + simulations (group-wise)

Marks Breakup

Homework/Assignments/Group works

Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks

Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks

Part B1: Individual experiment + report

Part B2: Group-wise simulations + report + demo

Part B1 + Part B2 - 50 Marks

Part C: Self-assessment + peer activities -- 5 marks

Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit.

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(More information will be given in the first class)

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Permitted Materials during Tests

Different conditions for different test/exams.

Conditions will be informed 1 week before the exams/test.

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/ ; Wireshark learn page: https://www.wireshark.org/#learnWS ; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/ ; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands on course for self motivated students. Students work in a group of two (max 3) usually. Students are expected to have sufficient programming and hardware development skills. The course is project oriented thus students are expected to deliver a working model with demonstrations. Preference will be given to Q5 students. Q1 students are discouraged unless otherwise explicitly allowed. ----- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. ----- This course is done in a group of two (max 3). Instructor/TA will help connecting a potential groupmate in the first class using Brightspace/Feedback Fruits. ----- Activities: How to in Groups? ----- Part 1: Seminar We guide the groups to select a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID19 related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their own responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/raspberry pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki based networks or Open testbeds projects. In these cases, online group activities are possible. NOTE: In some extreme cases, Instructor/TA may allow one person project after assessing the situation and after discussing with the student. ----- Supply of Hardware: ----- 1. Instructor/TA is able to provide take home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.	
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.	
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.	
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).	
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	
Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc.	

Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project.

Part 1: 30% of the whole marks;

Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment.

Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference.

Report will not have individual component, however, Q/A may have individual component.

Please NOTE: There would be no resit since this is a hands-on course being executed in a team.

As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Tags

Circuits

Group work

Programming

Project

Software

maximum aantal deelnemers Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics".	
Course Contents	We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming. The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non-trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (10% of grade each, 2 pages each) 2) Final report (10 % of grade, 4 pages) 3) Final project demonstration (70% of grade) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. There is no resit for this course. repair option: A new but related project will be given for the Report (W), Oral(0) and Demonstration (IP).	
Enrolment / Application	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) 1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4160	Blockchain Engineering	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	In this course you will learn all aspects of blockchain technology, including tamper-proof data structures, digital identities, transitive trust, fault tolerance, distributed consensus, smart contracts and applications. Ledgers and blockchains are an emerging technology with the potential to radically improve financial transactions, supply-chain flows, transactions in general, and distributed databases. The first three weeks of the course will provide a fast-paced introduction to Bitcoin, Ethereum, and TrustChain developed by TUDelft itself. The main component in this course is a team-based complex engineering project. This course is designed for computer scientists to understand blockchain technology and to produce significant hands-on experience. To provide a deep understanding of blockchain technology and understand why it is special you need to experience first-hand how it operates at a detailed technical level. Students design, implement, and test their own independent project in teams of 3-5 students. Students can choose from a pool of possible project ideas. This course requires you to like software engineering. Topics covered: -Blockchain basics and evolution Bitcoin 1st generation, smart contract generation, future 3rd generation (trust or trust in math) -identity and transitive trust Authentication and security primitives, tamper-proof identities, trust models, MITM attacks, Sybil attacks, and TrustChain by TUDelft -Consensus models Proof-of-work, permissioned, Proof-of-stake, Corda no-global-consensus, TUDelft bottom-up fast consensus model -Smart Contract pro/con encrypted data, Bitcoin scripts, Ethereum execution model, Hyperledger + Docker issues, Corda Jar file approach, Tezos difficult to use, powerful technology, vision of the future: trusted verified execution -Markets and exchanges Disruption by open markets, winner-takes-all, and multi-sided market platforms, Uber, Airbnb, 22 years of eBay, Silk Road, honesty among drug dealers, the role of trust in markets, P2P exchange markets -Decentralized Autonomous Organization, novel method to collaborate and organise any economic activity -Web3: how the Internet can be re-decentralised using DAOs Within this 2023 edition "the Delft DAO" will be prominently featured. TUDelft achieved a world-first in DAO research. We devised a full end-to-end proof-of-principle of a DAO which is capable of 0) near unbounded scalability 1) controlling money 2) democratic decision making and 3) continuous sustained self-evolution. This course provides you with the knowledge to work with this advanced technology. After this course you will have a firm grasp on the current operational blockchain-based systems, realistic view of real-world applications that may be built on top of ledger technology. You will be able to reason and discuss the open challenges and questions that still need to be resolved. This course is a key course for distributed systems students.	
Study Goals	After this course students are able to design and engineer complex blockchain-based systems. Students are able to describe blockchain technology, the various consensus model, smart contracts, markets, and relation to existing database technology. Student are able to setup a new architecture for blockchain applications.	
Education Method	This course consists of four 2-hour lectures. Each lecture is followed by a 4-hour homework period in the same week focused on understanding the background material. In week 1 you will form teams and initiate work on your blockchain engineering project. A list of projects to select from will be provided at the start of this course.	
Literature and Study Materials	Online course textbook: Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction by Narayanan, Bonneau, Felten, Miller and Goldfeder.	
Prerequisites	It is highly recommended to follow this course (see remarks): Security and Cryptography (Q1) Distributed Algorithms (Q2)	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) You will write down your project results in a single-page report, markdown format. You will be graded on your open source efforts located on Github and single-page report. Your grade will be expressed on a scale of 0 to 10. Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). This class has a limited capacity (50). If there is a larger number of enrollments than the capacity of the class, students will be assigned to their preferred blockchain engineering project based on their background, engineering experience level, and match to the course goals. Students who followed Security and Cryptography (Q1) and are also enrolled in Distributed Algorithms (Q2) will have priority for placement. Mathematics students are exempt from this, if they can show some minimal software development experience (e.g. Github profile). Finally, students with a Grade Point Average of 8.0 or higher are eligible for the challenging scientific projects, resulting in a research paper. These project receive intense guidance, but have no capacity limits.	

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: $0.2 * \text{Assignments} + 0.8 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4430	Network Security	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	The final grade of the course consists of the following components: Assignments (Pass/Fail) Final project (Report + Presentation) (weighting 100%) Final grade calculation: $P/F * \text{Assignments} + 1 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable, resit presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4016	Antenna Systems	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	Y. Aslan	
Instructor	Ir. P.J. Aubry	
Co-responsible for assignments	Ir. P.J.F. Swart	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4288, ET4595	
Expected prior knowledge	EE4C05	
Course Contents	The aim of this course is to introduce the students to the Antenna Techniques and Antenna Systems that are instrumental for applications such as telecommunications, remote sensing and radar systems. After introducing basic antenna concepts the basic antenna parameters are defined. A detailed description of the antenna gain, directivity, polarization, radiation pattern, bandwidth, beamwidth, efficiency, temperature, effective length, effective area and input impedance is given. The reflection and radiation from a dipole are detailed and analysed. Microstrip antenna is introduced and its performance is analysed. The radiation from linear, circular and planar array antennas is discussed in detail while stressing their importance in radar and remote sensing systems. The beamforming in arrays and antenna array synthesis are studied. Characterisation of Antenna System performance is discussed and derivation of Antenna System specifications from overall wireless system performance requirements is studied. Some basic concepts about antenna measurements techniques are introduced. The theoretical studies are complimented by computer simulations and measurements.	
Study Goals	By the end of this course, you should be able to: LO1: Determine (theoretically and experimentally) the fundamental antenna system performance metrics (such as directivity, gain, bandwidth etc.). LO2: Design basic antenna element/array structures (i.e. linear wire, microstrip, aperture) for the given criteria, by using CST and MATLAB tools. LO3: Assess the impact of a given antenna on the wireless system performance (in terms of the signal and noise power). LO4: Develop a suitable antenna model for a given application with specified requirements (e.g. communication, broadcasting, radar).	
Education Method	Lectures, Instructions, computer simulations, antenna measurements	
Literature and Study Materials	Course books (Balanis and Orfanidis), Lectures handouts, simulation and measurement assignments.	
Assessment	Written exam (problems solving) plus home works, simulation and measurement assignments In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4396	Mobile Networks	5
Responsible Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Not formally required, but recommended to have the basic knowledge taught in 'Fundamentals of Wireless Communications' (ET4358). If you didn't take this course, key aspects will be repeated during the class. In case you are lacking a comfortable background, talk to me in the first week of class and I'll help you catch up.	
Course Contents	The course addresses the technology and management of contemporary and future mobile cellular communication networks. With a focus on radio access, as it dominates both the performance and cost aspects of mobile networking, the course addresses 5G, 4G (LTE/LTE-A), 3G (UMTS/HSPA) and 2G (GSM/GPRS/EDGE) technologies, all of which are operational network technologies. An outlook to 6G will also be given. Fundamental aspects of cellular networking are explained and discussed, including e.g. the intrinsic tradeoffs between coverage, capacity and quality and key aspects that influence performance (traffic, propagation, technology, deployment, spectrum. Cellular network planning, dimensioning and performance analysis are treated, as well as radio resource management mechanisms (adaptive beamforming, channel-aware single/multi-user scheduling, handover control, link adaptation, admission, congestion control,) and the current trend towards (potentially AI/ML-based) self-management. Cutting-edge radio features including single/multi-user massive MIMO beamforming, which is at the heart of 5G, are also covered. Considering the on-going technological developments in this exciting domain, the course material is updated every year. The principal lecturer has a 25+-year track record of research and consultancy in the field. The course is enriched with notes about actual deployments, historical anecdotes, popular misconceptions, fascinating factoids, illustrative demonstrations and (potentially) guest appearances.	
Study Goals	The objective of the course is to provide the student with a thorough understanding of contemporary mobile cellular networking technologies, their fundamental properties and tradeoffs, as well as the key challenges and approaches related to network dimensioning, planning and (self-)optimization towards efficient provisioning of coverage, capacity and quality.	
Education Method	Lectures, demonstrations, in-class discussions, assignment(s).	
Literature and Study Materials	Primarily: the lecture slides! For the enthusiastic student, supplementary background materials are recommended in the form of e.g. (white/scientific) papers and book chapters. A number of books on wireless/mobile communications in general or about 2/3/4/5G technologies specifically are recommended as background material; specifics will be given in class / in the lecture slides. As said, this is merely recommended for the most eager students. In principle, the lecture slides cover all relevant content.	
Assessment	Closed-book exam, either written or oral (to be determined). And one (may be two) assignments. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

EE4605	Integrated Circuits for RF/Wireless Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Responsible Instructor	M. Spirito	
Instructor	Ing. M.J. Pelk	
Contact Hours / Week x/x/x/x	0,3,0,0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Common core course EE4C13. Other useful courses to consider as part of your preparation for RF design (not mandatory) are EE4C05 and EE4C10.	
Course Contents	In this course, you will learn to understand and design the circuit blocks of wireless systems. You can design and analyze circuits/ICs for wireless systems when completed. To do so, fundamental knowledge of RF/microwave design is required. To achieve such an insight, we discuss; optimum bias, device scaling, circuit configurations, and harmonic matching techniques for active devices (BJTs & FETS) operating at high frequencies to reach optimum gain, noise, linearity, and efficiency. The outline of the course is as follows; - Design of RF/microwave amplifiers, stability, gain, and impedance matching (HW1a) - Design of RF/microwave hybrids and transmission line-based passives, as well as on-chip (high Q) passive components; capacitors, varactors, inductors, transformers, and transmission lines (HW1b) - LNA design for simultaneous noise & impedance matching and LNA noise cancellation techniques (HW2). - Analysis and design techniques for circuit linearity enhancements (HW3) - RF mixer design for linearity - Design of energy-efficient & wideband power amplifiers & transmitters (HW4). A detailed course description addressing all topics covered in this lecture series can be found on Brightspace. Note that this EE4605, Integrated Circuits for RF/Wireless Applications course is part of the RF/wireless track and is preceded by EE4C13, Wireless Systems for Electrical Engineering Applications, which discusses wireless applications at the system level. This course (EE4605) can be followed up by ET4371 (strongly recommended), Advanced Digital Wireless transceivers, which addresses the latest trends in RF/wireless design which aim to fully utilize the high integration and signal processing strengths of advanced CMOS. Background reading for the various topics can be found in the recommended books given in the Literature and Study Materials section. This course will have three to four homework (to be handed in) that address the design of the various circuit blocks (RF amplifier & microwave passives, LNA, mixer, PA/transmitter) and their sub-components. For this homework, use will be made of Keysights ADS Advanced Design System and Cadence. Aspects of packaging, CAD, and RF testing relevant to wireless / mm-wave design are also emphasized.	
Study Goals	Examination and grading: Your overall grade will be based on your performance in the homework and a written exam. - Understand how to (optimally) design RF/wireless components and circuits. This includes: - How to design, dimension, and layout passive components like: PCB and on-chip transmission lines, capacitors, inductors, and transformers. - How to bias and scale active devices while understanding their basic operation and equivalent models. - How to stabilize active devices while maintaining their RF performance. - How to design matching networks for maximum gain, low noise, and high linearity. - How to design for maximum efficiency, bandwidth, ACPR, and lowest EVM. - Know how to use RF/Microwave design tools like Keysights ADS, Momentum (in addition to Cadence) to design RF integrated circuits and systems. - Understand the RF/microwave-specific data representations and analysis tools. - To understand the full design cycle from RF/wireless concept, RFIC design, to the final testing. - Being able to analyze and design integrated RF transceiver front-end circuits, like LNAs, mixers, and PAs.	
Education Method	Lectures + design related homeworks	
Literature and Study Materials	Hand-outs + recommended background books * Microwave Transistor Amplifiers: Analysis and Design (2nd edition) by Gonzalez, Guillermo * RF Microelectronics (2nd edition), by Behzad Razavi * Microwave Circuit Design, using Linear and Nonlinear Techniques, by George D. Vendelin, Ulrich L Rohde, Anthony M. Pavio * RF Power Amplifiers for wireless Communications (2nd edition), by Steve C. Cripps.	
Assessment	RF design related homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	More details are available on Bright Space. To participate in this course, please enroll.	

EE4620	Spectral Domain Methods in Electromagnetics	4
Responsible Instructor	Dr. D. Cavallo	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System	
Course Contents	1) Green's functions of stratified media (6 hours) 2) Dominant spectral contributions and far-field (4 hours) 2) Surface waves and leaky waves (6 hours) 3) Artificial dielectric materials (4 hours) 4) Spectral representations for planar Tx-lines (4 hours) 5) Connected arrays (10 hours)	
Study Goals	The students will be able to - apply spectral domain methods for the analysis of a number of electromagnetic problems (planar transmission lines, artificial dielectrics, stratified media problems) - create functional MATLAB scripts that implement the studied methods - present and critically analyze the results obtained with the MATLAB codes	
Education Method	- Lectures through PowerPoint slide projection - Practical MATLAB sessions with implementation of the analysis methods - Analysis of the results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	Home assignments (50% weight on the final grade) Final project presentation (50% weight on the final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4630	Telecommunication Network Architectures	3
Responsible Instructor	Ir. R.A.C.J. Noldus	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course presents the fundamentals of cellular mobile networks, including network architecture and services, with a focus on end-to-end and core network aspects. The course forms a companion course for Cellular Networks Radio Access Networks (ET4396). Students will be introduced to the concepts circuit switched and packet switched, the two fundamental communication methodologies that form the backbone of the mobile network. The course aims to educate students on the evolution of the network from 2G to 5G, whereby for each network generation ('G'), the new architectural principles are studied. For example, we discuss the GSM core network (2G), GPRS (2.5G), UMTS (3G), Evolved Packet Core (EPS; 4G), the IP Multimedia Subsystem (IMS), Multimedia telephony (MMTel), Intelligent Networks (IN), Circuit switched fallback (CSFB), Voice over LTE (VoLTE), Voice call continuity (VCC), Voice over Wifi and 5G network architecture. For the 5G network architecture, the concept of 'network slicing' will be presented, which is used for flexible resource allocation, for Internet of Things (IoT) applications and for 'verticals' such as Automotive, Smart Energy, Media & Entertainment, Vehicle communication and other forms of machine-type communication. The course also deals with the concept of virtualization and cloud computing. Networks are increasingly built as a collection of virtual network functions (VNF), deployed on industry standard computing platforms, comprising standard hardware and standard OS. Keywords: - Network architecture and network protocols - Core network, Radio access network (RAN) - Circuit switched, Packet switched - Location/mobility management/roaming - Security (authentication, ciphering,) - QoS/QoE/bearer architecture, QCI definitions, service management, policy control - IMS/VoLTE/Messaging/Voice-over-Wifi/Enriched calling - Virtualization, Cloud computing, Software Defined Networks (SDN) - Network slicing - Multi access Edge Computing (MEC) - Intelligent Networks - Verticals/applications, including e.g. Automotive, Smart Energy, IoT, Machine-to-machine communication - Web Real-time Communication (WebRTC)	
Study Goals	After this course, students are able to explain the meaning, working and rationale behind key cellular network concepts (such as those given by list of keywords above). Furthermore, students will be able to understand the challenges and associated trade-offs in the deployment and management of cellular networks. Students will also be able to analyse network architecture and evaluate network architecture on its merits or on its drawbacks.	
Education Method	This course consists of 8 classes with each class consisting of 2 hours. After the 4th lecture, a mid-term rehearsal takes place; and after the 8th lecture, an end-term rehearsal take place. These rehearsals help the students in preparing for the exam. The students are expected to study the lecture material after every lecture and to ask questions the next lecture, on any topic that requires further explanation. Homework will be provided after each lecture. Students may at any time interact with the lecturer via mail, to ask clarification on any topic of the course.	
Literature and Study Materials	Slides Articles and book chapters	
Assessment	Examination, closed book. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4670	PV Materials Processing and Characterization	4
Responsible Instructor	Prof.dr. A.W. Weeber	
Instructor	Dr. L. Mazzarella	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3 Different, to be announced	
Course Language	English	
Course Contents	The course is a follow-up of PV Technologies. Details of processing and characterization techniques of PV absorber materials and the solar cells functional coating will be discussed. Examples of lecture topics are advanced light management structures, novel surface passivating layers and contacting schemes. Background of deposition technologies such as Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD) and Atomic Layer Deposition (ALD) will be given. Furthermore, working principles of special coatings and its functionality when applied to different solar cell concepts, including tandem devices, will be presented. This course is an elective one following the solar profile. PV Basics and PV Technologies should have been concluded successfully.	
Study Goals	After this course students are able to understand processing mechanisms, material properties, and functionality and working principle of the novel functional coating applied to different solar cell concepts. Students are expected to be able to explain these working principles and have a solid background on different deposition technologies and characterization techniques, and will be able to describe the effect on the solar cells output. Students will be able to analyse and evaluate recent technology developments and the background of it, and summarize them during an oral presentation for the entire group. As we can see, these learning objectives are specific about the content, and they are phrased in terms of observable behaviour: understand, explain, evaluate, present and describe.	
Education Method	This course consists of eight 2-hour lectures. In the same week a 2 hour work session / instruction will be organized for a deeper understanding of the technologies discussed in the lectures. During the whole course the students will work on assignments which need to be submitted as reports. All assignments/reports should be assessed as sufficiently good (no specific mark). In week 9 (final schedule will be given at the beginning of the course) you will present results of a literature study on a special topic covered by the course for the entire group.	
Literature and Study Materials	Book (not obligatory, expensive): Photovoltaic Solar Energy: from fundamentals to applications by Angèle Reinders et al. Material from online course: PVCDROM, PV-manufacturing.org Slides during lectures Articles (referred to in lectures)	
Assessment	The final grade consists of the following parts: presentation (team work) (50% of final grade); written exam (individual) (50%); Both marks will be expressed on a scale of 1 to 10 and should be 5.80 or higher. Reports on the assignments shall be completed as well, and assessed as sufficiently good (no specific mark). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
maximum aantal deelnemers	Expected number of student: Maximum of 30. Use Brightspace and inform our secretary Mark Vielvoije (m.vielvoije@tudelft.nl) for attending the course.	

EE4675	Object classification with radar	4
Responsible Instructor	Dr. F. Fioranelli	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Advised: ET4169, specifically knowledge of microwave and radar basics such as measurements of range and velocity and related basic signal processing.	
Course Contents	While the idea of gaining information about the surrounding environment using electromagnetic waves is a relatively mature concept (a radar essentially!), a tremendous amount of research activities have recently coupled the analysis of data from EM sensors with techniques that can teach them to recognise automatically objects of interest. What natural or manmade objects are moving around, and what are they doing? Can we infer something about their behaviour or intention from such movements? These are an example of the big questions which data and clever signal processing from EM sensors can address. Applications range from the analysis of human gait and movements to the discrimination of birds vs Unmanned Aerial Vehicles (drones), to the identification of the subtle gestures people produce to interact with smart objects, amongst others. All these applications have in common two main areas of expertise: one has to do with understanding and processing the information encoded into the sensors signals (radar signal processing and data format); one has to do with the techniques to interpret this information automatically to make decisions (we could call this applied machine learning). In this course, we aim to provide a comprehensive introduction and explanation of both areas from an engineering application perspective, with the objective to learn how to use these tools for real-life applications in radar engineering. The course is broadly divided into two parts: 1. Part 1 will build on fundamentals of radar & microwave sensing courses to investigate the concept of micro-Doppler signatures, focusing on its modelling, the signal processing for its characterisation, and its relationships with the physical movement of objects and therefore their identity. Can we understand from the EM sensors data how the different parts of human bodies or manmade objects move? 2. Part 2 will introduce machine learning techniques for extraction of information from sensors data (feature extraction), for their selection, and for classification of objects of interest. Conventional supervised learning techniques will be presented (for example Nearest Neighbours and Support Vector Machine classifiers), along with some of the more recent data-driven approaches based on neural networks. The diversity of formats of the radar data for classification will also be considered, including micro-Doppler signatures, High Resolution Range Profiles, radar cubes. In both parts, students will be exposed to concrete examples and data directly taken from current activities at TU Delft, so that the research element is well embedded in the teaching.	
Study Goals	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The aim of this course is to introduce students to techniques to classify and identify objects of interest mixing machine learning and signal processing applied to data from radar sensors. By the end of this course, students will be able to: -- Apply relevant signal processing techniques to EM sensors and radar data to extract micro-Doppler signatures -- Compare the different domains of radar data that can be exploited for objects classification, and the convenience of using one rather than the other in different situations -- Analyse machine learning techniques for feature extraction and classification applied to radar sensing -- Propose possible approaches for given classification problems based on radar sensors data	
Education Method	The course consists of class lectures and practical activities in the laboratory where possible or following the lectures. Laboratory activities may take place at the existing facilities available for the MS3 (Microwave Sensing Signals & Systems) section.	
Course Relations	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). This course is intended as part of the Master program in Electrical Engineering, primarily for students in the WICOS and S&S tracks, but with expected interest also from students from the civil engineering faculty (CITG), mechanical/robotics engineering (3ME), and aerospace (LR), where activities in remote sensing and autonomous robotics have seen recent significant interest in radar sensing and radar-based classification. This course is envisaged to cover new topics, currently not present in the courses related to radar sensing on offer. In this regard the course is complementary to the existing courses of ET4169 (Radar I) providing the fundamentals of radar sensing and radar as an instrument, and ET4175 (Radar II) providing advanced information mostly focused on the theoretical basis of processing for radar detection and tracking. Therefore, this proposed new course fills a gap in presenting advanced content on the topic of radar-based classification and the related approaches, widening the offer of topics available to our students and contributing to their more well-rounded education. This new course can fit well with the new proposed order of the radar courses": EE4565 in Q2 (fundamentals of electromagnetic scattering and propagation, ET4169 in Q3 (basics of radar for broad audience of students), this course in Q4 (more specialised course in classification based on radar), ET4175 (more specialised course in advanced theory for radar detection and tracking). The new course also fits with the wider research strategy of the department, whereby classification and machine learning intersect the three overarching research themes of Next generation sensing and communication, Health and wellbeing, and Autonomous sensor systems. It is therefore expected that this course will enhance the research capabilities of students in view of their thesis projects, and benefit their employability afterwards. Specifically, radar-based classification for situational awareness is a timely topic, relevant to automotive (NXP), defence/security (Thales NL, TNO), robotics and HCI (Google has recently introduced a radar, Soli, within their phones for gesture-based interaction), healthcare (contactless estimation of vital signs). As part of the educational activities in this course, it is envisaged to have at least one guest lecture on relevant topic from an external stakeholder (for example from Thales NL and TNO).	
Literature and Study Materials	-Lecture slides and pre-class activities provided by the instructors -Selected chapters of the following books. They should all be available as e-book via the TU Delft Library. ---Richards, Mark A. Scheer, James A. Holm, William A.. (2010). Principles of Modern Radar, Volume I - Basic Principles. SciTech Publishing. ---Chen, Victor. Micro-Doppler Effect in Radar, Artech House, 2011. ---Chen, Victor C.; Tahmouh, David; Miceli, William J.. (2014). Radar Micro-Doppler Signatures: Processing and Applications. Institution of Engineering and Technology. ---Chen, Victor C.; Ling, Hao. (2002). Time-Frequency Transforms for Radar Imaging and Signal Analysis. Artech House. ---Gurbuz, Sevgi Zubeyde. (2021). Deep Neural Network Design for Radar Applications. Institution of Engineering and	

Assessment

Technology.

---Fioranelli, Francesco; Griffiths, Hugh; Ritchie, Matthew; Balleri, Alessio. (2020). Micro-Doppler Radar and its Applications. Institution of Engineering and Technology.

---Geron Aurelien. (2019). Hands-on machine learning with scikit-learn, keras, and tensorflow : concepts, tools, and techniques to build intelligent systems (Second). O'Reilly Media.

-Selected published research articles available in IEEE Xplore (to be updated year by year, as novel research is generated - these are accessible via TU Delft Library)

The final grade of the course consists of the following components:

-a written exam with questions (that can be performed online in Brightspace if needed).

-a report on practical/project activities (which may include collection and/or processing of some radar data, and implementation of a complete radar classification pipeline).

The written exam is worth 75% of the final grade and the report(s) on practical activities 25%.

As repair possibility the written exam has one resit opportunity.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor analysis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: $0.5 * \text{Oral discussion 1} + 0.5 * \text{Oral discussion 2}$. Both partial grades must be sufficient, i.e., a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4725	Quasi Optical Systems	5
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. D. Cavallo	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is required knowledge of - Electromagnetics - Transmission lines - Antennas	
Course Contents	The exam 'AP3701 Sub-mm and terahertz physics and applications' is recommended. This course includes the 5 topics listed hereafter: 1) Spectral domain Green's function and Far Field evaluation (4 hours) 2) Physical Optics for reflector and lens antennas (8 hours) 3) Focal Plane Arrays and Fourier Optics (4 hours) 4) Diffraction from half-plane (2 hours) 5) Periodic structures and arrays (8 hours)	
Study Goals	The student will be able to 1. derive and use the spectral Green's function method for EM problems (diffraction, optics, evaluation of far-fields); 2. apply electromagnetic methods (physical optics, Fourier optics) for quasi-optical systems and analyze focal plane arrays; 3. apply the mathematical tools for describing the behavior of periodic arrays;	
Education Method	Course slides, support material, exercises and examples in Matlab, homework assignments (all available in brightspace).	
Assessment	Exam 33% from Matlab exercises during the course (one assignment per instruction) 33% from a 15min individual oral presentation (mandatory) 33% from oral examination (mandatory) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4730	High Frequency Wireless Architectures	3
Responsible Instructor	Dr.ir. M. Alonso Del Pino	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - Advanced Electromagnetics - RF Wireless Systems - Quasi-Optical Systems	
Course Contents	1) Time-domain systems with photo-conductive antennas 2) Line of Sight communication systems with directive antennas 3) Imaging systems at high frequencies	
Study Goals	The students will be able to - Understand time-domain sensing systems and photoconductive pulsed source modeling techniques - Analyze spectroscopic measurements obtained on a time-domain system - Understand high-frequency communication scenarios and their trade-offs - Apply quasi-optical system concepts on over-the-air high-frequency antenna measurements - Understand high-frequency imaging systems and their trade-offs - Apply quasi-optical system concepts on a real high-frequency imaging system	
Education Method	- Lectures through PowerPoint slides - Practical laboratory sessions - Analysis of the experimental results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	The final mark will be the average between: 50% from the reports of each laboratory session 50% from a final presentation and QA session In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4745	Superconducting Astronomical Instrumentation	5
Responsible Instructor	Dr. A. Endo	
Instructor	Prof.dr.ir. J.J.A. Baselmans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Successful completion of the following courses is recommended but **not required**: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System - AP3701 Submm and terahertz physics and applications	
Summary	Astronomical signal detection is one of the most extreme areas of electrical engineering, involving scales ranging from nanometers to gigaparsecs in length, microseconds to gigayears in time, and $>10^{40}$ W to zeptowatts in power. Indeed, the quest to detect the faintest signals from the most distant objects in our Universe involves a continuous push in what human can observe. Exposure to such radical problems should expand the scope and imagination of enthusiastic Delft students who do not wish to stay within the limitation of current technology. Detection of light, or electromagnetic waves, is one of the most fundamental methods of sensing, yet the correct understanding of the interaction of light and matter (the detector) requires careful analysis, especially when it comes to detecting signals that are so faint that the quantum nature of light become significant. Through the first part of the course, students will learn fundamental concepts for describing the process of the emission and detection of light from astronomical sources, for example basic astronomical emission processes including black-body radiation, photon bunching, Noise Equivalent Power and noise temperature. One of the earliest yet still most active areas of application of superconductors is astronomical detectors. Nowadays, the extreme sensitivity of superconducting circuits is also being actively utilized in the field of quantum computation. Indeed, one of the most promising architecture for a quantum computer involves the integration of large numbers of superconducting microwave resonators, which were originally inspired by microwave kinetic inductance detectors (taught in great depth in this course.) Students will learn the basics of superconductivity that is relevant for microwave circuits, and get an introduction to a commonly used EM simulator Sonnet lite, helping them also visualize the functionality of distributed element circuits and the effect of superconductivity. Finally, as practical applications we will take microwave kinetic inductance detectors as an example of an incoherent detector, and hot electron bolometer mixers as an example of a coherent detector. TU Delft, in close collaboration with SRON Netherlands Insitute for Space Reasearch, are world-leading in the technology of both of these detectors, hence the students can interact with the true state-of-the art of superconducting astronomical detectors with ample live examples.	
Course Contents	1) Introduction to Millimeter-Submillimeter astronomy 2) Superconducting detectors and instruments (mm-submm wave to visible light) 3) Observation techniques	
Study Goals	The students will be able to - Explain and perform calculations regarding the fundamental physics and sensitivity in detecting photons from astronomical sources - Analyze and design simple superconducting transmission line circuits - Explain the working principle and performance (limits) of incoherent and coherent superconducting detectors, especially hot electron bolometers and kinetic inductance detectors.	
Education Method	- Lectures through PowerPoint slide projection - Exercises on calculation problems - Exercises using a free EM solver Sonnet Lite	
Assessment	The final mark will be based on the homework assignments, and a final written exam.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE5020	Sensor Signal and Data Processing	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge obtained for instance via the course Detection and Estimation (ET4386) is usefull though not required.	
Course Contents	In this course a variety of modern signal and data processing techniques will be presented along with their application in sensor systems in general and surveillance (radar) systems in particular. Topics that will be presented include: detecting the presence and estimating the state of a single moving object based upon sensor measurements, modeling and estimation with multiple models, rejection of outliers in the measurements, the extension to a time-varying number of objects, the use of multiple sensors, and optimizing the resources of the sensor(s). Techniques that will be presented include: (extended) Kalman filtering, Interacting Multiple Model (IMM) filtering, non-linear filters such as the particle filter, Sequential Likelihood Ratio Testing (SLRT), Multiple Hypothesis Tracking (MHT), track-before-detect, data fusion concepts.	
Study Goals	The main goal of this course is to teach students to apply basic (and more advanced) detection and estimation theory to design algorithmic solutions to (complex) signal and data processing problems	
Education Method	Lectures, instruction lectures, home exercises	
Literature and Study Materials	Lecture notes, and for each topic specific background and overview literature will be provided.	
Assessment	Project work plus an oral exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and probability theory, as being treated in B.Sc. programmes like Electrical Engineering, Computer Science, and Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The general overall objectives of this course are that the students can - reproduce the error control concepts as presented in the lecture notes; - explain the meaning and usefulness of these concepts; - apply the concepts to transmission and storage systems, including basic calculations, at the level of the exercises in the lecture notes and previous exams. More specifically, the students are able to - discuss the basic trade-offs between efficiency, reliability, and complexity in designing coding schemes for transmission and storage systems; - apply finite fields theory in order to construct (cyclic) codes; - encode data into code sequences that are resistant to a certain number of random and/or burst errors; - decode received sequences using (iterative) hard or soft decision techniques.	
Education Method	Lectures; expected workload is 20 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 32 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber, available on Brightspace. Also previous exams as well as demos are available via Brightspace.	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Actual course information available on Brightspace.	

ET4034	Telecom Business Architectures and Models	4
Responsible Instructor	Dr. E.F.M. van Boven	
Responsible for assignments	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Summary	Edgar van Boven, (KPN, TUD EEMCS)-- Past, present & future of communications -- Assignment 1 Eric Smeitink (KPN, TUD EEMCS)--Strategy, Spectrum & 5G Ton van der Knaap, (Stedin)-- What, why and how of business architecture; experiences from a grid operator and a telecom operator compared Samuel Pronk, (Krauthammer)-- Real Options-- Assignment 2 Ramin Hekmat, (KPN)-- Security and privacy in contemporary communication networks-- Assignment 3 Wouter de Vries (KPN)-- Artificial Intelligent Processes for People Tim Daeleman, (Prodapt Consulting)and Erik Lemmens (KPN)-- Digital Telecom Platforms Jose Almodovar, (TNO)--Standardization of Mobile Communications Frank Mertz (KPN)--5G Networks Developments and Opportunities Remco Helwerda (KPN) -- Digitalization Marco van der Pal, (Cap Gemini)--The IoT landscape architecture Ruud van de Bovenkamp (KPN)-- Fiber-to-the-Home Access Networks -- Assignment 4 Mark Dirksen (KPN)-- Business Continuity Management Pieter Veenstra, (Titanium)-- Service Architecture and 5G Roaming Security Edgar van Boven (KPN, TUD EEMCS)-- Value Case pitch practicing -- Assignment 5	
Course Contents	The essence of the course The ET4034 course Telecom, Architectures and Business Models was designed in 2001 and scheduled since 2002. The aim of the course is giving students from any faculty of the Delft University of Technology the opportunity to learn about a large variety of aspects of the telecom domain, not only its technologies, but also the diverse and challenging daily practice of organisations active in the communications sector. The course is provided by a team of 15 lecturers interactively sharing their knowledge about communications infrastructure in the context of overarching societal trends, service & network architecture, long term developments and new technology evolving from standardization initiatives. As a consequence of continuous innovation and societal developments, the course content is being updated each year including a well-considered choice of central themes that enhance and connect the course lectures. In 2022, the themes Internet of Things, Artificial Intelligence and 5G were highlighted, to be updated in 2023.	
Course Contents Continuation	EE5010 Internships or ET4399 projects in telecom related industries.	
Study Goals	Overview and understanding the Communications Sector, Services, Telecom infrastructures and Value Cases serving society	
Education Method	Online lectures, one powerpoint pitch presentation given by student teams and four off-line written assignments in teams of 2 or 3 students	
Literature and Study Materials	See information on Brightspace.tudelft.nl	
Reader	Course lectures, background information and video recordings available on ET4034 Brightspace	
Prerequisites	Finalised BSc as ET4034 is meant for Master students. PhD students are also welcome and can obtain credits when successfully passing the course assignments via the Graduate School.	
Assessment	The examination of the course Together forming the basis for the examination, the ET4034 course contains five assignments of which four written documents and one opportunity to strengthen ones pitching skills presenting a self-developed Value Case to a live audience. Students are invited to work in couples or trio's on the five assignments. The four written assignments each have a weight factor 1 and the Value Case pitch presentation has a weight factor 0,25 in the end mark of the ET4034 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Not applicable, the course assignments are four documents written offline together with a course partner. The Value Case powerpoint presentation is a pitch that takes circa 7 minutes.	
Maximum number of participants	No limitation	
Self test	Not applicable	
maximum aantal deelnemers	No limitation	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4173	Introduction to UWB technology, systems and applications	4
Responsible Instructor	Prof.dr. O. Yarovy	
Instructor	O.A. Krasnov	
Responsible for assignments	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0,0,0,6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Ultra-wideband (UWB) technology is becoming of greater importance as the pressure for signal bandwidth increases. This course introduces students to UWB technology as it will be used in future communications, radar and navigation systems, and gives detailed attention to both UWB wireless systems and applications in the frequency (FD) and time (TD) domains. The course consists of three major parts. In Part 1 the fundamentals of UWB technology such as waveforms and spectra of short pulses, basic UWB signal processing, radiation and propagation of short electromagnetic pulses are discussed. In Part 2 of the course is dedicated to realizations of UWB technology and systems. General approach to UWB wireless system design is introduced. UWB system design implementation of the key facets of modern UWB systems in both the frequency (FD) and time (TD) domains is discussed. In Part 3 applications of UWB technology in telecommunications, radar, imaging and positioning are discussed. General discussion of the system-level trade offs of such UWB systems is complemented by a detailed analysis of practical systems designed and implemented at TU-Delft. Lectures in the course are complemented by laboratories, at which students get hands-on experience in UWB signal processing and get in touch with wireless applications of UWB systems.	
Study Goals	This course introduces students to UWB technology as it is used in telecommunications, radar and navigation systems, and gives detailed attention to UWB wireless systems. After finishing this course, students will understand UWB signal properties in time and frequency domains, impact of UWB signal properties on wireless system architecture and performance, become familiar with available UWB technologies and UWB wireless system designs, and understand advantages and limitations of UWB technology for different applications. Students will also get hands-on experience with UWB signals and systems.	
Education Method	The course consists of lectures, supervised laboratories with experimental UWB equipment and the final assignment. The course finishes with a symposium, at which the final assignments are presented.	
Literature and Study Materials	Lecture handouts; course books, research papers. M.-G. Di Benedetto, G. Giancola Understanding Ultra Wide Band Radio Fundamentals, Prentice Hall PTR, 2004, ISBN-0-13-148003-0; T. Zwick et al., Ultra-wideband RF system engineering, Cambridge University Press, 2013, ISBN 978-1-107-01555-5	
Assessment	The total mark is based on evaluations of laboratory assignments reports, final assignment essay and presentation at the closing symposium. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4175	Radar II: Theory and System Design	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/0/4 (Q5)	
Education Period	1	
Start Education	5	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4169 Microwaves, Radar and Remote Sensing. Specifically, knowledge is expected on: microwaves basics, radar, scattering mechanisms.	
Course Contents	Radar systems play an important role in today's world. Aviation could not exist without them. But also shipping, traffic and remote sensing extensively employ radar. In today's systems, signal processing is the central theme that enables all radar applications. However, thorough understanding of the physical and electrical engineering radar principles are essential to the successful radar engineer. "Radar Systems" provides an introduction to the design and operation of modern radar systems. Topics include radar principles and designs, the interaction of radar with the environment, i.e. clutter and detection of targets, subsequent signal processing and signal to noise topics. Furthermore waveform design driven by range and Doppler measurements, suppression of ambiguities are treated. Finally modern developments in active arrays and digital beamforming are discussed.	
Study Goals	- The aim of this course is to advance students' knowledge in microwave radar systems. The course will build on the introductory course Microwaves, Radar and Remote Sensing. - The participants will gain sufficient knowledge to understand design, operation and signal processing of modern radar systems and architectures, taking into account radar clutter and detection theory.	
Education Method	The course consists of 10 regular lectures and a 1/2 day excursion to TNO, The Hague or a guest lecture.	
Literature and Study Materials	Kingsley and Quegan: Understanding Radar Systems; Scitech, ISBN 1-891121-05-7 "Principles of Modern Radar: Basic Principles", Mark A. Richards (Editor), James A. Scheer (Editor), William A. Holm (Editor); ISBN-13: 978-1891121524 additional handouts/ reader	
Assessment	Oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	By appointment with Esther de Klerk (E.J.DeKlerk@tudelft.nl)	

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the fundamental problems that distributed systems face Understand the most important distributed algorithms that solve these problems Reason about the execution of distributed algorithms Program distributed algorithms Select and summarize relevant literature on distributed algorithms	
Education Method	Lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 75%). In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams. Paper summary (weighting 25%) Final grade calculation: $0.75 * \text{Written exam} + 0.25 * \text{Paper summary}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper summary: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None except the list of algorithms	
Special Information	Email: in4150-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Remarks	Lab work is 40 hrs.	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	J.A. Martinez Castaneda	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications.	
Education Method	Through assignments, students are expected to have the chance to work on the topics covered in the lectures. Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	-Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: $0.7 \cdot \text{Written exam} + 0.3 \cdot \text{Mandatory assignments}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable	

Exam Hours	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
	In case of in person examination at campus: The exam is closed book.
	A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side.
	In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

SC42101	Networked and Distributed Control Systems	4
Responsible Instructor	T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures.	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four assessments, i.e., written assignment reports, each representing 25% of the final grade. 2. For some assignments part of the grading is done via Peer Review (and in addition performing the peer evaluation will contribute to the assignment grade). 3. In order to complete the course, each assessment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Department	3mE Department Delft Center for Systems and Control	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested profiles WiCoS 2023	
Introduction 1	The tracks in EE typically have an informal sub-structure called profiles. A profile is a more or less canned study path. It typically prepares for a thesis subject in a certain area and typically with a specific professor or small set of professors of one section, overseeing that research area. For example, the Microelectronics track offers the Digital Systems profile, the RF/Analog Circuit Design profile, the Microsystems & Technology profile and the Biomedical Electronics profile. Other tracks offer different profiles. For the purposes of composing an IEP, a profile presents a short list of coherent courses (in all tiers) that together leads to a coherent program of study that prepare for a thesis topic in the area of the profile. As such, a profile can specify (at most) two from three courses in the common core, two from three courses in the track core, and at most 12 EC of additional specialization courses. Note, however, that profiles are nothing more than convenient selections of courses that go well together, and do not imply any formal restriction whatsoever. Based on individual interest and subject to approval of the IEP by the (tentative) thesis supervisor and the MSc coordinator and the responsible professor, a much wider variety of r programs is possible. Students are free to compose their individual program within the rules as specified in FIXME.

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Applied Electromagnetics 2023

EE4510	Advanced Electromagnetics	5
Responsible Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	1) Ohmic losses at high frequencies 2) Potentials and Green's functions 3) Image theorem 4) Fields radiated by sources in the far field region 5) Equivalence and reciprocity theorems 6) Antennas in transmission and reception 7) Green's function of infinite line source 3) Application case: quasi-analytic analysis of transmission lines at high frequencies	
Study Goals	1) EM analysis of wireless links Basic characterization/understanding of a wireless link between a transmitting and receiving antenna 2) EM analysis of transmission lines Basic understanding of the impact of radiation and ohmic losses at high frequencies	
Education Method	The course consists on 1) Theoretical lectures on the basic physical mechanisms and how to analyse them via EM analytical techniques 2) Practical Instructions to get familiar with such techniques 3) Study of an application case	
Assessment	The assessment of the course will be the following. The final mark will be the average between the home assignments, written exam and an oral exam (depending on the number of students). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4565	Propagation and Scattering of EM waves	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Instructor	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	ET4169,	
Expected prior knowledge	Basic knowledge of Electromagnetics, EE4C05 Electromagnetics	
Course Contents	The Propagation and Scattering of electromagnetic (EM) waves course provides an overview of the essential aspects of the physical layer for a wide range of applications that rely on electromagnetic waves for the information transmission and sensing (wireless communications, radar, and navigation). Both the propagation medium and environment change the characteristics of the electromagnetic waves and affects the quality of the wireless link. The course focuses on physical models of propagation and scattering phenomena with emphasis on practical estimations of these effects influence on operation and design of telecommunication systems. The following aspects will be covered in the course: * Overview of the radio spectrum, services, and frequency allocation, propagation/scattering environment, frequency band designation for EM waves * Characterization of propagation media * Antenna and noise concepts * EM waves interaction with material objects * EM wave scattering on an ensemble of discrete particles * EM wave scattering on a rough surface * EM wave scattering on a complex and distributed objects, the concept of radar cross-section as a characteristic observed with radar objects, * Direct transmission, * Power balance for communication and radar systems in cases of direct transmission, atmospheric gas attenuation, signal attenuation, and scattering in rain * Refraction - wave propagation through a stratified medium * Propagation near the Earth's surface * Diffraction on terrain, Fresnel zones, propagation over a plane/spherical Earth, practical propagation path analysis * Empirical Path Loss Models * Models of fading effects * Ionospheric propagation/scattering of electromagnetic waves Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	After this course students have to be able * to recognize and explain different mechanisms of electromagnetic waves (EMW) propagation/scattering and relations between them; * to assess different mechanisms of EMW propagation/scattering in relation to characteristics of EMW and media; * to estimate quantitatively the influence of the propagation and scattering mechanisms on trade-offs that have to be made when designing a wireless communication or sensing system. * to analyze real-world scenarios and identify the involved propagation and scattering mechanisms and their relevant aspects;	
Education Method	Lectures, instruction lectures, self-study recommended references, home exercises, practicum	
Literature and Study Materials	Levis, C.A. Radiowave propagation: physics and applications, Wiley, 2010 - 301 pages. ISBN-13: 978-0470542958 Barclay, L.W. Propagation of Radiowaves, IET, 1996, 2003. - 352 pages. ISBN-13: 978-0852961025	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: 10% of averaged grades for homeworks (max 1 point); 20% of averaged grades for practicum reports (max 2 points); 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Elektricity & Magnetism Information & Communication Practicals Signals and Systems Technology Telecommunication	

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

EE4620	Spectral Domain Methods in Electromagnetics	4
Responsible Instructor	Dr. D. Cavallo	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System	
Course Contents	1) Green's functions of stratified media (6 hours) 2) Dominant spectral contributions and far-field (4 hours) 2) Surface waves and leaky waves (6 hours) 3) Artificial dielectric materials (4 hours) 4) Spectral representations for planar Tx-lines (4 hours) 5) Connected arrays (10 hours)	
Study Goals	The students will be able to - apply spectral domain methods for the analysis of a number of electromagnetic problems (planar transmission lines, artificial dielectrics, stratified media problems) - create functional MATLAB scripts that implement the studied methods - present and critically analyze the results obtained with the MATLAB codes	
Education Method	- Lectures through PowerPoint slide projection - Practical MATLAB sessions with implementation of the analysis methods - Analysis of the results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	Home assignments (50% weight on the final grade) Final project presentation (50% weight on the final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4725	Quasi Optical Systems	5
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. D. Cavallo	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is required knowledge of - Electromagnetics - Transmission lines - Antennas	
Course Contents	The exam 'AP3701 Sub-mm and terahertz physics and applications' is recommended. This course includes the 5 topics listed hereafter: 1) Spectral domain Green's function and Far Field evaluation (4 hours) 2) Physical Optics for reflector and lens antennas (8 hours) 3) Focal Plane Arrays and Fourier Optics (4 hours) 4) Diffraction from half-plane (2 hours) 5) Periodic structures and arrays (8 hours)	
Study Goals	The student will be able to 1. derive and use the spectral Green's function method for EM problems (diffraction, optics, evaluation of far-fields); 2. apply electromagnetic methods (physical optics, Fourier optics) for quasi-optical systems and analyze focal plane arrays; 3. apply the mathematical tools for describing the behavior of periodic arrays;	
Education Method	Course slides, support material, exercises and examples in Matlab, homework assignments (all available in brightspace).	
Assessment	Exam 33% from Matlab exercises during the course (one assignment per instruction) 33% from a 15min individual oral presentation (mandatory) 33% from oral examination (mandatory) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4730	High Frequency Wireless Architectures	3
Responsible Instructor	Dr.ir. M. Alonso Del Pino	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - Advanced Electromagnetics - RF Wireless Systems - Quasi-Optical Systems	
Course Contents	1) Time-domain systems with photo-conductive antennas 2) Line of Sight communication systems with directive antennas 3) Imaging systems at high frequencies	
Study Goals	The students will be able to - Understand time-domain sensing systems and photoconductive pulsed source modeling techniques - Analyze spectroscopic measurements obtained on a time-domain system - Understand high-frequency communication scenarios and their trade-offs - Apply quasi-optical system concepts on over-the-air high-frequency antenna measurements - Understand high-frequency imaging systems and their trade-offs - Apply quasi-optical system concepts on a real high-frequency imaging system	
Education Method	- Lectures through PowerPoint slides - Practical laboratory sessions - Analysis of the experimental results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	The final mark will be the average between: 50% from the reports of each laboratory session 50% from a final presentation and QA session In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4745	Superconducting Astronomical Instrumentation	5
Responsible Instructor	Dr. A. Endo	
Instructor	Prof.dr.ir. J.J.A. Baselmans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Successful completion of the following courses is recommended but **not required**: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System - AP3701 Submm and terahertz physics and applications	
Summary	Astronomical signal detection is one of the most extreme areas of electrical engineering, involving scales ranging from nanometers to gigaparsecs in length, microseconds to gigayears in time, and $>10^{40}$ W to zeptowatts in power. Indeed, the quest to detect the faintest signals from the most distant objects in our Universe involves a continuous push in what human can observe. Exposure to such radical problems should expand the scope and imagination of enthusiastic Delft students who do not wish to stay within the limitation of current technology. Detection of light, or electromagnetic waves, is one of the most fundamental methods of sensing, yet the correct understanding of the interaction of light and matter (the detector) requires careful analysis, especially when it comes to detecting signals that are so faint that the quantum nature of light become significant. Through the first part of the course, students will learn fundamental concepts for describing the process of the emission and detection of light from astronomical sources, for example basic astronomical emission processes including black-body radiation, photon bunching, Noise Equivalent Power and noise temperature. One of the earliest yet still most active areas of application of superconductors is astronomical detectors. Nowadays, the extreme sensitivity of superconducting circuits is also being actively utilized in the field of quantum computation. Indeed, one of the most promising architecture for a quantum computer involves the integration of large numbers of superconducting microwave resonators, which were originally inspired by microwave kinetic inductance detectors (taught in great depth in this course.) Students will learn the basics of superconductivity that is relevant for microwave circuits, and get an introduction to a commonly used EM simulator Sonnet lite, helping them also visualize the functionality of distributed element circuits and the effect of superconductivity. Finally, as practical applications we will take microwave kinetic inductance detectors as an example of an incoherent detector, and hot electron bolometer mixers as an example of a coherent detector. TU Delft, in close collaboration with SRON Netherlands Institute for Space Research, are world-leading in the technology of both of these detectors, hence the students can interact with the true state-of-the art of superconducting astronomical detectors with ample live examples.	
Course Contents	1) Introduction to Millimeter-Submillimeter astronomy 2) Superconducting detectors and instruments (mm-submm wave to visible light) 3) Observation techniques	
Study Goals	The students will be able to - Explain and perform calculations regarding the fundamental physics and sensitivity in detecting photons from astronomical sources - Analyze and design simple superconducting transmission line circuits - Explain the working principle and performance (limits) of incoherent and coherent superconducting detectors, especially hot electron bolometers and kinetic inductance detectors.	
Education Method	- Lectures through PowerPoint slide projection - Exercises on calculation problems - Exercises using a free EM solver Sonnet Lite	
Assessment	The final mark will be based on the homework assignments, and a final written exam.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovyj	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Astronomical Instrumentation 2023

AP3701	Submm and terahertz physics and applications	3
Responsible Instructor	Dr. A.J.L. Adam	
Responsible Instructor	Dr. J.R. Gao	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	The course is intended for MSc students from different disciplines, including those from Applied Physics, Electrical Engineering, and Space Engineering, but it is also open to interested PhD students.	
Expected prior knowledge	Electromagnetism, Any introduction of Optics, and Introduction of solid-state Physics. If you miss some of them, we will also add necessary background knowledge during the course	
Course Contents	The submm and terahertz region of the electromagnetic spectrum, which spans from 100 GHz to 10 THz, has attracted a lot of attention in recent years due to the developments in new detection and generation techniques and due to the applications for astronomic observations in space and the potential applications as an imaging technique for security, communications and biomedical applications. The Submm&THz frequency range is rapidly emerging which allows unearthing new science and technology. We wish to provide a course in which students can get an introduction and learn the basic physics & principle, and will also be updated with the latest developments. The course will focus on the topics like Photonics, Time-Domain Spectroscopy, Antenna/Waveguide, terahertz detectors and sources, THz Space application. Here the titles of all the different lectures given: 1. Introduction to THz. Light Interaction with Matter 2. THz photonics (THz fibers, Photonic crystals & Metamaterials. High THz fields. Nonlinear interaction of THz waves with matter) 3. THz Imaging (THz Time Domain spectroscopy, Transmission, Reflection, Time of Flight Tomography, Near field, Tomography) 4. THz Sensing (Application to Chemistry, Security, Biology & Medicine; Pump-probe experiment) 5. Submm & THz Coupling (Waveguide, Si Lens, Antennas) 6. Sources 7. Nano-detectors and cameras 8. Submm & THz Sensing instruments from space (as a few examples) Additionally: A case study and design of an space instrument by a subgroup of students (Adam/Gao) (To be announced)	
Study Goals	Understand submm & terahertz electromagnetic radiation and propagation fundamentals, and the basics of typical terahertz detectors and sources. Distinguish different terahertz antennas with regard to their advantages and disadvantages for terahertz radiations. Explain the working principle of THz Time-Domain Spectroscopy. Build practical experience on a THz imaging setup and a heterodyne receiver system. Highlight recent advances of THz research and development from the academic and astronomic sectors. Summarize state-of-the-art THz applications, including space applications.	
Education Method	The course will be given as following: 1) one 2x45 min lecture per week 2) Home work every week on the topics following the lecture. The homework counts 30% of the grade. 3) Design an instrument concept and address a specific topics with groups and present 4) a written exam on all topics discussed during the lectures and homework We based our lectures on current research and science on THz technologies	
Literature and Study Materials	The lectures will be based on a few selected chapters of different books. For Gao's lectures: Chapter 5, 6 and 7 in the book of "Terahertz Astronomy" by Christopher K. Walker. For Aurele's lectures: Terahertz Technology, Volume 77, A. Rostami, H. Rasooli, and H. Baghban, An Overview of the Technological and Scientific Achievements of the Terahertz, pp. 189. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011. (http://link.springer.com/chapter/10.1007/978-3-642-15793-6_1) After each lecture, the teachers will provide also the powerpoint slides used during the lectures.	
Assessment	20% from the homework 30 % from case/project study report/presentation; 50 % from the written exam	
Permitted Materials during Tests		
Elective	Yes	
Tags	Building & Spatial Development Physics Projects Technology	

EE4510	Advanced Electromagnetics	5
Responsible Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	1) Ohmic losses at high frequencies 2) Potentials and Green's functions 3) Image theorem 4) Fields radiated by sources in the far field region 5) Equivalence and reciprocity theorems 6) Antennas in transmission and reception 7) Green's function of infinite line source 3) Application case: quasi-analytic analysis of transmission lines at high frequencies	
Study Goals	1) EM analysis of wireless links Basic characterization/understanding of a wireless link between a transmitting and receiving antenna 2) EM analysis of transmission lines Basic understanding of the impact of radiation and ohmic losses at high frequencies	
Education Method	The course consists on 1) Theoretical lectures on the basic physical mechanisms and how to analyse them via EM analytical techniques 2) Practical Instructions to get familiar with such techniques 3) Study of an application case	
Assessment	The assessment of the course will be the following. The final mark will be the average between the home assignments, written exam and an oral exam (depending on the number of students). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4565	Propagation and Scattering of EM waves	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Instructor	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	ET4169,	
Expected prior knowledge	Basic knowledge of Electromagnetics, EE4C05 Electromagnetics	
Course Contents	The Propagation and Scattering of electromagnetic (EM) waves course provides an overview of the essential aspects of the physical layer for a wide range of applications that rely on electromagnetic waves for the information transmission and sensing (wireless communications, radar, and navigation). Both the propagation medium and environment change the characteristics of the electromagnetic waves and affects the quality of the wireless link. The course focuses on physical models of propagation and scattering phenomena with emphasis on practical estimations of these effects influence on operation and design of telecommunication systems. The following aspects will be covered in the course: * Overview of the radio spectrum, services, and frequency allocation, propagation/scattering environment, frequency band designation for EM waves * Characterization of propagation media * Antenna and noise concepts * EM waves interaction with material objects * EM wave scattering on an ensemble of discrete particles * EM wave scattering on a rough surface * EM wave scattering on a complex and distributed objects, the concept of radar cross-section as a characteristic observed with radar objects, * Direct transmission, * Power balance for communication and radar systems in cases of direct transmission, atmospheric gas attenuation, signal attenuation, and scattering in rain * Refraction - wave propagation through a stratified medium * Propagation near the Earth's surface * Diffraction on terrain, Fresnel zones, propagation over a plane/spherical Earth, practical propagation path analysis * Empirical Path Loss Models * Models of fading effects * Ionospheric propagation/scattering of electromagnetic waves Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	After this course students have to be able * to recognize and explain different mechanisms of electromagnetic waves (EMW) propagation/scattering and relations between them; * to assess different mechanisms of EMW propagation/scattering in relation to characteristics of EMW and media; * to estimate quantitatively the influence of the propagation and scattering mechanisms on trade-offs that have to be made when designing a wireless communication or sensing system. * to analyze real-world scenarios and identify the involved propagation and scattering mechanisms and their relevant aspects;	
Education Method	Lectures, instruction lectures, self-study recommended references, home exercises, practicum	
Literature and Study Materials	Levis, C.A. Radiowave propagation: physics and applications, Wiley, 2010 - 301 pages. ISBN-13: 978-0470542958 Barclay, L.W. Propagation of Radiowaves, IET, 1996, 2003. - 352 pages. ISBN-13: 978-0852961025	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: 10% of averaged grades for homeworks (max 1 point); 20% of averaged grades for practicum reports (max 2 points); 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Elektricity & Magnetism Information & Communication Practicals Signals and Systems Technology Telecommunication	

EE4620	Spectral Domain Methods in Electromagnetics	4
Responsible Instructor	Dr. D. Cavallo	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System	
Course Contents	1) Green's functions of stratified media (6 hours) 2) Dominant spectral contributions and far-field (4 hours) 2) Surface waves and leaky waves (6 hours) 3) Artificial dielectric materials (4 hours) 4) Spectral representations for planar Tx-lines (4 hours) 5) Connected arrays (10 hours)	
Study Goals	The students will be able to - apply spectral domain methods for the analysis of a number of electromagnetic problems (planar transmission lines, artificial dielectrics, stratified media problems) - create functional MATLAB scripts that implement the studied methods - present and critically analyze the results obtained with the MATLAB codes	
Education Method	- Lectures through PowerPoint slide projection - Practical MATLAB sessions with implementation of the analysis methods - Analysis of the results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	Home assignments (50% weight on the final grade) Final project presentation (50% weight on the final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4725	Quasi Optical Systems	5
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. D. Cavallo	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is required knowledge of - Electromagnetics - Transmission lines - Antennas	
Course Contents	The exam 'AP3701 Sub-mm and terahertz physics and applications' is recommended. This course includes the 5 topics listed hereafter: 1) Spectral domain Green's function and Far Field evaluation (4 hours) 2) Physical Optics for reflector and lens antennas (8 hours) 3) Focal Plane Arrays and Fourier Optics (4 hours) 4) Diffraction from half-plane (2 hours) 5) Periodic structures and arrays (8 hours)	
Study Goals	The student will be able to 1. derive and use the spectral Green's function method for EM problems (diffraction, optics, evaluation of far-fields); 2. apply electromagnetic methods (physical optics, Fourier optics) for quasi-optical systems and analyze focal plane arrays; 3. apply the mathematical tools for describing the behavior of periodic arrays;	
Education Method	Course slides, support material, exercises and examples in Matlab, homework assignments (all available in brightspace).	
Assessment	Exam 33% from Matlab exercises during the course (one assignment per instruction) 33% from a 15min individual oral presentation (mandatory) 33% from oral examination (mandatory) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4730	High Frequency Wireless Architectures	3
Responsible Instructor	Dr.ir. M. Alonso Del Pino	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - Advanced Electromagnetics - RF Wireless Systems - Quasi-Optical Systems	
Course Contents	1) Time-domain systems with photo-conductive antennas 2) Line of Sight communication systems with directive antennas 3) Imaging systems at high frequencies	
Study Goals	The students will be able to - Understand time-domain sensing systems and photoconductive pulsed source modeling techniques - Analyze spectroscopic measurements obtained on a time-domain system - Understand high-frequency communication scenarios and their trade-offs - Apply quasi-optical system concepts on over-the-air high-frequency antenna measurements - Understand high-frequency imaging systems and their trade-offs - Apply quasi-optical system concepts on a real high-frequency imaging system	
Education Method	- Lectures through PowerPoint slides - Practical laboratory sessions - Analysis of the experimental results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	The final mark will be the average between: 50% from the reports of each laboratory session 50% from a final presentation and QA session In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4745	Superconducting Astronomical Instrumentation	5
Responsible Instructor	Dr. A. Endo	
Instructor	Prof.dr.ir. J.J.A. Baselmans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Successful completion of the following courses is recommended but **not required**: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System - AP3701 Submm and terahertz physics and applications	
Summary	Astronomical signal detection is one of the most extreme areas of electrical engineering, involving scales ranging from nanometers to gigaparsecs in length, microseconds to gigayears in time, and $>10^{40}$ W to zeptowatts in power. Indeed, the quest to detect the faintest signals from the most distant objects in our Universe involves a continuous push in what human can observe. Exposure to such radical problems should expand the scope and imagination of enthusiastic Delft students who do not wish to stay within the limitation of current technology. Detection of light, or electromagnetic waves, is one of the most fundamental methods of sensing, yet the correct understanding of the interaction of light and matter (the detector) requires careful analysis, especially when it comes to detecting signals that are so faint that the quantum nature of light become significant. Through the first part of the course, students will learn fundamental concepts for describing the process of the emission and detection of light from astronomical sources, for example basic astronomical emission processes including black-body radiation, photon bunching, Noise Equivalent Power and noise temperature. One of the earliest yet still most active areas of application of superconductors is astronomical detectors. Nowadays, the extreme sensitivity of superconducting circuits is also being actively utilized in the field of quantum computation. Indeed, one of the most promising architecture for a quantum computer involves the integration of large numbers of superconducting microwave resonators, which were originally inspired by microwave kinetic inductance detectors (taught in great depth in this course.) Students will learn the basics of superconductivity that is relevant for microwave circuits, and get an introduction to a commonly used EM simulator Sonnet lite, helping them also visualize the functionality of distributed element circuits and the effect of superconductivity. Finally, as practical applications we will take microwave kinetic inductance detectors as an example of an incoherent detector, and hot electron bolometer mixers as an example of a coherent detector. TU Delft, in close collaboration with SRON Netherlands Insitute for Space Reasearch, are world-leading in the technology of both of these detectors, hence the students can interact with the true state-of-the art of superconducting astronomical detectors with ample live examples.	
Course Contents	1) Introduction to Millimeter-Submillimeter astronomy 2) Superconducting detectors and instruments (mm-submm wave to visible light) 3) Observation techniques	
Study Goals	The students will be able to - Explain and perform calculations regarding the fundamental physics and sensitivity in detecting photons from astronomical sources - Analyze and design simple superconducting transmission line circuits - Explain the working principle and performance (limits) of incoherent and coherent superconducting detectors, especially hot electron bolometers and kinetic inductance detectors.	
Education Method	- Lectures through PowerPoint slide projection - Exercises on calculation problems - Exercises using a free EM solver Sonnet Lite	
Assessment	The final mark will be based on the homework assignments, and a final written exam.	
In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.		
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4289	Integrated Circuits and MEMS Technology	4
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Instructor	Dr. C.M.F. Viellard-Boutry	
Instructor	Dr.ing. H.W. van Zeijl	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics (EE 4585)	
Course Contents	This lecture is an introduction in the fabrication technologies used for Integrated Circuits (ICs) and Micro Electro-Mechanical Systems (MEMS). The aim of this lecture is to build a bridge between the physical world (the device physics) and the electronic world (the device characteristics). In this way, students gain more insight in the critical aspects and challenges in IC fabrication. The series of lectures starts with the discussion of basic microelectronic devices with a brief review of the physics involved. The associated basic fabrication technology, required to fabricate these devices is also discussed. After that, the mainstream IC fabrication technology (CMOS) will be discussed in more detail. In separate chapters, silicon crystal growth, epitaxy, ion implantation, chemical vapor deposition, etching and photolithography will be introduced. Their physical motivation and the impact on the device characteristics will be discussed. Finally, silicon bulk and surface micromachining technologies are introduced to illustrate the potential of 3D micro structuring in the development of Microsystems or MEMS. These IC compatible technologies are employed to realize multi-function systems for many scientific and industrial application areas.	
Study Goals	You know the definition of the basic process steps to fabricate a device (IC or MEMS) You can explain the basic physics and chemistry of these fabrication steps in an IC /MEMS process flow You can analyse the relationship between the series of process steps necessary to build the device and the device performance. You can evaluate how device requirements are translated into a process flow You are able to propose solutions to specific device technological questions	
Education Method	lectures	
Literature and Study Materials	R.C. Jager introduction to Microelectronic Fabrication & course slides	
Assessment	written exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. S. Vollebregt	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Microwave Sensing and Radar 2023	
Introduction 1	The profile Microwave Sensing focuses on education on physical and signal processing foundations of remote sensing exploiting electromagnetic waves in a number of societal important applications. It covers propagation and scattering of electromagnetic waves by natural and man-made objects, principles & designs of electromagnetic sensors for near-, intermediate- and far-field sensing, as well as the physical model based algorithms and methods that enable the detection and parameter estimation of objects from measured electromagnetic fields. Special attention is given to such areas as array antenna systems and radar front-ends (beamforming in conventional phased arrays and distributed MIMO systems, imaging), waveform agility (adaptive selection of the most efficient sensing waveforms such as e.g. LFM, OFDM, UWB and space-time coded sequences), signal and data processing algorithms for advanced object detection, classification and tracking. The application areas considered in this profile range from (phased array) Radar and distributed Radar systems in far-field to microwave vision in intermediate- and near-fields including medical imaging (High-field MRI). Responsible professor: Prof. A. Yarovoy
Introduction 2	Compulsory EE4C05 Electromagnetics EE4565 Propagation and Scattering of EM waves ET4169 Microwaves, Radar & Remote Sensing ET4175 Radar Systems Suggested ET4386 Estimation and Detection EE4600 RF / Wireless Concepts and Systems EE4510 Advanced Electromagnetics EE4016 Antenna Systems EE5020 Sensor Signal and Data Processing ET4173 Introduction to UWB technology, systems and applications EE4620 Spectral Domain Methods in Electromagnetics EE4595 Wavefield Imaging ET4288 Applied Electromagnetic Analysis in Wireless, Microwave and Radar Engineering

EE4016	Antenna Systems	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	Y. Aslan	
Instructor	Ir. P.J. Aubry	
Co-responsible for assignments	Ir. P.J.F. Swart	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4288, ET4595	
Expected prior knowledge	EE4C05	
Course Contents	The aim of this course is to introduce the students to the Antenna Techniques and Antenna Systems that are instrumental for applications such as telecommunications, remote sensing and radar systems. After introducing basic antenna concepts the basic antenna parameters are defined. A detailed description of the antenna gain, directivity, polarization, radiation pattern, bandwidth, beamwidth, efficiency, temperature, effective length, effective area and input impedance is given. The reflection and radiation from a dipole are detailed and analysed. Microstrip antenna is introduced and its performance is analysed. The radiation from linear, circular and planar array antennas is discussed in detail while stressing their importance in radar and remote sensing systems. The beamforming in arrays and antenna array synthesis are studied. Characterisation of Antenna System performance is discussed and derivation of Antenna System specifications from overall wireless system performance requirements is studied. Some basic concepts about antenna measurements techniques are introduced. The theoretical studies are complimented by computer simulations and measurements.	
Study Goals	By the end of this course, you should be able to: LO1: Determine (theoretically and experimentally) the fundamental antenna system performance metrics (such as directivity, gain, bandwidth etc.). LO2: Design basic antenna element/array structures (i.e. linear wire, microstrip, aperture) for the given criteria, by using CST and MATLAB tools. LO3: Assess the impact of a given antenna on the wireless system performance (in terms of the signal and noise power). LO4: Develop a suitable antenna model for a given application with specified requirements (e.g. communication, broadcasting, radar).	
Education Method	Lectures, Instructions, computer simulations, antenna measurements	
Literature and Study Materials	Course books (Balanis and Orfanidis), Lectures handouts, simulation and measurement assignments.	
Assessment	Written exam (problems solving) plus home works, simulation and measurement assignments In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4510	Advanced Electromagnetics	5
Responsible Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	1) Ohmic losses at high frequencies 2) Potentials and Green's functions 3) Image theorem 4) Fields radiated by sources in the far field region 5) Equivalence and reciprocity theorems 6) Antennas in transmission and reception 7) Green's function of infinite line source 3) Application case: quasi-analytic analysis of transmission lines at high frequencies	
Study Goals	1) EM analysis of wireless links Basic characterization/understanding of a wireless link between a transmitting and receiving antenna 2) EM analysis of transmission lines Basic understanding of the impact of radiation and ohmic losses at high frequencies	
Education Method	The course consists on 1) Theoretical lectures on the basic physical mechanisms and how to analyse them via EM analytical techniques 2) Practical Instructions to get familiar with such techniques 3) Study of an application case	
Assessment	The assessment of the course will be the following. The final mark will be the average between the home assignments, written exam and an oral exam (depending on the number of students). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4565	Propagation and Scattering of EM waves	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Instructor	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	ET4169,	
Expected prior knowledge	Basic knowledge of Electromagnetics, EE4C05 Electromagnetics	
Course Contents	The Propagation and Scattering of electromagnetic (EM) waves course provides an overview of the essential aspects of the physical layer for a wide range of applications that rely on electromagnetic waves for the information transmission and sensing (wireless communications, radar, and navigation). Both the propagation medium and environment change the characteristics of the electromagnetic waves and affects the quality of the wireless link. The course focuses on physical models of propagation and scattering phenomena with emphasis on practical estimations of these effects influence on operation and design of telecommunication systems. The following aspects will be covered in the course: * Overview of the radio spectrum, services, and frequency allocation, propagation/scattering environment, frequency band designation for EM waves * Characterization of propagation media * Antenna and noise concepts * EM waves interaction with material objects * EM wave scattering on an ensemble of discrete particles * EM wave scattering on a rough surface * EM wave scattering on a complex and distributed objects, the concept of radar cross-section as a characteristic observed with radar objects, * Direct transmission, * Power balance for communication and radar systems in cases of direct transmission, atmospheric gas attenuation, signal attenuation, and scattering in rain * Refraction - wave propagation through a stratified medium * Propagation near the Earth's surface * Diffraction on terrain, Fresnel zones, propagation over a plane/spherical Earth, practical propagation path analysis * Empirical Path Loss Models * Models of fading effects * Ionospheric propagation/scattering of electromagnetic waves Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	After this course students have to be able * to recognize and explain different mechanisms of electromagnetic waves (EMW) propagation/scattering and relations between them; * to assess different mechanisms of EMW propagation/scattering in relation to characteristics of EMW and media; * to estimate quantitatively the influence of the propagation and scattering mechanisms on trade-offs that have to be made when designing a wireless communication or sensing system. * to analyze real-world scenarios and identify the involved propagation and scattering mechanisms and their relevant aspects;	
Education Method	Lectures, instruction lectures, self-study recommended references, home exercises, practicum	
Literature and Study Materials	Levis, C.A. Radiowave propagation: physics and applications, Wiley, 2010 - 301 pages. ISBN-13: 978-0470542958 Barclay, L.W. Propagation of Radiowaves, IET, 1996, 2003. - 352 pages. ISBN-13: 978-0852961025	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: 10% of averaged grades for homeworks (max 1 point); 20% of averaged grades for practicum reports (max 2 points); 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Elektricity & Magnetism Information & Communication Practicals Signals and Systems Technology Telecommunication	

EE4620	Spectral Domain Methods in Electromagnetics	4
Responsible Instructor	Dr. D. Cavallo	
Instructor	Prof.dr.ing. A. Neto	
Instructor	Dr. N. Llombart Juan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The successful completion of the following courses is required: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System	
Course Contents	1) Green's functions of stratified media (6 hours) 2) Dominant spectral contributions and far-field (4 hours) 2) Surface waves and leaky waves (6 hours) 3) Artificial dielectric materials (4 hours) 4) Spectral representations for planar Tx-lines (4 hours) 5) Connected arrays (10 hours)	
Study Goals	The students will be able to - apply spectral domain methods for the analysis of a number of electromagnetic problems (planar transmission lines, artificial dielectrics, stratified media problems) - create functional MATLAB scripts that implement the studied methods - present and critically analyze the results obtained with the MATLAB codes	
Education Method	- Lectures through PowerPoint slide projection - Practical MATLAB sessions with implementation of the analysis methods - Analysis of the results	
Literature and Study Materials	Mandatory: course slides/handouts Optional: recommended IEEE Transaction articles for consultation and more in-depth study	
Assessment	Home assignments (50% weight on the final grade) Final project presentation (50% weight on the final grade) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4675	Object classification with radar	4
Responsible Instructor	Dr. F. Fioranelli	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Advised: ET4169, specifically knowledge of microwave and radar basics such as measurements of range and velocity and related basic signal processing.	
Course Contents	While the idea of gaining information about the surrounding environment using electromagnetic waves is a relatively mature concept (a radar essentially!), a tremendous amount of research activities have recently coupled the analysis of data from EM sensors with techniques that can teach them to recognise automatically objects of interest. What natural or manmade objects are moving around, and what are they doing? Can we infer something about their behaviour or intention from such movements? These are an example of the big questions which data and clever signal processing from EM sensors can address. Applications range from the analysis of human gait and movements to the discrimination of birds vs Unmanned Aerial Vehicles (drones), to the identification of the subtle gestures people produce to interact with smart objects, amongst others. All these applications have in common two main areas of expertise: one has to do with understanding and processing the information encoded into the sensors signals (radar signal processing and data format); one has to do with the techniques to interpret this information automatically to make decisions (we could call this applied machine learning). In this course, we aim to provide a comprehensive introduction and explanation of both areas from an engineering application perspective, with the objective to learn how to use these tools for real-life applications in radar engineering. The course is broadly divided into two parts: 1. Part 1 will build on fundamentals of radar & microwave sensing courses to investigate the concept of micro-Doppler signatures, focusing on its modelling, the signal processing for its characterisation, and its relationships with the physical movement of objects and therefore their identity. Can we understand from the EM sensors data how the different parts of human bodies or manmade objects move? 2. Part 2 will introduce machine learning techniques for extraction of information from sensors data (feature extraction), for their selection, and for classification of objects of interest. Conventional supervised learning techniques will be presented (for example Nearest Neighbours and Support Vector Machine classifiers), along with some of the more recent data-driven approaches based on neural networks. The diversity of formats of the radar data for classification will also be considered, including micro-Doppler signatures, High Resolution Range Profiles, radar cubes. In both parts, students will be exposed to concrete examples and data directly taken from current activities at TU Delft, so that the research element is well embedded in the teaching.	
Study Goals	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The aim of this course is to introduce students to techniques to classify and identify objects of interest mixing machine learning and signal processing applied to data from radar sensors. By the end of this course, students will be able to: -- Apply relevant signal processing techniques to EM sensors and radar data to extract micro-Doppler signatures -- Compare the different domains of radar data that can be exploited for objects classification, and the convenience of using one rather than the other in different situations -- Analyse machine learning techniques for feature extraction and classification applied to radar sensing -- Propose possible approaches for given classification problems based on radar sensors data	
Education Method	The course consists of class lectures and practical activities in the laboratory where possible or following the lectures. Laboratory activities may take place at the existing facilities available for the MS3 (Microwave Sensing Signals & Systems) section.	
Course Relations	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). This course is intended as part of the Master program in Electrical Engineering, primarily for students in the WICOS and S&S tracks, but with expected interest also from students from the civil engineering faculty (CITG), mechanical/robotics engineering (3ME), and aerospace (LR), where activities in remote sensing and autonomous robotics have seen recent significant interest in radar sensing and radar-based classification. This course is envisaged to cover new topics, currently not present in the courses related to radar sensing on offer. In this regard the course is complementary to the existing courses of ET4169 (Radar I) providing the fundamentals of radar sensing and radar as an instrument, and ET4175 (Radar II) providing advanced information mostly focused on the theoretical basis of processing for radar detection and tracking. Therefore, this proposed new course fills a gap in presenting advanced content on the topic of radar-based classification and the related approaches, widening the offer of topics available to our students and contributing to their more well-rounded education. This new course can fit well with the new proposed order of the radar courses": EE4565 in Q2 (fundamentals of electromagnetic scattering and propagation, ET4169 in Q3 (basics of radar for broad audience of students), this course in Q4 (more specialised course in classification based on radar), ET4175 (more specialised course in advanced theory for radar detection and tracking). The new course also fits with the wider research strategy of the department, whereby classification and machine learning intersect the three overarching research themes of Next generation sensing and communication, Health and wellbeing, and Autonomous sensor systems. It is therefore expected that this course will enhance the research capabilities of students in view of their thesis projects, and benefit their employability afterwards. Specifically, radar-based classification for situational awareness is a timely topic, relevant to automotive (NXP), defence/security (Thales NL, TNO), robotics and HCI (Google has recently introduced a radar, Soli, within their phones for gesture-based interaction), healthcare (contactless estimation of vital signs). As part of the educational activities in this course, it is envisaged to have at least one guest lecture on relevant topic from an external stakeholder (for example from Thales NL and TNO).	
Literature and Study Materials	-Lecture slides and pre-class activities provided by the instructors -Selected chapters of the following books. They should all be available as e-book via the TU Delft Library. ---Richards, Mark A. Scheer, James A. Holm, William A.. (2010). Principles of Modern Radar, Volume I - Basic Principles. SciTech Publishing. ---Chen, Victor. Micro-Doppler Effect in Radar, Artech House, 2011. ---Chen, Victor C.; Tahmouh, David; Miceli, William J.. (2014). Radar Micro-Doppler Signatures: Processing and Applications. Institution of Engineering and Technology. ---Chen, Victor C.; Ling, Hao. (2002). Time-Frequency Transforms for Radar Imaging and Signal Analysis. Artech House. ---Gurbuz, Sevgi Zubeyde. (2021). Deep Neural Network Design for Radar Applications. Institution of Engineering and	

Assessment

Technology.

---Fioranelli, Francesco; Griffiths, Hugh; Ritchie, Matthew; Balleri, Alessio. (2020). Micro-Doppler Radar and its Applications. Institution of Engineering and Technology.

---Geron Aurelien. (2019). Hands-on machine learning with scikit-learn, keras, and tensorflow : concepts, tools, and techniques to build intelligent systems (Second). O'Reilly Media.

-Selected published research articles available in IEEE Xplore (to be updated year by year, as novel research is generated - these are accessible via TU Delft Library)

The final grade of the course consists of the following components:

-a written exam with questions (that can be performed online in Brightspace if needed).

-a report on practical/project activities (which may include collection and/or processing of some radar data, and implementation of a complete radar classification pipeline).

The written exam is worth 75% of the final grade and the report(s) on practical activities 25%.

As repair possibility the written exam has one resit opportunity.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE5020	Sensor Signal and Data Processing	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge obtained for instance via the course Detection and Estimation (ET4386) is usefull though not required.	
Course Contents	In this course a variety of modern signal and data processing techniques will be presented along with their application in sensor systems in general and surveillance (radar) systems in particular. Topics that will be presented include: detecting the presence and estimating the state of a single moving object based upon sensor measurements, modeling and estimation with multiple models, rejection of outliers in the measurements, the extension to a time-varying number of objects, the use of multiple sensors, and optimizing the resources of the sensor(s). Techniques that will be presented include: (extended) Kalman filtering, Interacting Multiple Model (IMM) filtering, non-linear filters such as the particle filter, Sequential Likelihood Ratio Testing (SLRT), Multiple Hypothesis Tracking (MHT), track-before-detect, data fusion concepts.	
Study Goals	The main goal of this course is to teach students to apply basic (and more advanced) detection and estimation theory to design algorithmic solutions to (complex) signal and data processing problems	
Education Method	Lectures, instruction lectures, home exercises	
Literature and Study Materials	Lecture notes, and for each topic specific background and overview literature will be provided.	
Assessment	Project work plus an oral exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovyj	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4173	Introduction to UWB technology, systems and applications	4
Responsible Instructor	Prof.dr. O. Yarovy	
Instructor	O.A. Krasnov	
Responsible for assignments	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0,0,0,6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Ultra-wideband (UWB) technology is becoming of greater importance as the pressure for signal bandwidth increases. This course introduces students to UWB technology as it will be used in future communications, radar and navigation systems, and gives detailed attention to both UWB wireless systems and applications in the frequency (FD) and time (TD) domains. The course consists of three major parts. In Part 1 the fundamentals of UWB technology such as waveforms and spectra of short pulses, basic UWB signal processing, radiation and propagation of short electromagnetic pulses are discussed. In Part 2 of the course is dedicated to realizations of UWB technology and systems. General approach to UWB wireless system design is introduced. UWB system design implementation of the key facets of modern UWB systems in both the frequency (FD) and time (TD) domains is discussed. In Part 3 applications of UWB technology in telecommunications, radar, imaging and positioning are discussed. General discussion of the system-level trade offs of such UWB systems is complemented by a detailed analysis of practical systems designed and implemented at TU-Delft. Lectures in the course are complemented by laboratories, at which students get hands-on experience in UWB signal processing and get in touch with wireless applications of UWB systems.	
Study Goals	This course introduces students to UWB technology as it is used in telecommunications, radar and navigation systems, and gives detailed attention to UWB wireless systems. After finishing this course, students will understand UWB signal properties in time and frequency domains, impact of UWB signal properties on wireless system architecture and performance, become familiar with available UWB technologies and UWB wireless system designs, and understand advantages and limitations of UWB technology for different applications. Students will also get hands-on experience with UWB signals and systems.	
Education Method	The course consists of lectures, supervised laboratories with experimental UWB equipment and the final assignment. The course finishes with a symposium, at which the final assignments are presented.	
Literature and Study Materials	Lecture handouts; course books, research papers. M.-G. Di Benedetto, G. Giancola Understanding Ultra Wide Band Radio Fundamentals, Prentice Hall PTR, 2004, ISBN-0-13-148003-0; T. Zwick et al., Ultra-wideband RF system engineering, Cambridge University Press, 2013, ISBN 978-1-107-01555-5	
Assessment	The total mark is based on evaluations of laboratory assignments reports, final assignment essay and presentation at the closing symposium. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4175	Radar II: Theory and System Design	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/0/4 (Q5)	
Education Period	1	
Start Education	5	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4169 Microwaves, Radar and Remote Sensing. Specifically, knowledge is expected on: microwaves basics, radar, scattering mechanisms.	
Course Contents	Radar systems play an important role in today's world. Aviation could not exist without them. But also shipping, traffic and remote sensing extensively employ radar. In today's systems, signal processing is the central theme that enables all radar applications. However, thorough understanding of the physical and electrical engineering radar principles are essential to the successful radar engineer. "Radar Systems" provides an introduction to the design and operation of modern radar systems. Topics include radar principles and designs, the interaction of radar with the environment, i.e. clutter and detection of targets, subsequent signal processing and signal to noise topics. Furthermore waveform design driven by range and Doppler measurements, suppression of ambiguities are treated. Finally modern developments in active arrays and digital beamforming are discussed.	
Study Goals	- The aim of this course is to advance students' knowledge in microwave radar systems. The course will build on the introductory course Microwaves, Radar and Remote Sensing. - The participants will gain sufficient knowledge to understand design, operation and signal processing of modern radar systems and architectures, taking into account radar clutter and detection theory.	
Education Method	The course consists of 10 regular lectures and a 1/2 day excursion to TNO, The Hague or a guest lecture.	
Literature and Study Materials	Kingsley and Quegan: Understanding Radar Systems; Scitech, ISBN 1-891121-05-7 "Principles of Modern Radar: Basic Principles", Mark A. Richards (Editor), James A. Scheer (Editor), William A. Holm (Editor); ISBN-13: 978-1891121524 additional handouts/ reader	
Assessment	Oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	By appointment with Esther de Klerk (E.J.DeKlerk@tudelft.nl)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Physical Layer Signal Processing for Communication 2023	
Introduction 1	The profile Signal Processing for Communications focuses on physical layer aspects of (wireless) communication systems, i.e., how to transport information through a physical radio channel. It covers the fundamental issues of modulation, coding, channel estimation, equalization, and detection of the transmitted data. Important aspects are further coping with hardware deficiencies (e.g. carrier offset recovery), the exploitation of multiple antennas (MIMO systems), as well as how to exploit the channel efficiently (channel statistics) using modern modulation schemes like spread spectrum and Orthogonal Frequency Division Multiplexing (OFDM) and Ultra Wide-Band (UWB) signals. Current research topics focus also on using the communication signal for localization. Applications are indoor localization, sensor networks, and satellite swarm navigation Responsible professors: Prof. A.J. van der Veen Prof. G.J.T. Leus
Introduction 2	Compulsory EE4C03 Statistical Digital Signal Processing and Modeling ET4358 Wireless Communications ET4386 Estimation and Detection Suggested EE4530 Applied Convex Optimization EE4740 "Data Compression: Entropy and Sparsity Perspectives" EE4396 Mobile Networks EE4685 Machine Learning, a Bayesian Perspective EE4715 Array processing

EE4396	Mobile Networks	5
Responsible Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Not formally required, but recommended to have the basic knowledge taught in 'Fundamentals of Wireless Communications' (ET4358). If you didn't take this course, key aspects will be repeated during the class. In case you are lacking a comfortable background, talk to me in the first week of class and I'll help you catch up.	
Course Contents	The course addresses the technology and management of contemporary and future mobile cellular communication networks. With a focus on radio access, as it dominates both the performance and cost aspects of mobile networking, the course addresses 5G, 4G (LTE/LTE-A), 3G (UMTS/HSPA) and 2G (GSM/GPRS/EDGE) technologies, all of which are operational network technologies. An outlook to 6G will also be given. Fundamental aspects of cellular networking are explained and discussed, including e.g. the intrinsic tradeoffs between coverage, capacity and quality and key aspects that influence performance (traffic, propagation, technology, deployment, spectrum. Cellular network planning, dimensioning and performance analysis are treated, as well as radio resource management mechanisms (adaptive beamforming, channel-aware single/multi-user scheduling, handover control, link adaptation, admission, congestion control,) and the current trend towards (potentially AI/ML-based) self-management. Cutting-edge radio features including single/multi-user massive MIMO beamforming, which is at the heart of 5G, are also covered. Considering the on-going technological developments in this exciting domain, the course material is updated every year. The principal lecturer has a 25+-year track record of research and consultancy in the field. The course is enriched with notes about actual deployments, historical anecdotes, popular misconceptions, fascinating factoids, illustrative demonstrations and (potentially) guest appearances.	
Study Goals	The objective of the course is to provide the student with a thorough understanding of contemporary mobile cellular networking technologies, their fundamental properties and tradeoffs, as well as the key challenges and approaches related to network dimensioning, planning and (self-)optimization towards efficient provisioning of coverage, capacity and quality.	
Education Method	Lectures, demonstrations, in-class discussions, assignment(s).	
Literature and Study Materials	Primarily: the lecture slides! For the enthusiastic student, supplementary background materials are recommended in the form of e.g. (white/scientific) papers and book chapters. A number of books on wireless/mobile communications in general or about 2/3/4/5G technologies specifically are recommended as background material; specifics will be given in class / in the lecture slides. As said, this is merely recommended for the most eager students. In principle, the lecture slides cover all relevant content.	
Assessment	Closed-book exam, either written or oral (to be determined). And one (may be two) assignments. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor analysis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: $0.5 * \text{Oral discussion 1} + 0.5 * \text{Oral discussion 2}$. Both partial grades must be sufficient, i.e., a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Network Architectures and Services 2023	
Introduction 1	By specializing into the Network Architectures and Services profile, you focus your future education and research on the broad area of complex networks. These days, complex networks are omnipresent society, and range from man-made infrastructures such as digital communications and energy networks, to biological, brain, social, and economic networks. By choosing this profile you create for yourself the opportunity to: - build theoretical skills in the field of Network Science that help you model, analyse, and optimize complex networks through topological, spectral and functional studies, as well as stochastic models; and - build engineering skills pertinent to the Digital Communications industry that help you plan and design robust network architectures and state-of-the-art communication protocols, as well as manage strategic challenges for network operators; and conduct an internship or final thesis at one of our industrial or research partners. If you have questions, please contact the following individuals from the NAS research group: Prof. P.F.A. Van Mieghem dr. Maksim Kitsak
Introduction 2	Compulsory - EE4C06 Networking - ET4396 Mobile Networks - CS4195 Modeling and Data Analysis in Complex Networks - IN4341 Performance Analysis Suggested - ET4358 Fundamentals of Wireless Communications - EE4530 Applied Convex Optimization - EE4740 "Data Compression: Entropy and Sparsity Perspectives" - EE4630 Telecommunication Network Architectures - ET4034 Telecom Business Architectures and Models - CESE4050 Measuring and Simulating the Internet - WI4525TU Monte Carlo simulation and stochastic processes - CS4220 Machine Learning 1 - CESE4045 High-performance data networking

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
Study Goals	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment (and hence 100% of the grade) will be based on an exam that covers both the theory from the slides as well as the content from the reader. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0 (Lectures(2x2h), Pract)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Strongly advised: (Advanced) Networking course (e.g., CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved).	

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: $0.2 * \text{Assignments} + 0.8 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: $1 * \text{Written exam}$ A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4396	Mobile Networks	5
Responsible Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Not formally required, but recommended to have the basic knowledge taught in 'Fundamentals of Wireless Communications' (ET4358). If you didn't take this course, key aspects will be repeated during the class. In case you are lacking a comfortable background, talk to me in the first week of class and I'll help you catch up.	
Course Contents	The course addresses the technology and management of contemporary and future mobile cellular communication networks. With a focus on radio access, as it dominates both the performance and cost aspects of mobile networking, the course addresses 5G, 4G (LTE/LTE-A), 3G (UMTS/HSPA) and 2G (GSM/GPRS/EDGE) technologies, all of which are operational network technologies. An outlook to 6G will also be given. Fundamental aspects of cellular networking are explained and discussed, including e.g. the intrinsic tradeoffs between coverage, capacity and quality and key aspects that influence performance (traffic, propagation, technology, deployment, spectrum. Cellular network planning, dimensioning and performance analysis are treated, as well as radio resource management mechanisms (adaptive beamforming, channel-aware single/multi-user scheduling, handover control, link adaptation, admission, congestion control,) and the current trend towards (potentially AI/ML-based) self-management. Cutting-edge radio features including single/multi-user massive MIMO beamforming, which is at the heart of 5G, are also covered. Considering the on-going technological developments in this exciting domain, the course material is updated every year. The principal lecturer has a 25+-year track record of research and consultancy in the field. The course is enriched with notes about actual deployments, historical anecdotes, popular misconceptions, fascinating factoids, illustrative demonstrations and (potentially) guest appearances.	
Study Goals	The objective of the course is to provide the student with a thorough understanding of contemporary mobile cellular networking technologies, their fundamental properties and tradeoffs, as well as the key challenges and approaches related to network dimensioning, planning and (self-)optimization towards efficient provisioning of coverage, capacity and quality.	
Education Method	Lectures, demonstrations, in-class discussions, assignment(s).	
Literature and Study Materials	Primarily: the lecture slides! For the enthusiastic student, supplementary background materials are recommended in the form of e.g. (white/scientific) papers and book chapters. A number of books on wireless/mobile communications in general or about 2/3/4/5G technologies specifically are recommended as background material; specifics will be given in class / in the lecture slides. As said, this is merely recommended for the most eager students. In principle, the lecture slides cover all relevant content.	
Assessment	Closed-book exam, either written or oral (to be determined). And one (may be two) assignments. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4630	Telecommunication Network Architectures	3
Responsible Instructor	Ir. R.A.C.J. Noldus	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course presents the fundamentals of cellular mobile networks, including network architecture and services, with a focus on end-to-end and core network aspects. The course forms a companion course for Cellular Networks Radio Access Networks (ET4396). Students will be introduced to the concepts circuit switched and packet switched, the two fundamental communication methodologies that form the backbone of the mobile network. The course aims to educate students on the evolution of the network from 2G to 5G, whereby for each network generation ('G'), the new architectural principles are studied. For example, we discuss the GSM core network (2G), GPRS (2.5G), UMTS (3G), Evolved Packet Core (EPS; 4G), the IP Multimedia Subsystem (IMS), Multimedia telephony (MMTel), Intelligent Networks (IN), Circuit switched fallback (CSFB), Voice over LTE (VoLTE), Voice call continuity (VCC), Voice over Wifi and 5G network architecture. For the 5G network architecture, the concept of 'network slicing' will be presented, which is used for flexible resource allocation, for Internet of Things (IoT) applications and for 'verticals' such as Automotive, Smart Energy, Media & Entertainment, Vehicle communication and other forms of machine-type communication. The course also deals with the concept of virtualization and cloud computing. Networks are increasingly built as a collection of virtual network functions (VNF), deployed on industry standard computing platforms, comprising standard hardware and standard OS. Keywords: - Network architecture and network protocols - Core network, Radio access network (RAN) - Circuit switched, Packet switched - Location/mobility management/roaming - Security (authentication, ciphering,) - QoS/QoE/bearer architecture, QCI definitions, service management, policy control - IMS/VoLTE/Messaging/Voice-over-Wifi/Enriched calling - Virtualization, Cloud computing, Software Defined Networks (SDN) - Network slicing - Multi access Edge Computing (MEC) - Intelligent Networks - Verticals/applications, including e.g. Automotive, Smart Energy, IoT, Machine-to-machine communication - Web Real-time Communication (WebRTC)	
Study Goals	After this course, students are able to explain the meaning, working and rationale behind key cellular network concepts (such as those given by list of keywords above). Furthermore, students will be able to understand the challenges and associated trade-offs in the deployment and management of cellular networks. Students will also be able to analyse network architecture and evaluate network architecture on its merits or on its drawbacks.	
Education Method	This course consists of 8 classes with each class consisting of 2 hours. After the 4th lecture, a mid-term rehearsal takes place; and after the 8th lecture, an end-term rehearsal take place. These rehearsals help the students in preparing for the exam. The students are expected to study the lecture material after every lecture and to ask questions the next lecture, on any topic that requires further explanation. Homework will be provided after each lecture. Students may at any time interact with the lecturer via mail, to ask clarification on any topic of the course.	
Literature and Study Materials	Slides Articles and book chapters	
Assessment	Examination, closed book. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4034	Telecom Business Architectures and Models	4
Responsible Instructor	Dr. E.F.M. van Boven	
Responsible for assignments	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Summary	Edgar van Boven, (KPN, TUD EEMCS)-- Past, present & future of communications -- Assignment 1 Eric Smeitink (KPN, TUD EEMCS)--Strategy, Spectrum & 5G Ton van der Knaap, (Stedin)-- What, why and how of business architecture; experiences from a grid operator and a telecom operator compared Samuel Pronk, (Krauthammer)-- Real Options-- Assignment 2 Ramin Hekmat, (KPN)-- Security and privacy in contemporary communication networks-- Assignment 3 Wouter de Vries (KPN)-- Artificial Intelligent Processes for People Tim Daeleman, (Prodapt Consulting)and Erik Lemmens (KPN)-- Digital Telecom Platforms Jose Almodovar, (TNO)--Standardization of Mobile Communications Frank Mertz (KPN)--5G Networks Developments and Opportunities Remco Helwerda (KPN) -- Digitalization Marco van der Pal, (Cap Gemini)--The IoT landscape architecture Ruud van de Bovenkamp (KPN)-- Fiber-to-the-Home Access Networks -- Assignment 4 Mark Dirksen (KPN)-- Business Continuity Management Pieter Veenstra, (Titanium)-- Service Architecture and 5G Roaming Security Edgar van Boven (KPN, TUD EEMCS)-- Value Case pitch practicing -- Assignment 5	
Course Contents	The essence of the course The ET4034 course Telecom, Architectures and Business Models was designed in 2001 and scheduled since 2002. The aim of the course is giving students from any faculty of the Delft University of Technology the opportunity to learn about a large variety of aspects of the telecom domain, not only its technologies, but also the diverse and challenging daily practice of organisations active in the communications sector. The course is provided by a team of 15 lecturers interactively sharing their knowledge about communications infrastructure in the context of overarching societal trends, service & network architecture, long term developments and new technology evolving from standardization initiatives. As a consequence of continuous innovation and societal developments, the course content is being updated each year including a well-considered choice of central themes that enhance and connect the course lectures. In 2022, the themes Internet of Things, Artificial Intelligence and 5G were highlighted, to be updated in 2023.	
Course Contents Continuation	EE5010 Internships or ET4399 projects in telecom related industries.	
Study Goals	Overview and understanding the Communications Sector, Services, Telecom infrastructures and Value Cases serving society	
Education Method	Online lectures, one powerpoint pitch presentation given by student teams and four off-line written assignments in teams of 2 or 3 students	
Literature and Study Materials	See information on Brightspace.tudelft.nl	
Reader	Course lectures, background information and video recordings available on ET4034 Brightspace	
Prerequisites	Finalised BSc as ET4034 is meant for Master students. PhD students are also welcome and can obtain credits when successfully passing the course assignments via the Graduate School.	
Assessment	The examination of the course Together forming the basis for the examination, the ET4034 course contains five assignments of which four written documents and one opportunity to strengthen ones pitching skills presenting a self-developed Value Case to a live audience. Students are invited to work in couples or trio's on the five assignments. The four written assignments each have a weight factor 1 and the Value Case pitch presentation has a weight factor 0,25 in the end mark of the ET4034 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Not applicable, the course assignments are four documents written offline together with a course partner. The Value Case powerpoint presentation is a pitch that takes circa 7 minutes.	
Maximum number of participants	No limitation	
Self test	Not applicable	
maximum aantal deelnemers	No limitation	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queueing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formulary is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students are expected to have: superior knowledge and skill in elementary probability (details below); some experience working with Matlab, Python, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor. Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course should be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful. Testing yourself and doing a thorough review in the first few weeks is strongly advised; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.	
Course Contents	This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components: modelling and models (and some of their pluses and minuses), learning a number of stochastic processes and some of their properties, doing Monte Carlo simulations, analyzing the generated data and drawing conclusions from them. Contents and format of this course are intended to combine the following: 1. real (possibly stylized) problems as starting points; 2. enough about modelling to make participants aware of the impact of modelling decisions; 3. identification of a number of stochastic processes present in/related to models and problems; 4. (limited) treatment of underlying theory, guided by: (a) what is necessary to ensure that participants know how to do things right and (b) what enables them to do things smartly; 5. facts, properties, and theorems: (a) sincere attempt to understand where possible; (b) focus on practical implications/uses; (c) illustrations rather than proofs/theory; 6. useful tools and techniques from statistics, probability, and stochastic processes; 7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory; 8. providing as much opportunity as feasible for participants to work on/with problems from their own field. Tentative topics/contents list: Review basic probability: most as self-study, some in class, plus mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal process. Markov chains: discrete time; continuous time; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) SDEs.	
Study Goals	Learning objectives: after following the course, participants will be able to 1. in an iid setting, carry out Monte Carlo simulations; and 2. extract and collect, summarize, and interpret the data generated; and 3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?; 4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method; 5. for small-sized practical problems: build a suitable model and simulate it; 6. extract and collect required model parameters from a problem description or background; 7. recognize where modelling choices are made; 8. make elementary modelling decisions themselves; 9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion; 10. recognize/identify the covered stochastic processes in practical settings/problems; 11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models; 12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.	
Education Method	Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in R, Matlab, or Python (or other language). Spread over the semester, five homework assignments. I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time.	
Literature and Study Materials	Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes. For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge"). Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin. Lecture slides, notes, and additional references will be posted on Brightspace. A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I	

	prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab and R. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and this should be possible and arranged). In all cases: you should have the software installed and be familiar with its use.
Assessment	1. Probability test on knowledge and skills needed in/for the course (10%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed); counts for 70%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practicals Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Electives: Internet of Things 2023	
Introduction 1	Embedded Software Group believes that seeing is believing. We experiment with many types of wireless networks (let those be LTE, WiFi or sensors) and push the boundaries of what is possible with off-the-shelf and state-of-the-art wireless nodes. We also research new and exciting communication domains, including battery-less devices, visual light communication motes and wireless power transfer networks. We are a group that has a good connection with the industry, allowing our MSc students to do internships and thesis work outside. With us you will learn the bridge the networking theory with practice and will be able to understand how to analyze and prototype real networks (and communication protocols for such). You will also improve your programming skills be knowledgeable about network measurement and simulation. Responsible professor: Prof. K.G. Langendoen
Introduction 2	Compulsory - EE4C06 Networking - ET4358 Wireless Communications - ET4394 Wireless Networking - IN4254 Smart Phone Sensing Suggested - ET4388 Ad-hoc Networks - EE4630 Telecommunication Network Architectures - ET4396 Mobile Networks - CS4055 High-performance data networking - IN4398 Seminar Internet of Things - ET4173 Introduction to UWB technology, systems and applications - ET4285 Measuring and Simulating the Internet - ET4397IN Network Security - IN4341 Performance Analysis - EE5020 Sensor Signal and Data Processing

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
Study Goals	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment (and hence 100% of the grade) will be based on an exam that covers both the theory from the slides as well as the content from the reader. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0 (Lectures(2x2h), Pract)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Strongly advised: (Advanced) Networking course (e.g., CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved).	

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ----- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. ----- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) - o Definition of ad-hoc networks - o Comparison with infrastructure based systems - o Typical applications - o Advantages and challenges - o Radio technologies for ad-hoc networks - o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks - o Propagation models - o Topology models based on graph theory - o Degree and hopcount - o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) - o Introduction to MAC protocols - o Issues and design goals - o Classification - o Directional, multi-channel MAC protocols - o Energy efficiency in MAC protocols - o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) - o Flooding - o Node discovery, neighbour discovery - o Route establishment - o Topology maintenance, localisation - o Proactive, reactive and hybrid routing - o Typical protocols - o Energy efficiency in routing - o Broadcast and multicast - o Effects of mobility on connectivity and capacity - o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks - o Wireless sensor networks (Chapter 12 of the textbook and papers) - o Cooperation (Reference papers) - o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) - o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for	

textbooks, research papers, etc).
3. Some recent journal papers

4. Optional Reference Books

- 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms)
- 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library.
- 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library.
- 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer.
5. OPNET/ns-2 web pages, tutorials and video lectures

Books

Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoh, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.

Assessment

1. There will be totally three written/oral tests/examinations for this course (including the final exam).
2. The students will carry out a project in a group and submit a short report.
3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis.

Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class.

Breakup:

2 Tests + Final Exam = 55%

Project 40%

Self assessment + Reflection 3%

Activities on Feedback Fruits or any Online platform 2%

Resit: written/oral exam [+ carry over from projects].

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Changes due to COVID19 in 2021 is below;
in case there is a similar situation, we use the following methods.

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There are three chapters

Chapter 1: Network modelling

Chapter 2: MAC protocol

Chapter 3: Routing

+

Lab: Simple experiments using laptops (individual) + simulations (group-wise)

Marks Breakup

Homework/Assignments/Group works

Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks

Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks

Part B1: Individual experiment + report

Part B2: Group-wise simulations + report + demo

Part B1 + Part B2 - 50 Marks

Part C: Self-assessment + peer activities -- 5 marks

Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit.

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(More information will be given in the first class)

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Permitted Materials during Tests

Different conditions for different test/exams.

Conditions will be informed 1 week before the exams/test.

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/ ; Wireshark learn page: https://www.wireshark.org/#learnWS ; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/ ; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands on course for self motivated students. Students work in a group of two (max 3) usually. Students are expected to have sufficient programming and hardware development skills. The course is project oriented thus students are expected to deliver a working model with demonstrations. Preference will be given to Q5 students. Q1 students are discouraged unless otherwise explicitly allowed. ----- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. ----- This course is done in a group of two (max 3). Instructor/TA will help connecting a potential groupmate in the first class using Brightspace/Feedback Fruits. ----- Activities: How to in Groups? ----- Part 1: Seminar We guide the groups to select a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID19 related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their own responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/raspberry pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki based networks or Open testbeds projects. In these cases, online group activities are possible. NOTE: In some extreme cases, Instructor/TA may allow one person project after assessing the situation and after discussing with the student. ----- Supply of Hardware: ----- 1. Instructor/TA is able to provide take home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.	
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.	
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.	
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).	
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	
Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc.	

Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project.

Part 1: 30% of the whole marks;

Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment.

Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference.

Report will not have individual component, however, Q/A may have individual component.

Please NOTE: There would be no resit since this is a hands-on course being executed in a team.

As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Tags

Circuits

Group work

Programming

Project

Software

maximum aantal deelnemers Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics".	
Course Contents	We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming. The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non-trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (10% of grade each, 2 pages each) 2) Final report (10 % of grade, 4 pages) 3) Final project demonstration (70% of grade) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. There is no resit for this course. repair option: A new but related project will be given for the Report (W), Oral(0) and Demonstration (IP).	
Enrolment / Application	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) 1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4430	Network Security	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	The final grade of the course consists of the following components: Assignments (Pass/Fail) Final project (Report + Presentation) (weighting 100%) Final grade calculation: $P/F * \text{Assignments} + 1 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable, resit presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4396	Mobile Networks	5
Responsible Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Not formally required, but recommended to have the basic knowledge taught in 'Fundamentals of Wireless Communications' (ET4358). If you didn't take this course, key aspects will be repeated during the class. In case you are lacking a comfortable background, talk to me in the first week of class and I'll help you catch up.	
Course Contents	The course addresses the technology and management of contemporary and future mobile cellular communication networks. With a focus on radio access, as it dominates both the performance and cost aspects of mobile networking, the course addresses 5G, 4G (LTE/LTE-A), 3G (UMTS/HSPA) and 2G (GSM/GPRS/EDGE) technologies, all of which are operational network technologies. An outlook to 6G will also be given. Fundamental aspects of cellular networking are explained and discussed, including e.g. the intrinsic tradeoffs between coverage, capacity and quality and key aspects that influence performance (traffic, propagation, technology, deployment, spectrum. Cellular network planning, dimensioning and performance analysis are treated, as well as radio resource management mechanisms (adaptive beamforming, channel-aware single/multi-user scheduling, handover control, link adaptation, admission, congestion control,) and the current trend towards (potentially AI/ML-based) self-management. Cutting-edge radio features including single/multi-user massive MIMO beamforming, which is at the heart of 5G, are also covered. Considering the on-going technological developments in this exciting domain, the course material is updated every year. The principal lecturer has a 25+-year track record of research and consultancy in the field. The course is enriched with notes about actual deployments, historical anecdotes, popular misconceptions, fascinating factoids, illustrative demonstrations and (potentially) guest appearances.	
Study Goals	The objective of the course is to provide the student with a thorough understanding of contemporary mobile cellular networking technologies, their fundamental properties and tradeoffs, as well as the key challenges and approaches related to network dimensioning, planning and (self-)optimization towards efficient provisioning of coverage, capacity and quality.	
Education Method	Lectures, demonstrations, in-class discussions, assignment(s).	
Literature and Study Materials	Primarily: the lecture slides! For the enthusiastic student, supplementary background materials are recommended in the form of e.g. (white/scientific) papers and book chapters. A number of books on wireless/mobile communications in general or about 2/3/4/5G technologies specifically are recommended as background material; specifics will be given in class / in the lecture slides. As said, this is merely recommended for the most eager students. In principle, the lecture slides cover all relevant content.	
Assessment	Closed-book exam, either written or oral (to be determined). And one (may be two) assignments. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4630	Telecommunication Network Architectures	3
Responsible Instructor	Ir. R.A.C.J. Noldus	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course presents the fundamentals of cellular mobile networks, including network architecture and services, with a focus on end-to-end and core network aspects. The course forms a companion course for Cellular Networks Radio Access Networks (ET4396). Students will be introduced to the concepts circuit switched and packet switched, the two fundamental communication methodologies that form the backbone of the mobile network. The course aims to educate students on the evolution of the network from 2G to 5G, whereby for each network generation ('G'), the new architectural principles are studied. For example, we discuss the GSM core network (2G), GPRS (2.5G), UMTS (3G), Evolved Packet Core (EPS; 4G), the IP Multimedia Subsystem (IMS), Multimedia telephony (MMTel), Intelligent Networks (IN), Circuit switched fallback (CSFB), Voice over LTE (VoLTE), Voice call continuity (VCC), Voice over Wifi and 5G network architecture. For the 5G network architecture, the concept of 'network slicing' will be presented, which is used for flexible resource allocation, for Internet of Things (IoT) applications and for 'verticals' such as Automotive, Smart Energy, Media & Entertainment, Vehicle communication and other forms of machine-type communication. The course also deals with the concept of virtualization and cloud computing. Networks are increasingly built as a collection of virtual network functions (VNF), deployed on industry standard computing platforms, comprising standard hardware and standard OS. Keywords: - Network architecture and network protocols - Core network, Radio access network (RAN) - Circuit switched, Packet switched - Location/mobility management/roaming - Security (authentication, ciphering,) - QoS/QoE/bearer architecture, QCI definitions, service management, policy control - IMS/VoLTE/Messaging/Voice-over-Wifi/Enriched calling - Virtualization, Cloud computing, Software Defined Networks (SDN) - Network slicing - Multi access Edge Computing (MEC) - Intelligent Networks - Verticals/applications, including e.g. Automotive, Smart Energy, IoT, Machine-to-machine communication - Web Real-time Communication (WebRTC)	
Study Goals	After this course, students are able to explain the meaning, working and rationale behind key cellular network concepts (such as those given by list of keywords above). Furthermore, students will be able to understand the challenges and associated trade-offs in the deployment and management of cellular networks. Students will also be able to analyse network architecture and evaluate network architecture on its merits or on its drawbacks.	
Education Method	This course consists of 8 classes with each class consisting of 2 hours. After the 4th lecture, a mid-term rehearsal takes place; and after the 8th lecture, an end-term rehearsal take place. These rehearsals help the students in preparing for the exam. The students are expected to study the lecture material after every lecture and to ask questions the next lecture, on any topic that requires further explanation. Homework will be provided after each lecture. Students may at any time interact with the lecturer via mail, to ask clarification on any topic of the course.	
Literature and Study Materials	Slides Articles and book chapters	
Assessment	Examination, closed book. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE5020	Sensor Signal and Data Processing	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge obtained for instance via the course Detection and Estimation (ET4386) is usefull though not required.	
Course Contents	In this course a variety of modern signal and data processing techniques will be presented along with their application in sensor systems in general and surveillance (radar) systems in particular. Topics that will be presented include: detecting the presence and estimating the state of a single moving object based upon sensor measurements, modeling and estimation with multiple models, rejection of outliers in the measurements, the extension to a time-varying number of objects, the use of multiple sensors, and optimizing the resources of the sensor(s). Techniques that will be presented include: (extended) Kalman filtering, Interacting Multiple Model (IMM) filtering, non-linear filters such as the particle filter, Sequential Likelihood Ratio Testing (SLRT), Multiple Hypothesis Tracking (MHT), track-before-detect, data fusion concepts.	
Study Goals	The main goal of this course is to teach students to apply basic (and more advanced) detection and estimation theory to design algorithmic solutions to (complex) signal and data processing problems	
Education Method	Lectures, instruction lectures, home exercises	
Literature and Study Materials	Lecture notes, and for each topic specific background and overview literature will be provided.	
Assessment	Project work plus an oral exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4173	Introduction to UWB technology, systems and applications	4
Responsible Instructor	Prof.dr. O. Yarovyj	
Instructor	O.A. Krasnov	
Responsible for assignments	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0,0,0,6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Ultra-wideband (UWB) technology is becoming of greater importance as the pressure for signal bandwidth increases. This course introduces students to UWB technology as it will be used in future communications, radar and navigation systems, and gives detailed attention to both UWB wireless systems and applications in the frequency (FD) and time (TD) domains. The course consists of three major parts. In Part 1 the fundamentals of UWB technology such as waveforms and spectra of short pulses, basic UWB signal processing, radiation and propagation of short electromagnetic pulses are discussed. In Part 2 of the course is dedicated to realizations of UWB technology and systems. General approach to UWB wireless system design is introduced. UWB system design implementation of the key facets of modern UWB systems in both the frequency (FD) and time (TD) domains is discussed. In Part 3 applications of UWB technology in telecommunications, radar, imaging and positioning are discussed. General discussion of the system-level trade offs of such UWB systems is complemented by a detailed analysis of practical systems designed and implemented at TU-Delft. Lectures in the course are complemented by laboratories, at which students get hands-on experience in UWB signal processing and get in touch with wireless applications of UWB systems.	
Study Goals	This course introduces students to UWB technology as it is used in telecommunications, radar and navigation systems, and gives detailed attention to UWB wireless systems. After finishing this course, students will understand UWB signal properties in time and frequency domains, impact of UWB signal properties on wireless system architecture and performance, become familiar with available UWB technologies and UWB wireless system designs, and understand advantages and limitations of UWB technology for different applications. Students will also get hands-on experience with UWB signals and systems.	
Education Method	The course consists of lectures, supervised laboratories with experimental UWB equipment and the final assignment. The course finishes with a symposium, at which the final assignments are presented.	
Literature and Study Materials	Lecture handouts; course books, research papers. M.-G. Di Benedetto, G. Giancola Understanding Ultra Wide Band Radio Fundamentals, Prentice Hall PTR, 2004, ISBN-0-13-148003-0; T. Zwick et al., Ultra-wideband RF system engineering, Cambridge University Press, 2013, ISBN 978-1-107-01555-5	
Assessment	The total mark is based on evaluations of laboratory assignments reports, final assignment essay and presentation at the closing symposium.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queueing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formulary is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Thesis project 2023

Introduction 1

This is the thesis project, this being the concluding study unit of the program. The project amounts to 45 EC and involves a research or design task with sufficient academic level. The project may be executed within a research program at TU Delft, or in a suitable research institute or company. The project must be carried out with a systematic approach and should include all phases of a research or design project: analysis, modelling, implementation/construction and validation/evaluation. The student executes the thesis project independently, with guidance of a thesis supervisor and under the responsibility of the full professor who approved the IEP.

ET4300	Master Thesis	45
Responsible Instructor	Prof. M. Alavi	
Responsible Instructor	Dr. D. Cavallo	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	M. Babaie	
Responsible Instructor	Dr. A. Lekic	
Responsible Instructor	M.A. Kitsak	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MScEE and MScCE programs. It is an individual in-depth research or expert design assignment with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. The thesis project may be executed within a research program at TU Delft, or in a suitable research institute or company. The nominal duration of a thesis project is 30 weeks (45 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. With regard to the thesis project, the thesis supervisor may impose additional conditions on the IEP to be met before starting the thesis project. If desired, an internship or an extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in scope and a separate deliverable such that the thesis project can be clearly evaluated in isolation. The thesis project is governed by the EEMCS Graduation Policy. Below you find a short description of the different steps to be taken during the thesis work: 1. Orientation 2. Kick-off meeting 3. Midterm meeting 4. Green-light review 5. Graduation Day (defence, assessment and graduation ceremony)	
Course Contents Continuation	Declaration of Credits Obtained Before you can start your Masters Thesis Project you need to have a by the Board of Examiners approved Individual Exam Programme (IEP) and at least 60 credits of the Master's degree course work as stated in the IEP should be completed. You can prove this by a form called the Declaration of Credits Obtained (Verklaring behaalde studiepunten) (60 EC). You can request this form from the service desk in the EEMCS building 36 (Mekelweg 4) at the servicedesk with your campuscard or send an email to servicepuntEWI@tudelft.nl (use your with your TU Delft e-mail!) You have to show this form to your thesis advisor.	
Study Goals	The master thesis project serves to prove that the student has acquired the academic competencies of a Master of Science. The thesis project is an individual in-depth research or expert design assignment independently carried out by the student with a systematic approach including all phases of a research or design project: - problem formulation and planning of the work - analysis, - modeling, - implementation/construction, - validation/evaluation, - presentation of the obtained results, orally and in writing. The achievements and resultant thesis report should comprise original work carried out by the student.	
Education Method	Orientation After having obtained a DCO, the thesis work starts with a thesis proposal, which is defined together with the supervisor. The proposal, of at most 2 pages A4, is written according to the format which can be downloaded from the Brightspace page of ET4300. The thesis proposal contains: - a working title, - a preliminary problem definition, - intended outcomes of the thesis project, - intended milestones, - expected duration and (preliminary) date of the thesis defense. The (preliminary) thesis committee is also indicated on this proposal. If desired, an internship or extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in activity and a separate deliverable such that the thesis project can be clearly evaluated in isolation. This is indicated in the thesis proposal. The thesis related work done during an internship or extra project should be described in the thesis report (typically one chapter). Kick-off meeting The Orientation phase indicated above results in a thesis proposal written by the student. The objective of the kick-off meeting is the formal approval of this thesis proposal by the supervisor(s). In this meeting, the research question(s), research methodology and planning are presented by the student and discussed with the supervisor(s). After approval, the "Thesis Proposal form" is filled out. Midterm meeting After 3-4 months, the student presents (by an oral presentation and a written report) the work performed so far to the supervisor(s). On the basis of the work done, it is decided whether the goals, as formulated in the thesis proposal, are still feasible. If not, they are reconsidered and adapted. After the mid-term meeting, the student has a clear idea of how to proceed in order to finish the thesis in time. Thesis committee This committee is responsible for the assessment of the thesis work and the thesis presentation. In order to be able to defend the thesis, the student must fill out the thesis committee form, which needs to be signed by the responsible full or associate professor. This form needs to be handed in to the Board of Examiners of the study programme at least 3 months prior to the MSc thesis presentation (the defence). Green-light review In the final phase of the thesis project, there will be a green-light review. During this meeting it is decided whether the student is	

ready to proceed with the thesis defence, resulting in a "go/no-go". It is compulsory to have a (draft) thesis ready for the green-light review. In case of a "go", the "Final Examination form" is handed in, which will trigger a check on the fulfillment of all requirements for getting the MScEE or MScCE diploma.

Graduation day

The final step of the graduation process is the presentation and defence of the thesis work. The candidate may present and defend the thesis work only after having finished all courses as stated in the approved IEP. The defence consists of two parts. First, the work is presented in a public session in front of the thesis committee, family, friends and fellow students. The presentation is followed by a closed session where the candidate will defend the work in front of the thesis committee. Immediately following the assessment, provided that there is a positive outcome, the diploma ceremony starts and the diploma will be handed out.

Assessment

The thesis project is assessed by the thesis committee based on the actual thesis work carried out, the final report, the public presentation, and the closed defense session and based on the learning objectives of the thesis. The items assessed in the final grading are:

Quality of the work: research method/design, application/development of theory, interpretation of results, scientific significance;

Performance: critical attitude, creativity, initiative, interaction with peers/superiors, planning;

Report: content, structure, quality of writing, independence in writing;

Presentation & defense: content, form, performance, scientific discussion.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Track Signals & Systems 2023

Contact for students	coordinator-MSc-SS@tudelft.nl
Introduction 1	The MSc EE program in Signals and Systems is provided by several groups from the faculties EEMCS and 3mE that are involved in signal processing related research. The program takes 24 months and is fully conducted in English. It offers challenging high-level education and research to talented students holding a BSc degree in technology or science. Signal analysis and processing plays an important role in integrated electronic systems, being crucial to the acquisition of signals and the extraction and interpretation of the information the signals are presumed to carry. Signal processing is also used to interpret the nature of physical processes based on the observation of either a signal or the manner in which the process alters the characteristics of a signal. Electrical engineers who specialize in signals and systems design and develop electronic systems over a wide range of applications. Examples include the development of medical equipment, audio processing devices, wireless communication systems, radar and remote-sensing systems, large antenna-arrays for radio astronomy, and control systems. The Master of Science programme in Electrical Engineering at TU Delft offers the track Signals and Systems. In this MSc track, you will be trained to develop and apply theory and algorithms in the broad field of signal processing and system design. The programme offers you the possibility to specialize more specifically in any of several particular aspects of signal processing, including audio and acoustic signal processing, signal processing for communications, biomedical signal processing, signal processing for remote sensing and radar systems, and signal processing for control systems. When selecting courses and a graduation project, you can choose out of several specialisations. Graduation work can be done either at Delft University, or in industry and other (international) universities. Several laboratories are involved in the MSc program Signals and Systems: - Signal Processing Systems (faculty EEMCS), - Microwave Sensing, Signals & Systems (faculty EEMCS), - Bioelectronics (faculty EEMCS), - Delft Center for Systems and Control (faculty 3mE), The career prospects for Signals & Systems graduates are very good. There is a close connection between TU Delft and the Industry. During your studies you will get the opportunity to get in contact with high tech companies via internships and thesis projects, but also via the EEMCS recruitment days and the technical career fair where you can get your first job interviews. With their solid theoretical and engineering background, graduates of the program have a broad selection of career paths in both the academic and the industrial domains. The program provides an excellent foundation for careers in research, development and consultancy, as well as for commercial and management positions. Our graduates are working in mid-sized and large companies as researchers, developers and consultants. Many have built their careers in commercial and management positions. They have found employment opportunities in prominent firms that are active in the Netherlands (e.g. Philips, Bosch Security Systems, ASTRON, Ericsson, TNO and ASML), as well as in global firms, including hearing-aid producers Oticon and Siemens, mobile phone manufacturers (e.g. Nokia, Samsung and Ericsson) and other highly regarded firms (e.g. Google, NXP Semiconductors and Bang & Olufsen).
Program Goals	The fundamental goal of this MSc program is to train students to become independently thinking and broadly developed professionals that are able to solve future problems using novel concepts, by offering thorough fundamental knowledge and by involving them in cutting-edge research programs in a stimulating environment.

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Main core S&S 2023	
Introduction 1	This is the common core. It is common to all programs and contains two general engineering oriented courses being compulsory for all students. These courses are Profile Orientation and Academic skills and Systems Engineering. The common core further contains 9 fundamental courses introducing all students to the breadth of EE and CE. Of these 9 fundamentals, all students must select two courses. Common core courses are fundamental and general in nature, and should be attractive to a relevant fraction of the students in multiple tracks. These courses are programmed in the first quarter of the academic year. Typically, they have three parts. These are (1) an introductory part that might be seen as a quick repeat of the expected prerequisite knowledge of incoming students (or as a quick introduction if some knowledge is still missing), (2) the main body, and (3) a part that makes a specific connection to the different tracks in the program. This last part may actually be track-specific. Common core EE EE4C01 Profile Orientation and Academic Skills EE4C11 System Engineering Common core: Select 2 out of 9 EE4C03 Statistical Digital Signal Processing and Modeling EE4C04 Control Theory EE4C05 Electromagnetics EE4C06 Networking CESE4010 Advanced Computing Systems EE4C08 Measurement and Instrumentation EE4C10 Analog Circuit Design Fundamentals EE4C12 Machine Learning for Electrical Engineering Applications EE4C13 Wireless Systems for Electrical Engineering Applications When following the track Signals and Systems, it is mandatory to take the core course EE4C03 Statistical Digital Signal Processing and Modeling. Instead of the course EE4C04 Control System Design, students can also choose the course SC42015 Control Theory. It is not possible to include both courses on the ISP.

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Instructor	Ir. J.W. Peltenburg	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = $0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3)$ L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1] In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C01	Profile Orientation and Academic Skills	3
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Responsible Instructor	L.C. Schroten	
Instructor	Dr. C.A. Benitez Avila	
Instructor	L.C. Schroten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: basic knowledge as acquired with a BSc in Physics, EE, Material Science. Strongly advised: Information Literacy 1 and 2 (e.g. Master Programme Information Literacy 1 & 2 or Electrical Engineering Information Literacy 1 and 2, both available in Brightspace).	
Course Contents	- Introduction to the MScEE tracks (Microelectronics, Electrical Power Engineering, Signals & Systems, Wireless Communication and Sensing) - Introduction to the research and expertise of the research groups taking part in the selected master track - Knowledge of ethics and intellectual property right - Skills in academic writing - Skills in oral presentation of scientific topics	
Study Goals	With this course the students - obtain relevant information on the MScEE study and the chosen track, become aware of cultural differences and how this affects intercultural communication, form a network with fellow master EE students. - acquire an overview of the main challenges in and potential applications of electrical engineering - become familiar with research topics addressed within the EE department - master basic skills in oral presentation of scientific topics - master basic skills in academic writing - acquire basic principle of ethics and intellectual property right - are able to extract the main information from a scientific paper or report - are able to write a paper (mostly a literature review) on a scientific question	
Education Method	The course includes lectures, reading materials and exercises. During the course, students work on a writing assignment, an ethics case and give several presentations. Attendance is mandatory as all class activities are highly interactive and peer feedback is a necessity.	
Literature and Study Materials	TU Present & TU Write. Both are available in Brightspace.	
Assessment	The pass/fail for this course is determined by the evaluation of the final writing assignment, the oral presentation and attending the Ethics seminar(s). All three components should be evaluated with a pass to pass the whole EE4C01 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C04	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential Realization of transfer function models by state space descriptions, coordinate changes, canonical forms Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback Application of LQ regulator Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle Reference signal modeling, integral action for zero steady-state error. Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design and - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	P.P. Sundaramoorthy	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C13	Wireless Systems for Electrical Engineering Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	C. Gao	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	M. Babaie	
Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	M. Spirito	
Instructor	Prof. M. Alavi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE4605 'Integrated Circuits and Systems for Wireless Applications' and EE4371 'Advanced Wireless Transceivers.'	
Course Contents	In this course, you will learn about the commonly used architectures in wireless systems, with the requirements on their building blocks. The system key building blocks are addressed at their functional level regarding electrical properties, such as transfer/gain, noise, linearity, frequency selectivity, bandwidth, and other parameters. Physical constraints of practical components and circuits are used to identify the achievable signal quality, selectivity, sensitivity, and link budget of a wireless system. The lectures will explain the causes and impact of device/circuit non-linearities and noise on wireless systems' sensitivity and dynamic range. The course will introduce the concepts of frequency generation, modulation/demodulation, and up/down conversion. To teach you to work at RF frequencies with wireless systems, the principles of impedance matching are explained, and tools are provided to analyze/design/characterize RF (matching) circuitry/systems. The common part of the course also includes an overview of the figure of merits (FoMs) of various TX/Rx architectures/system blocks and the required instrumentation to characterize them. This course belongs to the RF/wireless profile and is an excellent preparation for the course; EE4605, Integrated Circuits for RF/Wireless Applications, which focuses on the design and realization of analog intensive building blocks for wireless transceivers as well as for ET4371, Advanced Digital Wireless transceivers which is addressing the latest trends in RF/wireless design aiming to fully utilize the integration and signal processing strengths of advanced CMOS. The course is completed by four short specialization topics which can be followed based on your personal preference, namely: (1) Introduction to wireless power transfer technology, (2) Beamforming basics: concepts and architectures, (3) Neural Networks for Wireless & Radar Systems, (4) Hands-on experience with RF characterization equipment on practical circuits. (The latter (4) has an attendee limit of 15 people.) Only one of these specialization topics is required to complete the overall course.	
Study Goals	Understand wireless system architectures with their subblocks to such a level that you can analyze their various performance parameters in terms of; link budget, sensitivity, spectral purity, P1dB, IP3, DR, SFDR, noise, phase noise, selectivity, EVM, APCR, and efficiency. In addition, you will master a basic understanding of S-parameters, impedance matching, and RF instrumentation.	
Education Method	This course will be delivered with complementary exercise lectures to familiarize students with the concepts and calculations described during the lectures.	
Literature and Study Materials	These lectures are (in part) based on the book of Behzad Razavi, RF microelectronics 2nd edition, Chapters 1 to 4.	
Assessment	Examination and grading: Your overall grade will be based on your performance in the written exam (80%), compulsory homework, and the mark obtained in your track-specific project (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year

Organization

Education

2023/2024

Electrical Engineering, Mathematics and Computer Science

Master Electrical Engineering

Track core S&S 2023	
Introduction 1	This is the track core, offering the first level of branching between tracks. The track cores typically contain some 6 courses, of which the students select three courses.
Introduction 2	Notice that when following the track Signals and Systems, it is mandatory to take the track core course ET4386 Estimation and Detection.

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovyj	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested specialisation courses S&S 2023

Introduction 1	This is the specialization space. In this tier, students have to select courses totalling to at least 24 EC . In essence, all of the MSc EE and MSc CE courses can be chosen as a specialization course, including those from the main core and the track core that were not yet chosen.
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AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morphological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: $((0.85 \cdot EX + 0.15) \cdot AS) / 2$ or without the bonus point from the practicals: $(EX + AS) / 2$ With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3232	Medical Imaging Signals and Systems	6
Responsible Instructor	Dr.ir. M.C. Goorden	
Instructor	Dr. D. Maresca	
Instructor	Dr. S.D. Weingärtner	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A thorough mathematical basis is essential to follow the class. This can for example include Linear Algebra I + II and Calculus I + II. An in-depth understanding of Fourier analysis and basic physical understanding of electromagnetism is necessary."	
Course Contents	The course Medical Imaging Signals and Systems covers the principles of advanced medical imaging modalities. Particularly, students will study the physics, acquisition techniques and signal processing underlying conventional X-ray, Computerized Tomography (CT), Single Photon Emission Computed Tomography (SPECT), Positron Emission Tomography (PET), Ultrasound (US) and Magnetic Resonance Imaging (MRI). Course Content: Introductory lecture with the history of medical imaging and its importance for modern medicine and biomedical research. Conventional X-ray, CT, SPECT and PET imaging (33% of content): Basic principles of planar X-ray imaging and scintigraphy (short); Advanced methods for image acquisition using CT, SPECT and PET; Mathematical techniques for image reconstruction, including filtered backprojection and iterative approaches; Issues of image quality, especially related to resolution, signal to noise ratio, quantitative and dynamic imaging. MR imaging (33% of content): Fundamentals of MRI: spin, Larmor frequency, T1, T2, T2* contrast (short); Spatial encoding: K-space; Image Quality: PSF, signal strength and noise level; Advanced MRI techniques such as diffusion, perfusion and flow imaging. Ultrasound imaging (33% of content): Physical and mathematical principles of ultrasound imaging, including probe design and advanced US imaging techniques such as harmonic imaging and non-linear reconstruction techniques for imaging tissue properties. The course will end with a mini-symposium on specialized imaging techniques. Therefore, recently developed medical imaging methods will be studied, going beyond the theory of the lectures. Students working in pairs will write a short essay on a selected technique and present it to the group.	
Study Goals	1. Acquire in-depth knowledge about the physics and image reconstruction underlying X-ray, CT, acoustical and magnetic resonance imaging; 2. Being able to solve elementary problems related to the theory mentioned in 1; 3. Being able to solve more advanced problems addressing the theory mentioned in 1 by combining mathematical skills and physical insight; 4. Able to acquire new knowledge about clinical a medical imaging technique; 5. Being able to present newly acquired knowledge about a medical imaging technique.	
Education Method	Lectures and an assignment.	
Literature and Study Materials	Book: Medical Imaging Signals & Systems, Jerry L. Prince, Jonathan Links. Upper Saddle River NJ: Prentice Hall, second edition, 2014, 544 pp, ISBN-13: 978-0132145183. We will provide Chapter 1 "Brief introduction to MRI physics" of the book: "Magnetic Resonance Image Reconstruction Theory, Methods, and Applications", 1st Edition - November 4, 2022, Editors: Mehmet Akcakaya, Mariya Ivanova Doneva, Claudia Prieto, Paperback ISBN: 9780128227268 , eBook ISBN: 9780128227466	
Assessment	Additional handouts wherever necessary. The course on Medical Imaging comes with an assignment for which you will need to hand in a paper and make a presentation. This is an obligatory part, for which you will get a grade from 1-10 (A). Additionally, there will be an exam also graded from 1-10 (B). The grades for both parts needs to be at least 6.0 in order to pass. The final grade is calculated by: $(A + B)/2$. It is not possible to take part in the exam if you have not made the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Pocket calculator.	
Elective	Yes	

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41100	Medical Devices: Designing for CE-marking	3
Responsible Instructor	A.J. Loeve	
Co-responsible for assignments	R. Straathof	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The assignments follow the line of the lectures (see course syllabus) and have to be completed using one of a selection of actual prototypes that were manufactured within the MISIT group. The assignments all tackle a part of the CE-marking process, such as risk analyses, redesign for cleanability, materials selection etc.(detailed info in the course syllabus on Brightspace).	
Study Goals	The assignments and final reports will be judged on: - Demonstration of concise line of reasoning and critical thinking - Effort to be complete - Originality of approach and proposed redesign, and contribution to minimize risks - Use of information sources - Written presentation The final presentation will be judged on: - Balance between main points and details - Liveliness - Clear message - Visuals - One's own reflection on the course and learned aspects	
Education Method	Each week, a different aspect is discussed related to the eventual goal of receiving permission to use a medical device on patients in the hospital. The lectures (one hour each week) are given by the course lecturers and guest lecturers affiliated to hospitals, research institutes and companies. After each lecture, there is some time for discussion and questions about the previously made assignment. The assignments are completed by teams of four (sometimes five, depending on the class size) students working in close collaboration. The final report consists of all assignments having been put together after processing the feedback. On the last course day, the student teams will have to present their final results and grade each other.	
Assessment	Final presentation (20%), assignments grades (40%) and final assignment report (40%)	
Remarks	Due to the extensive feedback cycles in this course, no more than 30 students will be admitted. The application and admission procedure will be communicated via the course's Brightspace page as soon as it opens (usually about 2 months before the start of the course).	
Department	3mE Department Biomechanical Engineering	

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ----- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. ----- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) - o Definition of ad-hoc networks - o Comparison with infrastructure based systems - o Typical applications - o Advantages and challenges - o Radio technologies for ad-hoc networks - o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks - o Propagation models - o Topology models based on graph theory - o Degree and hopcount - o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) - o Introduction to MAC protocols - o Issues and design goals - o Classification - o Directional, multi-channel MAC protocols - o Energy efficiency in MAC protocols - o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) - o Flooding - o Node discovery, neighbour discovery - o Route establishment - o Topology maintenance, localisation - o Proactive, reactive and hybrid routing - o Typical protocols - o Energy efficiency in routing - o Broadcast and multicast - o Effects of mobility on connectivity and capacity - o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks - o Wireless sensor networks (Chapter 12 of the textbook and papers) - o Cooperation (Reference papers) - o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) - o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for	

textbooks, research papers, etc).
3. Some recent journal papers

4. Optional Reference Books

- 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms)
- 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library.
- 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library.
- 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer.
5. OPNET/ns-2 web pages, tutorials and video lectures

Books

Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.

Assessment

1. There will be totally three written/oral tests/examinations for this course (including the final exam).
2. The students will carry out a project in a group and submit a short report.
3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis.

Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class.

Breakup:

2 Tests + Final Exam = 55%

Project 40%

Self assessment + Reflection 3%

Activities on Feedback Fruits or any Online platform 2%

Resit: written/oral exam [+ carry over from projects].

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Changes due to COVID19 in 2021 is below;
in case there is a similar situation, we use the following methods.

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There are three chapters

Chapter 1: Network modelling

Chapter 2: MAC protocol

Chapter 3: Routing

+

Lab: Simple experiments using laptops (individual) + simulations (group-wise)

Marks Breakup

Homework/Assignments/Group works

Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks

Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks

Part B1: Individual experiment + report

Part B2: Group-wise simulations + report + demo

Part B1 + Part B2 - 50 Marks

Part C: Self-assessment + peer activities -- 5 marks

Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit.

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(More information will be given in the first class)

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Permitted Materials during Tests

Different conditions for different test/exams.

Conditions will be informed 1 week before the exams/test.

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy).	
	Experience in programming and installing programming languages/libraries is required.	
	Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.	
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace.	
Assessment	Assignments are based on PyTorch: https://pytorch.org/ The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4016	Antenna Systems	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	Y. Aslan	
Instructor	Ir. P.J. Aubry	
Co-responsible for assignments	Ir. P.J.F. Swart	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4288, ET4595	
Expected prior knowledge	EE4C05	
Course Contents	The aim of this course is to introduce the students to the Antenna Techniques and Antenna Systems that are instrumental for applications such as telecommunications, remote sensing and radar systems. After introducing basic antenna concepts the basic antenna parameters are defined. A detailed description of the antenna gain, directivity, polarization, radiation pattern, bandwidth, beamwidth, efficiency, temperature, effective length, effective area and input impedance is given. The reflection and radiation from a dipole are detailed and analysed. Microstrip antenna is introduced and its performance is analysed. The radiation from linear, circular and planar array antennas is discussed in detail while stressing their importance in radar and remote sensing systems. The beamforming in arrays and antenna array synthesis are studied. Characterisation of Antenna System performance is discussed and derivation of Antenna System specifications from overall wireless system performance requirements is studied. Some basic concepts about antenna measurements techniques are introduced. The theoretical studies are complimented by computer simulations and measurements.	
Study Goals	By the end of this course, you should be able to: LO1: Determine (theoretically and experimentally) the fundamental antenna system performance metrics (such as directivity, gain, bandwidth etc.). LO2: Design basic antenna element/array structures (i.e. linear wire, microstrip, aperture) for the given criteria, by using CST and MATLAB tools. LO3: Assess the impact of a given antenna on the wireless system performance (in terms of the signal and noise power). LO4: Develop a suitable antenna model for a given application with specified requirements (e.g. communication, broadcasting, radar).	
Education Method	Lectures, Instructions, computer simulations, antenna measurements	
Literature and Study Materials	Course books (Balanis and Orfanidis), Lectures handouts, simulation and measurement assignments.	
Assessment	Written exam (problems solving) plus home works, simulation and measurement assignments In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

EE4650	Advanced Magnetic Resonance Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	A. Webb	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Vector algebra and vector calculus, linear circuit theory, signals and systems (Fourier transform, steady-state analysis), linear algebra and differential equations, electromagnetics (BSc level or higher)	
Course Contents	General introduction, history of MRI, general description of an MR scanner, general principles of MRI, applications and current research themes. General and brief description of spin, restriction to particles with spin $\frac{1}{2}$ (proton), equilibrium magnetization for spin $\frac{1}{2}$ particles (proton), the Bloch equation, general description of the gyromagnetic ratio, T1 and T2 relaxation times. Electromagnetic fields: Maxwells equations, the electromagnetic boundary conditions, the temporal and spatial Fourier transform, time-harmonic fields, the polarization state of an electromagnetic field (linear, circular, elliptical), Poyntings theorem (power balance), and reciprocity. Flipping the magnetization: received signal in terms of the electromagnetic receive field. Solutions of Blochs equation for various magnetic fields, the most important formula in MRI: formula for the Larmor frequency, chemical shift, scalar coupling, basics of 1-DNMR spectroscopy, elementary measurement sequences, spin-echo, inversion recovery, measuring T1 and T2, introducing T2*. Using gradients to spatially encode signals, k-space formalism, image reconstruction via multi-dimensional Fourier transforms, simple gradient echo and spin echo pulse sequences, three-dimensional imaging, angiography. Background knowledge for hardware design: Maxwells equations and quasi-static fields, the law of Biot-Savart, spherical harmonics. Interaction of RF with tissue as a function of frequency, different media/tissue types and the constitutive relations. RF coil and array design: design of RF coils and RF arrays, the Specific Absorption Rate (SAR). The electromagnetic field due to an electric dipole (small current-carrying piece of wire) and a magnetic dipole (small current-carrying loop). Gradient coil design and the static background field: Realizing gradient fields, Helmholtz and Maxwell coils, advanced gradient coil design using stream functions, consequences of nonuniform static background fields. Numerical modeling, software packages, body models, coil models. Measurements and imaging in practice, SNR and other practical aspects. Current/future research topics, parallel imaging, sparse sampling, electrical properties tomography	
Study Goals	After completing this course, the students will: 1. Understand the principles of magnetic resonance from both a quantum and classical mechanics point-of-view, 2. Understand the interaction of strong electromagnetic fields at high, medium and low frequencies, how these can be manipulated, and the safety features associated with these fields, 3. Understand how spatial information is encoded into magnetic resonance images, the origins of image artifacts, and the signal processing algorithms involved in image reconstruction, 4. Understand the relaxation phenomena associated with magnetic resonance, and how these are important in determining image contrast, 5. Understand how the different components of magnetic resonance systems are designed using Maxwells equations, spherical harmonics, and the Biot-Savart law, 6. Understand the current state-of-the-art of MRI hardware and image processing, and the new opportunities for future developments.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes. Will be made available via Brightspace.	
Assessment	Assignment and oral exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4675	Object classification with radar	4
Responsible Instructor	Dr. F. Fioranelli	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Advised: ET4169, specifically knowledge of microwave and radar basics such as measurements of range and velocity and related basic signal processing.	
Course Contents	While the idea of gaining information about the surrounding environment using electromagnetic waves is a relatively mature concept (a radar essentially!), a tremendous amount of research activities have recently coupled the analysis of data from EM sensors with techniques that can teach them to recognise automatically objects of interest. What natural or manmade objects are moving around, and what are they doing? Can we infer something about their behaviour or intention from such movements? These are an example of the big questions which data and clever signal processing from EM sensors can address. Applications range from the analysis of human gait and movements to the discrimination of birds vs Unmanned Aerial Vehicles (drones), to the identification of the subtle gestures people produce to interact with smart objects, amongst others. All these applications have in common two main areas of expertise: one has to do with understanding and processing the information encoded into the sensors signals (radar signal processing and data format); one has to do with the techniques to interpret this information automatically to make decisions (we could call this applied machine learning). In this course, we aim to provide a comprehensive introduction and explanation of both areas from an engineering application perspective, with the objective to learn how to use these tools for real-life applications in radar engineering. The course is broadly divided into two parts: 1. Part 1 will build on fundamentals of radar & microwave sensing courses to investigate the concept of micro-Doppler signatures, focusing on its modelling, the signal processing for its characterisation, and its relationships with the physical movement of objects and therefore their identity. Can we understand from the EM sensors data how the different parts of human bodies or manmade objects move? 2. Part 2 will introduce machine learning techniques for extraction of information from sensors data (feature extraction), for their selection, and for classification of objects of interest. Conventional supervised learning techniques will be presented (for example Nearest Neighbours and Support Vector Machine classifiers), along with some of the more recent data-driven approaches based on neural networks. The diversity of formats of the radar data for classification will also be considered, including micro-Doppler signatures, High Resolution Range Profiles, radar cubes. In both parts, students will be exposed to concrete examples and data directly taken from current activities at TU Delft, so that the research element is well embedded in the teaching.	
Study Goals	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The aim of this course is to introduce students to techniques to classify and identify objects of interest mixing machine learning and signal processing applied to data from radar sensors. By the end of this course, students will be able to: -- Apply relevant signal processing techniques to EM sensors and radar data to extract micro-Doppler signatures -- Compare the different domains of radar data that can be exploited for objects classification, and the convenience of using one rather than the other in different situations -- Analyse machine learning techniques for feature extraction and classification applied to radar sensing -- Propose possible approaches for given classification problems based on radar sensors data	
Education Method	The course consists of class lectures and practical activities in the laboratory where possible or following the lectures. Laboratory activities may take place at the existing facilities available for the MS3 (Microwave Sensing Signals & Systems) section.	
Course Relations	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). This course is intended as part of the Master program in Electrical Engineering, primarily for students in the WICOS and S&S tracks, but with expected interest also from students from the civil engineering faculty (CITG), mechanical/robotics engineering (3ME), and aerospace (LR), where activities in remote sensing and autonomous robotics have seen recent significant interest in radar sensing and radar-based classification. This course is envisaged to cover new topics, currently not present in the courses related to radar sensing on offer. In this regard the course is complementary to the existing courses of ET4169 (Radar I) providing the fundamentals of radar sensing and radar as an instrument, and ET4175 (Radar II) providing advanced information mostly focused on the theoretical basis of processing for radar detection and tracking. Therefore, this proposed new course fills a gap in presenting advanced content on the topic of radar-based classification and the related approaches, widening the offer of topics available to our students and contributing to their more well-rounded education. This new course can fit well with the new proposed order of the radar courses": EE4565 in Q2 (fundamentals of electromagnetic scattering and propagation, ET4169 in Q3 (basics of radar for broad audience of students), this course in Q4 (more specialised course in classification based on radar), ET4175 (more specialised course in advanced theory for radar detection and tracking). The new course also fits with the wider research strategy of the department, whereby classification and machine learning intersect the three overarching research themes of Next generation sensing and communication, Health and wellbeing, and Autonomous sensor systems. It is therefore expected that this course will enhance the research capabilities of students in view of their thesis projects, and benefit their employability afterwards. Specifically, radar-based classification for situational awareness is a timely topic, relevant to automotive (NXP), defence/security (Thales NL, TNO), robotics and HCI (Google has recently introduced a radar, Soli, within their phones for gesture-based interaction), healthcare (contactless estimation of vital signs). As part of the educational activities in this course, it is envisaged to have at least one guest lecture on relevant topic from an external stakeholder (for example from Thales NL and TNO).	
Literature and Study Materials	-Lecture slides and pre-class activities provided by the instructors -Selected chapters of the following books. They should all be available as e-book via the TU Delft Library. ---Richards, Mark A. Scheer, James A. Holm, William A.. (2010). Principles of Modern Radar, Volume I - Basic Principles. SciTech Publishing. ---Chen, Victor. Micro-Doppler Effect in Radar, Artech House, 2011. ---Chen, Victor C.; Tahmouh, David; Miceli, William J.. (2014). Radar Micro-Doppler Signatures: Processing and Applications. Institution of Engineering and Technology. ---Chen, Victor C.; Ling, Hao. (2002). Time-Frequency Transforms for Radar Imaging and Signal Analysis. Artech House. ---Gurbuz, Sevgi Zubeyde. (2021). Deep Neural Network Design for Radar Applications. Institution of Engineering and	

Assessment

Technology.

---Fioranelli, Francesco; Griffiths, Hugh; Ritchie, Matthew; Balleri, Alessio. (2020). Micro-Doppler Radar and its Applications. Institution of Engineering and Technology.

---Geron Aurelien. (2019). Hands-on machine learning with scikit-learn, keras, and tensorflow : concepts, tools, and techniques to build intelligent systems (Second). O'Reilly Media.

-Selected published research articles available in IEEE Xplore (to be updated year by year, as novel research is generated - these are accessible via TU Delft Library)

The final grade of the course consists of the following components:

-a written exam with questions (that can be performed online in Brightspace if needed).

-a report on practical/project activities (which may include collection and/or processing of some radar data, and implementation of a complete radar classification pipeline).

The written exam is worth 75% of the final grade and the report(s) on practical activities 25%.

As repair possibility the written exam has one resit opportunity.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor analysis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: $0.5 * \text{Oral discussion 1} + 0.5 * \text{Oral discussion 2}$. Both partial grades must be sufficient, i.e., a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4750	Tensor Networks for Green AI and Signal Processing	4
Responsible Instructor	B. Hunyadi	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	2/0/0/0 lectures and 4/0/0/0 computer sessions	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Course aims and objectives This course provides a solid mathematical foundation to tensor networks and discusses their use in machine learning and signal processing. Tensors are multi-dimensional generalization of matrices. Real-life data often comes in high dimensions, such as of a video stream, or multichannel EEG recorded from multiple patients under various conditions. Tensor-based tools are crucial to efficiently represent and process this data. Alternative (matrix-based) solutions artificially segment such high-dimensional data into shorter one- or two-dimensional arrays, causing information loss by destroying correlations in these data. The key idea of tensor networks is to write a tensor in terms of much smaller tensors that represent these correlations among the different dimensions. On one hand, such representations can reveal hidden structure in the data and can even lead to decompositions with physically interpretable terms. Moreover, in this way, the original tensor never needs to be kept explicitly in memory. This latter aspect becomes especially interesting considering the ever-increasing dataset sizes brought on by recent advances in sensor and imaging technology. Computations based on tensor networks enable faster training and inference of AI models based on such large datasets. Therefore, tensor networks facilitate the shift towards the direction of Green AI. Green AI aims to decrease the environmental footprint of AI computation and increase the inclusivity of AI research by becoming more independent of expensive computer hardware needed for training. Course contents and description The first introductory lecture gives a motivation to the use of tensor networks. Various compelling examples from the teachers own research experience will be given. Next, the basic notation and the use of tensor diagrams will be presented, and important matrix and tensor operations will be explained, such as various products (inner, outer, Kronecker, Khatri-Rao, etc.), norms, reshaping (matricization, vectorization). These operations and concepts are crucial to understand how matrix decompositions generalize to tensor methods. This leads to the core of the course content. The most well-known tensor network is the singular value decomposition (SVD) of a matrix. The following two lectures present how the SVD can be generalized to tensors in two different ways, one leading to the multi-linear singular value decomposition (MLSVD), and the other leading to the canonical polyadic decomposition (CPD). The properties, computation and applications of both generalizations will be discussed. The third tensor network structure that generalizes the matrix SVD are tensor trains (TT). Their properties, computation and applications will also be discussed. TTs are particularly well-suited to represent large-scale vectors and matrices as most of the common linear algebra operations can be done with TTs. Finally, the last two lectures focus on the application of tensors to various signal processing (e.g. blind source separation, data fusion) and machine learning problems (e.g. kernel methods). For this, we will introduce advanced concepts such as different tensorization techniques, constrained and coupled decompositions. Throughout the course, various applications will be presented. From the field of biomedical signal processing this includes epileptic seizure localization in EEG, fusion of simultaneous EEG-fMRI, blind deconvolution of brain responses in functional ultrasound during visual information processing or prostate cancer detection using ultrasound. The course explains how tensor networks pave the way towards Green AI. This is done by showing how machine learning methods like kernel machines, artificial neural networks, and Gaussian processes can be made more efficient and tractable for big data and higher dimensional problems.	
Course Contents Continuation	Educational method: The course consists of 8 2-hour lectures of theory, including examples and applications. There will be 4 2-hour computer lab sessions that discuss practical implementation and applications. These lab sessions help student complete their computer assignments, which are an important part of their assessment. The examples in the assignments are taken e.g. from biomedical applications (ECG, EEG data), computer vision, etc.	
Study Goals	After the course, the student is able to: - Explain different tensor networks (TN) - Implement basic tensor networks - Formulate a basic data science / signal processing problem as a TN - Make informed choices in the practical use of TNs, such as rank or initialization method. - Formulate research questions for their own (master thesis level) research or independent literature study in the field and its applications, i.e. tensor-based (biomedical) signal processing, image analysis, control, etc.	
Education Method	8 lectures and 4 computer sessions	
Literature and Study Materials	The course covers chapters Part I and selected chapters of Part II of the book below. Further applications and examples will be provided from recent scientific literature and the teachers own research experience. Andrzej Cichocki, Namgil Lee, Ivan Oseledets, Anh-Huy Phan, Qibin Zhao and Danilo P. Mandic (2016), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 1 Low-Rank Tensor Decompositions", Foundations and Trends® in Machine Learning: Vol. 9: No. 4-5, pp 249-429. http://dx.doi.org/10.1561/22000000059 Andrzej Cichocki, Anh-Huy Phan, Qibin Zhao, Namgil Lee, Ivan Oseledets, Masashi Sugiyama and Danilo P. Mandic (2017), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 2 Applications and Future Perspectives", Foundations and Trends® in Machine Learning: Vol. 9: No. 6, pp 431-673. http://dx.doi.org/10.1561/22000000067	
Assessment	The students will complete 4 assignments. Each assignment will be discussed during the corresponding computer lab session. The first 3 assignments are common for all students, and they are completed individually. The final assignment is executed in pairs, and the topic is chosen from a list of several options, depending on the interest of the students. The students will submit their code as well as a report about the assignments. The scores of the assignments contribute to final grade with the following weights: 5% / 10% / 10% / 75% and students need to pass each individual assignment. The final assignment will also be defended with an individual oral examination, where theoretical questions related to the project will be asked. In case of a retake a new final assignment has to be completed. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Artificial intelligence	
Co-Instructor	Dr.ir. K. Batselier	

EE5020	Sensor Signal and Data Processing	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge obtained for instance via the course Detection and Estimation (ET4386) is usefull though not required.	
Course Contents	In this course a variety of modern signal and data processing techniques will be presented along with their application in sensor systems in general and surveillance (radar) systems in particular. Topics that will be presented include: detecting the presence and estimating the state of a single moving object based upon sensor measurements, modeling and estimation with multiple models, rejection of outliers in the measurements, the extension to a time-varying number of objects, the use of multiple sensors, and optimizing the resources of the sensor(s). Techniques that will be presented include: (extended) Kalman filtering, Interacting Multiple Model (IMM) filtering, non-linear filters such as the particle filter, Sequential Likelihood Ratio Testing (SLRT), Multiple Hypothesis Tracking (MHT), track-before-detect, data fusion concepts.	
Study Goals	The main goal of this course is to teach students to apply basic (and more advanced) detection and estimation theory to design algorithmic solutions to (complex) signal and data processing problems	
Education Method	Lectures, instruction lectures, home exercises	
Literature and Study Materials	Lecture notes, and for each topic specific background and overview literature will be provided.	
Assessment	Project work plus an oral exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and probability theory, as being treated in B.Sc. programmes like Electrical Engineering, Computer Science, and Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The general overall objectives of this course are that the students can - reproduce the error control concepts as presented in the lecture notes; - explain the meaning and usefulness of these concepts; - apply the concepts to transmission and storage systems, including basic calculations, at the level of the exercises in the lecture notes and previous exams. More specifically, the students are able to - discuss the basic trade-offs between efficiency, reliability, and complexity in designing coding schemes for transmission and storage systems; - apply finite fields theory in order to construct (cyclic) codes; - encode data into code sequences that are resistant to a certain number of random and/or burst errors; - decode received sequences using (iterative) hard or soft decision techniques.	
Education Method	Lectures; expected workload is 20 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 32 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber, available on Brightspace. Also previous exams as well as demos are available via Brightspace.	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Actual course information available on Brightspace.	

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4173	Introduction to UWB technology, systems and applications	4
Responsible Instructor	Prof.dr. O. Yarovy	
Instructor	O.A. Krasnov	
Responsible for assignments	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0,0,0,6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Ultra-wideband (UWB) technology is becoming of greater importance as the pressure for signal bandwidth increases. This course introduces students to UWB technology as it will be used in future communications, radar and navigation systems, and gives detailed attention to both UWB wireless systems and applications in the frequency (FD) and time (TD) domains. The course consists of three major parts. In Part 1 the fundamentals of UWB technology such as waveforms and spectra of short pulses, basic UWB signal processing, radiation and propagation of short electromagnetic pulses are discussed. In Part 2 of the course is dedicated to realizations of UWB technology and systems. General approach to UWB wireless system design is introduced. UWB system design implementation of the key facets of modern UWB systems in both the frequency (FD) and time (TD) domains is discussed. In Part 3 applications of UWB technology in telecommunications, radar, imaging and positioning are discussed. General discussion of the system-level trade offs of such UWB systems is complemented by a detailed analysis of practical systems designed and implemented at TU-Delft. Lectures in the course are complemented by laboratories, at which students get hands-on experience in UWB signal processing and get in touch with wireless applications of UWB systems.	
Study Goals	This course introduces students to UWB technology as it is used in telecommunications, radar and navigation systems, and gives detailed attention to UWB wireless systems. After finishing this course, students will understand UWB signal properties in time and frequency domains, impact of UWB signal properties on wireless system architecture and performance, become familiar with available UWB technologies and UWB wireless system designs, and understand advantages and limitations of UWB technology for different applications. Students will also get hands-on experience with UWB signals and systems.	
Education Method	The course consists of lectures, supervised laboratories with experimental UWB equipment and the final assignment. The course finishes with a symposium, at which the final assignments are presented.	
Literature and Study Materials	Lecture handouts; course books, research papers. M.-G. Di Benedetto, G. Giancola Understanding Ultra Wide Band Radio Fundamentals, Prentice Hall PTR, 2004, ISBN-0-13-148003-0; T. Zwick et al., Ultra-wideband RF system engineering, Cambridge University Press, 2013, ISBN 978-1-107-01555-5	
Assessment	The total mark is based on evaluations of laboratory assignments reports, final assignment essay and presentation at the closing symposium. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4175	Radar II: Theory and System Design	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/0/4 (Q5)	
Education Period	1	
Start Education	5	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4169 Microwaves, Radar and Remote Sensing. Specifically, knowledge is expected on: microwaves basics, radar, scattering mechanisms.	
Course Contents	Radar systems play an important role in today's world. Aviation could not exist without them. But also shipping, traffic and remote sensing extensively employ radar. In today's systems, signal processing is the central theme that enables all radar applications. However, thorough understanding of the physical and electrical engineering radar principles are essential to the successful radar engineer. "Radar Systems" provides an introduction to the design and operation of modern radar systems. Topics include radar principles and designs, the interaction of radar with the environment, i.e. clutter and detection of targets, subsequent signal processing and signal to noise topics. Furthermore waveform design driven by range and Doppler measurements, suppression of ambiguities are treated. Finally modern developments in active arrays and digital beamforming are discussed.	
Study Goals	- The aim of this course is to advance students' knowledge in microwave radar systems. The course will build on the introductory course Microwaves, Radar and Remote Sensing. - The participants will gain sufficient knowledge to understand design, operation and signal processing of modern radar systems and architectures, taking into account radar clutter and detection theory.	
Education Method	The course consists of 10 regular lectures and a 1/2 day excursion to TNO, The Hague or a guest lecture.	
Literature and Study Materials	Kingsley and Quegan: Understanding Radar Systems; Scitech, ISBN 1-891121-05-7 "Principles of Modern Radar: Basic Principles", Mark A. Richards (Editor), James A. Scheer (Editor), William A. Holm (Editor); ISBN-13: 978-1891121524 additional handouts/ reader	
Assessment	Oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	By appointment with Esther de Klerk (E.J.DeKlerk@tudelft.nl)	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	J.A. Martinez Castaneda	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications.	
Education Method	Through assignments, students are expected to have the chance to work on the topics covered in the lectures. Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	-Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: $0.7 \cdot \text{Written exam} + 0.3 \cdot \text{Mandatory assignments}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable	

Exam Hours	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
	In case of in person examination at campus: The exam is closed book.
	A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side.
	In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measurement; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: a. Discuss a suitable visualization for a given medical problem. b. Implement the most suitable solution. c. Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: $0.4 * \text{Written exam} + 0.6 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course.	
Co-Instructor	In the first lecture, details on the evaluation and practical information on the course will be given. Prof.dr. E. Eisemann	

IN4314	Seminar Selected Topics in Multimedia Computing	5
Responsible Instructor	P.S. Cesar Garcia	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	signal (image, audio) processing, pattern recognition, networking and distributed systems	
Course Contents	Through all the exciting recent advances in digital media technology and the rapid growth of social media platforms, multimedia content is increasingly embedded in our daily lives, gaining enormous potential in improving the traditional educational, professional, business, communication and entertainment processes. To be able to use this potential for transferring these processes into user-centric interactive multimedia applications, technology is required that can help us access, deliver, enrich and share rich-media content. This course provides insight into the state-of-the-art cross-disciplinary research efforts related to the development of such technology. The topics covered by the course include, but are not limited to, multimedia systems (transport and delivery, telepresence and VR, mobile), multimedia experiences (Quality of Experience, Collaboration), and multimedia engagement (emotional and social signals and social multimedia).	
Study Goals	To become acquainted with the state-of-the-art research and development activities in the field of Multimedia Computing, and to become an expert in one particular "hot topic", such that they are able to identify the "knowledge gap" (i.e., the place in which more research is needed in order to advance the state of the art).	
Education Method	readings, seminar discussions, presentations, survey paper	
Literature and Study Materials	Readings, possibly including video lectures.	
Assessment	The final grade of the course consists of the following components: Presentation on a pre-existing survey (weighting 10%) Writing own survey on a new topic (weighting 65%) Presentation on own survey topic (weighting 25%) Final grade calculation: $0.1 * \text{Presentation pre-existing survey} + 0.65 * \text{Writing own survey} + 0.25 * \text{Presentation own survey}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation on a pre-existing survey: resit Writing own survey on a new topic: re-submit deliverable Presentation on own survey topic: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory Fundamentals of Coding in Python and MatLab	
Course Contents	Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems (direct and inverse) * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control * Examples of applications	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: Practical assignment and written exam, open book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity for a resit if they fail the written examination. * The groups that are graded between 5 and 5.75 in an assignment, will have two weeks to integrate the assessors feedback. If this is done in a satisfactory way, they can be rewarded with a mark that is not higher than 6/10. The exact instruction for the addition will be announced on Brightspace.	
Tags	Artificial intelligence Complex Mathematics Intensive Linear Algebra Mathematics Modelling Programming Project Proofs	
Department	3mE Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42015	Control Theory	6
Responsible Instructor	T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15% grade) and written final examination (85% grade) will determine the total final grade. The mid-term result is only valid for the academic year when it was obtained. For the resit examination there will be a written examination (100% grade) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Department	3mE Department Delft Center for Systems and Control	

SC42035	Integration Project Systems and Control	5
Responsible Instructor	D. Boskos	
Instructor	Dr. L. Laurenti	
Instructor	Ir. S.C. Bregman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	SC42015 Control Theory SC42155 Modeling of Dynamical Systems SC42095 Control Engineering SC42025 Filtering and Identification	
Course Contents	The course is based on practical laboratory sessions, in which students will gain hands-on experience with the application of control theory to real-world systems. Matlab and Simulink are used as the primary software environment for the design, analysis, and real-time implementation of the algorithms. Students work in groups of two in the lab, with a setup of their choice. The setups include for instance the rotary pendulum, helicopter model, inverted pendulum on a cart, rotational pendulum, reaction wheel pendulum, and magnetic levitation system.	
Study Goals	The goal of this course is to integrate and apply the theoretical knowledge gained in the compulsory courses of the Systems and Control curriculum. These courses cover the subjects of Control Theory, Physical Modeling, Filtering and Identification. The concepts and tools to be used include mechanistic modeling (based on principles like mass balances, Lagrange equations, etc.), filtering and estimation (e.g., Kalman filtering), linear control design and performance analysis, and system identification in open and closed loop. It is assumed that the students already know these concepts or are able to look them up in the literature. At the end of the course, the students should be able to: - Develop a mathematical model of a real-world dynamical system - Build a simulation model of the system in Matlab/Simulink - Identify the model parameters - Design a state estimator to recover unmeasured system states - Design controllers that regulate the derived system model - Integrate their acquired control engineering knowledge in laboratory setups	
Education Method	Project, demonstration	
Assessment	There is no written exam. The final grade is determined on the basis of a written report (50% of the grade), the final discussion of the results with the lecturers (50%) and the performance during the entire course period (rounding off the grade).	
Remarks	Old course code: SC52035	
Department	3mE Department Delft Center for Systems and Control	

SC42075	Modeling and Control of Hybrid Systems	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	core Systems & Control topics (including state-space models, discrete-time and continuous-time models, stability, linearization), optimization basics (linear programming), Matlab	
Course Contents	General introduction, examples of hybrid systems and motivation Modeling frameworks (automata, hybrid automata, piecewise-affine systems, complementarity systems, mixed logic dynamical systems, Petri nets) Properties and analysis of hybrid systems (well-posedness, Zeno behavior, stability, liveness, safety, ...) Control of hybrid systems (switching controllers, model predictive control) Verification and tools	
Study Goals	Recent technological innovations have caused a considerable interest in the study of dynamical processes of a mixed continuous and discrete nature. Such processes are called hybrid systems and are characterized by the interaction of time-continuous models (governed by differential or difference equations) on the one hand, and logic rules and discrete-event systems (described by, e.g., automata, finite state machines, etc.) on the other. A hybrid system also arises in practice when continuous physical processes are controlled via embedded software that intrinsically has a finite number of states only (e.g., on/off control). Recent interest in hybrid systems is stimulated by developments in nonlinear control theory, intelligent control, adaptive control, and computer science. The purpose of the course is to introduce a variety of hybrid systems modeling, analysis and control techniques.	
Education Method	lectures + assignment Depending on the corona measures, the lectures will be taught on-campus or online.	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study. written exam (individual, closed-book, no calculators, counts for 60% of the final marks) + assignment (group assignment, assessed through written report, counts for 40% of the final marks) In case of unforeseen circumstances or measures resulting from COVID-19, the exam will be organized as an open-book take-home exam. Important: partial marks for exam or assignment do not carry over from one academic year to the next	
Department	3mE Department Delft Center for Systems and Control	
Contact	Questions are preferably asked via the Discussion forum on Brightspace, or (for questions on the material of the lectures) during the online or on-campus lectures	

SC42125	Model Predictive Control	4
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Model Predictive Control (MPC) is perhaps the most effective optimal control strategy for constrained dynamical systems. The basic concept of MPC is to exploit a dynamic model to forecast the system behavior, and optimize the forecast to determine the control input as the best decision at the current time. With emphasis on constrained linear systems, the course will present the theoretical fundamentals of MPC, such as Lyapunov stability, optimality and robustness, and its computational methods, such as quadratic programming.	
Study Goals	- Derive state prediction matrices from discrete-time linear models - Design the cost function, state and input constraints - Design MPC controllers with guaranteed recursive feasibility and asymptotic stability via appropriate terminal cost and constraint set - Formulate and solve constrained-linear-quadratic MPC problems via quadratic programming - Implement and simulate closed-loop systems controlled by MPC on Matlab - Design MPC controllers with integral action for reference tracking	
Education Method	Lectures, instructions.	
Assessment	Final project and oral exam.	
Elective	Yes	
Tags	Mathematics Matlab Optimalisation	
Department	3mE Department Delft Center for Systems and Control	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Free elective courses S&S 2023

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Suggested profiles S&S 2023	
Introduction 1	The tracks in EE typically have an informal sub-structure called profiles. A profile is a more or less canned study path. It typically prepares for a thesis subject in a certain area and typically with a specific professor or small set of professors of one section, overseeing that research area. For example, the Microelectronics track offers the Digital Systems profile, the RF/Analog Circuit Design profile, the Microsystems & Technology profile and the Biomedical Electronics profile. Other tracks offer different profiles. For the purposes of composing an IEP, a profile presents a short list of coherent courses (in all tiers) that together leads to a coherent program of study that prepare for a thesis topic in the area of the profile. As such, a profile can specify (at most) two from three courses in the common core, two from three courses in the track core, and at most 12 EC of additional specialization courses. Note, however, that profiles are nothing more than convenient selections of courses that go well together, and do not imply any formal restriction whatsoever. Based on individual interest and subject to approval of the IEP by the (tentative) thesis supervisor and the MSc coordinator and the responsible professor, a much wider variety of r programs is possible. Students are free to compose their individual program within the rules as specified in FIXME.

Array Processing 2023	
Introduction 1	Array processing focuses on the processing, modelling and analysis of signals that are acquired by arrays of sensors. Although the application areas, the necessary domain knowledge and type of sensors (e.g., antennas, microphones, electrodes, etc.) can be rather diverse, the underlying fundamentals and methodology that forms the basis for the different applications shows many similarities. The different application areas that consider the use of array processing are therefore gathered in this profile called array processing and are therefore rather diverse. It includes, among other things: - Signal processing for communication (Prof. dr. ir. G.J.T. Leus: G.J.T.Leus@tudelft.nl) - Audio and acoustic signal processing (Dr. ir. R. C. Hendriks & Dr. J. Martinez: r.c.hendriks@tudelft.nl/ J.A.MartinezCastaneda@tudelft.nl) - biomedical signal processing (Dr. ir. R. C. Hendriks & Prof. dr. ir. G.J.T. Leus: r.c.hendriks@tudelft.nl / G.J.T.Leus@tudelft.nl) - Astronomy (Prof. dr. ir. A. J. van der Veen: a.j.vanderveen@tudelft.nl) - Array processing for Radar (Prof. dr. ir. A. J. van der Veen & Prof. A. Yarovoy: a.j.vanderveen@tudelft.nl / A.Yarovoy@tudelft.nl)

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4540	Distributed Signal Processing	5
Responsible Instructor	Prof.dr.ir. R. Heusdens	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In the course Distributed Signal Processing, attention will be paid to decentralised signal processing techniques. Due to the explosion in size and complexity of modern datasets, it is increasingly important to be able to solve problems with a very large number of features or training examples. In industry, this trend has been referred to as Big Data, and it has had a significant impact in areas as varied as machine learning, internet applications, computational biology, medicine, finance, marketing, journalism, network analysis, weather forecast, telecommunication, and logistics. As a result, both the decentralised collection or storage of these datasets as well as accompanying distributed solution methods are either necessary or at least highly desirable. Many problems of interest in the above mentioned areas can be posed in the framework of convex optimisation. Even in cases where the cost function is non-convex, for example in algorithms that operate in high-dimensional spaces or that train non-linear models such as tensor models and deep neural networks, convex optimisation is used to solve (convex) relaxed optimisation problems, as non-convex problems are often NP-hard to solve. Algorithms for distributed signal processing can, in many cases, be analysed in a unified manner using the abstraction of monotone operators and fixed-point theory. We, therefore, will focus on these topics in more detail during the course and use this to solve certain problems in a distributed way. We will analyse the algorithms not only in terms of convergence rate, but will consider the sensitivity to transmission failures and privacy-related issues as well. We will consider the following topics: Distributed consensus algorithms * synchronous/asynchronous averaging * Randomized gossip * Geographic gossip * Gossip with eavesdropping * Sum-weight averaging Distributed optimisation * Jacobi and Gauss-Seidel algorithm * (Sub)gradient descent * Dual ascent, dual decomposition * Method of multipliers * Alternating direction method of multipliers (ADMM) * Primal-dual method of multipliers (PDMM) Monotone operator theory * Monotone operators * Fixed-point iterations * Forward-backward splitting * Peaceman-Rachford splitting * Douglas-Rachford splitting Privacy and security * Differential privacy * Secure multi-party computation * Subspace perturbation	
Study Goals	The student * is able to explain the use of convex optimization in decentralized signal processing * is able to discuss the structure and principles of convex optimization techniques * is able to discuss primal and dual properties of optimization algorithms * is able to implement state-of-the-art convex optimization algorithms for decentralized signal processing * is able to explain the use of monotone operator theory in the design of distributed optimization algorithms * is able to determine the convergence rate of distributed algorithms	
Education Method	lectures, mini projects + hands-on exercises	
Assessment	Oral examination + evaluation of the mini projects	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor analysis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: $0.5 * \text{Oral discussion 1} + 0.5 * \text{Oral discussion 2}$. Both partial grades must be sufficient, i.e., a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4750	Tensor Networks for Green AI and Signal Processing	4
Responsible Instructor	B. Hunyadi	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	2/0/0/0 lectures and 4/0/0/0 computer sessions	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Course aims and objectives This course provides a solid mathematical foundation to tensor networks and discusses their use in machine learning and signal processing. Tensors are multi-dimensional generalization of matrices. Real-life data often comes in high dimensions, such as of a video stream, or multichannel EEG recorded from multiple patients under various conditions. Tensor-based tools are crucial to efficiently represent and process this data. Alternative (matrix-based) solutions artificially segment such high-dimensional data into shorter one- or two-dimensional arrays, causing information loss by destroying correlations in these data. The key idea of tensor networks is to write a tensor in terms of much smaller tensors that represent these correlations among the different dimensions. On one hand, such representations can reveal hidden structure in the data and can even lead to decompositions with physically interpretable terms. Moreover, in this way, the original tensor never needs to be kept explicitly in memory. This latter aspect becomes especially interesting considering the ever-increasing dataset sizes brought on by recent advances in sensor and imaging technology. Computations based on tensor networks enable faster training and inference of AI models based on such large datasets. Therefore, tensor networks facilitate the shift towards the direction of Green AI. Green AI aims to decrease the environmental footprint of AI computation and increase the inclusivity of AI research by becoming more independent of expensive computer hardware needed for training. Course contents and description The first introductory lecture gives a motivation to the use of tensor networks. Various compelling examples from the teachers own research experience will be given. Next, the basic notation and the use of tensor diagrams will be presented, and important matrix and tensor operations will be explained, such as various products (inner, outer, Kronecker, Khatri-Rao, etc.), norms, reshaping (matricization, vectorization). These operations and concepts are crucial to understand how matrix decompositions generalize to tensor methods. This leads to the core of the course content. The most well-known tensor network is the singular value decomposition (SVD) of a matrix. The following two lectures present how the SVD can be generalized to tensors in two different ways, one leading to the multi-linear singular value decomposition (MLSVD), and the other leading to the canonical polyadic decomposition (CPD). The properties, computation and applications of both generalizations will be discussed. The third tensor network structure that generalizes the matrix SVD are tensor trains (TT). Their properties, computation and applications will also be discussed. TTs are particularly well-suited to represent large-scale vectors and matrices as most of the common linear algebra operations can be done with TTs. Finally, the last two lectures focus on the application of tensors to various signal processing (e.g. blind source separation, data fusion) and machine learning problems (e.g. kernel methods). For this, we will introduce advanced concepts such as different tensorization techniques, constrained and coupled decompositions. Throughout the course, various applications will be presented. From the field of biomedical signal processing this includes epileptic seizure localization in EEG, fusion of simultaneous EEG-fMRI, blind deconvolution of brain responses in functional ultrasound during visual information processing or prostate cancer detection using ultrasound. The course explains how tensor networks pave the way towards Green AI. This is done by showing how machine learning methods like kernel machines, artificial neural networks, and Gaussian processes can be made more efficient and tractable for big data and higher dimensional problems.	
Course Contents Continuation	Educational method: The course consists of 8 2-hour lectures of theory, including examples and applications. There will be 4 2-hour computer lab sessions that discuss practical implementation and applications. These lab sessions help student complete their computer assignments, which are an important part of their assessment. The examples in the assignments are taken e.g. from biomedical applications (ECG, EEG data), computer vision, etc.	
Study Goals	After the course, the student is able to: - Explain different tensor networks (TN) - Implement basic tensor networks - Formulate a basic data science / signal processing problem as a TN - Make informed choices in the practical use of TNs, such as rank or initialization method. - Formulate research questions for their own (master thesis level) research or independent literature study in the field and its applications, i.e. tensor-based (biomedical) signal processing, image analysis, control, etc.	
Education Method	8 lectures and 4 computer sessions	
Literature and Study Materials	The course covers chapters Part I and selected chapters of Part II of the book below. Further applications and examples will be provided from recent scientific literature and the teachers own research experience. Andrzej Cichocki, Namgil Lee, Ivan Oseledets, Anh-Huy Phan, Qibin Zhao and Danilo P. Mandic (2016), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 1 Low-Rank Tensor Decompositions", Foundations and Trends® in Machine Learning: Vol. 9: No. 4-5, pp 249-429. http://dx.doi.org/10.1561/22000000059 Andrzej Cichocki, Anh-Huy Phan, Qibin Zhao, Namgil Lee, Ivan Oseledets, Masashi Sugiyama and Danilo P. Mandic (2017), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 2 Applications and Future Perspectives", Foundations and Trends® in Machine Learning: Vol. 9: No. 6, pp 431-673. http://dx.doi.org/10.1561/22000000067	
Assessment	The students will complete 4 assignments. Each assignment will be discussed during the corresponding computer lab session. The first 3 assignments are common for all students, and they are completed individually. The final assignment is executed in pairs, and the topic is chosen from a list of several options, depending on the interest of the students. The students will submit their code as well as a report about the assignments. The scores of the assignments contribute to final grade with the following weights: 5% / 10% / 10% / 75% and students need to pass each individual assignment. The final assignment will also be defended with an individual oral examination, where theoretical questions related to the project will be asked. In case of a retake a new final assignment has to be completed. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Artificial intelligence	
Co-Instructor	Dr.ir. K. Batselier	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Biomedical Signal Processing 2023

Responsible Program Employee	Prof.dr.ir. W.A. Serdijn
Introduction 1	Biomedical Signal Processing focuses on signal processing for the acquisition, modeling, analysis and processing of biomedical signals. These can be natural signals from the body as with the signals measured in the heart, or brain (EEG, ECG) multichannel signals from imaging systems (e.g. ultrasound, MRI or photo-acoustic imaging), or modeling of systems like the human hearing or the cardiac system. Contact persons: Dr. ir. R. F. Remis (R.F.Remis@tudelft.nl) Dr. B. Hunyadi (B.Hunyadi@tudelft.nl) Dr. ir. R. C. Hendriks (R.C.Hendriks@tudelft.nl) Prof. G.J.T. Leus (G.J.T.Leus@tudelft.nl) Dr. J.H.G. Dauwels (J.H.G.Dauwels@tudelft.nl) The following list of specialization courses is suggested:

AP3232	Medical Imaging Signals and Systems	6
Responsible Instructor	Dr.ir. M.C. Goorden	
Instructor	Dr. D. Maresca	
Instructor	Dr. S.D. Weingärtner	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A thorough mathematical basis is essential to follow the class. This can for example include Linear Algebra I + II and Calculus I + II. An in-depth understanding of Fourier analysis and basic physical understanding of electromagnetism is necessary."	
Course Contents	The course Medical Imaging Signals and Systems covers the principles of advanced medical imaging modalities. Particularly, students will study the physics, acquisition techniques and signal processing underlying conventional X-ray, Computerized Tomography (CT), Single Photon Emission Computed Tomography (SPECT), Positron Emission Tomography (PET), Ultrasound (US) and Magnetic Resonance Imaging (MRI). Course Content: Introductory lecture with the history of medical imaging and its importance for modern medicine and biomedical research. Conventional X-ray, CT, SPECT and PET imaging (33% of content): Basic principles of planar X-ray imaging and scintigraphy (short); Advanced methods for image acquisition using CT, SPECT and PET; Mathematical techniques for image reconstruction, including filtered backprojection and iterative approaches; Issues of image quality, especially related to resolution, signal to noise ratio, quantitative and dynamic imaging. MR imaging (33% of content): Fundamentals of MRI: spin, Larmor frequency, T1, T2, T2* contrast (short); Spatial encoding: K-space; Image Quality: PSF, signal strength and noise level; Advanced MRI techniques such as diffusion, perfusion and flow imaging. Ultrasound imaging (33% of content): Physical and mathematical principles of ultrasound imaging, including probe design and advanced US imaging techniques such as harmonic imaging and non-linear reconstruction techniques for imaging tissue properties. The course will end with a mini-symposium on specialized imaging techniques. Therefore, recently developed medical imaging methods will be studied, going beyond the theory of the lectures. Students working in pairs will write a short essay on a selected technique and present it to the group.	
Study Goals	1. Acquire in-depth knowledge about the physics and image reconstruction underlying X-ray, CT, acoustical and magnetic resonance imaging; 2. Being able to solve elementary problems related to the theory mentioned in 1; 3. Being able to solve more advanced problems addressing the theory mentioned in 1 by combining mathematical skills and physical insight; 4. Able to acquire new knowledge about clinical a medical imaging technique; 5. Being able to present newly acquired knowledge about a medical imaging technique.	
Education Method	Lectures and an assignment.	
Literature and Study Materials	Book: Medical Imaging Signals & Systems, Jerry L. Prince, Jonathan Links. Upper Saddle River NJ: Prentice Hall, second edition, 2014, 544 pp, ISBN-13: 978-0132145183. We will provide Chapter 1 "Brief introduction to MRI physics" of the book: "Magnetic Resonance Image Reconstruction Theory, Methods, and Applications", 1st Edition - November 4, 2022, Editors: Mehmet Akcakaya, Mariya Ivanova Doneva, Claudia Prieto, Paperback ISBN: 9780128227268 , eBook ISBN: 9780128227466	
Assessment	Additional handouts wherever necessary. The course on Medical Imaging comes with an assignment for which you will need to hand in a paper and make a presentation. This is an obligatory part, for which you will get a grade from 1-10 (A). Additionally, there will be an exam also graded from 1-10 (B). The grades for both parts needs to be at least 6.0 in order to pass. The final grade is calculated by: $(A + B)/2$. It is not possible to take part in the exam if you have not made the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Pocket calculator.	
Elective	Yes	

AP3531	Acoustical Imaging	6
Responsible Instructor	Dr. D. Maresca	
Instructor	Dr.ing. P.L.M.J. van Neer	
Instructor	Dr.ir. D.J. Verschuur	
Instructor	Dr.ir. M.D. Verweij	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	B.Sc.-level knowledge on wave phenomena and wave theory B.Sc.-level knowledge on Fourier transforms and signal processing. Basic knowledge of programming in Matlab or Python	
Course Contents	Applications of acoustical imaging in medical imaging, seismic imaging and non-destructive testing of construction materials. Review of Fourier and Radon transforms. The 3D acoustic wave equation and its properties. Derivation of the Kirchhoff and Rayleigh integral for acoustic waves. Forward and inverse wave field extrapolation in homogeneous and inhomogeneous media and its use in acoustical imaging. Resolution of the acoustic images and estimation of background propagation velocities. Medical ultrasound: basic physical principles, principles of ultrasound applications in medicine and neuroscience, Doppler imaging, ultrasound contrast agents. Ultrasound transducers and beamforming. The non-linear wave equation and harmonic imaging. Medical ultrasound imaging using contrast agents. Inverse scattering theory and full waveform inversion. Invited medical lectures on ultrasound imaging in cardiology and neurosurgery.	
Study Goals	Thorough understanding of the principles underlying acoustical imaging. Physical aspects and practical insights in the properties of wavefields in a medium. Understand the basic methodologies for geophysical and medical imaging, their resemblances and differences. Acquire comprehension of the potential and limits of current imaging techniques and the value of non-linear inversion techniques. Use theoretical concepts and bring them into practice.	
Education Method	Oral lectures and mandatory homework assignments in Matlab or Python.	
Literature and Study Materials	- Handouts of lecture notes - Book: The Principles of Quantitative Acoustical Imaging, by A. Gisolf and D.J. Verschuur, EAGE Publications BV (for sale at secretary office of section MI/CI, room F268)	
Assessment	Examination after successful completion of homework assignments. Oral exam if nr of students <20, otherwise written exam. For the homework assignments there is only pass/no-pass	
Remarks	The lectures are given by several lecturers: Dr. Eric Verschuur, Dr. Martin Verweij, Dr. David Maresca and Dr. Paul van Neer. Invited lecturers will cover specific topics, including medical lectures.	
Elective	Yes	
Tags	Image processing Matlab Physics	

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41095	Medical Devices: Past, Present, Future	3
Responsible Instructor	Dr. D. Dodou	
Instructor	Prof.dr. J.J. van den Dobbelsteen	
Instructor	Prof.dr. J. Dankelman	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. History of surgery 2. Minimally invasive techniques 3. Single-port procedures & NOTES 4. Percutaneous interventions 5. Robotic surgery 6. In vivo robots 7. Medical safety 8. Maintenance of surgical instruments	
Study Goals	The student will be able to describe the function, effectiveness, and challenges of surgical instruments in minimally invasive procedures. More specifically, the student will be able to: 1. illustrate advantages and disadvantages of minimally invasive surgery. 1. illustrate advantages and disadvantages of single-port procedures and NOTES. 3. describe instruments used during minimally invasive surgery, single-port procedures, and NOTES. 4. formulate criteria that surgical instruments have to fulfill in order to prevent sterilization problems. 5. indicate problems with quality control of surgical instruments. 6. describe the fields where robotic surgery is used and indicate the possibilities and limitations of the use of robotics. 7. classify technologies developed to train surgeons outside the operating theater. 8. discuss safety issues in surgery. 9. select information from the literature related to a certain issue and evaluate new developments.	
Education Method	Lectures. 4 hours per week	
Assessment	Homework assignments	
Department	3mE Department Biomechanical Engineering	

EE4555	Active Implantable Biomedical Microsystems	5
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Responsible Instructor	Dr.ir. C. Strydis	
Responsible Instructor	Dr. V. Giagka	
Responsible Instructor	T.M. Lopes Marta da Costa	
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none none	
Course Language	English	
Required for	- MSc Biomedical Engineering, Medical Devices track; - MSc Electrical Engineering, Biomedical Devices profile; - MSc thesis projects in the Section Bioelectronics.	
Expected prior knowledge	It is recommended but not required to have completed/attended the following courses: 1. Anatomy and Physiology (BM41055) 2. Fundamentals of Biomedical Engineering (BM41140) 3. Bioelectricity (ET4130) 4. Themes in Biomedical Electronics (ET4127) 5. Sensors and actuators (ET4257) 6. Measurement and Instrumentation (EE4C08) 7. Systems Engineering (EE4C02)	
Course Contents	Active Implantable Biomedical Microsystems (AIBM) in the human body are now becoming more widely acceptable and available since the development of cardiac pacemakers. Today, miniaturized wireless implantable systems are changing the face of biomedical research and clinical practices through the development of cochlear implants, neuroprostheses, braincomputer interfaces, deep organ pressure sensors, precise drug delivery units, bioelectronic medicine and electroceuticals. With a greater life expectancy and an increasing demand for medical healthcare to guarantee a sufficient health-related quality of life, there is a greater demand on technology and electrical and biomedical engineers to develop active implantable systems for a wide variety of medical diagnostics, treatments and therapies. This course provides a background on the application of AIBMs, an introduction to the latest techniques used and examples of existing medical devices. It can be used as a guide to the design of AIBMs and also as a reference for existing medical devices. The course is aimed at MSc students in Electrical Engineering (tracks Microelectronics, and Signals&Systems) and in Biomedical Engineering (track Medical Devices) who have an interest in the research and design of AIBMs. The following topics are treated: 1. Electrophysiology 2. Electrodes, optodes, sensors and actuators 3. Electrode characterization and modeling 4. Electronic system design 5. Stimulation electronics 6. Front-end electronics (signal conditioning, amplifiers, filters, ADCs, DACs, etc.) 7. Wireless data transfer 8. Energy harvesting and power delivery 9. Applications: cochlear implants and cardiac pacemakers 10. Digital design 11. Biocompatibility, materials and fabrication considerations 12. Assembly, system integration and packaging 13. Application: spinal cord stimulator 14. Reliability and security 15. Risk management and/or 16. Ethics	
Study Goals	After successful completion of the course, you: 1. Know, understand and can apply (3) the functionality, construction and application of active implantable biomedical microsystems (AIBMs) and their sub-systems and parts; 2. Know, understand and can apply (3) the fundamental principles and approaches for designing an implantable system; 3. Know, understand, can apply, analyse and can reason about (evaluate; 5) the design requirements, application-specific implementation challenges, and appropriate design choices of existing AIBMs, incl. cardiac pacemakers, cochlear implants and spinal cord stimulators; 4. Know and understand (2) the most important implantability, biocompatibility, reliability, safety and security issues of AIBMs; 5. Know and understand (2) the most important regulatory and clinical constraints of AIBMs 6. Can reason (5) about future developments of AIBMs, a.o. electroceuticals and bioelectronic medicine; NB. The numbers correspond with Bloom's taxonomy.	
Education Method	1. Lectures 2. Self study 3. Case study 4. Group project	
Literature and Study Materials	1. a collection of book chapters, thesis chapters and review/overview papers; 2. handouts of the slides used during the lectures, 3. video recordings of the lectures all made available via the online learning system (Brightspace). 4. scientific papers (IEEE Xplore, PubMed); 5. patents; 6. user manuals; 7. patient manuals; 8. manufacturer websites; 9. the Internet in general, to be consulted by the students themselves.	
Assessment	The student has to: - Form a group (6 students/group) in the first week, based on affinity with the topics below. - As a group, perform a case study of a complete device, which will be announced in Wk. 4.2.	

- Investigate 6 topics:
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- Prepare an annotated slide set on the first 3 topics in Wk. 4.5
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Max. 12 slides (4 per group of 2 students), using a prescribed template
 - *To be uploaded on Brightspace before Wk 4.6
 - *To be commented on in written form and handed in by another group during the 1st lecture in Wk. 4.7
 - *Sub-group (of 2 students) on topic A comment on topic B; B -> C; C -> A
- Prepare a group presentation on the second 3 topics:
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- One student, to be randomly selected by the teachers on the day of the presentation, will do the presentation (max. 12 minutes) for the entire group in Wk. 4.10 or 4.11.
- Other groups will discuss the presentations (4 minutes) and ask questions and will comment briefly (max. 2 minutes per group / presentation) on all presentations

The teachers will:

- Organize/manage the course
- Give the lectures
- Provide relevant material and background knowledge
- Define the case study
- Give the templates for the annotated slides and the comments
- Grade the slide sets per group of 2 students
- Grade the comments on the slide sets per group of 2 students
- Return the commented slide sets to the authors
- Buy the T-shirts to wear during the final presentations
- Grade the presentations per group of 6 students
- Grade the comments made on the presentations per group of 6 students
- Decide on the final marks

The final mark is based on the grade received for the annotated and commented slide sets (37.5% + 12.5% (bonus/malus), respectively), and the grade received for the group presentation (45%) and the comments made on another group presentation (5%).

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4650	Advanced Magnetic Resonance Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	A. Webb	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Vector algebra and vector calculus, linear circuit theory, signals and systems (Fourier transform, steady-state analysis), linear algebra and differential equations, electromagnetics (BSc level or higher)	
Course Contents	General introduction, history of MRI, general description of an MR scanner, general principles of MRI, applications and current research themes. General and brief description of spin, restriction to particles with spin $\frac{1}{2}$ (proton), equilibrium magnetization for spin $\frac{1}{2}$ particles (proton), the Bloch equation, general description of the gyromagnetic ratio, T1 and T2 relaxation times. Electromagnetic fields: Maxwells equations, the electromagnetic boundary conditions, the temporal and spatial Fourier transform, time-harmonic fields, the polarization state of an electromagnetic field (linear, circular, elliptical), Poyntings theorem (power balance), and reciprocity. Flipping the magnetization: received signal in terms of the electromagnetic receive field. Solutions of Blochs equation for various magnetic fields, the most important formula in MRI: formula for the Larmor frequency, chemical shift, scalar coupling, basics of 1-DNMR spectroscopy, elementary measurement sequences, spin-echo, inversion recovery, measuring T1 and T2, introducing T2*. Using gradients to spatially encode signals, k-space formalism, image reconstruction via multi-dimensional Fourier transforms, simple gradient echo and spin echo pulse sequences, three-dimensional imaging, angiography. Background knowledge for hardware design: Maxwells equations and quasi-static fields, the law of Biot-Savart, spherical harmonics. Interaction of RF with tissue as a function of frequency, different media/tissue types and the constitutive relations. RF coil and array design: design of RF coils and RF arrays, the Specific Absorption Rate (SAR). The electromagnetic field due to an electric dipole (small current-carrying piece of wire) and a magnetic dipole (small current-carrying loop). Gradient coil design and the static background field: Realizing gradient fields, Helmholtz and Maxwell coils, advanced gradient coil design using stream functions, consequences of nonuniform static background fields. Numerical modeling, software packages, body models, coil models. Measurements and imaging in practice, SNR and other practical aspects. Current/future research topics, parallel imaging, sparse sampling, electrical properties tomography	
Study Goals	After completing this course, the students will: 1. Understand the principles of magnetic resonance from both a quantum and classical mechanics point-of-view, 2. Understand the interaction of strong electromagnetic fields at high, medium and low frequencies, how these can be manipulated, and the safety features associated with these fields, 3. Understand how spatial information is encoded into magnetic resonance images, the origins of image artifacts, and the signal processing algorithms involved in image reconstruction, 4. Understand the relaxation phenomena associated with magnetic resonance, and how these are important in determining image contrast, 5. Understand how the different components of magnetic resonance systems are designed using Maxwells equations, spherical harmonics, and the Biot-Savart law, 6. Understand the current state-of-the-art of MRI hardware and image processing, and the new opportunities for future developments.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes. Will be made available via Brightspace.	
Assessment	Assignment and oral exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor analysis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: $0.5 * \text{Oral discussion 1} + 0.5 * \text{Oral discussion 2}$. Both partial grades must be sufficient, i.e., a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4750	Tensor Networks for Green AI and Signal Processing	4
Responsible Instructor	B. Hunyadi	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	2/0/0/0 lectures and 4/0/0/0 computer sessions	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Course aims and objectives This course provides a solid mathematical foundation to tensor networks and discusses their use in machine learning and signal processing. Tensors are multi-dimensional generalization of matrices. Real-life data often comes in high dimensions, such as of a video stream, or multichannel EEG recorded from multiple patients under various conditions. Tensor-based tools are crucial to efficiently represent and process this data. Alternative (matrix-based) solutions artificially segment such high-dimensional data into shorter one- or two-dimensional arrays, causing information loss by destroying correlations in these data. The key idea of tensor networks is to write a tensor in terms of much smaller tensors that represent these correlations among the different dimensions. On one hand, such representations can reveal hidden structure in the data and can even lead to decompositions with physically interpretable terms. Moreover, in this way, the original tensor never needs to be kept explicitly in memory. This latter aspect becomes especially interesting considering the ever-increasing dataset sizes brought on by recent advances in sensor and imaging technology. Computations based on tensor networks enable faster training and inference of AI models based on such large datasets. Therefore, tensor networks facilitate the shift towards the direction of Green AI. Green AI aims to decrease the environmental footprint of AI computation and increase the inclusivity of AI research by becoming more independent of expensive computer hardware needed for training. Course contents and description The first introductory lecture gives a motivation to the use of tensor networks. Various compelling examples from the teachers own research experience will be given. Next, the basic notation and the use of tensor diagrams will be presented, and important matrix and tensor operations will be explained, such as various products (inner, outer, Kronecker, Khatri-Rao, etc.), norms, reshaping (matricization, vectorization). These operations and concepts are crucial to understand how matrix decompositions generalize to tensor methods. This leads to the core of the course content. The most well-known tensor network is the singular value decomposition (SVD) of a matrix. The following two lectures present how the SVD can be generalized to tensors in two different ways, one leading to the multi-linear singular value decomposition (MLSVD), and the other leading to the canonical polyadic decomposition (CPD). The properties, computation and applications of both generalizations will be discussed. The third tensor network structure that generalizes the matrix SVD are tensor trains (TT). Their properties, computation and applications will also be discussed. TTs are particularly well-suited to represent large-scale vectors and matrices as most of the common linear algebra operations can be done with TTs. Finally, the last two lectures focus on the application of tensors to various signal processing (e.g. blind source separation, data fusion) and machine learning problems (e.g. kernel methods). For this, we will introduce advanced concepts such as different tensorization techniques, constrained and coupled decompositions. Throughout the course, various applications will be presented. From the field of biomedical signal processing this includes epileptic seizure localization in EEG, fusion of simultaneous EEG-fMRI, blind deconvolution of brain responses in functional ultrasound during visual information processing or prostate cancer detection using ultrasound. The course explains how tensor networks pave the way towards Green AI. This is done by showing how machine learning methods like kernel machines, artificial neural networks, and Gaussian processes can be made more efficient and tractable for big data and higher dimensional problems.	
Course Contents Continuation	Educational method: The course consists of 8 2-hour lectures of theory, including examples and applications. There will be 4 2-hour computer lab sessions that discuss practical implementation and applications. These lab sessions help student complete their computer assignments, which are an important part of their assessment. The examples in the assignments are taken e.g. from biomedical applications (ECG, EEG data), computer vision, etc.	
Study Goals	After the course, the student is able to: - Explain different tensor networks (TN) - Implement basic tensor networks - Formulate a basic data science / signal processing problem as a TN - Make informed choices in the practical use of TNs, such as rank or initialization method. - Formulate research questions for their own (master thesis level) research or independent literature study in the field and its applications, i.e. tensor-based (biomedical) signal processing, image analysis, control, etc.	
Education Method	8 lectures and 4 computer sessions	
Literature and Study Materials	The course covers chapters Part I and selected chapters of Part II of the book below. Further applications and examples will be provided from recent scientific literature and the teachers own research experience. Andrzej Cichocki, Namgil Lee, Ivan Oseledets, Anh-Huy Phan, Qibin Zhao and Danilo P. Mandic (2016), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 1 Low-Rank Tensor Decompositions", Foundations and Trends® in Machine Learning: Vol. 9: No. 4-5, pp 249-429. http://dx.doi.org/10.1561/22000000059 Andrzej Cichocki, Anh-Huy Phan, Qibin Zhao, Namgil Lee, Ivan Oseledets, Masashi Sugiyama and Danilo P. Mandic (2017), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 2 Applications and Future Perspectives", Foundations and Trends® in Machine Learning: Vol. 9: No. 6, pp 431-673. http://dx.doi.org/10.1561/22000000067	
Assessment	The students will complete 4 assignments. Each assignment will be discussed during the corresponding computer lab session. The first 3 assignments are common for all students, and they are completed individually. The final assignment is executed in pairs, and the topic is chosen from a list of several options, depending on the interest of the students. The students will submit their code as well as a report about the assignments. The scores of the assignments contribute to final grade with the following weights: 5% / 10% / 10% / 75% and students need to pass each individual assignment. The final assignment will also be defended with an individual oral examination, where theoretical questions related to the project will be asked. In case of a retake a new final assignment has to be completed. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Artificial intelligence	
Co-Instructor	Dr.ir. K. Batselier	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measurement; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: - Discuss a suitable visualization for a given medical problem. - Implement the most suitable solution. - Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: $0.4 * \text{Written exam} + 0.6 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eiseemann	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Signal Processing for Radar Systems and Remote Sensing 2023	
Introduction 1	Signal Processing for Remote Sensing and Radar Systems focuses on microwave system design, analogue and digital processing of microwave signals, signal processing for typical remote sensing applications (antenna array pattern synthesis; space-time adaptive signal processing in ultra-wideband arrays; detection and feature extraction; tracking and navigation; classification, and imaging) that are performed at either a single node or via distributed sensing. Contact persons: Prof. A. Yarovoy (A.Yarovoy@tudelft.nl) Prof. A.J. van der Veen (A.J.vanderVeen@tudelft.nl) The following list of specialization courses is suggested:

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4540	Distributed Signal Processing	5
Responsible Instructor	Prof.dr.ir. R. Heusdens	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In the course Distributed Signal Processing, attention will be paid to decentralised signal processing techniques. Due to the explosion in size and complexity of modern datasets, it is increasingly important to be able to solve problems with a very large number of features or training examples. In industry, this trend has been referred to as Big Data, and it has had a significant impact in areas as varied as machine learning, internet applications, computational biology, medicine, finance, marketing, journalism, network analysis, weather forecast, telecommunication, and logistics. As a result, both the decentralised collection or storage of these datasets as well as accompanying distributed solution methods are either necessary or at least highly desirable. Many problems of interest in the above mentioned areas can be posed in the framework of convex optimisation. Even in cases where the cost function is non-convex, for example in algorithms that operate in high-dimensional spaces or that train non-linear models such as tensor models and deep neural networks, convex optimisation is used to solve (convex) relaxed optimisation problems, as non-convex problems are often NP-hard to solve. Algorithms for distributed signal processing can, in many cases, be analysed in a unified manner using the abstraction of monotone operators and fixed-point theory. We, therefore, will focus on these topics in more detail during the course and use this to solve certain problems in a distributed way. We will analyse the algorithms not only in terms of convergence rate, but will consider the sensitivity to transmission failures and privacy-related issues as well. We will consider the following topics: Distributed consensus algorithms * synchronous/asynchronous averaging * Randomized gossip * Geographic gossip * Gossip with eavesdropping * Sum-weight averaging Distributed optimisation * Jacobi and Gauss-Seidel algorithm * (Sub)gradient descent * Dual ascent, dual decomposition * Method of multipliers * Alternating direction method of multipliers (ADMM) * Primal-dual method of multipliers (PDMM) Monotone operator theory * Monotone operators * Fixed-point iterations * Forward-backward splitting * Peaceman-Rachford splitting * Douglas-Rachford splitting Privacy and security * Differential privacy * Secure multi-party computation * Subspace perturbation	
Study Goals	The student * is able to explain the use of convex optimization in decentralized signal processing * is able to discuss the structure and principles of convex optimization techniques * is able to discuss primal and dual properties of optimization algorithms * is able to implement state-of-the-art convex optimization algorithms for decentralized signal processing * is able to explain the use of monotone operator theory in the design of distributed optimization algorithms * is able to determine the convergence rate of distributed algorithms	
Education Method	lectures, mini projects + hands-on exercises	
Assessment	Oral examination + evaluation of the mini projects In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4565	Propagation and Scattering of EM waves	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Instructor	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	ET4169,	
Expected prior knowledge	Basic knowledge of Electromagnetics, EE4C05 Electromagnetics	
Course Contents	The Propagation and Scattering of electromagnetic (EM) waves course provides an overview of the essential aspects of the physical layer for a wide range of applications that rely on electromagnetic waves for the information transmission and sensing (wireless communications, radar, and navigation). Both the propagation medium and environment change the characteristics of the electromagnetic waves and affects the quality of the wireless link. The course focuses on physical models of propagation and scattering phenomena with emphasis on practical estimations of these effects influence on operation and design of telecommunication systems. The following aspects will be covered in the course: * Overview of the radio spectrum, services, and frequency allocation, propagation/scattering environment, frequency band designation for EM waves * Characterization of propagation media * Antenna and noise concepts * EM waves interaction with material objects * EM wave scattering on an ensemble of discrete particles * EM wave scattering on a rough surface * EM wave scattering on a complex and distributed objects, the concept of radar cross-section as a characteristic observed with radar objects, * Direct transmission, * Power balance for communication and radar systems in cases of direct transmission, atmospheric gas attenuation, signal attenuation, and scattering in rain * Refraction - wave propagation through a stratified medium * Propagation near the Earth's surface * Diffraction on terrain, Fresnel zones, propagation over a plane/spherical Earth, practical propagation path analysis * Empirical Path Loss Models * Models of fading effects * Ionospheric propagation/scattering of electromagnetic waves Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	After this course students have to be able * to recognize and explain different mechanisms of electromagnetic waves (EMW) propagation/scattering and relations between them; * to assess different mechanisms of EMW propagation/scattering in relation to characteristics of EMW and media; * to estimate quantitatively the influence of the propagation and scattering mechanisms on trade-offs that have to be made when designing a wireless communication or sensing system. * to analyze real-world scenarios and identify the involved propagation and scattering mechanisms and their relevant aspects;	
Education Method	Lectures, instruction lectures, self-study recommended references, home exercises, practicum	
Literature and Study Materials	Levis, C.A. Radiowave propagation: physics and applications, Wiley, 2010 - 301 pages. ISBN-13: 978-0470542958 Barclay, L.W. Propagation of Radiowaves, IET, 1996, 2003. - 352 pages. ISBN-13: 978-0852961025	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: 10% of averaged grades for homeworks (max 1 point); 20% of averaged grades for practicum reports (max 2 points); 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Elektricity & Magnetism Information & Communication Practicals Signals and Systems Technology Telecommunication	

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

EE4675	Object classification with radar	4
Responsible Instructor	Dr. F. Fioranelli	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Advised: ET4169, specifically knowledge of microwave and radar basics such as measurements of range and velocity and related basic signal processing.	
Course Contents	While the idea of gaining information about the surrounding environment using electromagnetic waves is a relatively mature concept (a radar essentially!), a tremendous amount of research activities have recently coupled the analysis of data from EM sensors with techniques that can teach them to recognise automatically objects of interest. What natural or manmade objects are moving around, and what are they doing? Can we infer something about their behaviour or intention from such movements? These are an example of the big questions which data and clever signal processing from EM sensors can address. Applications range from the analysis of human gait and movements to the discrimination of birds vs Unmanned Aerial Vehicles (drones), to the identification of the subtle gestures people produce to interact with smart objects, amongst others. All these applications have in common two main areas of expertise: one has to do with understanding and processing the information encoded into the sensors signals (radar signal processing and data format); one has to do with the techniques to interpret this information automatically to make decisions (we could call this applied machine learning). In this course, we aim to provide a comprehensive introduction and explanation of both areas from an engineering application perspective, with the objective to learn how to use these tools for real-life applications in radar engineering. The course is broadly divided into two parts: 1. Part 1 will build on fundamentals of radar & microwave sensing courses to investigate the concept of micro-Doppler signatures, focusing on its modelling, the signal processing for its characterisation, and its relationships with the physical movement of objects and therefore their identity. Can we understand from the EM sensors data how the different parts of human bodies or manmade objects move? 2. Part 2 will introduce machine learning techniques for extraction of information from sensors data (feature extraction), for their selection, and for classification of objects of interest. Conventional supervised learning techniques will be presented (for example Nearest Neighbours and Support Vector Machine classifiers), along with some of the more recent data-driven approaches based on neural networks. The diversity of formats of the radar data for classification will also be considered, including micro-Doppler signatures, High Resolution Range Profiles, radar cubes. In both parts, students will be exposed to concrete examples and data directly taken from current activities at TU Delft, so that the research element is well embedded in the teaching.	
Study Goals	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The aim of this course is to introduce students to techniques to classify and identify objects of interest mixing machine learning and signal processing applied to data from radar sensors. By the end of this course, students will be able to: -- Apply relevant signal processing techniques to EM sensors and radar data to extract micro-Doppler signatures -- Compare the different domains of radar data that can be exploited for objects classification, and the convenience of using one rather than the other in different situations -- Analyse machine learning techniques for feature extraction and classification applied to radar sensing -- Propose possible approaches for given classification problems based on radar sensors data	
Education Method	The course consists of class lectures and practical activities in the laboratory where possible or following the lectures. Laboratory activities may take place at the existing facilities available for the MS3 (Microwave Sensing Signals & Systems) section.	
Course Relations	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). This course is intended as part of the Master program in Electrical Engineering, primarily for students in the WICOS and S&S tracks, but with expected interest also from students from the civil engineering faculty (CITG), mechanical/robotics engineering (3ME), and aerospace (LR), where activities in remote sensing and autonomous robotics have seen recent significant interest in radar sensing and radar-based classification. This course is envisaged to cover new topics, currently not present in the courses related to radar sensing on offer. In this regard the course is complementary to the existing courses of ET4169 (Radar I) providing the fundamentals of radar sensing and radar as an instrument, and ET4175 (Radar II) providing advanced information mostly focused on the theoretical basis of processing for radar detection and tracking. Therefore, this proposed new course fills a gap in presenting advanced content on the topic of radar-based classification and the related approaches, widening the offer of topics available to our students and contributing to their more well-rounded education. This new course can fit well with the new proposed order of the radar courses": EE4565 in Q2 (fundamentals of electromagnetic scattering and propagation, ET4169 in Q3 (basics of radar for broad audience of students), this course in Q4 (more specialised course in classification based on radar), ET4175 (more specialised course in advanced theory for radar detection and tracking). The new course also fits with the wider research strategy of the department, whereby classification and machine learning intersect the three overarching research themes of Next generation sensing and communication, Health and wellbeing, and Autonomous sensor systems. It is therefore expected that this course will enhance the research capabilities of students in view of their thesis projects, and benefit their employability afterwards. Specifically, radar-based classification for situational awareness is a timely topic, relevant to automotive (NXP), defence/security (Thales NL, TNO), robotics and HCI (Google has recently introduced a radar, Soli, within their phones for gesture-based interaction), healthcare (contactless estimation of vital signs). As part of the educational activities in this course, it is envisaged to have at least one guest lecture on relevant topic from an external stakeholder (for example from Thales NL and TNO).	
Literature and Study Materials	-Lecture slides and pre-class activities provided by the instructors -Selected chapters of the following books. They should all be available as e-book via the TU Delft Library. ---Richards, Mark A. Scheer, James A. Holm, William A.. (2010). Principles of Modern Radar, Volume I - Basic Principles. SciTech Publishing. ---Chen, Victor. Micro-Doppler Effect in Radar, Artech House, 2011. ---Chen, Victor C.; Tahmouh, David; Miceli, William J.. (2014). Radar Micro-Doppler Signatures: Processing and Applications. Institution of Engineering and Technology. ---Chen, Victor C.; Ling, Hao. (2002). Time-Frequency Transforms for Radar Imaging and Signal Analysis. Artech House. ---Gurbuz, Sevgi Zubeyde. (2021). Deep Neural Network Design for Radar Applications. Institution of Engineering and	

Assessment	Technology. ---Fioranelli, Francesco; Griffiths, Hugh; Ritchie, Matthew; Balleri, Alessio. (2020). Micro-Doppler Radar and its Applications. Institution of Engineering and Technology. ---Geron Aurelien. (2019). Hands-on machine learning with scikit-learn, keras, and tensorflow : concepts, tools, and techniques to build intelligent systems (Second). O'Reilly Media. -Selected published research articles available in IEEE Xplore (to be updated year by year, as novel research is generated - these are accessible via TU Delft Library)
	The final grade of the course consists of the following components: -a written exam with questions (that can be performed online in Brightspace if needed). -a report on practical/project activities (which may include collection and/or processing of some radar data, and implementation of a complete radar classification pipeline). The written exam is worth 75% of the final grade and the report(s) on practical activities 25%. As repair possibility the written exam has one resit opportunity. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE5020	Sensor Signal and Data Processing	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge obtained for instance via the course Detection and Estimation (ET4386) is usefull though not required.	
Course Contents	In this course a variety of modern signal and data processing techniques will be presented along with their application in sensor systems in general and surveillance (radar) systems in particular. Topics that will be presented include: detecting the presence and estimating the state of a single moving object based upon sensor measurements, modeling and estimation with multiple models, rejection of outliers in the measurements, the extension to a time-varying number of objects, the use of multiple sensors, and optimizing the resources of the sensor(s). Techniques that will be presented include: (extended) Kalman filtering, Interacting Multiple Model (IMM) filtering, non-linear filters such as the particle filter, Sequential Likelihood Ratio Testing (SLRT), Multiple Hypothesis Tracking (MHT), track-before-detect, data fusion concepts.	
Study Goals	The main goal of this course is to teach students to apply basic (and more advanced) detection and estimation theory to design algorithmic solutions to (complex) signal and data processing problems	
Education Method	Lectures, instruction lectures, home exercises	
Literature and Study Materials	Lecture notes, and for each topic specific background and overview literature will be provided.	
Assessment	Project work plus an oral exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4173	Introduction to UWB technology, systems and applications	4
Responsible Instructor	Prof.dr. O. Yarovy	
Instructor	O.A. Krasnov	
Responsible for assignments	Ir. P.J. Aubry	
Contact Hours / Week x/x/x/x	0,0,0,6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Ultra-wideband (UWB) technology is becoming of greater importance as the pressure for signal bandwidth increases. This course introduces students to UWB technology as it will be used in future communications, radar and navigation systems, and gives detailed attention to both UWB wireless systems and applications in the frequency (FD) and time (TD) domains. The course consists of three major parts. In Part 1 the fundamentals of UWB technology such as waveforms and spectra of short pulses, basic UWB signal processing, radiation and propagation of short electromagnetic pulses are discussed. In Part 2 of the course is dedicated to realizations of UWB technology and systems. General approach to UWB wireless system design is introduced. UWB system design implementation of the key facets of modern UWB systems in both the frequency (FD) and time (TD) domains is discussed. In Part 3 applications of UWB technology in telecommunications, radar, imaging and positioning are discussed. General discussion of the system-level trade offs of such UWB systems is complemented by a detailed analysis of practical systems designed and implemented at TU-Delft. Lectures in the course are complemented by laboratories, at which students get hands-on experience in UWB signal processing and get in touch with wireless applications of UWB systems.	
Study Goals	This course introduces students to UWB technology as it is used in telecommunications, radar and navigation systems, and gives detailed attention to UWB wireless systems. After finishing this course, students will understand UWB signal properties in time and frequency domains, impact of UWB signal properties on wireless system architecture and performance, become familiar with available UWB technologies and UWB wireless system designs, and understand advantages and limitations of UWB technology for different applications. Students will also get hands-on experience with UWB signals and systems.	
Education Method	The course consists of lectures, supervised laboratories with experimental UWB equipment and the final assignment. The course finishes with a symposium, at which the final assignments are presented.	
Literature and Study Materials	Lecture handouts; course books, research papers. M.-G. Di Benedetto, G. Giancola Understanding Ultra Wide Band Radio Fundamentals, Prentice Hall PTR, 2004, ISBN-0-13-148003-0; T. Zwick et al., Ultra-wideband RF system engineering, Cambridge University Press, 2013, ISBN 978-1-107-01555-5	
Assessment	The total mark is based on evaluations of laboratory assignments reports, final assignment essay and presentation at the closing symposium. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4175	Radar II: Theory and System Design	4
Responsible Instructor	Dr.ir. J.N. Driessen	
Contact Hours / Week x/x/x/x	0/0/0/4 (Q5)	
Education Period	1	
Start Education	5	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4169 Microwaves, Radar and Remote Sensing. Specifically, knowledge is expected on: microwaves basics, radar, scattering mechanisms.	
Course Contents	Radar systems play an important role in today's world. Aviation could not exist without them. But also shipping, traffic and remote sensing extensively employ radar. In today's systems, signal processing is the central theme that enables all radar applications. However, thorough understanding of the physical and electrical engineering radar principles are essential to the successful radar engineer. "Radar Systems" provides an introduction to the design and operation of modern radar systems. Topics include radar principles and designs, the interaction of radar with the environment, i.e. clutter and detection of targets, subsequent signal processing and signal to noise topics. Furthermore waveform design driven by range and Doppler measurements, suppression of ambiguities are treated. Finally modern developments in active arrays and digital beamforming are discussed.	
Study Goals	- The aim of this course is to advance students' knowledge in microwave radar systems. The course will build on the introductory course Microwaves, Radar and Remote Sensing. - The participants will gain sufficient knowledge to understand design, operation and signal processing of modern radar systems and architectures, taking into account radar clutter and detection theory.	
Education Method	The course consists of 10 regular lectures and a 1/2 day excursion to TNO, The Hague or a guest lecture.	
Literature and Study Materials	Kingsley and Quegan: Understanding Radar Systems; Scitech, ISBN 1-891121-05-7 "Principles of Modern Radar: Basic Principles", Mark A. Richards (Editor), James A. Scheer (Editor), William A. Holm (Editor); ISBN-13: 978-1891121524 additional handouts/ reader	
Assessment	Oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	By appointment with Esther de Klerk (E.J.DeKlerk@tudelft.nl)	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Systems and Control for EE 2023

Introduction 1

Systems and Control focuses on the development of mathematical models of complex dynamic systems, as well as on the use of these models to optimize/control their behavior in feed-forward and feedback configurations. The strength of feedback control rests in its capacity for dealing with uncertainty. This makes it possible to design low-complexity controllers for achieving high performance in controlled complex systems

Contact persons:
Dr. M. Mazo Espinosa (M.Mazo@tudelft.nl)

The following list of specialization courses is suggested:

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4540	Distributed Signal Processing	5
Responsible Instructor	Prof.dr.ir. R. Heusdens	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In the course Distributed Signal Processing, attention will be paid to decentralised signal processing techniques. Due to the explosion in size and complexity of modern datasets, it is increasingly important to be able to solve problems with a very large number of features or training examples. In industry, this trend has been referred to as Big Data, and it has had a significant impact in areas as varied as machine learning, internet applications, computational biology, medicine, finance, marketing, journalism, network analysis, weather forecast, telecommunication, and logistics. As a result, both the decentralised collection or storage of these datasets as well as accompanying distributed solution methods are either necessary or at least highly desirable. Many problems of interest in the above mentioned areas can be posed in the framework of convex optimisation. Even in cases where the cost function is non-convex, for example in algorithms that operate in high-dimensional spaces or that train non-linear models such as tensor models and deep neural networks, convex optimisation is used to solve (convex) relaxed optimisation problems, as non-convex problems are often NP-hard to solve. Algorithms for distributed signal processing can, in many cases, be analysed in a unified manner using the abstraction of monotone operators and fixed-point theory. We, therefore, will focus on these topics in more detail during the course and use this to solve certain problems in a distributed way. We will analyse the algorithms not only in terms of convergence rate, but will consider the sensitivity to transmission failures and privacy-related issues as well. We will consider the following topics: Distributed consensus algorithms * synchronous/asynchronous averaging * Randomized gossip * Geographic gossip * Gossip with eavesdropping * Sum-weight averaging Distributed optimisation * Jacobi and Gauss-Seidel algorithm * (Sub)gradient descent * Dual ascent, dual decomposition * Method of multipliers * Alternating direction method of multipliers (ADMM) * Primal-dual method of multipliers (PDMM) Monotone operator theory * Monotone operators * Fixed-point iterations * Forward-backward splitting * Peaceman-Rachford splitting * Douglas-Rachford splitting Privacy and security * Differential privacy * Secure multi-party computation * Subspace perturbation	
Study Goals	The student * is able to explain the use of convex optimization in decentralized signal processing * is able to discuss the structure and principles of convex optimization techniques * is able to discuss primal and dual properties of optimization algorithms * is able to implement state-of-the-art convex optimization algorithms for decentralized signal processing * is able to explain the use of monotone operator theory in the design of distributed optimization algorithms * is able to determine the convergence rate of distributed algorithms	
Education Method	lectures, mini projects + hands-on exercises	
Assessment	Oral examination + evaluation of the mini projects In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

RO47002	Machine Learning for Robotics	5
Responsible Instructor	Dr. J. Kober	
Instructor	Dr. H. Caesar	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a broad overview of machine learning techniques and their practical application in robotics. Intro to Machine Learning and python - Machine Learning in Robotics - Understanding the goal of machine learning, fundamental problems, high-level overview. Visualizing your data - Supervised vs Unsupervised vs Reinforcement learning - Model-driven vs Data-driven: White box, Gray box, Black box - Prior knowledge vs Unstructured. Feature extraction, linear regression + Deep Learning. Interpretability Hands-on Machine Learning - Example machine learning project in Robotics - Binary classification, decision boundaries, using logistic regression/SVM (one parameter) - How to perform a classification experiment. Dataset splitting, learning curves, metrics, comparing results Regression & Data collection - Regression and Data Collection in Robotics - Regression methods, least squares fitting - Overfitting, cross validation, regularization - Collecting (noisy) data, labelling data, outliers - High-dimensional data, data augmentation - Hyper-parameter optimization (grid search vs random search) Classification - Classification in Robotics - Parametric vs Non-parametric classifiers - Logistic regression - Decision tree, forest - Bayesian classification: Bayes' rule, naive Bayes, Gaussian Mixture Model - k-nearest neighbour - SVM, kernel-SVM, dual problem - Multi-class classification, metrics (confusion matrix), class imbalance Unsupervised Learning - Unsupervised Learning in Robotics - Clustering: K-means, Gaussian Mixture Model, DBSCAN - Dimensionality reduction: PCA, Local Linear Embedding (LLE) Neural Networks - Neural Networks in Robotics - Multi-Layer Perceptron, gradient descent - Neural Networks, and deep learning Advanced machine learning - Outlook: Vanishing gradient problem, Dropout, Optimizers, Data augmentation, AutoEncoder - Outlook: Time Series (RNN, LSTM) - Outlook: Reinforcement learning	
Study Goals	Knowledge, insight, judgement and skills: After completing this course, students will be able to - explain the role of ML in robotics, and describe possible use cases, and when not to use ML - explain the main concepts in ML (overfitting, training and validation, classification vs regression, cross validation, regularization, parametrics vs. non-parametrics, supervised/unsupervised/reinforcement learning) - explain the principles of well-established ML methods (e.g. Logistic regression, SVM, RFs, Neural Networks) - collect and preprocess data (normalization, feature selection, augmentation, splitting) for ML - analyze a ML problem, consider available data and computational resources, and select the appropriate ML methods to apply - perform experiments, evaluate the results, and draw sound conclusions - present the findings of a ML experiment clearly in a report using graphs and scientific English - write python code to visualize data and implement ML methods using popular python frameworks, such as pandas, numpy, matplotlib, jupyter notebook, sklearn, keras Transferable and interpersonal skills: After completing this course, students will be able to - collaborate and coordinate with a lab partner on coding and reporting tasks	
Education Method	Contact hours - Flipped classroom: pre-recorded lectures (2 hours per week) + classroom discussion (2 hours per week) - Q&A on practica (2 hours per week) Formative feedback - Quizzes -- Individual -- On material of previous lecture(s) -- Via BrightSpace at set time slots - Practica -- Pairwise collaboration -- Submission via BrightSpace -- Counts for knock-out criteria -- No late submission -- No resit Deadlines - For each practicum, the deadline is approx. one week after the practicum is published.	
Literature and Study Materials	Study materials - Slides (released shortly before lectures) - Book: "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems" 2nd edition, Aurélien Géron	

Prerequisites	Suggested literature - "Pattern Recognition and Machine Learning", C. Bishop - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Hastie, Tibshirani, Friedman TU Delft BSc Mechanical Engineering curriculum, or equivalent. Especially familiarity with - Programming in python - Linear Algebra - Probability theory
Assessment	Final assignment (30%) - Pairwise collaboration - Late submission: -1 grade point per day - No resit - Deadline: Will be released after the 6th practicum, giving the students approx. two weeks to read the assignment, collect the necessary data, perform their experiments, and write their report Exam (70%) - Individual - Written closed-book exam - Exam in Q1, resit in Q2 Knock-out criteria - At least 4 practicum grades need to be a pass (of current or previous academic year) - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6
Department	3mE Department Cognitive Robotics

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory Fundamentals of Coding in Python and MatLab	
Course Contents	Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems (direct and inverse) * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control * Examples of applications	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: Practical assignment and written exam, open book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity for a resit if they fail the written examination. * The groups that are graded between 5 and 5.75 in an assignment, will have two weeks to integrate the assessors feedback. If this is done in a satisfactory way, they can be rewarded with a mark that is not higher than 6/10. The exact instruction for the addition will be announced on Brightspace.	
Tags	Artificial intelligence Complex Mathematics Intensive Linear Algebra Mathematics Modelling Programming Project Proofs	
Department	3mE Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42015	Control Theory	6
Responsible Instructor	T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15% grade) and written final examination (85% grade) will determine the total final grade. The mid-term result is only valid for the academic year when it was obtained. For the resit examination there will be a written examination (100% grade) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Department	3mE Department Delft Center for Systems and Control	

SC42025	Filtering & Identification	6
Responsible Instructor	Dr. M. Kok	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The objective of this course is to apply filtering and identification methods. The focus is both on deriving these methods and on using these methods on real-life data sequences. The course focuses on stochastic and recursive least squares, on (extended) Kalman filtering, on prediction error methods and on subspace identification. The course also includes a review of specific topics from linear algebra, dynamical system theory and statistics, that are relevant for filtering and system identification.	
Study Goals	At the end of the course you should be able to: 1) Apply filtering and identification methods to solve previously unseen estimation problems. As sub-learning goals you should be able to: a. Relate basic linear algebra, signal processing techniques and probability theory to physical properties of a (dynamical) system. b. Derive the solution of the stochastic and recursive least squares problems from both a Bayesian perspective and in terms of unbiased / minimum variance properties. c. Derive the Kalman filter from both a Bayesian perspective and in terms of unbiased / minimum variance properties. d. Derive subspace and prediction error identification methods. e. Motivate the steps that should be taken in the derivations in b-d. 2) Apply filtering and identification methods to estimate the state and identify the model using real-life data sequences. As sub-learning goals you should be able to: a. Implement a (stochastic / recursive) least squares problem, an (extended) Kalman filter and identification methods in Matlab. b. Apply and motivate practical aspects of filtering and identification.	
Education Method	Lectures 0/6/0/0	
Literature and Study Materials	Book Filtering and System Identification: A Least Squares Approach by Michel Verhaegen and Vincent Verdult. ISBN: 13-9780521875127	
Assessment	Written exam (open book) and practical exercises. If needed due to covid-19 restrictions, the exam will be online.	
Remarks	The software package Matlab is needed to solve the practical exercise.	
Department	3mE Department Delft Center for Systems and Control	

SC42056	Optimization for Systems and Control	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	basic experience with Matlab	
Course Contents	Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field. How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control. More specifically, the following topics will be treated: * Linear Programming * Quadratic Programming * Nonlinear Optimization * Constraints in Nonlinear Optimization * Convex Optimization * Global Optimization * Matlab Optimization Toolbox * Multi-Objective Optimization * Integer Optimization	
Study Goals	After this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem. They should have insight into the basic operation of some of the most important and relevant optimization algorithms. They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach. Finally, they should be able to solve the resulting optimization problem in practice.	
Education Method	lectures + 2 assignments Depending on the corona measures, the lectures will be taught on-campus or online.	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study. exam (individual, counts for 75% of the final marks) + 2 assignments (group assignments, assessed through written reports, count for 25% of the final marks) Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next. Depending on the of measures resulting from COVID-19, the prescribed exam will either be an open-book online exam, or if the opportunity arises for examinations on campus, a closed-book written exam (fully closed-book, no calculators nor formulas sheets allowed).	
Permitted Materials during Tests	none	
Department	3mE Department Delft Center for Systems and Control	
Contact	questions are preferably asked via the Discussion forum on Brightspace or (for questions on the material of the lectures) during the online or on-campus lectures	

SC42075	Modeling and Control of Hybrid Systems	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	core Systems & Control topics (including state-space models, discrete-time and continuous-time models, stability, linearization), optimization basics (linear programming), Matlab	
Course Contents	General introduction, examples of hybrid systems and motivation Modeling frameworks (automata, hybrid automata, piecewise-affine systems, complementarity systems, mixed logic dynamical systems, Petri nets) Properties and analysis of hybrid systems (well-posedness, Zeno behavior, stability, liveness, safety, ...) Control of hybrid systems (switching controllers, model predictive control) Verification and tools	
Study Goals	Recent technological innovations have caused a considerable interest in the study of dynamical processes of a mixed continuous and discrete nature. Such processes are called hybrid systems and are characterized by the interaction of time-continuous models (governed by differential or difference equations) on the one hand, and logic rules and discrete-event systems (described by, e.g., automata, finite state machines, etc.) on the other. A hybrid system also arises in practice when continuous physical processes are controlled via embedded software that intrinsically has a finite number of states only (e.g., on/off control). Recent interest in hybrid systems is stimulated by developments in nonlinear control theory, intelligent control, adaptive control, and computer science. The purpose of the course is to introduce a variety of hybrid systems modeling, analysis and control techniques.	
Education Method	lectures + assignment Depending on the corona measures, the lectures will be taught on-campus or online.	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study. written exam (individual, closed-book, no calculators, counts for 60% of the final marks) + assignment (group assignment, assessed through written report, counts for 40% of the final marks) In case of unforeseen circumstances or measures resulting from COVID-19, the exam will be organized as an open-book take-home exam.	
Department	Important: partial marks for exam or assignment do not carry over from one academic year to the next 3mE Department Delft Center for Systems and Control	
Contact	Questions are preferably asked via the Discussion forum on Brightspace, or (for questions on the material of the lectures) during the online or on-campus lectures	

SC42101	Networked and Distributed Control Systems	4
Responsible Instructor	T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures.	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four assessments, i.e., written assignment reports, each representing 25% of the final grade. 2. For some assignments part of the grading is done via Peer Review (and in addition performing the peer evaluation will contribute to the assignment grade). 3. In order to complete the course, each assessment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Department	3mE Department Delft Center for Systems and Control	

SC42125	Model Predictive Control	4
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Model Predictive Control (MPC) is perhaps the most effective optimal control strategy for constrained dynamical systems. The basic concept of MPC is to exploit a dynamic model to forecast the system behavior, and optimize the forecast to determine the control input as the best decision at the current time. With emphasis on constrained linear systems, the course will present the theoretical fundamentals of MPC, such as Lyapunov stability, optimality and robustness, and its computational methods, such as quadratic programming.	
Study Goals	- Derive state prediction matrices from discrete-time linear models - Design the cost function, state and input constraints - Design MPC controllers with guaranteed recursive feasibility and asymptotic stability via appropriate terminal cost and constraint set - Formulate and solve constrained-linear-quadratic MPC problems via quadratic programming - Implement and simulate closed-loop systems controlled by MPC on Matlab - Design MPC controllers with integral action for reference tracking	
Education Method	Lectures, instructions.	
Assessment	Final project and oral exam.	
Elective	Yes	
Tags	Mathematics Matlab Optimalisation	
Department	3mE Department Delft Center for Systems and Control	

SC42130	Fault Diagnosis and Fault Tolerant Control	4
Responsible Instructor	Dr. R. Ferrari	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Control Systems Design, System Identification, Signal Processing, Statistics and Probability	
Summary	In critical systems such as transportation systems, power plants, distribution networks or life-support systems it is not enough to design a control system that meets the required performance levels. It is also of paramount importance to guarantee that such requirements, or a subset of them, are met also during non ideal conditions such as in the presence of physical failures or cyber attacks. Most of all, safety must be guaranteed at all times. This can be obtained by designing into the control system additional functionalities that will allow to detect the occurrence of faults or cyber attacks, and deploy adequate countermeasures, which may include the reconfiguration of the control system.	
Course Contents	Faults and failures in dynamical systems; Components and Services model; Availability and Reliability; FTA (Fault tree Analysis); FMEA (Failure Modes and Effects Analysis); Signal based methods for Diagnosis; Model based methods for Diagnosis; Residuals computation and evaluation; Learning of fault model; Fault Tolerant Control (Model Matching, Virtual Sensor and Actuator, Adaptive Control)	
Study Goals	after the course the student should be able to: 1) analyse the structure of a system and identify main components, the services provided by each and carry out a Fault Modes and Effects Analysis; 2) analyse and model faults or cyber attacks that can occur in a given dynamical system; 3) design an algorithm for detecting, isolate and identify them; 4) design a policy for reconfiguring the control system in order to accommodate them.	
Education Method	Flipped classroom: video lectures online, in-class exercises, homework	
Literature and Study Materials	Recorded lectures, Slides, Example exercises, Suggested textbook: Blanke, M., Kinnaert, M., Lunze, J., Staroswiecki, M., & Schröder, J. (2006). Diagnosis and fault-tolerant control (Vol. 691). Berlin: Springer.	
Assessment	Homework 30%, final exam (written) 70%	
Elective	Yes	
Tags	Mathematics Matlab Mechatronics Modelling Numeric Methods Process Signals and Systems	
Department	3mE Department Delft Center for Systems and Control	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Thesis project 2023

Introduction 1

This is the thesis project, this being the concluding study unit of the program. The project amounts to 45 EC and involves a research or design task with sufficient academic level. The project may be executed within a research program at TU Delft, or in a suitable research institute or company. The project must be carried out with a systematic approach and should include all phases of a research or design project: analysis, modelling, implementation/construction and validation/evaluation. The student executes the thesis project independently, with guidance of a thesis supervisor and under the responsibility of the full professor who approved the IEP.

ET4300	Master Thesis	45
Responsible Instructor	Prof. M. Alavi	
Responsible Instructor	Dr. D. Cavallo	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	M. Babaie	
Responsible Instructor	Dr. A. Lekic	
Responsible Instructor	M.A. Kitsak	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MScEE and MScCE programs. It is an individual in-depth research or expert design assignment with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. The thesis project may be executed within a research program at TU Delft, or in a suitable research institute or company. The nominal duration of a thesis project is 30 weeks (45 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. With regard to the thesis project, the thesis supervisor may impose additional conditions on the IEP to be met before starting the thesis project. If desired, an internship or an extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in scope and a separate deliverable such that the thesis project can be clearly evaluated in isolation. The thesis project is governed by the EEMCS Graduation Policy. Below you find a short description of the different steps to be taken during the thesis work: 1. Orientation 2. Kick-off meeting 3. Midterm meeting 4. Green-light review 5. Graduation Day (defence, assessment and graduation ceremony)	
Course Contents Continuation	Declaration of Credits Obtained Before you can start your Masters Thesis Project you need to have a by the Board of Examiners approved Individual Exam Programme (IEP) and at least 60 credits of the Master's degree course work as stated in the IEP should be completed. You can prove this by a form called the Declaration of Credits Obtained (Verklaring behaalde studiepunten) (60 EC). You can request this form from the service desk in the EEMCS building 36 (Mekelweg 4) at the servicedesk with your campuscard or send an email to servicepuntEWI@tudelft.nl (use your with your TU Delft e-mail!) You have to show this form to your thesis advisor.	
Study Goals	The master thesis project serves to prove that the student has acquired the academic competencies of a Master of Science. The thesis project is an individual in-depth research or expert design assignment independently carried out by the student with a systematic approach including all phases of a research or design project: - problem formulation and planning of the work - analysis, - modeling, - implementation/construction, - validation/evaluation, - presentation of the obtained results, orally and in writing. The achievements and resultant thesis report should comprise original work carried out by the student.	
Education Method	Orientation After having obtained a DCO, the thesis work starts with a thesis proposal, which is defined together with the supervisor. The proposal, of at most 2 pages A4, is written according to the format which can be downloaded from the Brightspace page of ET4300. The thesis proposal contains: - a working title, - a preliminary problem definition, - intended outcomes of the thesis project, - intended milestones, - expected duration and (preliminary) date of the thesis defense. The (preliminary) thesis committee is also indicated on this proposal. If desired, an internship or extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in activity and a separate deliverable such that the thesis project can be clearly evaluated in isolation. This is indicated in the thesis proposal. The thesis related work done during an internship or extra project should be described in the thesis report (typically one chapter). Kick-off meeting The Orientation phase indicated above results in a thesis proposal written by the student. The objective of the kick-off meeting is the formal approval of this thesis proposal by the supervisor(s). In this meeting, the research question(s), research methodology and planning are presented by the student and discussed with the supervisor(s). After approval, the "Thesis Proposal form" is filled out. Midterm meeting After 3-4 months, the student presents (by an oral presentation and a written report) the work performed so far to the supervisor(s). On the basis of the work done, it is decided whether the goals, as formulated in the thesis proposal, are still feasible. If not, they are reconsidered and adapted. After the mid-term meeting, the student has a clear idea of how to proceed in order to finish the thesis in time. Thesis committee This committee is responsible for the assessment of the thesis work and the thesis presentation. In order to be able to defend the thesis, the student must fill out the thesis committee form, which needs to be signed by the responsible full or associate professor. This form needs to be handed in to the Board of Examiners of the study programme at least 3 months prior to the MSc thesis presentation (the defence). Green-light review In the final phase of the thesis project, there will be a green-light review. During this meeting it is decided whether the student is	

ready to proceed with the thesis defence, resulting in a "go/no-go". It is compulsory to have a (draft) thesis ready for the green-light review. In case of a "go", the "Final Examination form" is handed in, which will trigger a check on the fulfillment of all requirements for getting the MScEE or MScCE diploma.

Graduation day

The final step of the graduation process is the presentation and defence of the thesis work. The candidate may present and defend the thesis work only after having finished all courses as stated in the approved IEP. The defence consists of two parts. First, the work is presented in a public session in front of the thesis committee, family, friends and fellow students. The presentation is followed by a closed session where the candidate will defend the work in front of the thesis committee. Immediately following the assessment, provided that there is a positive outcome, the diploma ceremony starts and the diploma will be handed out.

Assessment

The thesis project is assessed by the thesis committee based on the actual thesis work carried out, the final report, the public presentation, and the closed defense session and based on the learning objectives of the thesis. The items assessed in the final grading are:

Quality of the work: research method/design, application/development of theory, interpretation of results, scientific significance;

Performance: critical attitude, creativity, initiative, interaction with peers/superiors, planning;

Report: content, structure, quality of writing, independence in writing;

Presentation & defense: content, form, performance, scientific discussion.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested Free elective space 2023

Introduction 1

The fourth tier is the free electives space. In this tier, students can select any MSc level course from (theoretically) any field. Fill up until total becomes 120 EC, should need no more than 15 EC

This space can be used to provide deeper specialization or to broaden the scope of your knowledge by taking more fundamental courses, possibly from other tracks or fields. This space can also be used for entrepreneurship courses, language courses, presentation courses or other skills-courses and the like. To a limited extend (at most 10 EC), this space can also be used to address deficits in prerequisite knowledge by taking BSc-level courses. Such courses in the MSc are called homologation courses. At most 6 EC of these homologation courses can be language courses.

The free elective space can also be used for an internship (min. 9 EC, max. 15 EC) or an extra project (max. 15 EC). The latter offers students the opportunity to enhance lab skills (such as a specific training), academic skills (such as writing a paper for publishing or participating in a conference with a poster presentation or paper) or for studying special topics (such as capita selecta with (guest) lecturers).

These courses as well as additional main core and track core courses are a-priori acceptable for the specialization space in a student's IEP, and students determine their IEP (including the specialization courses) in consultation with the signatories of their IEP (i.e. with the master coordinator and responsible professor). Note that the signatories can decide not to accept certain combinations of courses if there is too much overlap between them. Also note that it is possible to mix courses from different suggested profiles when it is accepted by the signatories of the IEP.

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

General Elective Courses and Academic Skills 2023

AP3841	Systems Engineering Company Project	15
Responsible Instructor	Dr. M. Sovago	
Contact Hours / Week x/x/x/x	Scheduled full-time in Q1.	
Education Period	1 4	
Start Education	4	
Exam Period	1	
Course Language	English	
Course Contents	The Systems Engineering (Physics Design) company project is specially designed for TU Delft Master students at Applied Sciences and EEMCS Faculty in their second year. Systems Engineering is a must-have skill for every applied science and engineering student. Systems Engineering is an interdisciplinary field of engineering and engineering management that focuses on how to design and manage complex systems over their life cycles. The brain behind every complex system is the Systems Engineering; and that can be you! Here you will develop creative and innovative thinking, gain experience while working with an industrial partner, collaborate in teams and develop strong personal skills. The project includes working in an industrial project defined by a company. It is similar to an R&D internship with an advantage of working in a motivating team. You will be supervised by an academic (systems engineering and personal skills development) and an industrial part-ner (guidance project). The programme focusses on finding your role in a multi-disciplinary team and team building, entrepreneurship, presentation and writing skills, cultural intelligence, self- and team-reflection. Join us for this fun project!	
Study Goals	Participating in this project will ensure you a must-have skills and competences luggage, pre-paring you for starting your future career. At the end of this project, you will be able to: 1. Apply your applied science and engineering knowledge to solve systems engineering prob-blems 2. Have experience with industrial, real-life systems engineering and research cases 3. Think creatively and innovative. Develop entrepreneurial skills 4. Effectively collaborate in a multi-cultural and multi-disciplinary team 5. Present and write your results in a clear and appealing manner 6. Critically reflect on your performance and interaction with your team	
Education Method	Project team work with end presentation and end report.	
Prerequisites	Passing the preparatory Systems Engineering (AP3831, Q3-Q4) course is required to be admitted to the Systems Engineering Company Project.	
Assessment	Assessment is based on the final report, final presentation and the project defence during the final presentation for the academic committee (staff TU Delft) and the industrial supervisor(s). The evaluation will be done by an evaluation committee and is divided over: Theory and concepts (40%) = quality of systems engineering Implementation (30%) = group performance, entrepreneurship, use of creativity tools, innovativeness, concrete results Communication (30%) = progress meetings, project management, report, presentations	
Enrolment / Application	In order to enrol for the project, please send an email to: Maria Sovago m.sovago@tudelft.nl. Please indicate your name, faculty, student number and the top 2 preferred projects you would like to work on. We will distribute you in a team depending on your preferences and background.	
Studyload/Week	The project takes place on September - half-November. This is a full-time activity (40h/week). Please follow the BrightSpace for the exact dates and planning.	
Schedule	Systems Engineering Course AP3831: February-June Q3+Q4 In order to participate to the Physics Design Pro0ject, students from Applied Physics should enrol and follow first the Systems Engineering course AP3831. Systems Engineering Industrial Project, September -half-November (2.5 months) All participating Master's students will work in teams for a full 2.5 months to solve an industrial systems engineering problem. You will work both at the company and at TU Delft, depending on the company case (to be establish during the kick-off meeting in June). Physical meetings as well as online teamwork will take place, depending on the situation. Final Presentation and Final Report Systems Engineering company project: half-November The teams will present the results and their report to the evaluation committee (TU Delft staff) and to the industrial partner. The date will be established together by the team and the company. Please follow the BrightSpace for the exact dates and planning.	
Location	You will work both at the company and at TU Delft, depending on the company case (to be establish during the kick-off meeting in June). Physical meetings as well as online teamwork will take place, depending on the situation. You will be meeting regularly with your supervisors. You can always contact Maria Sovago for questions: m.sovago@tudelft.nl; 061 375 8335; office at building 22, F.232.	

EE5010	Internship 10-15 ECTS	10
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Course Coordinator	M.J.W. Smits	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Master students of Electrical Engineering can follow EE5010 as part of their free electives. The internship period should be from 8 weeks (for which you will receive 10 credits) up to three months (for which you will receive 15 credits). Internships longer than 3 months will NOT be approved! The internship should be of sufficient level to be part of your MSc curriculum performed at a company or institute, either in the Netherlands or abroad. An internship in a university outside the Netherlands can be accepted. An internship at TU Delft will not be accepted. When carrying out Internship EE5010 at a company or institute, the following conditions apply: The student participates in the company or institute culture in general, including sharing the company-established balance between on-site and online work. The student takes part in the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...).	
Study Goals	There are three types of educational goals for the Internship: I. Profession related aspects Insight into the profession of the electrical engineer Insight into the duties and the responsibility of the engineer in a company II. Social-psychological aspects Learning to operate in another culture (both within and outside the company) Gaining insight into the position of the company with respect to comparable companies III. Professional knowledge and skills related aspects Learning to apply the gained knowledge and skills in a situation different from the (own) university situation Learning to acquire new knowledge and skills that are necessary to meet the requirements for the Internship	
Education Method	Detailed information on projects, procedures and application is available on the Brightspace course EE5010. The internship application form should be submitted to the Internship Office prior (!) to starting Internship EE. This application form contains a project description that the student writes with input from the company supervisor and a TU Delft supervisor. As a student, you are responsible to find a TU Delft supervisor, with appropriate expertise. He/she needs to be part of the scientific staff within the Faculty of Electrical Engineering, Mathematics and Computer Science, this means they need to be a Professor, Assistant Professor or Associate Professor. Your TU Delft supervisor is responsible for approving the project description before the start of your internship. After you finalize your internship, both company and TU Delft supervisor need to independently judge your internship report. A completed internship application form, submitted to the EEMCS internship office, indicates that you can do the internship for study credit	
Literature and Study Materials	Internship regulations: programme Electrical Engineering, Embedded Systems and Computer Engineering. Enroll in Brightspace course EE5010 for more information about the Internship. You can find the step-by-step guide explaining all procedures, forms and internship offers. For additional information e.g. about grants and funding for internships abroad enroll in Brightspace course Study Abroad EEMCS.	
Prerequisites	In order to ensure that students start the projects with a solid skill set, students can apply for the Internship after the third quarter of the first year of the MSc EE programme. In order to start the Internship the following requirements must be fulfilled: 1. The Individual Exam Programme (IEP) must have been submitted to and approved by the Board of Examiners. 2. You have obtained 45EC before the start of your Internship. Detailed information on projects, procedures and application is available on the Brightspace course EE5010. The internship application form should be submitted to the Internship Office prior (!) to starting Internship.	
Assessment	Pass/Fail (no grade) For the assessment you need to submit: The company assessment form The TU Delft assessment form An internship and evaluation report. You can find guidelines for the report on Brightspace. In order to give the company supervisor sufficient time to read and check the report, yet to ensure that students receive timely feedback and final grade, your company supervisor is asked to assess your internship at the end of your internship period, within two weeks after submission of the reports, and send his assessment form to the TU Delft supervisor. Your TU Delft supervisor needs to assess your internship within two weeks after receiving the assessment of the company supervisor. The TU Delft supervisor will then send the assessment form (including the company supervisor assessment form) to the EEMCS Internship Office; internship-eemcs@tudelft.nl. The assessment form can be found on Brightspace EE5010. Step by step: - During your internship period you write the scientific report. - At the end of your internship, you have up to one week to finalize your evaluation report. - At the end of your internship, you send the scientific report to your company supervisor. - The company supervisor sends his assessment within 2 weeks after receiving the scientific report to the TU Delft supervisor. - You send both reports to the internship office.	

- The TU Delft supervisor sends his assessment within 2 weeks after receiving the report and the company assessment and send both assessments to the internship office.

The assessment is based on the Company supervisor Assessment Form, the TU Delft supervisor Assessment Form and the Internship Report and will be graded with a pass or fail.

Your mark will be processed as soon as the Internship Office EEMCS has received:

Two digital assessment forms signed by the supervisors and send by the TU Delft supervisor to the internship office (you need to provide your supervisors with the assessment forms yourself. You can find them on Brightspace EE5010).

A digital copy of both reports sent by you to the internship office

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

ET4300-I	Thesis Preparation at Industry	0
Responsible Instructor	L.M. Ramirez Elizondo	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The Preparation at Industry can be used as a preparation for the thesis project in a company. It should be clear that there is a separation between the preparation and the actual thesis. Note: the Preparation at Industry can only be taken as part of the free electives. The supervisor needs to be part of the scientific staff within the Faculty of Electrical Engineering, Mathematics and Computer Science, this means they need to be a Full Professor, Assistant Professor or Associate Professor.	
Study Goals	To be formulated by the student in the Preparation proposal (see education method) and to be approved beforehand by the supervisor and the MSc coordinator.	
Education Method	Detailed information is available on Brightspace organization for the Thesis.	
Prerequisites	Before the start of the Preparation, a proposal of the Preparation at Industry needs to be provided according to a specified format. The format of this proposal can be downloaded from the Brightspace page of the EE Thesis organization (eemcs-msc-thesis-ee). The following information is required: name of the TUD supervisor, proposed number of credits, a motivation, learning goals and a description of the planned activities and output. This proposal needs to be approved by the TUD supervisor, and the MSc coordinator. This proposal must be send together with the three-party agreement, which you can find on the student portal (EEMCS Student portal > Education > Graduation (MSc) > Thesis at a company) to thesis-eemcs@tudelft.nl for approval and signature by the Dean.	
Assessment	After the assignment, a final report needs to be produced. The Board of Examiners may ask to see the report to check the study load and quality at the time of the final MSc exam. The Preparation at Industry and the resulting report is evaluated and graded by the supervisor. The grade is Pass/Fail. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4399	Extra Project (max. 15EC)	15
Responsible Instructor	L.M. Ramirez Elizondo	
Course Coordinator	S.M. Alavi	
Course Coordinator	M. Babaie	
Course Coordinator	J. Dong	
Course Coordinator	Dr. A. Lekic	
Course Coordinator	Dr. D. Cavallo	
Course Coordinator	Dr.ir. R. Vismara	
Course Coordinator	M.A. Kitsak	
Course Coordinator	R.T. Rajan	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This Extra Project offers students the opportunity to enhance lab skills (such as a specific training), academic skills (such as writing a paper for publishing or participating in a conference with a posterpresentation or paper) or for studying special topics (such as capita selecta with (guest) lecturers). Note: the Extra Project can only be taken as part of the free electives. The supervisor needs to be part of the scientific staff within the Faculty of Electrical Engineering, Mathematics and Computer Science, this means they need to be a Full Professor, Assistant Professor or Associate Professor. For the MSc SET programme they should be on the Green list.	
Study Goals	To be formulated by the student in the Extra Project proposal (see education method) and to be approved beforehand by the supervisor and the MSc coordinator.	
Education Method	Before the start of the Extra Project, a proposal of the project needs to be provided according to a specified format; the format of this Extra Project proposal can be downloaded from the Brightspace page of the Extra Project. The following information is required: name of the TUD supervisor, proposed number of credits, a motivation for the extra project, learning goals and a description of the planned activities and output. This project proposal needs to be approved by the TUD supervisor, and for EE and CE the MSc coordinator, and for SET the director of studies. When submitting the Individual Exam Programme to the Board of Examiners, the Extra Project proposal needs to be attached. After the assignment, a final report needs to be produced. The Board of Examiners may ask to see the report to check the study load and quality at the time of the final MSc exam. At most one Extra Project may be performed and it can only be taken in the free elective space of the MSc programme.	
Assessment	The Extra Project and the resulting report is evaluated and graded by the supervisor. The grade is Pass/Fail. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15
Responsible Instructor	Dr. P.K. Murukannaiah	
Responsible Instructor	J. Zatarain Salazar	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	This course is aimed at students who have already obtained at least 20EC computer science/artificial intelligence background. Having a solid shared pool of knowledge on AI enables the students embarking on this course to work in a team on an AI challenge at a master level. Prospective students, for example, have completed an AI minor, or have a good amount of AI knowledge from their bachelor (e.g., Mechanical Engineering).	
Course Contents	Artificial Intelligence (AI) is an interdisciplinary challenge. This course aims to provide students with a unique experience to learn, develop and apply AI to a real-world challenge, working in an interdisciplinary project team. During the course, the project team executes AI research, and/or produces AI solutions, that address real-world problems. Students collaborate in a group of 4-6 students, ideally, of diverse backgrounds (including Computer Science). Mini-tutorials will be organized during the course to provide students with knowledge and skill-based input required to successfully complete their project. The topics of the mini-tutorials will be based on the initiative from the project supervisors and other AI researchers, e.g., from the TU Delft AI Labs. The course will include a kick-off, mini-tutorials, and lab sessions with project supervisors.	
Study Goals	Upon finishing this course, the student should be able to: 1. position an AI challenge in the broader literature, e.g., by performing a literature review on the AI challenge in a domain; 2. set-up a research strategy for an interdisciplinary AI project; 3. develop and deploy AI solutions according to a research strategy; 4. effectively collaborate with the team and supervisor; 5. evaluate and reflect on their research / developed AI solution; 6. assess the impact of deploying AI-based solutions and interventions on individuals, organizations, and society; and 7. report and present their research on a scientific level.	
Education Method	The students may work and discuss in groups. Each student is responsible for gathering, sharing and implementing information within the overall project. This allows for collaboration and communication as well as self-directed learning. At least once a week, the students meet with their supervisor. Peer feedback as well as feedback from the supervisor is organized on essential steps in the research project, which include: 1. Project proposal 2. Mid-term review, and 3. A scientific report, describing the problem, methods, results, and conclusion 4. A scientific presentation The mini-tutorials are organized as electives, i.e., students identify which tutorials contribute to their assigned task and field of expertise within the group project.	
Assessment	The scientific report and the project presentation are assessed by the project supervisor and an external examiner. The assessment based on the clarity of presentation, the soundness of the methods used, and the conclusions drawn. Resit next quarter: A repair option is allowed only for students that received a fail mark (<5.8). Overall, the resit mark will be capped to 5.8.	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Laboratories 2023

ET4icp	IC-Technology Lab	2
Responsible Instructor	Dr.ir. S. Vollebregt	
Contact Hours / Week x/x/x/x	28 hrs.	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics as in EE4585	
Course Contents	During seven half days the students, in groups of 3, will have the opportunity to gain experience in the process- and device-simulation of a MOSFET transistor. They will perform several processing steps in the Else Kooi Lab clean room and will measure and characterize the fabricated device. Enrollment through Brightspace, max course capacity is 36 students!	
Study Goals	After this course, students are able to: 1. Explain the different process steps of the IC fabrication process 2. Perform calculations on the oxide thickness of silicon after thermal oxidation, and the implantation depth and diffusion depth of dopants 3. Explain the relation between threshold voltage and implantations, and relate this to the measured threshold voltages from transistors 4. Select dopant atoms and masking material for silicon devices based on the simulations performed in the course 5. Identify the different implanted regions and junctions in cross-sections obtained from doping profile simulations 6. Argue on the specific advantages and disadvantages of wet and dry etching as used during the lab course. 7. Analyse the measurements obtained from different test structures described in the course guide 8. Determine the doping levels from sheet resistance measurements	
Education Method	Lab course	
Literature and Study Materials	Lecture slides and a course manual which will be supplied during the compulsory lecture	
Assessment	Group report. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	28 hrs, first session is scheduled, after that on individual appointment. Students have to attend all parts of the course. The course can only accommodate 30 students.	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Business Engineering & Management 2023

TPM416A	Turning Technology into Business	6
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup!	
Study Goals	This course aims to equip students with a strong conceptual foundation to an active understanding of two domains: (1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations (2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.	
Education Method	7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching. Final Presentations in February.	
Literature and Study Materials	Will be provided during the course	
Assessment	Each group writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation. The report makes up 80% of the final grade. This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review).	
Enrolment / Application	MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1. Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Targetgroup	Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Courses in Control Systems Engineering 2023

ME46085-23	Mechatronic System Design	4
Responsible Instructor	Dr. S.H. Hossein Nia Kani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The following are strictly required to be able to follow the course: - Basics of Multi-body Dynamics, including the ability to form equations of motion. - Basics of Electromagnetism, particularly Maxwell's equations. - Fundamentals of signals and systems (Laplace transforms, transfer function, bode plot, Nyquist stability). It is expected that students are able to use MATLAB or are interested in spending time to learn it while passing the course. This is required to complete the assignments.	
Course Contents	Course Contents Mechatronic system design deals with the design of controlled motion systems by the integration of functional elements from a multitude of disciplines. It starts with thinking how the required function can be realised by the combination of different subsystems according to a Systems Engineering approach. It should be noted that the control principles used in this course place a strong emphasis on frequency domain methods by linearising the system at its working point with the use of Bode- and Nyquist plots. The main reason for this emphasis is the strong focus in other control related courses on (non-linear) time domain related methods while linearised frequency domain related methods are still dominantly applied in the industry. A mechatronic engineer should be able to work with both methods and use them where appropriate. Some supporting disciplines, like power-electronics and electromechanics, are not part of the BSc program of mechanical engineers. For this reason this course introduces these disciplines in connection with mechanical dynamics and PID-motion control principles to realise an optimally designed motion system. The target application for the lectures are motion systems that combine high speed movements with extreme precision. The course covers the following three main subjects: 1: Dynamics of motion systems in the time and frequency domain, including analytical frequency transfer functions that are represented in Bode and Nyquist plots. 2: Electromechanical actuators, mainly based on the electromagnetic Lorentz principle. Reluctance force and piezoelectric actuators will be shortly presented to complete the overview. 3: Motion control in the frequency domain with PID and model-based feedforward control-principles that effectively deal with the mechanical dynamic anomalies (resonances and eigenmodes) of the plant. The other relevant discipline, electronics and position measurement systems is dealt with in another course: ME46005, Physics and measurement. The most important educational element that will be addressed is the necessary knowledge of the physical phenomena that act on motion systems, to be able to critically judge results obtained with simulation software. The lectures challenge the capability of students to match simulation models with reality, to translate a real system into a sufficiently simplified dynamic model and use the derived dynamic properties to design a suitable, practically realisable controller. This course increases the understanding what a position control system does in reality in terms of virtual mechanical properties like stiffness and damping that are added to the mechanical plant by a closed loop feedback controller. It is shown how a motion system can be analysed and modelled top-down with approximating (scalar and linearised) calculations by hand, giving a sufficient feel of the problem to make valuable concept design decisions in an early stage. With this method students learn to work more efficiently by starting their design with a quick and dirty global analysis to prove feasibility or direct further detailed modelling in specific problem areas.	
Study Goals	Can analyse and derive improvements to the dynamic behaviour of an actuator-driven mechanical structure with maximum 6th order plant dynamics (incl actuator and amplifier) by means of Bode and Nyquist plots. Can select and calculate a single-axis functional electromagnetic actuator for a given specification, working according to the Lorentz and reluctance force generation principle. Can select a proper amplifier to drive the above mentioned electromagnetic actuator. Can understand the basic concept of piezoelectric actuators and their application in mechatronic systems Can identify and apply feedforward and feedback P-, PD- or PID-motion controller settings for a given plant, consisting of a dynamically realistic power amplifier, electromagnetic actuator and mechanical structure with an ideal sensor, to achieve a stable system targeting a specified maximum bandwidth or disturbance rejection.	
Education Method	Lectures are given in 12*2 lecture hours, with presentations on the theory and practice of active-controlled motion systems. In addition, students will work on assignments to implement the theory learned in the course.	
Literature and Study Materials	Only the book is allowed at the examination. Therefore, no printed copy of the first edition, no computers, e-readers, smartphones, or other items are permitted. However, small notes written on the pages of the book are allowed.	
Books	The Design of High Performance Mechatronics - 3rd Revised Edition by R. Munnig Schmidt, G. Schitter, A. Rankers and Jan Van Eijk	
Assessment	The Final Assignment will account for 50% of the final grade. The remaining 50% of the score will be based on an open-book written examination. The examination will consist of questions covering the above-mentioned study goals.	
Contact	s.h.hosseinniakani@tudelft.nl	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory Fundamentals of Coding in Python and MatLab	
Course Contents	Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems (direct and inverse) * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control * Examples of applications	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: Practical assignment and written exam, open book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity for a resit if they fail the written examination. * The groups that are graded between 5 and 5.75 in an assignment, will have two weeks to integrate the assessors feedback. If this is done in a satisfactory way, they can be rewarded with a mark that is not higher than 6/10. The exact instruction for the addition will be announced on Brightspace.	
Tags	Artificial intelligence Complex Mathematics Intensive Lineair Algebra Mathematics Modelling Programming Project Proofs	
Department	3mE Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42025	Filtering & Identification	6
Responsible Instructor	Dr. M. Kok	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The objective of this course is to apply filtering and identification methods. The focus is both on deriving these methods and on using these methods on real-life data sequences. The course focuses on stochastic and recursive least squares, on (extended) Kalman filtering, on prediction error methods and on subspace identification. The course also includes a review of specific topics from linear algebra, dynamical system theory and statistics, that are relevant for filtering and system identification.	
Study Goals	At the end of the course you should be able to: 1) Apply filtering and identification methods to solve previously unseen estimation problems. As sub-learning goals you should be able to: a. Relate basic linear algebra, signal processing techniques and probability theory to physical properties of a (dynamical) system. b. Derive the solution of the stochastic and recursive least squares problems from both a Bayesian perspective and in terms of unbiased / minimum variance properties. c. Derive the Kalman filter from both a Bayesian perspective and in terms of unbiased / minimum variance properties. d. Derive subspace and prediction error identification methods. e. Motivate the steps that should be taken in the derivations in b-d. 2) Apply filtering and identification methods to estimate the state and identify the model using real-life data sequences. As sub-learning goals you should be able to: a. Implement a (stochastic / recursive) least squares problem, an (extended) Kalman filter and identification methods in Matlab. b. Apply and motivate practical aspects of filtering and identification.	
Education Method	Lectures 0/6/0/0	
Literature and Study Materials	Book Filtering and System Identification: A Least Squares Approach by Michel Verhaegen and Vincent Verdult. ISBN: 13-9780521875127	
Assessment	Written exam (open book) and practical exercises. If needed due to covid-19 restrictions, the exam will be online.	
Remarks	The software package Matlab is needed to solve the practical exercise.	
Department	3mE Department Delft Center for Systems and Control	

SC42075	Modeling and Control of Hybrid Systems	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	core Systems & Control topics (including state-space models, discrete-time and continuous-time models, stability, linearization), optimization basics (linear programming), Matlab	
Course Contents	General introduction, examples of hybrid systems and motivation Modeling frameworks (automata, hybrid automata, piecewise-affine systems, complementarity systems, mixed logic dynamical systems, Petri nets) Properties and analysis of hybrid systems (well-posedness, Zeno behavior, stability, liveness, safety, ...) Control of hybrid systems (switching controllers, model predictive control) Verification and tools	
Study Goals	Recent technological innovations have caused a considerable interest in the study of dynamical processes of a mixed continuous and discrete nature. Such processes are called hybrid systems and are characterized by the interaction of time-continuous models (governed by differential or difference equations) on the one hand, and logic rules and discrete-event systems (described by, e.g., automata, finite state machines, etc.) on the other. A hybrid system also arises in practice when continuous physical processes are controlled via embedded software that intrinsically has a finite number of states only (e.g., on/off control). Recent interest in hybrid systems is stimulated by developments in nonlinear control theory, intelligent control, adaptive control, and computer science. The purpose of the course is to introduce a variety of hybrid systems modeling, analysis and control techniques.	
Education Method	lectures + assignment	
	Depending on the corona measures, the lectures will be taught on-campus or online.	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study. written exam (individual, closed-book, no calculators, counts for 60% of the final marks) + assignment (group assignment, assessed through written report, counts for 40% of the final marks) In case of unforeseen circumstances or measures resulting from COVID-19, the exam will be organized as an open-book take-home exam.	
Department	Important: partial marks for exam or assignment do not carry over from one academic year to the next	
Contact	3mE Department Delft Center for Systems and Control	
	Questions are preferably asked via the Discussion forum on Brightspace, or (for questions on the material of the lectures) during the online or on-campus lectures	

Year

Organization

Education

2023/2024

Electrical Engineering, Mathematics and Computer Science

Master Electrical Engineering

Language courses and skills 2023	
Introduction 1	Up to 6 credits may be spent on language courses. These may only be chosen if required. Placement tests showing the necessity to take one or more of these courses must be taken and submitted to the master coordinator.

TPM301B	Spoken English for Academic Purposes Intermediate	2
Module Manager	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/3/0/3	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.	
Study Goals	The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.	
Education Method	Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to feel (more) confident when communicating with others at the university.	
Assessment	When you have taken our Placement Test you will be recommended a language level, either Intermediate or Advanced. With this recommendation you can sign up for both of the courses offered at that level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM301B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. The course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc students	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/0/3/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for either of the courses offered at that level.	
Assessment	For this course you need a recommendation to take courses at the Advanced level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.	
Special Information	Enrolment is through Brightspace.	
Remarks	The course will only be offered when there is sufficient demand. If there are not enough enrollments, the course may be cancelled.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will self-study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final exam + a speaking test. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

WM1116TU	Dutch Elementary 2	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)	
Expected prior knowledge	A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final test + a speaking test.. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1117TU	Dutch Intermediate 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.	
Expected prior knowledge	A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Intermediate 1 (WM1117TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test + speaking test. Contact the secretary's office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether this course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as AI tools	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book: Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	This is a pass/fail course. There is no final exam. Assessment is based on the six written assignments and on the in-class discussions and presentations.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Free elective courses SS 2023

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Entrepreneurship and Innovation Education 2023

MOT9611	Project Entrepreneurship Thesis related	5
Module Manager	A. Boru	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Instructor	Dr. J.R. Ortt	
Instructor	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0,5/0,5/0,5/0,5	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	Students enrolled in the Master Annotation Entrepreneurship program.	
Expected prior knowledge	BEfore entering the entrepreneurship related project thesis, we expect you have passed the following courses: - MoT9610 - Entrepreneurship Basics Course - MoT9612 - Business Development Lab, short	
Course Contents	The specific content of the Project Entrepreneurship Thesis Related is to be defined together with the student and his/her supervisors in advance off the thesis period. However, a few guidelines for the structure of the work to be undertaken are: 1. The thesis theme should be directly related to the subject of the students Master Thesis. The relevance of the Project Entrepreneurship Thesis Related to the main Master Thesis should be clear; 2. The Project Entrepreneurship Thesis Related is a separate report that reflects a) the commercial and/or societal value of the technology, product or service which is subject of study in the master thesis, and b) a thorough investigation of the business model and organization by which the technology, product or service is brought to the market. The main goal is to articulate the identification and appropriation of the commercial and societal value of the research topic central to the master thesis project. This report can address several issues. Some examples: An exploration of the business opportunities of a technology or product/ service innovation and the possible partners that need to be included to bring it to commercial exploitation; An exploration of the possible market applications and product/service forms of a technology including a business model analyses of how to exploit it; An exploration of introduction strategies for a new high-tech product and the role of various business partners and stakeholders in the business model; A business model specification for a new high-tech product; A business plan for a new high-tech product; An investigation of the funding opportunities for the business model to improve and develop the technology further; 3. The project should state clearly whether it focusses on 'Starting your own Company' or 'new entrepreneurial venture projects within established organisations'; 4. The subject of the Project Entrepreneurship Thesis Related should include a reflection on the ambition of the student related to entrepreneurship; 5. In the case of a focus on 'Starting your own Company', thesis themes should be directly linked to the context of starting a new company. Ideally, the student should place himself as the entrepreneur willing to establish a start-up with the technology at hand either right after graduation or later; 6. In the case of a focus on 'new entrepreneurial venture projects within established organisations', the thesis should be directly linked to the context of an existing company. Ideally, the entrepreneurship thesis should be performed within the company; 7. Before commencing the Project Entrepreneurship Thesis Related project, students need to prepare a research proposal, including the activities and planning of work to be developed for the phases of literature review, field research, analysis and reporting. This plan should be reviewed by the Entrepreneurship supervisor. Students will be assessed on basis of this planning and its execution. A template for this plan is available from the DCE supervisors.	
Study Goals	The Project Entrepreneurship Thesis Related aims to develop and apply the skills and knowledge on entrepreneurship of students enrolled in the Master Annotation Entrepreneurship Program. The student applies all the knowledge and skills that he/she learned during the Annotation Entrepreneurship to a business problem related to the overarching Master Thesis, and shows an entrepreneurial attitude in tackling this problem. The Project Entrepreneurship Thesis Related should be seen as a complement to the overarching Master Thesis. The underlying idea is to make a bridge between the technical/ design aspects, examined in the students Master Thesis, and the business aspects of implementing the technology/product/service into the market/ society. The ability to make this connection is at the heart of technology-based entrepreneurship. The thesis, like the whole annotation program, can be focused on two main areas: Starting your own Company and new entrepreneurial venture projects within established organisations. On completing the Project Entrepreneurship Thesis Related, students should be able to: - Understand the practical business aspects of bringing a technology/ product/ service to the market (within the setting of a start-up or established firm); - Estimate a (potential) market and derive pricing strategies from this analysis. - Draw up the cost structure of a product or service and translate this into the accompanying financial statements. - Further develop their insights into what their motivations, ambitions and objectives are as entrepreneurs (in either a corporate setting or in a new start-up); - Be able to report their findings in a well-readable and attractive format, which should be easy to understand for decision makers; - Present their findings in the form of a short 'elevator pitch', so that decision makers understand the value of the concept and are motivated to know more about it.	
Education Method	The following learning methods can be applied: - Literature review associated to the thesis theme; - Field research and analysis (both on the technical and business aspects of the technology);	

Assessment	- The application of skills and knowledge acquired in the electives related to entrepreneurship, in a real context; - Critical dialogues and discussions on the findings and analysis done by the student with both the Master Thesis supervisor (for the technical aspects) and the Project Entrepreneurship Thesis Related supervisor (for the business aspects). The assessment is composed of four main elements: 1. Planning and execution of the thesis; including the necessary phases for literature review, field research, analysis and reporting. The timing for the delivery of the plan should be agreed upon with the supervisor for the Entrepreneurship Annotation Final Thesis, who will also evaluate it. The evaluation follows the following: (i) quality of the work plan (ii) ability to live up to the work plan and (iii) ability to be autonomous and having own initiative. The student should prove what he has learned about his own insights into what his motivations, ambitions and objectives with respect to entrepreneurship are. 2. Research quality of the Entrepreneurship Thesis Related; this is provided by a report and evaluated by the dedicated supervisor on the Project Entrepreneurship Thesis Related. The evaluation should follow these criteria: (i) ambition level, (ii) structure and boundaries, (iii) academic level with respect to the entrepreneurship skills taught at Delft University of Technology, (iv) originality, (v) usability, (vi) relevance and (vii) reflection on the product. Additional criteria, include: (i) the line of argument, (ii) use of English and (iii) design of the report. Criteria can vary by faculty and master program. 3. Personal reflection; the student will be judged on his ability to understand the practical business aspects of bringing a technology to the market, and to analyze real-life business problems in the field of entrepreneurship and innovation. 4 Elevator pitch and/or PowerPoint presentation given by the student at the end of the final thesis, evaluated by the supervisor. The evaluation should follow these criteria: (i) performance, (ii) ability to clearly answer to questions and (iii) usage of English language.
Elective	Yes
Category	MSc level

TPM401A	Technology Entrepreneurship and Innovation	5
Module Manager	H. Khodaei	
Instructor	P.B.M. Vandekerckhove	
Instructor	Dr.ing. V.E. Scholten	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/2/2/0	
Education Period	1 2 3	
Start Education	1 2 3	
Exam Period	1 2 3	
Course Language	English	
Summary	This course offers theoretical fundamentals of entrepreneurship and technology innovation. You create a case study of an existing start-up analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an high-tech start-up to academic research on the topics of technology entrepreneurship and innovation.	
Course Contents	Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, dynamics and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to:	
	LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Describe and apply the high-tech start-up growth process. LO 4 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 5 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 6 Analyse an existing high-tech start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations will be facilitated in an online mode (for example through zoom or Skype). Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)	
Enrolment / Application	Enrollment via Brightspace	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft is needed.	
Category	MSc level	

TPM411A	Idea to Start-up IT & AI	5
Module Manager	Dr. T.L. Dolkens	
Instructor	Dr. T.L. Dolkens	
Instructor	E. Çelik	
Responsible for assignments	Dr. T.L. Dolkens	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	This course used to carry the name MOT9612 - Business Development Lab. In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using computer sciences. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. On this journey, you will be helped by guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.	
Course Contents	You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large. To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.	
Study Goals	Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies. Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product. Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries. Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	You will work in teams which are formed in the first sessions of the course. Topics are discussed using case studies. You will present your work, provide peer-feedback as well and receive advise from various coaches. The course ends with a jury pitch and a two-pager of the business proposition. Team coaching with industry experts will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Books	The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2013.	
Assessment	The assessment of this course is Covid-19 proof. It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and peer evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Quantum Technology 2022

AP3432	Quantum Hardware 1 - Theoretical Concepts	4
Responsible Instructor	C.K. Andersen	
Instructor	M.F. Rimbach-Russ	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Undergraduate electricity and magnetism; AP3421 Fundamentals of quantum information; Bachelor Quantum Mechanics	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to control, couple, transmit and read out the fragile state of quantum systems with great precision, and in a technologically viable way. Quantum Hardware I is focused on teaching theoretical physics concepts for understanding this Hamiltonian engineering challenge in various quantum hardware platforms. The material will be taught using example systems such as superconducting qubits, atomic qubits, defect centers, and trapped-ion qubits.	
Study Goals	This course should enable you to: L01: Discuss single-qubit and two-qubit gate dynamics, qubit measurement, Rabi oscillations, dephasing & relaxation times on an abstract level. L02: Analyse quantum hardware systems by obtaining effective dynamics using approximations to the full Lindbladian of the system. L03: Evaluate performance of quantum hardware by identifying sources of noise and quantifying the effect of these. L04: Establish various quantum hardware systems relative to each other by identifying challenges and opportunities of the systems for quantum information processing.	
Education Method	The course will consist of weekly lectures and tutorial exercise sessions. Every week there will be hand-in assignments, which are factored in the final evaluation of the course.	
Course Relations	This course forms part of the curriculum on Quantum Technologies offered at TU Delft, which at present consists of: AP3421 Fundamentals of quantum information CS4090 Quantum communication and cryptography AP3432 Quantum Hardware 1 - Theoretical Concepts AP3442 Quantum Hardware 2 - Experimental State of the Art EE4575 Electronics for quantum computation Other Related Courses: AP3112 Quantum Optics and Lasers	
Literature and Study Materials	In this course, we will make use of: 1) 0Lecture notes And some selected material from the textbooks: a) Nielsen and Chuang, Quantum computation and information, Cambridge University Press. b) Gerry and Knight, "Introductory Quantum Optics", Cambridge University Press	
Assessment	Both hand-in assignments during the course and a final written exam will contribute to the final assessment of the course. Specifically, hand-in assignments will contribute 30% and the final written exam 70% of the final grade. Feedback on the hand-in assignments will be provided on a weekly basis and should help prepare students for the final exam.	
Continuing Courses	AP3442 Quantum Hardware 2 - Experimental State of the Art AP3472 Modeling of Superconducting Devices AP3662 Special Topics in Quantum Technology	

AP3442	Quantum Hardware 2 - Experimental State of the Art	4
Responsible Instructor	Prof.dr.ir. L.M.K. Vandersypen	
Instructor	Prof.dr.ir. R. Hanson	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course requires an understanding of undergraduate electricity and magnetism and the following concepts from quantum information theory: - Hamiltonian of a spin-1/2 particle and of a harmonic oscillator, and tensor products - quantum states (pure and mixed states, including density matrices) - unitary evolution (generated from the above Hamiltonians) - standard quantum operations (X, Y, Z rotations, Hadamard, CNOT, CPhase, SWAP, ...) - decoherence and non-unitary operations (e.g. through the product operator formalism) - simple quantum circuits and algorithms (e.g. teleportation, Deutsch-Jozsa algorithm, Grover's algorithm) - basics of quantum error correction (3-qubit bit flip/phase flip code, notion of fault-tolerance threshold) - simple quantum communication protocols (e.g. BB84, entanglement swapping for quantum repeaters) - quantum measurement (projective measurement in different bases)	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to initialize, control, couple, transmit and read out the fragile state of quantum systems with great precision and in a technologically viable way. While Quantum Hardware I is focused on teaching the underpinning theoretical tools, Quantum Hardware II will give you an overview of the experimental state-of-the-art. You will learn about several promising approaches for realizing quantum hardware, and critically assess the strengths and weaknesses of each approach. You will also get insight in the conceptual similarities and differences between the various technologies. Specifically, the course will cover general concepts and considerations of qubit hardware, trapped ions, superconducting circuits, quantum dots, impurities, cold atoms, photonic circuits, single-photon sources, single-photon detectors and quantum repeaters.	
Study Goals	- To acquire a good understanding of the requirements of quantum hardware for computing, communication and sensing, both at the conceptual level and at the practical level. - To acquire conceptual insight in the operation, opportunities, and challenges of various qubit realisations. - To obtain a good overview of the state-of-the-art.	
Education Method	In this course, lectures are combined with homework assignments as well as student presentations of recent research papers.	
Literature and Study Materials	Review papers to accompany the lecture slides.	
Assessment	Grades will be established based on homework assignments (40%), paper presentations (25%), and a final exam (35%). The lowest homework grade does not count. This means you can miss one homework without having points deducted. Two missed homeworks will affect your grade, regardless of why you missed them. In unusual cases, a retake exam--which will be oral--will be offered by appointment.	
Continuing Courses	This course forms part of the QuTech Academy curriculum on Quantum Technologies offered at TU Delft, which at present consists of: - AP3421 Fundamentals of quantum information - CS4090 Quantum communication and cryptography - AP3432 Quantum Hardware 1 - Theoretical Concepts - AP3442 Quantum Hardware 2 - Experimental State of the Art - EE4575 Electronics for quantum computation	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - Insight into the differences between classical and quantum communication and cryptography - Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: - Slides - Review Articles Auxilliary: - Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: - Homework (weighting 70%) - Final Project (weighting 30%) Final grade calculation: $0.7 * \text{Homework} + 0.3 * \text{Final Project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: - Homework: re-submit deliverable - Final Project: re-submit deliverable	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). - Abstract - Adventurous - Algorithmics - Challenging - Group Dynamics/Project Organisation - Information & Communication - Integrated - Intensive - Involved - Lineair Algebra - Mathematics - Physics - Quantum - Signals - Technology - Telecommunication	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

QIST4400	Quantum Computer Architecture	5
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr. S. Feld	
Instructor	Dr.ir. T.H. Oosterkamp	
Instructor	Dr. R. Ishihara	
Contact Hours / Week x/x/x/x	0/0/4/0/	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The course will assume knowledge of fundamentals of quantum information. Basic knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce the students the overall system of a quantum computer, focusing on the classical hardware and software infrastructure required to build a quantum computer together with the quantum hardware. Covered topics include: - Quantum computer architecture; quantum languages; compilers; QISA; microarchitecture. - Quantum simulator (QX simulator); - Quantum error correction: quantum error correction codes; encoding; logical operations. - Mapping of quantum circuits; placement of qubits; scheduling of operations; routing of qubits; - Classical hardware for quantum computing: classical controller and quantum processor; qubit hardware and the need for cryogenic electronics; - Receiver and transmitter architectures for controlling qubits: frequency up/down-conversion, modulation schemes, frequency generation. - Hardware building blocks: amplifiers, analog-to-digital converters, digital-to-analog converters, digital processing, FPGA - Cryo-CMOS: device characteristics, state-of-the-art, design principles.	
Study Goals	By the end of this course, students are able to: - Explain the different layers that are required to build a quantum computer. - Discuss the basic functioning of quantum error correction (QEC). - Carry out the mapping step of simple quantum circuits onto specific physical device layouts. - Interpret the results of different quantum circuits written in language QASM and executed using simulator QX, including the following: Basic quantum algorithms; Fundamental QEC codes; Surface code, especially for a Ninja star. - Identify and describe the different hardware blocks in the classical controller in a quantum computer. - Derive the specifications of the individual components in a classical controller from the requirements of the whole controller. - Compare different hardware blocks based on their specifications and can select the appropriate component for target functionality and specifications. - Describe the main features and advantages/disadvantages of the adoption of cryogenic electronics in general and cryo-CMOS specifically.	
Education Method	- Slide decks including small exercises - Videos including quizzes - Interactive lectures including activities - Homework assignments - Paper reading including processing its content	
Assessment	The final grade consists of two parts: - Part 1: Weekly homework, will be submitted electronically. Individual submission. - Part 2: Final assignment, preparation of presentation of 20 minutes with 10 minutes Q&A. Assignment will be handled by pairs of two, there will be individual grades. The global course grade is an average of each parts grade (50% part 1 + 50% part 2). Students are allowed to omit one homework. There is no resit.	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Other 2023

EE4700	Modelling, Algorithms and Data Structures	5
Responsible Instructor	Dr. S. Feld	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- Basics of digital technology - Boolean logic and reasoning - General concepts of programming languages	
Course Contents	To solve a given electrical engineering (EE) problem, we usually start with developing an appropriate model of such a problem. Once the problem is modeled using an appropriate data structure, one can focus on solving the modeled problem instead of the EE problem itself. Choosing the right data structure is therefore important for generating efficient and optimal solutions. Solutions for such problems consist usually of algorithms. Analyzing the algorithms in terms of efficiency, time complexity, required memory, etc. is therefore an important aspect when dealing with Electrical Engineering problems. This course deals with these three important aspects of solving electrical engineering problems, namely modeling, algorithms and data structures. It will cover the following topics: sets and logic; proofs; functions, sequences and relations; algorithms and complexity; introduction to number theory; counting methods; recurrence relations; graph theory; trees; network models.	
Study Goals	By the end of the course, the students are able to: - LO1: Create models for specific problems - LO2: Proofs a) Classify different proof methods b) Use them to prove theorems - LO3: Algorithms a) Develop algorithmic solutions for given problems b) Analyse those algorithms - LO4: Apply counting to discrete problems - LO5: Employ graph theory to solve some problems	
Education Method	Lectures	
Literature and Study Materials	- R. Johnsonbaugh, Discrete Mathematics, Pearson Prentice Hall, 7th Edition. - Lecture slides and other material, such as scientific papers, will be distributed.	
Assessment	Written exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Specialisation BD 2023	
Responsible Program	Prof.dr.ir. W.A. Serdijn
Employee	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Main core BD 2023

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Instructor	Ir. J.W. Peltenburg	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = $0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3)$ L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1] In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C01	Profile Orientation and Academic Skills	3
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Responsible Instructor	L.C. Schroten	
Instructor	Dr. C.A. Benitez Avila	
Instructor	L.C. Schroten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: basic knowledge as acquired with a BSc in Physics, EE, Material Science. Strongly advised: Information Literacy 1 and 2 (e.g. Master Programme Information Literacy 1 & 2 or Electrical Engineering Information Literacy 1 and 2, both available in Brightspace).	
Course Contents	- Introduction to the MScEE tracks (Microelectronics, Electrical Power Engineering, Signals & Systems, Wireless Communication and Sensing) - Introduction to the research and expertise of the research groups taking part in the selected master track - Knowledge of ethics and intellectual property right - Skills in academic writing - Skills in oral presentation of scientific topics	
Study Goals	With this course the students - obtain relevant information on the MScEE study and the chosen track, become aware of cultural differences and how this affects intercultural communication, form a network with fellow master EE students. - acquire an overview of the main challenges in and potential applications of electrical engineering - become familiar with research topics addressed within the EE department - master basic skills in oral presentation of scientific topics - master basic skills in academic writing - acquire basic principle of ethics and intellectual property right - are able to extract the main information from a scientific paper or report - are able to write a paper (mostly a literature review) on a scientific question	
Education Method	The course includes lectures, reading materials and exercises. During the course, students work on a writing assignment, an ethics case and give several presentations. Attendance is mandatory as all class activities are highly interactive and peer feedback is a necessity.	
Literature and Study Materials	TU Present & TU Write. Both are available in Brightspace.	
Assessment	The pass/fail for this course is determined by the evaluation of the final writing assignment, the oral presentation and attending the Ethics seminar(s). All three components should be evaluated with a pass to pass the whole EE4C01 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C03	Statistical Digital Signal Processing and Modeling	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Signals and Systems, eg, Laplace, Fourier and z-transforms; random processes; linear algebra; experience with Matlab	
Summary	This is a second course in discrete-time signal processing. It provides a comprehensive treatment of signal processing algorithms for modeling discrete-time signals, designing optimum filters, estimation of the power spectrum of a random process, and implementing adaptive filters. These are important topics that are frequently encountered in professional engineering, and major applications such as digital communication, array processing, and multimedia (speech and audio processing, image processing). The course provides a framework that connects signal models to filter structures, formulates filter design as an optimization problem, solved in turn via linear algebra techniques applied to structured matrices. The connections between these topics are strong, and provide insights that can also be used in other disciplines.	
Course Contents	The course treats: background in DSP, linear algebra and random processes; linear prediction, parametric methods such as Pade approximation, Prony's method and ARMA models; the Yule-Walker equations; Wiener and Kalman filtering; spectrum estimation (nonparametric and parametric), frequency estimation (Pisarenko, MUSIC algorithm); adaptive filtering (LMS, RLS). Part of the course is a track-specific take-home matlab assignment which addresses a practical modeling/estimation/tracking problem in power engineering, radar, speech processing, or bio signal processing.	
Study Goals	You will have acquired the fundamentals of advanced discrete-time signal processing, both from deterministic and stochastic signal processing viewpoints. Specifically, you can model discrete-time signals in various ways (pole-zero, all-pole, FIR, ARMA), you can estimate power spectra and frequency components in various ways (direct and via parametric models), you can design optimum filters (Wiener and Kalman), and you have a basic understanding of adaptive filtering (LMS, RLS algorithm). You can implement and test these algorithms in Matlab, and can indicate examples where these algorithms are used in engineering practice.	
Education Method	Lectures; take-home lab assignment with reporting	
Computer Use	Matlab (take-home exercises and compulsory lab assignment)	
Course Relations	This course complements ET 4152 Estimation and Detection and ET 4715 Array Processing; it is a good preparation for EE 4685 Machine Learning - A Bayesian Perspective.	
Literature and Study Materials	Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons, New York, 1996. ISBN: 0-471 59431-8	
Assessment	Written test, which counts for 80% of the final grade. The track-specific matlab assignment results in a report which is pass/fail (need to pass), furthermore it is graded and counts for 20% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	The exam is open book: you can use the book by Hayes and the course slides (no other materials allowed).	

EE4C04	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential Realization of transfer function models by state space descriptions, coordinate changes, canonical forms Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback Application of LQ regulator Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle Reference signal modeling, integral action for zero steady-state error. Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design and - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C05	Electromagnetics	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	M. Spirito	
Responsible Instructor	Prof.dr.ing. A. Neto	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	Dr. D. Cavallo	
Instructor	Prof.dr. O. Yarovy	
Instructor	M. Spirito	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Course Contents PART 1 (4 EC) 1. Fundamentals of E.M. (Andrea Neto) Maxwell Equations Constitutive relations (2 hours) Poynting Theorem Time/Freq. (2 hours) Boundary Conditions (2 hours) Exercises (2 hours) 2. Transmission lines (Andrea Neto) Telegrapher equation (2-hours) Standing waves, impedance transfer (2 hours) Exercises (2 hours) 3. Solutions of Maxwell Equations (Alexander Yarovoy) Radiated field and Hertz dipole (2 hours) Plane waves (2 hours) Scattering from interfaces (2 hours) Guided wave propagation (2 hours) Exercises (2 hours) 4. Fundamentals of RF (Marco Spirito) S-parameters and Smith Chart (2-hours) Practical implementation of Tx-lines (2 hours) Exercises (2 hours) PART 2: Assignment (1 EC) All interested Stake-holders for the course will propose assignments related to their specialization.	
Study Goals	The new E.M. aims to provide a general and basic introduction to the fundaments of the E.M. field theory that forms the basis of Electrical Engineering. After the course the student will be able to: - Visualize the relationship between the scalar quantities used in Electrical Engineering, voltages and currents, and the corresponding vector field quantities, electric and magnetic fields. - Qualitatively understand the properties of the lumped component circuit elements models which are valid at low frequency, based on their intuitive visualization of quasi static E.M. quasi static fields. - Understand the fundamental mechanisms associated to losses in modern circuit designs: eddy currents, skin effect, mismatch and radiation - Quantify the properties, propagation constants and characteristic impedances, of circuits which depend on their length in terms of the investigated wavelength. - Qualitatively understand the radiation properties of circuits, and the mutual coupling between antennas or circuits After the course the student - will master the introduced theoretical concepts in the Electromagnetics that necessary for understanding electrical circuits in various frequency regimes - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.	
Education Method	The course includes formal lectures and exercises in a rapport 3:1	
Literature and Study Materials	Lecture notes and the book from Orfanidis " Electromagnetic Waves and Antennas" (from the internet)	
Assessment	The assessment will be based on a written exam and a poster presentation on the topics covered during the assignment. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	F. Simjanoski	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C10	Analog Circuit Design Fundamentals	5
Responsible Instructor	Dr.drs. Q. Fan	
Responsible Instructor	F. Sebastiano	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to analog design: motivation of analog design; introduction to integrated circuits and CMOS. of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Transistor device physics: brief review of MOSFET device physics; MOSFET static and dynamic characteristics; device large-signal behavior, small-signal model. Single-stage amplifiers: common-source, common-gate, common-gate amplifier stages, including large-signal behavior, small-signal behavior, input/output impedance. Differential amplifiers: large-signal and small-signal analysis; differential-mode and common-mode behavior. Current mirrors and biasing: basic current mirrors; cascode current mirrors; current mirror as active load of a differential pair; biasing of common-source, common-gate and common-drain stages; biasing of differential pairs. Frequency response of amplifiers: frequency response of common-source, common-gate and common-drain stages; frequency response of differential pairs. Noise: review of basic noise principles (amplitude distribution, power spectral density, signal-to-noise ratio), type of noise in electronic devices; input- and output-referred noise; noise in single-stage amplifiers, current mirrors and differential pairs. Beside the above-mentioned common part, the students will be able to select a track-specific part among several proposed options.	
Study Goals	Students will be able to identify and describe the following fundamental aspects of analog circuit design: - o Basic physics of CMOS transistors - o Small-signal and large-signal models and behavior of CMOS transistors - o Working principle of basic transistor circuits, such as single-stage CMOS amplifier topologies (common-source, common-gate, common-drain), differential pair and current mirrors - o Biasing techniques for simple transistor circuits. - o Source of noise in CMOS circuits and their physical origin. - o Effect of noise on the performance of analog circuits - o The definition and the relevance of the frequency response of analog circuits Students will be able to perform the following analysis using analytical and approximate calculation methods: - o Calculate the gain, the input/output impedance of single-transistor stages, basic and cascaded current mirrors and differential pairs with resistive or active loads. - o Calculate the noise performance (input- and output-referred noise, signal-to-noise ratio) of basic analog circuits (single-transistor stages, differential pairs, current mirrors). - o Calculate the frequency response of basic analog circuits (single-transistor stages, differential pairs, current mirrors). Students will be able to simulate the transistor-level description of basic analog circuits using a Spice-like simulator. In particular, they will be able to simulate and evaluate the following: - o DC operation - o Frequency response - o Noise performance - o Transient behavior Students will be able to describe the trade-offs involved in the design of a basic analog circuit, such as a current mirror and a single transistor amplifier, and be able to apply such trade-offs in the design of those circuits. Students will be able to design simple analog circuits (a current mirror or a single-transistor amplifier, including the bias circuit) for a given specification and will be able to verify their design by both analytical calculation/estimation and circuit simulations.	
Education Method	Lectures + homeworks	
Literature and Study Materials	The main material for self-study will be Razavi's "Design of Analog CMOS Integrated Circuits" (2nd edition). Next to that, extra material will be made available via Brightspace. Books: B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education; 2 edition (January 20, 2016).	
Assessment	Homework (on both the common part and the track-specific part) and written on-line exam. Final grade computed as follows: 30% homework of the common part, 20% for the track-specific homework, 50% final exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	P.P. Sundaramoorthy	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (50%), project (30%), homework (20%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C13	Wireless Systems for Electrical Engineering Applications	5
Responsible Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	C. Gao	
Instructor	Y. Aslan	
Instructor	J. Dong	
Instructor	M. Babaie	
Instructor	Dr.ing. L.C.N. de Vreede	
Instructor	M. Spirito	
Instructor	Prof. M. Alavi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE4605 'Integrated Circuits and Systems for Wireless Applications' and EE4371 'Advanced Wireless Transceivers.'	
Course Contents	In this course, you will learn about the commonly used architectures in wireless systems, with the requirements on their building blocks. The system key building blocks are addressed at their functional level regarding electrical properties, such as transfer/gain, noise, linearity, frequency selectivity, bandwidth, and other parameters. Physical constraints of practical components and circuits are used to identify the achievable signal quality, selectivity, sensitivity, and link budget of a wireless system. The lectures will explain the causes and impact of device/circuit non-linearities and noise on wireless systems' sensitivity and dynamic range. The course will introduce the concepts of frequency generation, modulation/demodulation, and up/down conversion. To teach you to work at RF frequencies with wireless systems, the principles of impedance matching are explained, and tools are provided to analyze/design/characterize RF (matching) circuitry/systems. The common part of the course also includes an overview of the figure of merits (FoMs) of various TX/Rx architectures/system blocks and the required instrumentation to characterize them. This course belongs to the RF/wireless profile and is an excellent preparation for the course; EE4605, Integrated Circuits for RF/Wireless Applications, which focuses on the design and realization of analog intensive building blocks for wireless transceivers as well as for ET4371, Advanced Digital Wireless transceivers which is addressing the latest trends in RF/wireless design aiming to fully utilize the integration and signal processing strengths of advanced CMOS. The course is completed by four short specialization topics which can be followed based on your personal preference, namely: (1) Introduction to wireless power transfer technology, (2) Beamforming basics: concepts and architectures, (3) Neural Networks for Wireless & Radar Systems, (4) Hands-on experience with RF characterization equipment on practical circuits. (The latter (4) has an attendee limit of 15 people.) Only one of these specialization topics is required to complete the overall course.	
Study Goals	Understand wireless system architectures with their subblocks to such a level that you can analyze their various performance parameters in terms of; link budget, sensitivity, spectral purity, P1dB, IP3, DR, SFDR, noise, phase noise, selectivity, EVM, APCR, and efficiency. In addition, you will master a basic understanding of S-parameters, impedance matching, and RF instrumentation.	
Education Method	This course will be delivered with complementary exercise lectures to familiarize students with the concepts and calculations described during the lectures.	
Literature and Study Materials	These lectures are (in part) based on the book of Behzad Razavi, RF microelectronics 2nd edition, Chapters 1 to 4.	
Assessment	Examination and grading: Your overall grade will be based on your performance in the written exam (80%), compulsory homework, and the mark obtained in your track-specific project (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested profiles BD 2022	
Responsible Program	Prof.dr.ir. W.A. Serdijn
Employee	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

BioSemi 2023

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Track core BD Bio Semi 2023

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Literature and Study Materials	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The assessment is based on a written examination and non-mandatory assignments. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the six remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4289	Integrated Circuits and MEMS Technology	4
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Instructor	Dr. C.M.F. Viellard-Boutry	
Instructor	Dr.ing. H.W. van Zeijl	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics (EE 4585)	
Course Contents	This lecture is an introduction in the fabrication technologies used for Integrated Circuits (ICs) and Micro Electro-Mechanical Systems (MEMS). The aim of this lecture is to build a bridge between the physical world (the device physics) and the electronic world (the device characteristics). In this way, students gain more insight in the critical aspects and challenges in IC fabrication. The series of lectures starts with the discussion of basic microelectronic devices with a brief review of the physics involved. The associated basic fabrication technology, required to fabricate these devices is also discussed. After that, the mainstream IC fabrication technology (CMOS) will be discussed in more detail. In separate chapters, silicon crystal growth, epitaxy, ion implantation, chemical vapor deposition, etching and photolithography will be introduced. Their physical motivation and the impact on the device characteristics will be discussed. Finally, silicon bulk and surface micromachining technologies are introduced to illustrate the potential of 3D micro structuring in the development of Microsystems or MEMS. These IC compatible technologies are employed to realize multi-function systems for many scientific and industrial application areas.	
Study Goals	You know the definition of the basic process steps to fabricate a device (IC or MEMS) You can explain the basic physics and chemistry of these fabrication steps in an IC /MEMS process flow You can analyse the relationship between the series of process steps necessary to build the device and the device performance. You can evaluate how device requirements are translated into a process flow You are able to propose solutions to specific device technological questions	
Education Method	lectures	
Literature and Study Materials	R.C. Jager introduction to Microelectronic Fabrication & course slides	
Assessment	written exam In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. S. Vollebregt	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Profile core BD BioSemi 2023

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41095	Medical Devices: Past, Present, Future	3
Responsible Instructor	Dr. D. Dodou	
Instructor	Prof.dr. J.J. van den Dobbelsteen	
Instructor	Prof.dr. J. Dankelman	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. History of surgery 2. Minimally invasive techniques 3. Single-port procedures & NOTES 4. Percutaneous interventions 5. Robotic surgery 6. In vivo robots 7. Medical safety 8. Maintenance of surgical instruments	
Study Goals	The student will be able to describe the function, effectiveness, and challenges of surgical instruments in minimally invasive procedures. More specifically, the student will be able to: 1. illustrate advantages and disadvantages of minimally invasive surgery. 1. illustrate advantages and disadvantages of single-port procedures and NOTES. 3. describe instruments used during minimally invasive surgery, single-port procedures, and NOTES. 4. formulate criteria that surgical instruments have to fulfill in order to prevent sterilization problems. 5. indicate problems with quality control of surgical instruments. 6. describe the fields where robotic surgery is used and indicate the possibilities and limitations of the use of robotics. 7. classify technologies developed to train surgeons outside the operating theater. 8. discuss safety issues in surgery. 9. select information from the literature related to a certain issue and evaluate new developments.	
Education Method	Lectures. 4 hours per week	
Assessment	Homework assignments	
Department	3mE Department Biomechanical Engineering	

EE4555	Active Implantable Biomedical Microsystems	5
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Responsible Instructor	Dr.ir. C. Strydis	
Responsible Instructor	Dr. V. Giagka	
Responsible Instructor	T.M. Lopes Marta da Costa	
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none none	
Course Language	English	
Required for	- MSc Biomedical Engineering, Medical Devices track; - MSc Electrical Engineering, Biomedical Devices profile; - MSc thesis projects in the Section Bioelectronics.	
Expected prior knowledge	It is recommended but not required to have completed/attended the following courses: 1. Anatomy and Physiology (BM41055) 2. Fundamentals of Biomedical Engineering (BM41140) 3. Bioelectricity (ET4130) 4. Themes in Biomedical Electronics (ET4127) 5. Sensors and actuators (ET4257) 6. Measurement and Instrumentation (EE4C08) 7. Systems Engineering (EE4C02)	
Course Contents	Active Implantable Biomedical Microsystems (AIBM) in the human body are now becoming more widely acceptable and available since the development of cardiac pacemakers. Today, miniaturized wireless implantable systems are changing the face of biomedical research and clinical practices through the development of cochlear implants, neuroprostheses, braincomputer interfaces, deep organ pressure sensors, precise drug delivery units, bioelectronic medicine and electroceuticals. With a greater life expectancy and an increasing demand for medical healthcare to guarantee a sufficient health-related quality of life, there is a greater demand on technology and electrical and biomedical engineers to develop active implantable systems for a wide variety of medical diagnostics, treatments and therapies. This course provides a background on the application of AIBMs, an introduction to the latest techniques used and examples of existing medical devices. It can be used as a guide to the design of AIBMs and also as a reference for existing medical devices. The course is aimed at MSc students in Electrical Engineering (tracks Microelectronics, and Signals&Systems) and in Biomedical Engineering (track Medical Devices) who have an interest in the research and design of AIBMs. The following topics are treated: 1. Electrophysiology 2. Electrodes, optodes, sensors and actuators 3. Electrode characterization and modeling 4. Electronic system design 5. Stimulation electronics 6. Front-end electronics (signal conditioning, amplifiers, filters, ADCs, DACs, etc.) 7. Wireless data transfer 8. Energy harvesting and power delivery 9. Applications: cochlear implants and cardiac pacemakers 10. Digital design 11. Biocompatibility, materials and fabrication considerations 12. Assembly, system integration and packaging 13. Application: spinal cord stimulator 14. Reliability and security 15. Risk management and/or 16. Ethics	
Study Goals	After successful completion of the course, you: 1. Know, understand and can apply (3) the functionality, construction and application of active implantable biomedical microsystems (AIBMs) and their sub-systems and parts; 2. Know, understand and can apply (3) the fundamental principles and approaches for designing an implantable system; 3. Know, understand, can apply, analyse and can reason about (evaluate; 5) the design requirements, application-specific implementation challenges, and appropriate design choices of existing AIBMs, incl. cardiac pacemakers, cochlear implants and spinal cord stimulators; 4. Know and understand (2) the most important implantability, biocompatibility, reliability, safety and security issues of AIBMs; 5. Know and understand (2) the most important regulatory and clinical constraints of AIBMs 6. Can reason (5) about future developments of AIBMs, a.o. electroceuticals and bioelectronic medicine; NB. The numbers correspond with Bloom's taxonomy.	
Education Method	1. Lectures 2. Self study 3. Case study 4. Group project	
Literature and Study Materials	1. a collection of book chapters, thesis chapters and review/overview papers; 2. handouts of the slides used during the lectures, 3. video recordings of the lectures all made available via the online learning system (Brightspace). 4. scientific papers (IEEE Xplore, PubMed); 5. patents; 6. user manuals; 7. patient manuals; 8. manufacturer websites; 9. the Internet in general, to be consulted by the students themselves.	
Assessment	The student has to: - Form a group (6 students/group) in the first week, based on affinity with the topics below. - As a group, perform a case study of a complete device, which will be announced in Wk. 4.2.	

- Investigate 6 topics:
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- Prepare an annotated slide set on the first 3 topics in Wk. 4.5
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Max. 12 slides (4 per group of 2 students), using a prescribed template
 - *To be uploaded on Brightspace before Wk 4.6
 - *To be commented on in written form and handed in by another group during the 1st lecture in Wk. 4.7
 - *Sub-group (of 2 students) on topic A comment on topic B; B -> C; C -> A
- Prepare a group presentation on the second 3 topics:
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- One student, to be randomly selected by the teachers on the day of the presentation, will do the presentation (max. 12 minutes) for the entire group in Wk. 4.10 or 4.11.
- Other groups will discuss the presentations (4 minutes) and ask questions and will comment briefly (max. 2 minutes per group / presentation) on all presentations

The teachers will:

- Organize/manage the course
- Give the lectures
- Provide relevant material and background knowledge
- Define the case study
- Give the templates for the annotated slides and the comments
- Grade the slide sets per group of 2 students
- Grade the comments on the slide sets per group of 2 students
- Return the commented slide sets to the authors
- Buy the T-shirts to wear during the final presentations
- Grade the presentations per group of 6 students
- Grade the comments made on the presentations per group of 6 students
- Decide on the final marks

The final mark is based on the grade received for the annotated and commented slide sets (37.5% + 12.5% (bonus/malus), respectively), and the grade received for the group presentation (45%) and the comments made on another group presentation (5%).

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested free electives BioSemi 2023

BM41141	Fundamentals of Biomedical Engineering	5
Responsible Instructor	N. Tümer	
Instructor	Ir. A. Sakes	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, there will be: 1) lectures, through which you will become familiar with the field of biomedical engineering (BME) and the stakeholders playing essential roles within, 2) workshops on engineering role play, team dynamics, peer-review, and debate. Through these workshops, you will practice some of the mentioned skills, e.g., being able to work in teams, and having good communication skills, 3) two assignments: (1) self-reflective journey (mostly individual), and (2) challenge-based assignment (group work). These two assignments are connected with each other. In the first assignment, you will explore who you are and what your role would be in relation to your future career in the BME field. In the second assignment, you will have a challenging context. Together with your team, you will define a solution to a challenging problem considering your engineering roles.	
Study Goals	Knowledge, insight, and skills By the end of this course, you should be able to: 1. Define the problems that need to be addressed in the health field, considering crucial societal aspects, and the roles biomedical engineers and stakeholders can play in solving these issues. 2. Apply physical and engineering terms and principles in building a strategy within the multidisciplinary context of biomedical engineering. 3. Outline the steps needed to be taken in order to validate or implement biomedical techniques, devices, and methods in the health system. Transferable skills By the end of this course, you should be able to: 4. Analyse the impact and consequences of your work when intervening in the field of biomedical engineering. 5. Develop evaluative judgment on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback. 6. Reflect on your own interests, role, and personal development in relation to your future professional self in the field of biomedical engineering.	
Education Method	Lectures, workshops and assignments	
Assessment	Exam (40%) and assignments (60%)	
Department	3mE Department Biomechanical Engineering	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4260	Microsystem Integration	4
Responsible Instructor	T. Manzanque Garcia	
Instructor	K.M. Dowling	
Instructor	F. Arroyo Cardoso	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Instrumentation (EE4C08) Integrated circuits and MEMS technology (ET4289) Sensors & Actuators (ET4257)	
Course Contents	A microsystem is a complete instrument on a chip that can involve different domains of physics (electrical, mechanical, magnetic, etc.). The challenges associated with the combination of elements of different nature, including transducers and circuits into a single-chip integrated system, are more than compensated by the opportunities this concept offers in a wide range of applications. In this course, general concepts related to microsystems as sensors are first discussed. Secondly, approaches for multi-domain modeling are introduced and applied to the modelling electro-mechanical systems. Then, different transduction mechanisms, that allow for the conversion of signals between different physical domains, are introduced. Finally, all the modelling concepts are extended to the design of microsystem-based sensors. Throughout the course, the software package Comsol is introduced and used in several tutorials. The course includes theory (~8 lectures), Comsol tutorials, and tutorials for the realization of a final project, in which a microsystem is designed and modelled in groups.	
Study Goals	By the end of the course, you will be able to: - Identify parts of microsystems and their function. - Identify equations that govern the behaviour of different microsystem components, from different domains of physics. - Use electrical domain models for multidomain microsystems. - Use the finite element method (FEM) to model multidomain microsystems, using Comsol. - Analyze the dynamic behaviour of linear microsystems. - Design microsystems that meet given specifications, using both electrical domain modelling and FEM.	
Education Method	Lectures, tutorials and hands-on practical sessions with Comsol.	
Computer Use	The student requires access to computer systems for carrying out the Comsol tutorials and final assignment.	
Literature and Study Materials	Literature plus lecture notes.	
Assessment	The final grade of the course consists of the following components: - Final assignment report (50%) - Oral exam (50%) Final grade calculation: $(0.5 * \text{Final assignment report} + 0.5 * \text{Oral exam})$ In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4277	Microelectronics Reliability	4
Responsible Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Prof.dr. G.Q. Zhang	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course aims to provide the students with a thorough understanding of the reliability issues involved in electronic components and systems. The following subjects will be treated. - Basic reliability definition and lifetime distributions. - Reliability prediction methods. - Physical failure mechanisms in electronic components - Package related failures - Reliability screening and reliability testing - Failure analysis methods - Reliability data handling - Design considerations - System reliability	
Study Goals	After the course the students: - should see reliability as a basic requirement that should receive attention throughout a products complete lifecycle: specification, design, production, exploitation and disposal. - should have a sound understanding of the physical background of failures and the reliability test and failure analysis methods available. - should be able to make a proper reliability prediction based on available reliability (test) data. - should be able to evaluate reliability and availability figures of complex systems based on component reliability data. - should be able to make the proper design and maintenance choices to optimize these figures.	
Education Method	Lecture	
Assessment	written exam or project (project only possible when homework assignments are passed with average grade 6 or more) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4391	Advanced Microelectronics packaging	3
Responsible Instructor	Prof.dr. G.Q. Zhang	
Responsible Instructor	M. Mastrangeli	
Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Dr.ir. R.H. Poelma	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This is a multi-disciplinary course, dealing with scientifically challenging and economically important technology issues. MSc students from but not limited to electrical engineering, mechanical engineering, material science and physics are encouraged to attend.	
Course Contents	1. Introduction of micro/nanoelectronics packaging - Basics and state of the art of advanced front-end and back-end technologies - International roadmaps and strategic research agenda - Major application trends 2. Interconnect technologies 3. MEMS & sensor packaging 4. 3D and wafer-level system integration 5. Application specific packaging -1 (IC packaging, etc.) 6. Application specific packaging -2 (LED, etc) 7. Application specific packaging -3 (RF, Power, automotive, healthcare, etc.)	
Course Contents Continuation	As the bridge between devices and various multi-functional electronics systems, microelectronic system integration and packaging control more than 90% of the size, 60% of the cost, and largely the system performance and reliability. It is one of the most fascinating and rapidly developing technology and business fields in semiconductors, playing a dominant role in the development of future micro/nanoelectronics and systems.	
Study Goals	At the end of the course, the student should be able to: 1) Describe and assess basics and state-of-the-art in microelectronic packaging and reliability; 2) Describe and assess application-specific packaging and associated requirements, constraints and solutions for reliability; 3) Design, implement and characterize a standalone electronic and/or microelectromechanical system.	
Education Method	Lectures and field visit.	
Assessment	Assessment is based on mini-research projects, conducted preferably in teams. The projects are proposed by the instructors, and can also be proposed in agreement with the instructors by the students based on the content of the course or personal interests. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ing. H.W. van Zeijl	
Co-Instructor	M. Mastrangeli	

ET4icp	IC-Technology Lab	2
Responsible Instructor	Dr.ir. S. Vollebregt	
Contact Hours / Week x/x/x/x	28 hrs.	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Semiconductor Device Physics as in EE4585	
Course Contents	During seven half days the students, in groups of 3, will have the opportunity to gain experience in the process- and device-simulation of a MOSFET transistor. They will perform several processing steps in the Else Kooi Lab clean room and will measure and characterize the fabricated device. Enrollment through Brightspace, max course capacity is 36 students!	
Study Goals	After this course, students are able to: 1. Explain the different process steps of the IC fabrication process 2. Perform calculations on the oxide thickness of silicon after thermal oxidation, and the implantation depth and diffusion depth of dopants 3. Explain the relation between threshold voltage and implantations, and relate this to the measured threshold voltages from transistors 4. Select dopant atoms and masking material for silicon devices based on the simulations performed in the course 5. Identify the different implanted regions and junctions in cross-sections obtained from doping profile simulations 6. Argue on the specific advantages and disadvantages of wet and dry etching as used during the lab course. 7. Analyse the measurements obtained from different test structures described in the course guide 8. Determine the doping levels from sheet resistance measurements	
Education Method	Lab course	
Literature and Study Materials	Lecture slides and a course manual which will be supplied during the compulsory lecture	
Assessment	Group report. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	28 hrs, first session is scheduled, after that on individual appointment. Students have to attend all parts of the course. The course can only accommodate 30 students.	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (5ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

BioCAS 2023

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Track core BD BioCAS 2023

EE4520	Analog CMOS design I	3
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr.ir. K. Bult	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE4C10	
Course Contents	The goal of this course is to introduce students to the design of basic analog circuits in modern CMOS technology. Students will learn about the operation of MOSFETs, how they can be used to realize basic analog circuits and how to verify the performance of these circuits with the help of Spice-like circuit simulators (LTSpice will be used during the course). Covered topics: - DC, AC and transient analysis. - Opamps and feedback: effect of feedback on impedances; frequency behavior; time-domain behavior & stability; asymptotes & feedback; signal integrity. - The MOS transistor: regions of operation; simplified model for hand-calculations; second-order effects. - Single-Stage Amplifiers: DC behavior; AC Behavior; transient behavior; fundamental limitations. - Cascoding: effect on DC gain; effetc of MOS ft. - Noise: origins & properties, noise in circuit design. - Distortion: calculation procedure; symmetrical systems; distortion and feedback.	
Study Goals	- Students can explain the different region of operation of MOS transistors; they can apply the simplified model to CMOS circuit analysis and assess the effect of second-order effects. - Student can perform hand-calculation for AC, DC and transient analysis of CMOS circuits - Students can derive the performance of simple CMOS circuits (such as single-stage amplifiers) in terms of amplification, noise and distortion. - Students can design and optimize simple CMOS circuits (single-stage amplifiers, cacoded amplifiers). - Student can derive performance of CMOS circuits by means of a Spice-like circuit simulator and compare the results to simplified hand-calculation.	
Education Method	Lectures and hands-on sessions.	
Literature and Study Materials	B. Razavi: Design of Analog CMOS Integrated Circuits	
Reader	A reader will be provided	
Assessment	Final grade = homework (20%) + final project (80%)	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Profile core BD BioCAS 2023

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41095	Medical Devices: Past, Present, Future	3
Responsible Instructor	Dr. D. Dodou	
Instructor	Prof.dr. J.J. van den Dobbelsteen	
Instructor	Prof.dr. J. Dankelman	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. History of surgery 2. Minimally invasive techniques 3. Single-port procedures & NOTES 4. Percutaneous interventions 5. Robotic surgery 6. In vivo robots 7. Medical safety 8. Maintenance of surgical instruments	
Study Goals	The student will be able to describe the function, effectiveness, and challenges of surgical instruments in minimally invasive procedures. More specifically, the student will be able to: 1. illustrate advantages and disadvantages of minimally invasive surgery. 1. illustrate advantages and disadvantages of single-port procedures and NOTES. 3. describe instruments used during minimally invasive surgery, single-port procedures, and NOTES. 4. formulate criteria that surgical instruments have to fulfill in order to prevent sterilization problems. 5. indicate problems with quality control of surgical instruments. 6. describe the fields where robotic surgery is used and indicate the possibilities and limitations of the use of robotics. 7. classify technologies developed to train surgeons outside the operating theater. 8. discuss safety issues in surgery. 9. select information from the literature related to a certain issue and evaluate new developments.	
Education Method	Lectures. 4 hours per week	
Assessment	Homework assignments	
Department	3mE Department Biomechanical Engineering	

EE4555	Active Implantable Biomedical Microsystems	5
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Responsible Instructor	Dr.ir. C. Strydis	
Responsible Instructor	Dr. V. Giagka	
Responsible Instructor	T.M. Lopes Marta da Costa	
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none none	
Course Language	English	
Required for	- MSc Biomedical Engineering, Medical Devices track; - MSc Electrical Engineering, Biomedical Devices profile; - MSc thesis projects in the Section Bioelectronics.	
Expected prior knowledge	It is recommended but not required to have completed/attended the following courses: 1. Anatomy and Physiology (BM41055) 2. Fundamentals of Biomedical Engineering (BM41140) 3. Bioelectricity (ET4130) 4. Themes in Biomedical Electronics (ET4127) 5. Sensors and actuators (ET4257) 6. Measurement and Instrumentation (EE4C08) 7. Systems Engineering (EE4C02)	
Course Contents	Active Implantable Biomedical Microsystems (AIBM) in the human body are now becoming more widely acceptable and available since the development of cardiac pacemakers. Today, miniaturized wireless implantable systems are changing the face of biomedical research and clinical practices through the development of cochlear implants, neuroprostheses, braincomputer interfaces, deep organ pressure sensors, precise drug delivery units, bioelectronic medicine and electroceuticals. With a greater life expectancy and an increasing demand for medical healthcare to guarantee a sufficient health-related quality of life, there is a greater demand on technology and electrical and biomedical engineers to develop active implantable systems for a wide variety of medical diagnostics, treatments and therapies. This course provides a background on the application of AIBMs, an introduction to the latest techniques used and examples of existing medical devices. It can be used as a guide to the design of AIBMs and also as a reference for existing medical devices. The course is aimed at MSc students in Electrical Engineering (tracks Microelectronics, and Signals&Systems) and in Biomedical Engineering (track Medical Devices) who have an interest in the research and design of AIBMs. The following topics are treated: 1. Electrophysiology 2. Electrodes, optodes, sensors and actuators 3. Electrode characterization and modeling 4. Electronic system design 5. Stimulation electronics 6. Front-end electronics (signal conditioning, amplifiers, filters, ADCs, DACs, etc.) 7. Wireless data transfer 8. Energy harvesting and power delivery 9. Applications: cochlear implants and cardiac pacemakers 10. Digital design 11. Biocompatibility, materials and fabrication considerations 12. Assembly, system integration and packaging 13. Application: spinal cord stimulator 14. Reliability and security 15. Risk management and/or 16. Ethics	
Study Goals	After successful completion of the course, you: 1. Know, understand and can apply (3) the functionality, construction and application of active implantable biomedical microsystems (AIBMs) and their sub-systems and parts; 2. Know, understand and can apply (3) the fundamental principles and approaches for designing an implantable system; 3. Know, understand, can apply, analyse and can reason about (evaluate; 5) the design requirements, application-specific implementation challenges, and appropriate design choices of existing AIBMs, incl. cardiac pacemakers, cochlear implants and spinal cord stimulators; 4. Know and understand (2) the most important implantability, biocompatibility, reliability, safety and security issues of AIBMs; 5. Know and understand (2) the most important regulatory and clinical constraints of AIBMs 6. Can reason (5) about future developments of AIBMs, a.o. electroceuticals and bioelectronic medicine; NB. The numbers correspond with Bloom's taxonomy.	
Education Method	1. Lectures 2. Self study 3. Case study 4. Group project	
Literature and Study Materials	1. a collection of book chapters, thesis chapters and review/overview papers; 2. handouts of the slides used during the lectures, 3. video recordings of the lectures all made available via the online learning system (Brightspace). 4. scientific papers (IEEE Xplore, PubMed); 5. patents; 6. user manuals; 7. patient manuals; 8. manufacturer websites; 9. the Internet in general, to be consulted by the students themselves.	
Assessment	The student has to: - Form a group (6 students/group) in the first week, based on affinity with the topics below. - As a group, perform a case study of a complete device, which will be announced in Wk. 4.2.	

- Investigate 6 topics:
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- Prepare an annotated slide set on the first 3 topics in Wk. 4.5
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Max. 12 slides (4 per group of 2 students), using a prescribed template
 - *To be uploaded on Brightspace before Wk 4.6
 - *To be commented on in written form and handed in by another group during the 1st lecture in Wk. 4.7
 - *Sub-group (of 2 students) on topic A comment on topic B; B -> C; C -> A
- Prepare a group presentation on the second 3 topics:
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- One student, to be randomly selected by the teachers on the day of the presentation, will do the presentation (max. 12 minutes) for the entire group in Wk. 4.10 or 4.11.
- Other groups will discuss the presentations (4 minutes) and ask questions and will comment briefly (max. 2 minutes per group / presentation) on all presentations

The teachers will:

- Organize/manage the course
- Give the lectures
- Provide relevant material and background knowledge
- Define the case study
- Give the templates for the annotated slides and the comments
- Grade the slide sets per group of 2 students
- Grade the comments on the slide sets per group of 2 students
- Return the commented slide sets to the authors
- Buy the T-shirts to wear during the final presentations
- Grade the presentations per group of 6 students
- Grade the comments made on the presentations per group of 6 students
- Decide on the final marks

The final mark is based on the grade received for the annotated and commented slide sets (37.5% + 12.5% (bonus/malus), respectively), and the grade received for the group presentation (45%) and the comments made on another group presentation (5%).

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested free electives BioCAS 2023

BM41141	Fundamentals of Biomedical Engineering	5
Responsible Instructor	N. Tümer	
Instructor	Ir. A. Sakes	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, there will be: 1) lectures, through which you will become familiar with the field of biomedical engineering (BME) and the stakeholders playing essential roles within, 2) workshops on engineering role play, team dynamics, peer-review, and debate. Through these workshops, you will practice some of the mentioned skills, e.g., being able to work in teams, and having good communication skills, 3) two assignments: (1) self-reflective journey (mostly individual), and (2) challenge-based assignment (group work). These two assignments are connected with each other. In the first assignment, you will explore who you are and what your role would be in relation to your future career in the BME field. In the second assignment, you will have a challenging context. Together with your team, you will define a solution to a challenging problem considering your engineering roles.	
Study Goals	Knowledge, insight, and skills By the end of this course, you should be able to: 1. Define the problems that need to be addressed in the health field, considering crucial societal aspects, and the roles biomedical engineers and stakeholders can play in solving these issues. 2. Apply physical and engineering terms and principles in building a strategy within the multidisciplinary context of biomedical engineering. 3. Outline the steps needed to be taken in order to validate or implement biomedical techniques, devices, and methods in the health system. Transferable skills By the end of this course, you should be able to: 4. Analyse the impact and consequences of your work when intervening in the field of biomedical engineering. 5. Develop evaluative judgment on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback. 6. Reflect on your own interests, role, and personal development in relation to your future professional self in the field of biomedical engineering.	
Education Method	Lectures, workshops and assignments	
Assessment	Exam (40%) and assignments (60%)	
Department	3mE Department Biomechanical Engineering	

	- setup design considerations for signal handling capability and power efficiency and take adequate design decisions - setup design considerations for frequency response and take adequate design decisions - setup design considerations for weak distortion and take adequate design decisions - setup design considerations for biasing and design and implement proper biasing circuitry - evaluate the performance versus requirements of the final design and propose options for improvement - select proper models and design verification tools for each design step - write a design report in which all design steps are motivated and the performance is evaluated and explained.
Education Method	Lectures and Colstructions: The systematic electronics design method will be illustrated during the lectures and colstructions with a design example made in SLiCAP. The design example closely relates to the hearing loop system designed during the BSc course EE3C11. This BSc design was based on operational amplifiers as controllers. In this course, the focus will be on the design of the controllers as transistor circuits. During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design. The design steps are documented in a report using SLiCAP. The design report can be efficiently used as a basis for the exam, which is organized as a design review for the design team.
Computer Use	A computer with SliCAP and the free circuit simulator LT-Spice is necessary. Help for installing and using SLiCAP will be available during the lectures. SLiCAP is an acronym for Symbolic Linear Circuit Analysis Program.
Literature and Study Materials	See: https://analog-electronics.tudelft.nl/SLiCAP.html Book (also used during BSc course EE3C11): Structured Electronic Design Edition downloadable from: https://analog-electronics.tudelft.nl
Assessment	During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design used in every lecture to illustrate the application of the theory from the lectures. At the end of the lecture period, the student team makes an appointment review of their design. The design review consists of two sessions: 1) A design preview, where the students present and defend their design in a relaxed atmosphere. They can use the feedback to make improvements. 2) A design review, where the students present and defend their design again. Each team member has to give substantial input during the session. After this session, each student is graded individually. The design report, created with SLiCAP during the lectures by the team, can be efficiently used as a basis for the review. If there are logistic or personal reasons that make it impossible for a student to join a design team, it is possible to opt for an individual design review (oral exam). Please contact the educators about this. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4615	Digital IC Design II	3
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will consolidate the knowledge acquired in EE4610 by practicing the design of an advanced complex digital VLSI design, where high-level specifications are translated to transistor sizes, going through several abstraction levels. At each level, the opportunities and limitations of the physical implementation will be considered for finding better solutions and tradeoffs, by means optimization and the search of new topologies, while keeping complexity to a minimum to achieve the desired functionality. Parasitics and the other effects deriving from the analog behavior of digital circuits will be taken into account using four-corner analysis, as well as simplifications and modeling at each abstraction level and other semi-automatic tools.	
Study Goals	The main goal of the course is to give students the tools to successfully design complex, scalable circuits and systems. The students will also learn how to work in and coordinate teams of designers around a complex project from high-level specifications to the actual transistor-level implementation and post-layout verification.	
Education Method	The course will entirely be based in the lab where the target design will be implemented by students in groups of 2-3. Only based on students/groups wish, the final design will be laid out in a commercial CMOS process and possibly fabricated and tested.	
Assessment	The course will be graded based on the target design. The following criteria will be taken into account: 1. Functionality achieved; 2. High-level specifications met; 3. Four-corner analysis completed; Bonus points will be given to those students that will exceed the specifications and create a robust, reproducible design. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4252	Analog Integrated Circuit Design	4
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	This course is recommended for students intending to take: - Structured Electronic Design Laboratory (ET8011MSC), - Analog CMOS design II (EE4525), - Oversampled data converters (ET4278), - Nyquist-rate data converters (ET4369), - Introduction to Power Conversion Technology (ET4382). This course is obligatory for students that wish to do their final project in the area of Biomedical Circuits (and Systems) (BioCAS).	
Expected prior knowledge	This course assumes (though it is not obligatory) that you have completed your BSc studies in electrical engineering and taken our MSc courses Analog Circuit Design Fundamentals (EE4C10) and Structured Electronic Design (EE4109) Having attended Measurement and Instrumentation (EE4C08) and Analog CMOS Design I (EE4520) is a plus.	
Course Contents	An intermediate course in analog circuit synthesis for microelectronic designers. Topics include: - review of analog design basics (i.e., frequency response, feedback and stability, noise analysis); - MOS models for short-channel modern devices; - gm/ID design methodology for modern analog circuits; - physical layout (e.g., device matching) for robust analog circuits; - design of voltage sources ranging from simple voltage dividers to high-performance bandgaps, and - current source implementations from a single resistor to high-quality references based on negative-feedback structures; - biasing techniques; - high-performance analog filters, from impulse response or transfer function down to circuit-level design; - quality aspects, such as accuracy, noise and impedance levels are treated within the context of power and area consumption requirements.	
Study Goals	To know, understand and be able to analyze and design (synthesize): - high-performance analog integrated filters - voltage references - bandgap references - current sources - amplifiers in all IC technologies (CMOS, bipolar, ...) and regions of operation (weak, moderate and strong inversion; saturation and triode/conduction)	
Education Method	Interactive lectures, flipped classroom (discussing lecture recordings and answering questions), homework assignments, class project.	
Literature and Study Materials	Reader, course notes, and video recordings of the lectures - all available via Brightspace.	
Assessment	Written examination (70%; the other 30% comes from a successfully completed class project). Students are allowed to bring 1 handwritten piece (A4) of paper and a pocket calculator to the exam. The results of the homework assignments may add up to one bonus point to the exam, only for the exam in April. The re-sit written exam in June will count for 100% of the final grade. It is not possible to re-sit the class project. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4278	Over-Sampled Data Converters	4
Responsible Instructor	Prof.dr. K.A.A. Makinwa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10), Analog CMOS Design I (ET4295) and Nyquist-rate Analog-to-Digital Converters (ET4369). A good knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce students to the design of over-sampled converters. Students will learn about the basic functionality of sigma-delta converters, their architecture, how to simulate and design these architectures and how to map them to actual circuit blocks. Covered topics: - First-order sigma-delta: oversampling; noise-shaping; NTF/STF; decimation; overload; leakage. - Second-order sigma delta: linear model; stability; quantizer model; overload stability. - Sigma-delta architectures: bandpass; continuous-time; multi-bit; cascaded/MASH; feedback; feedforward; local feedback - Building blocks: SC integrator; RC integrator; loopfilter; current DAC; SC DAC; clock jitter - Case studies.	
Study Goals	- Students can explain the basic functionality of sigma delta modulators (over-sampling, noise-shaping, decimation, stability). - Students can describe the circuit implementations of various building blocks (integrators, DAC, quantizer, feed-forward and feedback coefficients). - Students can identify and compare the advantages and disadvantages of different architectures for sigma delta modulators (feedback, feed-forward, MASH, multi-bit, single-bit, continuous time and discrete time modulators). - Students can derive the STF and NTF from the linear model of a sigma delta modulator. - Students can design and simulate the architecture of a sigma delta modulator (continuous and discrete time, first and second order modulators, single-bit and multi-bit quantizer, MASH). - Students can simulate the effect of non-idealities of various blocks (such integrators limited gain, DAC non-linearity) on the performance of a sigma delta modulator. - Students can propose a (non-optimized) circuit implementation of a sigma delta modulator. - Students can identify the main techniques used in state-of-the-art designs of sigma delta modulators and assess their weakness and strengths. - Students can propose improvements to existing designs of sigma delta modulators.	
Education Method	Lectures and practical sessions	
Literature and Study Materials	hand-outs provided during the lectures; optional reference: R. Schreier, S. Pavan, and G.C.Temes, "Understanding Delta-Sigma Data Converters", Second Edition, published by John Wiley & Sons Inc., New Jersey, 2017.	
Assessment	Homework (30%), a final assignment discussed during the oral exam (50%), and an oral exam (20%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Textbook	
Remarks	Guest Instructor: Dr.M.Bolat kale, NXP Semiconductors	

ET4351	Digital VLSI Systems on Chip	4
Responsible Instructor	Dr.ing. C.P. Frenkel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of digital design (logic gates, combinational logic and sequential logic, finite state machines). Knowledge of hardware description languages (Verilog or VHDL), including how to write proper synthesizable code, is necessary. Previous exposure to microprocessor architecture is helpful but not necessary.	
Course Contents	Today's large-scale digital integrated circuits and systems-on-a-chip, such as modern general processing units (GPUs), can >50 billion transistors. Two things were instrumental in enabling this massive scalability: electronic design automation (EDA) tools and intellectual property (IP) blocks allowing for hierarchical design. In this course, you will learn how to use such EDA tools and IP blocks to go from the register-transfer level (RTL) description of your digital design to a final layout that is clean for tapeout and meets your specifications. To do so, you will also learn about power/performance/area tradeoff optimization, hardware-algorithm co-design, and memory hierarchy. These tools and concepts will be practiced and applied in a project, in which you will explore and apply your own design choices to meet a well-defined design target in a digital system-on-chip.	
Study Goals	After successful completion of the course, you will be able to: LO1. Design modern digital systems-on-a-chip with automated synthesis and place-and-route tools by - identifying the key design targets and tradeoffs, - selecting a proper IP to meet design targets, - translating the design constraints and tradeoffs to tool commands, - applying the tools to meet design targets. LO2. Optimize, in a rigorous and creative manner, the power/performance/area tradeoff associated with a given application use case. LO3. Link algorithmic requirements to hardware constraints. LO4. Extract the key performance metrics of a digital design by - verifying that simulation results can be trusted and that the design is clean for tapeout, - communicating these results effectively and rigorously, - discussing their implications within a broader societal context.	
Education Method	Lectures, labs	
Literature and Study Materials	* Course material (slides will be provided) * (Recommended) Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, by Erik Brunvand, Addison-Wesley, ISBN-10: 0321547993	
Assessment	The final course grade consists of two parts: * Final design report (50%) * Individual written exam (50%) The results of the two parts must have been obtained within the same academic year, and students must have a grade of 5.8 or higher for both the final design report and the individual written exam to pass the course. In case either of the two parts is failed, a resit must be taken in the form of an individual written exam that will account for 100% of the grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	C. Gao	

ET4369	Nyquist-Rate Data Converters	4
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the courses: Analog Circuit Design Fundamentals (EE4C10) and Analog CMOS Design I (ET4295)	
Course Contents	Introduction, sampling theory, quantization, and reconstruction; Digital-to-analog conversion: resistive, capacitive, and current steering; Sampling circuits: theory, non-idealities, and advanced techniques; Comparators: circuits, non-idealities, and design examples; Analog-to-digital conversion: flash, SAR, pipeline, time-interleaving.	
Study Goals	By the end of this course, you will be able to: - Explain the fundamentals of sampling, quantisation and reconstruction in data converters - Analyse the effect of non-idealities in sampling circuits and discuss advanced techniques to deal with them (bottom plate sampling, bootstrapping, oversampling, etc.) - Analyse the architectures and circuits commonly used in D/A interfaces (capacitive, resistive, current steering) and discuss their pros and cons - Analyse the architectures and circuits commonly used in A/D interfaces (flash, SAR, pipeline, time-interleaving) and discuss their pros and cons - read, understand and critically analyse state-of-the-art IEEE papers on Nyquist-rate data converters - Select the appropriate ADC architecture for a given set of performance specifications	
Education Method	4 hours of lectures plus a 4 hour tutorial	
Literature and Study Materials	MJM Pelgrom: Analog-to-Digital Conversion, 3rd Ed. Springer 2017	
Reader	Pdf versions of the slides and the homework will be provided	
Assessment	Weekly homework (formative assessment) + oral exam based on a selected ADC publication (summative assessment for 100% of the final grade). Resit oral exam based on a selected ADC publication possible in Q4 for 100% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (5ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

BioSP 2023

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Track core BD BioSP 2023

EE4595	Wavefield Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Linear algebra (four fundamental subspaces of a matrix, eigenvalue decomposition, singular value decomposition), signals and systems (the Fourier transform), and at least an elementary (BSc) course on electromagnetic or acoustic wave propagation.	
Course Contents	In many sensing and imaging problems (geophysical exploration, radio astronomy, nondestructive testing, medical X-ray tomography, etc.) one is interested in information about an object that is not directly accessible and possibly obscured by a surrounding medium. Waves, fields, or a stream of particles may be used as probes to obtain this information. By measuring the fields or particles after they have traveled through an object, the goal is to retrieve the location of this object or its internal constitution or both. This may be realized via sophisticated imaging and inversion algorithms that process the signals as measured by the receivers. These algorithms are often computationally intensive especially for 3D real-world problems. Moreover, inverse problems are usually ill-posed and solutions may be very sensitive to noise or other perturbations of the data. Developing efficient and robust imaging and inversion algorithms is therefore an extremely challenging problem. In this course, we present a general overview of this exciting research field. We follow a multidisciplinary approach and introduce various imaging principles, methods, and applications in a uniform framework. The commonalities that exist between different problems in different fields are emphasized and both linear and nonlinear state-of-the-art solution techniques are discussed.	
Study Goals	After a successful completion of this course, you will have an overview of the many different imaging and inversion problems that exist today. You will know the general physical principles behind image formation, and you will know how to retrieve the medium parameters of a penetrable object from measured data. You will also know the difference between an inverse source and inverse scattering problem and how different imaging techniques are related to Born-type linearization of the nonlinear inverse scattering problem. You will be familiar with different linear and nonlinear solution methods such filtered backpropagation, the linear and nonlinear conjugate gradient (CG) method, Contrast Source Inversion, and Newton's method and its variants.	
Education Method	Lectures	
Computer Use	Matlab	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes available on Brightspace.	
Assessment	Assignment and oral. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

BD BioSP 2023 (select 1 out of 4)

EE4530	Applied Convex Optimization	5
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	B. Hunyadi	
Instructor	A. Natali	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	[Theory] A good knowledge of calculus and linear algebra is required, a working knowledge of numerical methods is recommended. [Homeworks] Programming skills in Matlab are required.	
Course Contents	[Draft/Subject to minor changes] Week 1 - Introduction to mathematical optimization [Ch. 1 and 9 and notes] Week 2 - Introduction to convexity [Ch. 2 and 3] Week 3 - Convex optimization problems [Ch. 4] Week 4 - Lagrangian duality [Ch. 5 and 10] Week 5 - Applications in EE [TBD] Week 6 - Interior point methods [Ch. 11] Week 7 - Convex relaxations [notes]	
Study Goals	To be able to understand the key concepts in convex optimization To be able to formulate engineering problems into convex optimization problems and be able to analyze their analytical properties To be able to solve convex optimization problems using Matlab tools To be able to apply the learned key concepts to solve new optimization problems efficiently	
Education Method	Class lectures with slide presentations. Students are supposed to hand-in homeworks and pass a written examination.	
Literature and Study Materials	The course book is: Boyd-Vandenberghe, "Convex Optimization", Cambridge University Press. It can be found online.	
Books	Another useful book is: Nesterov, "Introductory Lectures on Convex Optimization - A Basic Course", Kluwer Academic Publishers. It can be found online.	
Assessment	Homework sets and written examination. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. - Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. - Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. - Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. - Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. - Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. - Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the written exam it is allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Profile core BD BioSP 2023

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41095	Medical Devices: Past, Present, Future	3
Responsible Instructor	Dr. D. Dodou	
Instructor	Prof.dr. J.J. van den Dobbelsteen	
Instructor	Prof.dr. J. Dankelman	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. History of surgery 2. Minimally invasive techniques 3. Single-port procedures & NOTES 4. Percutaneous interventions 5. Robotic surgery 6. In vivo robots 7. Medical safety 8. Maintenance of surgical instruments	
Study Goals	The student will be able to describe the function, effectiveness, and challenges of surgical instruments in minimally invasive procedures. More specifically, the student will be able to: 1. illustrate advantages and disadvantages of minimally invasive surgery. 1. illustrate advantages and disadvantages of single-port procedures and NOTES. 3. describe instruments used during minimally invasive surgery, single-port procedures, and NOTES. 4. formulate criteria that surgical instruments have to fulfill in order to prevent sterilization problems. 5. indicate problems with quality control of surgical instruments. 6. describe the fields where robotic surgery is used and indicate the possibilities and limitations of the use of robotics. 7. classify technologies developed to train surgeons outside the operating theater. 8. discuss safety issues in surgery. 9. select information from the literature related to a certain issue and evaluate new developments.	
Education Method	Lectures. 4 hours per week	
Assessment	Homework assignments	
Department	3mE Department Biomechanical Engineering	

EE4555	Active Implantable Biomedical Microsystems	5
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Responsible Instructor	Dr.ir. C. Strydis	
Responsible Instructor	Dr. V. Giagka	
Responsible Instructor	T.M. Lopes Marta da Costa	
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none none	
Course Language	English	
Required for	- MSc Biomedical Engineering, Medical Devices track; - MSc Electrical Engineering, Biomedical Devices profile; - MSc thesis projects in the Section Bioelectronics.	
Expected prior knowledge	It is recommended but not required to have completed/attended the following courses: 1. Anatomy and Physiology (BM41055) 2. Fundamentals of Biomedical Engineering (BM41140) 3. Bioelectricity (ET4130) 4. Themes in Biomedical Electronics (ET4127) 5. Sensors and actuators (ET4257) 6. Measurement and Instrumentation (EE4C08) 7. Systems Engineering (EE4C02)	
Course Contents	Active Implantable Biomedical Microsystems (AIBM) in the human body are now becoming more widely acceptable and available since the development of cardiac pacemakers. Today, miniaturized wireless implantable systems are changing the face of biomedical research and clinical practices through the development of cochlear implants, neuroprostheses, braincomputer interfaces, deep organ pressure sensors, precise drug delivery units, bioelectronic medicine and electroceuticals. With a greater life expectancy and an increasing demand for medical healthcare to guarantee a sufficient health-related quality of life, there is a greater demand on technology and electrical and biomedical engineers to develop active implantable systems for a wide variety of medical diagnostics, treatments and therapies. This course provides a background on the application of AIBMs, an introduction to the latest techniques used and examples of existing medical devices. It can be used as a guide to the design of AIBMs and also as a reference for existing medical devices. The course is aimed at MSc students in Electrical Engineering (tracks Microelectronics, and Signals&Systems) and in Biomedical Engineering (track Medical Devices) who have an interest in the research and design of AIBMs. The following topics are treated: 1. Electrophysiology 2. Electrodes, optodes, sensors and actuators 3. Electrode characterization and modeling 4. Electronic system design 5. Stimulation electronics 6. Front-end electronics (signal conditioning, amplifiers, filters, ADCs, DACs, etc.) 7. Wireless data transfer 8. Energy harvesting and power delivery 9. Applications: cochlear implants and cardiac pacemakers 10. Digital design 11. Biocompatibility, materials and fabrication considerations 12. Assembly, system integration and packaging 13. Application: spinal cord stimulator 14. Reliability and security 15. Risk management and/or 16. Ethics	
Study Goals	After successful completion of the course, you: 1. Know, understand and can apply (3) the functionality, construction and application of active implantable biomedical microsystems (AIBMs) and their sub-systems and parts; 2. Know, understand and can apply (3) the fundamental principles and approaches for designing an implantable system; 3. Know, understand, can apply, analyse and can reason about (evaluate; 5) the design requirements, application-specific implementation challenges, and appropriate design choices of existing AIBMs, incl. cardiac pacemakers, cochlear implants and spinal cord stimulators; 4. Know and understand (2) the most important implantability, biocompatibility, reliability, safety and security issues of AIBMs; 5. Know and understand (2) the most important regulatory and clinical constraints of AIBMs 6. Can reason (5) about future developments of AIBMs, a.o. electroceuticals and bioelectronic medicine; NB. The numbers correspond with Bloom's taxonomy.	
Education Method	1. Lectures 2. Self study 3. Case study 4. Group project	
Literature and Study Materials	1. a collection of book chapters, thesis chapters and review/overview papers; 2. handouts of the slides used during the lectures, 3. video recordings of the lectures all made available via the online learning system (Brightspace). 4. scientific papers (IEEE Xplore, PubMed); 5. patents; 6. user manuals; 7. patient manuals; 8. manufacturer websites; 9. the Internet in general, to be consulted by the students themselves.	
Assessment	The student has to: - Form a group (6 students/group) in the first week, based on affinity with the topics below. - As a group, perform a case study of a complete device, which will be announced in Wk. 4.2.	

- Investigate 6 topics:
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- Prepare an annotated slide set on the first 3 topics in Wk. 4.5
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Max. 12 slides (4 per group of 2 students), using a prescribed template
 - *To be uploaded on Brightspace before Wk 4.6
 - *To be commented on in written form and handed in by another group during the 1st lecture in Wk. 4.7
 - *Sub-group (of 2 students) on topic A comment on topic B; B -> C; C -> A
- Prepare a group presentation on the second 3 topics:
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- One student, to be randomly selected by the teachers on the day of the presentation, will do the presentation (max. 12 minutes) for the entire group in Wk. 4.10 or 4.11.
- Other groups will discuss the presentations (4 minutes) and ask questions and will comment briefly (max. 2 minutes per group / presentation) on all presentations

The teachers will:

- Organize/manage the course
- Give the lectures
- Provide relevant material and background knowledge
- Define the case study
- Give the templates for the annotated slides and the comments
- Grade the slide sets per group of 2 students
- Grade the comments on the slide sets per group of 2 students
- Return the commented slide sets to the authors
- Buy the T-shirts to wear during the final presentations
- Grade the presentations per group of 6 students
- Grade the comments made on the presentations per group of 6 students
- Decide on the final marks

The final mark is based on the grade received for the annotated and commented slide sets (37.5% + 12.5% (bonus/malus), respectively), and the grade received for the group presentation (45%) and the comments made on another group presentation (5%).

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested free electives BioSP 2023

AP3232	Medical Imaging Signals and Systems	6
Responsible Instructor	Dr.ir. M.C. Goorden	
Instructor	Dr. D. Maresca	
Instructor	Dr. S.D. Weingärtner	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A thorough mathematical basis is essential to follow the class. This can for example include Linear Algebra I + II and Calculus I + II. An in-depth understanding of Fourier analysis and basic physical understanding of electromagnetism is necessary."	
Course Contents	The course Medical Imaging Signals and Systems covers the principles of advanced medical imaging modalities. Particularly, students will study the physics, acquisition techniques and signal processing underlying conventional X-ray, Computerized Tomography (CT), Single Photon Emission Computed Tomography (SPECT), Positron Emission Tomography (PET), Ultrasound (US) and Magnetic Resonance Imaging (MRI). Course Content: Introductory lecture with the history of medical imaging and its importance for modern medicine and biomedical research. Conventional X-ray, CT, SPECT and PET imaging (33% of content): Basic principles of planar X-ray imaging and scintigraphy (short); Advanced methods for image acquisition using CT, SPECT and PET; Mathematical techniques for image reconstruction, including filtered backprojection and iterative approaches; Issues of image quality, especially related to resolution, signal to noise ratio, quantitative and dynamic imaging. MR imaging (33% of content): Fundamentals of MRI: spin, Larmor frequency, T1, T2, T2* contrast (short); Spatial encoding: K-space; Image Quality: PSF, signal strength and noise level; Advanced MRI techniques such as diffusion, perfusion and flow imaging. Ultrasound imaging (33% of content): Physical and mathematical principles of ultrasound imaging, including probe design and advanced US imaging techniques such as harmonic imaging and non-linear reconstruction techniques for imaging tissue properties. The course will end with a mini-symposium on specialized imaging techniques. Therefore, recently developed medical imaging methods will be studied, going beyond the theory of the lectures. Students working in pairs will write a short essay on a selected technique and present it to the group.	
Study Goals	1. Acquire in-depth knowledge about the physics and image reconstruction underlying X-ray, CT, acoustical and magnetic resonance imaging; 2. Being able to solve elementary problems related to the theory mentioned in 1; 3. Being able to solve more advanced problems addressing the theory mentioned in 1 by combining mathematical skills and physical insight; 4. Able to acquire new knowledge about clinical a medical imaging technique; 5. Being able to present newly acquired knowledge about a medical imaging technique.	
Education Method	Lectures and an assignment.	
Literature and Study Materials	Book: Medical Imaging Signals & Systems, Jerry L. Prince, Jonathan Links. Upper Saddle River NJ: Prentice Hall, second edition, 2014, 544 pp, ISBN-13: 978-0132145183. We will provide Chapter 1 "Brief introduction to MRI physics" of the book: "Magnetic Resonance Image Reconstruction Theory, Methods, and Applications", 1st Edition - November 4, 2022, Editors: Mehmet Akcakaya, Mariya Ivanova Doneva, Claudia Prieto, Paperback ISBN: 9780128227268 , eBook ISBN: 9780128227466	
Assessment	Additional handouts wherever necessary. The course on Medical Imaging comes with an assignment for which you will need to hand in a paper and make a presentation. This is an obligatory part, for which you will get a grade from 1-10 (A). Additionally, there will be an exam also graded from 1-10 (B). The grades for both parts needs to be at least 6.0 in order to pass. The final grade is calculated by: $(A + B)/2$. It is not possible to take part in the exam if you have not made the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Pocket calculator.	
Elective	Yes	

BM41141	Fundamentals of Biomedical Engineering	5
Responsible Instructor	N. Tümer	
Instructor	Ir. A. Sakes	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, there will be: 1) lectures, through which you will become familiar with the field of biomedical engineering (BME) and the stakeholders playing essential roles within, 2) workshops on engineering role play, team dynamics, peer-review, and debate. Through these workshops, you will practice some of the mentioned skills, e.g., being able to work in teams, and having good communication skills, 3) two assignments: (1) self-reflective journey (mostly individual), and (2) challenge-based assignment (group work). These two assignments are connected with each other. In the first assignment, you will explore who you are and what your role would be in relation to your future career in the BME field. In the second assignment, you will have a challenging context. Together with your team, you will define a solution to a challenging problem considering your engineering roles.	
Study Goals	Knowledge, insight, and skills By the end of this course, you should be able to: 1. Define the problems that need to be addressed in the health field, considering crucial societal aspects, and the roles biomedical engineers and stakeholders can play in solving these issues. 2. Apply physical and engineering terms and principles in building a strategy within the multidisciplinary context of biomedical engineering. 3. Outline the steps needed to be taken in order to validate or implement biomedical techniques, devices, and methods in the health system. Transferable skills By the end of this course, you should be able to: 4. Analyse the impact and consequences of your work when intervening in the field of biomedical engineering. 5. Develop evaluative judgment on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback. 6. Reflect on your own interests, role, and personal development in relation to your future professional self in the field of biomedical engineering.	
Education Method	Lectures, workshops and assignments	
Assessment	Exam (40%) and assignments (60%)	
Department	3mE Department Biomechanical Engineering	

EE4650	Advanced Magnetic Resonance Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	A. Webb	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Vector algebra and vector calculus, linear circuit theory, signals and systems (Fourier transform, steady-state analysis), linear algebra and differential equations, electromagnetics (BSc level or higher)	
Course Contents	General introduction, history of MRI, general description of an MR scanner, general principles of MRI, applications and current research themes. General and brief description of spin, restriction to particles with spin $\frac{1}{2}$ (proton), equilibrium magnetization for spin $\frac{1}{2}$ particles (proton), the Bloch equation, general description of the gyromagnetic ratio, T1 and T2 relaxation times. Electromagnetic fields: Maxwells equations, the electromagnetic boundary conditions, the temporal and spatial Fourier transform, time-harmonic fields, the polarization state of an electromagnetic field (linear, circular, elliptical), Poyntings theorem (power balance), and reciprocity. Flipping the magnetization: received signal in terms of the electromagnetic receive field. Solutions of Blochs equation for various magnetic fields, the most important formula in MRI: formula for the Larmor frequency, chemical shift, scalar coupling, basics of 1-DNMR spectroscopy, elementary measurement sequences, spin-echo, inversion recovery, measuring T1 and T2, introducing T2*. Using gradients to spatially encode signals, k-space formalism, image reconstruction via multi-dimensional Fourier transforms, simple gradient echo and spin echo pulse sequences, three-dimensional imaging, angiography. Background knowledge for hardware design: Maxwells equations and quasi-static fields, the law of Biot-Savart, spherical harmonics. Interaction of RF with tissue as a function of frequency, different media/tissue types and the constitutive relations. RF coil and array design: design of RF coils and RF arrays, the Specific Absorption Rate (SAR). The electromagnetic field due to an electric dipole (small current-carrying piece of wire) and a magnetic dipole (small current-carrying loop). Gradient coil design and the static background field: Realizing gradient fields, Helmholtz and Maxwell coils, advanced gradient coil design using stream functions, consequences of nonuniform static background fields. Numerical modeling, software packages, body models, coil models. Measurements and imaging in practice, SNR and other practical aspects. Current/future research topics, parallel imaging, sparse sampling, electrical properties tomography	
Study Goals	After completing this course, the students will: 1. Understand the principles of magnetic resonance from both a quantum and classical mechanics point-of-view, 2. Understand the interaction of strong electromagnetic fields at high, medium and low frequencies, how these can be manipulated, and the safety features associated with these fields, 3. Understand how spatial information is encoded into magnetic resonance images, the origins of image artifacts, and the signal processing algorithms involved in image reconstruction, 4. Understand the relaxation phenomena associated with magnetic resonance, and how these are important in determining image contrast, 5. Understand how the different components of magnetic resonance systems are designed using Maxwells equations, spherical harmonics, and the Biot-Savart law, 6. Understand the current state-of-the-art of MRI hardware and image processing, and the new opportunities for future developments.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes. Will be made available via Brightspace.	
Assessment	Assignment and oral exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor analysis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: $0.5 * \text{Oral discussion 1} + 0.5 * \text{Oral discussion 2}$. Both partial grades must be sufficient, i.e., a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measurement; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: - Discuss a suitable visualization for a given medical problem. - Implement the most suitable solution. - Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: $0.4 * \text{Written exam} + 0.6 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eismann	

SC42025	Filtering & Identification	6
Responsible Instructor	Dr. M. Kok	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The objective of this course is to apply filtering and identification methods. The focus is both on deriving these methods and on using these methods on real-life data sequences. The course focuses on stochastic and recursive least squares, on (extended) Kalman filtering, on prediction error methods and on subspace identification. The course also includes a review of specific topics from linear algebra, dynamical system theory and statistics, that are relevant for filtering and system identification.	
Study Goals	At the end of the course you should be able to: 1) Apply filtering and identification methods to solve previously unseen estimation problems. As sub-learning goals you should be able to: a. Relate basic linear algebra, signal processing techniques and probability theory to physical properties of a (dynamical) system. b. Derive the solution of the stochastic and recursive least squares problems from both a Bayesian perspective and in terms of unbiased / minimum variance properties. c. Derive the Kalman filter from both a Bayesian perspective and in terms of unbiased / minimum variance properties. d. Derive subspace and prediction error identification methods. e. Motivate the steps that should be taken in the derivations in b-d. 2) Apply filtering and identification methods to estimate the state and identify the model using real-life data sequences. As sub-learning goals you should be able to: a. Implement a (stochastic / recursive) least squares problem, an (extended) Kalman filter and identification methods in Matlab. b. Apply and motivate practical aspects of filtering and identification.	
Education Method	Lectures 0/6/0/0	
Literature and Study Materials	Book Filtering and System Identification: A Least Squares Approach by Michel Verhaegen and Vincent Verdult. ISBN: 13-9780521875127	
Assessment	Written exam (open book) and practical exercises. If needed due to covid-19 restrictions, the exam will be online.	
Remarks	The software package Matlab is needed to solve the practical exercise.	
Department	3mE Department Delft Center for Systems and Control	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (5ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Specialisation European Wind Energy Master - Electrical Power Systems within the track Electrical Power Engineering 2023

Introduction 1

All EWEM tracks consist of a two-year joint programme of 120 ECTS and are characterised by:

- * General introduction to Wind Energy: held at DTU, it involves the course of Wind Turbine Technology and Aerodynamics (10 ECTS), which is meant to give the student an overview of the current wind energy technology as well as of the historical, economical and political aspects related to wind energy. Further core and elective courses depend on the chosen track.
- * Internship (6-9 ECTS): compulsory for students of the Wind Physics track only, elective for students of all other tracks. The internship can be carried out at a EWEM associated partner or at a different institution, provided that the content of the internship is accepted by the university awarding the ECTS. The internship gives the student the opportunity to evaluate professional standards with respect to sustainable development, project and risk management, value management, health and safety management, knowledge management, business operation and organisational culture, policies and processes.
- * M.Sc. thesis (30-45 ECTS): the thesis work can be carried out at any EWEM associated partner and must be jointly supervised by supervisors from the two partners awarding the diploma for the specific track. The topic may involve a third party and is agreed upon by the student together with the two degree awarding partners, verifying feasibility and quality of the research questions and methodology. The graduation project can be mono or multidisciplinary and is worth 30 ECTS for Wind Physics students. For all other students the graduation project is worth 45 ECTS, whereby 15 ECTS are earned by the student during the 3rd semester through literature review, definition of the research question and methodology and by selection of the tools, whereas 30 ECTS are accomplished during the 4th semester through thesis development and reporting.

The Electric Power Systems track provides the students with technical knowledge about the components and systems necessary for the integration of renewable energy within the power system.

Profile of the Electric Power Systems graduate:

- * Fundamental knowledge of the physical components of power systems and electronics as well as of new designs of HVDC systems and control theory
- * Technical knowledge of the components and systems required for renewable energy integration within the power system
- Competences in modelling and analysis of composite systems with respect to both their steady state and dynamics
- * Mastery of electrical power systems design and its components with special attention on reliability, protection as well as operation and maintenance in offshore applications.

This track gives the student a combination of standard power system analysis skills and of more specific ones related to the electric power systems used in offshore applications. Within the field of electric power systems the student can further chose to follow different profiles, namely electric power systems for offshore wind farms, off-shore grids or technical and economic integration aspects of intermittent generation.

ET4108	Transients in Power Systems	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	EE2E21, ET4107 - knowledge of symmetrical component is highly recommended	
Course Contents	This course deals with the rapid phenomena in power systems caused by switching actions and atmospheric discharges. Some of the topics covered are circuit breakers, arc-circuit interaction, short-line fault, interruption of small inductive capacitive currents, travelling waves, atmospheric discharges, simulation of transients, circuit breaker testing.	
Course Contents Continuation	The subject deals with power system transients. It is divided in a few sections the goals of which are the students to be introduced by the basic concepts of switching transients and transient analysis as well as travelling wave phenomena. The subject also deals with the current interruption phenomena and circuit breakers as well as circuit breaker testing and insulation coordination. Additional material on Ferroresonance and surge arrester selections is being studied as a supplement to this project. All results are validated by computer simulations in Electromagnetic Transient Program (EMTP), which is used as a tool for detailed analysis of different cases like switching small inductive and capacitive currents, arc interruption, thermal and dielectric breakdown of the circuit breakers, travelling wave phenomena. Each chapter is also supported by several simplified examples (which are solvable on the brightspace)	
Study Goals	After following this course the student should be able to: - model power system components for transient analysis by means of lumped elements. - to understand the current interruption process and gained insight in the testing of switchgear in the High Power Lab. - to use symmetrical components during transient processes in a network, especially when one pole of the circuit breaker opens and the other two remain in closed position. - to understand under which circumstances the worst case scenario (highest, steepest TRV) takes place. - to be able to assess the behavior of the lightning stroke when arriving and junction points connecting one or more components with different characteristic impedance Based on aforementioned learning objectives, the student should be able to simulate at least simplified cases of switching and lighting transient phenomena in Electromagnetic Transient Program ATP-EMTP or other EMTP-based software environment.	
Education Method	Lectures (*). Students should spent approximately ten hours per week in order to achieve the learning goals well. Each chapter, if suitable, is also supported by solvable examples on the board and simplified/detailed simulation cases performed in Electromagnetic Transient Program (EMTP) environment. Students also get opportunity to obtain the EMTP royalty free software and to learn EMTP so that they can use it as a tool for different project cases or M.Sc thesis. The subject also provides opportunity for extra projects in which students may obtain credits depending on the amount of work done. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	L. vander Sluis, Transients in Power System, Wiley, ISBN 0 471 486 396. JA Martinez-Velasco, M Popov, Circuit breakers: Power System Transients Parameter determination, CRC Press, ISBN-13: 978-1420065299, pp. 447-556.	
Assessment	Additional material on ferroresonance phenomena will be provided by the lecturer Oral (closed book examination) which can be taken as written in case of more than 10 students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular transient case that occurs in practice. The amount of work required for particular problem solutions is in line with the number of points. The exam is passed when the student scores more than 57 %. Resit of the examination is possible according to the examination schedule. The content of the course material is divided into 9 chapters. All chapters weight equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

EWEM Electives/Specialisation Courses 2023

AE4W09	Wind Turbine Design	5
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control The use of standards for load calculations Other optional material (tbd)	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): - AE4W13 Site conditions for wind turbine design - AE4135 Rotor / wake aerodynamics - AE4W21-14 Wind turbine aeroelasticity - AE4W31 Floating offshore wind energy - ET4117 Electrical machines and drives - OE44135 Offshore wind support structures	

EE4111	High-Voltage DC	4
Responsible Instructor	Prof.ir. P.T.M. Vaessen	
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	ET4103 High Voltage Technology (High Voltage Constructions)	
Course Contents	This course presents the physical challenges faced by HVDC power components. Electrical insulation behaves quite differently under DC voltage compared to AC. This will influence the entire cycle from design to testing. Knowledge and insight of electric field behavior under DC stress is given theoretically and illustrated by practical applications from power industry (hvdc cable, converter transformers, bushings). The course contents are: Introduction to HVDC Electric fields under different DC conditions Surface charge, origin, effect and measurement Space charge, origin, effect and measurement Breakdown mechanisms Partial discharges and DC tests	
Study Goals	Note:the information provided in this page may change depending on the developments around the coronavirus By the end of the course students will be able to: - Know the difference between the electrical behavior of AC and DC insulation - Calculate the charge distribution and field strength in different situations - Elaborate on the behavior of materials under DC stress - Formulate design requirement for DC insulation - Describe the differences in partial discharges between AC and DC - Describe methods for surface and space charge measurements - Comment on DC testing	
Education Method	Lectures Demonstration(s)	
Literature and Study Materials	book: F.H. Kreuger. Industrial DC High Voltage. Delft University Press, 1995. hand-outs at lectures	
Assessment	Written examination (80% final written exam, 20% course project) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4114	Power System Protection and Control	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Instructor	Dr. A. Lekic	
Contact Hours / Week x/x/x/x	0,0,0,4 (32 hr. in total)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course deals with the important aspects of power system protection with respect to fault detection and isolation and it is organized as follows: - System grounding - types, characteristics, and the effects on fault current behavior - Instrument transformers - Fundamentals of power system protection - Overcurrent protection - Transformer protection - Distance protection - Fault locators based on point-of-wave and synchrophasor measurements - Testing of power system protection - Control strategies for the type-3 and type-4 wind turbines - Control of HVDC-based connections	
Study Goals	After the course, the student can learn: - typical grounding practices and how the grounding system influences applied protection principles - important aspects of current transformers and procedures and guidelines on how to choose current transformers for particular protection function - the effect of the saturation and DC offset on the choice of the instrument transformer - how the microprocessor protection operates - the difference between overcurrent, differential, and distance protection and the advantages/disadvantages of each - protection of electric grids with inverter-based generation (fault currents, relay performances) - control strategies for type-4 and type-3 wind turbines - HVDC MMC (full/half bridge) control for monopolar and bipolar connection - to use RTDS for different protection and MTDC operation scenarios The students should also be able to: - deal at least with basic settings of overcurrent protection - deal with the choice of primary and backup protection for simplified networks consisting of a generator, a transformer, and a line. - understand how protective relays are being tested by RTDS and Omicron	
Education Method	Lectures, contact hours (*). The students are expected to spend about 8h a week on this exam. The students will work on the experimental verification of the studied protection and control approaches using RTDS.	
Literature and Study Materials	G. Ziegler: Numerical Distance Protection: Principles and Applications G. Ziegler: Numerical Differential Protection: Principles and Applications Network Protection and Automation Guide, Protective Relays, Measurements and Control, General Electric S. Du, A. Dekka, B. Wu, N. Zargari, Modular multilevel converters: analysis, control, and applications, John Wiley & Sons, 2017 R. Wachal, A. Jindal, S. Denetiere, H. Saad, HVDC Converters in a HVDC Grid, December, 2014	
Assessment	Oral (closed book examination) can be taken as written in case more than ten students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular protection function that occurs in practice or a solvable problem that is in line with what has been studied during the course. The work required for particular problem solutions aligns with the number of points. The exam is passed when the student scores more than 57 %. Resit is possible according to the examination schedule. The content of the course material is divided into 8 chapters. All chapters weigh equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Remarks Consulting hours according to the appointment	

EE4375	Finite Element Modeling for Electrical Energy Applications	4
Responsible Instructor	Dr. D.J.P. Lahaye	
Contact Hours / Week x/x/x/x	0/0/2/0 and 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	We expect the following prior knowledge 1) linear algebra: concepts of vector, matrix, matrix-vector multiplication, solving linear systems, computing eigenvalues and eigenvectors of a matrix; 2) calculus of functions of single variable: concepts of derivative of a function, finite difference approximation, integral of a function, quadrature, ordinary differential equations; 3) calculus of functions of more variables: concepts of partial derivative of a function, surface and line integral, quadrature, partial differential equations; 4) electric and magnetic fields: concepts of electric field, magnetic field, Maxwell equations, electrical machines and transformers;	
Course Contents	This course consists of the following three blocks: 1) Block 1 (Week 1 and Week 2): In this block, the finite difference method for the Poisson equation on the interval (Week 1) and the square (Week 2) will be discussed. The differential equation and the boundary conditions will be discretized on a mesh by a finite difference scheme. This will result in linear equations for the grid unknowns. A stencil notation for these linear equations will be introduced. A linear system will be formed and solved. The discrete solution will be visualized on the mesh. Properties of the discretization method and the linear system will be discussed. 2) Block 2 (Week 3 and Week 4): In this block, the Galerkin finite element method for the one-dimensional (Week 3) and two-dimensional (Week 4) Poisson problem will be introduced. The problem in so-called strong form will be transformed into the problem in so-called weak or variational form. The problem in weak form will then be discretized in space using Lagrangian basis or shape functions. A linear system represents the Poisson equation on a discrete level will be assembled and solved. The finite difference and finite element solution methods will be compared. 3) Block 3 (Week 5, Week 6 and Week 7): In this block, the application of the finite element will be extended to treat more complex problems. Partial differential equations for the scalar and vector potential for the magnetic flux or field will be derived from the Maxwell equations. More complex geometries that represent transformers, actuators or rotating extrical machines will be meshed using unstructured triangular meshes. The Maxwell equations will be discretized on these meshes. The discrete linear system will be assembled by a element-by-element procedure and solved. The computed vector potential solution will be processed to compute the magnetic flux and the magnetic field.	
Study Goals	After successfully completing the course, the student will be able to: * formulate various boundary value problems for the one-dimensional and two-dimensional Poisson equation; * derive these boundary value problems from the Maxwell equations for the magnetic field by introducing the scalar and vector potential; * apply the finite difference and finite element method to solve these boundary value problems on a scale of problems of increasing complexity; * state properties of the finite difference and finite element method applied to Poisson boundary value problems; * describe the implementation of the finite difference and the finite element method in software; * give a physical interpretation of the numerical solution obtained for magnetic field problems;	
Education Method	Weekly lectures will be complemented by weekly practical lab sessions and three home work assignments. During the lab sessions, algorithms discussed in the lectures will be implemented, tested and further refined. Each homework assignment will correspond to a course block. The set of three assignment will form a portfolio.	
Reader	see http://ta.twi.tudelft.nl/nw/users/domenico/electro_fem/electro_fem.html	
Assessment	During the course, the student will build a portfolio of three home work assignments, each correspond to a block in the course. The objective of these assignments is to guide the students towards the implementation of a proper finite element code. Each assignment consist of an elective and a compulsory part. This portfolio will be assessed during the oral examination. During this examination, the student will be assessed on the theoretical foundation of finite difference and finite element method, on the application of these methods and their implementation in software. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4515	Advanced Power Electronics	4
Responsible Instructor	Dr. Z. Qin	
Instructor	Dr. T. Batista Soeiro	
Contact Hours / Week x/x/x/x	0/0/0/3 + lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is expected that the student has taken all the previous courses in the power electronics teaching line: ET4119, ET4116, ET4145	
Course Contents	The focus of this course is on providing the theoretical background of various design related aspects of power converters not covered in the previous courses. The topics covered include: switching behavior and loss modelling, soft switching vs hard switching, passive components, thermal management and control. The lab projects integrate the knowledge from this course and the preceding courses in a design assignment of designing, building and testing an efficient and small power converter.	
Study Goals	After this course, the students will be able to: - design a transformer/inductor - design the control - design the cooling - analyse and improve the efficiency - design, build, and test a converter with small size and high efficiency	
Education Method	Lectures and lab project	
Literature and Study Materials	Lecture handouts N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition.	
Assessment	projects (80%) and oral exam (20%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4536	DC and AC Microgrids	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	L.M. Ramirez Elizondo	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatziaargyriou	
Assessment	Written exam 70%, Assignments 30% In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTS) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

EE4665	Uncertainty modelling and risk assessment in electrical power systems	4
Responsible Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ir. J.L. Rueda Torres	
Co-responsible for assignments	E. Sharifnia	
Co-responsible for assignments	K. Bölat	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Elementary knowledge of probability theory and statistics (discrete and continuous probability distributions, expectation/variance/standard deviation, quantiles, least squares regression). Basic programming skills (examples code samples will be provided in Python, but other languages may be used for hand-in exercises). Working knowledge of power system structure, components and actors.	
Course Contents	This course is aimed at MSc EE students in the EPE track and MSc SET students with the Power profile. This course considers the challenges faced by operators when operating and planning electrical power systems in the face of uncertainty balancing cost, reliability and environmental objectives. The course specifically focuses on quantitative modelling of uncertainty and performing quantitative risk assessments to support decision making. The course starts with a description of uncertainties that are important to power system operation and planning, and common metrics for system performance. Case studies in power system reliability and system capacity planning are introduced. In this context, various approaches to modelling uncertainties are introduced. Historical data is used to create models for component reliability. Students will learn how to compute system level reliability measures from bottom-up reliability models, using enumeration approaches and Monte Carlo sampling. Special attention is given to the estimation sample variation using the normal assumption and bootstrap approaches. Finally, the impact of risk metrics and decision criteria on decision making is discussed, thereby giving students insight into the whole chain from measurements to models to model-based decision making.	
Study Goals	After completing this course, students will be able to: 1) Calculate commonly used reliability measures and interpret their values in the context of Western European power systems. 2) Construct and use Markov models for component reliability. 3) Follow prescribed steps to construct reliability models from failure and repair data. 4) Calculate the reliability of model systems on the basis of constituent component models. 5) Implement Monte Carlo sampling and bootstrapping methods to estimate risk measures and the accuracy of such estimates. 6) Compare model representations of uncertainty, and illustrate these in the context of power system reliability and planning. 7) Evaluate different grid development options using quantitative risk assessment and discuss how a decision for one option depends on the choice of risk measure and decision criterion.	
Education Method	Lectures (8x 2 hours); Computer labs (8x 2 hours); additional exercises for self study.	
Literature and Study Materials	1) Book, available as e-book through TU Delft (no additional cost) or physical copy: Bart W. Tuinema, José L. Rueda Torres, Alexandru I. Stefanov, Francisco M. Gonzalez-Longatt, and Mart A. M. van der Meijden, "Probabilistic Reliability Analysis of Power Systems," Springer Nature Switzerland AG 2020, May 2020. DOI: https://doi.org/10.1007/978-3-030-43498-4 . 2) Supplementary notes, videos, selected papers, reports and book chapters, available through Brightspace.	
Assessment	Completion with 'pass' mark of a small group assignment and computer lab exercise (with basic report) is required. The final grade is determined by a written exam (80%) and computer exercise with accompanying report (20%). A minimum mark of 5.0 must be attained for each component to pass. Component marks are given on a scale 0-10, rounded to the nearest 0.1 increment; the final grade is rounded to the nearest 0.5 increment. One resit opportunity is provided for the written exam. One repair opportunity is offered for the pass/fail assignments (two weeks after the initial deadline, unless agreed otherwise). A repair opportunity will also be offered for the graded exercise if the initial mark was between 4.0 and 5.7 (out of 10), with a deadline 3 weeks after the initial grade is announced. A maximum mark of 6.0 out of 10 can be obtained for the exercise through this route. If a student passes only part of the course in this academic year, all parts need to be retaken. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Energy Modelling Programming Stochastics	
Maximum number of participants	30.	

EE4755	Reliable Power Electronic Components and Systems	5
Responsible Instructor	A. Shekhar	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Following this course is only possible with course ET4119 or SET3095 in advance	
Course Contents	Understanding of Power electronic components: diodes, BJT's, MOSFET's, thyristors, IGBT's; Gate drive circuits; High frequency inductors; Semiconductor's thermal management and heatsinks. The course also deals with conducted electromagnetic interference in power electronic circuits. Finally, the course will discuss component reliability and its corresponding influence on the power electronic systems	
Study Goals	THEORY: Understanding the theoretical and practical operation of power electronics components (semiconductors and passives) based on the physical structure, the V-I characteristics, time and thermal behavior. PRACTICAL: Analysis of Boost-type Converter and the Design of components, including semiconductors, inductor and capacitor. The ability to perform (differential) voltage-, current and temperature measurements on these components via a remote controlled test setup and explain these measurements at the hand of theory. Students will be given a demo on reliability tests using power cycler in the PEARL lab and asked to discuss their observations on the obtained test result data.	
Education Method	Lectures, practicals	
Literature and Study Materials	TEXT BOOK: Mohan, Undeland and Robbins, Power Electronics, Converters, Applications and Design, ISBN:978-0-471-22693-2, 824 pages, 3rd edition, John Wiley & Sons, 2003. ADDITIONAL READING: Provided via Brightspace DATA SHEETS: Provided via Brightspace and hard-copy in the lab. HARDWARE: Provided in the lab sessions.	
Assessment	Written EXAM (40%) DESIGN REPORT of your inductors DUO PRESENTATION on design- and practical results (20%) FINAL REPORT on practical (40%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4113	Power System Dynamics	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Required: ET4107 Power System Analysis II (Master SET / Master EE / Master EWEM) SET3065 Intelligent Electrical Power Grids (Master SET / Master EE / Master EWEM) Optional: ET4116 Power Electronics (Master SET / Master EE / Master EWEM) ET4117 Electrical Machines and Drives (Master SET / Master EE / Master EWEM)	
Parts	Theory lectures: Lecture 1: Introduction to power system stability and synchronous generator Lecture 2: Controllers of the synchronous generator Lecture 3: The voltage source converter Lecture 4: Controllers of the voltage source converter Lecture 5: Applications of voltage source converter in electrical power systems Lecture 6: Rotor angle stability: transient stability and small-signal stability Lecture 7: Small-disturbance (small-signal) rotor angle stability (measurement based methods) Lab sessions: Session 1: Tutorial for PowerFactory (Single-Machine Infinite Bus system) Session 2: Modelling of an interconnected system (steady-state performance) Session 3: Dynamic models - Part 1 Session 4: Dynamic models - Part 2 Session 5: Long-term voltage stability assessment (including tutorial for Python) Session 6: Rotor angle stability assessment Session 7: Sensitivity analysis	
Course Contents	The course "Power Systems Dynamics" is devoted to the study behaviour of electrical power systems in non-steady state. Knowledge of the dynamic behaviour of electrical power systems and their components is important in order to predict whether the system (or a part thereof) remains in a stable state after a disturbance. The time frame of observation ranges from milliseconds to several minutes, depending on the kind of disturbance and the influence of applied controls. For each time frame, special models of the network components are derived. In this course, the model of the synchronous generator, the voltage source converter, and their associated control systems receive special attention due to their dominant influence on the dynamic behaviour of the power system. In the computer assignments, the theory is put into practice by means of software based simulations in DigSILENT PowerFactory as well as scripting in Python and Matlab. Subject to investigation are: modelling of synchronous generator and voltage source converter for RMS simulations, frequency and voltage controls, model-based and measurement-based eigenvalue analysis, wide-area monitoring, damping control, and RMS simulation of interconnected power systems.	
Study Goals	At the end of the course the students will be able to: - Describe the modeling aspects of power system components for computer aided simulation of electro-mechanical transients - Apply and interpret stability criteria for the analysis of systems dynamic behavior - Create a model of a power system in a simulation software package - Assess the system performance following typical disturbances	
Education Method	Lectures and computer assignments. Contact hours (8 weeks: 7 theory lectures, 7 lab sessions): 2 hours/theory lecture and 4 hours/lab session. Total: 42 hours. Self-study: 70 hours (9 hours/week).	
Computer Use	DigSILENT PowerFactory, Python, Matlab	
Literature and Study Materials	P. Kundur, Power System Stability and Control, Mc.Graw Hill, inc. 1994, ISBN-0-07-035958.	
Assessment	Lab report and final exam/re-exam including a theoretical questions component and a computer-based component. The computer-based part of the test requires familiarity with the models built during the lab assignments. The students are expected to run time-domain simulations and give insight into the observed phenomena. The final grade comprises: 1) Exam or re-exam (5 points). 2) Lab report and implementations (5 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Digital exam in the exam period of Q4. Duration: 3 hours. The resit is in the exam period of summer period. Duration: 3 hours.	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Study load: 112 hours.	
maximum aantal deelnemers	Limited to 25 students due to software availability	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalair - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteresis modulation - carrier modulation with feedback - scalar and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET4121	A.C. Machines	4
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	Control of Electrical Drives (ET4291)	
Expected prior knowledge	Electrical machines and drives (ET4117)	
Course Contents	The following subjects are discussed: - winding function and inductances; - time and spatial harmonics; - skin effect in electrical machines; - phasor, space vector and transformation in electrical machines; - the salient pole synchronous machine; - modeling and simulation of the dynamic performance of DC and AC machines.	
Study Goals	After following this course the students should be able to - describe and calculate the steady-state performance of AC machines; - calculate spatial harmonics and inductances of given winding distributions; - derive the dynamic models of DC and AC machines using space vectors; - model and simulate the dynamic performance of DC and AC machines; - evaluate the limitations of the dynamic models.	
Education Method	Lecture / Computer Simulation / Discussion	
Literature and Study Materials	- A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill. - lecture notes on Brightspace	
Assessment	Oral exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	- Presence at the computer simulation session is obliged. - Submission of assignments on Brightspace is obliged.	

ET4291	Control of Electrical Drives	5
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	practicum 0/0/0/4 vanaf week 2, instructie 0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Course Contents	- Dynamic modelling of DC and AC Machines supplied with power electronic converters. - Modelling of mechanical systems and thermal effects in drives. - Modelling and simulation of digital control of electrical drives. - Controller design for cascaded motor controller. - Scalar and vector control, field weakening control. - Practical design of an electrical drive system.	
Study Goals	Students should be able to: - Model the dynamic behaviour of DC and AC machines using state space representation. - Integrate ideal power electronic converter models and cascade control systems in the drive model. - Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive. - Calculate the steady state performance of DC and AC drives. - Implement scalar control system and vector control system for induction machines. - Implement vector control system for synchronous machines. - Implement field weakening control for DC and AC drives. - Implement the dynamic simulation models in MATLAB/Simulink. - Explain the practical design of electrical drives. - Evaluate the performance of an electrical drive system. - Design the controllers in electrical drive.	
Education Method	Lecture / Self-study / Simulation practical / Discussion	
Literature and Study Materials	- Rik De Doncker, Duco W.J. Pille, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: https://www.springer.com/gp/book/9789400701816 - Self-study materials on Brightspace.	
Reader	"Reader Control of Electrical Drives ET4291 Version maart 2009"	
Prerequisites	Electrical Machines and drives (ET4117) Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Assessment	Individual assignment and oral exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Note: In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	

ET4379	Photovoltaic Lab Course	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Responsible Instructor	Dr.ir. R. Vismara	
Contact Hours / Week x/x/x/x	4/0/0/0 Q1=Q5! Also possible online in Q6, Q7, Q8!	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The aim is to offer you an advanced practical course on solar cells and photovoltaic systems. The following areas of practical training will be part of the course: (1) Characteristics of artificial and natural light (2) Characterization of the performance of lab-scale solar cells (3) Behavior of photovoltaic modules in different temperature and shading conditions (4) Charge and discharge properties of batteries (5) Performance of power electronic components (charge controllers, inverters, maximum power point trackers) (6) Operation of entire photovoltaic systems	
Study Goals	(1) Experiment with photovoltaic systems and all of their components under lab and real-life conditions. (2) Describe the working principle behind each component of a photovoltaic system. (3) Analyze the operation of all the elements of photovoltaic systems and of the system as a whole. (4) Compare the performance of different types of components of photovoltaic systems.	
Education Method	Laboratory sessions	
Literature and Study Materials	Manuals and templates are available on BrightSpace.	
Prerequisites	This course is meant for students who have already passed at least the PV Basics course (ET4376) and preferably also PV Technologies (ET4377) and PV Systems (ET4378). This course is also meant for 1st-year masters students with a bachelor degree in electrical engineering.	
Assessment	The assessment is based on reporting results from the individual tasks, using an Excel template provided on Brightspace. Two-person teams will be formed at the plenary session, the attendance of which is compulsory. The final grade is per team. The final grade is the average of the six technical reports. A sufficient grade (greater than or equal to 5.8) in all six technical reports is required to complete the course successfully. In case of failure of a technical report, you can use buffer sessions on week 1.7 and 1.8 to repeat the experiments. --- A virtual version of this course is made available for students to attend in Q2, Q3, and Q4 of the academic year 2023-2024. Each of the six tasks of the on-campus course will run in the internet browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, using numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to pass the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. Attending one of these instances of the course, no teams will be formed and the final grade is per person. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	The on-campus course can accommodate up to 48 students. Anyone interested in attending this course must apply during Q4 of the immediately previous academic year. A selection will then take place and, before the beginning of the instance in Q5, the line-up of students will be announced. Student(s) who failed on-campus in Q5 can re-take the virtual version of the course. There is no limit to the number of students who can attend the course's virtual version.	
Remarks	This practical course will be given on campus only in Q1. Conversely, a virtual version will be available for students to attend in Q2, Q3, and Q4. The enrolment is open on Brightspace from Q1 of the academic year 2023-2024.	

ET8020	High Voltage Testing and Diagnostics	4
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Instructor	Ir. A.G.A. Lathouwers	
Instructor	C.S. Engelbrecht	
Practical Coordinator	I.M. Splinter	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The quality of the insulation of high voltage components (assets) e.g. transformers, cables, switch-gear, generators, is of large importance for the availability and reliability of power supply. Prior to service operation and in accordance with international standards all these components are heavily tested at different voltage stresses (HVDC, AC, Impulse). During the service life of 30 or more years local insulation defects (on mm-scale) occurring as a result of aging and degradation may produce a total blackout e.g. of a 3 km long power cable connection. To provide during continuous service high reliability level, insulation condition assessment by means of advanced and non-destructive diagnostics is very important. In particular, by means of advanced sensor techniques, signal processing, knowledge rules, the condition of the components has to be continuously or periodically monitored. The liberalization of energy sector, requests from the network providers an economic responsible technical management of the assets. In particular, based on the actual condition, the maintenance strategies have to be implemented. This course, supported by demonstrations, learns about the different voltage generation, diagnostics and monitoring techniques, weakest link statistics, knowledge rules generation, diagnosis support systems for asset management are shown.	
Study Goals	To obtain insight in the testing, diagnostics of HV components and be able to understand the processes of insulation-coordination and quality insurance by maintenance strategies. Within power utility, to be able to recognize and to understand the AM processes and to be able to implement methods to control these processes.	
Education Method	Lectures, demonstrations, practical	
Literature and Study Materials	Kreuger, F.H., Industrial High Voltage Vol. 4, 5, 6; ISBN 90-6275-562-3 (available for purchase at the course); Hand-outs (available at the course) Weibull analysis: handouts replace Ch.6 R. Ross, Reliability Analysis for Asset Management of Electric Power Grids, Wiley-IEEE, 2019, ISBN 978-1-119-12517-4, DOI:10.1002/9781119125204.	
Assessment	Oral, closed book, hand outs about Weibull Analysis- substituting Ch.6, and hand-outs about Asset Management Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to LO1: Describe the functions and topologies of cyber-physical energy systems LO2: Implement data driven modelling and numerical simulation of energy systems LO3: Calculate variable power flows based on optimization/probability theory LO4: Synthesize functional requirements of energy management systems LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1: The energy transition and future energy systems (Peter) 2: Analysis of steady-state performance in variable operating conditions (Jose) 3: Assessment and control of power system dynamic performance (Jose) 4: Lab 1 Digital and data driven modelling of interconnected systems 5: Lab 2 Simulation and impact analysis of renewables, storage, and demand 6: Data acquisition and communication for energy automation (Peter) 7: Artificial intelligence-based functionalities for situational awareness (Pedro & Jochen) 8: Cyber security of smart grids (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The final grade of the course consists of the following components: - Assignment reports (in time!) (50%) - Multiple choice Computer Exam (50%) In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Final grade calculation: $(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Assignment reports (in time!): resubmit deliverable - Multiple choice Computer Exam: computer resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Suggested profiles EWEM 2023

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

High voltage 2023

EE4111	High-Voltage DC	4
Responsible Instructor	Prof.ir. P.T.M. Vaessen	
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	ET4103 High Voltage Technology (High Voltage Constructions)	
Course Contents	This course presents the physical challenges faced by HVDC power components. Electrical insulation behaves quite differently under DC voltage compared to AC. This will influence the entire cycle from design to testing. Knowledge and insight of electric field behavior under DC stress is given theoretically and illustrated by practical applications from power industry (hvdc cable, converter transformers, bushings). The course contents are: Introduction to HVDC Electric fields under different DC conditions Surface charge, origin, effect and measurement Space charge, origin, effect and measurement Breakdown mechanisms Partial discharges and DC tests	
Study Goals	Note:the information provided in this page may change depending on the developments around the coronavirus By the end of the course students will be able to: - Know the difference between the electrical behavior of AC and DC insulation - Calculate the charge distribution and field strength in different situations - Elaborate on the behavior of materials under DC stress - Formulate design requirement for DC insulation - Describe the differences in partial discharges between AC and DC - Describe methods for surface and space charge measurements - Comment on DC testing	
Education Method	Lectures Demonstration(s)	
Literature and Study Materials	book: F.H. Kreuger. Industrial DC High Voltage. Delft University Press, 1995. hand-outs at lectures	
Assessment	Written examination (80% final written exam, 20% course project) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4114	Power System Protection and Control	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Instructor	Dr. A. Lekic	
Contact Hours / Week x/x/x/x	0,0,0,4 (32 hr. in total)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course deals with the important aspects of power system protection with respect to fault detection and isolation and it is organized as follows: - System grounding - types, characteristics, and the effects on fault current behavior - Instrument transformers - Fundamentals of power system protection - Overcurrent protection - Transformer protection - Distance protection - Fault locators based on point-of-wave and synchrophasor measurements - Testing of power system protection - Control strategies for the type-3 and type-4 wind turbines - Control of HVDC-based connections	
Study Goals	After the course, the student can learn: - typical grounding practices and how the grounding system influences applied protection principles - important aspects of current transformers and procedures and guidelines on how to choose current transformers for particular protection function - the effect of the saturation and DC offset on the choice of the instrument transformer - how the microprocessor protection operates - the difference between overcurrent, differential, and distance protection and the advantages/disadvantages of each - protection of electric grids with inverter-based generation (fault currents, relay performances) - control strategies for type-4 and type-3 wind turbines - HVDC MMC (full/half bridge) control for monopolar and bipolar connection - to use RTDS for different protection and MTDC operation scenarios The students should also be able to: - deal at least with basic settings of overcurrent protection - deal with the choice of primary and backup protection for simplified networks consisting of a generator, a transformer, and a line. - understand how protective relays are being tested by RTDS and Omicron	
Education Method	Lectures, contact hours (*). The students are expected to spend about 8h a week on this exam. The students will work on the experimental verification of the studied protection and control approaches using RTDS.	
Literature and Study Materials	G. Ziegler: Numerical Distance Protection: Principles and Applications G. Ziegler: Numerical Differential Protection: Principles and Applications Network Protection and Automation Guide, Protective Relays, Measurements and Control, General Electric S. Du, A. Dekka, B. Wu, N. Zargari, Modular multilevel converters: analysis, control, and applications, John Wiley & Sons, 2017 R. Wachal, A. Jindal, S. Denetiere, H. Saad, HVDC Converters in a HVDC Grid, December, 2014	
Assessment	Oral (closed book examination) can be taken as written in case more than ten students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular protection function that occurs in practice or a solvable problem that is in line with what has been studied during the course. The work required for particular problem solutions aligns with the number of points. The exam is passed when the student scores more than 57 %. Resit is possible according to the examination schedule. The content of the course material is divided into 8 chapters. All chapters weigh equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Remarks Consulting hours according to the appointment	

EE4375	Finite Element Modeling for Electrical Energy Applications	4
Responsible Instructor	Dr. D.J.P. Lahaye	
Contact Hours / Week x/x/x/x	0/0/2/0 and 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	We expect the following prior knowledge 1) linear algebra: concepts of vector, matrix, matrix-vector multiplication, solving linear systems, computing eigenvalues and eigenvectors of a matrix; 2) calculus of functions of single variable: concepts of derivative of a function, finite difference approximation, integral of a function, quadrature, ordinary differential equations; 3) calculus of functions of more variables: concepts of partial derivative of a function, surface and line integral, quadrature, partial differential equations; 4) electric and magnetic fields: concepts of electric field, magnetic field, Maxwell equations, electrical machines and transformers;	
Course Contents	This course consists of the following three blocks: 1) Block 1 (Week 1 and Week 2): In this block, the finite difference method for the Poisson equation on the interval (Week 1) and the square (Week 2) will be discussed. The differential equation and the boundary conditions will be discretized on a mesh by a finite difference scheme. This will result in linear equations for the grid unknowns. A stencil notation for these linear equations will be introduced. A linear system will be formed and solved. The discrete solution will be visualized on the mesh. Properties of the discretization method and the linear system will be discussed. 2) Block 2 (Week 3 and Week 4): In this block, the Galerkin finite element method for the one-dimensional (Week 3) and two-dimensional (Week 4) Poisson problem will be introduced. The problem in so-called strong form will be transformed into the problem in so-called weak or variational form. The problem in weak form will then be discretized in space using Lagrangian basis or shape functions. A linear system represents the Poisson equation on a discrete level will be assembled and solved. The finite difference and finite element solution methods will be compared. 3) Block 3 (Week 5, Week 6 and Week 7): In this block, the application of the finite element will be extended to treat more complex problems. Partial differential equations for the scalar and vector potential for the magnetic flux or field will be derived from the Maxwell equations. More complex geometries that represent transformers, actuators or rotating extrical machines will be meshed using unstructured triangular meshes. The Maxwell equations will be discretized on these meshes. The discrete linear system will be assembled by a element-by-element procedure and solved. The computed vector potential solution will be processed to compute the magnetic flux and the magnetic field.	
Study Goals	After successfully completing the course, the student will be able to: * formulate various boundary value problems for the one-dimensional and two-dimensional Poisson equation; * derive these boundary value problems from the Maxwell equations for the magnetic field by introducing the scalar and vector potential; * apply the finite difference and finite element method to solve these boundary value problems on a scale of problems of increasing complexity; * state properties of the finite difference and finite element method applied to Poisson boundary value problems; * describe the implementation of the finite difference and the finite element method in software; * give a physical interpretation of the numerical solution obtained for magnetic field problems;	
Education Method	Weekly lectures will be complemented by weekly practical lab sessions and three home work assignments. During the lab sessions, algorithms discussed in the lectures will be implemented, tested and further refined. Each homework assignment will correspond to a course block. The set of three assignment will form a portfolio.	
Reader	see http://ta.twi.tudelft.nl/nw/users/domenico/electro_fem/electro_fem.html	
Assessment	During the course, the student will build a portfolio of three home work assignments, each correspond to a block in the course. The objective of these assignments is to guide the students towards the implementation of a proper finite element code. Each assignment consist of an elective and a compulsory part. This portfolio will be assessed during the oral examination. During this examination, the student will be assessed on the theoretical foundation of finite difference and finite element method, on the application of these methods and their implementation in software. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTs) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

ET4108	Transients in Power Systems	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	EE2E21, ET4107 - knowledge of symmetrical component is highly recommended	
Course Contents	This course deals with the rapid phenomena in power systems caused by switching actions and atmospheric discharges. Some of the topics covered are circuit breakers, arc-circuit interaction, short-line fault, interruption of small inductive capacitive currents, travelling waves, atmospheric discharges, simulation of transients, circuit breaker testing.	
Course Contents Continuation	The subject deals with power system transients. It is divided in a few sections the goals of which are the students to be introduced by the basic concepts of switching transients and transient analysis as well as travelling wave phenomena. The subject also deals with the current interruption phenomena and circuit breakers as well as circuit breaker testing and insulation coordination. Additional material on Ferroresonance and surge arrester selections is being studied as a supplement to this project. All results are validated by computer simulations in Electromagnetic Transient Program (EMTP), which is used as a tool for detailed analysis of different cases like switching small inductive and capacitive currents, arc interruption, thermal and dielectric breakdown of the circuit breakers, travelling wave phenomena. Each chapter is also supported by several simplified examples (which are solvable on the brightspace)	
Study Goals	After following this course the student should be able to: - model power system components for transient analysis by means of lumped elements. - to understand the current interruption process and gained insight in the testing of switchgear in the High Power Lab. - to use symmetrical components during transient processes in a network, especially when one pole of the circuit breaker opens and the other two remain in closed position. - to understand under which circumstances the worst case scenario (highest, steepest TRV) takes place. - to be able to assess the behavior of the lightning stroke when arriving and junction points connecting one or more components with different characteristic impedance Based on aforementioned learning objectives, the student should be able to simulate at least simplified cases of switching and lighting transient phenomena in Electromagnetic Transient Program ATP-EMTP or other EMTP-based software environment.	
Education Method	Lectures (*). Students should spent approximately ten hours per week in order to achieve the learning goals well. Each chapter, if suitable, is also supported by solvable examples on the board and simplified/detailed simulation cases performed in Electromagnetic Transient Program (EMTP) environment. Students also get opportunity to obtain the EMTP royalty free software and to learn EMTP so that they can use it as a tool for different project cases or M.Sc thesis. The subject also provides opportunity for extra projects in which students may obtain credits depending on the amount of work done. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	L. vander Sluis, Transients in Power System, Wiley, ISBN 0 471 486 396. JA Martinez-Velasco, M Popov, Circuit breakers: Power System Transients Parameter determination, CRC Press, ISBN-13: 978-1420065299, pp. 447-556.	
Assessment	Additional material on ferroresonance phenomena will be provided by the lecturer Oral (closed book examination) which can be taken as written in case of more than 10 students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular transient case that occurs in practice. The amount of work required for particular problem solutions is in line with the number of points. The exam is passed when the student scores more than 57 %. Resit of the examination is possible according to the examination schedule. The content of the course material is divided into 9 chapters. All chapters weight equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalair - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteretic modulation - carrier modulation with feedback - scalair and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET8020	High Voltage Testing and Diagnostics	4
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Instructor	Ir. A.G.A. Lathouwers	
Instructor	C.S. Engelbrecht	
Practical Coordinator	I.M. Splinter	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The quality of the insulation of high voltage components (assets) e.g. transformers, cables, switch-gear, generators, is of large importance for the availability and reliability of power supply. Prior to service operation and in accordance with international standards all these components are heavily tested at different voltage stresses (HVDC, AC, Impulse). During the service life of 30 or more years local insulation defects (on mm-scale) occurring as a result of aging and degradation may produce a total blackout e.g. of a 3 km long power cable connection. To provide during continuous service high reliability level, insulation condition assessment by means of advanced and non-destructive diagnostics is very important. In particular, by means of advanced sensor techniques, signal processing, knowledge rules, the condition of the components has to be continuously or periodically monitored. The liberalization of energy sector, requests from the network providers an economic responsible technical management of the assets. In particular, based on the actual condition, the maintenance strategies have to be implemented. This course, supported by demonstrations, learns about the different voltage generation, diagnostics and monitoring techniques, weakest link statistics, knowledge rules generation, diagnosis support systems for asset management are shown.	
Study Goals	To obtain insight in the testing, diagnostics of HV components and be able to understand the processes of insulation-coordination and quality insurance by maintenance strategies. Within power utility, to be able to recognize and to understand the AM processes and to be able to implement methods to control these processes.	
Education Method	Lectures, demonstrations, practical	
Literature and Study Materials	Kreuger, F.H., Industrial High Voltage Vol. 4, 5, 6; ISBN 90-6275-562-3 (available for purchase at the course); Hand-outs (available at the course) Weibull analysis: handouts replace Ch.6 R. Ross, Reliability Analysis for Asset Management of Electric Power Grids, Wiley-IEEE, 2019, ISBN 978-1-119-12517-4, DOI:10.1002/9781119125204.	
Assessment	Oral, closed book, hand outs about Weibull Analysis- substituting Ch.6, and hand-outs about Asset Management Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Electronics and Electrical Machines 2023

EE4375	Finite Element Modeling for Electrical Energy Applications	4
Responsible Instructor	Dr. D.J.P. Lahaye	
Contact Hours / Week x/x/x/x	0/0/2/0 and 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	We expect the following prior knowledge 1) linear algebra: concepts of vector, matrix, matrix-vector multiplication, solving linear systems, computing eigenvalues and eigenvectors of a matrix; 2) calculus of functions of single variable: concepts of derivative of a function, finite difference approximation, integral of a function, quadrature, ordinary differential equations; 3) calculus of functions of more variables: concepts of partial derivative of a function, surface and line integral, quadrature, partial differential equations; 4) electric and magnetic fields: concepts of electric field, magnetic field, Maxwell equations, electrical machines and transformers;	
Course Contents	This course consists of the following three blocks: 1) Block 1 (Week 1 and Week 2): In this block, the finite difference method for the Poisson equation on the interval (Week 1) and the square (Week 2) will be discussed. The differential equation and the boundary conditions will be discretized on a mesh by a finite difference scheme. This will result in linear equations for the grid unknowns. A stencil notation for these linear equations will be introduced. A linear system will be formed and solved. The discrete solution will be visualized on the mesh. Properties of the discretization method and the linear system will be discussed. 2) Block 2 (Week 3 and Week 4): In this block, the Galerkin finite element method for the one-dimensional (Week 3) and two-dimensional (Week 4) Poisson problem will be introduced. The problem in so-called strong form will be transformed into the problem in so-called weak or variational form. The problem in weak form will then be discretized in space using Lagrangian basis or shape functions. A linear system represents the Poisson equation on a discrete level will be assembled and solved. The finite difference and finite element solution methods will be compared. 3) Block 3 (Week 5, Week 6 and Week 7): In this block, the application of the finite element will be extended to treat more complex problems. Partial differential equations for the scalar and vector potential for the magnetic flux or field will be derived from the Maxwell equations. More complex geometries that represent transformers, actuators or rotating extrical machines will be meshed using unstructured triangular meshes. The Maxwell equations will be discretized on these meshes. The discrete linear system will be assembled by a element-by-element procedure and solved. The computed vector potential solution will be processed to compute the magnetic flux and the magnetic field.	
Study Goals	After successfully completing the course, the student will be able to: * formulate various boundary value problems for the one-dimensional and two-dimensional Poisson equation; * derive these boundary value problems from the Maxwell equations for the magnetic field by introducing the scalar and vector potential; * apply the finite difference and finite element method to solve these boundary value problems on a scale of problems of increasing complexity; * state properties of the finite difference and finite element method applied to Poisson boundary value problems; * describe the implementation of the finite difference and the finite element method in software; * give a physical interpretation of the numerical solution obtained for magnetic field problems;	
Education Method	Weekly lectures will be complemented by weekly practical lab sessions and three home work assignments. During the lab sessions, algorithms discussed in the lectures will be implemented, tested and further refined. Each homework assignment will correspond to a course block. The set of three assignment will form a portfolio.	
Reader	see http://ta.twi.tudelft.nl/nw/users/domenico/electro_fem/electro_fem.html	
Assessment	During the course, the student will build a portfolio of three home work assignments, each correspond to a block in the course. The objective of these assignments is to guide the students towards the implementation of a proper finite element code. Each assignment consist of an elective and a compulsory part. This portfolio will be assessed during the oral examination. During this examination, the student will be assessed on the theoretical foundation of finite difference and finite element method, on the application of these methods and their implementation in software. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4515	Advanced Power Electronics	4
Responsible Instructor	Dr. Z. Qin	
Instructor	Dr. T. Batista Soeiro	
Contact Hours / Week x/x/x/x	0/0/0/3 + lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is expected that the student has taken all the previous courses in the power electronics teaching line: ET4119, ET4116, ET4145	
Course Contents	The focus of this course is on providing the theoretical background of various design related aspects of power converters not covered in the previous courses. The topics covered include: switching behavior and loss modelling, soft switching vs hard switching, passive components, thermal management and control. The lab projects integrate the knowledge from this course and the preceding courses in a design assignment of designing, building and testing an efficient and small power converter.	
Study Goals	After this course, the students will be able to: - design a transformer/inductor - design the control - design the cooling - analyse and improve the efficiency - design, build, and test a converter with small size and high efficiency	
Education Method	Lectures and lab project	
Literature and Study Materials	Lecture handouts N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition.	
Assessment	projects (80%) and oral exam (20%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4755	Reliable Power Electronic Components and Systems	5
Responsible Instructor	A. Shekhar	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Following this course is only possible with course ET4119 or SET3095 in advance	
Course Contents	Understanding of Power electronic components: diodes, BJT's, MOSFET's, thyristors, IGBT's; Gate drive circuits; High frequency inductors; Semiconductor's thermal management and heatsinks. The course also deals with conducted electromagnetic interference in power electronic circuits. Finally, the course will discuss component reliability and its corresponding influence on the power electronic systems	
Study Goals	THEORY: Understanding the theoretical and practical operation of power electronics components (semiconductors and passives) based on the physical structure, the V-I characteristics, time and thermal behavior. PRACTICAL: Analysis of Boost-type Converter and the Design of components, including semiconductors, inductor and capacitor. The ability to perform (differential) voltage-, current and temperature measurements on these components via a remote controlled test setup and explain these measurements at the hand of theory. Students will be given a demo on reliability tests using power cycler in the PEARL lab and asked to discuss their observations on the obtained test result data.	
Education Method	Lectures, practicals	
Literature and Study Materials	TEXT BOOK: Mohan, Undeland and Robbins, Power Electronics, Converters, Applications and Design, ISBN:978-0-471-22693-2, 824 pages, 3rd edition, John Wiley & Sons, 2003. ADDITIONAL READING: Provided via Brightspace DATA SHEETS: Provided via Brightspace and hard-copy in the lab. HARDWARE: Provided in the lab sessions.	
Assessment	Written EXAM (40%) DESIGN REPORT of your inductors DUO PRESENTATION on design- and practical results (20%) FINAL REPORT on practical (40%) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalair - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteric modulation - carrier modulation with feedback - scalair and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Responsible Instructor	H. Vahedi	
Instructor	G.R. Chandra Mouli	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4121	A.C. Machines	4
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	Control of Electrical Drives (ET4291)	
Expected prior knowledge	Electrical machines and drives (ET4117)	
Course Contents	The following subjects are discussed: - winding function and inductances; - time and spatial harmonics; - skin effect in electrical machines; - phasor, space vector and transformation in electrical machines; - the salient pole synchronous machine; - modeling and simulation of the dynamic performance of DC and AC machines.	
Study Goals	After following this course the students should be able to - describe and calculate the steady-state performance of AC machines; - calculate spatial harmonics and inductances of given winding distributions; - derive the dynamic models of DC and AC machines using space vectors; - model and simulate the dynamic performance of DC and AC machines; - evaluate the limitations of the dynamic models.	
Education Method	Lecture / Computer Simulation / Discussion	
Literature and Study Materials	- A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill. - lecture notes on Brightspace	
Assessment	Oral exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	- Presence at the computer simulation session is obliged. - Submission of assignments on Brightspace is obliged.	

ET4291	Control of Electrical Drives	5
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	practicum 0/0/0/4 vanaf week 2, instructie 0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Course Contents	- Dynamic modelling of DC and AC Machines supplied with power electronic converters. - Modelling of mechanical systems and thermal effects in drives. - Modelling and simulation of digital control of electrical drives. - Controller design for cascaded motor controller. - Scalar and vector control, field weakening control. - Practical design of an electrical drive system.	
Study Goals	Students should be able to: - Model the dynamic behaviour of DC and AC machines using state space representation. - Integrate ideal power electronic converter models and cascade control systems in the drive model. - Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive. - Calculate the steady state performance of DC and AC drives. - Implement scalar control system and vector control system for induction machines. - Implement vector control system for synchronous machines. - Implement field weakening control for DC and AC drives. - Implement the dynamic simulation models in MATLAB/Simulink. - Explain the practical design of electrical drives. - Evaluate the performance of an electrical drive system. - Design the controllers in electrical drive.	
Education Method	Lecture / Self-study / Simulation practical / Discussion	
Literature and Study Materials	- Rik De Doncker, Duco W.J. Pule, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: https://www.springer.com/gp/book/9789400701816 - Self-study materials on Brightspace.	
Reader	"Reader Control of Electrical Drives ET4291 Version maart 2009"	
Prerequisites	Electrical Machines and drives (ET4117) Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Assessment	Individual assignment and oral exam	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Smart AC Grids and DC Grids 2023

EE4536	DC and AC Microgrids	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	L.M. Ramirez Elizondo	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatziaargyriou	
Assessment	Written exam 70%, Assignments 30% In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTS) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

ET4108	Transients in Power Systems	4
Responsible Instructor	Prof.dr.ir. M. Popov	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	EE2E21, ET4107 - knowledge of symmetrical component is highly recommended	
Course Contents	This course deals with the rapid phenomena in power systems caused by switching actions and atmospheric discharges. Some of the topics covered are circuit breakers, arc-circuit interaction, short-line fault, interruption of small inductive capacitive currents, travelling waves, atmospheric discharges, simulation of transients, circuit breaker testing.	
Course Contents Continuation	The subject deals with power system transients. It is divided in a few sections the goals of which are the students to be introduced by the basic concepts of switching transients and transient analysis as well as travelling wave phenomena. The subject also deals with the current interruption phenomena and circuit breakers as well as circuit breaker testing and insulation coordination. Additional material on Ferroresonance and surge arrester selections is being studied as a supplement to this project. All results are validated by computer simulations in Electromagnetic Transient Program (EMTP), which is used as a tool for detailed analysis of different cases like switching small inductive and capacitive currents, arc interruption, thermal and dielectric breakdown of the circuit breakers, travelling wave phenomena. Each chapter is also supported by several simplified examples (which are solvable on the brightspace)	
Study Goals	After following this course the student should be able to: - model power system components for transient analysis by means of lumped elements. - to understand the current interruption process and gained insight in the testing of switchgear in the High Power Lab. - to use symmetrical components during transient processes in a network, especially when one pole of the circuit breaker opens and the other two remain in closed position. - to understand under which circumstances the worst case scenario (highest, steepest TRV) takes place. - to be able to assess the behavior of the lightning stroke when arriving and junction points connecting one or more components with different characteristic impedance Based on aforementioned learning objectives, the student should be able to simulate at least simplified cases of switching and lighting transient phenomena in Electromagnetic Transient Program ATP-EMTP or other EMTP-based software environment.	
Education Method	Lectures (*). Students should spent approximately ten hours per week in order to achieve the learning goals well. Each chapter, if suitable, is also supported by solvable examples on the board and simplified/detailed simulation cases performed in Electromagnetic Transient Program (EMTP) environment. Students also get opportunity to obtain the EMTP royalty free software and to learn EMTP so that they can use it as a tool for different project cases or M.Sc thesis. The subject also provides opportunity for extra projects in which students may obtain credits depending on the amount of work done. * The way of how the lectures are conducted and the study guide information may change depending on the developments around the corona virus.	
Literature and Study Materials	L. vander Sluis, Transients in Power System, Wiley, ISBN 0 471 486 396. JA Martinez-Velasco, M Popov, Circuit breakers: Power System Transients Parameter determination, CRC Press, ISBN-13: 978-1420065299, pp. 447-556.	
Assessment	Additional material on ferroresonance phenomena will be provided by the lecturer Oral (closed book examination) which can be taken as written in case of more than 10 students take the exam. Students get several questions and problems. The questions can be short questions or the student is asked to explain a process related to a particular transient case that occurs in practice. The amount of work required for particular problem solutions is in line with the number of points. The exam is passed when the student scores more than 57 %. Resit of the examination is possible according to the examination schedule. The content of the course material is divided into 9 chapters. All chapters weight equally. Hence, the exam may cover questions and solvable problems/exercises related to all chapters. The student passes when he/she obtains more than 57% of the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4113	Power System Dynamics	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Required: ET4107 Power System Analysis II (Master SET / Master EE / Master EWEM) SET3065 Intelligent Electrical Power Grids (Master SET / Master EE / Master EWEM) Optional: ET4116 Power Electronics (Master SET / Master EE / Master EWEM) ET4117 Electrical Machines and Drives (Master SET / Master EE / Master EWEM)	
Parts	Theory lectures: Lecture 1: Introduction to power system stability and synchronous generator Lecture 2: Controllers of the synchronous generator Lecture 3: The voltage source converter Lecture 4: Controllers of the voltage source converter Lecture 5: Applications of voltage source converter in electrical power systems Lecture 6: Rotor angle stability: transient stability and small-signal stability Lecture 7: Small-disturbance (small-signal) rotor angle stability (measurement based methods) Lab sessions: Session 1: Tutorial for PowerFactory (Single-Machine Infinite Bus system) Session 2: Modelling of an interconnected system (steady-state performance) Session 3: Dynamic models - Part 1 Session 4: Dynamic models - Part 2 Session 5: Long-term voltage stability assessment (including tutorial for Python) Session 6: Rotor angle stability assessment Session 7: Sensitivity analysis	
Course Contents	The course "Power Systems Dynamics" is devoted to the study behaviour of electrical power systems in non-steady state. Knowledge of the dynamic behaviour of electrical power systems and their components is important in order to predict whether the system (or a part thereof) remains in a stable state after a disturbance. The time frame of observation ranges from milliseconds to several minutes, depending on the kind of disturbance and the influence of applied controls. For each time frame, special models of the network components are derived. In this course, the model of the synchronous generator, the voltage source converter, and their associated control systems receive special attention due to their dominant influence on the dynamic behaviour of the power system. In the computer assignments, the theory is put into practice by means of software based simulations in DigSILENT PowerFactory as well as scripting in Python and Matlab. Subject to investigation are: modelling of synchronous generator and voltage source converter for RMS simulations, frequency and voltage controls, model-based and measurement-based eigenvalue analysis, wide-area monitoring, damping control, and RMS simulation of interconnected power systems.	
Study Goals	At the end of the course the students will be able to: - Describe the modeling aspects of power system components for computer aided simulation of electro-mechanical transients - Apply and interpret stability criteria for the analysis of systems dynamic behavior - Create a model of a power system in a simulation software package - Assess the system performance following typical disturbances	
Education Method	Lectures and computer assignments. Contact hours (8 weeks: 7 theory lectures, 7 lab sessions): 2 hours/theory lecture and 4 hours/lab session. Total: 42 hours. Self-study: 70 hours (9 hours/week).	
Computer Use	DigSILENT PowerFactory, Python, Matlab	
Literature and Study Materials	P. Kundur, Power System Stability and Control, Mc.Graw Hill, inc. 1994, ISBN-0-07-035958.	
Assessment	Lab report and final exam/re-exam including a theoretical questions component and a computer-based component. The computer-based part of the test requires familiarity with the models built during the lab assignments. The students are expected to run time-domain simulations and give insight into the observed phenomena. The final grade comprises: 1) Exam or re-exam (5 points). 2) Lab report and implementations (5 points). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Digital exam in the exam period of Q4. Duration: 3 hours. The resit is in the exam period of summer period. Duration: 3 hours.	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Study load: 112 hours.	
maximum aantal deelnemers	Limited to 25 students due to software availability	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalair - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteric modulation - carrier modulation with feedback - scalair and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Responsible Instructor	H. Vahedi	
Instructor	G.R. Chandra Mouli	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to LO1: Describe the functions and topologies of cyber-physical energy systems LO2: Implement data driven modelling and numerical simulation of energy systems LO3: Calculate variable power flows based on optimization/probability theory LO4: Synthesize functional requirements of energy management systems LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1: The energy transition and future energy systems (Peter) 2: Analysis of steady-state performance in variable operating conditions (Jose) 3: Assessment and control of power system dynamic performance (Jose) 4: Lab 1 Digital and data driven modelling of interconnected systems 5: Lab 2 Simulation and impact analysis of renewables, storage, and demand 6: Data acquisition and communication for energy automation (Peter) 7: Artificial intelligence-based functionalities for situational awareness (Pedro & Jochen) 8: Cyber security of smart grids (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The final grade of the course consists of the following components: - Assignment reports (in time!) (50%) - Multiple choice Computer Exam (50%) In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Final grade calculation: $(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Assignment reports (in time!): resubmit deliverable - Multiple choice Computer Exam: computer resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Free electives 2023

EE5015	Internship EWEM	6
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	x/x/x/x	
Start Education	1	
Course Language	English	
Expected prior knowledge	A general guideline for the start of the Internship is: the studies are finished apart from the MSc-project and maybe one or two courses. When this condition is met the Internship Officer is able to declare that the outgoing student will practically work on the level of an engineer. Hence, the company may expect something and the student can gain further experience at a rather high level.	
Course Contents	Master students of Electrical Engineering, EWEM can follow EE5015 as part of their free electives. The internship period should be 4 weeks (for which you will receive 6 credits). You can do a longer internship, but you will not receive additional credit points. The internship should be of sufficient level to be part of your MSc curriculum performed at a company or institute, either in the Netherlands or abroad. An internship in a university outside the Netherlands can be accepted. An internship at TU Delft will not be accepted. When carrying out Internship at a company or institute, the following conditions apply: The student joins in with the company or institute culture in general, including sharing the company-established balance between on-site and online work. The student joins in with the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...).	
Study Goals	There are three types of educational goals for the Internship: I. Profession related aspects Insight into the profession of the electrical engineer Insight into the duties and the responsibility of the engineer in a company II. Social-psychological aspects Learning to move in another culture (both within and outside the company) Dealing with another language during a longer period Gaining insight into the position of the company with respect to comparable companies III. Professional knowledge and skills related aspects Learning to apply the gained knowledge and skills in a situation different from the (own) university situation Learning to acquire new knowledge and skills that are necessary to meet the requirements for the Internship.	
Education Method	Detailed information on projects, procedures and application is available on the Brightspace course EE5010. The internship application form should be submitted to the Internship Office prior (!) to starting Internship EE. This application form contains a project description that the student writes with input from the company supervisor and a TU Delft supervisor. A completed internship application form, submitted to the EEMCS internship office, indicates that you can do the internship for study credit. Please, do not forget to include it in your IEP to prevent graduation delay.	
Literature and Study Materials	See: Internship regulations: programme Electrical Engineering, Embedded Systems and Computer Engineering. Enrol in Brightspace course EE5010 for more information about the Internship. You can find the step-by-step guide explaining all procedures, forms and internship offers. For additional information e.g. about grants and funding for internships abroad enrol in Brightspace course Study Abroad EEMCS.	
Prerequisites	A general guideline for the start of the Internship is: the studies are finished apart from the MSc-project and maybe one or two courses. When this condition is met the Internship Officer is able to declare that the outgoing student will practically work on the level of an engineer. Hence, the company may expect something and the student can gain further experience at a rather high level.	
Assessment	Pass/Fail (no grade) For the assessment you need to submit: The company assessment form The TU Delft assessment form An internship report. You can find guidelines for the report in the student manual on Brightspace. Your company supervisor needs to assess your internship at the end of your internship period, but within three months after the start of the internship, and send his assessment form to the TU Delft supervisor. Your TU Delft supervisor needs to assess your internship (within two weeks) after the end of your internship. The TU Delft supervisor will then send the assessment form (including the company supervisor review form) to Internship Office EEMCS. The review form can be found on Brightspace EE5010. The supervisors have to email the completed form to internship-eemcs@tudelft.nl. The assessment is based on the Company supervisor Assessment Form, the TU Delft supervisor Assessment Form and the Internship Report. The documents have to be submitted to the Internship Officer within two months after finishing the Internship. Within one month after submitting the different documents, the student will be informed about the credit points that are granted. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will self-study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final exam + a speaking test. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

Year	2023/2024
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Electrical Engineering

Thesis project 2023

Introduction 1

This is the thesis project, this being the concluding study unit of the program. The project amounts to 45 EC and involves a research or design task with sufficient academic level. The project may be executed within a research program at TU Delft, or in a suitable research institute or company. The project must be carried out with a systematic approach and should include all phases of a research or design project: analysis, modelling, implementation/construction and validation/evaluation. The student executes the thesis project independently, with guidance of a thesis supervisor and under the responsibility of the full professor who approved the IEP.

ET4300	Master Thesis	45
Responsible Instructor	Prof. M. Alavi	
Responsible Instructor	Dr. D. Cavallo	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	M. Babaie	
Responsible Instructor	Dr. A. Lekic	
Responsible Instructor	M.A. Kitsak	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MScEE and MScCE programs. It is an individual in-depth research or expert design assignment with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. The thesis project may be executed within a research program at TU Delft, or in a suitable research institute or company. The nominal duration of a thesis project is 30 weeks (45 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. With regard to the thesis project, the thesis supervisor may impose additional conditions on the IEP to be met before starting the thesis project. If desired, an internship or an extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in scope and a separate deliverable such that the thesis project can be clearly evaluated in isolation. The thesis project is governed by the EEMCS Graduation Policy. Below you find a short description of the different steps to be taken during the thesis work: 1. Orientation 2. Kick-off meeting 3. Midterm meeting 4. Green-light review 5. Graduation Day (defence, assessment and graduation ceremony)	
Course Contents Continuation	Declaration of Credits Obtained Before you can start your Masters Thesis Project you need to have a by the Board of Examiners approved Individual Exam Programme (IEP) and at least 60 credits of the Master's degree course work as stated in the IEP should be completed. You can prove this by a form called the Declaration of Credits Obtained (Verklaring behaalde studiepunten) (60 EC). You can request this form from the service desk in the EEMCS building 36 (Mekelweg 4) at the servicedesk with your campuscard or send an email to servicepuntEWI@tudelft.nl (use your with your TU Delft e-mail!) You have to show this form to your thesis advisor.	
Study Goals	The master thesis project serves to prove that the student has acquired the academic competencies of a Master of Science. The thesis project is an individual in-depth research or expert design assignment independently carried out by the student with a systematic approach including all phases of a research or design project: - problem formulation and planning of the work - analysis, - modeling, - implementation/construction, - validation/evaluation, - presentation of the obtained results, orally and in writing. The achievements and resultant thesis report should comprise original work carried out by the student.	
Education Method	Orientation After having obtained a DCO, the thesis work starts with a thesis proposal, which is defined together with the supervisor. The proposal, of at most 2 pages A4, is written according to the format which can be downloaded from the Brightspace page of ET4300. The thesis proposal contains: - a working title, - a preliminary problem definition, - intended outcomes of the thesis project, - intended milestones, - expected duration and (preliminary) date of the thesis defense. The (preliminary) thesis committee is also indicated on this proposal. If desired, an internship or extra project can be used as a preparation for the thesis project. If so, there should be a clear separation in activity and a separate deliverable such that the thesis project can be clearly evaluated in isolation. This is indicated in the thesis proposal. The thesis related work done during an internship or extra project should be described in the thesis report (typically one chapter). Kick-off meeting The Orientation phase indicated above results in a thesis proposal written by the student. The objective of the kick-off meeting is the formal approval of this thesis proposal by the supervisor(s). In this meeting, the research question(s), research methodology and planning are presented by the student and discussed with the supervisor(s). After approval, the "Thesis Proposal form" is filled out. Midterm meeting After 3-4 months, the student presents (by an oral presentation and a written report) the work performed so far to the supervisor(s). On the basis of the work done, it is decided whether the goals, as formulated in the thesis proposal, are still feasible. If not, they are reconsidered and adapted. After the mid-term meeting, the student has a clear idea of how to proceed in order to finish the thesis in time. Thesis committee This committee is responsible for the assessment of the thesis work and the thesis presentation. In order to be able to defend the thesis, the student must fill out the thesis committee form, which needs to be signed by the responsible full or associate professor. This form needs to be handed in to the Board of Examiners of the study programme at least 3 months prior to the MSc thesis presentation (the defence). Green-light review In the final phase of the thesis project, there will be a green-light review. During this meeting it is decided whether the student is	

ready to proceed with the thesis defence, resulting in a "go/no-go". It is compulsory to have a (draft) thesis ready for the green-light review. In case of a "go", the "Final Examination form" is handed in, which will trigger a check on the fulfillment of all requirements for getting the MScEE or MScCE diploma.

Graduation day

The final step of the graduation process is the presentation and defence of the thesis work. The candidate may present and defend the thesis work only after having finished all courses as stated in the approved IEP. The defence consists of two parts. First, the work is presented in a public session in front of the thesis committee, family, friends and fellow students. The presentation is followed by a closed session where the candidate will defend the work in front of the thesis committee. Immediately following the assessment, provided that there is a positive outcome, the diploma ceremony starts and the diploma will be handed out.

Assessment

The thesis project is assessed by the thesis committee based on the actual thesis work carried out, the final report, the public presentation, and the closed defense session and based on the learning objectives of the thesis. The items assessed in the final grading are:

Quality of the work: research method/design, application/development of theory, interpretation of results, scientific significance;

Performance: critical attitude, creativity, initiative, interaction with peers/superiors, planning;

Report: content, structure, quality of writing, independence in writing;

Presentation & defense: content, form, performance, scientific discussion.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

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